General Aptitude (GA)

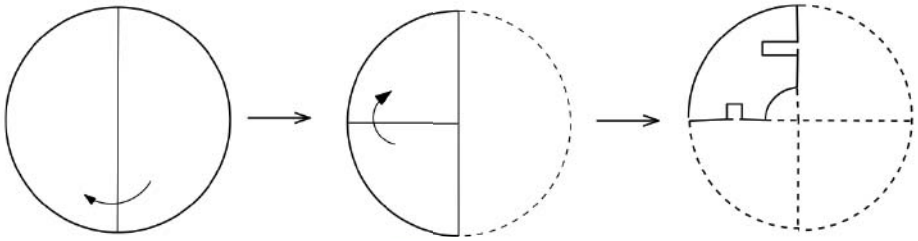
Q.1 – Q.5 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

Q.1	The ratio of boys to girls in a class is 7 to 3. Among the options below, an acceptable value for the total number of students in the class is:
(A)	21
(B)	37
(C)	50
(D)	73

Q.2	A polygon is convex if, for every pair of points, P and Q belonging to the polygon, the line segment PQ lies completely inside or on the polygon. Which one of the following is NOT a convex polygon?
(A)	
(B)	
(C)	
(D)	

Q.3	Consider the following sentences: (i) Everybody in the class is prepared for the exam. (ii) Babu invited Danish to his home because he enjoys playing chess. Which of the following is the CORRECT observation about the above two sentences?
(A)	(i) is grammatically correct and (ii) is unambiguous
(B)	(i) is grammatically incorrect and (ii) is unambiguous
(C)	(i) is grammatically correct and (ii) is ambiguous
(D)	(i) is grammatically incorrect and (ii) is ambiguous

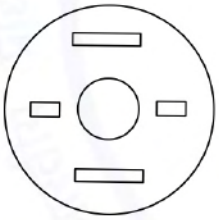
Q.4



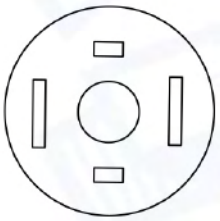
A circular sheet of paper is folded along the lines in the directions shown. The paper, after being punched in the final folded state as shown and unfolded in the reverse order of folding, will look like _____.

(A)

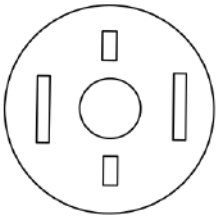
(B)



(C)



(D)



Q.5	_____ is to *surgery* as *writer* is to _____ Which one of the following options maintains a similar logical relation in the above sentence?
(A)	Plan, outline
(B)	Hospital, library
(C)	Doctor, book
(D)	Medicine, grammar

Q. 6 – Q. 10 Multiple Choice Question (MCQ), carry TWO marks each (for each wrong answer: – 2/3).

Q.6	We have 2 rectangular sheets of paper, M and N, of dimensions 6 cm x 1 cm each. Sheet M is rolled to form an open cylinder by bringing the short edges of the sheet together. Sheet N is cut into equal square patches and assembled to form the largest possible closed cube. Assuming the ends of the cylinder are closed, the ratio of the volume of the cylinder to that of the cube is _____
(A)	$\frac{\pi}{2}$
(B)	$\frac{3}{\pi}$
(C)	$\frac{9}{\pi}$
(D)	3π

Q.7

Items	Cost (₹)	Profit %	Marked Price (₹)
P	5,400	---	5,860
Q	---	25	10,000

Details of prices of two items P and Q are presented in the above table. The ratio of cost of item P to cost of item Q is 3:4. Discount is calculated as the difference between the marked price and the selling price. The profit percentage is calculated as the ratio of the difference between selling price and cost, to the cost $\left(\text{Profit } \% = \frac{\text{Selling price} - \text{Cost}}{\text{Cost}} \times 100\right)$.

The discount on item Q, as a percentage of its marked price, is _____

(A) 25

(B) 12.5

(C) 10

(D) 5

Q.8	There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag. The probability that at least two chocolates are identical is _____
(A)	0.3024
(B)	0.4235
(C)	0.6976
(D)	0.8125

Q.9	Given below are two statements 1 and 2, and two conclusions I and II. Statement 1: All bacteria are microorganisms. Statement 2: All pathogens are microorganisms. Conclusion I: Some pathogens are bacteria. Conclusion II: All pathogens are not bacteria. Based on the above statements and conclusions, which one of the following options is logically CORRECT?
(A)	Only conclusion I is correct
(B)	Only conclusion II is correct
(C)	Either conclusion I or II is correct.
(D)	Neither conclusion I nor II is correct.

Q.10	Some people suggest anti-obesity measures (AOM) such as displaying calorie information in restaurant menus. Such measures sidestep addressing the core problems that cause obesity: poverty and income inequality. Which one of the following statements summarizes the passage?
(A)	The proposed AOM addresses the core problems that cause obesity.
(B)	If obesity reduces, poverty will naturally reduce, since obesity causes poverty.
(C)	AOM are addressing the core problems and are likely to succeed.
(D)	AOM are addressing the problem superficially.

Computer Science and Information Technology (CS, Set-1)

Q.1 – Q.10 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

Q.1	Suppose that L_1 is a regular language and L_2 is a context-free language. Which one of the following languages is NOT necessarily context-free?
(A)	$L_1 \cap L_2$
(B)	$L_1 \cdot L_2$
(C)	$L_1 - L_2$
(D)	$L_1 \cup L_2$

Q.2	Let P be an array containing n integers. Let t be the lowest upper bound on the number of comparisons of the array elements, required to find the minimum and maximum values in an arbitrary array of n elements. Which one of the following choices is correct?
(A)	$t > 2n - 2$
(B)	$t > 3\left\lceil \frac{n}{2} \right\rceil$ and $t \leq 2n - 2$
(C)	$t > n$ and $t \leq 3\left\lceil \frac{n}{2} \right\rceil$
(D)	$t > \lceil \log_2(n) \rceil$ and $t \leq n$

Q.3	Consider the following three functions. $f_1 = 10^n \quad f_2 = n^{\log n} \quad f_3 = n^{\sqrt{n}}$ Which one of the following options arranges the functions in the increasing order of asymptotic growth rate?
(A)	f_3, f_2, f_1
(B)	f_2, f_1, f_3
(C)	f_1, f_2, f_3
(D)	f_2, f_3, f_1

Q.4	Consider the following statements. S_1 : The sequence of procedure calls corresponds to a preorder traversal of the activation tree. S_2 : The sequence of procedure returns corresponds to a postorder traversal of the activation tree. Which one of the following options is correct?
(A)	S_1 is true and S_2 is false
(B)	S_1 is false and S_2 is true
(C)	S_1 is true and S_2 is true
(D)	S_1 is false and S_2 is false

Q.5	Consider the following statements. S_1 : Every SLR(1) grammar is unambiguous but there are certain unambiguous grammars that are not SLR(1). S_2 : For any context-free grammar, there is a parser that takes at most $O(n^3)$ time to parse a string of length n. Which one of the following options is correct?
(A)	S_1 is true and S_2 is false
(B)	S_1 is false and S_2 is true
(C)	S_1 is true and S_2 is true
(D)	S_1 is false and S_2 is false

Q.6	Let the representation of a number in base 3 be 210. What is the hexadecimal representation of the number?
(A)	15
(B)	21
(C)	D2
(D)	528

Q.7	Let p and q be two propositions. Consider the following two formulae in propositional logic. S_1 : $(\neg p \wedge (p \vee q)) \rightarrow q$ S_2 : $q \rightarrow (\neg p \wedge (p \vee q))$ Which one of the following choices is correct?
(A)	Both S_1 and S_2 are tautologies.
(B)	S_1 is a tautology but S_2 is not a tautology.
(C)	S_1 is not a tautology but S_2 is a tautology.
(D)	Neither S_1 nor S_2 is a tautology.

Q.8	Consider the following two statements. S_1 : Destination MAC address of an ARP reply is a broadcast address. S_2 : Destination MAC address of an ARP request is a broadcast address. Which one of the following choices is correct?
(A)	Both S_1 and S_2 are true.
(B)	S_1 is true and S_2 is false.
(C)	S_1 is false and S_2 is true.
(D)	Both S_1 and S_2 are false.

Q.9	Consider the following array. 23 | 32 | 45 | 69 | 72 | 73 | 89 | 97 Which algorithm out of the following options uses the least number of comparisons (among the array elements) to sort the above array in ascending order?
(A)	Selection sort
(B)	Mergesort
(C)	Insertion sort
(D)	Quicksort using the last element as pivot

Q.10	A binary search tree T contains n distinct elements. What is the time complexity of picking an element in T that is smaller than the maximum element in T?
(A)	$\Theta(n \log n)$
(B)	$\Theta(n)$
(C)	$\Theta(\log n)$
(D)	$\Theta(1)$

Q.11 – Q.15 Multiple Select Question (MSQ), carry ONE mark each (no negative marks).

Q.11	In the context of operating systems, which of the following statements is/are correct with respect to paging?
(A)	Paging helps solve the issue of external fragmentation.
(B)	Page size has no impact on internal fragmentation.
(C)	Paging incurs memory overheads.
(D)	Multi-level paging is necessary to support pages of different sizes.

Q.12	Let $\langle M \rangle$ denote an encoding of an automaton M. Suppose that $\Sigma = \{0, 1\}$. Which of the following languages is/are NOT recursive?
(A)	$L = \{\langle M \rangle \mid$ M is a DFA such that $L(M) = \emptyset\}$
(B)	$L = \{\langle M \rangle \mid$ M is a DFA such that $L(M) = \Sigma^*\}$
(C)	$L = \{\langle M \rangle \mid$ M is a PDA such that $L(M) = \emptyset\}$
(D)	$L = \{\langle M \rangle \mid$ M is a PDA such that $L(M) = \Sigma^*\}$

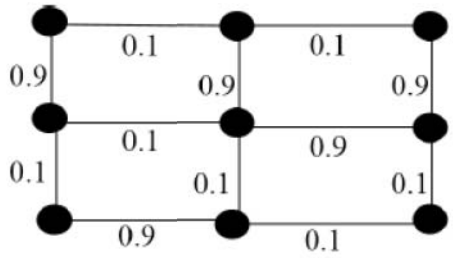
Q.13	Suppose a database system crashes again while recovering from a previous crash. Assume checkpointing is not done by the database either during the transactions or during recovery. Which of the following statements is/are correct?
(A)	The same undo and redo list will be used while recovering again.
(B)	The system cannot recover any further.
(C)	All the transactions that are already undone and redone will not be recovered again.
(D)	The database will become inconsistent.

Q.14	Which of the following standard C library functions will *always* invoke a system call when executed from a single-threaded process in a UNIX/Linux operating system?
(A)	`exit`
(B)	`malloc`
(C)	`sleep`
(D)	`strlen`

Q.15	Consider a linear list based directory implementation in a file system. Each directory is a list of nodes, where each node contains the file name along with the file metadata, such as the list of pointers to the data blocks. Consider a given directory `foo`. Which of the following operations will necessarily require a full scan of `foo` for successful completion?
(A)	Creation of a new file in `foo`
(B)	Deletion of an existing file from `foo`
(C)	Renaming of an existing file in `foo`
(D)	Opening of an existing file in `foo`

Q.16 – Q.25 Numerical Answer Type (NAT), carry ONE mark each (no negative marks).

Q.16	In an undirected connected planar graph G, there are eight vertices and five faces. The number of edges in G is _____.

Q.17	Consider the following undirected graph with edge weights as shown:  The number of minimum-weight spanning trees of the graph is _____.

Q.18	The lifetime of a component of a certain type is a random variable whose probability density function is exponentially distributed with parameter 2. For a randomly picked component of this type, the probability that its lifetime exceeds the expected lifetime (rounded to 2 decimal places) is _____ .

Q.19	There are 6 jobs with distinct difficulty levels, and 3 computers with distinct processing speeds. Each job is assigned to a computer such that: - The fastest computer gets the toughest job and the slowest computer gets the easiest job. - Every computer gets at least one job. The number of ways in which this can be done is _____.

Q.20	Consider the following expression. $$\lim_{x \to -3} \frac{\sqrt{2x+22}-4}{x+3}$$ The value of the above expression (rounded to 2 decimal places) is _____.

Q.21	Consider the following sequence of operations on an empty stack.
	push(54); push(52); pop(); push(55); push(62); s = pop();
	Consider the following sequence of operations on an empty queue.
	enqueue(21); enqueue(24); dequeue(); enqueue(28); enqueue(32); q = dequeue();
	The value of s + q is _____.

Q.22	Consider a computer system with a byte-addressable primary memory of size 2^{32} bytes. Assume the computer system has a direct-mapped cache of size 32 KB (1 KB = 2^{10} bytes), and each cache block is of size 64 bytes. The size of the tag field is _____ bits.

Q.23	A relation $r(A, B)$ in a relational database has 1200 tuples. The attribute A has integer values ranging from 6 to 20, and the attribute B has integer values ranging from 1 to 20. Assume that the attributes A and B are independently distributed. The estimated number of tuples in the output of $\sigma_{(A>10)\vee(B=18)}(r)$ is _____.

Q.24	Consider the following representation of a number in IEEE 754 single-precision floating point format with a bias of 127. $S : 1 \qquad E : 10000001 \qquad F : 11110000000000000000000$ Here S, E and F denote the sign, exponent and fraction components of the floating point representation. The decimal value corresponding to the above representation (rounded to 2 decimal places) is _____.

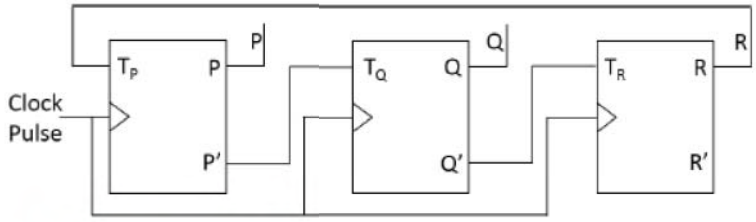
Q.25	Three processes arrive at time zero with CPU bursts of 16, 20 and 10 milliseconds. If the scheduler has prior knowledge about the length of the CPU bursts, the minimum achievable average waiting time for these three processes in a non-preemptive scheduler (rounded to nearest integer) is _____ milliseconds.

Q.26 – Q.39 Multiple Choice Question (MCQ), carry TWO mark each (for each wrong answer: – 2/3).

Q.26	Consider the following grammar (that admits a series of declarations, followed by expressions) and the associated syntax directed translation (SDT) actions, given as pseudo-code: $P \rightarrow D^* E^*$ $D \rightarrow$ `int ID` {record that ID.lexeme is of type `int`} $D \rightarrow$ `bool ID` {record that ID.lexeme is of type `bool`} $E \rightarrow E_1 + E_2$ {check that E_1.type = E_2.type = `int`; set E.type := `int`} $E \rightarrow !E_1$ {check that E_1.type = `bool`; set E.type := `bool`} $E \rightarrow$ `ID` {set E.type := `int`} With respect to the above grammar, which one of the following choices is correct?
(A)	The actions can be used to correctly type-check any syntactically correct program.
(B)	The actions can be used to type-check syntactically correct integer variable declarations and integer expressions.
(C)	The actions can be used to type-check syntactically correct boolean variable declarations and boolean expressions.
(D)	The actions will lead to an infinite loop.

Q.27	The following relation records the age of 500 employees of a company, where *empNo* (indicating the employee number) is the key: $empAge(\underline{empNo}, age)$ Consider the following relational algebra expression: $\Pi_{empNo}(empAge \bowtie_{(age>age1)} \rho_{empNo1,\, age1}(empAge))$ What does the above expression generate?
(A)	Employee numbers of only those employees whose age is the maximum.
(B)	Employee numbers of only those employees whose age is more than the age of exactly one other employee.
(C)	Employee numbers of all employees whose age is not the minimum.
(D)	Employee numbers of all employees whose age is the minimum.

Q.28 Consider a 3-bit counter, designed using T flip-flops, as shown below:



Assuming the initial state of the counter given by PQR as 000, what are the next three states?

(A) 011, 101, 000

(B) 001, 010, 111

(C) 011, 101, 111

(D) 001, 010, 000

Q.29 Assume that a 12-bit Hamming codeword consisting of 8-bit data and 4 check bits is $d_8d_7d_6d_5c_8d_4d_4d_3d_2c_4d_1c_2c_1$, where the data bits and the check bits are given in the following tables:

Data bits							
d_8	d_7	d_6	d_5	d_4	d_3	d_2	d_1
1	1	0	x	0	1	0	1

Check bits			
c_8	c_4	c_2	c_1
y	0	1	0

Which one of the following choices gives the correct values of x and y?

(A) x is 0 and y is 0.

(B) x is 0 and y is 1.

(C) x is 1 and y is 0.

(D) x is 1 and y is 1.

Q.30	Consider the following recurrence relation. $$T(n) = \begin{cases} T(n/2) + T(2n/5) + 7n & \text{if } n > 0 \\ 1 & \text{if } n = 0 \end{cases}$$ Which one of the following options is correct?
(A)	$T(n) = \Theta(n^{5/2})$
(B)	$T(n) = \Theta(n \log n)$
(C)	$T(n) = \Theta(n)$
(D)	$T(n) = \Theta((\log n)^{5/2})$

Q.31 Consider the following context-free grammar where the set of terminals is $\{a, b, c, d, f\}$.

$$\mathrm{S} \rightarrow d\,a\,\mathrm{T} \mid \mathrm{R}\,f$$
$$\mathrm{T} \rightarrow a\,\mathrm{S} \mid b\,a\,\mathrm{T} \mid \epsilon$$
$$\mathrm{R} \rightarrow c\,a\,\mathrm{T}\,\mathrm{R} \mid \epsilon$$

The following is a partially-filled LL(1) parsing table.

	a	b	c	d	f	\$
S			(1)	$\mathrm{S} \rightarrow d\,a\,\mathrm{T}$	(2)	
T	$\mathrm{T} \rightarrow a\,\mathrm{S}$	$\mathrm{T} \rightarrow b\,a\,\mathrm{T}$	(3)		$\mathrm{T} \rightarrow \epsilon$	(4)
R			$\mathrm{R} \rightarrow c\,a\,\mathrm{TR}$		$\mathrm{R} \rightarrow \epsilon$	

Which one of the following choices represents the correct combination for the numbered cells in the parsing table ("blank" denotes that the corresponding cell is empty)?

(A) (1) $\mathrm{S} \rightarrow \mathrm{R}\,f$ (2) $\mathrm{S} \rightarrow \mathrm{R}\,f$ (3) $\mathrm{T} \rightarrow \epsilon$ (4) $\mathrm{T} \rightarrow \epsilon$

(B) (1) blank (2) $\mathrm{S} \rightarrow \mathrm{R}\,f$ (3) $\mathrm{T} \rightarrow \epsilon$ (4) $\mathrm{T} \rightarrow \epsilon$

(C) (1) $\mathrm{S} \rightarrow \mathrm{R}\,f$ (2) blank (3) blank (4) $\mathrm{T} \rightarrow \epsilon$

(D) (1) blank (2) $\mathrm{S} \rightarrow \mathrm{R}\,f$ (3) blank (4) blank

Q.32	Let $r_i(z)$ and $w_i(z)$ denote read and write operations respectively on a data item z by a transaction T_i. Consider the following two schedules. $S_1: \; r_1(x)\; r_1(y)\; r_2(x)\; r_2(y)\; w_2(y)\; w_1(x)$ $S_2: \; r_1(x)\; r_2(x)\; r_2(y)\; w_2(y)\; r_1(y)\; w_1(x)$ Which one of the following options is correct?
(A)	S_1 is conflict serializable, and S_2 is not conflict serializable.
(B)	S_1 is not conflict serializable, and S_2 is conflict serializable.
(C)	Both S_1 and S_2 are conflict serializable.
(D)	Neither S_1 nor S_2 is conflict serializable.

Q.33	Consider the relation $R(P, Q, S, T, X, Y, Z, W)$ with the following functional dependencies. $PQ \rightarrow X;\; P \rightarrow YX;\; Q \rightarrow Y;\; Y \rightarrow ZW$ Consider the decomposition of the relation R into the constituent relations according to the following two decomposition schemes. $D_1: \quad R = [(P, Q, S, T);\; (P, T, X);\; (Q, Y);\; (Y, Z, W)]$ $D_2: \quad R = [(P, Q, S);\; (T, X);\; (Q, Y);\; (Y, Z, W)]$ Which one of the following options is correct?
(A)	D_1 is a lossless decomposition, but D_2 is a lossy decomposition.
(B)	D_1 is a lossy decomposition, but D_2 is a lossless decomposition.
(C)	Both D_1 and D_2 are lossless decompositions.
(D)	Both D_1 and D_2 are lossy decompositions.

Q.34	Let G be a group of order 6, and H be a subgroup of G such that $1 < \|H\| < 6$. Which one of the following options is correct?
(A)	Both G and H are always cyclic.
(B)	G may not be cyclic, but H is always cyclic.
(C)	G is always cyclic, but H may not be cyclic.
(D)	Both G and H may not be cyclic.

Q.35	Consider the two statements. S_1 : There exist random variables X and Y such that $\left(\mathbb{E}\left[\left(X-\mathbb{E}(X)\right)\left(Y-\mathbb{E}(Y)\right)\right]\right)^2 > \texttt{Var}[X]\,\texttt{Var}[Y]$ S_2 : For all random variables X and Y, $\texttt{Cov}[X,Y] = \mathbb{E}\left[\left\|X-\mathbb{E}[X]\right\|\left\|Y-\mathbb{E}[Y]\right\|\right]$ Which one of the following choices is correct?
(A)	Both S_1 and S_2 are true.
(B)	S_1 is true, but S_2 is false.
(C)	S_1 is false, but S_2 is true.
(D)	Both S_1 and S_2 are false.

Q.36	Let $G = (V, E)$ be an undirected unweighted connected graph. The *diameter* of G is defined as: $$\text{diam}(G) = \max_{u,v \in V} \{\text{the length of shortest path between } u \text{ and } v\}$$ Let M be the adjacency matrix of G. Define graph G_2 on the same set of vertices with adjacency matrix N, where $$N_{ij} = \begin{cases} 1 & \text{if } M_{ij} > 0 \text{ or } P_{ij} > 0, \text{ where } P = M^2 \\ 0 & \text{otherwise} \end{cases}$$ Which one of the following statements is true?
(A)	$\text{diam}(G_2) \leq \lceil \text{diam}(G)/2 \rceil$
(B)	$\lceil \text{diam}(G)/2 \rceil < \text{diam}(G_2) < \text{diam}(G)$
(C)	$\text{diam}(G_2) = \text{diam}(G)$
(D)	$\text{diam}(G) < \text{diam}(G_2) \leq 2\ \text{diam}(G)$

Q.37 Consider the following ANSI C program.

```
#include <stdio.h>
int main()
{
    int i, j, count;
    count = 0;
    i = 0;
    for (j = -3; j <= 3; j++)
    {
        if ((j >= 0) && (i++))
            count = count + j;
    }
    count = count + i;
    printf("%d", count);
    return 0;
}
```

Which one of the following options is correct?

(A) The program will not compile successfully.

(B) The program will compile successfully and output 10 when executed.

(C) The program will compile successfully and output 8 when executed.

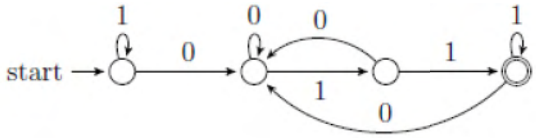
(D) The program will compile successfully and output 13 when executed.

Q.38 Consider the following language.

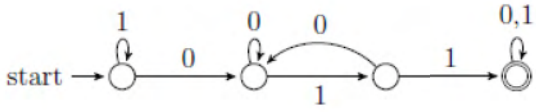
$$L = \{w \in \{0,1\}^* \mid w \text{ ends with the substring } 011\}$$

Which one of the following deterministic finite automata accepts L?

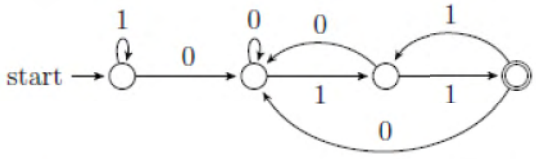
(A)



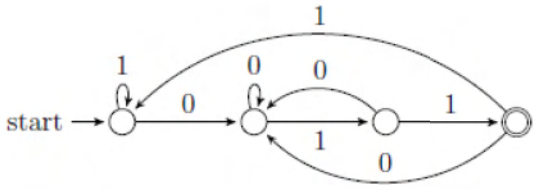
(B)



(C)



(D)



Q.39	For a Turing machine M, $\langle M \rangle$ denotes an encoding of M. Consider the following two languages. $L_1 = \{\langle M \rangle \mid M \text{ takes more than 2021 steps on all inputs}\}$ $L_2 = \{\langle M \rangle \mid M \text{ takes more than 2021 steps on some input}\}$ Which one of the following options is correct?
(A)	Both L_1 and L_2 are decidable.
(B)	L_1 is decidable and L_2 is undecidable.
(C)	L_1 is undecidable and L_2 is decidable.
(D)	Both L_1 and L_2 are undecidable.

Q.40 – Q.47 Multiple Select Question (MSQ), carry TWO mark each (no negative marks).

Q.40	Define R_n to be the maximum amount earned by cutting a rod of length n meters into one or more pieces of integer length and selling them. For `i > 0`, let `p[i]` denote the selling price of a rod whose length is `i` meters. Consider the array of prices: $\texttt{p[1]} = 1, \texttt{p[2]} = 5, \texttt{p[3]} = 8, \texttt{p[4]} = 9, \texttt{p[5]} = 10, \texttt{p[6]} = 17, \texttt{p[7]} = 18$ Which of the following statements is/are correct about R_7?
(A)	$R_7 = 18$
(B)	$R_7 = 19$
(C)	R_7 is achieved by three different solutions.
(D)	R_7 cannot be achieved by a solution consisting of three pieces.

Q.41	An *articulation point* in a connected graph is a vertex such that removing the vertex and its incident edges disconnects the graph into two or more connected components. Let T be a DFS tree obtained by doing DFS in a connected undirected graph G. Which of the following options is/are correct?
(A)	Root of T can never be an articulation point in G.
(B)	Root of T is an articulation point in G if and only if it has 2 or more children.
(C)	A leaf of T can be an articulation point in G.
(D)	If u is an articulation point in G such that x is an ancestor of u in T and y is a descendent of u in T, then all paths from x to y in G must pass through u.

Q.42	Consider the following Boolean expression. $$F = (X + Y + Z)(\overline{X} + Y)(\overline{Y} + Z)$$ Which of the following Boolean expressions is/are equivalent to $\overline{F}$ (complement of F)?
(A)	$(\overline{X} + \overline{Y} + \overline{Z})(X + \overline{Y})(Y + \overline{Z})$
(B)	$X\overline{Y} + \overline{Z}$
(C)	$(X + \overline{Z})(\overline{Y} + \overline{Z})$
(D)	$X\,\overline{Y} + Y\,\overline{Z} + \overline{X}\,\overline{Y}\,\overline{Z}$

Q.43	A relation R is said to be *circular* if aRb and bRc together imply cRa. Which of the following options is/are correct?
(A)	If a relation S is reflexive and symmetric, then S is an equivalence relation.
(B)	If a relation S is circular and symmetric, then S is an equivalence relation.
(C)	If a relation S is reflexive and circular, then S is an equivalence relation.
(D)	If a relation S is transitive and circular, then S is an equivalence relation.

Q.44	A TCP server application is programmed to listen on port number P on host S. A TCP client is connected to the TCP server over the network. Consider that while the TCP connection was active, the server machine S crashed and rebooted. Assume that the client does not use the TCP keepalive timer. Which of the following behaviors is/are possible?
(A)	If the client was waiting to receive a packet, it may wait indefinitely.
(B)	The TCP server application on S can listen on P after reboot.
(C)	If the client sends a packet after the server reboot, it will receive a RST segment.
(D)	If the client sends a packet after the server reboot, it will receive a FIN segment.

Q.45	Consider two hosts P and Q connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes, and between R and Q is 820 bytes. A TCP segment of size 1400 bytes was transferred from P to Q through R, with IP identification value as `0x1234`. Assume that the IP header size is 20 bytes. Further, the packet is allowed to be fragmented, i.e., `Don't Fragment (DF)` flag in the IP header is *not* set by P. Which of the following statements is/are correct?
(A)	Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes.
(B)	If the second fragment is lost, R will resend the fragment with the IP identification value `0x1234`.
(C)	If the second fragment is lost, P is required to resend the whole TCP segment.
(D)	TCP destination port can be determined by analysing *only* the second fragment.

Q.46 Consider the following pseudocode, where `S` is a semaphore initialized to 5 in line#2 and `counter` is a shared variable initialized to 0 in line#1. Assume that the increment operation in line#7 is *not* atomic.

```
1. int counter = 0;
2. Semaphore S = init(5);
3. void parop(void)
4. {
5.      wait(S);
6.      wait(S);
7.      counter++;
8.      signal(S);
9.      signal(S);
10.}
```

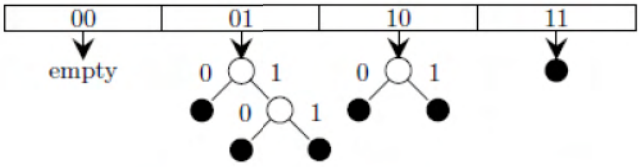
If five threads execute the function `parop` concurrently, which of the following program behavior(s) is/are possible?

(A) The value of `counter` is 5 after all the threads successfully complete the execution of `parop`.

(B) The value of `counter` is 1 after all the threads successfully complete the execution of `parop`.

(C) The value of `counter` is 0 after all the threads successfully complete the execution of `parop`.

(D) There is a deadlock involving all the threads.

Q.47	Consider a *dynamic* hashing approach for 4-bit integer keys: 1. There is a main hash table of size 4. 2. The 2 least significant bits of a key is used to index into the main hash table. 3. Initially, the main hash table entries are empty. 4. Thereafter, when more keys are hashed into it, to resolve collisions, the set of all keys corresponding to a main hash table entry is organized as a binary tree that grows on demand. 5. First, the 3rd least significant bit is used to divide the keys into left and right subtrees. 6. To resolve more collisions, each node of the binary tree is further sub-divided into left and right subtrees based on the 4th least significant bit. 7. A split is done only if it is needed, i.e., only when there is a collision. Consider the following state of the hash table.  Which of the following sequences of key insertions can cause the above state of the hash table (assume the keys are in decimal notation)?
(A)	5, 9, 4, 13, 10, 7
(B)	9, 5, 10, 6, 7, 1
(C)	10, 9, 6, 7, 5, 13
(D)	9, 5, 13, 6, 10, 14

Q.48 – Q.55 Numerical Answer Type (NAT), carry TWO mark each (no negative marks).

Q.48

Consider the following ANSI C function:

```
int SimpleFunction(int Y[], int n, int x)
{
  int total = Y[0], loopIndex;
  for (loopIndex = 1; loopIndex <= n - 1; loopIndex++)
          total = x * total + Y[loopIndex];
  return total;
}
```

Let Z be an array of 10 elements with Z[i]=1, for all i such that $0 \leq i \leq 9$. The value returned by SimpleFunction(Z, 10, 2) is _____.

Q.49

Consider the sliding window flow-control protocol operating between a sender and a receiver over a full-duplex error-free link. Assume the following:

- The time taken for processing the data frame by the receiver is negligible.
- The time taken for processing the acknowledgement frame by the sender is negligible.
- The sender has infinite number of frames available for transmission.
- The size of the data frame is 2,000 bits and the size of the acknowledgement frame is 10 bits.
- The link data rate in each direction is 1 Mbps ($= 10^6$ bits per second).
- One way propagation delay of the link is 100 milliseconds.

The minimum value of the sender's window size in terms of the number of frames, (rounded to the nearest integer) needed to achieve a link utilization of 50% is _____.

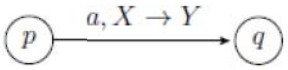
Q.50

Consider the following C code segment:

```
a = b + c;
e = a + 1;
d = b + c;
f = d + 1;
g = e + f;
```

In a compiler, this code segment is represented internally as a directed acyclic graph (DAG). The number of nodes in the DAG is _____.

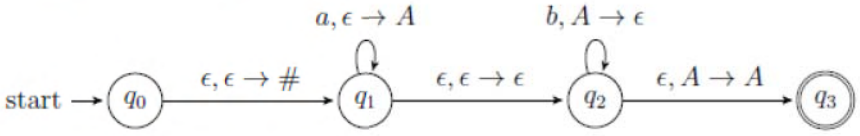
Q.51 In a pushdown automaton $P = (Q, \Sigma, \Gamma, \delta, q_0, F)$, a transition of the form,



where $p, q \in Q$, $a \in \Sigma \cup \{\epsilon\}$, and $X, Y \in \Gamma \cup \{\epsilon\}$, represents

$$(q, Y) \in \delta(p, a, X).$$

Consider the following pushdown automaton over the input alphabet $\Sigma = \{a, b\}$ and stack alphabet $\Gamma = \{\#, A\}$.



The number of strings of length 100 accepted by the above pushdown automaton is _____.

Q.52 Consider the following matrix.

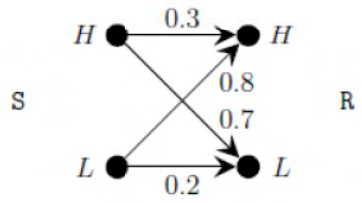
$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

The largest eigenvalue of the above matrix is _____.

Q.53 A five-stage pipeline has stage delays of 150, 120, 150, 160 and 140 nanoseconds. The registers that are used between the pipeline stages have a delay of 5 nanoseconds each.
The total time to execute 100 independent instructions on this pipeline, assuming there are no pipeline stalls, is _____ nanoseconds.

Q.54

A sender (S) transmits a signal, which can be one of the two kinds: H and L with probabilities 0.1 and 0.9 respectively, to a receiver (R).
In the graph below, the weight of edge (u, v) is the probability of receiving v when u is transmitted, where $u, v \in \{H, L\}$. For example, the probability that the received signal is L given the transmitted signal was H, is 0.7.



If the received signal is H, the probability that the transmitted signal was H (rounded to 2 decimal places) is _____.

Q.55

Consider the following instruction sequence where registers R1, R2 and R3 are general purpose and MEMORY[X] denotes the content at the memory location X.

Instruction	Semantics	Instruction Size (bytes)
MOV R1, (5000)	R1 ← MEMORY[5000]	4
MOV R2, (R3)	R2 ← MEMORY[R3]	4
ADD R2, R1	R2 ← R1+R2	2
MOV (R3), R2	MEMORY[R3] ← R2	4
INC R3	R3 ← R3+1	2
DEC R1	R1 ← R1-1	2
BNZ 1004	Branch if not zero to the given absolute address	2
HALT	Stop	1

Assume that the content of the memory location 5000 is 10, and the content of the register R3 is 3000. The content of each of the memory locations from 3000 to 3010 is 50. The instruction sequence starts from the memory location 1000. All the numbers are in decimal format. Assume that the memory is byte addressable.

After the execution of the program, the content of memory location 3010 is _____.

END OF QUESTION PAPER

GATE 2021 Answer Key for Computer Science and Information Technology (CS - 1)

Graduate Aptitude Test in Engineering (GATE 2021)

Subject/Paper: Computer Science and Information Technology (CS - 1)

Q. No.	Session	Question Type MCQ/MSQ/NAT	Section Name	Answer Key/Range	Marks	Negative Marks
1	5	MCQ	GA	C	1	1/3
2	5	MCQ	GA	A	1	1/3
3	5	MCQ	GA	C	1	1/3
4	5	MCQ	GA	A	1	1/3
5	5	MCQ	GA	C	1	1/3
6	5	MCQ	GA	C	2	2/3
7	5	MCQ	GA	C	2	2/3
8	5	MCQ	GA	C	2	2/3
9	5	MCQ	GA	C OR D	2	2/3
10	5	MCQ	GA	D	2	2/3
1	5	MCQ	CS	C	1	1/3
2	5	MCQ	CS	C	1	1/3
3	5	MCQ	CS	D	1	1/3
4	5	MCQ	CS	C	1	1/3
5	5	MCQ	CS	C	1	1/3
6	5	MCQ	CS	A	1	1/3
7	5	MCQ	CS	B	1	1/3
8	5	MCQ	CS	C	1	1/3
9	5	MCQ	CS	C	1	1/3
10	5	MCQ	CS	D	1	1/3
11	5	MSQ	CS	A; C	1	0

GATE 2021 Answer Key for Computer Science and Information Technology (CS - 1)

Q. No.	Session	Question Type MCQ/MSQ/NAT	Section Name	Answer Key/Range	Marks	Negative Marks
12	5	MSQ	CS	D	1	0
13	5	MSQ	CS	A	1	0
14	5	MSQ	CS	A; C	1	0
15	5	MSQ	CS	A; C	1	0
16	5	NAT	CS	11 to 11	1	0
17	5	NAT	CS	3 to 3	1	0
18	5	NAT	CS	0.35 to 0.39	1	0
19	5	NAT	CS	65 to 65	1	0
20	5	NAT	CS	0.25 to 0.25	1	0
21	5	NAT	CS	86 to 86	1	0
22	5	NAT	CS	17 to 17	1	0
23	5	NAT	CS	819 to 820 OR 205 to 205	1	0
24	5	NAT	CS	-7.75 to -7.75	1	0
25	5	NAT	CS	12 to 12	1	0
26	5	MCQ	CS	B	2	2/3
27	5	MCQ	CS	C	2	2/3
28	5	MCQ	CS	A	2	2/3
29	5	MCQ	CS	A	2	2/3
30	5	MCQ	CS	C	2	2/3
31	5	MCQ	CS	A	2	2/3
32	5	MCQ	CS	B	2	2/3
33	5	MCQ	CS	A	2	2/3
34	5	MCQ	CS	B	2	2/3

GATE 2021 Answer Key for Computer Science and Information Technology (CS - 1)

Q. No.	Session	Question Type MCQ/MSQ/NAT	Section Name	Answer Key/Range	Marks	Negative Marks
35	5	MCQ	CS	D	2	2/3
36	5	MCQ	CS	A	2	2/3
37	5	MCQ	CS	B	2	2/3
38	5	MCQ	CS	D	2	2/3
39	5	MCQ	CS	A	2	2/3
40	5	MSQ	CS	A; C	2	0
41	5	MSQ	CS	B	2	0
42	5	MSQ	CS	B; C; D	2	0
43	5	MSQ	CS	C	2	0
44	5	MSQ	CS	A; B; C	2	0
45	5	MSQ	CS	A; C	2	0
46	5	MSQ	CS	A; B; D	2	0
47	5	MSQ	CS	C	2	0
48	5	NAT	CS	1023 to 1023	2	0
49	5	NAT	CS	50 to 52	2	0
50	5	NAT	CS	6 to 6	2	0
51	5	NAT	CS	50 to 50	2	0
52	5	NAT	CS	3 to 3	2	0
53	5	NAT	CS	17160 to 17160	2	0
54	5	NAT	CS	0.04 to 0.04	2	0
55	5	NAT	CS	50 to 50	2	0

Graduate Aptitude Test in Engineering 2017

Question Paper Name: Computer Science and Information Technology 11th Feb 2017 session 2
Subject Name: Computer Science and Information Technology
Duration: 180
Total Marks: 100

Organizing Institute:

Indian Institute of Technology Roorkee

Question Number : 1 Correct : 1 Wrong : -0.33

The representation of the value of a 16-bit unsigned integer X in hexadecimal number system is BCA9. The representation of the value of X in octal number system is

(A) 571244 (B) 736251 (C) 571247 (D) 136251

Question Number : 2 Correct : 1 Wrong : -0.33

Match the following:

(P) `static char var;`	(i) Sequence of memory locations to store addresses
(Q) `m = malloc(10);` `m = NULL;`	(ii) A variable located in data section of memory
(R) `char *ptr[10];`	(iii) Request to allocate a CPU register to store data
(S) `register int var1;`	(iv) A lost memory which cannot be freed

(A) P → (ii), Q → (iv), R → (i), S → (iii)
(B) P → (ii), Q → (i), R → (iv), S → (iii)
(C) P → (ii), Q → (iv), R → (iii), S → (i)
(D) P → (iii), Q → (iv), R → (i), S → (ii)

Question Number : 3 Correct : 1 Wrong : -0.33

Match the algorithms with their time complexities:

Algorithm	Time complexity
(P) Towers of Hanoi with n disks	(i) $\Theta(n^2)$
(Q) Binary search given n sorted numbers	(ii) $\Theta(n \log n)$
(R) Heap sort given n numbers at the worst case	(iii) $\Theta(2^n)$
(S) Addition of two $n \times n$ matrices	(iv) $\Theta(\log n)$

(A) P → (iii), Q → (iv), R → (i), S → (ii)
(B) P → (iv), Q → (iii), R → (i), S → (ii)
(C) P → (iii), Q → (iv), R → (ii), S → (i)
(D) P → (iv), Q → (iii), R → (ii), S → (i)

Question Number : 4 **Correct : 1 Wrong : -0.33**

Let L_1, L_2 be any two context-free languages and R be any regular language. Then which of the following is/are CORRECT?

I. $L_1 \cup L_2$ is context-free.
II. $\overline{L_1}$ is context-free.
III. $L_1 - R$ is context-free.
IV. $L_1 \cap L_2$ is context-free.

(A) I, II and IV only (B) I and III only (C) II and IV only (D) I only

Question Number : 5 **Correct : 1 Wrong : -0.33**

Match the following according to input (from the left column) to the compiler phase (in the right column) that processes it:

(P) Syntax tree	(i) Code generator
(Q) Character stream	(ii) Syntax analyzer
(R) Intermediate representation	(iii) Semantic analyzer
(S) Token stream	(iv) Lexical analyzer

(A) P → (ii), Q → (iii), R → (iv), S → (i)
(B) P → (ii), Q → (i), R → (iii), S → (iv)
(C) P → (iii), Q → (iv), R → (i), S → (ii)
(D) P → (i), Q → (iv), R → (ii), S → (iii)

Question Number : 6 **Correct : 1 Wrong : -0.33**

Which of the following statements about parser is/are CORRECT?

I. Canonical LR is more powerful than SLR.
II. SLR is more powerful than LALR.
III. SLR is more powerful than Canonical LR.

(A) I only (B) II only (C) III only (D) II and III only

Question Number : 7 **Correct : 1 Wrong : -0.33**

Which of the following is/are shared by all the threads in a process?

I. Program counter
II. Stack
III. Address space
IV. Registers

(A) I and II only (B) III only (C) IV only (D) III and IV only

Question Number : 8 **Correct : 1 Wrong : -0.33**

In a file allocation system, which of the following allocation scheme(s) can be used if no external fragmentation is allowed?

I. Contiguous
II. Linked
III. Indexed

(A) I and III only (B) II only (C) III only (D) II and III only

Question Number : 9 **Correct : 1 Wrong : -0.33**

Consider the following statements about the routing protocols, Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) in an IPv4 network.

I: RIP uses distance vector routing
II: RIP packets are sent using UDP
III: OSPF packets are sent using TCP
IV: OSPF operation is based on link-state routing

Which of the statements above are CORRECT?

(A) I and IV only
(B) I, II and III only
(C) I, II and IV only
(D) II, III and IV only

Question Number : 10 **Correct : 1 Wrong : -0.33**

If $f(x) = R\ sin\left(\frac{\pi x}{2}\right) + S$, $f'\left(\frac{1}{2}\right) = \sqrt{2}$ and $\int_0^1 f(x)dx = \frac{2R}{\pi}$, then the constants R and S are, respectively

(A) $\frac{2}{\pi}$ and $\frac{16}{\pi}$

(B) $\frac{2}{\pi}$ and 0

(C) $\frac{4}{\pi}$ and 0

(D) $\frac{4}{\pi}$ and $\frac{16}{\pi}$

Question Number : 11 **Correct : 1 Wrong : -0.33**

Let p, q, r denote the statements "*It is raining*", "*It is cold*", and "*It is pleasant*", respectively. Then the statement "*It is not raining and it is pleasant, and it is not pleasant only if it is raining and it is cold*" is represented by

(A) $(\neg p \wedge r) \wedge (\neg r \rightarrow (p \wedge q))$

(B) $(\neg p \wedge r) \wedge ((p \wedge q) \rightarrow \neg r)$

(C) $(\neg p \wedge r) \vee ((p \wedge q) \rightarrow \neg r)$

(D) $(\neg p \wedge r) \vee (r \rightarrow (p \wedge q))$

Question Number : 12 **Correct : 1 Wrong : -0.33**

Given the following binary number in 32-bit (single precision) IEEE-754 format:

00111110011011010000000000000000

The decimal value closest to this floating-point number is

(A) 1.45×10^1 (B) 1.45×10^{-1} (C) 2.27×10^{-1} (D) 2.27×10^1

Question Number : 13 **Correct : 1 Wrong : -0.33**

A circular queue has been implemented using a singly linked list where each node consists of a value and a single pointer pointing to the next node. We maintain exactly two external pointers **FRONT** and **REAR** pointing to the front node and the rear node of the queue, respectively. Which of the following statements is/are CORRECT for such a circular queue, so that insertion and deletion operations can be performed in $O(1)$ time?

I. Next pointer of front node points to the rear node.
II. Next pointer of rear node points to the front node.

(A) I only

(B) II only

(C) Both I and II

(D) Neither I nor II

Question Number : 14 **Correct : 1 Wrong : -0.33**

Consider the following function implemented in C:

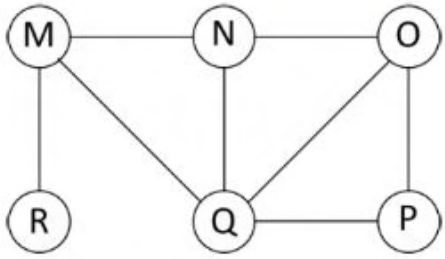
```
void printxy(int x, int y) {
        int *ptr;
        x = 0;
        ptr = &x;
        y = *ptr;
        *ptr = 1;
        printf("%d,%d",x,y);
}
```

The output of invoking `printxy(1,1)` is

(A) 0,0 (B) 0,1 (C) 1,0 (D) 1,1

Question Number : 15 **Correct : 1 Wrong : -0.33**

The Breadth First Search (BFS) algorithm has been implemented using the queue data structure. Which one of the following is a possible order of visiting the nodes in the graph below?



(A) MNOPQR
(B) NQMPOR
(C) QMNROP
(D) POQNMR

Question Number : 16 **Correct : 1 Wrong : -0.33**

Identify the language generated by the following grammar, where S is the start variable.

$$S \rightarrow XY$$
$$X \rightarrow aX \mid a$$
$$Y \rightarrow aYb \mid \epsilon$$

(A) $\{a^m b^n \mid m \geq n,\ n > 0\}$
(B) $\{a^m b^n \mid m \geq n,\ n \geq 0\}$
(C) $\{a^m b^n \mid m > n,\ n \geq 0\}$
(D) $\{a^m b^n \mid m > n,\ n > 0\}$

Question Number : 17 **Correct : 1 Wrong : -0.33**

An ER model of a database consists of entity types A and B. These are connected by a relationship R which does not have its own attribute. Under which one of the following conditions, can the relational table for R be merged with that of A?

(A) Relationship R is one-to-many and the participation of A in R is total.

(B) Relationship R is one-to-many and the participation of A in R is partial.

(C) Relationship R is many-to-one and the participation of A in R is total.

(D) Relationship R is many-to-one and the participation of A in R is partial.

Question Number : 18 **Correct : 1 Wrong : -0.33**

Consider socket API on a Linux machine that supports connected UDP sockets. A connected UDP socket is a UDP socket on which **connect** function has already been called. Which of the following statements is/are CORRECT?

I. A connected UDP socket can be used to communicate with multiple peers simultaneously.

II. A process can successfully call **connect** function again for an already connected UDP socket.

(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Question Number : 19 Correct : 1 Wrong : 0

Consider the following tables T1 and T2.

T1

P	Q
2	2
3	8
7	3
5	8
6	9
8	5
9	8

T2

R	S
2	2
8	3
3	2
9	7
5	7
7	2

In table T1, **P** is the primary key and **Q** is the foreign key referencing **R** in table T2 with on-delete cascade and on-update cascade. In table T2, **R** is the primary key and **S** is the foreign key referencing **P** in table T1 with on-delete set NULL and on-update cascade. In order to delete record $\langle 3, 8 \rangle$ from table T1, the number of additional records that need to be deleted from table T1 is _____.

Question Number : 20 Correct : 1 Wrong : 0

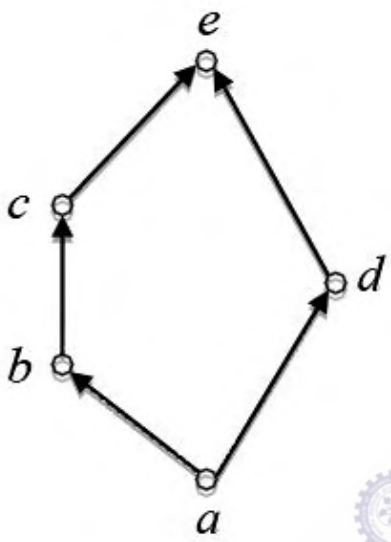
The maximum number of IPv4 router addresses that can be listed in the record route (RR) option field of an IPv4 header is _____.

Question Number : 21 **Correct : 1 Wrong : 0**

Consider the set $X=\{a,b,c,d,e\}$ under the partial ordering

$R=\{(a,a),(a,b),(a,c),(a,d),(a,e),(b,b),(b,c),(b,e),(c,c),(c,e),(d,d),(d,e),(e,e)\}$.

The Hasse diagram of the partial order (X,R) is shown below.



The minimum number of ordered pairs that need to be added to R to make (X,R) a lattice is _____ .

Question Number : 22 **Correct : 1 Wrong : 0**

Let $P=\begin{bmatrix}1 & 1 & -1\\ 2 & -3 & 4\\ 3 & -2 & 3\end{bmatrix}$ and $Q=\begin{bmatrix}-1 & -2 & -1\\ 6 & 12 & 6\\ 5 & 10 & 5\end{bmatrix}$ be two matrices.

Then the rank of $P+Q$ is _____.

Question Number : 23 **Correct : 1 Wrong : 0**

G is an undirected graph with n vertices and 25 edges such that each vertex of G has degree at least 3. Then the maximum possible value of n is _____.

Question Number : 24 **Correct : 1 Wrong : 0**

Consider a quadratic equation $x^2 - 13x + 36 = 0$ with coefficients in a base b. The solutions of this equation in the same base b are $x = 5$ and $x = 6$. Then $b =$ _____.

Question Number : 25 **Correct : 1 Wrong : 0**

The minimum possible number of states of a deterministic finite automaton that accepts the regular language $L = \{w_1 a w_2 \mid w_1, w_2 \in \{a, b\}^*, |w_1| = 2, |w_2| \geq 3\}$ is _____.

Question Number : 26 **Correct : 2 Wrong : -0.66**

P and Q are considering to apply for a job. The probability that P applies for the job is $\frac{1}{4}$, the probability that P applies for the job given that Q applies for the job is $\frac{1}{2}$, and the probability that Q applies for the job given that P applies for the job is $\frac{1}{3}$. Then the probability that P does not apply for the job given that Q does not apply for the job is

(A) $\frac{4}{5}$ (B) $\frac{5}{6}$ (C) $\frac{7}{8}$ (D) $\frac{11}{12}$

Question Number : 27 **Correct : 2 Wrong : -0.66**

If w, x, y, z are Boolean variables, then which one of the following is INCORRECT?

(A) $\overline{wx + w(x + y) + x(x + y)} = x + wy$
(B) $w\bar{x}(y + \bar{z}) + \bar{w}x = \bar{w} + x + \bar{y}z$
(C) $(w\bar{x}(y + x\bar{z}) + \bar{w}\bar{x})y = x\bar{y}$
(D) $(w + y)(wxy + wyz) = wxy + wyz$

Question Number : 28 **Correct : 2 Wrong : -0.66**

Given $f(w,x,y,z) = \sum_m(0,1,2,3,7,8,10) + \sum_d(5,6,11,15)$, where d represents the *don't-care* condition in Karnaugh maps. Which of the following is a minimum product-of-sums (POS) form of $f(w,x,y,z)$?

(A) $f = (\bar{w} + \bar{z})(\bar{x} + z)$
(B) $f = (\bar{w} + z)(x + z)$
(C) $f = (w + z)(\bar{x} + z)$
(D) $f = (w + \bar{z})(\bar{x} + z)$

Question Number : 29 **Correct : 2 Wrong : -0.66**

In a two-level cache system, the access times of L_1 and L_2 caches are 1 and 8 clock cycles, respectively. The miss penalty from the L_2 cache to main memory is 18 clock cycles. The miss rate of L_1 cache is twice that of L_2. The average memory access time (AMAT) of this cache system is 2 cycles. The miss rates of L_1 and L_2 respectively are:

(A) 0.111 and 0.056
(B) 0.056 and 0.111
(C) 0.0892 and 0.1784
(D) 0.1784 and 0.0892

Question Number : 30 **Correct : 2 Wrong : -0.66**

Consider the recurrence function

$$T(n) = \begin{cases} 2T(\sqrt{n}) + 1, & n > 2 \\ 2, & 0 < n \leq 2 \end{cases}$$

Then $T(n)$ in terms of Θ notation is

(A) $\Theta(\log \log n)$
(B) $\Theta(\log n)$
(C) $\Theta(\sqrt{n})$
(D) $\Theta(n)$

Question Number : 31 **Correct : 2 Wrong : -0.66**

For any discrete random variable X, with probability mass function $P(X=j)=p_j,\ p_j \geq 0,\ j \in \{0,\ldots,N\}$, and $\sum_{j=0}^{N} p_j = 1$, define the polynomial function $g_X(z)=\sum_{j=0}^{N} p_j z^j$. For a certain discrete random variable Y, there exists a scalar $\beta \in [0,1]$ such that $g_Y(z)=(1-\beta+\beta z)^N$. The expectation of Y is

(A) $N\beta(1-\beta)$

(B) $N\beta$

(C) $N(1-\beta)$

(D) Not expressible in terms of N and β alone

Question Number : 32 **Correct : 2 Wrong : -0.66**

Consider the following expression grammar G :

```
E -> E - T | T
T -> T + F | F
F -> (E) | id
```

Which of the following grammars is not left recursive, but is equivalent to G?

(A)
```
E -> E - T | T
T -> T + F | F
F -> (E) | id
```

(B)
```
E -> TE'
E' -> -TE' | ϵ
T -> T + F | F
F -> (E) | id
```

(C)
```
E -> TX
X -> -TX |ϵ
T -> FY
Y -> +FY |ϵ
F -> (E) | id
```

(D)
```
E -> TX | (TX)
X -> -TX | +TX | ϵ
T -> id
```

Question Number : 33 Correct : 2 Wrong : -0.66

A system shares 9 tape drives. The current allocation and maximum requirement of tape drives for three processes are shown below:

Process	Current Allocation	Maximum Requirement
P1	3	7
P2	1	6
P3	3	5

Which of the following best describes current state of the system?

(A) Safe, Deadlocked
(B) Safe, Not Deadlocked
(C) Not Safe, Deadlocked
(D) Not Safe, Not Deadlocked

Question Number : 34 Correct : 2 Wrong : -0.66

Consider a binary code that consists of only four valid codewords as given below:

$$00000, 01011, 10101, 11110$$

Let the minimum Hamming distance of the code be p and the maximum number of erroneous bits that can be corrected by the code be q. Then the values of p and q are

(A) p=3 and q=1
(B) p=3 and q=2
(C) p=4 and q=1
(D) p=4 and q=2

Question Number : 35 Correct : 2 Wrong : -0.66

Consider two hosts X and Y, connected by a single direct link of rate 10^6 bits/sec. The distance between the two hosts is 10,000 km and the propagation speed along the link is 2×10^8 m/sec. Host X sends a file of 50,000 bytes as one large message to host Y continuously. Let the transmission and propagation delays be p milliseconds and q milliseconds, respectively. Then the values of p and q are

(A) p=50 and q=100
(B) p=50 and q=400
(C) p=100 and q=50
(D) p=400 and q=50

Question Number : 36 **Correct : 2 Wrong : -0.66**

The pre-order traversal of a binary search tree is given by 12, 8, 6, 2, 7, 9, 10, 16, 15, 19, 17, 20. Then the post-order traversal of this tree is:

(A) 2, 6, 7, 8, 9, 10, 12, 15, 16, 17, 19, 20
(B) 2, 7, 6, 10, 9, 8, 15, 17, 20, 19, 16, 12
(C) 7, 2, 6, 8, 9, 10, 20, 17, 19, 15, 16, 12
(D) 7, 6, 2, 10, 9, 8, 15, 16, 17, 20, 19, 12

Question Number : 37 **Correct : 2 Wrong : -0.66**

Consider the C program fragment below which is meant to divide `x` by `y` using repeated subtractions. The variables `x`, `y`, `q` and `r` are all `unsigned int`.

```
while (r >= y) {
  r = r - y;
  q = q + 1;
}
```

Which of the following conditions on the variables `x`, `y`, `q` and `r` before the execution of the fragment will ensure that the loop terminates in a state satisfying the condition `x == (y*q + r)`?

(A) `(q == r) && (r == 0)`

(B) `(x > 0) && (r == x) && (y > 0)`

(C) `(q == 0) && (r == x) && (y > 0)`

(D) `(q == 0) && (y > 0)`

Question Number : 38 **Correct : 2 Wrong : -0.66**

Consider the following C function.

```
int fun(int n) {
        int i, j;
        for(i = 1; i <= n; i++) {
                for(j = 1; j < n; j += i) {
                        printf(" %d %d",i,j);
                }
        }
}
```

Time complexity of `fun` in terms of Θ notation is

(A) $\Theta(n\sqrt{n})$ (B) $\Theta(n^2)$ (C) $\Theta(n \log n)$ (D) $\Theta(n^2 \log n)$

Question Number : 39 **Correct : 2 Wrong : -0.66**

Let δ denote the transition function and $\hat{\delta}$ denote the extended transition function of the ϵ-NFA whose transition table is given below:

δ	ϵ	a	b
$\rightarrow q_0$	$\{q_2\}$	$\{q_1\}$	$\{q_0\}$
q_1	$\{q_2\}$	$\{q_2\}$	$\{q_3\}$
q_2	$\{q_0\}$	$\emptyset$	$\emptyset$
q_3	$\emptyset$	$\emptyset$	$\{q_2\}$

Then $\hat{\delta}(q_2, aba)$ is

(A) $\emptyset$ (B) $\{q_0, q_1, q_3\}$ (C) $\{q_0, q_1, q_2\}$ (D) $\{q_0, q_2, q_3\}$

Question Number : 40 **Correct : 2 Wrong : -0.66**

Consider the following languages.

$L_1 = \{a^p \mid p \text{ is a prime number}\}$
$L_2 = \{a^n b^m c^{2m} \mid n \geq 0, m \geq 0\}$
$L_3 = \{a^n b^n c^{2n} \mid n \geq 0\}$
$L_4 = \{a^n b^n \mid n \geq 1\}$

Which of the following are CORRECT?

I. L_1 is context-free but not regular.
II. L_2 is not context-free.
III. L_3 is not context-free but recursive.
IV. L_4 is deterministic context-free.

(A) I, II and IV only (B) II and III only (C) I and IV only (D) III and IV only

Question Number : 41 Correct : 2 Wrong : -0.66

Let $L(R)$ be the language represented by regular expression R. Let $L(G)$ be the language generated by a context free grammar G. Let $L(M)$ be the language accepted by a Turing machine M. Which of the following decision problems are undecidable?

I. Given a regular expression R and a string w, is $w \in L(R)$?
II. Given a context-free grammar G, is $L(G) = \emptyset$?
III. Given a context-free grammar G, is $L(G) = \Sigma^*$ for some alphabet Σ ?
IV. Given a Turing machine M and a string w, is $w \in L(M)$?

(A) I and IV only (B) II and III only (C) II, III and IV only (D) III and IV only

Question Number : 42 Correct : 2 Wrong : -0.66

The next state table of a 2-bit saturating up-counter is given below.

Q_1	Q_0	Q_1^+	Q_0^+
0	0	0	1
0	1	1	0
1	0	1	1
1	1	1	1

The counter is built as a synchronous sequential circuit using T flip-flops. The expressions for T_1 and T_0 are

(A) $T_1 = Q_1 Q_0$, $T_0 = \bar{Q}_1 \bar{Q}_0$

(B) $T_1 = \bar{Q}_1 Q_0$, $T_0 = \bar{Q}_1 + \bar{Q}_0$

(C) $T_1 = Q_1 + Q_0$, $T_0 = \bar{Q}_1 + \bar{Q}_0$

(D) $T_1 = \bar{Q}_1 Q_0$, $T_0 = Q_1 + Q_0$

Question Number : 43 **Correct : 2 Wrong : 0**

Consider the following snippet of a C program. Assume that `swap(&x, &y)` exchanges the contents of `x` and `y`.

```
int main() {
      int array[] = {3, 5, 1, 4, 6, 2};
      int done = 0;
      int i;

      while (done == 0) {
            done = 1;
            for (i=0; i<=4; i++) {
                  if (array[i] < array[i+1]) {
                        swap(&array[i], &array[i+1]);
                        done = 0;
                  }
            }
            for (i=5; i>=1; i--) {
                  if (array[i] > array[i-1]) {
                        swap(&array[i], &array[i-1]);
                        done = 0;
                  }
            }

      }
      printf("%d",array[3]);
}
```

The output of the program is _____.

Question Number : 44 Correct : 2 Wrong : 0

Two transactions T_1 and T_2 are given as

$$T_1: r_1(X)w_1(X)r_1(Y)w_1(Y)$$
$$T_2: r_2(Y)w_2(Y)r_2(Z)w_2(Z)$$

where $r_i(V)$ denotes a *read* operation by transaction T_i on a variable V and $w_i(V)$ denotes a *write* operation by transaction T_i on a variable V. The total number of conflict serializable schedules that can be formed by T_1 and T_2 is _____.

Question Number : 45 Correct : 2 Wrong : 0

The read access times and the hit ratios for different caches in a memory hierarchy are as given below.

Cache	Read access time (in nanoseconds)	Hit ratio
I-cache	2	0.8
D-cache	2	0.9
L2-cache	8	0.9

The read access time of main memory is 90 nanoseconds. Assume that the caches use the referred-word-first read policy and the write back policy. Assume that all the caches are direct mapped caches. Assume that the dirty bit is always 0 for all the blocks in the caches. In execution of a program, 60% of memory reads are for instruction fetch and 40% are for memory operand fetch. The average read access time in nanoseconds (up to 2 decimal places) is _____.

Question Number : 46 **Correct : 2 Wrong : 0**

Consider the following database table named *top_scorer*.

top_scorer

player	country	goals
Klose	Germany	16
Ronaldo	Brazil	15
G Müller	Germany	14
Fontaine	France	13
Pelé	Brazil	12
Klinsmann	Germany	11
Kocsis	Hungary	11
Batistuta	Argentina	10
Cubillas	Peru	10
Lato	Poland	10
Lineker	England	10
T Müller	Germany	10
Rahn	Germany	10

Consider the following SQL query:

```
SELECT ta.player FROM top_scorer AS ta
WHERE ta.goals >ALL (SELECT tb.goals
            FROM top_scorer AS tb
            WHERE tb.country = 'Spain')
AND ta.goals >ANY (SELECT tc.goals
            FROM top_scorer AS tc
            WHERE tc.country = 'Germany')
```

The number of tuples returned by the above SQL query is _____.

Question Number : 47 **Correct : 2 Wrong : 0**

If the ordinary generating function of a sequence $\{a_n\}_{n=0}^{\infty}$ is $\frac{1+z}{(1-z)^3}$, then $a_3 - a_0$ is equal to _____.

Question Number : 48 Correct : 2 Wrong : 0

If a random variable X has a Poisson distribution with mean 5, then the expectation $E[(X+2)^2]$ equals _____.

Question Number : 49 Correct : 2 Wrong : 0

In a B+ tree, if the search-key value is 8 bytes long, the block size is 512 bytes and the block pointer size is 2 bytes, then the maximum order of the B+ tree is _____.

Question Number : 50 Correct : 2 Wrong : 0

A message is made up entirely of characters from the set $X = \{P, Q, R, S, T\}$. The table of probabilities for each of the characters is shown below:

Character	Probability
P	0.22
Q	0.34
R	0.17
S	0.19
T	0.08
Total	1.00

If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded message in bits is _____.

Question Number : 51 Correct : 2 Wrong : 0

Consider the set of processes with arrival time (in milliseconds), CPU burst time (in milliseconds), and priority (0 is the highest priority) shown below. None of the processes have I/O burst time.

Process	Arrival Time	Burst Time	Priority
P_1	0	11	2
P_2	5	28	0
P_3	12	2	3
P_4	2	10	1
P_5	9	16	4

The average waiting time (in milliseconds) of all the processes using preemptive priority scheduling algorithm is _____.

Question Number : 52 Correct : 2 Wrong : 0

If the characteristic polynomial of a 3×3 matrix M over $\mathbb{R}$ (the set of real numbers) is $\lambda^3 - 4\lambda^2 + a\lambda + 30$, $a \in \mathbb{R}$, and one eigenvalue of M is 2, then the largest among the absolute values of the eigenvalues of M is _____.

Question Number : 53 Correct : 2 Wrong : 0

Consider a machine with a byte addressable main memory of 2^{32} bytes divided into blocks of size 32 bytes. Assume that a direct mapped cache having 512 cache lines is used with this machine. The size of the tag field in bits is _____.

Question Number : 54 **Correct : 2 Wrong : 0**

Consider the following C Program.

```
#include<stdio.h>
int main() {
        int m = 10;
        int n, n1;
        n = ++m;
        n1 = m++;
        n--;
        --n1;
        n -= n1;
        printf("%d", n);
        return 0;
}
```

The output of the program is _____.

Question Number : 55 **Correct : 2 Wrong : 0**

Consider the following C Program.

```
#include<stdio.h>
#include<string.h>
int main() {
        char* c = "GATECSIT2017";
        char* p = c;
        printf("%d", (int)strlen(c+2[p]-6[p]-1));
        return 0;
}
```

The output of the program is _____.

Question Number : 56 **Correct : 1 Wrong : -0.33**

Choose the option with words that are not synonyms.

(A) aversion, dislike
(B) luminous, radiant
(C) plunder, loot
(D) yielding, resistant

Question Number : 57 **Correct : 1 Wrong : -0.33**

Saturn is _____ to be seen on a clear night with the naked eye.

(A) enough bright (B) bright enough (C) as enough bright (D) bright as enough

Question Number : 58 **Correct : 1 Wrong : -0.33**

There are five buildings called V, W, X, Y and Z in a row (not necessarily in that order). V is to the West of W. Z is to the East of X and the West of V. W is to the West of Y. Which is the building in the middle?

(A) V (B) W (C) X (D) Y

Question Number : 59 **Correct : 1 Wrong : -0.33**

A test has twenty questions worth 100 marks in total. There are two types of questions. Multiple choice questions are worth 3 marks each and essay questions are worth 11 marks each. How many multiple choice questions does the exam have?

(A) 12 (B) 15 (C) 18 (D) 19

Question Number : 60 **Correct : 1 Wrong : -0.33**

There are 3 red socks, 4 green socks and 3 blue socks. You choose 2 socks. The probability that they are of the same colour is

(A) 1/5 (B) 7/30 (C) 1/4 (D) 4/15

Question Number : 61 **Correct : 2 Wrong : -0.66**

"We lived in a culture that denied any merit to literary works, considering them important only when they were handmaidens to something seemingly more urgent – namely ideology. This was a country where all gestures, even the most private, were interpreted in political terms."

The author's belief that ideology is not as important as literature is revealed by the word:

(A) 'culture' (B) 'seemingly' (C) 'urgent' (D) 'political'

Question Number : 62 **Correct : 2 Wrong : -0.66**

There are three boxes. One contains apples, another contains oranges and the last one contains both apples and oranges. All three are known to be incorrectly labelled. If you are permitted to open just one box and then pull out and inspect only one fruit, which box would you open to determine the contents of all three boxes?

(A) The box labelled 'Apples'
(B) The box labelled 'Apples and Oranges'
(C) The box labelled 'Oranges'
(D) Cannot be determined

Question Number : 63 **Correct : 2 Wrong : -0.66**

X is a 30 digit number starting with the digit 4 followed by the digit 7. Then the number X^3 will have

(A) 90 digits (B) 91 digits (C) 92 digits (D) 93 digits

Question Number : 64 **Correct : 2 Wrong : -0.66**

The number of roots of $e^x + 0.5x^2 - 2 = 0$ in the range $[-5, 5]$ is

(A) 0 (B) 1 (C) 2 (D) 3

Question Number : 65 **Correct : 2 Wrong : -0.66**

An air pressure contour line joins locations in a region having the same atmospheric pressure. The following is an air pressure contour plot of a geographical region. Contour lines are shown at 0.05 bar intervals in this plot.

If the possibility of a thunderstorm is given by how fast air pressure rises or drops over a region, which of the following regions is most likely to have a thunderstorm?

(A) P (B) Q (C) R (D) S

Q. No.	Type	Section	Key	Marks
1	MCQ	CS-2	D	1
2	MCQ	CS-2	A	1
3	MCQ	CS-2	C	1
4	MCQ	CS-2	B	1
5	MCQ	CS-2	C	1
6	MCQ	CS-2	A	1
7	MCQ	CS-2	B	1
8	MCQ	CS-2	D	1
9	MCQ	CS-2	C	1
10	MCQ	CS-2	C	1
11	MCQ	CS-2	A	1
12	MCQ	CS-2	C	1
13	MCQ	CS-2	B	1
14	MCQ	CS-2	C	1
15	MCQ	CS-2	D	1
16	MCQ	CS-2	C	1
17	MCQ	CS-2	C	1
18	MCQ	CS-2	B	1
19	NAT	CS-2	0.0 to 0.0	1
20	NAT	CS-2	9.0 to 9.0	1
21	NAT	CS-2	-0.01 to 0.01	1
22	NAT	CS-2	2.0 to 2.0	1
23	NAT	CS-2	16.0 to 16.0	1
24	NAT	CS-2	8.0 to 8.0	1
25	NAT	CS-2	8.0 to 8.0	1
26	MCQ	CS-2	A	2
27	MCQ	CS-2	C	2
28	MCQ	CS-2	A	2
29	MCQ	CS-2	A	2
30	MCQ	CS-2	B	2
31	MCQ	CS-2	B	2
32	MCQ	CS-2	C	2
33	MCQ	CS-2	B	2
34	MCQ	CS-2	A	2
35	MCQ	CS-2	D	2
36	MCQ	CS-2	B	2

37	MCQ	CS-2	C	2
38	MCQ	CS-2	C	2
39	MCQ	CS-2	C	2
40	MCQ	CS-2	D	2
41	MCQ	CS-2	D	2
42	MCQ	CS-2	B	2
43	NAT	CS-2	3.0 to 3.0	2
44	NAT	CS-2	54.0 to 54.0	2
45	NAT	CS-2	4.72 to 4.72	2
46	NAT	CS-2	7.0 to 7.0	2
47	NAT	CS-2	15.0 to 15.0	2
48	NAT	CS-2	54.0 to 54.0	2
49	NAT	CS-2	52.0 to 52.0	2
50	NAT	CS-2	225.0 to 225.0	2
51	NAT	CS-2	29.0 to 29.0	2
52	NAT	CS-2	5.0 to 5.0	2
53	NAT	CS-2	18.0 to 18.0	2
54	NAT	CS-2	0.0 to 0.0	2
55	NAT	CS-2	2.0 to 2.0	2
56	MCQ	GA	D	1
57	MCQ	GA	B	1
58	MCQ	GA	A	1
59	MCQ	GA	B	1
60	MCQ	GA	D	1
61	MCQ	GA	B	2
62	MCQ	GA	B	2
63	MCQ	GA	A	2
64	MCQ	GA	C	2
65	MCQ	GA	C	2

2007
CS: Computer Science and Engineering

Duration : Three Hours Maximum Marks :150

Read the following instructions carefully.

1. This question paper contains 85 objective type questions. Q.1 to Q.20 carry **one** mark each and Q.21 to Q.85 carry **two** marks each.
2. Attempt all the questions.
3. Questions must be answered on **O**bjective **R**esponse **S**heet (**ORS**) by darkening the appropriate bubble (marked A, B, C, D) using HB pencil against the question number on the left hand side of the **ORS**. **Each question has only one correct answer.** In case you wish to change an answer, erase the old answer completely.
4. Wrong answers will carry NEGATIVE marks. In Q.1 to Q.20, **0.25** mark will be deducted for each wrong answer. In Q.21 to Q.76, Q.78, Q.80, Q.82 and in Q.84, **0.5** mark will be deducted for each wrong answer. However, there is no negative marking in Q.77, Q.79, Q.81, Q.83 and in Q.85. More than one answer bubbled against a question will be taken as an incorrect response. Unattempted questions will not carry any marks.
5. Write your registration number, your name and name of the examination centre at the specified locations on the right half of the **ORS**.
6. Using HB pencil, darken the appropriate bubble under each digit of your registration number and the letters corresponding to your paper code.
7. Calculator is allowed in the examination hall.
8. Charts, graph sheets or tables are NOT allowed in the examination hall.
9. Rough work can be done on the question paper itself. Additionally blank pages are given at the end of the question paper for rough work.
10. This question paper contains **24** printed pages including pages for rough work. Please check all pages and report, if there is any discrepancy.

Q. 1 – Q. 20 carry one mark each.

Q.1 Consider the following two statements about the function $f(x)=|x|$:

P. $f(x)$ is continuous for all real values of x

Q. $f(x)$ is differentiable for all real values of x

Which of the following is **TRUE**?

(A) P is true and Q is false.
(B) P is false and Q is true.
(C) Both P and Q are true.
(D) Both P and Q are false.

Q.2 Let S be a set of n elements. The number of ordered pairs in the largest and the smallest equivalence relations on S are

(A) n and n (B) n^2 and n (C) n^2 and 0 (D) n and 1

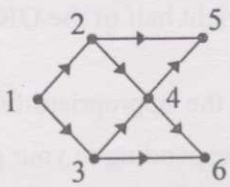
Q.3 What is the maximum number of different Boolean functions involving n Boolean variables?

(A) n^2 (B) 2^n (C) 2^{2^n} (D) 2^{n^2}

Q.4 Let G be the non-planar graph with the minimum possible number of edges. Then G has

(A) 9 edges and 5 vertices
(B) 9 edges and 6 vertices
(C) 10 edges and 5 vertices
(D) 10 edges and 6 vertices

Q.5 Consider the DAG with V = {1,2,3,4,5,6}, shown below.



Which of the following is **NOT** a topological ordering?

(A) 1 2 3 4 5 6 (B) 1 3 2 4 5 6
(C) 1 3 2 4 6 5 (D) 3 2 4 1 6 5

Q.6 Which of the following problems is undecidable?

(A) Membership problem for CFGs.
(B) Ambiguity problem for CFGs.
(C) Finiteness problem for FSAs.
(D) Equivalence problem for FSAs.

Q.7 Which of the following is **TRUE**?

(A) Every subset of a regular set is regular.
(B) Every finite subset of a non-regular set is regular.
(C) The union of two non-regular sets is not regular.
(D) Infinite union of finite sets is regular.

Q.8 How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates?

(A) 7 (B) 8 (C) 9 (D) 10

Q.9 Consider the following Boolean function of four variables:

$$f(w,x,y,z) = \sum(1,3,4,6,9,11,12,14)$$

The function is

(A) independent of one variable.
(B) independent of two variables.
(C) independent of three variables.
(D) dependent on all the variables.

Q.10 Consider a 4-way set associative cache consisting of 128 lines with a line size of 64 words. The CPU generates a 20-bit address of a word in main memory. The number of bits in the TAG, LINE and WORD fields are respectively:

(A) 9, 6, 5 (B) 7, 7, 6 (C) 7, 5, 8 (D) 9, 5, 6

Q.11 Consider a disk pack with 16 surfaces, 128 tracks per surface and 256 sectors per track. 512 bytes of data are stored in a bit serial manner in a sector. The capacity of the disk pack and the number of bits required to specify a particular sector in the disk are respectively:

(A) 256 Mbyte, 19 bits
(B) 256 Mbyte, 28 bits
(C) 512 Mbyte, 20 bits
(D) 64 Gbyte, 28 bits

Q.12 The height of a binary tree is the maximum number of edges in any root to leaf path. The maximum number of nodes in a binary tree of height h is:

(A) $2^h - 1$ (B) $2^{h-1} - 1$ (C) $2^{h+1} - 1$ (D) 2^{h+1}

Q.13 The maximum number of binary trees that can be formed with three unlabeled nodes is:

(A) 1 (B) 5 (C) 4 (D) 3

Q.14 Which of the following sorting algorithms has the lowest worst-case complexity?

(A) Merge sort
(B) Bubble sort
(C) Quick sort
(D) Selection sort

Q.15 Consider the following segment of C-code:

```
int j, n;
  j = 1;
  while (j <= n)
      j = j*2;
```

The number of comparisons made in the execution of the loop for any $n > 0$ is:

(A) $\lceil \log_2 n \rceil + 1$ (B) n (C) $\lceil \log_2 n \rceil$ (D) $\lfloor \log_2 n \rfloor + 1$

Q.16 Group 1 contains some CPU scheduling algorithms and Group 2 contains some applications. Match entries in Group 1 to entries in Group 2.

Group 1	Group 2
P. Gang Scheduling	1. Guaranteed Scheduling
Q. Rate Monotonic Scheduling	2. Real-time Scheduling
R. Fair Share Scheduling	3. Thread Scheduling

(A) P-3; Q-2; R-1
(B) P-1; Q-2; R-3
(C) P-2; Q-3; R-1
(D) P-1; Q-3; R-2

Q.17 Consider the following statements about user level threads and kernel level threads. Which one of the following statements is **FALSE**?

(A) Context switch time is longer for kernel level threads than for user level threads.
(B) User level threads do not need any hardware support.
(C) Related kernel level threads can be scheduled on different processors in a multi-processor system.
(D) Blocking one kernel level thread blocks all related threads.

Q.18 Which one of the following is a top-down parser?

(A) Recursive descent parser.
(B) Operator precedence parser.
(C) An LR(k) parser.
(D) An LALR(k) parser.

Q.19 In Ethernet when Manchester encoding is used, the bit rate is:

(A) Half the baud rate.
(B) Twice the baud rate.
(C) Same as the baud rate.
(D) None of the above.

Q.20 Which one of the following uses UDP as the transport protocol?

(A) HTTP (B) Telnet (C) DNS (D) SMTP

Q. 21 to Q. 75 carry two marks each.

Q.21 How many different non-isomorphic Abelian groups of order 4 are there?

(A) 2 (B) 3 (C) 4 (D) 5

Q.22 Let *Graph(x)* be a predicate which denotes that x is a graph. Let *Connected(x)* be a predicate which denotes that x is connected. Which of the following first order logic sentences **DOES NOT** represent the statement: "*Not every graph is connected*"?

(A) $\neg\forall x\,(Graph(x) \Rightarrow Connected(x))$
(B) $\exists x\,(Graph(x) \wedge \neg Connected(x))$
(C) $\neg\forall x\,(\neg Graph(x) \vee Connected(x))$
(D) $\forall x\,(Graph(x) \Rightarrow \neg Connected(x))$

Q.23 Which of the following graphs has an Eulerian circuit?

(A) Any k-regular graph where k is an even number.
(B) A complete graph on 90 vertices.
(C) The complement of a cycle on 25 vertices.
(D) None of the above.

Q.24 Suppose we uniformly and randomly select a permutation from the 20! permutations of $1,2,3,\ldots,20$. What is the probability that 2 appears at an earlier position than any other even number in the selected permutation?

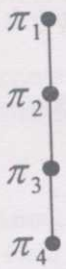
(A) $\frac{1}{2}$
(B) $\frac{1}{10}$
(C) $\frac{9!}{20!}$
(D) None of the above.

Q.25 Let A be a 4×4 matrix with eigenvalues $-5,-2,1,4$. Which of the following is an eigenvalue of $\begin{bmatrix} A & I \\ I & A \end{bmatrix}$, where I is the 4×4 identity matrix?

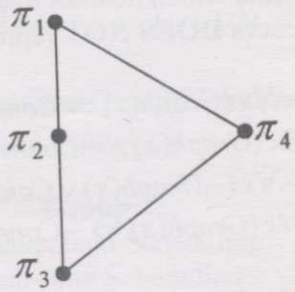
(A) -5 (B) -7 (C) 2 (D) 1

Q.26 · Consider the set $S=\{a,b,c,d\}$. Consider the following 4 partitions π_1,π_2,π_3,π_4 on S: $\pi_1=\{\overline{abcd}\}, \pi_2=\{\overline{ab},\overline{cd}\}, \pi_3=\{\overline{abc},\overline{d}\}, \pi_4=\{\overline{a},\overline{b},\overline{c},\overline{d}\}$. Let $\prec$ be the partial order on the set of partitions $S'=\{\pi_1,\pi_2,\pi_3,\pi_4\}$ defined as follows: $\pi_i \prec \pi_j$ if and only if π_i refines π_j. The poset digram for $(S',\prec)$ is

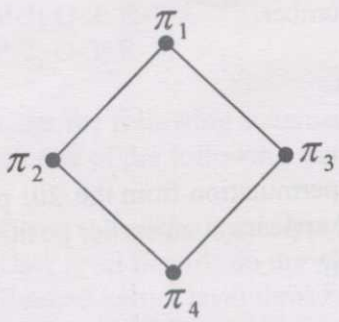
(A)



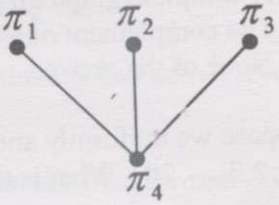
(B)



(C)



(D)



Q.27 Consider the set of (column) vectors defined by $X=\{x\in R^3 \mid x_1+x_2+x_3=0$, where $x^T=[x_1,x_2,x_3]^T\}$. Which of the following is **TRUE**?

(A) $\{[1,-1,0]^T,[1,0,-1]^T\}$ is a basis for the subspace X.

(B) $\{[1,-1,0]^T,[1,0,-1]^T\}$ is a linearly independent set, but it does not span X and therefore is not a basis of X.

(C) X is not a subspace of R^3.

(D) None of the above.

Q.28 Consider the series $x_{n+1}=\frac{x_n}{2}+\frac{9}{8x_n}$, $x_0=0.5$ obtained from the Newton-Raphson method. The series converges to

(A) 1.5 (B) $\sqrt{2}$ (C) 1.6 (D) 1.4

Q.29 A minimum state deterministic finite automaton accepting the language $L=\{w \mid w\in\{0,1\}^*$, number of 0s and 1s in w are divisible by 3 and 5, respectively$\}$ has

(A) 15 states (B) 11 states (C) 10 states (D) 9 states

Q.30 The language $L = \{ 0^i21^i \mid i \geq 0 \}$ over the alphabet {0, 1, 2} is

(A) not recursive.
(B) is recursive and is a deterministic CFL.
(C) is a regular language.
(D) is not a deterministic CFL but a CFL.

Q.31 Which of the following languages is regular?

(A) $\left\{ ww^R \middle| w \in \{0,1\}^+ \right\}$

(B) $\left\{ ww^R x \middle| x, w \in \{0,1\}^+ \right\}$

(C) $\left\{ wxw^R \middle| x, w \in \{0,1\}^+ \right\}$

(D) $\left\{ xww^R \middle| x, w \in \{0,1\}^+ \right\}$

Q.32 Let $f(w,x,y,z) = \sum(0,4,5,7,8,9,13,15)$. Which of the following expressions are **NOT** equivalent to f ?

(P) $x'y'z'+w'xy'+wy'z+xz$
(Q) $w'y'z'+wx'y'+xz$
(R) $w'y'z'+wx'y'+xyz+xy'z$
(S) $x'y'z'+wx'y'+w'y$

(A) P only
(B) Q and S
(C) R and S
(D) S only

Q.33 Define the connective * for the Boolean variables X and Y as: $X * Y = XY + X'Y'$. Let $Z = X * Y$. Consider the following expressions P, Q and R.

$P: X = Y * Z$ $\quad Q: Y = X * Z$ $\quad R: X * Y * Z = 1$

Which of the following is **TRUE**?

(A) Only P and Q are valid.
(B) Only Q and R are valid.
(C) Only P and R are valid.
(D) All P, Q, R are valid.

Q.34 Suppose only one multiplexer and one inverter are allowed to be used to implement any Boolean function of n variables. What is the minimum size of the multiplexer needed?

(A) 2^n line to 1 line
(B) 2^{n+1} line to 1 line
(C) 2^{n-1} line to 1 line
(D) 2^{n-2} line to 1 line

Q.35 In a look-ahead carry generator, the carry generate function G_i and the carry propagate function P_i for inputs A_i and B_i are given by:

$$P_i = A_i \oplus B_i \text{ and } G_i = A_i B_i$$

The expressions for the sum bit S_i and the carry bit C_{i+1} of the look-ahead carry adder are given by:

$$S_i = P_i \oplus C_i \text{ and } C_{i+1} = G_i + P_i C_i, \text{ where } C_0 \text{ is the input carry.}$$

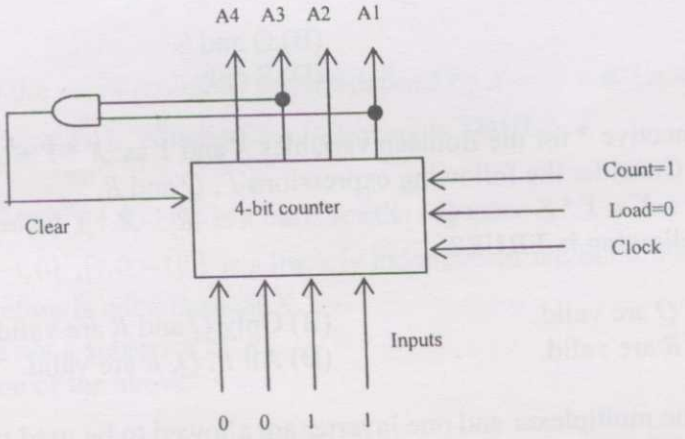
Consider a two-level logic implementation of the look-ahead carry generator. Assume that all P_i and G_i are available for the carry generator circuit and that the AND and OR gates can have any number of inputs. The number of AND gates and OR gates needed to implement the look-ahead carry generator for a 4-bit adder with S_3, S_2, S_1, S_0 and C_4 as its outputs are respectively:

(A) 6, 3 (B) 10, 4 (C) 6, 4 (D) 10, 5

Q.36 The control signal functions of a 4-bit binary counter are given below (where X is "don't care"):

Clear	Clock	Load	Count	Function
1	X	X	X	Clear to 0
0	X	0	0	No change
0	↑	1	X	Load input
0	↑	0	1	Count next

The counter is connected as follows:



Assume that the counter and gate delays are negligible. If the counter starts at 0, then it cycles through the following sequence:

(A) 0, 3, 4 (B) 0, 3, 4, 5
(C) 0, 1, 2, 3, 4 (D) 0, 1, 2, 3, 4, 5

Q.37 Consider a pipelined processor with the following four stages:

IF: Instruction Fetch
ID: Instruction Decode and Operand Fetch
EX: Execute
WB: Write Back

The IF, ID and WB stages take one clock cycle each to complete the operation. The number of clock cycles for the EX stage depends on the instruction. The ADD and SUB instructions need 1 clock cycle and the MUL instruction needs 3 clock cycles in the EX stage. Operand forwarding is used in the pipelined processor. What is the number of clock cycles taken to complete the following sequence of instructions?

ADD	R2, R1, R0	R2 ← R1 + R0
MUL	R4, R3, R2	R4 ← R3 * R2
SUB	R6, R5, R4	R6 ← R5 – R4

(A) 7 (B) 8 (C) 10 (D) 14

Q.38 The following postfix expression with single digit operands is evaluated using a stack:

8 2 3 ^ / 2 3 * + 5 1 * –

Note that ^ is the exponentiation operator. The top two elements of the stack after the first * is evaluated are:

(A) 6,1 (B) 5,7 (C) 3,2 (D) 1,5

Q.39 The inorder and preorder traversal of a binary tree are

d b e a f c g and a b d e c f g, respectively.

The postorder traversal of the binary tree is

(A) d e b f g c a
(B) e d b g f c a
(C) e d b f g c a
(D) d e f g b c a

Q.40 Consider a hash table of size seven, with starting index zero, and a hash function $(3x+4) \bmod 7$. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1,3,8,10 is inserted into the table using closed hashing? Note that – denotes an empty location in the table.

(A) 8,–,–,–,–,–,10
(B) 1,8,10,–,–,–,3
(C) 1,–,–,–,–,–,3
(D) 1,10,8,–,–,–,3

Q.41 In an unweighted, undirected connected graph, the shortest path from a node S to every other node is computed most efficiently, in terms of *time complexity*, by

(A) Dijkstra's algorithm starting from S.
(B) Warshall's algorithm.
(C) performing a DFS starting from S.
(D) performing a BFS starting from S.

Q.42 Consider the following C function:

```
int f(int n)
{static int r=0;
 if (n <= 0) return 1;
 if (n > 3)
    {r = n;
     return f(n-2)+2;
    }
 return f(n-1)+r;
}
```

What is the value of `f(5)`?

(A) 5 (B) 7 (C) 9 (D) 18

Q.43 A complete n-ary tree is a tree in which each node has n children or no children. Let I be the number of internal nodes and L be the number of leavés in a complete n-ary tree. If $L = 41$, and $I = 10$, what is the value of n?

(A) 3 (B) 4 (C) 5 (D) 6

Q.44 In the following C function, let $n \geq m$.

```
int gcd(n,m)
{
  if(n%m == 0) return m;
  n = n%m;
  return gcd(m,n);
}
```

How many recursive calls are made by this function?

(A) $\Theta(\log_2 n)$ (B) $\Omega(n)$ (C) $\Theta(\log_2 \log_2 n)$ (D) $\Theta(\sqrt{n})$

Q.45 What is the *time complexity* of the following recursive function:

```
int DoSomething (int n){
   if (n <= 2)
      return 1;
   else
       return(DoSomething(floor(sqrt(n))) + n);
}
```

(A) $\Theta(n^2)$ (B) $\Theta(n \log_2 n)$ (C) $\Theta(\log_2 n)$ (D) $\Theta(\log_2 \log_2 n)$

Q.46 Consider the following C program segment where CellNode represents a node in a binary tree:

```
struct CellNode {
  struct CellNode *leftChild;
   int element;
  struct CellNode *rightChild;
};
int GetValue(struct CellNode *ptr) {
 int value = 0;
 if (ptr != NULL) {
    if ((ptr->leftChild == NULL) &&
          (ptr->rightChild == NULL))
       value = 1;
    else
       value = value + GetValue(ptr->leftChild)
                     + GetValue(ptr->rightChild);
 }
return(value);
}
```

The value returned by GetValue when a pointer to the root of a binary tree is passed as its argument is:

(A) the number of nodes in the tree.
(B) the number of internal nodes in the tree.
(C) the number of leaf nodes in the tree.
(D) the height of the tree.

Q.47 Consider the process of inserting an element into a *Max Heap*, where the *Max Heap* is represented by an *array*. Suppose we perform a binary search on the path from the new leaf to the root to find the position for the newly inserted element, the number of *comparisons* performed is:

(A) $\Theta(\log_2 n)$
(B) $\Theta(\log_2 \log_2 n)$
(C) $\Theta(n)$
(D) $\Theta(n \log_2 n)$

Q.48 Which of the following is **TRUE** about formulae in Conjunctive Normal Form?

(A) For any formula, there is a truth assignment for which at least half the clauses evaluate to true.
(B) For any formula, there is a truth assignment for which all the clauses evaluate to true.
(C) There is a formula such that for each truth assignment, at most one-fourth of the clauses evaluate to true.
(D) None of the above.

Q.49 Let w be the minimum weight among all edge weights in an undirected connected graph. Let e be a specific edge of weight w. Which of the following is **FALSE**?

(A) There is a minimum spanning tree containing e.
(B) If e is not in a minimum spanning tree T, then in the cycle formed by adding e to T, all edges have the same weight.
(C) Every minimum spanning tree has an edge of weight w.
(D) e is present in every minimum spanning tree.

Q.50 An array of n numbers is given, where n is an even number. The maximum as well as the minimum of these n numbers needs to be determined. Which of the following is **TRUE** about the number of comparisons needed?

(A) At least $2n-c$ comparisons, for some constant c, are needed.
(B) At most $1.5n-2$ comparisons are needed.
(C) At least $n\log_2 n$ comparisons are needed.
(D) None of the above.

Q.51 Consider the following C code segment:

```
int IsPrime(n)
{
  int i,n;
  for(i=2;i <= sqrt(n);i++)
    if(n%i == 0)
       {printf("Not Prime\n"); return 0;}
  return 1;
}
```

Let $T(n)$ denote the number of times the *for* loop is executed by the program on input n. Which of the following is **TRUE**?

(A) $T(n) = O(\sqrt{n})$ and $T(n) = \Omega(\sqrt{n})$
(B) $T(n) = O(\sqrt{n})$ and $T(n) = \Omega(1)$
(C) $T(n) = O(n)$ and $T(n) = \Omega(\sqrt{n})$
(D) None of the above.

Q.52 Consider the grammar with non-terminals $N = \{S, C, S_1\}$, terminals $T = \{a, b, i, t, e\}$, with S as the start symbol, and the following set of rules:

$$S \rightarrow iCtSS_1 \mid a$$
$$S_1 \rightarrow eS \mid \varepsilon$$
$$C \rightarrow b$$

The grammar is **NOT** LL(1) because:

(A) it is left recursive.
(B) it is right recursive.
(C) it is ambiguous.
(D) it is not context-free.

Q.53 Consider the following two statements:

P: Every regular grammar is LL(1)
Q: Every regular set has a LR(1) grammar

Which of the following is **TRUE**?

(A) Both P and Q are true.
(B) P is true and Q is false.
(C) P is false and Q is true.
(D) Both P and Q are false.

Q.54 In a simplified computer the instructions are:

OP R_j,R_i - Performs R_j OP R_i and stores the result in register R_i.
OP *m*,R_i - Performs *val* OP R_i and stores the result in R_i. *val* denotes the content of memory location *m*.
MOV *m*,R_i - Moves the content of memory location *m* to register R_i.
MOV R_i,*m* - Moves the content of register R_i to memory location *m*.

The computer has only two registers, and OP is either ADD or SUB. Consider the following basic block:

$t_1 = a + b$
$t_2 = c + d$
$t_3 = e - t_2$
$t_4 = t_1 - t_3$

Assume that all operands are initially in memory. The final value of the computation should be in memory. What is the minimum number of MOV instructions in the code generated for this basic block?

(A) 2
(B) 3
(C) 5
(D) 6

Q.55 An operating system uses Shortest Remaining Time first (SRT) process scheduling algorithm. Consider the arrival times and execution times for the following processes:

Process	Execution time	Arrival time
P1	20	0
P2	25	15
P3	10	30
P4	15	45

What is the total waiting time for process P2?

(A) 5
(B) 15
(C) 40
(D) 55

Q.56 A virtual memory system uses First In First Out (FIFO) page replacement policy and allocates a fixed number of frames to a process. Consider the following statements:

P: Increasing the number of page frames allocated to a process sometimes increases the page fault rate.
Q: Some programs do not exhibit locality of reference.

Which one of the following is **TRUE**?

(A) Both P and Q are true, and Q is the reason for P.
(B) Both P and Q are true, but Q is not the reason for P.
(C) P is false, but Q is true.
(D) Both P and Q are false.

Q.57 A single processor system has three resource types X, Y, and Z, which are shared by three processes. There are 5 units of each resource type. Consider the following scenario, where the column ***alloc*** denotes the number of units of each resource type allocated to each process, and the column ***request*** denotes the number of units of each resource type requested by a process in order to complete execution. Which of these processes will finish **LAST**?

	alloc	*request*
	X Y Z	X Y Z
P0	1 2 1	1 0 3
P1	2 0 1	0 1 2
P2	2 2 1	1 2 0

(A) P0
(B) P1
(C) P2
(D) None of the above, since the system is in a deadlock.

Q.58 Two processes, P1 and P2, need to access a critical section of code. Consider the following synchronization construct used by the processes:

```
/* P1 */
while (true) {
   wants1 = true;
   while (wants2==true);
   /* Critical
      Section */
   wants1=false;
}
/* Remainder section */
```

```
/* P2 */
while (true) {
   wants2 = true;
   while (wants1==true);
   /* Critical
      Section */
   wants2=false;
}
/* Remainder section */
```

Here, `wants1` and `wants2` are shared variables, which are initialized to `false`. Which one of the following statements is **TRUE** about the above construct?

(A) It does not ensure mutual exclusion.
(B) It does not ensure bounded waiting.
(C) It requires that processes enter the critical section in strict alternation.
(D) It does not prevent deadlocks, but ensures mutual exclusion.

Q.59 Information about a collection of students is given by the relation **studInfo**(studId, name, sex). The relation **enroll**(studId, courseId) gives which student has enrolled for (or taken) what course(s). Assume that every course is taken by at least one male and at least one female student. What does the following relational algebra expression represent?

$$\Pi_{courseId}((\Pi_{studId}(\sigma_{sex = \text{"female"}}(studInfo)) \times \Pi_{courseId}(enroll)) - enroll)$$

(A) Courses in which all the female students are enrolled.
(B) Courses in which a proper subset of female students are enrolled.
(C) Courses in which only male students are enrolled.
(D) None of the above.

Q.60 Consider the relation **employee**(name, sex, supervisorName) with *name* as the key. *supervisorName* gives the name of the supervisor of the employee under consideration. What does the following Tuple Relational Calculus query produce?

$$\{e.name \mid employee(e) \wedge (\forall x)[\neg employee(x) \vee x.supervisorName \neq e.name \vee x.sex = \text{"male"}]\}$$

(A) Names of employees with a male supervisor.
(B) Names of employees with no immediate male subordinates.
(C) Names of employees with no immediate female subordinates.
(D) Names of employees with a female supervisor.

Q.61 Consider the table **employee**(empId, name, department, salary) and the two queries Q_1, Q_2 below. Assuming that department 5 has more than one employee, and we want to find the employees who get higher salary than anyone in the department 5, which one of the statements is **TRUE** for any arbitrary employee table?

Q_1: Select e.empId
From employee e
Where not exists
(Select * From employee s Where s.department = "5" and s.salary >= e.salary)

Q_2: Select e.empId
From employee e
Where e.salary > Any
(Select distinct salary From employee s Where s.department = "5")

(A) Q_1 is the correct query.
(B) Q_2 is the correct query.
(C) Both Q_1 and Q_2 produce the same answer.
(D) Neither Q_1 nor Q_2 is the correct query.

Q.62 Which one of the following statements is **FALSE**?

(A) Any relation with two attributes is in BCNF.
(B) A relation in which every key has only one attribute is in 2NF.
(C) A prime attribute can be transitively dependent on a key in a 3NF relation.
(D) A prime attribute can be transitively dependent on a key in a BCNF relation.

Q.63 The order of a leaf node in a B^+-tree is the maximum number of (value, data record pointer) pairs it can hold. Given that the block size is 1K bytes, data record pointer is 7 bytes long, the value field is 9 bytes long and a block pointer is 6 bytes long, what is the order of the leaf node?

(A) 63 (B) 64 (C) 67 (D) 68

Q.64 Consider the following schedules involving two transactions. Which one of the following statements is **TRUE**?

S_1: $r_1(X)$; $r_1(Y)$; $r_2(X)$; $r_2(Y)$; $w_2(Y)$; $w_1(X)$
S_2: $r_1(X)$; $r_2(X)$; $r_2(Y)$; $w_2(Y)$; $r_1(Y)$; $w_1(X)$

(A) Both S_1 and S_2 are conflict serializable.
(B) S_1 is conflict serializable and S_2 is not conflict serializable.
(C) S_1 is not conflict serializable and S_2 is conflict serializable.
(D) Both S_1 and S_2 are not conflict serializable.

Q.65 There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that **ONLY** one station transmits in a given time slot?

(A) $np(1-p)^{n-1}$ (B) $(1-p)^{n-1}$ (C) $p(1-p)^{n-1}$ (D) $1-(1-p)^{n-1}$

Q.66 In a token ring network the transmission speed is 10^7 bps and the propagation speed is 200 metres/μs. The 1-bit delay in this network is equivalent to:

(A) 500 metres of cable.
(B) 200 metres of cable.
(C) 20 metres of cable.
(D) 50 metres of cable.

Q.67 The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

(A) 62 subnets and 262142 hosts.
(B) 64 subnets and 262142 hosts.
(C) 62 subnets and 1022 hosts.
(D) 64 subnets and 1024 hosts.

Q.68 The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted is:

(A) 11001001000 (B) 11001001011 (C) 11001010 (D) 110010010011

Q.69 The distance between two stations M and N is L kilometres. All frames are K bits long. The propagation delay per kilometre is t seconds. Let R bits/second be the channel capacity. Assuming that processing delay is negligible, the *minimum* number of bits for the sequence number field in a frame for maximum utilization, when the *sliding window protocol* is used, is:

(A) $\left\lceil \log_2 \frac{2LtR + 2K}{K} \right\rceil$

(B) $\left\lceil \log_2 \frac{2LtR}{K} \right\rceil$

(C) $\left\lceil \log_2 \frac{2LtR + K}{K} \right\rceil$

(D) $\left\lceil \log_2 \frac{2LtR + K}{2K} \right\rceil$

Q.70 Match the following:

P. SMTP	1. Application layer
Q. BGP	2. Transport layer
R. TCP	3. Data link layer
S. PPP	4. Network layer
	5. Physical layer

(A) P-2, Q-1, R-3, S-5
(B) P-1, Q-4, R-2, S-3
(C) P-1, Q-4, R-2, S-5
(D) P-2, Q-4, R-1, S-3

Common Data Questions

Common Data for Questions 71,72,73:

Consider the following program segment. Here R1, R2 and R3 are the general purpose registers.

	Instruction	Operation	Instruction size (no. of words)
	MOV R1, (3000)	R1 ← M[3000]	2
LOOP:	MOV R2, (R3)	R2 ← M[R3]	1
	ADD R2, R1	R2 ← R1 + R2	1
	MOV (R3), R2	M[R3] ← R2	1
	INC R3	R3 ← R3 + 1	1
	DEC R1	R1 ← R1 − 1	1
	BNZ LOOP	Branch on not zero	2
	HALT	Stop	1

Assume that the content of memory location 3000 is 10 and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2010 is 100. The program is loaded from the memory location 1000. All the numbers are in decimal.

Q.71 Assume that the memory is word addressable. The number of memory references for accessing the data in executing the program completely is

(A) 10 (B) 11 (C) 20 (D) 21

Q.72 Assume that the memory is word addressable. After the execution of this program, the content of memory location 2010 is

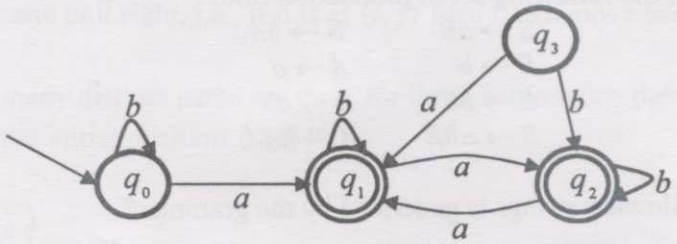
(A) 100 (B) 101 (C) 102 (D) 110

Q.73 Assume that the memory is byte addressable and the word size is 32 bits. If an interrupt occurs during the execution of the instruction "INC R3", what return address will be pushed on to the stack?

(A) 1005 (B) 1020 (C) 1024 (D) 1040

Common Data for Questions 74, 75:

Consider the following Finite State Automaton:



Q.74 The language accepted by this automaton is given by the regular expression

(A) $b^*ab^*ab^*ab^*$ (B) $(a+b)^*$ (C) $b^*a(a+b)^*$ (D) $b^*ab^*ab^*$

Q.75 The minimum state automaton equivalent to the above FSA has the following number of states

(A) 1 (B) 2 (C) 3 (D) 4

Linked Answer Questions: Q.76 to Q.85 carry two marks each.

Statement for Linked Answer Questions 76 & 77:

Suppose the letters a,b,c,d,e,f have probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}$, respectively.

Q.76 Which of the following is the Huffman code for the letters a,b,c,d,e,f ?

(A) 0,10,110,1110,11110,11111
(B) 11,10,011,010,001,000
(C) 11,10,01,001,0001,0000
(D) 110,100,010,000,001,111

Q.77 What is the average length of the correct answer to Q.76?

(A) 3 (B) 2.1875 (C) 2.25 (D) 1.9375

Statement for Linked Answer Questions 78 & 79:

Consider the CFG with $\{S, A, B\}$ as the non-terminal alphabet, $\{a, b\}$ as the terminal alphabet, S as the start symbol and the following set of production rules:

$S \rightarrow aB$ $\quad$ $S \rightarrow bA$
$B \rightarrow b$ $\quad$ $A \rightarrow a$
$B \rightarrow bS$ $\quad$ $A \rightarrow aS$
$B \rightarrow aBB$ $\quad$ $A \rightarrow bAA$

Q.78 Which of the following strings is generated by the grammar?

(A) *aaaabb* (B) *aabbbb* (C) *aabbab* (D) *abbbba*

Q.79 For the correct answer string to Q.78, how many derivation trees are there?

(A) 1 (B) 2 (C) 3 (D) 4

Statement for Linked Answer Questions 80 & 81:

Consider a machine with a byte addressable main memory of 2^{16} bytes. Assume that a direct mapped data cache consisting of 32 lines of 64 bytes each is used in the system. A 50 × 50 two-dimensional array of bytes is stored in the main memory starting from memory location 1100H. Assume that the data cache is initially empty. The complete array is accessed twice. Assume that the contents of the data cache do not change in between the two accesses.

Q.80 How many data cache misses will occur in total?

(A) 48 (B) 50 (C) 56 (D) 59

Q.81 Which of the following lines of the data cache will be replaced by new blocks in accessing the array for the second time?

(A) line 4 to line 11
(B) line 4 to line 12
(C) line 0 to line 7
(D) line 0 to line 8

Statement for Linked Answer Questions 82 & 83:

A process has been allocated 3 page frames. Assume that none of the pages of the process are available in the memory initially. The process makes the following sequence of page references (reference string): **1, 2, 1, 3, 7, 4, 5, 6, 3, 1**

Q.82 If optimal page replacement policy is used, how many page faults occur for the above reference string?

(A) 7 (B) 8 (C) 9 (D) 10

Q.83 Least Recently Used (LRU) page replacement policy is a practical approximation to optimal page replacement. For the above reference string, how many more page faults occur with LRU than with the optimal page replacement policy?

(A) 0 (B) 1 (C) 2 (D) 3

Statement for Linked Answer Questions 84 & 85:

Suppose that a robot is placed on the Cartesian plane. At each step it is allowed to move either one unit up or one unit right, i.e., if it is at (i, j) then it can move to either $(i+1, j)$ or $(i, j+1)$.

Q.84 How many distinct paths are there for the robot to reach the point $(10,10)$ starting from the initial position $(0,0)$?

(A) $\binom{20}{10}$

(B) 2^{20}

(C) 2^{10}

(D)None of the above

Q.85 Suppose that the robot is not allowed to traverse the line segment from (4,4) to (5,4). With this constraint, how many distinct paths are there for the robot to reach (10,10) starting from (0,0)?

(A) 2^9

(B) 2^{19}

(C) $\binom{8}{4} \times \binom{11}{5}$

(D) $\binom{20}{10} - \binom{8}{4} \times \binom{11}{5}$

END OF THE QUESTION PAPER

CS : COMPUTER SCIENCE AND ENGINEERING

Duration : Three Hours *Maximum Marks* :150

Read the following instructions carefully

1. This question paper contains **24** printed pages including pages for rough work. Please check all pages and report discrepancy, if any.

2. Write your registration number, your name and name of the examination centre at the specified locations on the right half of the ORS.

3. Using HB pencil, darken the appropriate bubble under each digit of your registration number and the letters corresponding to your paper code.

4. All the questions in this question paper are of objective type.

5. Questions must be answered on **O**bjective **R**esponse **S**heet (**ORS**) by darkening the appropriate bubble (marked A, B, C, D) using HB pencil against the question number on the left hand side of the ORS. **Each question has only one correct answer**. In case you wish to change an answer, erase the old answer completely. More than one answer bubbled against a question will be treated as a wrong answer.

6. Questions 1 through 20 are 1-mark questions and questions 21 through 85 are 2-mark questions.

7. Questions 71 through 73 is one set of common data questions, questions 74 and 75 is another pair of common data questions. The question pairs (76, 77), (78, 79), (80, 81), (82, 83) and (84, 85) are questions with linked answers. The answer to the second question of the above pairs will depend on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is un-attempted, then the answer to the second question in the pair will not be evaluated.

8. Un-attempted questions will carry zero marks.

9. **NEGATIVE MARKING**: For Q.1 to Q.20, ***0.25*** mark will be deducted for each wrong answer. For Q.21 to Q.75, ***0.5*** mark will be deducted for each wrong answer. For the pairs of questions with linked answers, there will be negative marks only for wrong answer to the first question, i.e. for Q.76, Q.78, Q.80, Q.82 and Q.84, ***0.5*** mark will be deducted for each wrong answer. There is no negative marking for Q.77, Q.79, Q.81, Q.83 and Q.85.

10. Calculator **without data connectivity** is allowed in the examination hall.

11. Charts, graph sheets and tables are NOT allowed in the examination hall.

12. Rough work can be done on the question paper itself. Additional blank pages are given at the end of the question paper for rough work.

Q. 1 – Q. 20 carry one mark each.

Q.1 $\lim_{x\to\infty}\frac{x-\sin x}{x+\cos x}$ equals

(A) 1 (B) -1 (C) ∞ (D) $-\infty$

Q.2 If P, Q, R are subsets of the universal set U, then $(P\cap Q\cap R)\cup(P^c\cap Q\cap R)\cup Q^c\cup R^c$ is

(A) $Q^c\cup R^c$ (B) $P\cup Q^c\cup R^c$ (C) $P^c\cup Q^c\cup R^c$ (D) U

Q.3 The following system of equations

$$x_1 + x_2 + 2x_3 = 1$$
$$x_1 + 2x_2 + 3x_3 = 2$$
$$x_1 + 4x_2 + \alpha x_3 = 4$$

has a unique solution. The only possible value(s) for α is/are

(A) 0 (B) either 0 or 1 (C) one of 0, 1 or -1 (D) any real number

Q.4 In the IEEE floating point representation the hexadecimal value 0x00000000 corresponds to

(A) the normalized value 2^{-127}
(B) the normalized value 2^{-126}
(C) the normalized value +0
(D) the special value +0

Q.5 In the Karnaugh map shown below, X denotes a don't care term. What is the minimal form of the function represented by the Karnaugh map?

cd \ ab	00	01	11	10
00	1	1		1
01	X			
11	X			
10	1	1		X

(A) $\bar{b}.\bar{d}+\bar{a}.\bar{d}$ (B) $\bar{a}.\bar{b}+\bar{b}.\bar{d}+\bar{a}.b.\bar{d}$
(C) $\bar{b}.\bar{d}+\bar{a}.b.\bar{d}$ (D) $\bar{a}.\bar{b}+\bar{b}.\bar{d}+\bar{a}.\bar{d}$

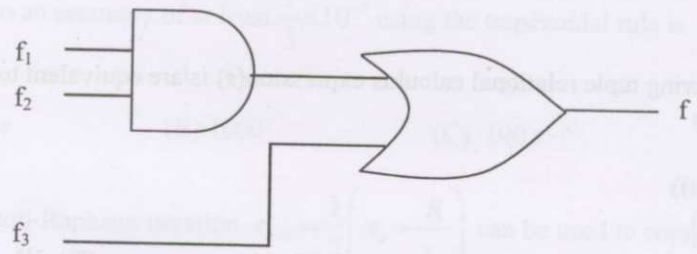
Q.6 Let r denote number system radix. The only value(s) of r that satisfy the equation $\sqrt{121_r}=11_r$ is/are

(A) decimal 10 (B) decimal 11
(C) decimal 10 and 11 (D) any value > 2

Q.7 The most efficient algorithm for finding the number of connected components in an undirected graph on n vertices and m edges has time complexity

(A) $\Theta(n)$ (B) $\Theta(m)$ (C) $\Theta(m+n)$ (D) $\Theta(mn)$

Q.8 Given f_1, f_3 and f in canonical sum of products form (in decimal) for the circuit



$f_1 = \Sigma\, m\,(4, 5, 6, 7, 8)$
$f_3 = \Sigma\, m\,(1, 6, 15)$
$f = \Sigma\, m\,(1, 6, 8, 15)$
then f_2 is

(A) $\Sigma\, m\,(4, 6)$ (B) $\Sigma\, m\,(4, 8)$ (C) $\Sigma\, m\,(6, 8)$ (D) $\Sigma\, m\,(4, 6, 8)$

Q.9 Which of the following is true for the language $\{a^p \mid p \text{ is a prime}\}$?

(A) It is not accepted by a Turing Machine
(B) It is regular but not context-free
(C) It is context-free but not regular
(D) It is neither regular nor context-free, but accepted by a Turing machine

Q.10 Which of the following are decidable?

I. Whether the intersection of two regular languages is infinite
II. Whether a given context-free language is regular
III. Whether two push-down automata accept the same language
IV. Whether a given grammar is context-free

(A) I and II (B) I and IV (C) II and III (D) II and IV

Q.11 Which of the following describes a handle (as applicable to LR-parsing) appropriately?

(A) It is the position in a sentential form where the next shift or reduce operation will occur
(B) It is a non-terminal whose production will be used for reduction in the next step
(C) It is a production that may be used for reduction in a future step along with a position in the sentential form where the next shift or reduce operation will occur
(D) It is the production *p* that will be used for reduction in the next step along with a position in the sentential form where the right hand side of the production may be found

Q.12 Some code optimizations are carried out on the intermediate code because

(A) they enhance the portability of the compiler to other target processors
(B) program analysis is more accurate on intermediate code than on machine code
(C) the information from dataflow analysis cannot otherwise be used for optimization
(D) the information from the front end cannot otherwise be used for optimization

Q.13 If L and $\overline{L}$ are recursively enumerable then L is

(A) regular (B) context-free
(C) context-sensitive (D) recursive

Q.14 What is the maximum size of data that the application layer can pass on to the TCP layer below?

(A) Any size
(B) 2^{16} bytes – size of TCP header
(C) 2^{16} bytes
(D) 1500 bytes

Q.15 Which of the following tuple relational calculus expression(s) is/are equivalent to $\forall t \in r(P(t))$?

I. $\neg\exists\, t \in r(P(t))$
II. $\exists\, t \notin r(P(t))$
III. $\neg\exists\, t \in r(\neg P(t))$
IV. $\exists\, t \notin r(\neg P(t))$

(A) I only (B) II only (C) III only (D) III and IV only

Q.16 A clustering index is defined on the fields which are of type

(A) non-key and ordering
(B) non-key and non-ordering
(C) key and ordering
(D) key and non-ordering

Q.17 Which of the following system calls results in the sending of SYN packets?

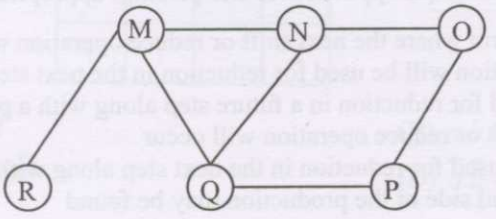
(A) socket (B) bind (C) listen (D) connect

Q.18 Which combination of the integer variables `x`, `y` and `z` makes the variable `a` get the value 4 in the following expression?

```
a = ( x > y ) ? (( x > z ) ? x : z ) : (( y > z ) ? y : z)
```

(A) x = 3, y = 4, z = 2
(B) x = 6, y = 5, z = 3
(C) x = 6, y = 3, z = 5
(D) x = 5, y = 4, z = 5

Q.19 The Breadth First Search algorithm has been implemented using the queue data structure. One possible order of visiting the nodes of the following graph is



(A) MNOPQR (B) NQMPOR (C) QMNPRO (D) QMNPOR

Q.20 The data blocks of a very large file in the Unix file system are allocated using

(A) contiguous allocation
(B) linked allocation
(C) indexed allocation
(D) an extension of indexed allocation

Q. 21 to Q.75 carry two marks each.

Q.21 The minimum number of equal length subintervals needed to approximate $\int_1^2 xe^x dx$ to an accuracy of at least $\frac{1}{3}\times 10^{-6}$ using the trapezoidal rule is

(A) $1000e$ (B) 1000 (C) $100e$ (D) 100

Q.22 The Newton-Raphson iteration $x_{n+1} = \frac{1}{2}\left(x_n + \frac{R}{x_n}\right)$ can be used to compute the

(A) square of R (B) reciprocal of R (C) square root of R (D) logarithm of R

Q.23 Which of the following statements is true for every planar graph on n vertices?

(A) The graph is connected
(B) The graph is Eulerian
(C) The graph has a vertex-cover of size at most $3n/4$
(D) The graph has an independent set of size at least $n/3$

Q.24 Let $P = \sum_{\substack{1\le i\le 2k \\ i \text{ odd}}} i$ and $Q = \sum_{\substack{1\le i\le 2k \\ i \text{ even}}} i$, where k is a positive integer. Then

(A) $P = Q - k$ (B) $P = Q + k$ (C) $P = Q$ (D) $P = Q + 2k$

Q.25 A point on a curve is said to be an extremum if it is a local minimum or a local maximum. The number of distinct extrema for the curve $3x^4 - 16x^3 + 24x^2 + 37$ is

(A) 0 (B) 1 (C) 2 (D) 3

Q.26 If P, Q, R are Boolean variables, then

$$\left(P+\bar{Q}\right)\left(P.\bar{Q}+P.R\right)\left(\bar{P}.\bar{R}+\bar{Q}\right)$$

simplifies to

(A) $P.\bar{Q}$ (B) $P.\bar{R}$ (C) $P.\bar{Q}+R$ (D) $P.\bar{R}+Q$

Q.27 Aishwarya studies either computer science or mathematics everyday. If she studies computer science on a day, then the probability that she studies mathematics the next day is 0.6. If she studies mathematics on a day, then the probability that she studies computer science the next day is 0.4. Given that Aishwarya studies computer science on Monday, what is the probability that she studies computer science on Wednesday?

(A) 0.24 (B) 0.36 (C) 0.4 (D) 0.6

Q.28 How many of the following matrices have an eigenvalue 1?

$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 \\ 1 & -1 \end{bmatrix}$

(A) one (B) two (C) three (D) four

Q.29 Let X be a random variable following normal distribution with mean +1 and variance 4. Let Y be another normal variable with mean -1 and variance unknown. If $P(X \leq -1) = P(Y \geq 2)$, the standard deviation of Y is

(A) 3 (B) 2 (C) $\sqrt{2}$ (D) 1

Q.30 Let *fsa* and *pda* be two predicates such that *fsa(x)* means *x* is a finite state automaton, and *pda(y)* means that *y* is a pushdown automaton. Let *equivalent* be another predicate such that *equivalent(a,b)* means *a* and *b* are equivalent. Which of the following first order logic statements represents the following:

Each finite state automaton has an equivalent pushdown automaton.

(A) $(\forall x\ fsa(x)) \Rightarrow (\exists y\ pda(y) \wedge equivalent\ (x, y))$
(B) $\sim\forall y\ (\exists x\ fsa(x) \Rightarrow pda\ (y) \wedge equivalent\ (x, y))$
(C) $\forall x\ \exists y\ (fsa(x) \wedge pda(y) \wedge equivalent\ (x, y))$
(D) $\forall x\ \exists y\ (fsa(y) \wedge pda(x) \wedge equivalent\ (x, y))$

Q.31 P and Q are two propositions. Which of the following logical expressions are equivalent?

I. $P \vee \sim Q$
II. $\sim(\sim P \wedge Q)$
III. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge \sim Q)$
IV. $(P \wedge Q) \vee (P \wedge \sim Q) \vee (\sim P \wedge Q)$

(A) Only I and II (B) Only I, II and III
(C) Only I, II and IV (D) All of I, II, III and IV

Q.32 For a magnetic disk with concentric circular tracks, the seek latency is not linearly proportional to the seek distance due to

(A) non-uniform distribution of requests
(B) arm starting and stopping inertia
(C) higher capacity of tracks on the periphery of the platter
(D) use of unfair arm scheduling policies

Q.33 Which of the following is/are true of the auto-increment addressing mode?

I. It is useful in creating self-relocating code
II. If it is included in an Instruction Set Architecture, then an additional ALU is required for effective address calculation
III. The amount of increment depends on the size of the data item accessed

(A) I only (B) II only (C) III only (D) II and III only

Q.34 Which of the following must be true for the RFE (Return from Exception) instruction on a general purpose processor?

I. It must be a trap instruction
II. It must be a privileged instruction
III. An exception cannot be allowed to occur during execution of an RFE instruction

(A) I only (B) II only (C) I and II only (D) I, II and III only

Q.35 For inclusion to hold between two cache levels L1 and L2 in a multi-level cache hierarchy, which of the following are necessary?

I. L1 must be a write-through cache
II. L2 must be a write-through cache
III. The associativity of L2 must be greater than that of L1
IV. The L2 cache must be at least as large as the L1 cache

(A) IV only (B) I and IV only (C) I, III and IV only (D) I, II, III and IV

Q.36 Which of the following are NOT true in a pipelined processor?

I. Bypassing can handle all RAW hazards.
II. Register renaming can eliminate all register carried WAR hazards.
III. Control hazard penalties can be eliminated by dynamic branch prediction.

(A) I and II only (B) I and III only (C) II and III only (D) I, II and III

Q.37 The use of multiple register windows with overlap causes a reduction in the number of memory accesses for

I. function locals and parameters
II. register saves and restores
III. instruction fetches

(A) I only (B) II only (C) III only (D) I, II and III

Q.38 In an instruction execution pipeline, the earliest that the data TLB (Translation Lookaside Buffer) can be accessed is

(A) before effective address calculation has started
(B) during effective address calculation
(C) after effective address calculation has completed
(D) after data cache lookup has completed

Q.39 Consider the following functions:

$f(n) = 2^n$

$g(n) = n!$

$h(n) = n^{\log n}$

Which of the following statements about the asymptotic behaviour of $f(n), g(n)$, and $h(n)$ is true?

(A) $f(n) = O(g(n)); g(n) = O(h(n))$ (B) $f(n) = \Omega(g(n)); g(n) = O(h(n))$
(C) $g(n) = O(f(n)); h(n) = O(f(n))$ (D) $h(n) = O(f(n)); g(n) = \Omega(f(n))$

Q.40 The minimum number of comparisons required to determine if an integer appears more than $n/2$ times in a sorted array of n integers is

(A) $\Theta(n)$ (B) $\Theta(\log n)$ (C) $\Theta(\log^* n)$ (D) $\Theta(1)$

Q.41 A B-tree of order 4 is built from scratch by 10 successive insertions. What is the maximum number of node splitting operations that may take place?

(A) 3 (B) 4 (C) 5 (D) 6

Q.42 G is a graph on n vertices and $2n$-2 edges. The edges of G can be partitioned into two edge-disjoint spanning trees. Which of the following is NOT true for G?

(A) For every subset of k vertices, the induced subgraph has at most $2k$-2 edges
(B) The minimum cut in G has at least two edges
(C) There are two edge-disjoint paths between every pair of vertices
(D) There are two vertex-disjoint paths between every pair of vertices

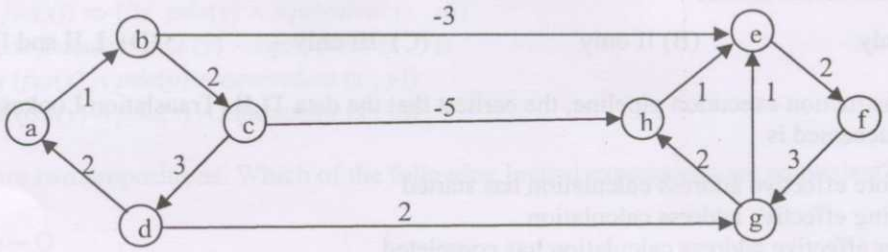
Q.43 Consider the Quicksort algorithm. Suppose there is a procedure for finding a pivot element which splits the list into two sub-lists each of which contains at least one-fifth of the elements. Let $T(n)$ be the number of comparisons required to sort n elements. Then

(A) $T(n) \le 2T(n/5) + n$ (B) $T(n) \le T(n/5) + T(4n/5) + n$
(C) $T(n) \le 2T(4n/5) + n$ (D) $T(n) \le 2T(n/2) + n$

Q.44 The subset-sum problem is defined as follows: Given a set S of n positive integers and a positive integer W, determine whether there is a subset of S whose elements sum to W.
An algorithm Q solves this problem in $O(nW)$ time. Which of the following statements is false?

(A) Q solves the subset-sum problem in polynomial time when the input is encoded in unary
(B) Q solves the subset-sum problem in polynomial time when the input is encoded in binary
(C) The subset sum problem belongs to the class NP
(D) The subset sum problem is NP-hard

Q.45



Dijkstra's single source shortest path algorithm when run from vertex a in the above graph, computes the correct shortest path distance to

(A) only vertex a
(B) only vertices a, e, f, g, h
(C) only vertices a, b, c, d
(D) all the vertices

Q.46 You are given the postorder traversal, P, of a binary search tree on the n elements $1, 2, \ldots, n$. You have to determine the unique binary search tree that has P as its postorder traversal. What is the time complexity of the most efficient algorithm for doing this?

(A) $\Theta(\log n)$
(B) $\Theta(n)$
(C) $\Theta(n \log n)$
(D) none of the above, as the tree cannot be uniquely determined.

Q.47 We have a binary heap on n elements and wish to insert n more elements (not necessarily one after another) into this heap. The total time required for this is

(A) $\Theta(\log n)$ (B) $\Theta(n)$ (C) $\Theta(n \log n)$ (D) $\Theta(n^2)$

Q.48 Which of the following statements is false?

(A) Every NFA can be converted to an equivalent DFA
(B) Every non-deterministic Turing machine can be converted to an equivalent deterministic Turing machine
(C) Every regular language is also a context-free language
(D) Every subset of a recursively enumerable set is recursive

Q.49 Given below are two finite state automata (→ indicates the start state and F indicates a final state)

Y:

	a	b
→ 1	1	2
2 (F)	2	1

Z:

	a	b
→ 1	2	2
2 (F)	1	1

Which of the following represents the product automaton Z ×Y?

(A)

	a	b
→ P	S	R
Q	R	S
R (F)	Q	P
S	Q	P

(B)

	a	b
→ P	S	Q
Q	R	S
R (F)	Q	P
S	P	Q

(C)

	a	b
→ P	Q	S
Q	R	S
R (F)	Q	P
S	Q	P

(D)

	a	b
→ P	S	Q
Q	S	R
R (F)	Q	P
S	Q	P

Q.50 Which of the following statements are true?

I. Every left-recursive grammar can be converted to a right-recursive grammar and vice-versa
II. All ε-productions can be removed from any context-free grammar by suitable transformations
III. The language generated by a context-free grammar all of whose productions are of the form $X \rightarrow w$ or $X \rightarrow wY$ (where, w is a string of terminals and Y is a non-terminal), is always regular
IV. The derivation trees of strings generated by a context-free grammar in Chomsky Normal Form are always binary trees

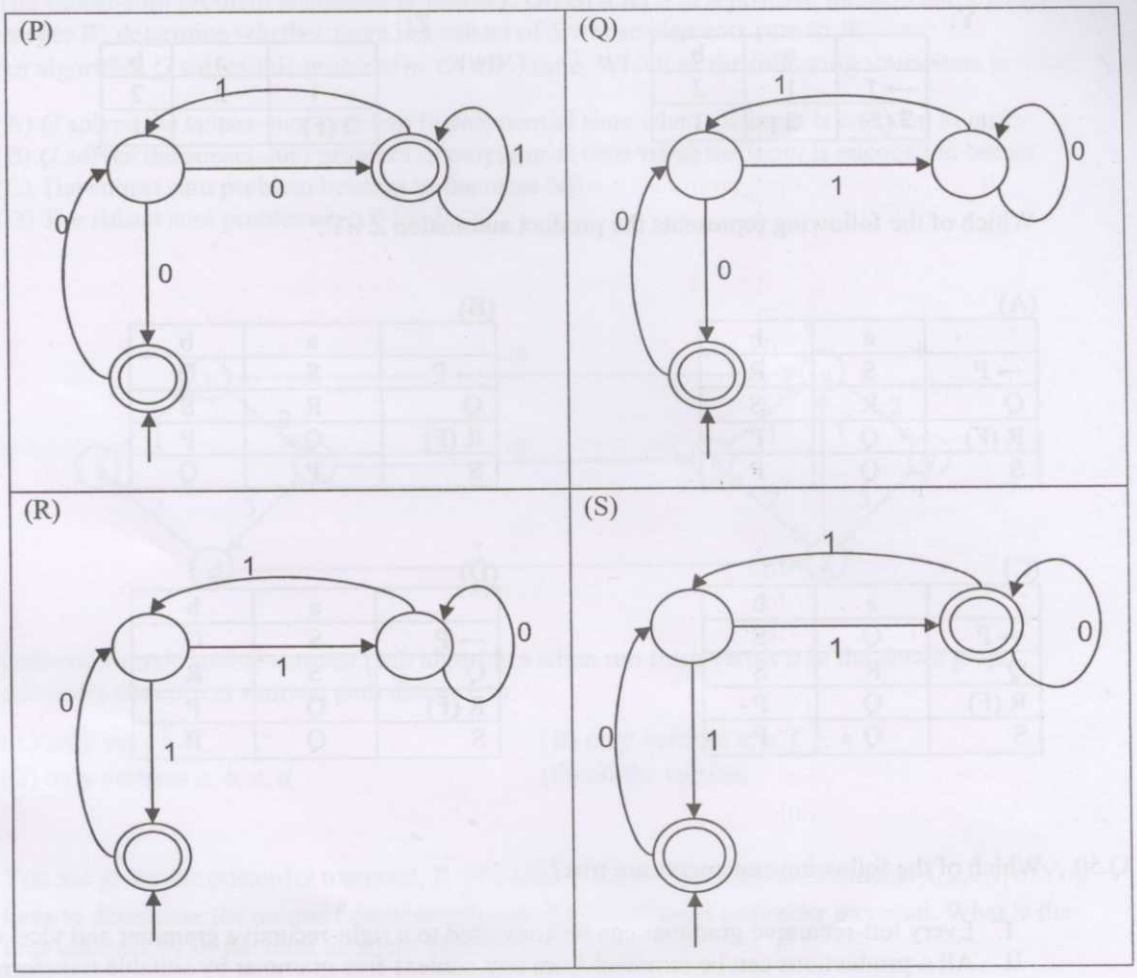
(A) I, II, III and IV
(B) II, III and IV only
(C) I, III and IV only
(D) I, II and IV only

Q.51 Match the following:

(E) Checking that identifiers are declared before their use	(P) $L = \{a^n b^m c^n d^m \mid n \geq 1, m \geq 1\}$
(F) Number of formal parameters in the declaration of a function agrees with the number of actual parameters in a use of that function	(Q) $X \rightarrow X\,b\,X \mid X\,c\,X \mid d\,X\,f \mid g$
(G) Arithmetic expressions with matched pairs of parentheses	(R) $L = \{ wcw \mid w \in (a \mid b)^* \}$
(H) Palindromes	(S) $X \rightarrow b\,X\,b \mid c\,X\,c \mid \varepsilon$

(A) E – P, F – R, G – Q, H – S
(B) E – R, F – P, G – S, H – Q
(C) E – R, F – P, G – Q, H – S
(D) E – P, F – R, G – S, H – Q

Q.52 Match the following NFAs with the regular expressions they correspond to.



1. $\varepsilon + 0(01^*1+00)^*01^*$
2. $\varepsilon + 0(10^*1+00)^*0$
3. $\varepsilon + 0(10^*1+10)^*1$
4. $\varepsilon + 0(10^*1+10)^*10^*$

(A) P – 2, Q – 1, R – 3, S – 4
(B) P – 1, Q – 3, R – 2, S – 4
(C) P – 1, Q – 2, R – 3, S – 4
(D) P – 3, Q – 2, R – 1, S – 4

Q.53 Which of the following are regular sets?

I. $\{a^n b^{2m} \mid n \geq 0, m \geq 0\}$
II. $\{a^n b^m \mid n = 2m\}$
III. $\{a^n b^m \mid n \neq m\}$
IV. $\{xcy \mid x,y \in \{a,b\}^*\}$

(A) I and IV only (B) I and III only (C) I only (D) IV only

Q.54 Which of the following are true?

I. A programming language which does not permit global variables of any kind and has no nesting of procedures/functions, but permits recursion can be implemented with static storage allocation
II. Multi-level access link (or display) arrangement is needed to arrange activation records only if the programming language being implemented has nesting of procedures/functions
III. Recursion in programming languages cannot be implemented with dynamic storage allocation
IV. Nesting of procedures/functions and recursion require a dynamic heap allocation scheme and cannot be implemented with a stack-based allocation scheme for activation records
V. Programming languages which permit a function to return a function as its result cannot be implemented with a stack-based storage allocation scheme for activation records

(A) II and V only (B) I, III and IV only (C) I, II and V only (D) II, III and V only

Q.55 An LALR(1) parser for a grammar G can have shift-reduce (S-R) conflicts if and only if

(A) the SLR(1) parser for G has S-R conflicts
(B) the LR(1) parser for G has S-R conflicts
(C) the LR(0) parser for G has S-R conflicts
(D) the LALR(1) parser for G has reduce-reduce conflicts

Q.56 In the slow start phase of the TCP congestion control algorithm, the size of the congestion window

(A) does not increase (B) increases linearly
(C) increases quadratically (D) increases exponentially

Q.57 If a class B network on the Internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

(A) 1022 (B) 1023 (C) 2046 (D) 2047

Q.58 A computer on a 10Mbps network is regulated by a token bucket. The token bucket is filled at a rate of 2Mbps. It is initially filled to capacity with 16 Megabits. What is the maximum duration for which the computer can transmit at the full 10Mbps?

(A) 1.6 seconds (B) 2 seconds (C) 5 seconds (D) 8 seconds

Q.59 A client process P needs to make a TCP connection to a server process S. Consider the following situation: the server process S executes a `socket()`, a `bind()` and a `listen()` system call in that order, following which it is preempted. Subsequently, the client process P executes a `socket()` system call followed by `connect()` system call to connect to the server process S. The server process has not executed any `accept()` system call. Which one of the following events could take place?

(A) `connect()` system call returns successfully
(B) `connect()` system call blocks
(C) `connect()` system call returns an error
(D) `connect()` system call results in a core dump

Q.60 What is printed by the following C program?

```
int f(int x, int *py, int **ppz)
{
    int y,z;
    **ppz += 1; z = *ppz;
    *py += 2; y = *py;
    x += 3;
    return x+y+z;
}
void main()
{
    int c, *b, **a;
    c = 4; b = &c; a = &b;
    printf("%d",f(c,b,a));
}
```

(A) 18 (B) 19 (C) 21 (D) 22

Q.61 Choose the correct option to fill ?1 and ?2 so that the program below prints an input string in reverse order. Assume that the input string is terminated by a newline character.

```
void reverse(void){
  int c;
  if(?1) reverse();
  ?2
}
main(){
  printf("Enter Text"); printf("\n");
  reverse(); printf("\n");
}
```

(A) ?1 is (getchar() != '\n')
?2 is getchar(c);

(B) ?1 is (c = getchar()) != '\n')
?2 is getchar(c);

(C) ?1 is (c != '\n')
?2 is putchar(c);

(D) ?1 is ((c = getchar()) != '\n')
?2 is putchar(c);

Q.62 The following C function takes a singly-linked list of integers as a parameter and rearranges the elements of the list. The function is called with the list containing the integers 1,2,3,4,5,6,7 in the given order. What will be the contents of the list after the function completes execution?

```
struct node {
  int value;
  struct node *next;
};
void rearrange (struct node *list){
  struct node *p, *q;
  int temp;
  if (!list || !list -> next) return;
  p = list; q = list -> next;
  while (q) {
    temp = p -> value; p -> value = q -> value;
    q -> value = temp; p = q -> next;
    q = p ? p -> next : 0;
  }
}
```

(A) 1,2,3,4,5,6,7 (B) 2,1,4,3,6,5,7
(C) 1,3,2,5,4,7,6 (D) 2,3,4,5,6,7,1

Q.63 The P and V operations on counting semaphores, where s is a counting semaphore, are defined as follows:

P(s): s = s – 1;
if s < 0 then wait;

V(s): s = s + 1;
if s <= 0 then wakeup a process waiting on s;

Assume that P_b and V_b, the wait and signal operations on binary semaphores are provided. Two binary semaphores x_b and y_b are used to implement the semaphore operations P(s) and V(s) as follows:

```
P(s):   Pb(xb);
        s = s - 1;
        if (s < 0) {
          Vb(xb);
          Pb(yb);
        }
        else Vb(xb);

V(s):   Pb(xb);
        s = s + 1;
        if (s <= 0) Vb(yb);
        Vb(xb);
```

The initial values of x_b and y_b are respectively

(A) 0 and 0 (B) 0 and 1 (C) 1 and 0 (D) 1 and 1

Q.64 Which of the following statements about synchronous and asynchronous I/O is NOT true?

(A) An ISR is invoked on completion of I/O in synchronous I/O but not in asynchronous I/O
(B) In both synchronous and asynchronous I/O, an ISR (Interrupt Service Routine) is invoked after completion of the I/O
(C) A process making a synchronous I/O call waits until I/O is complete, but a process making an asynchronous I/O call does not wait for completion of the I/O
(D) In the case of synchronous I/O, the process waiting for the completion of I/O is woken up by the ISR that is invoked after the completion of I/O

Q.65 Which of the following is NOT true of deadlock prevention and deadlock avoidance schemes?

(A) In deadlock prevention, the request for resources is always granted if the resulting state is safe
(B) In deadlock avoidance, the request for resources is always granted if the resulting state is safe
(C) Deadlock avoidance is less restrictive than deadlock prevention
(D) Deadlock avoidance requires knowledge of resource requirements *a priori*

Q.66 A process executes the following code

```
for(i=0; i<n; i++) fork();
```

The total number of child processes created is

(A) n (B) $2^n - 1$ (C) 2^n (D) $2^{n+1} - 1$

Q.67 A processor uses 36 bit physical addresses and 32 bit virtual addresses, with a page frame size of 4 Kbytes. Each page table entry is of size 4 bytes. A three level page table is used for virtual-to-physical address translation, where the virtual address is used as follows

- bits 30-31 are used to index into the first level page table,
- bits 21-29 are used to index into the second level page table,
- bits 12-20 are used to index into the third level page table, and
- bits 0-11 are used as offset within the page.

The number of bits required for addressing the next level page table (or page frame) in the page table entry of the first, second and third level page tables are respectively

(A) 20, 20 and 20 (B) 24, 24 and 24 (C) 24, 24 and 20 (D) 25, 25 and 24

Q.68 Let R and S be two relations with the following schema

R(P, Q, R1, R2, R3)
S (P, Q, S1, S2)

where {P,Q} is the key for both schemas. Which of the following queries are equivalent?

I. $\Pi_P (R \bowtie S)$
II. $\Pi_P (R) \bowtie \Pi_P (S)$
III. $\Pi_P (\Pi_{P,Q} (R) \cap \Pi_{P,Q} (S))$
IV. $\Pi_P (\Pi_{P,Q} (R) - (\Pi_{P,Q} (R) - \Pi_{P,Q} (S)))$

(A) Only I and II (B) Only I and III
(C) Only I , II and III (D) Only I, III and IV

Q.69 Consider the following relational schemes for a library database:

Book (Title, Author, Catalog_no, Publisher, Year, Price)
Collection (Title, Author, Catalog_no)

with the following functional dependencies:

I. Title Author → Catalog_no
II. Catalog_no → Title Author Publisher Year
III. Publisher Title Year → Price

Assume {Author,Title} is the key for both schemes. Which of the following statements is true?

(A) Both Book and Collection are in BCNF
(B) Both Book and Collection are in 3NF only
(C) Book is in 2NF and Collection is in 3NF
(D) Both Book and Collection are in 2NF only

Q.70 Consider a file of 16384 records. Each record is 32 bytes long and its key field is of size 6 bytes. The file is ordered on a non-key field, and the file organization is unspanned. The file is stored in a file system with block size 1024 bytes, and the size of a block pointer is 10 bytes. If the secondary index is built on the key field of the file, and a multi-level index scheme is used to store the secondary index, the number of first-level and second-level blocks in the multi-level index are respectively

(A) 8 and 0 (B) 128 and 6 (C) 256 and 4 (D) 512 and 5

Common Data Questions

Common Data for Questions 71,72 and 73:

Consider a machine with a 2-way set associative data cache of size 64 Kbytes and block size 16 bytes. The cache is managed using 32 bit virtual addresses and the page size is 4 Kbytes. A program to be run on this machine begins as follows:

```
double ARR [1024] [1024];
int i, j ;

 /* Initialize array ARR to 0.0 */
  for (i = 0; i < 1024; i++)
    for (j = 0; j < 1024; j++)
      ARR [i] [j] = 0.0;
```

The size of double is 8Bytes. Array ARR is located in memory starting at the beginning of virtual page 0xFF000 and stored in row major order. The cache is initially empty and no pre-fetching is done. The only data memory references made by the program are those to array ARR.

Q.71 The total size of the tags in the cache directory is

(A) 32 Kbits (B) 34 Kbits (C) 64 Kbits (D) 68 Kbits

Q.72 Which of the following array elements has the same cache index as ARR [0] [0]?

(A) ARR [0] [4] (B) ARR [4] [0] (C) ARR [0] [5] (D) ARR [5] [0]

Q.73 The cache hit ratio for this initialization loop is

(A) 0% (B) 25% (C) 50% (D) 75%

Common Data for Questions 74 and 75:

Consider the following C functions:

```
int f1 ( int n )
{
  if (n == 0 || n == 1)
    return n;
  else
    return (2*f1(n-1) + 3*f1(n-2));
}

int f2 ( int n )
{
  int i;
  int X[N], Y[N], Z[N];

  X[0] = Y[0] = Z[0] = 0;
  X[1] = 1; Y[1] = 2; Z[1] = 3;
  for (i = 2; i <= n; i++){
    X[i] = Y[i-1] + Z[i-2];
    Y[i] = 2 * X[i];
    Z[i] = 3 * X[i];
  }
  return X[n];
}
```

Q.74 The running time of `f1(n)` and `f2(n)` are

(A) $\Theta(n)$ and $\Theta(n)$ (B) $\Theta(2^n)$ and $\Theta(n)$

(C) $\Theta(n)$ and $\Theta(2^n)$ (D) $\Theta(2^n)$ and $\Theta(2^n)$

Q.75 `f1(8)` and `f2(8)` return the values

(A) 1661 and 1640 (B) 59 and 59 (C) 1640 and 1640 (D) 1640 and 1661

Linked Answer Questions: Q.76 to Q.85 carry two marks each.

Statement for Linked Answer Questions 76 and 77:

Delayed branching can help in the handling of control hazards.

Q.76 For all delayed conditional branch instructions, irrespective of whether the condition evaluates to true or false,

(A) the instruction following the conditional branch instruction in memory is executed
(B) the first instruction in the fall through path is executed
(C) the first instruction in the taken path is executed
(D) the branch takes longer to execute than any other instruction

Q.77 The following code is to run on a pipelined processor with one branch delay slot:

I1: ADD R2 ← R7 + R8
I2: SUB R4 ← R5 – R6
I3: ADD R1 ← R2 + R3
I4: STORE Memory[R4] ← R1
BRANCH to Label if R1 == 0

Which of the instructions I1, I2, I3 or I4 can legitimately occupy the delay slot without any other program modification?

(A) I1 (B) I2 (C) I3 (D) I4

Statement for Linked Answer Questions 78 and 79:

Let x_n denote the number of binary strings of length n that contain no consecutive 0*s*.

Q.78 Which of the following recurrences does x_n satisfy?

(A) $x_n = 2x_{n-1}$ (B) $x_n = x_{\lfloor n/2 \rfloor} + 1$ (C) $x_n = x_{\lfloor n/2 \rfloor} + n$ (D) $x_n = x_{n-1} + x_{n-2}$

Q.79 The value of x_5 is

(A) 5 (B) 7 (C) 8 (D) 16

Statement for Linked Answer Questions 80 and 81:

The subset-sum problem is defined as follows. Given a set of n positive integers, $S = \{a_1, a_2, a_3, \ldots, a_n\}$, and a positive integer W, is there a subset of S whose elements sum to W? A dynamic program for solving this problem uses a 2-dimensional Boolean array, X, with n rows and W+1 columns. $X[i, j]$, $1 \le i \le n$, $0 \le j \le W$, is TRUE if and only if there is a subset of $\{a_1, a_2, \ldots, a_i\}$ whose elements sum to j.

Q.80 Which of the following is valid for $2 \le i \le n$ and $a_i \le j \le W$?

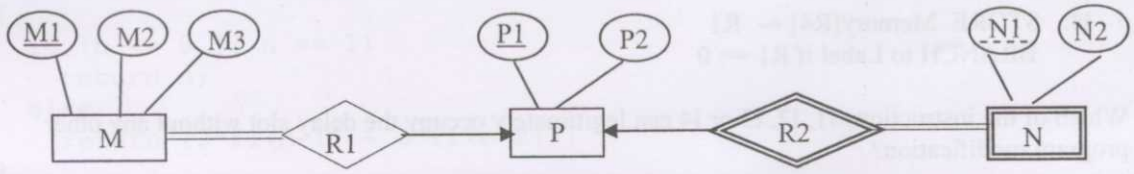
(A) $X[i, j] = X[i-1, j] \vee X[i, j-a_i]$
(B) $X[i, j] = X[i-1, j] \vee X[i-1, j-a_i]$
(C) $X[i, j] = X[i-1, j] \wedge X[i, j-a_i]$
(D) $X[i, j] = X[i-1, j] \wedge X[i-1, j-a_i]$

Q.81 Which entry of the array X, if TRUE, implies that there is a subset whose elements sum to W?

(A) $X[1, W]$ (B) $X[n, 0]$ (C) $X[n, W]$ (D) $X[n-1, n]$

Statement for Linked Answer Questions 82 and 83:

Consider the following ER diagram



Q.82 The minimum number of tables needed to represent M,N,P,R1,R2 is

(A) 2 (B) 3 (C) 4 (D) 5

Q.83 Which of the following is a correct attribute set for one of the tables for the correct answer to the above question?

(A) {M1, M2, M3, P1} (B) {M1, P1, N1, N2}
(C) {M1, P1, N1} (D) {M1, P1}

Statement for Linked Answer Questions 84 and 85:

Consider the following C program that attempts to locate an element x in an array Y[] using binary search. The program is erroneous.

```
1.    f(int Y[10], int x){
2.        int i, j, k;
3.        i = 0; j = 9;
4.        do {
5.             k = (i+j)/2;
6.             if (Y[k] < x) i = k; else j = k;
7.           } while ((Y[k] != x) && (i < j));
8.        if (Y[k] == x) printf("x is in the array");
9.        else printf("x is not in the array");
10.   }
```

Q.84 On which of the following contents of Y and x does the program fail?

(A) Y is [1 2 3 4 5 6 7 8 9 10] and x < 10
(B) Y is [1 3 5 7 9 11 13 15 17 19] and x < 1
(C) Y is [2 2 2 2 2 2 2 2 2 2] and x > 2
(D) Y is [2 4 6 8 10 12 14 16 18 20] and 2 < x < 20 and x is even

Q.85 The correction needed in the program to make it work properly is

(A) change line 6 to: `if (Y[k] < x) i = k+1; else j = k-1;`
(B) change line 6 to: `if (Y[k] < x) i = k-1; else j = k+1;`
(C) change line 6 to: `if (Y[k] <= x) i = k; else j = k;`
(D) change line 7 to: `} while ((Y[k] == x) && (i < j));`

END OF THE QUESTION PAPER

CS : COMPUTER SCIENCE & INFORMATION TECHNOLOGY

Duration: Three Hours *Maximum Marks* :100

Read the following instructions carefully.

1. This question paper contains **16** printed pages including pages for rough work. Please check all pages and report discrepancy, if any.

2. Write your registration number, your name and name of the examination centre at the specified locations on the right half of the **O**ptical **R**esponse **S**heet (**ORS**).

3. Using HB pencil, darken the appropriate bubble under each digit of your registration number and the letters corresponding to your paper code.

4. All questions in this paper are of objective type.

5. Questions must be answered on **O**ptical **R**esponse **S**heet (**ORS**) by darkening the appropriate bubble (marked A, B, C, D) using HB pencil against the question number on the left hand side of the ORS. **Each question has only one correct answer**. In case you wish to change an answer, erase the old answer completely. More than one answer bubbled against a question will be treated as an incorrect response.

6. There are a total of 60 questions carrying 100 marks. Questions 1 through 20 are 1-mark questions, questions 21 through 60 are 2-mark questions.

7. Questions 51 through 56 (3 pairs) are common data questions and question pairs (57, 58) and (59, 60) are linked answer questions. The answer to the second question of the above 2 pairs depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is un-attempted, then the answer to the second question in the pair will not be evaluated.

8. Un-attempted questions will carry zero marks.

9. Wrong answers will carry NEGATIVE marks. For Q.1 to Q.20, ⅓ mark will be deducted for each wrong answer. For Q. 21 to Q. 56, ⅔ mark will be deducted for each wrong answer. The question pairs (Q.57, Q.58), and (Q.59, Q.60) are questions with linked answers. There will be negative marks only for wrong answer to the first question of the linked answer question pair i.e. for Q.57 and Q.59, ⅔ mark will be deducted for each wrong answer. There is no negative marking for Q.58 and Q.60.

10. Calculator (without data connectivity) is allowed in the examination hall.

11. Charts, graph sheets or tables are NOT allowed in the examination hall.

12. Rough work can be done on the question paper itself. Additionally, blank pages are given at the end of the question paper for rough work.

Q. 1 – Q. 20 carry one mark each.

Q.1 Which one of the following is NOT necessarily a property of a Group ?

(A) Commutativity
(B) Associativity
(C) Existence of inverse for every element
(D) Existence of identity

Q.2 What is the chromatic number of an n-vertex simple connected graph which does not contain any odd length cycle ? Assume $n \geq 2$.

(A) 2 (B) 3 (C) $n-1$ (D) n

Q.3 Which one of the following is TRUE for any simple connected undirected graph with more than 2 vertices ?

(A) No two vertices have the same degree.
(B) At least two vertices have the same degree.
(C) At least three vertices have the same degree.
(D) All vertices have the same degree.

Q.4 Consider the binary relation $R = \{(x, y), (x, z), (z, x), (z, y)\}$ on the set $\{x, y, z\}$. Which one of the following is TRUE ?

(A) R is symmetric but NOT antisymmetric.
(B) R is NOT symmetric but antisymmetric.
(C) R is both symmetric and antisymmetric.
(D) R is neither symmetric nor antisymmetric.

Q.5 $(1217)_8$ is equivalent to

(A) $(1217)_{16}$ (B) $(028F)_{16}$ (C) $(2297)_{10}$ (D) $(0B17)_{16}$

Q.6 What is the minimum number of gates required to implement the Boolean function (AB + C) if we have to use only 2-input NOR gates ?

(A) 2 (B) 3 (C) 4 (D) 5

Q.7 How many 32K × 1 RAM chips are needed to provide a memory capacity of 256 K-bytes ?

(A) 8 (B) 32 (C) 64 (D) 128

Q.8 A CPU generally handles an interrupt by executing an interrupt service routine

(A) as soon as an interrupt is raised.
(B) by checking the interrupt register at the end of fetch cycle.
(C) by checking the interrupt register after finishing the execution of the current instruction.
(D) by checking the interrupt register at fixed time intervals.

Q.9 In which one of the following page replacement policies, Belady's anomaly may occur ?

(A) FIFO (B) Optimal (C) LRU (D) MRU

Q.10 The essential content(s) in each entry of a page table is/are

(A) virtual page number.
(B) page frame number.
(C) both virtual page number and page frame number.
(D) access right information.

Q.11 What is the number of swaps required to sort n elements using selection sort, in the worst case ?

(A) $\Theta(n)$ (B) $\Theta(n \log n)$ (C) $\Theta(n^2)$ (D) $\Theta(n^2 \log n)$

Q.12 $S \rightarrow aSa \mid bSb \mid a \mid b$

The language generated by the above grammar over the alphabet {a, b} is the set of

(A) all palindromes.
(B) all odd length palindromes.
(C) strings that begin and end with the same symbol.
(D) all even length palindromes.

Q.13 Which of the following statement(s) is/are correct regarding Bellman-Ford shortest path algorithm ?

P. Always finds a negative weighted cycle, if one exists.
Q. Finds whether any negative weighted cycle is reachable from the source.

(A) P only (B) Q only (C) both P and Q (D) neither P nor Q

Q.14 Let π_A be a problem that belongs to the class NP. Then which one of the following is TRUE ?

(A) There is no polynomial time algorithm for π_A.
(B) If π_A can be solved deterministically in polynomial time, then P = NP.
(C) If π_A is NP-hard, then it is NP-complete.
(D) π_A may be undecidable.

Q.15 Which one of the following languages over the alphabet {0, 1} is described by the regular expression: $(0 + 1)^*0(0 + 1)^*0(0 + 1)^*$?

(A) The set of all strings containing the substring 00.
(B) The set of all strings containing at most two 0's.
(C) The set of all strings containing at least two 0's.
(D) The set of all strings that begin and end with either 0 or 1.

Q.16 Which one of the following is FALSE ?

(A) There is a unique minimal DFA for every regular language.
(B) Every NFA can be converted to an equivalent PDA.
(C) Complement of every context-free language is recursive.
(D) Every nondeterministic PDA can be converted to an equivalent deterministic PDA.

Q.17 Match all items in **Group 1** with correct options from those given in **Group 2**.

Group 1	Group 2
P. Regular expression	1. Syntax analysis
Q. Pushdown automata	2. Code generation
R. Dataflow analysis	3. Lexical analysis
S. Register allocation	4. Code Optimization

(A) P – 4, Q – 1, R – 2, S – 3 (B) P – 3, Q – 1, R – 4, S – 2
(C) P – 3, Q – 4, R – 1, S – 2 (D) P – 2, Q – 1, R – 4, S – 3

Q.18 Consider the program below:

```
#include <stdio.h>
int fun(int n, int *f_p) {
  int t, f;
  if (n <= 1) {
    *f_p = 1 ;
    return 1 ;
  }
  t = fun (n-1, f_p) ;
  f = t + *f_p ;
  *f_p = t ;
  return f ;
}
int main() {
  int x = 15;
  printf ("%d\n", fun(5, &x));
  return 0;
}
```

The value printed is :

(A) 6 (B) 8 (C) 14 (D) 15

Q.19 The coupling between different modules of a software is categorized as follows :

I. Content coupling
II. Common coupling
III. Control coupling
IV. Stamp coupling
V. Data coupling

Coupling between modules can be ranked in the order of strongest (least desirable) to weakest (most desirable) as follows :

(A) I – II – III – IV – V (B) V – IV – III – II – I
(C) I – III – V – II – IV (D) IV – II – V – III – I

Q.20 Consider the HTML table definition given below :

```
<table border=1>
 <tr> <td rowspan=2> ab </td>
      <td colspan=2> cd </td>
 </tr>
 <tr> <td> ef </td>
      <td rowspan=2> gh </td>
 </tr>
 <tr> <td colspan=2> ik </td>
 </tr>
</table>
```

The number of rows in each column and the number of columns in each row are :

(A) $\langle 2,2,3 \rangle$ and $\langle 2,3,2 \rangle$ (B) $\langle 2,2,3 \rangle$ and $\langle 2,2,3 \rangle$
(C) $\langle 2,3,2 \rangle$ and $\langle 2,3,2 \rangle$ (D) $\langle 2,3,2 \rangle$ and $\langle 2,2,3 \rangle$

Q. 21 to Q. 60 carry two marks each.

Q.21 An unbalanced dice (with 6 faces, numbered from 1 to 6) is thrown. The probability that the face value is odd is 90% of the probability that the face value is even. The probability of getting any even numbered face is the same.

If the probability that the face is even given that it is greater than 3 is 0.75, which one of the following options is closest to the probability that the face value exceeds 3 ?

(A) 0.453 (B) 0.468 (C) 0.485 (D) 0.492

Q.22 For the composition table of a cyclic group shown below

*	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	b	a
d	d	c	a	b

Which one of the following choices is correct ?

(A) a, b are generators (B) b, c are generators
(C) c, d are generators (D) d, a are generators

Q.23 Which one of the following is the most appropriate logical formula to represent the statement :

"*Gold and silver ornaments are precious*"

The following notations are used :
$G(x)$: x is a gold ornament.
$S(x)$: x is a silver ornament.
$P(x)$: x is precious.

(A) $\forall x(P(x) \rightarrow (G(x) \wedge S(x)))$ (B) $\forall x((G(x) \wedge S(x)) \rightarrow P(x))$
(C) $\exists x((G(x) \wedge S(x)) \rightarrow P(x))$ (D) $\forall x((G(x) \vee S(x)) \rightarrow P(x))$

Q.24 The binary operation $\square$ is defined as follows :

P	Q	P $\square$ Q
T	T	T
T	F	T
F	T	F
F	F	T

Which one of the following is equivalent to P $\vee$ Q ?

(A) $\neg$Q $\square$ $\neg$ P (B) P $\square$ $\neg$Q (C) $\neg$P $\square$ Q (D) $\neg$P $\square$ $\neg$Q

Q.25 $\int_0^{\pi/4} (1-\tan x)/(1+\tan x)dx$

evaluates to

(A) 0 (B) 1 (C) ln 2 (D) ½ ln 2

Q.26 Consider the following well-formed formulae :

I. $\neg\forall x(P(x))$

II. $\neg\exists x(P(x))$

III. $\neg\exists x(\neg P(x))$

IV. $\exists x(\neg P(x))$

Which of the above are equivalent ?

(A) I and III (B) I and IV (C) II and III (D) II and IV

Q.27 Given the following state table of an FSM with two states A and B, one input and one output :

Present State A	Present State B	Input	Next State A	Next State B	Output
0	0	0	0	0	1
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	1	0	0
0	0	1	0	1	0
0	1	1	0	0	1
1	0	1	0	1	1
1	1	1	0	0	1

If the initial state is A = 0, B = 0, what is the minimum length of an input string which will take the machine to the state A = 0, B = 1 with Output = 1?

(A) 3 (B) 4 (C) 5 (D) 6

Q.28 Consider a 4 stage pipeline processor. The number of cycles needed by the four instructions I1, I2, I3, I4 in stages S1, S2, S3, S4 is shown below :

	S1	S2	S3	S4
I1	2	1	1	1
I2	1	3	2	2
I3	2	1	1	3
I4	1	2	2	2

What is the number of cycles needed to execute the following loop ?

for (i = 1 to 2) {I1; I2; I3; I4;}

(A) 16 (B) 23 (C) 28 (D) 30

Q.29 Consider a 4-way set associative cache (initially empty) with total 16 cache blocks. The main memory consists of 256 blocks and the request for memory blocks is in the following order :

0, 255, 1, 4, 3, 8, 133, 159, 216, 129, 63, 8, 48, 32, 73, 92, 155.

Which one of the following memory block will NOT be in cache if LRU replacement policy is used ?

(A) 3 (B) 8 (C) 129 (D) 216

Q.30 Consider a system with 4 types of resources R1 (3 units), R2 (2 units), R3 (3 units), R4 (2 units). A non-preemptive resource allocation policy is used. At any given instance, a request is not entertained if it cannot be completely satisfied. Three processes P1, P2, P3 request the resources as follows if executed independently.

Process P1:	Process P2:	Process P3:
t = 0: requests 2 units of R2	t = 0: requests 2 units of R3	t = 0: requests 1 unit of R4
t = 1: requests 1 unit of R3	t = 2: requests 1 unit of R4	t = 2: requests 2 units of R1
t = 3: requests 2 units of R1	t = 4: requests 1 unit of R1	t = 5: releases 2 units of R1
t = 5: releases 1 unit of R2 and 1 unit of R1	t = 6: releases 1 unit of R3	t = 7: requests 1 unit of R2
t = 7: releases 1 unit of R3	t = 8: Finishes	t = 8: requests 1 unit of R3
t = 8: requests 2 units of R4		t = 9: Finishes
t = 10: Finishes		

Which one of the following statements is TRUE if all three processes run concurrently starting at time t = 0 ?

(A) All processes will finish without any deadlock.
(B) Only P1 and P2 will be in deadlock.
(C) Only P1 and P3 will be in deadlock.
(D) All three processes will be in deadlock.

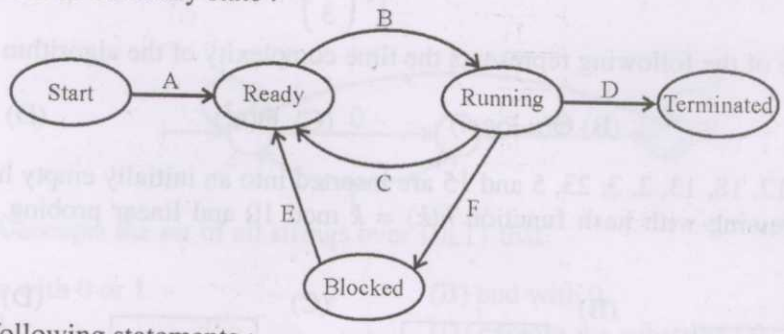
Q.31 Consider a disk system with 100 cylinders. The requests to access the cylinders occur in following sequence :

4, 34, 10, 7, 19, 73, 2, 15, 6, 20.

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and shortest seek time first policy is used ?

(A) 95 ms (B) 119 ms (C) 233 ms (D) 276 ms

Q.32 In the following process state transition diagram for a uniprocessor system, assume that there are always some processes in the ready state :



Now consider the following statements :

I. If a process makes a transition D, it would result in another process making transition A immediately.
II. A process P_2 in blocked state can make transition E while another process P_1 is in running state.
III. The OS uses preemptive scheduling.
IV. The OS uses non-preemptive scheduling.

Which of the above statements are TRUE ?

(A) I and II (B) I and III (C) II and III (D) II and IV

Q.33 The **enter_CS()** and **leave_CS()** functions to implement critical section of a process are realized using test-and-set instruction as follows :

```
void enter_CS(X)
{
        while (test-and-set(X));
}
void leave_CS(X)
{
        X=0;
}
```

In the above solution, X is a memory location associated with the CS and is initialized to 0. Now consider the following statements :

I. The above solution to CS problem is deadlock-free.
II. The solution is starvation free.
III. The processes enter CS in FIFO order.
IV. More than one process can enter CS at the same time.

Which of the above statements are TRUE ?

(A) I only (B) I and II (C) II and III (D) IV only

Q.34 A multilevel page table is preferred in comparison to a single level page table for translating virtual address to physical address because

(A) it reduces the memory access time to read or write a memory location.
(B) it helps to reduce the size of page table needed to implement the virtual address space of a process.
(C) it is required by the translation lookaside buffer.
(D) it helps to reduce the number of page faults in page replacement algorithms.

Q.35 The running time of an algorithm is represented by the following recurrence relation :

$$T(n) = \begin{cases} n & n \le 3 \\ T\left(\frac{n}{3}\right) + cn & otherwise \end{cases}$$

Which one of the following represents the time complexity of the algorithm ?

(A) $\Theta(n)$ (B) $\Theta(n \log n)$ (C) $\Theta(n^2)$ (D) $\Theta(n^2 \log n)$

Q.36 The keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function $h(k) = k$ mod 10 and linear probing. What is the resultant hash table ?

(A)

0	
1	
2	2
3	23
4	
5	15
6	
7	
8	18
9	

(B)

0	
1	
2	12
3	13
4	
5	5
6	
7	
8	18
9	

(C)

0	
1	
2	12
3	13
4	2
5	3
6	23
7	5
8	18
9	15

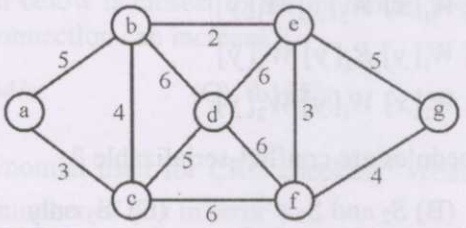
(D)

0	
1	
2	12, 2
3	13, 3, 23
4	
5	5, 15
6	
7	
8	18
9	

Q.37 What is the maximum height of any AVL-tree with 7 nodes ? Assume that the height of a tree with a single node is 0.

(A) 2 (B) 3 (C) 4 (D) 5

Q.38 Consider the following graph :



Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm ?

(A) (b, e) (e, f) (a, c) (b, c) (f, g) (c, d) (B) (b, e) (e, f) (a, c) (f, g) (b, c) (c, d)
(C) (b, e) (a, c) (e, f) (b, c) (f, g) (c, d) (D) (b, e) (e, f) (b, c) (a, c) (f, g) (c, d)

Q.39 In quick sort, for sorting n elements, the $(n/4)^{th}$ smallest element is selected as pivot using an $O(n)$ time algorithm. What is the worst case time complexity of the quick sort ?

(A) $\Theta(n)$ (B) $\Theta(n \log n)$ (C) $\Theta(n^2)$ (D) $\Theta(n^2 \log n)$

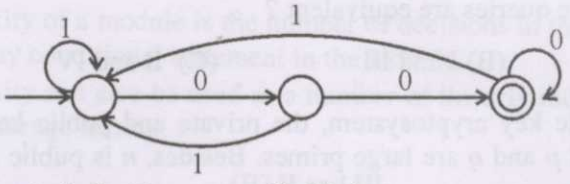
Q.40 Let $L = L_1 \cap L_2$, where L_1 and L_2 are languages as defined below :

$L_1 = \{a^m b^m c\, a^n b^n \mid m, n \geq 0\}$
$L_2 = \{a^i b^j c^k \mid i, j, k \geq 0\}$

Then L is

(A) not recursive. (B) regular.
(C) context-free but not regular. (D) recursively enumerable but not context-free.

Q.41



The above DFA accepts the set of all strings over {0, 1} that

(A) begin either with 0 or 1. (B) end with 0.
(C) end with 00. (D) contain the substring 00.

Q.42 Which of the following statements are TRUE ?

I. There exist parsing algorithms for some programming languages whose complexities are less than $\Theta(n^3)$.
II. A programming language which allows recursion can be implemented with static storage allocation.
III. No L-attributed definition can be evaluated in the framework of bottom-up parsing.
IV. Code improving transformations can be performed at both source language and intermediate code level.

(A) I and II (B) I and IV (C) III and IV (D) I, III and IV

Q.43 Consider two transactions T_1 and T_2, and four schedules S_1, S_2, S_3, S_4 of T_1 and T_2 as given below :

$T_1 : R_1[x]\ W_1[x]\ W_1[y]$

$T_2 : R_2[x]\ R_2[y]\ W_2[y]$

$S_1 : R_1[x]\ R_2[x]\ R_2[y]\ W_1[x]\ W_1[y]\ W_2[y]$

$S_2 : R_1[x]\ R_2[x]\ R_2[y]\ W_1[x]\ W_2[y]\ W_1[y]$

$S_3 : R_1[x]\ W_1[x]\ R_2[x]\ W_1[y]\ R_2[y]\ W_2[y]$

$S_4 : R_2[x]\ R_2[y]\ R_1[x]\ W_1[x]\ W_1[y]\ W_2[y]$

Which of the above schedules are conflict-serializable ?

(A) S_1 and S_2 (B) S_2 and S_3 (C) S_3 only (D) S_4 only

Q.44 The following key values are inserted into a B+ –tree in which order of the internal nodes is 3, and that of the leaf nodes is 2, in the sequence given below. The order of internal nodes is the maximum number of tree pointers in each node, and the order of leaf nodes is the maximum number of data items that can be stored in it. The B+ –tree is initially empty.

10, 3, 6, 8, 4, 2, 1

The maximum number of times leaf nodes would get split up as a result of these insertions is

(A) 2 (B) 3 (C) 4 (D) 5

Q.45 Let R and S be relational schemes such that $R = \{a, b, c\}$ and $S = \{c\}$. Now consider the following queries on the database :

I. $\pi_{R-S}(r) - \pi_{R-S}(\pi_{R-S}(r) \times s - \pi_{R-S,S}(r))$

II. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall u \in s(\exists v \in r(u = v[s] \wedge t = v\,[R-S]))\}$

III. $\{t \mid t \in \pi_{R-S}(r) \wedge \forall v \in r(\exists u \in s(u = v[s] \wedge t = v\,[R-S]))\}$

IV. Select $R.a$, $R.b$
from R, S
where $R.c = S.c$

Which of the above queries are equivalent ?

(A) I and II (B) I and III (C) II and IV (D) III and IV

Q.46 In the RSA public key cryptosystem, the private and public keys are (e, n) and (d, n) respectively, where $n = p * q$ and p and q are large primes. Besides, n is public and p and q are private. Let M be an integer such that $0 < M < n$ and $\phi(n) = (p-1)(q-1)$. Now consider the following equations.

I. $M' = M^e \bmod n$
$M = (M')^d \bmod n$

II. $ed \equiv 1 \bmod n$

III. $ed \equiv 1 \bmod \phi(n)$

IV. $M' = M^e \bmod \phi(n)$
$M = (M')^d \bmod \phi(n)$

Which of the above equations correctly represent RSA cryptosystem ?

(A) I and II (B) I and III (C) II and IV (D) III and IV

Q.47 While opening a TCP connection, the initial sequence number is to be derived using a time-of-day (ToD) clock that keeps running even when the host is down. The low order 32 bits of the counter of the ToD clock is to be used for the initial sequence numbers. The clock counter increments once per millisecond. The maximum packet lifetime is given to be 64s.

Which one of the choices given below is closest to the minimum permissible rate at which sequence numbers used for packets of a connection can increase ?

(A) 0.015/s (B) 0.064/s (C) 0.135/s (D) 0.327/s

Q.48 Let $G(x)$ be the generator polynomial used for CRC checking. What is the condition that should be satisfied by $G(x)$ to detect odd number of bits in error ?

(A) $G(x)$ contains more than two terms.

(B) $G(x)$ does not divide $1+x^k$, for any k not exceeding the frame length.

(C) $1+x$ is a factor of $G(x)$.

(D) $G(x)$ has an odd number of terms.

Q.49 Which of the following statements are TRUE ?

I. The context diagram should depict the system as a single bubble.
II. External entities should be identified clearly at all levels of DFDs.
III. Control information should not be represented in a DFD.
IV. A data store can be connected either to another data store or to an external entity.

(A) II and III (B) I, II and IV
(C) I and III (D) I, II and III

Q.50 Consider the following statements about the cyclomatic complexity of the control flow graph of a program module. Which of these are TRUE ?

I. The cyclomatic complexity of a module is equal to the maximum number of linearly independent circuits in the graph.
II. The cyclomatic complexity of a module is the number of decisions in the module plus one, where a decision is effectively any conditional statement in the module.
III. The cyclomatic complexity can also be used as a number of linearly independent paths that should be tested during path coverage testing.

(A) I and II (B) II and III
(C) I and III (D) I, II and III

Common Data Questions

Common Data for Questions 51 and 52:

A hard disk has 63 sectors per track, 10 platters each with 2 recording surfaces and 1000 cylinders. The address of a sector is given as a triple $\langle c,h,s \rangle$, where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0^{th} sector is addressed as $\langle 0,0,0 \rangle$, the 1^{st} sector as $\langle 0,0,1 \rangle$, and so on.

Q.51 The address $\langle 400,16,29 \rangle$ corresponds to sector number:

(A) 505035 (B) 505036 (C) 505037 (D) 505038

Q.52 The address of 1039^{th} sector is

(A) $\langle 0,15,31 \rangle$ (B) $\langle 0,16,30 \rangle$

(C) $\langle 0,16,31 \rangle$ (D) $\langle 0,17,31 \rangle$

Common Data for Questions 53 and 54:

A sub-sequence of a given sequence is just the given sequence with some elements (possibly none or all) left out. We are given two sequences `X[m]` and `Y[n]` of lengths m and n, respectively, with indexes of `X` and `Y` starting from 0.

Q.53 We wish to find the length of the longest common sub-sequence (LCS) of `X[m]` and `Y[n]` as *l*(m,n), where an incomplete recursive definition for the function *l(i,j)* to compute the length of the LCS of `X[m]` and `Y[n]` is given below :

```
l(i,j) = 0        , if either i=0 or j=0
       =  expr1 , if i,j>0 and X[i-1]=Y[j-1]
       =  expr2 , if i,j>0 and X[i-1]≠Y[j-1]
```

Which one of the following options is correct ?

(A) **expr1** ≡ *l*(i-1,j)+1

(B) **expr1** ≡ *l*(i,j-1)

(C) **expr2** ≡ max(*l*(i-1,j), *l*(i,j-1))

(D) **expr2** ≡ max(*l*(i-1,j-1), *l*(i,j))

Q.54 The values of *l(i,j)* could be obtained by dynamic programming based on the correct recursive definition of *l(i,j)* of the form given above, using an `array L[M,N]`, where `M = m + 1` and `N = n + 1`, such that `L[i,j] = l(i,j)`.

Which one the following statements would be TRUE regarding the dynamic programming solution for the recursive definition of *l(i,j)*?

(A) All elements of `L` should be initialized to 0 for the values of *l(i,j)* to be properly computed.

(B) The values of *l(i,j)* may be computed in a row major order or column major order of `L[M,N]`.

(C) The values of *l(i,j)* cannot be computed in either row major order or column major order of `L[M,N]`.

(D) `L[p,q]` needs to be computed before `L[r,s]` if either *p<r* or *q<s*.

Common Data for Questions 55 and 56:

Consider the following relational schema :

Suppliers(sid: integer, sname:string, city:string, street:string)

Parts(pid:integer, pname:string, color:string)

Catalog(sid:integer, pid:integer, cost:real)

Q.55 Consider the following relational query on the above database :

```
SELECT  S.sname
FROM    Suppliers S
WHERE  S.sid NOT IN ( SELECT C.sid
                      FROM   Catalog C
                      WHERE C.pid NOT IN ( SELECT P.pid
                                           FROM   Parts P
                                           WHERE P.color <> 'blue'))
```

Assume that relations corresponding to the above schema are not empty. Which one of the following is the correct interpretation of the above query ?

(A) Find the names of all suppliers who have supplied a non-blue part.
(B) Find the names of all suppliers who have not supplied a non-blue part.
(C) Find the names of all suppliers who have supplied only blue parts.
(D) Find the names of all suppliers who have not supplied only blue parts.

Q.56 Assume that, in the suppliers relation above, each supplier and each street within a city has a unique name, and (sname, city) forms a candidate key. No other functional dependencies are implied other than those implied by primary and candidate keys. Which one of the following is TRUE about the above schema ?

(A) The schema is in BCNF.
(B) The schema is in 3NF but not in BCNF.
(C) The schema is in 2NF but not in 3NF.
(D) The schema is not in 2NF.

Linked Answer Questions

Statement for Linked Answer Questions 57 and 58:

Frames of 1000 bits are sent over a 10^6 bps duplex link between two hosts. The propagation time is 25ms. Frames are to be transmitted into this link to maximally pack them in transit (within the link).

Q.57 What is the minimum number of bits (l) that will be required to represent the sequence numbers distinctly ? Assume that no time gap needs to be given between transmission of two frames.

(A) $l = 2$ (B) $l = 3$ (C) $l = 4$ (D) $l = 5$

Q.58 Suppose that the sliding window protocol is used with the sender window size of 2^l, where l is the number of bits identified in the earlier part and acknowledgements are always piggy backed. After sending 2^l frames, what is the minimum time the sender will have to wait before starting transmission of the next frame ? (Identify the closest choice ignoring the frame processing time.)

(A) 16ms (B) 18ms (C) 20ms (D) 22ms

Statement for Linked Answer Questions 59 and 60:

Consider a binary max-heap implemented using an array.

Q.59 Which one of the following array represents a binary max-heap ?

(A) { 25, 12, 16, 13, 10, 8, 14 }

(B) { 25, 14, 13, 16, 10, 8, 12 }

(C) { 25, 14, 16, 13, 10, 8, 12 }

(D) { 25, 14, 12, 13, 10, 8, 16 }

Q.60 What is the content of the array after two delete operations on the correct answer to the previous question ?

(A) { 14, 13, 12, 10, 8 }

(B) { 14, 12, 13, 8, 10 }

(C) { 14, 13, 8, 12, 10 }

(D) { 14, 13, 12, 8, 10 }

END OF THE QUESTION PAPER

CS : COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Duration: Three Hours *Maximum Marks*: 100

Read the following instructions carefully.

1. This question paper contains **24** pages including blank pages for rough work. Please check all pages and report discrepancy, if any.

2. Write your registration number, your name and name of the examination centre at the specified locations on the right half of the **Optical Response Sheet (ORS)**.

3. Using HB pencil, darken the appropriate bubble under each digit of your registration number and the letters corresponding to your paper code.

4. All questions in this paper are of objective type.

5. Questions must be answered on the **ORS** by darkening the appropriate bubble (marked A, B, C, D) using HB pencil against the question number on the left hand side of the ORS. **For each question darken the bubble of the correct answer**. In case you wish to change an answer, erase the old answer completely. More than one answer bubbled against a question will be treated as an incorrect response.

6. There are a total of 65 questions carrying 100 marks.

7. Questions Q.1 – Q.25 will carry 1-mark each, and questions Q.26 – Q.55 will carry 2-marks each.

8. Questions Q.48 – Q.51 (2 pairs) are common data questions and question pairs (Q.52, Q.53) and (Q.54, Q.55) are linked answer questions. The answer to the second question of the linked answer questions depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is un-attempted, then the answer to the second question in the pair will not be evaluated.

9. Questions Q.56 – Q.65 belong to General Aptitude (GA). Questions Q.56 – Q.60 will carry 1-mark each, and questions Q.61 – Q.65 will carry 2-marks each. The GA questions will begin on a fresh page starting from page **15**.

10. Un-attempted questions will carry zero marks.

11. Wrong answers will carry **NEGATIVE** marks. For Q.1 – Q.25 and Q.56 – Q.60, $\frac{1}{3}$ mark will be deducted for each wrong answer. For Q.26 – Q.51 and Q.61 – Q.65, $\frac{2}{3}$ mark will be deducted for each wrong answer. The question pairs (Q.52, Q.53), and (Q.54, Q.55) are questions with linked answers. There will be negative marks only for wrong answer to the first question of the linked answer question pair i.e. for Q.52 and Q.54, $\frac{2}{3}$ mark will be deducted for each wrong answer. There is no negative marking for Q.53 and Q.55.

12. Calculator (without data connectivity) is allowed in the examination hall.

13. Charts, graph sheets or tables are NOT allowed in the examination hall.

14. Rough work can be done on the question paper itself. Additionally, blank pages are provided at the end of the question paper for rough work.

Q.1 – Q.25 carry one mark each.

Q.1 Let $G = (V, E)$ be a graph. Define $\xi(G) = \sum_d i_d \times d$, where i_d is the number of vertices of degree d in G. If S and T are two different trees with $\xi(S) = \xi(T)$, then

(A) $|S| = 2|T|$ (B) $|S| = |T| - 1$ (C) $|S| = |T|$ (D) $|S| = |T| + 1$

Q.2 Newton-Raphson method is used to compute a root of the equation $x^2 - 13 = 0$ with 3.5 as the initial value. The approximation after one iteration is

(A) 3.575 (B) 3.677 (C) 3.667 (D) 3.607

Q.3 What is the possible number of reflexive relations on a set of 5 elements?

(A) 2^{10} (B) 2^{15} (C) 2^{20} (D) 2^{25}

Q.4 Consider the set $S = \{1, \omega, \omega^2\}$, where ω and ω^2 are cube roots of unity. If * denotes the multiplication operation, the structure $\{S, *\}$ forms

(A) a group (B) a ring (C) an integral domain (D) a field

Q.5 What is the value of $\lim_{n\to\infty}\left(1-\frac{1}{n}\right)^{2n}$?

(A) 0 (B) e^{-2} (C) $e^{-1/2}$ (D) 1

Q.6 The minterm expansion of $f(P,Q,R) = PQ + Q\overline{R} + P\overline{R}$ is

(A) $m_2 + m_4 + m_6 + m_7$ (B) $m_0 + m_1 + m_3 + m_5$

(C) $m_0 + m_1 + m_6 + m_7$ (D) $m_2 + m_3 + m_4 + m_5$

Q.7 A main memory unit with a capacity of 4 megabytes is built using 1M × 1-bit DRAM chips. Each DRAM chip has 1K rows of cells with 1K cells in each row. The time taken for a single refresh operation is 100 nanoseconds. The time required to perform one refresh operation on all the cells in the memory unit is

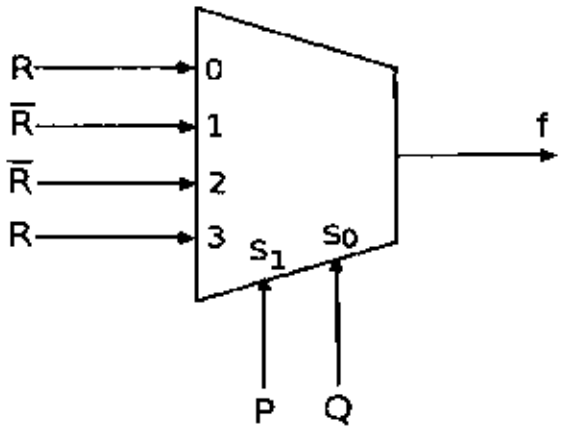
(A) 100 nanoseconds (B) $100*2^{10}$ nanoseconds

(C) $100*2^{20}$ nanoseconds (D) $3200*2^{20}$ nanoseconds

Q.8 P is a 16-bit signed integer. The 2's complement representation of P is $(F87B)_{16}$. The 2's complement representation of $8*P$ is

(A) $(C3D8)_{16}$ (B) $(187B)_{16}$ (C) $(F878)_{16}$ (D) $(987B)_{16}$

Q.9 The Boolean expression for the output f of the multiplexer shown below is



(A) $\overline{P \oplus Q \oplus R}$ (B) $P \oplus Q \oplus R$ (C) $P+Q+R$ (D) $\overline{P+Q+R}$

Q.10 In a binary tree with n nodes, every node has an odd number of descendants. Every node is considered to be its own descendant. What is the number of nodes in the tree that have exactly one child?

(A) 0 (B) 1 (C) $(n-1)/2$ (D) $n-1$

Q.11 What does the following program print?

```
#include <stdio.h>

void f(int *p, int *q){
       p = q;
      *p = 2;
}
int i=0, j=1;

int main(){
      f(&i, &j);
      printf("%d %d\n", i, j);
      return 0;
}
```

(A) 2 2 (B) 2 1 (C) 0 1 (D) 0 2

Q.12 Two alternative packages A and B are available for processing a database having 10^k records. Package A requires $0.0001\,n^2$ time units and package B requires $10\,n \log_{10} n$ time units to process n records. What is the smallest value of k for which package B will be preferred over A?

(A) 12 (B) 10 (C) 6 (D) 5

Q.13 Which data structure in a compiler is used for managing information about variables and their attributes?

(A) Abstract syntax tree
(B) Symbol table
(C) Semantic stack
(D) Parse table

Q.14 Which languages necessarily need heap allocation in the runtime environment?

(A) Those that support recursion.
(B) Those that use dynamic scoping.
(C) Those that allow dynamic data structures.
(D) Those that use global variables.

Q.15 One of the header fields in an IP datagram is the Time-to-Live (TTL) field. Which of the following statements best explains the need for this field?

(A) It can be used to prioritize packets.
(B) It can be used to reduce delays.
(C) It can be used to optimize throughput.
(D) It can be used to prevent packet looping.

Q.16 Which one of the following is not a client-server application?

(A) Internet chat (B) Web browsing
(C) E-mail (D) Ping

Q.17 Let L1 be a recursive language. Let L2 and L3 be languages that are recursively enumerable but not recursive. Which of the following statements is not necessarily true?

(A) L2 – L1 is recursively enumerable.
(B) L1 – L3 is recursively enumerable.
(C) L2 $\cap$ L3 is recursively enumerable.
(D) L2 $\cup$ L3 is recursively enumerable.

Q.18 Consider a B^+-tree in which the maximum number of keys in a node is 5. What is the minimum number of keys in any non-root node?

(A) 1 (B) 2 (C) 3 (D) 4

Q.19 A relational schema for a train reservation database is given below.

```
Passenger(pid, pname, age)
Reservation(pid, class, tid)
```

Table: Passenger

pid	pname	Age
0	'Sachin'	65
1	'Rahul'	66
2	'Sourav'	67
3	'Anil'	69

Table: Reservation

pid	class	tid
0	'AC'	8200
1	'AC'	8201
2	'SC'	8201
5	'AC'	8203
1	'SC'	8204
3	'AC'	8202

What pids are returned by the following SQL query for the above instance of the tables?

```
SELECT pid
FROM   Reservation
WHERE  class = 'AC' AND
       EXISTS (SELECT *
               FROM Passenger
               WHERE age > 65 AND
               Passenger.pid = Reservation.pid)
```

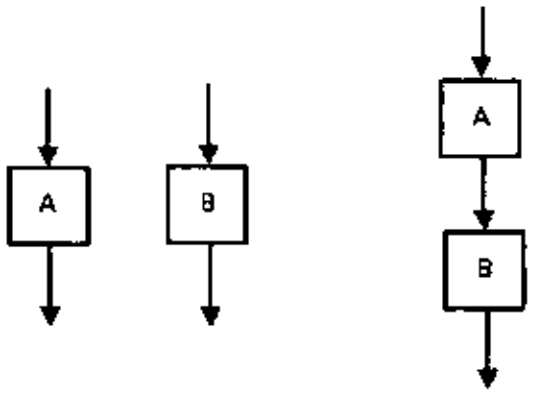
(A) 1, 0 (B) 1, 2 (C) 1, 3 (D) 1, 5

Q.20 Which of the following concurrency control protocols ensure both conflict serializability and freedom from deadlock?

I. 2-phase locking
II. Time-stamp ordering

(A) I only (B) II only (C) Both I and II (D) Neither I nor II

Q.21 The cyclomatic complexity of each of the modules A and B shown below is 10. What is the cyclomatic complexity of the sequential integration shown on the right hand side?



(A) 19 (B) 21 (C) 20 (D) 10

Q.22 What is the appropriate pairing of items in the two columns listing various activities encountered in a software life cycle?

P. Requirements Capture	1. Module Development and Integration
Q. Design	2. Domain Analysis
R. Implementation	3. Structural and Behavioral Modeling
S. Maintenance	4. Performance Tuning

(A) P-3 Q-2 R-4 S-1 (B) P-2 Q-3 R-1 S-4 (C) P-3 Q-2 R-1 S-4 (D) P-2 Q-3 R-4 S-1

Q.23 Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared boolean variables S1 and S2 are randomly assigned.

Method used by P1	Method used by P2

`while (S1 == S2);`
`Critical Section`
`S1 = S2;`
	`while (S1 != S2);`
	`Critical Section`
	`S2 = not(S1);`

Which one of the following statements describes the properties achieved?

(A) Mutual exclusion but not progress
(B) Progress but not mutual exclusion
(C) Neither mutual exclusion nor progress
(D) Both mutual exclusion and progress

Q.24 A system uses FIFO policy for page replacement. It has 4 page frames with no pages loaded to begin with. The system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order. How many page faults will occur?

(A) 196 (B) 192 (C) 197 (D) 195

Q.25 Which of the following statements are true?

I. Shortest remaining time first scheduling may cause starvation
II. Preemptive scheduling may cause starvation
III. Round robin is better than FCFS in terms of response time

(A) I only (B) I and III only (C) II and III only (D) I, II and III

Q.26 – Q.55 carry two marks each.

Q.26 Consider a company that assembles computers. The probability of a faulty assembly of any computer is p. The company therefore subjects each computer to a testing process. This testing process gives the correct result for any computer with a probability of q. What is the probability of a computer being declared faulty?

(A) $pq+(1-p)(1-q)$ (B) $(1-q)p$ (C) $(1-p)q$ (D) pq

Q.27 What is the probability that a divisor of 10^{99} is a multiple of 10^{96} ?

(A) 1/625 (B) 4/625 (C) 12/625 (D) 16/625

Q.28 The degree sequence of a simple graph is the sequence of the degrees of the nodes in the graph in decreasing order. Which of the following sequences can not be the degree sequence of any graph?

I. 7, 6, 5, 4, 4, 3, 2, 1
II. 6, 6, 6, 6, 3, 3, 2, 2
III. 7, 6, 6, 4, 4, 3, 2, 2
IV. 8, 7, 7, 6, 4, 2, 1, 1

(A) I and II (B) III and IV (C) IV only (D) II and IV

Q.29 Consider the following matrix

$$A=\begin{bmatrix} 2 & 3 \\ x & y \end{bmatrix}$$

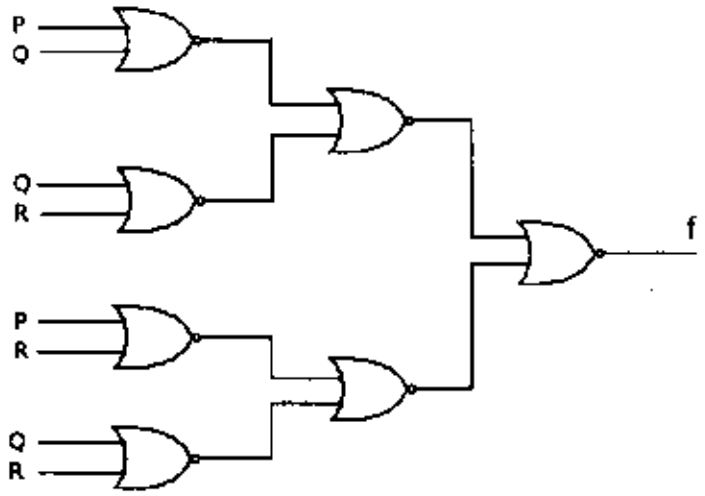
If the eigenvalues of A are 4 and 8, then

(A) $x = 4, y = 10$ (B) $x = 5, y = 8$ (C) $x = -3, y = 9$ (D) $x = -4, y = 10$

Q.30 Suppose the predicate $F(x,y,t)$ is used to represent the statement that person x can fool person y at time t. Which one of the statements below expresses best the meaning of the formula $\forall x \exists y \exists t(\neg F(x, y, t))$?

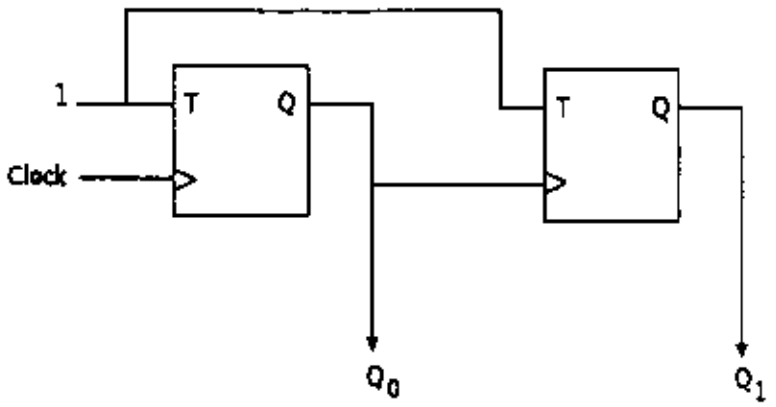
(A) Everyone can fool some person at some time.
(B) No one can fool everyone all the time.
(C) Everyone cannot fool some person all the time.
(D) No one can fool some person at some time.

Q.31 What is the boolean expression for the output f of the combinational logic circuit of NOR gates given below?



(A) $\overline{Q+R}$ (B) $\overline{P+Q}$ (C) $\overline{P+R}$ (D) $\overline{P+Q+R}$

Q.32 In the sequential circuit shown below, if the initial value of the output Q_1Q_0 is 00, what are the next four values of Q_1Q_0?



(A) 11,10,01,00 (B) 10,11,01,00 (C) 10,00,01,11 (D) 11,10,00,01

Q.33 A 5-stage pipelined processor has Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Write Operand (WO) stages. The IF, ID, OF and WO stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD and SUB instructions, 3 clock cycles for MUL instruction, and 6 clock cycles for DIV instruction respectively. Operand forwarding is used in the pipeline. What is the number of clock cycles needed to execute the following sequence of instructions?

Instruction	Meaning of instruction
I_0: MUL R_2, R_0, R_1	$R_2 \leftarrow R_0 * R_1$
I_1: DIV R_5, R_3, R_4	$R_5 \leftarrow R_3 / R_4$
I_2: ADD R_2, R_5, R_2	$R_2 \leftarrow R_5 + R_2$
I_3: SUB R_5, R_2, R_6	$R_5 \leftarrow R_2 - R_6$

(A) 13 (B) 15 (C) 17 (D) 19

Q.34 The weight of a sequence $a_0, a_1, \ldots, a_{n-1}$ of real numbers is defined as $a_0 + a_1/2 + \ldots + a_{n-1}/2^{n-1}$. A subsequence of a sequence is obtained by deleting some elements from the sequence, keeping the order of the remaining elements the same. Let X denote the maximum possible weight of a subsequence of $a_0, a_1, \ldots, a_{n-1}$ and Y the maximum possible weight of a subsequence of $a_1, a_2, \ldots, a_{n-1}$. Then X is equal to

(A) $\max(Y, a_0 + Y)$ (B) $\max(Y, a_0 + Y/2)$ (C) $\max(Y, a_0 + 2Y)$ (D) $a_0 + Y/2$

Q.35 What is the value printed by the following C program?

```
#include <stdio.h>

int f(int *a, int n)
{
        if (n <= 0) return 0;
        else if (*a % 2 == 0) return *a + f(a+1,n-1);
        else return *a - f(a+1,n-1);
}

int main( )
{
        int a[] = {12, 7, 13, 4, 11, 6};
        printf("%d", f(a,6));
        return 0;
 }
```

(A) –9 (B) 5 (C) 15 (D) 19

Q.36 The following C function takes a singly-linked list as input argument. It modifies the list by moving the last element to the front of the list and returns the modified list. Some part of the code is left blank.

```
typedef struct node {
        int value;
        struct node *next;
}     Node;

Node *move_to_front(Node *head) {
       Node *p, *q;
       if ((head = = NULL) || (head->next = = NULL)) return head;
       q = NULL; p = head;
       while (p->next != NULL) {
              q = p;
              p = p->next;
       }
       ____________________________________
       return head;
}
```

Choose the correct alternative to replace the blank line.

(A) `q = NULL; p->next = head; head = p;`
(B) `q->next = NULL; head = p; p->next = head;`
(C) `head = p; p->next = q; q->next = NULL;`
(D) `q->next = NULL; p->next = head; head = p;`

Q.37 The program below uses six temporary variables `a, b, c, d, e, f`.

```
a = 1
b = 10
c = 20
d = a + b
e = c + d
f = c + e
b = c + e
e = b + f
d = 5 + e
return d + f
```

Assuming that all operations take their operands from registers, what is the minimum number of registers needed to execute this program without spilling?

(A) 2 (B) 3 (C) 4 (D) 6

Q.38 The grammar S → aSa | bS | c is

(A) LL(1) but not LR(1)
(B) LR(1) but not LL(1)
(C) Both LL(1) and LR(1)
(D) Neither LL(1) nor LR(1)

Q.39 Let $L = \{ w \in (0 + 1)^* \mid w$ has even number of 1s$\}$, i.e. L is the set of all bit strings with even number of 1s. Which one of the regular expressions below represents L?

(A) (0*10*1)* (B) 0*(10*10*)* (C) 0*(10*1)*0* (D) 0*1(10*1)*10*

Q.40 Consider the languages $L1 = \{0^i1^j \mid i \neq j\}$, $L2 = \{0^i1^j \mid i = j\}$, $L3 = \{0^i1^j \mid i = 2j+1\}$, $L4 = \{0^i1^j \mid i \neq 2j\}$. Which one of the following statements is true?

(A) Only *L2* is context free.
(B) Only *L2* and *L3* are context free.
(C) Only *L1* and *L2* are context free.
(D) All are context free.

Q.41 Let w be any string of length n in $\{0,1\}^*$. Let L be the set of all substrings of w. What is the minimum number of states in a non-deterministic finite automaton that accepts L?

(A) $n-1$ (B) n (C) $n+1$ (D) 2^{n-1}

Q.42 Consider the following schedule for transactions T1, T2 and T3:

T1	T2	T3
Read (X)		
	Read (Y)	
		Read (Y)
	Write (Y)	
Write (X)		
		Write (X)
	Read (X)	
	Write (X)	

Which one of the schedules below is the correct serialization of the above?

(A) T1 → T3 → T2 (B) T2 → T1 → T3
(C) T2 → T3 → T1 (D) T3 → T1 → T2

Q.43 The following functional dependencies hold for relations $R(A, B, C)$ and $S(B, D, E)$:
$B \rightarrow A$,
$A \rightarrow C$
The relation R contains 200 tuples and the relation S contains 100 tuples. What is the maximum number of tuples possible in the natural join $R \bowtie S$?

(A) 100 (B) 200 (C) 300 (D) 2000

Q.44 The following program is to be tested for statement coverage:

```
begin
   if (a == b) {S1; exit;}
   else if (c == d) {S2;}
        else {S3; exit;}
   S4;
end
```

The test cases T1, T2, T3 and T4 given below are expressed in terms of the properties satisfied by the values of variables a, b, c and d. The exact values are not given.

```
T1: a, b, c and d are all equal
T2: a, b, c and d are all distinct
T3: a = b and c != d
T4: a != b and c = d
```

Which of the test suites given below ensures coverage of statements S1, S2, S3 and S4?

(A) T1, T2, T3 (B) T2, T4 (C) T3, T4 (D) T1, T2, T4

Q.45 The following program consists of 3 concurrent processes and 3 binary semaphores. The semaphores are initialized as S0=1, S1=0, S2=0.

Process P0	Process P1	Process P2

while (true) {
wait (S0);
print '0';
release (S1);
release (S2);
}
	wait (S1);
	release (S0);
		wait (S2)
		release (S0);

How many times will process P0 print '0'?

(A) At least twice (B) Exactly twice (C) Exactly thrice (D) Exactly once

Q.46 A system has n resources $R_0..R_{n-1}$, and k processes $P_0..P_{k-1}$. The implementation of the resource request logic of each process P_i is as follows:

```
if (i%2==0) {
    if (i<n) request R_i ;
    if (i+2 < n) request R_i+2;
}
else {
     if (i<n) request  R_n-i ;
     if (i-2<n) request  R_n-i ;;
}
```

In which one of the following situations is a deadlock possible?

(A) $n = 40, k = 26$ (B) $n = 21, k = 12$ (C) $n = 20, k = 10$ (D) $n = 41, k = 19$

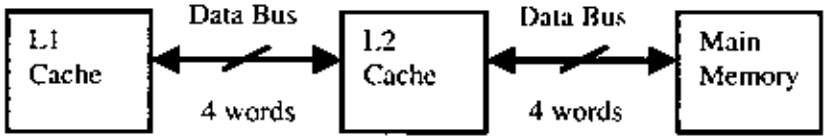
Q.47 Suppose computers A and B have IP addresses 10.105.1.113 and 10.105.1.91 respectively and they both use the same netmask N. Which of the values of N given below should not be used if A and B should belong to the same network?

(A) 255.255.255.0 (B) 255.255.255.128 (C) 255.255.255.192 (D) 255.255.255.224

Common Data Questions

Common Data for Questions 48 and 49:

A computer system has an L1 cache, an L2 cache, and a main memory unit connected as shown below. The block size in L1 cache is 4 words. The block size in L2 cache is 16 words. The memory access times are 2 nanoseconds, 20 nanoseconds and 200 nanoseconds for L1 cache, L2 cache and main memory unit respectively.



Q.48 When there is a miss in L1 cache and a hit in L2 cache, a block is transferred from L2 cache to L1 cache. What is the time taken for this transfer?

(A) 2 nanoseconds (B) 20 nanoseconds (C) 22 nanoseconds (D) 88 nanoseconds

Q.49 When there is a miss in both L1 cache and L2 cache, first a block is transferred from main memory to L2 cache, and then a block is transferred from L2 cache to L1 cache. What is the total time taken for these transfers?

(A) 222 nanoseconds (B) 888 nanoseconds
(C) 902 nanoseconds (D) 968 nanoseconds

Common Data for Questions 50 and 51:

Consider a complete undirected graph with vertex set {0,1,2,3,4}. Entry W_{ij} in the matrix W below is the weight of the edge $\{i,j\}$.

$$W = \begin{pmatrix} 0 & 1 & 8 & 1 & 4 \\ 1 & 0 & 12 & 4 & 9 \\ 8 & 12 & 0 & 7 & 3 \\ 1 & 4 & 7 & 0 & 2 \\ 4 & 9 & 3 & 2 & 0 \end{pmatrix}$$

Q.50 What is the minimum possible weight of a spanning tree T in this graph such that vertex 0 is a leaf node in the tree T?

(A) 7 (B) 8 (C) 9 (D) 10

Q.51 What is the minimum possible weight of a path P from vertex 1 to vertex 2 in this graph such that P contains at most 3 edges?

(A) 7 (B) 8 (C) 9 (D) 10

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

A hash table of length 10 uses open addressing with hash function $h(k) = k \bmod 10$, and linear probing. After inserting 6 values into an empty hash table, the table is as shown below.

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

Q.52 Which one of the following choices gives a possible order in which the key values could have been inserted in the table?

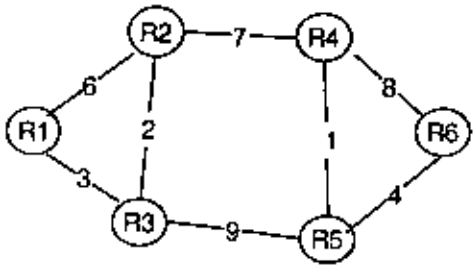
(A) 46, 42, 34, 52, 23, 33
(B) 34, 42, 23, 52, 33, 46
(C) 46, 34, 42, 23, 52, 33
(D) 42, 46, 33, 23, 34, 52

Q.53 How many different insertion sequences of the key values using the same hash function and linear probing will result in the hash table shown above?

(A) 10 (B) 20 (C) 30 (D) 40

Statement for Linked Answer Questions 54 and 55:

Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram.



Q.54 All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbour with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data?

(A) 4 (B) 3 (C) 2 (D) 1

Q.55 Suppose the weights of all unused links in the previous question are changed to 2 and the distance vector algorithm is used again until all routing tables stabilize. How many links will now remain unused?

(A) 0 (B) 1 (C) 2 (D) 3

General Aptitude (GA) Questions

Q.56 – Q.60 carry one mark each.

Q.56 *Choose the most appropriate word from the options given below to complete the following sentence:*
His rather casual remarks on politics _____ his lack of seriousness about the subject.

(A) masked
(B) belied
(C) betrayed
(D) suppressed

Q.57 *Which of the following options is the closest in meaning to the word below:*
Circuitous

(A) cyclic
(B) indirect
(C) confusing
(D) crooked

Q.58 *Choose the most appropriate word from the options given below to complete the following sentence:*
If we manage to _____ our natural resources, we would leave a better planet for our children.

(A) uphold
(B) restrain
(C) cherish
(D) conserve

Q.59 25 persons are in a room. 15 of them play hockey, 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is:

(A) 2 (B) 17 (C) 13 (D) 3

Q.60 *The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.*
Unemployed : Worker

(A) fallow : land
(B) unaware : sleeper
(C) wit : jester
(D) renovated : house

Q.61 – Q.65 carry two marks each.

Q.61 If 137 + 276 = 435 how much is 731 + 672?

(A) 534 (B) 1403 (C) 1623 (D) 1513

Q.62 Hari (H), Gita (G), Irfan (I) and Saira (S) are siblings (i.e. brothers and sisters). All were born on 1st January. The age difference between any two successive siblings (that is born one after another) is less than 3 years. Given the following facts:

i. Hari's age + Gita's age > Irfan's age + Saira's age.
ii. The age difference between Gita and Saira is 1 year. However, Gita is not the oldest and Saira is not the youngest.
iii. There are no twins.

In what order were they born (oldest first)?

(A) HSIG (B) SGHI (C) IGSH (D) IHSG

Q.63 **Modern warfare has changed from large scale clashes of armies to suppression of civilian populations. Chemical agents that do their work silently appear to be suited to such warfare; and regretfully, there exist people in military establishments who think that chemical agents are useful tools for their cause.**

Which of the following statements best sums up the meaning of the above passage:

(A) Modern warfare has resulted in civil strife.
(B) Chemical agents are useful in modern warfare.
(C) Use of chemical agents in warfare would be undesirable.
(D) People in military establishments like to use chemical agents in war.

Q.64 5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long will it take to build the wall?

(A) 20 days (B) 18 days (C) 16 days (D) 15 days

Q.65 Given digits 2, 2, 3, 3, 3, 4, 4, 4, 4 how many distinct 4 digit numbers greater than 3000 can be formed?

(A) 50 (B) 51 (C) 52 (D) 54

END OF THE QUESTION PAPER

Space for Rough Work

Space for Rough Work

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Space for Rough Work

2012 **Question Booklet Code →** **A**

CS : COMPUTER SCIENCE & INFORMATION TECHNOLOGY

Duration: Three Hours *Maximum Marks*: 100

Read the following instructions carefully.

1. Do not open the seal of the Question Booklet until you are asked to do so by the invigilator.
2. Take out the **O**ptical **R**esponse **S**heet (**ORS)** from this Question Booklet **without breaking the seal** and read the instructions printed on the ORS carefully. If you find that the Question Booklet Code printed at the right hand top corner of this page does not match with the Booklet Code on the **ORS**, exchange the booklet immediately with a new sealed Question Booklet.
3. On the right half of the **ORS,** using ONLY a **black ink ball point pen**, (i) darken the bubble corresponding to your test paper code and the appropriate bubble under each digit of your registration number and (ii) write your registration number, your name and name of the examination centre and put your signature at the specified location.
4. This Question Booklet contains **20** pages including blank pages for rough work. After you are permitted to open the seal, please check all pages and report discrepancies, if any, to the invigilator.
5. There are a total of 65 questions carrying 100 marks. All these questions are of objective type. Each question has only **one** correct answer. Questions must be answered on the left hand side of the **ORS** by darkening the appropriate bubble (marked A, B, C, D) using ONLY a **black ink ball point pen** against the question number. **For each question darken the bubble of the correct answer**. More than one answer bubbled against a question will be treated as an incorrect response.
6. Since bubbles darkened by the black ink ball point pen **cannot** be erased, candidates should darken the bubbles in the ORS **very carefully**.
7. Questions Q.1 – Q.25 carry 1 mark each. Questions Q.26 – Q.55 carry 2 marks each. The 2 marks questions include two pairs of common data questions and two pairs of linked answer questions. The answer to the second question of the linked answer questions depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is unattempted, then the answer to the second question in the pair will not be evaluated.
8. Questions Q.56 – Q.65 belong to General Aptitude (GA) section and carry a total of 15 marks. Questions Q.56 – Q.60 carry 1 mark each, and questions Q.61 – Q.65 carry 2 marks each.
9. Unattempted questions will result in zero mark and wrong answers will result in **NEGATIVE** marks. For all 1 mark questions, ⅓ mark will be deducted for each wrong answer. For all 2 marks questions, ⅔ mark will be deducted for each wrong answer. However, in the case of the linked answer question pair, there will be negative marks only for wrong answer to the first question and no negative marks for wrong answer to the second question.
10. Calculator is allowed whereas charts, graph sheets or tables are **NOT** allowed in the examination hall.
11. Rough work can be done on the question paper itself. Blank pages are provided at the end of the question paper for rough work.
12. Before the start of the examination, write your name and registration number in the space provided below using a black ink ball point pen.

Name								
Registration Number	**CS**							

Q. 1 – Q. 25 carry one mark each.

Q.1 Consider the following logical inferences.

I_1: If it rains then the cricket match will not be played.
The cricket match was played.
Inference: There was no rain.
I_2: If it rains then the cricket match will not be played.
It did not rain.
Inference: The cricket match was played.

Which of the following is **TRUE**?

(A) Both I_1 and I_2 are correct inferences
(B) I_1 is correct but I_2 is not a correct inference
(C) I_1 is not correct but I_2 is a correct inference
(D) Both I_1 and I_2 are not correct inferences

Q.2 Which of the following is **TRUE**?

(A) Every relation in 3NF is also in BCNF
(B) A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
(C) Every relation in BCNF is also in 3NF
(D) No relation can be in both BCNF and 3NF

Q.3 What will be the output of the following C program segment?

```
char inChar = 'A' ;
switch ( inChar ) {
case 'A' : printf ("Choice A\ n") ;
case 'B' :
case 'C' : printf ("Choice B") ;
case 'D' :
case 'E' :
default  : printf ( " No Choice" ) ; }
```

(A) No Choice
(B) Choice A
(C) Choice A
Choice B No Choice
(D) Program gives no output as it is erroneous

Q.4 Assuming P ≠ NP, which of the following is **TRUE**?

(A) NP-complete = NP
(B) NP-complete ∩ P = ∅
(C) NP-hard = NP
(D) P = NP-complete

Q.5 The worst case running time to search for an element in a balanced binary search tree with $n2^n$ elements is

(A) $\Theta(n \log n)$ (B) $\Theta(n2^n)$ (C) $\Theta(n)$ (D) $\Theta(\log n)$

Q.6 The truth table

X	Y	f (X, Y)
0	0	0
0	1	0
1	0	1
1	1	1

represents the Boolean function

(A) X (B) X + Y (C) $X \oplus Y$ (D) Y

Q.7 The decimal value 0.5 in IEEE single precision floating point representation has

(A) fraction bits of 000…000 and exponent value of 0
(B) fraction bits of 000…000 and exponent value of −1
(C) fraction bits of 100…000 and exponent value of 0
(D) no exact representation

Q.8 A process executes the code

```
fork();
fork();
fork();
```

The total number of **child** processes created is

(A) 3 (B) 4 (C) 7 (D) 8

Q.9 Consider the function $f(x) = \sin(x)$ in the interval $x \in [\pi/4, 7\pi/4]$. The number and location(s) of the local minima of this function are

(A) One, at $\pi/2$
(B) One, at $3\pi/2$
(C) Two, at $\pi/2$ and $3\pi/2$
(D) Two, at $\pi/4$ and $3\pi/2$

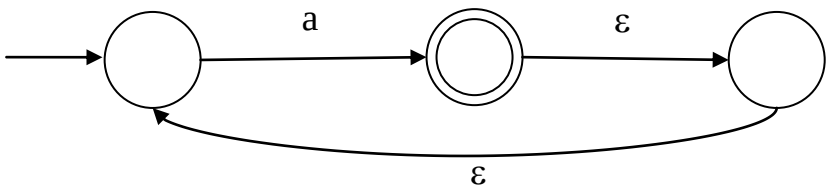
Q.10 The protocol data unit (PDU) for the application layer in the Internet stack is

(A) Segment (B) Datagram (C) Message (D) Frame

Q.11 Let A be the 2×2 matrix with elements $a_{11} = a_{12} = a_{21} = +1$ and $a_{22} = -1$. Then the eigenvalues of the matrix A^{19} are

(A) 1024 and −1024 (B) $1024\sqrt{2}$ and $-1024\sqrt{2}$
(C) $4\sqrt{2}$ and $-4\sqrt{2}$ (D) $512\sqrt{2}$ and $-512\sqrt{2}$

Q.12 What is the complement of the language accepted by the NFA shown below?
Assume $\Sigma = \{a\}$ and ε is the empty string.



(A) $\varnothing$ (B) $\{\varepsilon\}$ (C) a^* (D) $\{a, \varepsilon\}$

Q.13 What is the correct translation of the following statement into mathematical logic?

"Some real numbers are rational"

(A) $\exists x$ (real(x) $\vee$ rational(x))
(B) $\forall x$ (real(x) $\rightarrow$ rational(x))
(C) $\exists x$ (real(x) $\wedge$ rational(x))
(D) $\exists x$ (rational(x) $\rightarrow$ real(x))

Q.14 Given the basic ER and relational models, which of the following is **INCORRECT**?

(A) An attribute of an entity can have more than one value
(B) An attribute of an entity can be composite
(C) In a row of a relational table, an attribute can have more than one value
(D) In a row of a relational table, an attribute can have exactly one value or a NULL value

Q.15 Which of the following statements are **TRUE** about an SQL query?

P : An SQL query can contain a HAVING clause even if it does not have a GROUP BY clause
Q : An SQL query can contain a HAVING clause only if it has a GROUP BY clause
R : All attributes used in the GROUP BY clause must appear in the SELECT clause
S : Not all attributes used in the GROUP BY clause need to appear in the SELECT clause

(A) P and R (B) P and S (C) Q and R (D) Q and S

Q.16 The recurrence relation capturing the optimal execution time of the *Towers of Hanoi* problem with n discs is

(A) $T(n) = 2T(n-2) + 2$ (B) $T(n) = 2T(n-1) + n$
(C) $T(n) = 2T(n/2) + 1$ (D) $T(n) = 2T(n-1) + 1$

Q.17 Let G be a simple undirected planar graph on 10 vertices with 15 edges. If G is a connected graph, then the number of **bounded** faces in any embedding of G on the plane is equal to

(A) 3 (B) 4 (C) 5 (D) 6

Q.18 Let $W(n)$ and $A(n)$ denote respectively, the worst case and average case running time of an algorithm executed on an input of size n. Which of the following is **ALWAYS TRUE**?

(A) $A(n) = \Omega(W(n))$ (B) $A(n) = \Theta(W(n))$
(C) $A(n) = O(W(n))$ (D) $A(n) = o(W(n))$

Q.19 The amount of ROM needed to implement a 4 bit multiplier is

(A) 64 bits (B) 128 bits (C) 1 Kbits (D) 2 Kbits

Q.20 Register renaming is done in pipelined processors

(A) as an alternative to register allocation at compile time
(B) for efficient access to function parameters and local variables
(C) to handle certain kinds of hazards
(D) as part of address translation

Q.21 Consider a random variable X that takes values +1 and −1 with probability 0.5 each. The values of the cumulative distribution function $F(x)$ at $x = -1$ and +1 are

(A) 0 and 0.5 (B) 0 and 1 (C) 0.5 and 1 (D) 0.25 and 0.75

Q.22 Which of the following transport layer protocols is used to support electronic mail?

(A) SMTP (B) IP (C) TCP (D) UDP

Q.23 In the IPv4 addressing format, the number of networks allowed under Class C addresses is

(A) 2^{14} (B) 2^{7} (C) 2^{21} (D) 2^{24}

Q.24 Which of the following problems are decidable?

1) Does a given program ever produce an output?
2) If L is a context-free language, then, is $\overline{L}$ also context-free?
3) If L is a regular language, then, is $\overline{L}$ also regular?
4) If L is a recursive language, then, is $\overline{L}$ also recursive?

(A) 1, 2, 3, 4 (B) 1, 2 (C) 2, 3, 4 (D) 3, 4

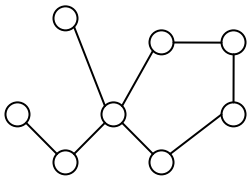
Q.25 Given the language $L = \{ab, aa, baa\}$, which of the following strings are in L^*?

1) *abaabaaabaa*
2) *aaaabaaaa*
3) *baaaaabaaaab*
4) *baaaaabaa*

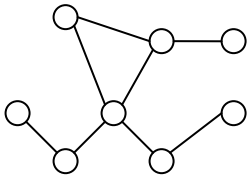
(A) 1, 2 and 3 (B) 2, 3 and 4
(C) 1, 2 and 4 (D) 1, 3 and 4

Q. 26 to Q. 55 carry two marks each.

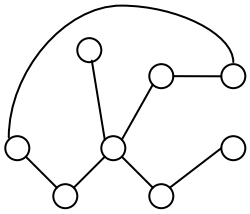
Q.26 Which of the following graphs is isomorphic to



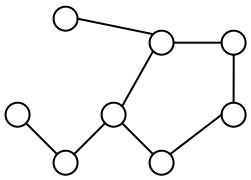
(A)



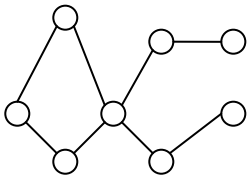
(B)



(C)



(D)



Q.27 Consider the following transactions with data items P and Q initialized to zero:

T_1:
```
read (P);
read (Q);
if  P = 0 then Q := Q + 1 ;
write (Q).
```

T_2:
```
read (Q);
read (P);
if  Q = 0 then P := P + 1 ;
write (P).
```

Any **non-serial** interleaving of **T_1** and **T_2** for concurrent execution leads to

(A) a serializable schedule
(B) a schedule that is not conflict serializable
(C) a conflict serializable schedule
(D) a schedule for which a precedence graph cannot be drawn

Q.28 The bisection method is applied to compute a zero of the function $f(x) = x^4 - x^3 - x^2 - 4$ in the interval [1,9]. The method converges to a solution after _____ iterations.

(A) 1 (B) 3 (C) 5 (D) 7

Q.29 Let G be a weighted graph with edge weights greater than one and G′ be the graph constructed by squaring the weights of edges in G. Let T and T′ be the minimum spanning trees of G and G′, respectively, with total weights t and t′. Which of the following statements is **TRUE**?

(A) T′ = T with total weight $t' = t^2$
(B) T′ = T with total weight $t' < t^2$
(C) T′ ≠ T but total weight $t' = t^2$
(D) None of the above

Q.30 What is the minimal form of the Karnaugh map shown below? Assume that **X** denotes a don't care term.

cd \ ab	00	01	11	10
00	**1**	**X**	**X**	**1**
01	**X**			**1**
11				
10	**1**			**X**

(A) $\bar{b}\bar{d}$ (B) $\bar{b}\bar{d}+\bar{b}\bar{c}$ (C) $\bar{b}\bar{d}+a\bar{b}\bar{c}d$ (D) $\bar{b}\bar{d}+\bar{b}\bar{c}+\bar{c}\bar{d}$

Q.31 Consider the 3 processes, P1, P2 and P3 shown in the table.

Process	**Arrival time**	**Time Units Required**
P1	0	5
P2	1	7
P3	3	4

The completion order of the 3 processes under the policies FCFS and RR2 (round robin scheduling with CPU quantum of 2 time units) are

(A) **FCFS:** P1, P2, P3 **RR2:** P1, P2, P3 (B) **FCFS:** P1, P3, P2 **RR2:** P1, P3, P2
(C) **FCFS:** P1, P2, P3 **RR2:** P1, P3, P2 (D) **FCFS:** P1, P3, P2 **RR2:** P1, P2, P3

Q.32 `Fetch_And_Add(X,i)` is an atomic Read-Modify-Write instruction that reads the value of memory location X, increments it by the value i, and returns the old value of X. It is used in the pseudocode shown below to implement a busy-wait lock. L is an unsigned integer shared variable initialized to 0. The value of 0 corresponds to lock being available, while any non-zero value corresponds to the lock being not available.

```
AcquireLock(L){
      while (Fetch_And_Add(L,1))
            L = 1;
}

ReleaseLock(L){
      L = 0;
}
```

This implementation

(A) fails as L can overflow
(B) fails as L can take on a non-zero value when the lock is actually available
(C) works correctly but may starve some processes
(D) works correctly without starvation

Q.33 Suppose a fair six-sided die is rolled once. If the value on the die is 1, 2, or 3, the die is rolled a second time. What is the probability that the sum total of values that turn up is at least 6?

(A) 10/21 (B) 5/12 (C) 2/3 (D) 1/6

Q.34 An Internet Service Provider (ISP) has the following chunk of CIDR-based IP addresses available with it: 245.248.128.0/20. The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B?

(A) 245.248.136.0/21 and 245.248.128.0/22
(B) 245.248.128.0/21 and 245.248.128.0/22
(C) 245.248.132.0/22 and 245.248.132.0/21
(D) 245.248.136.0/24 and 245.248.132.0/21

Q.35 Suppose a circular queue of capacity ($n-1$) elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variables, respectively. Initially, REAR = FRONT = 0. The conditions to detect *queue full* and *queue empty* are

(A) *full*: (REAR+1) mod n == FRONT
empty: REAR == FRONT

(B) *full*: (REAR+1) mod n == FRONT
empty: (FRONT+1) mod n == REAR

(C) *full*: REAR == FRONT
empty: (REAR+1) mod n == FRONT

(D) *full*: (FRONT+1) mod n == REAR
empty: REAR == FRONT

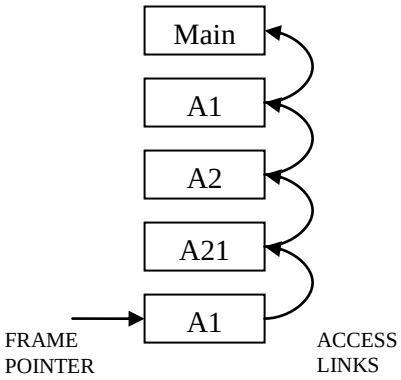
Q.36 Consider the program given below, in a block-structured pseudo-language with lexical scoping and nesting of procedures permitted.

```
Program main;
   Var ...

   Procedure A1;
      Var ...
      Call A2;
   End A1

   Procedure A2;
      Var ...

      Procedure A21;
         Var ...
         Call A1;
      End A21

      Call A21;
   End A2

   Call A1;
End main.
```

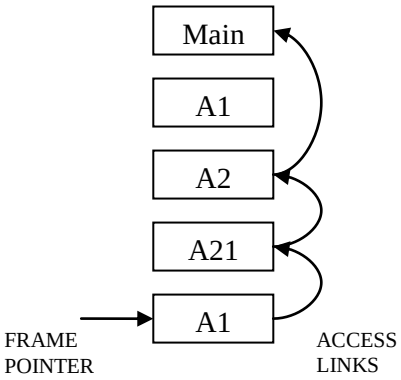
Consider the calling chain: Main → A1 → A2 → A21 → A1

The correct set of activation records along with their access links is given by

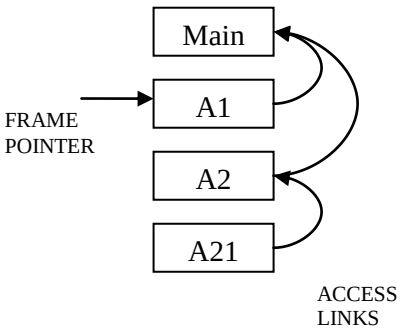
(A)



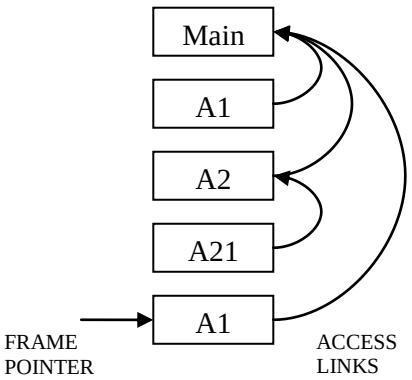
(B)



(C)



(D)



Q.37 How many onto (or surjective) functions are there from an n-element ($n \geq 2$) set to a 2-element set?

(A) 2^n (B) $2^n - 1$ (C) $2^n - 2$ (D) $2(2^n - 2)$

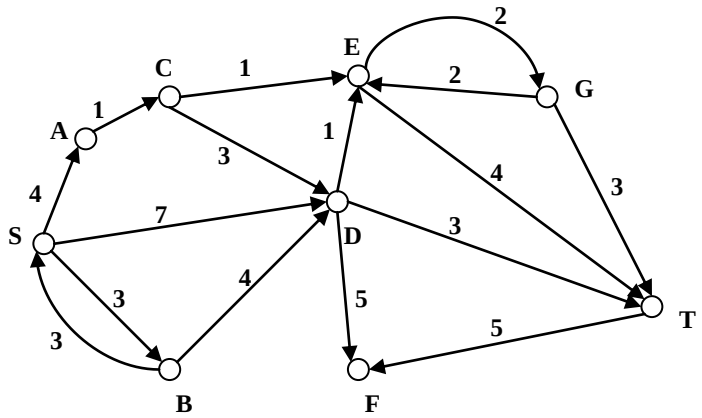
Q.38 Let G be a complete undirected graph on 6 vertices. If vertices of G are labeled, then the number of distinct cycles of length 4 in G is equal to

(A) 15 (B) 30 (C) 90 (D) 360

Q.39 A list of n strings, each of length n, is sorted into lexicographic order using the merge-sort algorithm. The worst case running time of this computation is

(A) O ($n \log n$) (B) O ($n^2 \log n$) (C) O ($n^2 + \log n$) (D) O (n^2)

Q.40 Consider the directed graph shown in the figure below. There are multiple shortest paths between vertices S and T. Which one will be reported by Dijkstra's shortest path algorithm? Assume that, in any iteration, the shortest path to a vertex v is updated only when a strictly shorter path to v is discovered.



(A) SDT (B) SBDT (C) SACDT (D) SACET

Q.41 A file system with 300 GByte disk uses a file descriptor with 8 direct block addresses, 1 indirect block address and 1 doubly indirect block address. The size of each disk block is 128 Bytes and the size of each disk block address is 8 Bytes. The maximum possible file size in this file system is

(A) 3 KBytes
(B) 35 KBytes
(C) 280 KBytes
(D) dependent on the size of the disk

Q.42 Consider the virtual page reference string

1, 2, 3, 2, 4, 1, 3, 2, 4, 1

on a demand paged virtual memory system running on a computer system that has main memory size of 3 page frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacement policy. Then

(A) OPTIMAL < LRU < FIFO (B) OPTIMAL < FIFO < LRU
(C) OPTIMAL = LRU (D) OPTIMAL = FIFO

Q.43 Suppose $R_1(\underline{A}, B)$ and $R_2(\underline{C}, D)$ are two relation schemas. Let r_1 and r_2 be the corresponding relation instances. B is a foreign key that refers to C in R_2. If data in r_1 and r_2 satisfy referential integrity constraints, which of the following is **ALWAYS TRUE**?

(A) $\prod_B(r_1) - \prod_C(r_2) = \varnothing$
(B) $\prod_C(r_2) - \prod_B(r_1) = \varnothing$
(C) $\prod_B(r_1) = \prod_C(r_2)$
(D) $\prod_B(r_1) - \prod_C(r_2) \neq \varnothing$

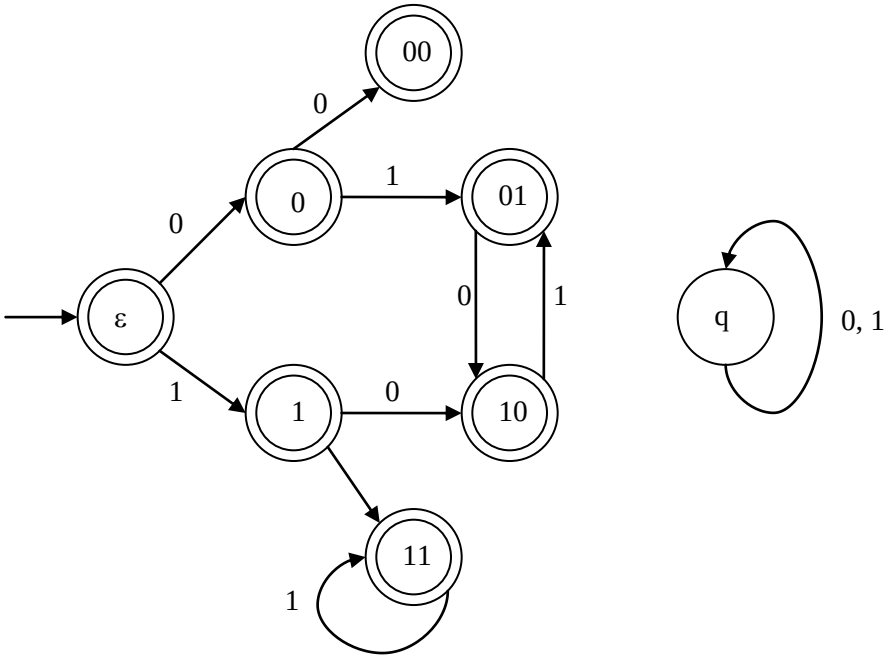
Q.44 Consider a source computer (S) transmitting a file of size 10^6 bits to a destination computer (D) over a network of two routers (R_1 and R_2) and three links (L_1, L_2, and L_3). L_1 connects S to R_1; L_2 connects R_1 to R_2; and L_3 connects R_2 to D. Let each link be of length 100 km. Assume signals travel over each link at a speed of 10^8 meters per second. Assume that the link bandwidth on each link is 1Mbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D?

(A) 1005 ms (B) 1010 ms (C) 3000 ms (D) 3003 ms

Q.45 Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

(A) 8 MSS (B) 14 MSS (C) 7 MSS (D) 12 MSS

Q.46 Consider the set of strings on {0,1} in which, *every substring of 3 symbols* has at most *two* zeros. For example, 001110 and 011001 are in the language, but 100010 is not. All strings of length less than 3 are also in the language. A partially completed DFA that accepts this language is shown below.



The missing arcs in the DFA are

(A)

	00	**01**	**10**	**11**	**q**
00	1	0			
01				1	
10	0				
11			0		

(B)

	00	**01**	**10**	**11**	**q**
00		0			1
01		1			
10				0	
11		0			

(C)

	00	**01**	**10**	**11**	**q**
00		1			0
01		1			
10			0		
11		0			

(D)

	00	**01**	**10**	**11**	**q**
00		1			0
01				1	
10	0				
11			0		

Q.47 The height of a tree is defined as the number of edges on the longest path in the tree. The function shown in the pseudocode below is invoked as `height(root)` to compute the height of a binary tree rooted at the tree pointer `root`.

```
int height (treeptr n)
{ if (n == NULL) return -1;
  if (n → left == NULL)
      if (n → right == NULL) return 0;

      else return [ B1 ] ;                          // Box 1

  else { h1 = height (n → left);
         if (n → right == NULL) return (1+h1);
         else { h2 = height (n → right);

                 return [ B2 ] ;                    // Box 2
               }
       }
}
```

The appropriate expressions for the two boxes B1 and B2 are

(A) B1: (1+height(n → right))
B2: (1+max(h1, h2))

(B) B1: (height(n → right))
B2: (1+max(h1,h2))

(C) B1: height(n → right)
B2: max(h1, h2)

(D) B1: (1+ height(n → right))
B2: max(h1, h2)

Common Data Questions

Common Data for Questions 48 and 49:

Consider the following C code segment.

```
int a, b, c = 0;
void prtFun(void);
main( )
{   static int a = 1;                /*  Line 1 */
    prtFun( );
    a += 1;
    prtFun( );
    printf(" \n %d  %d ", a, b);

}

void prtFun(void)
{   static int a = 2;               /*  Line 2 */
    int b = 1;
    a += ++b;
    printf(" \n %d  %d ", a, b);

}
```

Q.48 What output will be generated by the given code segment?

(A)		(B)		(C)		(D)	
3	1	4	2	4	2	3	1
4	1	6	1	6	2	5	2
4	2	6	1	2	0	5	2

Q.49 What output will be generated by the given code segment if:
Line 1 is replaced by **`auto int a = 1;`**
Line 2 is replaced by **`register int a = 2;`**

(A)		(B)		(C)		(D)	
3	1	4	2	4	2	4	2
4	1	6	1	6	2	4	2
4	2	6	1	2	0	2	0

Common Data for Questions 50 and 51:

Consider the following relations A, B and C:

A

Id	Name	Age
12	Arun	60
15	Shreya	24
99	Rohit	11

B

Id	Name	Age
15	Shreya	24
25	Hari	40
98	Rohit	20
99	Rohit	11

C

Id	Phone	Area
10	2200	02
99	2100	01

Q.50 How many tuples does the result of the following relational algebra expression contain? Assume that the schema of $A \cup B$ is the same as that of A.

$(A \cup B) \bowtie_{A.Id > 40 \vee C.Id < 15} C$

(A) 7 (B) 4 (C) 5 (D) 9

Q.51 How many tuples does the result of the following SQL query contain?

```
SELECT A.Id
FROM A
WHERE A.Age > ALL (SELECT B.Age
                   FROM B
                   WHERE B.Name = 'Arun')
```

(A) 4 (B) 3 (C) 0 (D) 1

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

For the grammar below, a partial LL(1) parsing table is also presented along with the grammar. Entries that need to be filled are indicated as **E1**, **E2**, and **E3**. ε is the empty string, $ indicates end of input, and, | separates alternate right hand sides of productions.

S → a A b B | b A a B | ε
A → S
B → S

	a	b	$
S	**E1**	**E2**	S → ε
A	A → S	A → S	error
B	B → S	B → S	**E3**

Q.52 The FIRST and FOLLOW sets for the non-terminals A and B are

(A) FIRST(A) = {a, b, ε} = FIRST(B)
FOLLOW(A) = {a, b}
FOLLOW(B) = {a, b, $}

(B) FIRST(A) = {a, b, $}
FIRST(B) = {a, b, ε}
FOLLOW(A) = {a, b}
FOLLOW(B) = {$}

(C) FIRST(A) = {a, b, ε} = FIRST(B)
FOLLOW(A) = {a, b}
FOLLOW(B) = ∅

(D) FIRST(A) = {a, b} = FIRST(B)
FOLLOW(A) = {a, b}
FOLLOW(B) = {a, b}

Q.53 The appropriate entries for E1, E2, and E3 are

(A) E1: S → aAbB, A → S
E2: S → bAaB, B → S
E3: B → S

(B) E1: S → aAbB, S → ε
E2: S → bAaB, S → ε
E3: S → ε

(C) E1: S → aAbB, S → ε
E2: S → bAaB, S → ε
E3: B → S

(D) E1: A → S, S → ε
E2: B → S, S → ε
E3: B → S

Statement for Linked Answer Questions 54 and 55:

A computer has a 256 KByte, 4-way set associative, write back data cache with block size of 32 Bytes. The processor sends 32 bit addresses to the cache controller. Each cache tag directory entry contains, in addition to address tag, 2 valid bits, 1 modified bit and 1 replacement bit.

Q.54 The number of bits in the tag field of an address is

(A) 11 (B) 14 (C) 16 (D) 27

Q.55 The size of the cache tag directory is

(A) 160 Kbits (B) 136 Kbits (C) 40 Kbits (D) 32 Kbits

General Aptitude (GA) Questions

Q. 56 – Q. 60 carry one mark each.

Q.56 The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of ₹ 50 per unit. The number of units to be produced by the firm such that the profit is maximized is

(A) 5 (B) 10 (C) 15 (D) 25

Q.57 Choose the most appropriate alternative from the options given below to complete the following sentence:

Despite several _____ the mission succeeded in its attempt to resolve the conflict.

(A) attempts (B) setbacks (C) meetings (D) delegations

Q.58 Which one of the following options is the closest in meaning to the word given below?

Mitigate

(A) Diminish (B) Divulge (C) Dedicate (D) Denote

Q.59 Choose the grammatically **INCORRECT** sentence:

(A) They gave us the money back less the service charges of Three Hundred rupees.
(B) This country's expenditure is not less than that of Bangladesh.
(C) The committee initially asked for a funding of Fifty Lakh rupees, but later settled for a lesser sum.
(D) This country's expenditure on educational reforms is very less.

Q.60 Choose the most appropriate alternative from the options given below to complete the following sentence:

Suresh's dog is the one _____ was hurt in the stampede.

(A) that (B) which (C) who (D) whom

Q. 61 - Q. 65 carry two marks each.

Q.61 **Wanted Temporary, Part-time persons for the post of Field Interviewer to conduct personal interviews to collect and collate economic data. Requirements: High School-pass, must be available for Day, Evening and Saturday work. Transportation paid, expenses reimbursed.**

Which one of the following is the best inference from the above advertisement?

(A) Gender-discriminatory
(B) Xenophobic
(C) Not designed to make the post attractive
(D) Not gender-discriminatory

Q.62 A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

(A) 8 meters (B) 10 meters (C) 12 meters (D) 14 meters

Q.63 An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable.

The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

(A) 0.288 (B) 0.334 (C) 0.667 (D) 0.720

Q.64 Which of the following assertions are **CORRECT**?

P: Adding 7 to each entry in a list adds 7 to the mean of the list
Q: Adding 7 to each entry in a list adds 7 to the standard deviation of the list
R: Doubling each entry in a list doubles the mean of the list
S: Doubling each entry in a list leaves the standard deviation of the list unchanged

(A) P, Q (B) Q, R (C) P, R (D) R, S

Q.65 Given the sequence of terms, AD CG FK JP, the next term is

(A) OV (B) OW (C) PV (D) PW

END OF THE QUESTION PAPER

GATE 2012 - Answer Key - Paper : CS

Paper	Question no.	Code : A	Code : B	Code : C	Code : D
CS	1	B	C	B	D
CS	2	C	C	D	C
CS	3	Marks to All	C	C	C
CS	4	B	D	D	C
CS	5	C	D	C	C
CS	6	A	C	B	C
CS	7	B	D	A	D
CS	8	C	C	C	C
CS	9	D	C	B	D
CS	10	C	C	Marks to All	D
CS	11	D	C	C	C
CS	12	B	C	B	C
CS	13	C	D	C	B
CS	14	C	B	D	C
CS	15	C	C	C	D
CS	16	D	Marks to All	C	C
CS	17	D	B	C	D
CS	18	C	C	C	C
CS	19	D	A	D	B
CS	20	C	B	C	A
CS	21	C	C	D	C
CS	22	C	D	D	B
CS	23	C	C	C	Marks to All
CS	24	D	D	C	C
CS	25	C	B	C	B
CS	26	B	C	D	D
CS	27	B	Marks to All	A	A
CS	28	B	Marks to All	A	Marks to All
CS	29	Marks to All	D	B	A
CS	30	B	B	B	A
CS	31	C	B	C	B
CS	32	B	A	B	B
CS	33	B	A	Marks to All	D
CS	34	A	Marks to All	B	Marks to All
CS	35	A	A	B	Marks to All
CS	36	D	D	B	C
CS	37	C	B	D	D
CS	38	Marks to All	B	A	A
CS	39	Marks to All	B	Marks to All	A
CS	40	D	Marks to All	A	B

GATE 2012 - Answer Key - Paper : CS

Paper	Question no.	Code : A	Code : B	Code : C	Code : D
CS	41	B	B	A	B
CS	42	B	C	B	C
CS	43	A	B	B	B
CS	44	A	B	D	Marks to All
CS	45	Marks to All	A	Marks to All	B
CS	46	D	A	Marks to All	B
CS	47	A	D	C	B
CS	48	C	D	A	B
CS	49	D	C	B	A
CS	50	A	B	C	D
CS	51	B	A	D	C
CS	52	A	C	A	C
CS	53	C	A	C	A
CS	54	C	A	C	A
CS	55	A	C	A	C
CS	56	A	B	A	D
CS	57	B	A	D	Marks to All
CS	58	A	D	Marks to All	A
CS	59	D	Marks to All	A	B
CS	60	Marks to All	A	B	A
CS	61	D	B	B	C
CS	62	B	B	C	A
CS	63	B	C	A	D
CS	64	C	A	D	B
CS	65	A	D	B	B

SESSION - 1

CS : COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Duration: Three Hours *Maximum Marks*: 100

Read the following instructions carefully.

1. Do not open the seal of the Question Booklet until you are asked to do so by the invigilator.
2. Take out the **O**ptical **R**esponse **S**heet (**ORS)** from this Question Booklet **without breaking the seal** and read the instructions printed on the ORS carefully. If you find that either
   a. The Question Booklet Code printed at the right hand top corner of this page does not match with the Question Booklet Code at the right hand top corner of the **ORS** or
   b. The Question Paper Code preceding the Registration number on the **ORS** is not **CS**, then exchange the booklet immediately with a new sealed Question Booklet.
3. On the right hand side of the **ORS,** using ONLY a **black ink ballpoint pen**, (i) darken the appropriate bubble under each digit of your registration number and (ii) write your registration number, your name and name of the examination centre and put your signature at the specified location.
4. This Question Booklet contains **20** pages including blank pages for rough work. After you are permitted to open the seal, check all pages and report discrepancies, if any, to the invigilator.
5. There are a total of 65 questions carrying 100 marks. All these questions are of objective type. Each question has only **one** correct answer. Questions must be answered on the left hand side of the **ORS** by darkening the appropriate bubble (marked A, B, C, D) using ONLY a **black ink ballpoint pen** against the question number. **For each question darken the bubble of the correct answer**. More than one answer bubbled against a question will be treated as an incorrect response.
6. Since bubbles darkened by the black ink ballpoint pen **cannot** be erased, candidates should darken the bubbles in the ORS **very carefully**.
7. Questions Q.1 – Q.25 carry 1 mark each. Questions Q.26 – Q.55 carry 2 marks each. The 2 marks questions include two pairs of common data questions and two pairs of linked answer questions. The answer to the second question of the linked answer questions depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is not attempted, then the answer to the second question in the pair will not be evaluated.
8. Questions Q.56 – Q.65 belong to General Aptitude (GA) section and carry a total of 15 marks. Questions Q.56 – Q.60 carry 1 mark each, and questions Q.61 – Q.65 carry 2 marks each.
9. Questions not attempted will result in zero mark and wrong answers will result in **NEGATIVE** marks. For all 1 mark questions, ⅓ mark will be deducted for each wrong answer. For all 2 marks questions, ⅔ mark will be deducted for each wrong answer. However, in the case of the linked answer question pair, there will be negative marks only for wrong answer to the first question and no negative marks for wrong answer to the second question.
10. Calculator is allowed whereas charts, graph sheets or tables are **NOT** allowed in the examination hall.
11. Rough work can be done on the Question Booklet itself. Blank pages are provided at the end of the Question Booklet for rough work.
12. Before the start of the examination, write your name and registration number in the space provided below using a black ink ballpoint pen.

Name									
Registration Number	**CS**								

Q.1 to Q.25 carry one mark each.

Q.1 A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is **TRUE** about $\oplus$?

(A) Commutative but not associative
(B) Both commutative and associative
(C) Associative but not commutative
(D) Neither commutative nor associative

Q.2 Suppose p is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM, and p has a Poisson distribution with mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

(A) $8/(2e^3)$ (B) $9/(2e^3)$ (C) $17/(2e^3)$ (D) $26/(2e^3)$

Q.3 Which one of the following does **NOT** equal $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$?

(A) $\begin{vmatrix} 1 & x(x+1) & x+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \end{vmatrix}$

(B) $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \end{vmatrix}$

(C) $\begin{vmatrix} 0 & x-y & x^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$

(D) $\begin{vmatrix} 2 & x+y & x^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

Q.4 The smallest integer that can be represented by an 8-bit number in 2's complement form is

(A) -256 (B) -128 (C) -127 (D) 0

Q.5 In the following truth table, V = 1 if and only if the input is valid.

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

What function does the truth table represent?

(A) Priority encoder
(B) Decoder
(C) Multiplexer
(D) Demultiplexer

Q.6 Which one of the following is the tightest upper bound that represents the number of swaps required to sort n numbers using selection sort?

(A) $O(\log n)$ (B) $O(n)$ (C) $O(n \log n)$ (D) $O(n^2)$

Q.7 Which one of the following is the tightest upper bound that represents the time complexity of inserting an object into a binary search tree of n nodes?

(A) $O(1)$ (B) $O(\log n)$ (C) $O(n)$ (D) $O(n \log n)$

Q.8 Consider the languages $L_1 = \Phi$ and $L_2 = \{a\}$. Which one of the following represents $L_1 L_2^* \cup L_1^*$?

(A) {ϵ} (B) Φ (C) *a** (D) {ϵ, *a*}

Q.9 What is the maximum number of reduce moves that can be taken by a bottom-up parser for a grammar with no epsilon- and unit-production (i.e., of type A → ϵ and A → a) to parse a string with *n* tokens?

(A) *n*/2 (B) *n*-1 (C) 2*n*-1 (D) 2^n

Q.10 A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every *T* time units and decides the next process to schedule. Which one of the following is **TRUE** if the processes have no I/O operations and all arrive at time zero?

(A) This algorithm is equivalent to the first-come-first-serve algorithm.
(B) This algorithm is equivalent to the round-robin algorithm.
(C) This algorithm is equivalent to the shortest-job-first algorithm.
(D) This algorithm is equivalent to the shortest-remaining-time-first algorithm.

Q.11 Match the problem domains in **GROUP I** with the solution technologies in **GROUP II**.

GROUP I	**GROUP II**
(P) Service oriented computing	(1) Interoperability
(Q) Heterogeneous communicating systems	(2) BPMN
(R) Information representation	(3) Publish-find-bind
(S) Process description	(4) XML

(A) P-1, Q-2, R-3, S-4 (B) P-3, Q-4, R-2, S-1
(C) P-3, Q-1, R-4, S-2 (D) P-4, Q-3, R-2, S-1

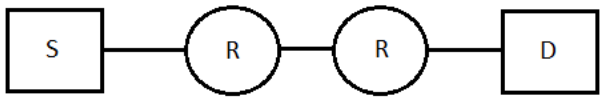
Q.12 The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are

(A) TCP, UDP, UDP and TCP
(B) UDP, TCP, TCP and UDP
(C) UDP, TCP, UDP and TCP
(D) TCP, UDP, TCP and UDP

Q.13 Using public key cryptography, X adds a digital signature σ to message M, encrypts <M, σ>, and sends it to Y, where it is decrypted. Which one of the following sequences of keys is used for the operations?

(A) Encryption: X's private key followed by Y's private key; Decryption: X's public key followed by Y's public key
(B) Encryption: X's private key followed by Y's public key; Decryption: X's public key followed by Y's private key
(C) Encryption: X's public key followed by Y's private key; Decryption: Y's public key followed by X's private key
(D) Encryption: X's private key followed by Y's public key; Decryption: Y's private key followed by X's public key

Q.14 Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.



(A) Network layer – 4 times and Data link layer – 4 times
(B) Network layer – 4 times and Data link layer – 3 times
(C) Network layer – 4 times and Data link layer – 6 times
(D) Network layer – 2 times and Data link layer – 6 times

Q.15 An index is clustered, if

(A) it is on a set of fields that form a candidate key.
(B) it is on a set of fields that include the primary key.
(C) the data records of the file are organized in the same order as the data entries of the index.
(D) the data records of the file are organized not in the same order as the data entries of the index.

Q.16 Three concurrent processes *X*, *Y*, and *Z* execute three different code segments that access and update certain shared variables. Process *X* executes the *P* operation (i.e., *wait*) on semaphores *a*, *b* and *c*; process *Y* executes the *P* operation on semaphores *b*, *c* and *d*; process *Z* executes the *P* operation on semaphores *c*, *d*, and *a* before entering the respective code segments. After completing the execution of its code segment, each process invokes the *V* operation (i.e., *signal*) on its three semaphores. All semaphores are binary semaphores initialized to one. Which one of the following represents a deadlock-free order of invoking the *P* operations by the processes?

(A) *X*: *P*(*a*)*P*(*b*)*P*(*c*) *Y*: *P*(*b*)*P*(*c*)*P*(*d*) *Z*: *P*(*c*)*P*(*d*)*P*(*a*)
(B) *X*: *P*(*b*)*P*(*a*)*P*(*c*) *Y*: *P*(*b*)*P*(*c*)*P*(*d*) *Z*: *P*(*a*)*P*(*c*)*P*(*d*)
(C) *X*: *P*(*b*)*P*(*a*)*P*(*c*) *Y*: *P*(*c*)*P*(*b*)*P*(*d*) *Z*: *P*(*a*)*P*(*c*)*P*(*d*)
(D) *X*: *P*(*a*)*P*(*b*)*P*(*c*) *Y*: *P*(*c*)*P*(*b*)*P*(*d*) *Z*: *P*(*c*)*P*(*d*)*P*(*a*)

Q.17 Which of the following statements is/are **FALSE**?

1. For every non-deterministic Turing machine, there exists an equivalent deterministic Turing machine.
2. Turing recognizable languages are closed under union and complementation.
3. Turing decidable languages are closed under intersection and complementation.
4. Turing recognizable languages are closed under union and intersection.

(A) 1 and 4 only (B) 1 and 3 only (C) 2 only (D) 3 only

Q.18 Which of the following statements are **TRUE**?

1. The problem of determining whether there exists a cycle in an undirected graph is in P.
2. The problem of determining whether there exists a cycle in an undirected graph is in NP.
3. If a problem A is NP-Complete, there exists a non-deterministic polynomial time algorithm to solve A.

(A) 1, 2 and 3 (B) 1 and 2 only (C) 2 and 3 only (D) 1 and 3 only

Q.19 What is the time complexity of Bellman-Ford single-source shortest path algorithm on a complete graph of n vertices?

(A) $\Theta(n^2)$ (B) $\Theta(n^2 \log n)$ (C) $\Theta(n^3)$ (D) $\Theta(n^3 \log n)$

Q.20 In a k-way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set are placed in sequence one after another. The lines in set s are sequenced before the lines in set (s+1). The main memory blocks are numbered 0 onwards. The main memory block numbered j must be mapped to any one of the cache lines from

(A) (j mod v) * k to (j mod v) * k + (k-1) (B) (j mod v) to (j mod v) + (k-1)
(C) (j mod k) to (j mod k) + (v-1) (D) (j mod k) * v to (j mod k) * v + (v-1)

Q.21 Which one of the following expressions does **NOT** represent exclusive NOR of x and y?

(A) $xy + x'y'$ (B) $x \oplus y'$ (C) $x' \oplus y$ (D) $x' \oplus y'$

Q.22 Which one of the following functions is continuous at $x = 3$?

(A) $f(x) = \begin{cases} 2, & \text{if } x = 3 \\ x-1, & \text{if } x > 3 \\ \dfrac{x+3}{3}, & \text{if } x < 3 \end{cases}$ (B) $f(x) = \begin{cases} 4, & \text{if } x = 3 \\ 8-x & \text{if } x \neq 3 \end{cases}$

(C) $f(x) = \begin{cases} x+3, & \text{if } x \leq 3 \\ x-4 & \text{if } x > 3 \end{cases}$ (D) $f(x) = \dfrac{1}{x^3 - 27}$, if $x \neq 3$

Q.23 Function f is known at the following points:

x	0	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4	2.7	3.0
$f(x)$	0	0.09	0.36	0.81	1.44	2.25	3.24	4.41	5.76	7.29	9.00

The value of $\int_0^3 f(x)\,dx$ computed using the trapezoidal rule is

(A) 8.983 (B) 9.003 (C) 9.017 (D) 9.045

Q.24 Consider an undirected random graph of eight vertices. The probability that there is an edge between a pair of vertices is 1/2. What is the expected number of unordered cycles of length three?

(A) 1/8 (B) 1 (C) 7 (D) 8

Q.25 Which of the following statements is/are **TRUE** for undirected graphs?

P: Number of odd degree vertices is even.
Q: Sum of degrees of all vertices is even.

(A) P only (B) Q only (C) Both P and Q (D) Neither P nor Q

Q.26 to Q.55 carry two marks each.

Q.26 The line graph *L(G)* of a simple graph *G* is defined as follows:

- There is exactly one vertex *v(e)* in *L(G)* for each edge *e* in *G*.
- For any two edges *e* and *e'* in *G*, *L(G)* has an edge between *v(e)* and *v(e')*, if and only if *e* and *e'* are incident with the same vertex in *G*.

Which of the following statements is/are **TRUE**?

(P) The line graph of a cycle is a cycle.
(Q) The line graph of a clique is a clique.
(R) The line graph of a planar graph is planar.
(S) The line graph of a tree is a tree.

(A) P only
(B) P and R only
(C) R only
(D) P, Q and S only

Q.27 What is the logical translation of the following statement?

"None of my friends are perfect."

(A) $\exists x(F(x) \wedge \neg P(x))$
(B) $\exists x(\neg F(x) \wedge P(x))$
(C) $\exists x(\neg F(x) \wedge \neg P(x))$
(D) $\neg \exists x(F(x) \wedge P(x))$

Q.28 Consider the following sequence of micro-operations.

MBR ← PC
MAR ← X
PC ← Y
Memory ← MBR

Which one of the following is a possible operation performed by this sequence?

(A) Instruction fetch
(B) Operand fetch
(C) Conditional branch
(D) Initiation of interrupt service

Q.29 Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each cylinder contains 64 sectors (0-63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is <cylinder no., surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

(A) 1281
(B) 1282
(C) 1283
(D) 1284

Q.30 The number of elements that can be sorted in $\Theta(\log n)$ time using heap sort is

(A) $\Theta(1)$
(B) $\Theta\left(\sqrt{\log n}\right)$
(C) $\Theta\left(\frac{\log n}{\log \log n}\right)$
(D) $\Theta(\log n)$

Q.31 Consider the following function:

```
int unknown(int n){
          int i, j, k=0;
          for (i=n/2; i<=n; i++)
              for (j=2; j<=n; j=j*2)
                  k = k + n/2;
          return (k);
}
```

The return value of the function is

(A) $\Theta(n^2)$ (B) $\Theta(n^2 \log n)$ (C) $\Theta(n^3)$ (D) $\Theta(n^3 \log n)$

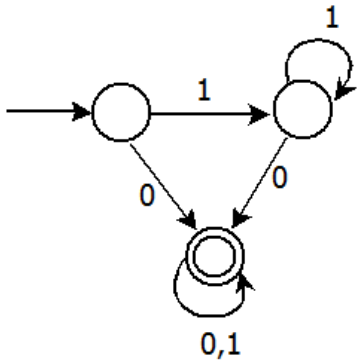
Q.32 Consider the following languages.

$$L_1 = \{0^p\, 1^q\, 0^r \mid p, q, r \geq 0\}$$
$$L_2 = \{0^p\, 1^q\, 0^r \mid p, q, r \geq 0,\ p \neq r\}$$

Which one of the following statements is **FALSE**?

(A) L_2 is context-free.
(B) $L_1 \cap L_2$ is context-free.
(C) Complement of L_2 is recursive.
(D) Complement of L_1 is context-free but not regular.

Q.33 Consider the DFA A given below.



Which of the following are **FALSE**?

1. Complement of $L(A)$ is context-free.
2. $L(A) = L((11*0+0)(0 + 1)*0*1*)$
3. For the language accepted by A, A is the minimal DFA.
4. A accepts all strings over {0, 1} of length at least 2.

(A) 1 and 3 only (B) 2 and 4 only (C) 2 and 3 only (D) 3 and 4 only

Q.34 A shared variable x, initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to memory, and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e., *wait*) on a counting semaphore S and invokes the V operation (i.e., *signal*) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?

(A) –2 (B) –1 (C) 1 (D) 2

Q.35 Consider the following relational schema.

Students(rollno: integer, sname: string)
Courses(courseno: integer, cname: string)
Registration(rollno: integer, courseno: integer, percent: real)

Which of the following queries are equivalent to this query in English?
"Find the distinct names of all students who score more than 90% in the course numbered 107"

(I) SELECT DISTINCT S.sname
FROM Students as S, Registration as R
WHERE R.rollno=S.rollno AND R.courseno=107 AND R.percent >90

(II) $\prod_{sname}(\sigma_{courseno=107 \wedge percent>90}$ (Registration $\bowtie$ Students))

(III) $\{T \mid \exists S \in$ Students, $\exists R \in$ Registration (S.rollno=R.rollno $\wedge$

R.courseno=107 $\wedge$ R.percent>90 $\wedge$T.sname=S.sname)}

(IV) $\{<S_N> \mid \exists S_R \exists R_P$ ($<S_R, S_N> \in$ Students $\wedge <S_R, 107, R_P> \in$ Registration $\wedge R_P>90)\}$

(A) I, II, III and IV
(B) I, II and III only
(C) I, II and IV only
(D) II, III and IV only

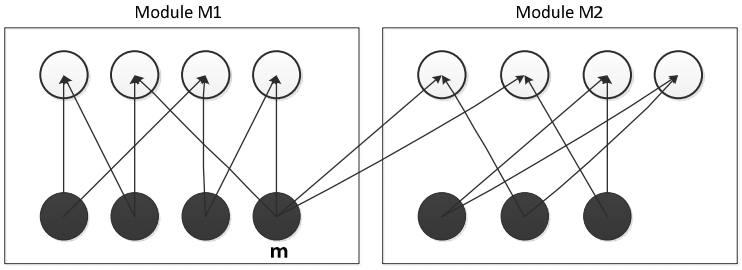
Q.36 Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

(A) 1 (B) 2 (C) 2.5 (D) 5

Q.37 In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are

(A) Last fragment, 2400 and 2789
(B) First fragment, 2400 and 2759
(C) Last fragment, 2400 and 2759
(D) Middle fragment, 300 and 689

Q.38 The following figure represents access graphs of two modules M1 and M2. The filled circles represent methods and the unfilled circles represent attributes. If method m is moved to module M2 keeping the attributes where they are, what can we say about the average cohesion and coupling between modules in the system of two modules?



(A) There is no change.
(B) Average cohesion goes up but coupling is reduced.
(C) Average cohesion goes down and coupling also reduces.
(D) Average cohesion and coupling increase.

Q.39 A certain computation generates two arrays `a` and `b` such that `a[i]=f(i)` for $0 \le i < n$ and b[i] = g (a[i])for $0 \le i < n$. Suppose this computation is decomposed into two concurrent processes *X* and *Y* such that *X* computes the array `a` and *Y* computes the array `b`. The processes employ two binary semaphores *R* and *S*, both initialized to zero. The array `a` is shared by the two processes. The structures of the processes are shown below.

Process *X*:
```
private i;
for (i=0; i<n; i++) {
   a[i] = f(i);
   ExitX(R, S);
}
```

Process *Y*:
```
private i;
for (i=0; i<n; i++) {
   EntryY(R, S);
   b[i] = g(a[i]);
}
```

Which one of the following represents the **CORRECT** implementations of `ExitX` and `EntryY`?

(A)
```
ExitX(R, S) {
   P(R);
   V(S);
}
EntryY(R, S) {
   P(S);
   V(R);
}
```

(B)
```
ExitX(R, S) {
   V(R);
   V(S);
}
EntryY(R, S) {
   P(R);
   P(S);
}
```

(C)
```
ExitX(R, S) {
   P(S);
   V(R);
}
EntryY(R, S) {
   V(S);
   P(R);
}
```

(D)
```
ExitX(R, S) {
   V(R);
   P(S);
}
EntryY(R, S) {
   V(S);
   P(R);
}
```

Q.40 Consider the following two sets of LR(1) items of an LR(1) grammar.

X → c**.**X, c/d	X → c**.**X, $
X → **.**cX, c/d	X → **.**cX, $
X → **.**d, c/d	X → **.**d, $

Which of the following statements related to merging of the two sets in the corresponding LALR parser is/are **FALSE**?

1. Cannot be merged since look aheads are different.
2. Can be merged but will result in S-R conflict.
3. Can be merged but will result in R-R conflict.
4. Cannot be merged since *goto* on *c* will lead to two different sets.

(A) 1 only (B) 2 only (C) 1 and 4 only (D) 1, 2, 3 and 4

Q.41 Which of the following is/are undecidable?

1. *G* is a CFG. Is $L(G) = \Phi$?
2. *G* is a CFG. Is $L(G) = \sum^*$?
3. *M* is a Turing machine. Is *L*(*M*) regular?
4. *A* is a DFA and *N* is an NFA. Is *L*(*A*) = *L*(*N*)?

(A) 3 only (B) 3 and 4 only (C) 1, 2 and 3 only (D) 2 and 3 only

Q.42 What is the return value of `f(p,p)`, if the value of `p` is initialized to 5 before the call? Note that the first parameter is passed by reference, whereas the second parameter is passed by value.

```
int f (int &x, int c) {
   c = c - 1;
   if (c==0) return 1;
   x = x + 1;
   return f(x,c) * x;
}
```

(A) 3024 (B) 6561 (C) 55440 (D) 161051

Q.43 The preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42. Which one of the following is the postorder traversal sequence of the same tree?

(A) 10, 20, 15, 23, 25, 35, 42, 39, 30 (B) 15, 10, 25, 23, 20, 42, 35, 39, 30
(C) 15, 20, 10, 23, 25, 42, 35, 39, 30 (D) 15, 10, 23, 25, 20, 35, 42, 39, 30

Q.44 Consider the following operation along with `Enqueue` and `Dequeue` operations on queues, where *k* is a global parameter.

```
MultiDequeue(Q){
      m = k
      while (Q is not empty) and (m > 0) {
            Dequeue(Q)
            m = m – 1
      }
}
```

What is the worst case time complexity of a sequence of *n* queue operations on an initially empty queue?

(A) $\Theta(n)$ (B) $\Theta(n + k)$ (C) $\Theta(nk)$ (D) $\Theta(n^2)$

Q.45 Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI and WO are 5 ns, 7 ns, 10 ns, 8 ns and 6 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 1 ns. A program consisting of 12 instructions I_1, I_2, I_3, …, I_{12} is executed in this pipelined processor. Instruction I_4 is the only branch instruction and its branch target is I_9. If the branch is taken during the execution of this program, the time (in ns) needed to complete the program is

(A) 132 (B) 165 (C) 176 (D) 328

Q.46 A RAM chip has a capacity of 1024 words of 8 bits each (1K × 8). The number of 2 × 4 decoders with enable line needed to construct a 16K × 16 RAM from 1K × 8 RAM is

(A) 4 (B) 5 (C) 6 (D) 7

Q.47 Which one of the following is **NOT** logically equivalent to $\neg\exists x(\forall y(\alpha)\wedge\forall z(\beta))$?

(A) $\forall x(\exists z(\neg\beta)\rightarrow\forall y(\alpha))$ (B) $\forall x(\forall z(\beta)\rightarrow\exists y(\neg\alpha))$
(C) $\forall x(\forall y(\alpha)\rightarrow\exists z(\neg\beta))$ (D) $\forall x(\exists y(\neg\alpha)\rightarrow\exists z(\neg\beta))$

Common Data Questions

Common Data for Questions 48 and 49:

The following code segment is executed on a processor which allows only register operands in its instructions. Each instruction can have atmost two source operands and one destination operand. Assume that all variables are dead after this code segment.

```
c = a + b;
d = c * a;
e = c + a;
x = c * c;
if (x > a) {
   y = a * a;
}
else {
   d = d * d;
   e = e * e;
}
```

Q.48 Suppose the instruction set architecture of the processor has only two registers. The only allowed compiler optimization is code motion, which moves statements from one place to another while preserving correctness. What is the minimum number of spills to memory in the compiled code?

(A) 0 (B) 1 (C) 2 (D) 3

Q.49 What is the minimum number of registers needed in the instruction set architecture of the processor to compile this code segment without any spill to memory? Do not apply any optimization other than optimizing register allocation.

(A) 3 (B) 4 (C) 5 (D) 6

Common Data for Questions 50 and 51:

The procedure given below is required to find and replace certain characters inside an input character string supplied in array `A`. The characters to be replaced are supplied in array `oldc`, while their respective replacement characters are supplied in array `newc`. Array `A` has a fixed length of five characters, while arrays `oldc` and `newc` contain three characters each. However, the procedure is flawed.

```
void find_and_replace (char *A, char *oldc, char *newc) {
        for (int i=0; i<5; i++)
            for (int j=0; j<3; j++)
                if (A[i] == oldc[j])  A[i] = newc[j];
 }
```

The procedure is tested with the following four test cases.

(1) `oldc = "abc", newc = "dab"` (2) `oldc = "cde", newc = "bcd"`
(3) `oldc = "bca", newc = "cda"` (4) `oldc = "abc", newc = "bac"`

Q.50 The tester now tests the program on all input strings of length five consisting of characters `'a'`, `'b'`, `'c'`, `'d'` and `'e'` with duplicates allowed. If the tester carries out this testing with the four test cases given above, how many test cases will be able to capture the flaw?

(A) Only one (B) Only two (C) Only three (D) All four

Q.51 If array `A` is made to hold the string `"abcde"`, which of the above four test cases will be successful in exposing the flaw in this procedure?

(A) None (B) 2 only
(C) 3 and 4 only (D) 4 only

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table (T_1), which occupies exactly one page. Each entry of T_1 stores the base address of a page of the second-level table (T_2). Each entry of T_2 stores the base address of a page of the third-level table (T_3). Each entry of T_3 stores a page table entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16-way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes.

Q.52 What is the size of a page in KB in this computer?

(A) 2 (B) 4 (C) 8 (D) 16

Q.53 What is the minimum number of page colours needed to guarantee that no two synonyms map to different sets in the processor cache of this computer?

(A) 2 (B) 4 (C) 8 (D) 16

Statement for Linked Answer Questions 54 and 55:

Relation R has eight attributes $ABCDEFGH$. Fields of R contain only atomic values.
$F=\{CH\rightarrow G,\ A\rightarrow BC,\ B\rightarrow CFH,\ E\rightarrow A,\ F\rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.

Q.54 How many candidate keys does the relation R have?

(A) 3 (B) 4 (C) 5 (D) 6

Q.55 The relation R is

(A) in 1NF, but not in 2NF.
(B) in 2NF, but not in 3NF.
(C) in 3NF, but not in BCNF.
(D) in BCNF.

General Aptitude (GA) Questions

Q.56 to Q.60 carry one mark each.

Q.56 Which one of the following options is the closest in meaning to the word given below?

Nadir

(A) Highest (B) Lowest (C) Medium (D) Integration

Q.57 Complete the sentence:
Universalism is to particularism as diffuseness is to _____.

(A) specificity (B) neutrality (C) generality (D) adaptation

Q.58 What will be the maximum sum of 44, 42, 40, ... ?

(A) 502 (B) 504 (C) 506 (D) 500

Q.59 Were you a bird, you _____ in the sky.

(A) would fly (B) shall fly (C) should fly (D) shall have flown

Q.60 Choose the grammatically **INCORRECT** sentence:

(A) He is of Asian origin.
(B) They belonged to Africa.
(C) She is an European.
(D) They migrated from India to Australia.

Q.61 to Q.65 carry two marks each.

Q.61 Find the sum of the expression

$$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+.....+\frac{1}{\sqrt{80}+\sqrt{81}}$$

(A) 7 (B) 8
(C) 9 (D) 10

Q.62 Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

(A) 13/90 (B) 12/90 (C) 78/90 (D) 77/90

Q.63 After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again, the spider failed but that did not deter it to refrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

Which one of the following assertions is best supported by the above information?

(A) Failure is the pillar of success.
(B) Honesty is the best policy.
(C) Life begins and ends with adventures.
(D) No adversity justifies giving up hope.

Q.64 A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

(A) 36 (B) 30 (C) 24 (D) 18

Q.65 The current erection cost of a structure is *Rs*. 13,200. If the labour wages per day increase by 1/5 of the current wages and the working hours decrease by 1/24 of the current period, then the new cost of erection in *Rs*. is

(A) 16,500 (B) 15,180 (C) 11,000 (D) 10,120

END OF THE QUESTION PAPER

SESSION - 2

CS : COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Duration: Three Hours *Maximum Marks*: 100

Read the following instructions carefully.

1. Do not open the seal of the Question Booklet until you are asked to do so by the invigilator.
2. Take out the **O**ptical **R**esponse **S**heet (**ORS)** from this Question Booklet **without breaking the seal** and read the instructions printed on the ORS carefully. If you find that either
   a. The Question Booklet Code printed at the right hand top corner of this page does not match with the Question Booklet Code at the right hand top corner of the **ORS** or
   b. The Question Paper Code preceding the Registration number on the **ORS** is not **CS**, then exchange the booklet immediately with a new sealed Question Booklet.
3. On the right hand side of the **ORS,** using ONLY a **black ink ballpoint pen**, (i) darken the appropriate bubble under each digit of your registration number and (ii) write your registration number, your name and name of the examination centre and put your signature at the specified location.
4. This Question Booklet contains **20** pages including blank pages for rough work. After you are permitted to open the seal, check all pages and report discrepancies, if any, to the invigilator.
5. There are a total of 65 questions carrying 100 marks. All these questions are of objective type. Each question has only **one** correct answer. Questions must be answered on the left hand side of the **ORS** by darkening the appropriate bubble (marked A, B, C, D) using ONLY a **black ink ballpoint pen** against the question number. **For each question darken the bubble of the correct answer**. More than one answer bubbled against a question will be treated as an incorrect response.
6. Since bubbles darkened by the black ink ballpoint pen **cannot** be erased, candidates should darken the bubbles in the ORS **very carefully**.
7. Questions Q.1 – Q.25 carry 1 mark each. Questions Q.26 – Q.55 carry 2 marks each. The 2 marks questions include two pairs of common data questions and two pairs of linked answer questions. The answer to the second question of the linked answer questions depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is not attempted, then the answer to the second question in the pair will not be evaluated.
8. Questions Q.56 – Q.65 belong to General Aptitude (GA) section and carry a total of 15 marks. Questions Q.56 – Q.60 carry 1 mark each, and questions Q.61 – Q.65 carry 2 marks each.
9. Questions not attempted will result in zero mark and wrong answers will result in **NEGATIVE** marks. For all 1 mark questions, ⅓ mark will be deducted for each wrong answer. For all 2 marks questions, ⅔ mark will be deducted for each wrong answer. However, in the case of the linked answer question pair, there will be negative marks only for wrong answer to the first question and no negative marks for wrong answer to the second question.
10. Calculator is allowed whereas charts, graph sheets or tables are **NOT** allowed in the examination hall.
11. Rough work can be done on the Question Booklet itself. Blank pages are provided at the end of the Question Booklet for rough work.
12. Before the start of the examination, write your name and registration number in the space provided below using a black ink ballpoint pen.

Name									
Registration Number	**CS**								

Q.1 to Q.25 carry one mark each.

Q.1 Consider an undirected random graph of eight vertices. The probability that there is an edge between a pair of vertices is 1/2. What is the expected number of unordered cycles of length three?

(A) 1/8 (B) 1 (C) 7 (D) 8

Q.2 Which of the following statements is/are **TRUE** for undirected graphs?

P: Number of odd degree vertices is even.
Q: Sum of degrees of all vertices is even.

(A) P only (B) Q only (C) Both P and Q (D) Neither P nor Q

Q.3 Function f is known at the following points:

x	0	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4	2.7	3.0
$f(x)$	0	0.09	0.36	0.81	1.44	2.25	3.24	4.41	5.76	7.29	9.00

The value of $\int_0^3 f(x)\,dx$ computed using the trapezoidal rule is

(A) 8.983 (B) 9.003 (C) 9.017 (D) 9.045

Q.4 Which one of the following functions is continuous at $x = 3$?

(A) $f(x)=\begin{cases} 2, & \text{if } x=3 \\ x-1, & \text{if } x>3 \\ \dfrac{x+3}{3}, & \text{if } x<3 \end{cases}$

(B) $f(x)=\begin{cases} 4, & \text{if } x=3 \\ 8-x & \text{if } x\neq 3 \end{cases}$

(C) $f(x)=\begin{cases} x+3, & \text{if } x\leq 3 \\ x-4 & \text{if } x>3 \end{cases}$

(D) $f(x)=\dfrac{1}{x^3-27}$, if $x \neq 3$

Q.5 Which one of the following expressions does **NOT** represent exclusive NOR of x and y?

(A) $xy+x'y'$ (B) $x\oplus y'$ (C) $x'\oplus y$ (D) $x'\oplus y'$

Q.6 In a k-way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set are placed in sequence one after another. The lines in set s are sequenced before the lines in set (s+1). The main memory blocks are numbered 0 onwards. The main memory block numbered j must be mapped to any one of the cache lines from

(A) (j mod v) * k to (j mod v) * k + (k-1) (B) (j mod v) to (j mod v) + (k-1)
(C) (j mod k) to (j mod k) + (v-1) (D) (j mod k) * v to (j mod k) * v + (v-1)

Q.7 What is the time complexity of Bellman-Ford single-source shortest path algorithm on a complete graph of n vertices?

(A) $\Theta(n^2)$ (B) $\Theta(n^2 \log n)$ (C) $\Theta(n^3)$ (D) $\Theta(n^3 \log n)$

Q.8 Which of the following statements are **TRUE**?

1. The problem of determining whether there exists a cycle in an undirected graph is in P.
2. The problem of determining whether there exists a cycle in an undirected graph is in NP.
3. If a problem A is NP-Complete, there exists a non-deterministic polynomial time algorithm to solve A.

(A) 1, 2 and 3 (B) 1 and 2 only (C) 2 and 3 only (D) 1 and 3 only

Q.9 Which of the following statements is/are **FALSE**?

1. For every non-deterministic Turing machine, there exists an equivalent deterministic Turing machine.
2. Turing recognizable languages are closed under union and complementation.
3. Turing decidable languages are closed under intersection and complementation.
4. Turing recognizable languages are closed under union and intersection.

(A) 1 and 4 only (B) 1 and 3 only (C) 2 only (D) 3 only

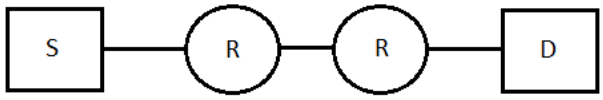
Q.10 Three concurrent processes *X*, *Y*, and *Z* execute three different code segments that access and update certain shared variables. Process *X* executes the *P* operation (i.e., *wait*) on semaphores *a*, *b* and *c*; process *Y* executes the *P* operation on semaphores *b*, *c* and *d*; process *Z* executes the *P* operation on semaphores *c*, *d*, and *a* before entering the respective code segments. After completing the execution of its code segment, each process invokes the *V* operation (i.e., *signal*) on its three semaphores. All semaphores are binary semaphores initialized to one. Which one of the following represents a deadlock-free order of invoking the *P* operations by the processes?

(A) *X*: *P*(*a*)*P*(*b*)*P*(*c*) *Y*: *P*(*b*)*P*(*c*)*P*(*d*) *Z*: *P*(*c*)*P*(*d*)*P*(*a*)
(B) *X*: *P*(*b*)*P*(*a*)*P*(*c*) *Y*: *P*(*b*)*P*(*c*)*P*(*d*) *Z*: *P*(*a*)*P*(*c*)*P*(*d*)
(C) *X*: *P*(*b*)*P*(*a*)*P*(*c*) *Y*: *P*(*c*)*P*(*b*)*P*(*d*) *Z*: *P*(*a*)*P*(*c*)*P*(*d*)
(D) *X*: *P*(*a*)*P*(*b*)*P*(*c*) *Y*: *P*(*c*)*P*(*b*)*P*(*d*) *Z*: *P*(*c*)*P*(*d*)*P*(*a*)

Q.11 An index is clustered, if

(A) it is on a set of fields that form a candidate key.
(B) it is on a set of fields that include the primary key.
(C) the data records of the file are organized in the same order as the data entries of the index.
(D) the data records of the file are organized not in the same order as the data entries of the index.

Q.12 Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.



(A) Network layer – 4 times and Data link layer – 4 times
(B) Network layer – 4 times and Data link layer – 3 times
(C) Network layer – 4 times and Data link layer – 6 times
(D) Network layer – 2 times and Data link layer – 6 times

Q.13 The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are

(A) TCP, UDP, UDP and TCP
(B) UDP, TCP, TCP and UDP
(C) UDP, TCP, UDP and TCP
(D) TCP, UDP, TCP and UDP

Q.14 Using public key cryptography, X adds a digital signature σ to message M, encrypts <M, σ>, and sends it to Y, where it is decrypted. Which one of the following sequences of keys is used for the operations?

(A) Encryption: X's private key followed by Y's private key; Decryption: X's public key followed by Y's public key
(B) Encryption: X's private key followed by Y's public key; Decryption: X's public key followed by Y's private key
(C) Encryption: X's public key followed by Y's private key; Decryption: Y's public key followed by X's private key
(D) Encryption: X's private key followed by Y's public key; Decryption: Y's private key followed by X's public key

Q.15 Match the problem domains in **GROUP I** with the solution technologies in **GROUP II**.

GROUP I	**GROUP II**
(P) Service oriented computing	(1) Interoperability
(Q) Heterogeneous communicating systems	(2) BPMN
(R) Information representation	(3) Publish-find-bind
(S) Process description	(4) XML

(A) P-1, Q-2, R-3, S-4 (B) P-3, Q-4, R-2, S-1
(C) P-3, Q-1, R-4, S-2 (D) P-4, Q-3, R-2, S-1

Q.16 A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every *T* time units and decides the next process to schedule. Which one of the following is **TRUE** if the processes have no I/O operations and all arrive at time zero?

(A) This algorithm is equivalent to the first-come-first-serve algorithm.
(B) This algorithm is equivalent to the round-robin algorithm.
(C) This algorithm is equivalent to the shortest-job-first algorithm.
(D) This algorithm is equivalent to the shortest-remaining-time-first algorithm.

Q.17 What is the maximum number of reduce moves that can be taken by a bottom-up parser for a grammar with no epsilon- and unit-production (i.e., of type A → ϵ and A → a) to parse a string with *n* tokens?

(A) *n*/2 (B) *n*-1 (C) 2*n*-1 (D) 2^n

Q.18 Consider the languages $L_1 = \Phi$ and $L_2 = \{a\}$. Which one of the following represents $L_1 L_2^* \cup L_1^*$?

(A) {ϵ} (B) Φ (C) *a** (D) {ϵ, *a*}

Q.19 Which one of the following is the tightest upper bound that represents the time complexity of inserting an object into a binary search tree of n nodes?

(A) O(1) (B) O(log n) (C) O(n) (D) O(n log n)

Q.20 Which one of the following is the tightest upper bound that represents the number of swaps required to sort n numbers using selection sort?

(A) O(log n) (B) O(n) (C) O(n log n) (D) O(n^2)

Q.21 In the following truth table, V = 1 if and only if the input is valid.

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

What function does the truth table represent?

(A) Priority encoder (B) Decoder
(C) Multiplexer (D) Demultiplexer

Q.22 The smallest integer that can be represented by an 8-bit number in 2's complement form is

(A) -256 (B) -128 (C) -127 (D) 0

Q.23 Which one of the following does **NOT** equal $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$?

(A) $\begin{vmatrix} 1 & x(x+1) & x+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \end{vmatrix}$ (B) $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \end{vmatrix}$

(C) $\begin{vmatrix} 0 & x-y & x^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$ (D) $\begin{vmatrix} 2 & x+y & x^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

Q.24 Suppose p is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM, and p has a Poisson distribution with mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

(A) $8/(2e^3)$ (B) $9/(2e^3)$ (C) $17/(2e^3)$ (D) $26/(2e^3)$

Q.25 A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is **TRUE** about $\oplus$?

(A) Commutative but not associative (B) Both commutative and associative
(C) Associative but not commutative (D) Neither commutative nor associative

Q.26 to Q.55 carry two marks each.

Q.26 Which one of the following is **NOT** logically equivalent to $\neg \exists x (\forall y(\alpha) \wedge \forall z(\beta))$?

(A) $\forall x(\exists z(\neg\beta) \rightarrow \forall y(\alpha))$ (B) $\forall x(\forall z(\beta) \rightarrow \exists y(\neg\alpha))$
(C) $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg\beta))$ (D) $\forall x(\exists y(\neg\alpha) \rightarrow \exists z(\neg\beta))$

Q.27 A RAM chip has a capacity of 1024 words of 8 bits each (1K × 8). The number of 2 × 4 decoders with enable line needed to construct a 16K × 16 RAM from 1K × 8 RAM is

(A) 4 (B) 5 (C) 6 (D) 7

Q.28 Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI and WO are 5 ns, 7 ns, 10 ns, 8 ns and 6 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 1 ns. A program consisting of 12 instructions I_1, I_2, I_3, …, I_{12} is executed in this pipelined processor. Instruction I_4 is the only branch instruction and its branch target is I_9. If the branch is taken during the execution of this program, the time (in ns) needed to complete the program is

(A) 132 (B) 165 (C) 176 (D) 328

Q.29 Consider the following operation along with `Enqueue` and `Dequeue` operations on queues, where *k* is a global parameter.

```
MultiDequeue(Q){
      m = k
      while (Q is not empty) and (m > 0) {
            Dequeue(Q)
            m = m – 1
      }
}
```

What is the worst case time complexity of a sequence of *n* queue operations on an initially empty queue?

(A) $\Theta(n)$ (B) $\Theta(n + k)$ (C) $\Theta(nk)$ (D) $\Theta(n^2)$

Q.30 The preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42. Which one of the following is the postorder traversal sequence of the same tree?

(A) 10, 20, 15, 23, 25, 35, 42, 39, 30 (B) 15, 10, 25, 23, 20, 42, 35, 39, 30
(C) 15, 20, 10, 23, 25, 42, 35, 39, 30 (D) 15, 10, 23, 25, 20, 35, 42, 39, 30

Q.31 What is the return value of `f(p,p)`, if the value of `p` is initialized to 5 before the call? Note that the first parameter is passed by reference, whereas the second parameter is passed by value.

```
int f (int &x, int c) {
   c = c - 1;
   if (c==0) return 1;
   x = x + 1;
   return f(x,c) * x;
}
```

(A) 3024 (B) 6561 (C) 55440 (D) 161051

Q.32 Which of the following is/are undecidable?

1. G is a CFG. Is $L(G) = \Phi$?
2. G is a CFG. Is $L(G) = \sum^*$?
3. M is a Turing machine. Is $L(M)$ regular?
4. A is a DFA and N is an NFA. Is $L(A) = L(N)$?

(A) 3 only (B) 3 and 4 only (C) 1, 2 and 3 only (D) 2 and 3 only

Q.33 Consider the following two sets of LR(1) items of an LR(1) grammar.

X → c**.**X, c/d	X → c**.**X, $
X → **.**cX, c/d	X → **.**cX, $
X → **.**d, c/d	X → **.**d, $

Which of the following statements related to merging of the two sets in the corresponding LALR parser is/are **FALSE**?

1. Cannot be merged since look aheads are different.
2. Can be merged but will result in S-R conflict.
3. Can be merged but will result in R-R conflict.
4. Cannot be merged since *goto* on *c* will lead to two different sets.

(A) 1 only (B) 2 only (C) 1 and 4 only (D) 1, 2, 3 and 4

Q.34 A certain computation generates two arrays `a` and `b` such that `a[i]=f(i)` for $0 \le i < n$ and b[i] = g (a[i])for $0 \le i < n$. Suppose this computation is decomposed into two concurrent processes X and Y such that X computes the array `a` and Y computes the array `b`. The processes employ two binary semaphores R and S, both initialized to zero. The array `a` is shared by the two processes. The structures of the processes are shown below.

Process *X*:
```
private i;
for (i=0; i<n; i++) {
   a[i] = f(i);
   ExitX(R, S);
}
```

Process *Y*:
```
private i;
for (i=0; i<n; i++) {
   EntryY(R, S);
   b[i] = g(a[i]);
}
```

Which one of the following represents the **CORRECT** implementations of `ExitX` and `EntryY`?

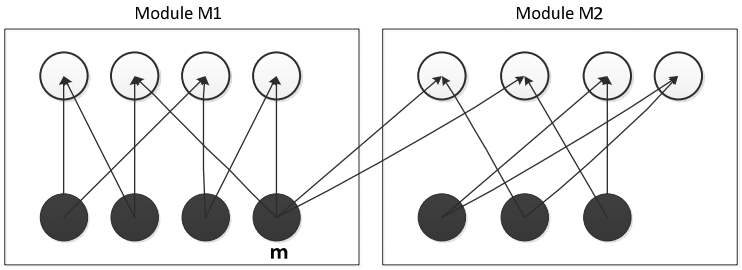
(A)
```
ExitX(R, S) {
   P(R);
   V(S);
}
EntryY(R, S) {
   P(S);
   V(R);
}
```

(B)
```
ExitX(R, S) {
   V(R);
   V(S);
}
EntryY(R, S) {
   P(R);
   P(S);
}
```

(C)
```
ExitX(R, S) {
   P(S);
   V(R);
}
EntryY(R, S) {
   V(S);
   P(R);
}
```

(D)
```
ExitX(R, S) {
   V(R);
   P(S);
}
EntryY(R, S) {
   V(S);
   P(R);
}
```

Q.35 The following figure represents access graphs of two modules M1 and M2. The filled circles represent methods and the unfilled circles represent attributes. If method m is moved to module M2 keeping the attributes where they are, what can we say about the average cohesion and coupling between modules in the system of two modules?



(A) There is no change.
(B) Average cohesion goes up but coupling is reduced.
(C) Average cohesion goes down and coupling also reduces.
(D) Average cohesion and coupling increase.

Q.36 In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are

(A) Last fragment, 2400 and 2789
(B) First fragment, 2400 and 2759
(C) Last fragment, 2400 and 2759
(D) Middle fragment, 300 and 689

Q.37 Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

(A) 1 (B) 2 (C) 2.5 (D) 5

Q.38 Consider the following relational schema.

Students(rollno: integer, sname: string)
Courses(courseno: integer, cname: string)
Registration(rollno: integer, courseno: integer, percent: real)

Which of the following queries are equivalent to this query in English?
"Find the distinct names of all students who score more than 90% in the course numbered 107"

(I) SELECT DISTINCT S.sname
FROM Students as S, Registration as R
WHERE R.rollno=S.rollno AND R.courseno=107 AND R.percent >90

(II) $\prod_{sname}(\sigma_{courseno=107 \wedge percent>90}$ (Registration $\bowtie$ Students))

(III) $\{T \mid \exists S \in$ Students, $\exists R \in$ Registration (S.rollno=R.rollno $\wedge$

R.courseno=107 $\wedge$ R.percent>90 $\wedge$T.sname=S.sname)}

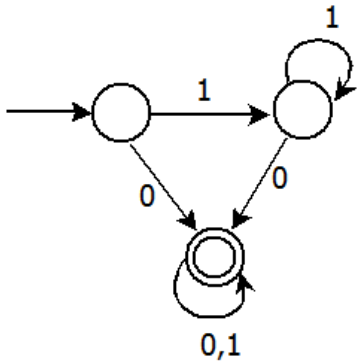
(IV) $\{<S_N> \mid \exists S_R \exists R_P$ ($<S_R, S_N> \in$ Students $\wedge <S_R, 107, R_P> \in$ Registration $\wedge R_P>90)\}$

(A) I, II, III and IV (B) I, II and III only
(C) I, II and IV only (D) II, III and IV only

Q.39 A shared variable x, initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the processes W and X reads x from memory, increments by one, stores it to memory, and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory, and then terminates. Each process before reading x invokes the P operation (i.e., *wait*) on a counting semaphore S and invokes the V operation (i.e., *signal*) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?

(A) –2 (B) –1 (C) 1 (D) 2

Q.40 Consider the DFA A given below.



Which of the following are **FALSE**?

1. Complement of $L(A)$ is context-free.
2. $L(A) = L((11*0+0)(0 + 1)*0*1*)$
3. For the language accepted by A, A is the minimal DFA.
4. A accepts all strings over {0, 1} of length at least 2.

(A) 1 and 3 only (B) 2 and 4 only (C) 2 and 3 only (D) 3 and 4 only

Q.41 Consider the following languages.

$$L_1 = \{0^p\, 1^q\, 0^r \mid p, q, r \geq 0\}$$
$$L_2 = \{0^p\, 1^q\, 0^r \mid p, q, r \geq 0,\ p \neq r\}$$

Which one of the following statements is **FALSE**?

(A) L_2 is context-free.
(B) $L_1 \cap L_2$ is context-free.
(C) Complement of L_2 is recursive.
(D) Complement of L_1 is context-free but not regular.

Q.42 Consider the following function:

```
int unknown(int n){
          int i, j, k=0;
          for (i=n/2; i<=n; i++)
              for (j=2; j<=n; j=j*2)
                  k = k + n/2;
          return (k);
}
```

The return value of the function is

(A) $\Theta(n^2)$ (B) $\Theta(n^2 \log n)$ (C) $\Theta(n^3)$ (D) $\Theta(n^3 \log n)$

Q.43 The number of elements that can be sorted in $\Theta(\log n)$ time using heap sort is

(A) $\Theta(1)$ (B) $\Theta\left(\sqrt{\log\ n}\right)$ (C) $\Theta\left(\frac{\log n}{\log\ \log\ n}\right)$ (D) $\Theta(\log n)$

Q.44 Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each cylinder contains 64 sectors (0-63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is <cylinder no., surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

(A) 1281 (B) 1282 (C) 1283 (D) 1284

Q.45 Consider the following sequence of micro-operations.

MBR ← PC
MAR ← X
PC ← Y
Memory ← MBR

Which one of the following is a possible operation performed by this sequence?

(A) Instruction fetch
(B) Operand fetch
(C) Conditional branch
(D) Initiation of interrupt service

Q.46 The line graph *L(G)* of a simple graph *G* is defined as follows:

- There is exactly one vertex *v(e)* in *L(G)* for each edge *e* in *G*.
- For any two edges *e* and *e'* in *G*, *L(G)* has an edge between *v(e)* and *v(e')*, if and only if *e* and *e'* are incident with the same vertex in *G*.

Which of the following statements is/are **TRUE**?

(P) The line graph of a cycle is a cycle.
(Q) The line graph of a clique is a clique.
(R) The line graph of a planar graph is planar.
(S) The line graph of a tree is a tree.

(A) P only
(B) P and R only
(C) R only
(D) P, Q and S only

Q.47 What is the logical translation of the following statement?

"None of my friends are perfect."

(A) $\exists x(F(x) \wedge \neg P(x))$
(B) $\exists x(\neg F(x) \wedge P(x))$
(C) $\exists x(\neg F(x) \wedge \neg P(x))$
(D) $\neg \exists x(F(x) \wedge P(x))$

Common Data Questions

Common Data for Questions 48 and 49:

The procedure given below is required to find and replace certain characters inside an input character string supplied in array A. The characters to be replaced are supplied in array oldc, while their respective replacement characters are supplied in array newc. Array A has a fixed length of five characters, while arrays oldc and newc contain three characters each. However, the procedure is flawed.

```
void find_and_replace (char *A, char *oldc, char *newc) {
        for (int i=0; i<5; i++)
            for (int j=0; j<3; j++)
                if (A[i] == oldc[j])  A[i] = newc[j];
 }
```

The procedure is tested with the following four test cases.

(1) oldc = "abc", newc = "dab" (2) oldc = "cde", newc = "bcd"
(3) oldc = "bca", newc = "cda" (4) oldc = "abc", newc = "bac"

Q.48 The tester now tests the program on all input strings of length five consisting of characters 'a', 'b', 'c', 'd' and 'e' with duplicates allowed. If the tester carries out this testing with the four test cases given above, how many test cases will be able to capture the flaw?

(A) Only one (B) Only two (C) Only three (D) All four

Q.49 If array A is made to hold the string "abcde", which of the above four test cases will be successful in exposing the flaw in this procedure?

(A) None (B) 2 only
(C) 3 and 4 only (D) 4 only

Common Data for Questions 50 and 51:

The following code segment is executed on a processor which allows only register operands in its instructions. Each instruction can have atmost two source operands and one destination operand. Assume that all variables are dead after this code segment.

```
c = a + b;
d = c * a;
e = c + a;
x = c * c;
if (x > a) {
   y = a * a;
}
else {
   d = d * d;
   e = e * e;
}
```

Q.50 Suppose the instruction set architecture of the processor has only two registers. The only allowed compiler optimization is code motion, which moves statements from one place to another while preserving correctness. What is the minimum number of spills to memory in the compiled code?

(A) 0 (B) 1 (C) 2 (D) 3

Q.51 What is the minimum number of registers needed in the instruction set architecture of the processor to compile this code segment without any spill to memory? Do not apply any optimization other than optimizing register allocation.

(A) 3 (B) 4 (C) 5 (D) 6

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

Relation R has eight attributes $ABCDEFGH$. Fields of R contain only atomic values.
$F=\{CH\rightarrow G,\ A\rightarrow BC,\ B\rightarrow CFH,\ E\rightarrow A,\ F\rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.

Q.52 How many candidate keys does the relation R have?

(A) 3 (B) 4 (C) 5 (D) 6

Q.53 The relation R is

(A) in 1NF, but not in 2NF.
(B) in 2NF, but not in 3NF.
(C) in 3NF, but not in BCNF.
(D) in BCNF.

Statement for Linked Answer Questions 54 and 55:

A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table (T_1), which occupies exactly one page. Each entry of T_1 stores the base address of a page of the second-level table (T_2). Each entry of T_2 stores the base address of a page of the third-level table (T_3). Each entry of T_3 stores a page table entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16-way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes.

Q.54 What is the size of a page in KB in this computer?

(A) 2 (B) 4 (C) 8 (D) 16

Q.55 What is the minimum number of page colours needed to guarantee that no two synonyms map to different sets in the processor cache of this computer?

(A) 2 (B) 4 (C) 8 (D) 16

General Aptitude (GA) Questions

Q.56 to Q.60 carry one mark each.

Q.56 Complete the sentence:
Universalism is to particularism as diffuseness is to _____.

(A) specificity (B) neutrality (C) generality (D) adaptation

Q.57 Were you a bird, you _____ in the sky.

(A) would fly (B) shall fly (C) should fly (D) shall have flown

Q.58 Which one of the following options is the closest in meaning to the word given below?

Nadir

(A) Highest (B) Lowest (C) Medium (D) Integration

Q.59 Choose the grammatically **INCORRECT** sentence:

(A) He is of Asian origin.
(B) They belonged to Africa.
(C) She is an European.
(D) They migrated from India to Australia.

Q.60 What will be the maximum sum of 44, 42, 40, ... ?

(A) 502 (B) 504 (C) 506 (D) 500

Q.61 to Q.65 carry two marks each.

Q.61 Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

(A) 13/90 (B) 12/90 (C) 78/90 (D) 77/90

Q.62 A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

(A) 36 (B) 30 (C) 24 (D) 18

Q.63 Find the sum of the expression

$$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+.....+\frac{1}{\sqrt{80}+\sqrt{81}}$$

(A) 7 (B) 8
(C) 9 (D) 10

Q.64 The current erection cost of a structure is *Rs*. 13,200. If the labour wages per day increase by 1/5 of the current wages and the working hours decrease by 1/24 of the current period, then the new cost of erection in *Rs*. is

(A) 16,500 (B) 15,180 (C) 11,000 (D) 10,120

Q.65 After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again, the spider failed but that did not deter it to refrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

Which one of the following assertions is best supported by the above information?

(A) Failure is the pillar of success.
(B) Honesty is the best policy.
(C) Life begins and ends with adventures.
(D) No adversity justifies giving up hope.

END OF THE QUESTION PAPER

SESSION - 3

CS : COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Duration: Three Hours *Maximum Marks*: 100

Read the following instructions carefully.

1. Do not open the seal of the Question Booklet until you are asked to do so by the invigilator.
2. Take out the **O**ptical **R**esponse **S**heet (**ORS)** from this Question Booklet **without breaking the seal** and read the instructions printed on the ORS carefully. If you find that either
   a. The Question Booklet Code printed at the right hand top corner of this page does not match with the Question Booklet Code at the right hand top corner of the **ORS** or
   b. The Question Paper Code preceding the Registration number on the **ORS** is not **CS**, then exchange the booklet immediately with a new sealed Question Booklet.
3. On the right hand side of the **ORS,** using ONLY a **black ink ballpoint pen**, (i) darken the appropriate bubble under each digit of your registration number and (ii) write your registration number, your name and name of the examination centre and put your signature at the specified location.
4. This Question Booklet contains **20** pages including blank pages for rough work. After you are permitted to open the seal, check all pages and report discrepancies, if any, to the invigilator.
5. There are a total of 65 questions carrying 100 marks. All these questions are of objective type. Each question has only **one** correct answer. Questions must be answered on the left hand side of the **ORS** by darkening the appropriate bubble (marked A, B, C, D) using ONLY a **black ink ballpoint pen** against the question number. **For each question darken the bubble of the correct answer**. More than one answer bubbled against a question will be treated as an incorrect response.
6. Since bubbles darkened by the black ink ballpoint pen **cannot** be erased, candidates should darken the bubbles in the ORS **very carefully**.
7. Questions Q.1 – Q.25 carry 1 mark each. Questions Q.26 – Q.55 carry 2 marks each. The 2 marks questions include two pairs of common data questions and two pairs of linked answer questions. The answer to the second question of the linked answer questions depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is not attempted, then the answer to the second question in the pair will not be evaluated.
8. Questions Q.56 – Q.65 belong to General Aptitude (GA) section and carry a total of 15 marks. Questions Q.56 – Q.60 carry 1 mark each, and questions Q.61 – Q.65 carry 2 marks each.
9. Questions not attempted will result in zero mark and wrong answers will result in **NEGATIVE** marks. For all 1 mark questions, ⅓ mark will be deducted for each wrong answer. For all 2 marks questions, ⅔ mark will be deducted for each wrong answer. However, in the case of the linked answer question pair, there will be negative marks only for wrong answer to the first question and no negative marks for wrong answer to the second question.
10. Calculator is allowed whereas charts, graph sheets or tables are **NOT** allowed in the examination hall.
11. Rough work can be done on the Question Booklet itself. Blank pages are provided at the end of the Question Booklet for rough work.
12. Before the start of the examination, write your name and registration number in the space provided below using a black ink ballpoint pen.

Name									
Registration Number	**CS**								

Q.1 to Q.25 carry one mark each.

Q.1 Using public key cryptography, X adds a digital signature σ to message M, encrypts <M, σ>, and sends it to Y, where it is decrypted. Which one of the following sequences of keys is used for the operations?

(A) Encryption: X's private key followed by Y's private key; Decryption: X's public key followed by Y's public key
(B) Encryption: X's private key followed by Y's public key; Decryption: X's public key followed by Y's private key
(C) Encryption: X's public key followed by Y's private key; Decryption: Y's public key followed by X's private key
(D) Encryption: X's private key followed by Y's public key; Decryption: Y's private key followed by X's public key

Q.2 Match the problem domains in **GROUP I** with the solution technologies in **GROUP II**.

GROUP I	**GROUP II**
(P) Service oriented computing	(1) Interoperability
(Q) Heterogeneous communicating systems	(2) BPMN
(R) Information representation	(3) Publish-find-bind
(S) Process description	(4) XML

(A) P-1, Q-2, R-3, S-4 (B) P-3, Q-4, R-2, S-1
(C) P-3, Q-1, R-4, S-2 (D) P-4, Q-3, R-2, S-1

Q.3 A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every *T* time units and decides the next process to schedule. Which one of the following is **TRUE** if the processes have no I/O operations and all arrive at time zero?

(A) This algorithm is equivalent to the first-come-first-serve algorithm.
(B) This algorithm is equivalent to the round-robin algorithm.
(C) This algorithm is equivalent to the shortest-job-first algorithm.
(D) This algorithm is equivalent to the shortest-remaining-time-first algorithm.

Q.4 What is the maximum number of reduce moves that can be taken by a bottom-up parser for a grammar with no epsilon- and unit-production (i.e., of type A → ϵ and A → a) to parse a string with *n* tokens?

(A) *n*/2 (B) *n*-1 (C) 2*n*-1 (D) 2^n

Q.5 Consider the languages $L_1 = \Phi$ and $L_2 = \{a\}$. Which one of the following represents $L_1 L_2^* \cup L_1^*$?

(A) {ϵ} (B) Φ (C) *a** (D) {ϵ, *a*}

Q.6 Which one of the following is the tightest upper bound that represents the time complexity of inserting an object into a binary search tree of *n* nodes?

(A) O(1) (B) O(log *n*) (C) O(*n*) (D) O(*n* log *n*)

Q.7 Which one of the following is the tightest upper bound that represents the number of swaps required to sort *n* numbers using selection sort?

(A) O(log *n*) (B) O(*n*) (C) O(*n* log *n*) (D) $O(n^2)$

Q.8 In the following truth table, V = 1 if and only if the input is valid.

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

What function does the truth table represent?

(A) Priority encoder (B) Decoder
(C) Multiplexer (D) Demultiplexer

Q.9 The smallest integer that can be represented by an 8-bit number in 2's complement form is

(A) -256 (B) -128 (C) -127 (D) 0

Q.10 Which one of the following does **NOT** equal $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$?

(A) $\begin{vmatrix} 1 & x(x+1) & x+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \end{vmatrix}$ (B) $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \end{vmatrix}$

(C) $\begin{vmatrix} 0 & x-y & x^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$ (D) $\begin{vmatrix} 2 & x+y & x^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

Q.11 Suppose *p* is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM, and *p* has a Poisson distribution with mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

(A) $8/(2e^3)$ (B) $9/(2e^3)$ (C) $17/(2e^3)$ (D) $26/(2e^3)$

Q.12 A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is **TRUE** about $\oplus$?

(A) Commutative but not associative (B) Both commutative and associative
(C) Associative but not commutative (D) Neither commutative nor associative

Q.13 Which of the following statements is/are **TRUE** for undirected graphs?

P: Number of odd degree vertices is even.
Q: Sum of degrees of all vertices is even.

(A) P only (B) Q only (C) Both P and Q (D) Neither P nor Q

Q.14 Consider an undirected random graph of eight vertices. The probability that there is an edge between a pair of vertices is 1/2. What is the expected number of unordered cycles of length three?

(A) 1/8 (B) 1 (C) 7 (D) 8

Q.15 Function f is known at the following points:

x	0	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4	2.7	3.0
$f(x)$	0	0.09	0.36	0.81	1.44	2.25	3.24	4.41	5.76	7.29	9.00

The value of $\int_0^3 f(x)\,dx$ computed using the trapezoidal rule is

(A) 8.983 (B) 9.003 (C) 9.017 (D) 9.045

Q.16 Which one of the following functions is continuous at $x = 3$?

(A) $f(x)=\begin{cases}2, & \text{if } x=3\\ x-1, & \text{if } x>3\\ \dfrac{x+3}{3}, & \text{if } x<3\end{cases}$ (B) $f(x)=\begin{cases}4, & \text{if } x=3\\ 8-x & \text{if } x\neq 3\end{cases}$

(C) $f(x)=\begin{cases}x+3, & \text{if } x\le 3\\ x-4 & \text{if } x>3\end{cases}$ (D) $f(x)=\dfrac{1}{x^3-27}$, if $x \neq 3$

Q.17 Which one of the following expressions does **NOT** represent exclusive NOR of x and y?

(A) $xy + x'y'$ (B) $x \oplus y'$ (C) $x' \oplus y$ (D) $x' \oplus y'$

Q.18 In a k-way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set are placed in sequence one after another. The lines in set s are sequenced before the lines in set (s+1). The main memory blocks are numbered 0 onwards. The main memory block numbered j must be mapped to any one of the cache lines from

(A) (j mod v) * k to (j mod v) * k + (k-1) (B) (j mod v) to (j mod v) + (k-1)
(C) (j mod k) to (j mod k) + (v-1) (D) (j mod k) * v to (j mod k) * v + (v-1)

Q.19 What is the time complexity of Bellman-Ford single-source shortest path algorithm on a complete graph of n vertices?

(A) $\Theta(n^2)$ (B) $\Theta(n^2 \log n)$ (C) $\Theta(n^3)$ (D) $\Theta(n^3 \log n)$

Q.20 Which of the following statements are **TRUE**?

1 The problem of determining whether there exists a cycle in an undirected graph is in P.
2. The problem of determining whether there exists a cycle in an undirected graph is in NP.
3. If a problem A is NP-Complete, there exists a non-deterministic polynomial time algorithm to solve A.

(A) 1, 2 and 3 (B) 1 and 2 only (C) 2 and 3 only (D) 1 and 3 only

Q.21 Which of the following statements is/are **FALSE**?

1. For every non-deterministic Turing machine, there exists an equivalent deterministic Turing machine.
2. Turing recognizable languages are closed under union and complementation.
3. Turing decidable languages are closed under intersection and complementation.
4. Turing recognizable languages are closed under union and intersection.

(A) 1 and 4 only (B) 1 and 3 only (C) 2 only (D) 3 only

Q.22 Three concurrent processes *X*, *Y*, and *Z* execute three different code segments that access and update certain shared variables. Process *X* executes the *P* operation (i.e., *wait*) on semaphores *a*, *b* and *c*; process *Y* executes the *P* operation on semaphores *b*, *c* and *d*; process *Z* executes the *P* operation on semaphores *c*, *d*, and *a* before entering the respective code segments. After completing the execution of its code segment, each process invokes the *V* operation (i.e., *signal*) on its three semaphores. All semaphores are binary semaphores initialized to one. Which one of the following represents a deadlock-free order of invoking the *P* operations by the processes?

(A) *X*: *P*(*a*)*P*(*b*)*P*(*c*) *Y*: *P*(*b*)*P*(*c*)*P*(*d*) *Z*: *P*(*c*)*P*(*d*)*P*(*a*)
(B) *X*: *P*(*b*)*P*(*a*)*P*(*c*) *Y*: *P*(*b*)*P*(*c*)*P*(*d*) *Z*: *P*(*a*)*P*(*c*)*P*(*d*)
(C) *X*: *P*(*b*)*P*(*a*)*P*(*c*) *Y*: *P*(*c*)*P*(*b*)*P*(*d*) *Z*: *P*(*a*)*P*(*c*)*P*(*d*)
(D) *X*: *P*(*a*)*P*(*b*)*P*(*c*) *Y*: *P*(*c*)*P*(*b*)*P*(*d*) *Z*: *P*(*c*)*P*(*d*)*P*(*a*)

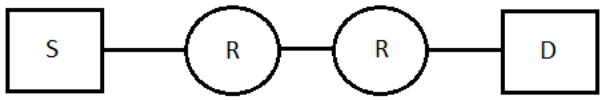
Q.23 An index is clustered, if

(A) it is on a set of fields that form a candidate key.
(B) it is on a set of fields that include the primary key.
(C) the data records of the file are organized in the same order as the data entries of the index.
(D) the data records of the file are organized not in the same order as the data entries of the index.

Q.24 The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are

(A) TCP, UDP, UDP and TCP
(B) UDP, TCP, TCP and UDP
(C) UDP, TCP, UDP and TCP
(D) TCP, UDP, TCP and UDP

Q.25 Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.



(A) Network layer – 4 times and Data link layer – 4 times
(B) Network layer – 4 times and Data link layer – 3 times
(C) Network layer – 4 times and Data link layer – 6 times
(D) Network layer – 2 times and Data link layer – 6 times

Q.26 to Q.55 carry two marks each.

Q.26 Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

(A) 1 (B) 2 (C) 2.5 (D) 5

Q.27 Consider the following relational schema.

Students(rollno: integer, sname: string)
Courses(courseno: integer, cname: string)
Registration(rollno: integer, courseno: integer, percent: real)

Which of the following queries are equivalent to this query in English?
"Find the distinct names of all students who score more than 90% in the course numbered 107"

(I) SELECT DISTINCT S.sname
FROM Students as S, Registration as R
WHERE R.rollno=S.rollno AND R.courseno=107 AND R.percent >90

(II) $\prod_{sname}(\sigma_{courseno=107 \wedge percent>90}$ (Registration $\bowtie$ Students))

(III) $\{T \mid \exists S \in$ Students, $\exists R \in$ Registration (S.rollno=R.rollno $\wedge$

R.courseno=107 $\wedge$ R.percent>90 $\wedge$T.sname=S.sname)}

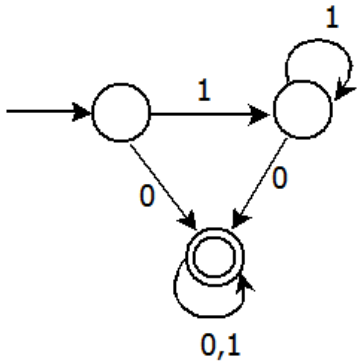
(IV) $\{<S_N> \mid \exists S_R \exists R_P$ ($<S_R, S_N> \in$ Students $\wedge <S_R, 107, R_P> \in$ Registration $\wedge R_P>90)\}$

(A) I, II, III and IV (B) I, II and III only
(C) I, II and IV only (D) II, III and IV only

Q.28 A shared variable *x*, initialized to zero, is operated on by four concurrent processes *W*, *X*, *Y*, *Z* as follows. Each of the processes *W* and *X* reads *x* from memory, increments by one, stores it to memory, and then terminates. Each of the processes *Y* and *Z* reads *x* from memory, decrements by two, stores it to memory, and then terminates. Each process before reading *x* invokes the *P* operation (i.e., *wait*) on a counting semaphore *S* and invokes the *V* operation (i.e., *signal*) on the semaphore *S* after storing *x* to memory. Semaphore *S* is initialized to two. What is the maximum possible value of *x* after all processes complete execution?

(A) –2 (B) –1 (C) 1 (D) 2

Q.29 Consider the DFA A given below.



Which of the following are **FALSE**?

1. Complement of $L(A)$ is context-free.
2. $L(A) = L((11*0+0)(0 + 1)*0*1*)$
3. For the language accepted by A, A is the minimal DFA.
4. A accepts all strings over {0, 1} of length at least 2.

(A) 1 and 3 only (B) 2 and 4 only (C) 2 and 3 only (D) 3 and 4 only

Q.30 Consider the following languages.

$$L_1 = \{0^p 1^q 0^r \mid p, q, r \geq 0\}$$
$$L_2 = \{0^p 1^q 0^r \mid p, q, r \geq 0,\ p \neq r\}$$

Which one of the following statements is **FALSE**?

(A) L_2 is context-free.
(B) $L_1 \cap L_2$ is context-free.
(C) Complement of L_2 is recursive.
(D) Complement of L_1 is context-free but not regular.

Q.31 Consider the following function:

```
int unknown(int n){
          int i, j, k=0;
          for (i=n/2; i<=n; i++)
              for (j=2; j<=n; j=j*2)
                  k = k + n/2;
          return (k);
}
```

The return value of the function is

(A) $\Theta(n^2)$ (B) $\Theta(n^2 \log n)$ (C) $\Theta(n^3)$ (D) $\Theta(n^3 \log n)$

Q.32 The number of elements that can be sorted in $\Theta(\log n)$ time using heap sort is

(A) $\Theta(1)$ (B) $\Theta\left(\sqrt{\log n}\right)$ (C) $\Theta\left(\frac{\log n}{\log \log n}\right)$ (D) $\Theta(\log n)$

Q.33 Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each cylinder contains 64 sectors (0-63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is <cylinder no., surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

(A) 1281 (B) 1282 (C) 1283 (D) 1284

Q.34 Consider the following sequence of micro-operations.

MBR ← PC
MAR ← X
PC ← Y
Memory ← MBR

Which one of the following is a possible operation performed by this sequence?

(A) Instruction fetch
(B) Operand fetch
(C) Conditional branch
(D) Initiation of interrupt service

Q.35 The line graph *L(G)* of a simple graph *G* is defined as follows:

- There is exactly one vertex *v(e)* in *L(G)* for each edge *e* in *G*.
- For any two edges *e* and *e'* in *G*, *L(G)* has an edge between *v(e)* and *v(e')*, if and only if *e* and *e'* are incident with the same vertex in *G*.

Which of the following statements is/are **TRUE**?

(P) The line graph of a cycle is a cycle.
(Q) The line graph of a clique is a clique.
(R) The line graph of a planar graph is planar.
(S) The line graph of a tree is a tree.

(A) P only
(B) P and R only
(C) R only
(D) P, Q and S only

Q.36 What is the logical translation of the following statement?

"None of my friends are perfect."

(A) $\exists x(F(x) \wedge \neg P(x))$
(B) $\exists x(\neg F(x) \wedge P(x))$
(C) $\exists x(\neg F(x) \wedge \neg P(x))$
(D) $\neg \exists x(F(x) \wedge P(x))$

Q.37 Which one of the following is **NOT** logically equivalent to $\neg \exists x(\forall y(\alpha) \wedge \forall z(\beta))$?

(A) $\forall x(\exists z(\neg \beta) \rightarrow \forall y(\alpha))$
(B) $\forall x(\forall z(\beta) \rightarrow \exists y(\neg \alpha))$
(C) $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg \beta))$
(D) $\forall x(\exists y(\neg \alpha) \rightarrow \exists z(\neg \beta))$

Q.38 A RAM chip has a capacity of 1024 words of 8 bits each (1K × 8). The number of 2 × 4 decoders with enable line needed to construct a 16K × 16 RAM from 1K × 8 RAM is

(A) 4 (B) 5 (C) 6 (D) 7

Q.39 Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI and WO are 5 ns, 7 ns, 10 ns, 8 ns and 6 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 1 ns. A program consisting of 12 instructions I_1, I_2, I_3, …, I_{12} is executed in this pipelined processor. Instruction I_4 is the only branch instruction and its branch target is I_9. If the branch is taken during the execution of this program, the time (in ns) needed to complete the program is

(A) 132 (B) 165 (C) 176 (D) 328

Q.40 Consider the following operation along with `Enqueue` and `Dequeue` operations on queues, where *k* is a global parameter.

```
MultiDequeue(Q){
      m = k
      while (Q is not empty) and (m > 0) {
            Dequeue(Q)
            m = m – 1
      }
}
```

What is the worst case time complexity of a sequence of *n* queue operations on an initially empty queue?

(A) $\Theta(n)$ (B) $\Theta(n + k)$ (C) $\Theta(nk)$ (D) $\Theta(n^2)$

Q.41 The preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42. Which one of the following is the postorder traversal sequence of the same tree?

(A) 10, 20, 15, 23, 25, 35, 42, 39, 30 (B) 15, 10, 25, 23, 20, 42, 35, 39, 30
(C) 15, 20, 10, 23, 25, 42, 35, 39, 30 (D) 15, 10, 23, 25, 20, 35, 42, 39, 30

Q.42 What is the return value of `f(p,p)`, if the value of `p` is initialized to 5 before the call? Note that the first parameter is passed by reference, whereas the second parameter is passed by value.

```
int f (int &x, int c) {
   c = c - 1;
   if (c==0) return 1;
   x = x + 1;
   return f(x,c) * x;
}
```

(A) 3024 (B) 6561 (C) 55440 (D) 161051

Q.43 Which of the following is/are undecidable?

1. *G* is a CFG. Is $L(G) = \Phi$?
2. *G* is a CFG. Is $L(G) = \sum^*$?
3. *M* is a Turing machine. Is $L(M)$ regular?
4. *A* is a DFA and *N* is an NFA. Is $L(A) = L(N)$?

(A) 3 only (B) 3 and 4 only (C) 1, 2 and 3 only (D) 2 and 3 only

Q.44 Consider the following two sets of LR(1) items of an LR(1) grammar.

X → c**.**X, c/d	X → c**.**X, $
X → **.**cX, c/d	X → **.**cX, $
X → **.**d, c/d	X → **.**d, $

Which of the following statements related to merging of the two sets in the corresponding LALR parser is/are **FALSE**?

1. Cannot be merged since look aheads are different.
2. Can be merged but will result in S-R conflict.
3. Can be merged but will result in R-R conflict.
4. Cannot be merged since *goto* on *c* will lead to two different sets.

(A) 1 only (B) 2 only (C) 1 and 4 only (D) 1, 2, 3 and 4

Q.45 A certain computation generates two arrays `a` and `b` such that `a[i]=f(i)` for $0 \leq i < n$ and b[i] = g (a[i])for $0 \leq i < n$. Suppose this computation is decomposed into two concurrent processes *X* and *Y* such that *X* computes the array `a` and *Y* computes the array `b`. The processes employ two binary semaphores *R* and *S*, both initialized to zero. The array `a` is shared by the two processes. The structures of the processes are shown below.

Process *X*:
```
private i;
for (i=0; i<n; i++) {
   a[i] = f(i);
   ExitX(R, S);
}
```

Process *Y*:
```
private i;
for (i=0; i<n; i++) {
   EntryY(R, S);
   b[i] = g(a[i]);
}
```

Which one of the following represents the **CORRECT** implementations of `ExitX` and `EntryY`?

(A)
```
ExitX(R, S) {
   P(R);
   V(S);
}
EntryY(R, S) {
   P(S);
   V(R);
}
```

(B)
```
ExitX(R, S) {
   V(R);
   V(S);
}
EntryY(R, S) {
   P(R);
   P(S);
}
```

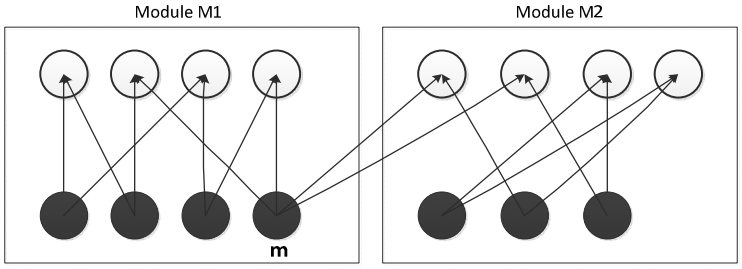
(C)
```
ExitX(R, S) {
   P(S);
   V(R);
}
EntryY(R, S) {
   V(S);
   P(R);
}
```

(D)
```
ExitX(R, S) {
   V(R);
   P(S);
}
EntryY(R, S) {
   V(S);
   P(R);
}
```

Q.46 In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are

(A) Last fragment, 2400 and 2789
(B) First fragment, 2400 and 2759
(C) Last fragment, 2400 and 2759
(D) Middle fragment, 300 and 689

Q.47 The following figure represents access graphs of two modules M1 and M2. The filled circles represent methods and the unfilled circles represent attributes. If method `m` is moved to module M2 keeping the attributes where they are, what can we say about the average cohesion and coupling between modules in the system of two modules?



(A) There is no change.
(B) Average cohesion goes up but coupling is reduced.
(C) Average cohesion goes down and coupling also reduces.
(D) Average cohesion and coupling increase.

Common Data Questions

Common Data for Questions 48 and 49:

The following code segment is executed on a processor which allows only register operands in its instructions. Each instruction can have atmost two source operands and one destination operand. Assume that all variables are dead after this code segment.

```
c = a + b;
d = c * a;
e = c + a;
x = c * c;
if (x > a) {
   y = a * a;
}
else {
   d = d * d;
   e = e * e;
}
```

Q.48 What is the minimum number of registers needed in the instruction set architecture of the processor to compile this code segment without any spill to memory? Do not apply any optimization other than optimizing register allocation.

(A) 3 (B) 4 (C) 5 (D) 6

Q.49 Suppose the instruction set architecture of the processor has only two registers. The only allowed compiler optimization is code motion, which moves statements from one place to another while preserving correctness. What is the minimum number of spills to memory in the compiled code?

(A) 0 (B) 1 (C) 2 (D) 3

Common Data for Questions 50 and 51:

The procedure given below is required to find and replace certain characters inside an input character string supplied in array `A`. The characters to be replaced are supplied in array `oldc`, while their respective replacement characters are supplied in array `newc`. Array `A` has a fixed length of five characters, while arrays `oldc` and `newc` contain three characters each. However, the procedure is flawed.

```
void find_and_replace (char *A, char *oldc, char *newc) {
        for (int i=0; i<5; i++)
            for (int j=0; j<3; j++)
                if (A[i] == oldc[j])  A[i] = newc[j];
 }
```

The procedure is tested with the following four test cases.

(1) `oldc = "abc", newc = "dab"` (2) `oldc = "cde", newc = "bcd"`
(3) `oldc = "bca", newc = "cda"` (4) `oldc = "abc", newc = "bac"`

Q.50 If array `A` is made to hold the string `"abcde"`, which of the above four test cases will be successful in exposing the flaw in this procedure?

(A) None (B) 2 only
(C) 3 and 4 only (D) 4 only

Q.51 The tester now tests the program on all input strings of length five consisting of characters `'a'`, `'b'`, `'c'`, `'d'` and `'e'` with duplicates allowed. If the tester carries out this testing with the four test cases given above, how many test cases will be able to capture the flaw?

(A) Only one (B) Only two (C) Only three (D) All four

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table (T_1), which occupies exactly one page. Each entry of T_1 stores the base address of a page of the second-level table (T_2). Each entry of T_2 stores the base address of a page of the third-level table (T_3). Each entry of T_3 stores a page table entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16-way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes.

Q.52 What is the size of a page in KB in this computer?

(A) 2 (B) 4 (C) 8 (D) 16

Q.53 What is the minimum number of page colours needed to guarantee that no two synonyms map to different sets in the processor cache of this computer?

(A) 2 (B) 4 (C) 8 (D) 16

Statement for Linked Answer Questions 54 and 55:

Relation R has eight attributes $ABCDEFGH$. Fields of R contain only atomic values.
$F=\{CH\rightarrow G, A\rightarrow BC, B\rightarrow CFH, E\rightarrow A, F\rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.

Q.54 How many candidate keys does the relation R have?

(A) 3 (B) 4 (C) 5 (D) 6

Q.55 The relation R is

(A) in 1NF, but not in 2NF. (B) in 2NF, but not in 3NF.
(C) in 3NF, but not in BCNF. (D) in BCNF.

General Aptitude (GA) Questions

Q.56 to Q.60 carry one mark each.

Q.56 Were you a bird, you _____ in the sky.

(A) would fly (B) shall fly (C) should fly (D) shall have flown

Q.57 Choose the grammatically **INCORRECT** sentence:

(A) He is of Asian origin.
(B) They belonged to Africa.
(C) She is an European.
(D) They migrated from India to Australia.

Q.58 Complete the sentence:
Universalism is to particularism as diffuseness is to _____.

(A) specificity (B) neutrality (C) generality (D) adaptation

Q.59 What will be the maximum sum of 44, 42, 40, ... ?

(A) 502 (B) 504 (C) 506 (D) 500

Q.60 Which one of the following options is the closest in meaning to the word given below?

Nadir

(A) Highest (B) Lowest (C) Medium (D) Integration

Q.61 to Q.65 carry two marks each.

Q.61 A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

(A) 36 (B) 30 (C) 24 (D) 18

Q.62 The current erection cost of a structure is *Rs*. 13,200. If the labour wages per day increase by 1/5 of the current wages and the working hours decrease by 1/24 of the current period, then the new cost of erection in *Rs*. is

(A) 16,500 (B) 15,180 (C) 11,000 (D) 10,120

Q.63 Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

(A) 13/90 (B) 12/90 (C) 78/90 (D) 77/90

Q.64 After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again, the spider failed but that did not deter it to refrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

Which one of the following assertions is best supported by the above information?

(A) Failure is the pillar of success.
(B) Honesty is the best policy.
(C) Life begins and ends with adventures.
(D) No adversity justifies giving up hope.

Q.65 Find the sum of the expression

$$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+.....+\frac{1}{\sqrt{80}+\sqrt{81}}$$

(A) 7 (B) 8
(C) 9 (D) 10

END OF THE QUESTION PAPER

SESSION - 4

CS : COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

Duration: Three Hours *Maximum Marks*: 100

Read the following instructions carefully.

1. Do not open the seal of the Question Booklet until you are asked to do so by the invigilator.
2. Take out the **O**ptical **R**esponse **S**heet (**ORS)** from this Question Booklet **without breaking the seal** and read the instructions printed on the ORS carefully. If you find that either
   a. The Question Booklet Code printed at the right hand top corner of this page does not match with the Question Booklet Code at the right hand top corner of the **ORS** or
   b. The Question Paper Code preceding the Registration number on the **ORS** is not **CS**, then exchange the booklet immediately with a new sealed Question Booklet.
3. On the right hand side of the **ORS,** using ONLY a **black ink ballpoint pen**, (i) darken the appropriate bubble under each digit of your registration number and (ii) write your registration number, your name and name of the examination centre and put your signature at the specified location.
4. This Question Booklet contains **20** pages including blank pages for rough work. After you are permitted to open the seal, check all pages and report discrepancies, if any, to the invigilator.
5. There are a total of 65 questions carrying 100 marks. All these questions are of objective type. Each question has only **one** correct answer. Questions must be answered on the left hand side of the **ORS** by darkening the appropriate bubble (marked A, B, C, D) using ONLY a **black ink ballpoint pen** against the question number. **For each question darken the bubble of the correct answer**. More than one answer bubbled against a question will be treated as an incorrect response.
6. Since bubbles darkened by the black ink ballpoint pen **cannot** be erased, candidates should darken the bubbles in the ORS **very carefully**.
7. Questions Q.1 – Q.25 carry 1 mark each. Questions Q.26 – Q.55 carry 2 marks each. The 2 marks questions include two pairs of common data questions and two pairs of linked answer questions. The answer to the second question of the linked answer questions depends on the answer to the first question of the pair. If the first question in the linked pair is wrongly answered or is not attempted, then the answer to the second question in the pair will not be evaluated.
8. Questions Q.56 – Q.65 belong to General Aptitude (GA) section and carry a total of 15 marks. Questions Q.56 – Q.60 carry 1 mark each, and questions Q.61 – Q.65 carry 2 marks each.
9. Questions not attempted will result in zero mark and wrong answers will result in **NEGATIVE** marks. For all 1 mark questions, ⅓ mark will be deducted for each wrong answer. For all 2 marks questions, ⅔ mark will be deducted for each wrong answer. However, in the case of the linked answer question pair, there will be negative marks only for wrong answer to the first question and no negative marks for wrong answer to the second question.
10. Calculator is allowed whereas charts, graph sheets or tables are **NOT** allowed in the examination hall.
11. Rough work can be done on the Question Booklet itself. Blank pages are provided at the end of the Question Booklet for rough work.
12. Before the start of the examination, write your name and registration number in the space provided below using a black ink ballpoint pen.

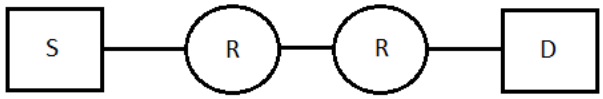
Name									
Registration Number	**CS**								

Q.1 to Q.25 carry one mark each.

Q.1 The transport layer protocols used for real time multimedia, file transfer, DNS and email, respectively are

(A) TCP, UDP, UDP and TCP
(B) UDP, TCP, TCP and UDP
(C) UDP, TCP, UDP and TCP
(D) TCP, UDP, TCP and UDP

Q.2 Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D.



(A) Network layer – 4 times and Data link layer – 4 times
(B) Network layer – 4 times and Data link layer – 3 times
(C) Network layer – 4 times and Data link layer – 6 times
(D) Network layer – 2 times and Data link layer – 6 times

Q.3 An index is clustered, if

(A) it is on a set of fields that form a candidate key.
(B) it is on a set of fields that include the primary key.
(C) the data records of the file are organized in the same order as the data entries of the index.
(D) the data records of the file are organized not in the same order as the data entries of the index.

Q.4 Three concurrent processes *X*, *Y*, and *Z* execute three different code segments that access and update certain shared variables. Process *X* executes the *P* operation (i.e., *wait*) on semaphores *a*, *b* and *c*; process *Y* executes the *P* operation on semaphores *b*, *c* and *d*; process *Z* executes the *P* operation on semaphores *c*, *d*, and *a* before entering the respective code segments. After completing the execution of its code segment, each process invokes the *V* operation (i.e., *signal*) on its three semaphores. All semaphores are binary semaphores initialized to one. Which one of the following represents a deadlock-free order of invoking the *P* operations by the processes?

(A) *X*: *P*(*a*)*P*(*b*)*P*(*c*) *Y*: *P*(*b*)*P*(*c*)*P*(*d*) *Z*: *P*(*c*)*P*(*d*)*P*(*a*)
(B) *X*: *P*(*b*)*P*(*a*)*P*(*c*) *Y*: *P*(*b*)*P*(*c*)*P*(*d*) *Z*: *P*(*a*)*P*(*c*)*P*(*d*)
(C) *X*: *P*(*b*)*P*(*a*)*P*(*c*) *Y*: *P*(*c*)*P*(*b*)*P*(*d*) *Z*: *P*(*a*)*P*(*c*)*P*(*d*)
(D) *X*: *P*(*a*)*P*(*b*)*P*(*c*) *Y*: *P*(*c*)*P*(*b*)*P*(*d*) *Z*: *P*(*c*)*P*(*d*)*P*(*a*)

Q.5 Which of the following statements is/are **FALSE**?

1. For every non-deterministic Turing machine, there exists an equivalent deterministic Turing machine.
2. Turing recognizable languages are closed under union and complementation.
3. Turing decidable languages are closed under intersection and complementation.
4. Turing recognizable languages are closed under union and intersection.

(A) 1 and 4 only (B) 1 and 3 only (C) 2 only (D) 3 only

Q.6 Which of the following statements are **TRUE**?

1. The problem of determining whether there exists a cycle in an undirected graph is in P.
2. The problem of determining whether there exists a cycle in an undirected graph is in NP.
3. If a problem A is NP-Complete, there exists a non-deterministic polynomial time algorithm to solve A.

(A) 1, 2 and 3 (B) 1 and 2 only (C) 2 and 3 only (D) 1 and 3 only

Q.7 What is the time complexity of Bellman-Ford single-source shortest path algorithm on a complete graph of n vertices?

(A) $\Theta(n^2)$ (B) $\Theta(n^2 \log n)$ (C) $\Theta(n^3)$ (D) $\Theta(n^3 \log n)$

Q.8 In a k-way set associative cache, the cache is divided into v sets, each of which consists of k lines. The lines of a set are placed in sequence one after another. The lines in set s are sequenced before the lines in set (s+1). The main memory blocks are numbered 0 onwards. The main memory block numbered j must be mapped to any one of the cache lines from

(A) (j mod v) * k to (j mod v) * k + (k-1) (B) (j mod v) to (j mod v) + (k-1)
(C) (j mod k) to (j mod k) + (v-1) (D) (j mod k) * v to (j mod k) * v + (v-1)

Q.9 Which one of the following expressions does **NOT** represent exclusive NOR of x and y?

(A) $xy + x'y'$ (B) $x \oplus y'$ (C) $x' \oplus y$ (D) $x' \oplus y'$

Q.10 Which one of the following functions is continuous at $x = 3$?

(A) $f(x) = \begin{cases} 2, & \text{if } x = 3 \\ x-1, & \text{if } x > 3 \\ \dfrac{x+3}{3}, & \text{if } x < 3 \end{cases}$

(B) $f(x) = \begin{cases} 4, & \text{if } x = 3 \\ 8-x & \text{if } x \neq 3 \end{cases}$

(C) $f(x) = \begin{cases} x+3, & \text{if } x \leq 3 \\ x-4 & \text{if } x > 3 \end{cases}$

(D) $f(x) = \dfrac{1}{x^3 - 27}$, if $x \neq 3$

Q.11 Function f is known at the following points:

x	0	0.3	0.6	0.9	1.2	1.5	1.8	2.1	2.4	2.7	3.0
$f(x)$	0	0.09	0.36	0.81	1.44	2.25	3.24	4.41	5.76	7.29	9.00

The value of $\int_0^3 f(x)\,dx$ computed using the trapezoidal rule is

(A) 8.983 (B) 9.003 (C) 9.017 (D) 9.045

Q.12 Which of the following statements is/are **TRUE** for undirected graphs?

P: Number of odd degree vertices is even.
Q: Sum of degrees of all vertices is even.

(A) P only (B) Q only (C) Both P and Q (D) Neither P nor Q

Q.13 Consider an undirected random graph of eight vertices. The probability that there is an edge between a pair of vertices is 1/2. What is the expected number of unordered cycles of length three?

(A) 1/8 (B) 1 (C) 7 (D) 8

Q.14 A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is **TRUE** about $\oplus$?

(A) Commutative but not associative
(B) Both commutative and associative
(C) Associative but not commutative
(D) Neither commutative nor associative

Q.15 Suppose p is the number of cars per minute passing through a certain road junction between 5 PM and 6 PM, and p has a Poisson distribution with mean 3. What is the probability of observing fewer than 3 cars during any given minute in this interval?

(A) $8/(2e^3)$ (B) $9/(2e^3)$ (C) $17/(2e^3)$ (D) $26/(2e^3)$

Q.16 Which one of the following does **NOT** equal $\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$?

(A) $\begin{vmatrix} 1 & x(x+1) & x+1 \\ 1 & y(y+1) & y+1 \\ 1 & z(z+1) & z+1 \end{vmatrix}$

(B) $\begin{vmatrix} 1 & x+1 & x^2+1 \\ 1 & y+1 & y^2+1 \\ 1 & z+1 & z^2+1 \end{vmatrix}$

(C) $\begin{vmatrix} 0 & x-y & x^2-y^2 \\ 0 & y-z & y^2-z^2 \\ 1 & z & z^2 \end{vmatrix}$

(D) $\begin{vmatrix} 2 & x+y & x^2+y^2 \\ 2 & y+z & y^2+z^2 \\ 1 & z & z^2 \end{vmatrix}$

Q.17 The smallest integer that can be represented by an 8-bit number in 2's complement form is

(A) -256 (B) -128 (C) -127 (D) 0

Q.18 In the following truth table, V = 1 if and only if the input is valid.

Inputs				Outputs		
D_0	D_1	D_2	D_3	X_0	X_1	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

What function does the truth table represent?

(A) Priority encoder
(B) Decoder
(C) Multiplexer
(D) Demultiplexer

Q.19 Which one of the following is the tightest upper bound that represents the number of swaps required to sort n numbers using selection sort?

(A) $O(\log n)$ (B) $O(n)$ (C) $O(n \log n)$ (D) $O(n^2)$

Q.20 Which one of the following is the tightest upper bound that represents the time complexity of inserting an object into a binary search tree of n nodes?

(A) $O(1)$ (B) $O(\log n)$ (C) $O(n)$ (D) $O(n \log n)$

Q.21 Consider the languages $L_1 = \Phi$ and $L_2 = \{a\}$. Which one of the following represents $L_1 L_2^* \cup L_1^*$?

(A) {ϵ} (B) Φ (C) *a** (D) {ϵ, *a*}

Q.22 What is the maximum number of reduce moves that can be taken by a bottom-up parser for a grammar with no epsilon- and unit-production (i.e., of type A → ϵ and A → a) to parse a string with *n* tokens?

(A) *n*/2 (B) *n*-1 (C) 2*n*-1 (D) 2^n

Q.23 A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every *T* time units and decides the next process to schedule. Which one of the following is **TRUE** if the processes have no I/O operations and all arrive at time zero?

(A) This algorithm is equivalent to the first-come-first-serve algorithm.
(B) This algorithm is equivalent to the round-robin algorithm.
(C) This algorithm is equivalent to the shortest-job-first algorithm.
(D) This algorithm is equivalent to the shortest-remaining-time-first algorithm.

Q.24 Match the problem domains in **GROUP I** with the solution technologies in **GROUP II**.

GROUP I	**GROUP II**
(P) Service oriented computing	(1) Interoperability
(Q) Heterogeneous communicating systems	(2) BPMN
(R) Information representation	(3) Publish-find-bind
(S) Process description	(4) XML

(A) P-1, Q-2, R-3, S-4 (B) P-3, Q-4, R-2, S-1
(C) P-3, Q-1, R-4, S-2 (D) P-4, Q-3, R-2, S-1

Q.25 Using public key cryptography, X adds a digital signature σ to message M, encrypts <M, σ>, and sends it to Y, where it is decrypted. Which one of the following sequences of keys is used for the operations?

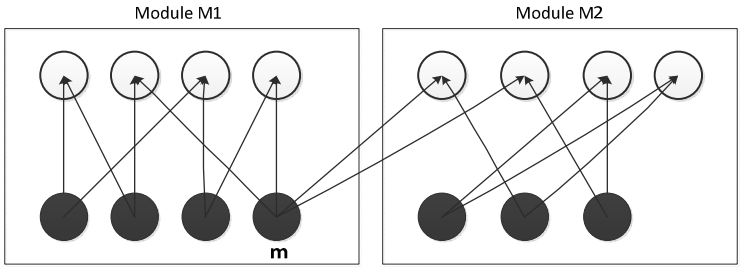
(A) Encryption: X's private key followed by Y's private key; Decryption: X's public key followed by Y's public key
(B) Encryption: X's private key followed by Y's public key; Decryption: X's public key followed by Y's private key
(C) Encryption: X's public key followed by Y's private key; Decryption: Y's public key followed by X's private key
(D) Encryption: X's private key followed by Y's public key; Decryption: Y's private key followed by X's public key

Q.26 to Q.55 carry two marks each.

Q.26 In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment offset value is 300. The position of the datagram, the sequence numbers of the first and the last bytes of the payload, respectively are

(A) Last fragment, 2400 and 2789
(B) First fragment, 2400 and 2759
(C) Last fragment, 2400 and 2759
(D) Middle fragment, 300 and 689

Q.27 The following figure represents access graphs of two modules M1 and M2. The filled circles represent methods and the unfilled circles represent attributes. If method `m` is moved to module M2 keeping the attributes where they are, what can we say about the average cohesion and coupling between modules in the system of two modules?



(A) There is no change.
(B) Average cohesion goes up but coupling is reduced.
(C) Average cohesion goes down and coupling also reduces.
(D) Average cohesion and coupling increase.

Q.28 A certain computation generates two arrays `a` and `b` such that `a[i]=f(i)` for $0 \le i < n$ and b[i] = g (a[i])for $0 \le i < n$. Suppose this computation is decomposed into two concurrent processes *X* and *Y* such that *X* computes the array `a` and *Y* computes the array `b`. The processes employ two binary semaphores *R* and *S*, both initialized to zero. The array `a` is shared by the two processes. The structures of the processes are shown below.

Process *X*:
```
private i;
for (i=0; i<n; i++) {
   a[i] = f(i);
   ExitX(R, S);
}
```

Process *Y*:
```
private i;
for (i=0; i<n; i++) {
   EntryY(R, S);
   b[i] = g(a[i]);
}
```

Which one of the following represents the **CORRECT** implementations of `ExitX` and `EntryY`?

(A)
```
ExitX(R, S) {
   P(R);
   V(S);
}
EntryY(R, S) {
   P(S);
   V(R);
}
```

(B)
```
ExitX(R, S) {
   V(R);
   V(S);
}
EntryY(R, S) {
   P(R);
   P(S);
}
```

(C)
```
ExitX(R, S) {
   P(S);
   V(R);
}
EntryY(R, S) {
   V(S);
   P(R);
}
```

(D)
```
ExitX(R, S) {
   V(R);
   P(S);
}
EntryY(R, S) {
   V(S);
   P(R);
}
```

Q.29 Consider the following two sets of LR(1) items of an LR(1) grammar.

X → c**.**X, c/d	X → c**.**X, $
X → **.**cX, c/d	X → **.**cX, $
X → **.**d, c/d	X → **.**d, $

Which of the following statements related to merging of the two sets in the corresponding LALR parser is/are **FALSE**?

1. Cannot be merged since look aheads are different.
2. Can be merged but will result in S-R conflict.
3. Can be merged but will result in R-R conflict.
4. Cannot be merged since *goto* on *c* will lead to two different sets.

(A) 1 only (B) 2 only (C) 1 and 4 only (D) 1, 2, 3 and 4

Q.30 Which of the following is/are undecidable?

1. *G* is a CFG. Is $L(G) = \Phi$?
2. *G* is a CFG. Is $L(G) = \sum^*$?
3. *M* is a Turing machine. Is *L*(*M*) regular?
4. *A* is a DFA and *N* is an NFA. Is *L*(*A*) = *L*(*N*)?

(A) 3 only (B) 3 and 4 only (C) 1, 2 and 3 only (D) 2 and 3 only

Q.31 What is the return value of `f(p,p)`, if the value of `p` is initialized to 5 before the call? Note that the first parameter is passed by reference, whereas the second parameter is passed by value.

```
int f (int &x, int c) {
   c = c - 1;
   if (c==0) return 1;
   x = x + 1;
   return f(x,c) * x;
}
```

(A) 3024 (B) 6561 (C) 55440 (D) 161051

Q.32 The preorder traversal sequence of a binary search tree is 30, 20, 10, 15, 25, 23, 39, 35, 42. Which one of the following is the postorder traversal sequence of the same tree?

(A) 10, 20, 15, 23, 25, 35, 42, 39, 30 (B) 15, 10, 25, 23, 20, 42, 35, 39, 30
(C) 15, 20, 10, 23, 25, 42, 35, 39, 30 (D) 15, 10, 23, 25, 20, 35, 42, 39, 30

Q.33 Consider an instruction pipeline with five stages without any branch prediction: Fetch Instruction (FI), Decode Instruction (DI), Fetch Operand (FO), Execute Instruction (EI) and Write Operand (WO). The stage delays for FI, DI, FO, EI and WO are 5 ns, 7 ns, 10 ns, 8 ns and 6 ns, respectively. There are intermediate storage buffers after each stage and the delay of each buffer is 1 ns. A program consisting of 12 instructions I_1, I_2, I_3, …, I_{12} is executed in this pipelined processor. Instruction I_4 is the only branch instruction and its branch target is I_9. If the branch is taken during the execution of this program, the time (in ns) needed to complete the program is

(A) 132 (B) 165 (C) 176 (D) 328

Q.34 Consider the following operation along with `Enqueue` and `Dequeue` operations on queues, where *k* is a global parameter.

```
MultiDequeue(Q){
      m = k
           while (Q is not empty) and (m > 0) {
           Dequeue(Q)
           m = m – 1
      }
}
```

What is the worst case time complexity of a sequence of *n* queue operations on an initially empty queue?

(A) $\Theta(n)$ (B) $\Theta(n + k)$ (C) $\Theta(nk)$ (D) $\Theta(n^2)$

Q.35 A RAM chip has a capacity of 1024 words of 8 bits each (1K × 8). The number of 2 × 4 decoders with enable line needed to construct a 16K × 16 RAM from 1K × 8 RAM is

(A) 4 (B) 5 (C) 6 (D) 7

Q.36 Which one of the following is **NOT** logically equivalent to $\neg\exists x(\forall y(\alpha)\wedge\forall z(\beta))$?

(A) $\forall x(\exists z(\neg\beta)\rightarrow\forall y(\alpha))$ (B) $\forall x(\forall z(\beta)\rightarrow\exists y(\neg\alpha))$
(C) $\forall x(\forall y(\alpha)\rightarrow\exists z(\neg\beta))$ (D) $\forall x(\exists y(\neg\alpha)\rightarrow\exists z(\neg\beta))$

Q.37 What is the logical translation of the following statement?

"None of my friends are perfect."

(A) $\exists x(F(x)\wedge\neg P(x))$ (B) $\exists x(\neg F(x)\wedge P(x))$
(C) $\exists x(\neg F(x)\wedge\neg P(x))$ (D) $\neg\exists x(F(x)\wedge P(x))$

Q.38 The line graph *L(G)* of a simple graph *G* is defined as follows:

- There is exactly one vertex *v(e)* in *L(G)* for each edge *e* in *G*.
- For any two edges *e* and *e'* in *G*, *L(G)* has an edge between *v(e)* and *v(e')*, if and only if *e* and *e'* are incident with the same vertex in *G*.

Which of the following statements is/are **TRUE**?

(P) The line graph of a cycle is a cycle.
(Q) The line graph of a clique is a clique.
(R) The line graph of a planar graph is planar.
(S) The line graph of a tree is a tree.

(A) P only (B) P and R only
(C) R only (D) P, Q and S only

Q.39 Consider the following sequence of micro-operations.

MBR ← PC
MAR ← X
PC ← Y
Memory ← MBR

Which one of the following is a possible operation performed by this sequence?

(A) Instruction fetch (B) Operand fetch
(C) Conditional branch (D) Initiation of interrupt service

Q.40 Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each cylinder contains 64 sectors (0-63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder-wise and the addressing format is <cylinder no., surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?

(A) 1281 (B) 1282 (C) 1283 (D) 1284

Q.41 The number of elements that can be sorted in $\Theta(\log n)$ time using heap sort is

(A) $\Theta(1)$ (B) $\Theta\left(\sqrt{\log\ n}\right)$ (C) $\Theta\left(\frac{\log n}{\log\ \log\ n}\right)$ (D) $\Theta(\log n)$

Q.42 Consider the following function:

```
int unknown(int n){
          int i, j, k=0;
          for (i=n/2; i<=n; i++)
              for (j=2; j<=n; j=j*2)
                  k = k + n/2;
          return (k);
}
```

The return value of the function is

(A) $\Theta(n^2)$ (B) $\Theta(n^2 \log n)$ (C) $\Theta(n^3)$ (D) $\Theta(n^3 \log n)$

Q.43 Consider the following languages.

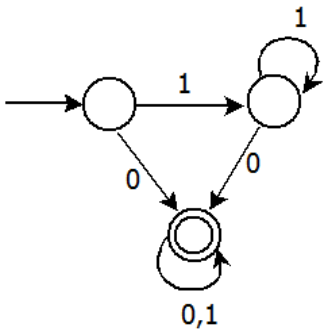
$L_1 = \{0^p\ 1^q\ 0^r \mid p, q, r \geq 0\}$

$L_2 = \{0^p\ 1^q\ 0^r \mid p, q, r \geq 0,\ p \neq r\}$

Which one of the following statements is **FALSE**?

(A) L_2 is context-free.
(B) $L_1 \cap L_2$ is context-free.
(C) Complement of L_2 is recursive.
(D) Complement of L_1 is context-free but not regular.

Q.44 Consider the DFA *A* given below.



Which of the following are **FALSE**?

1. Complement of $L(A)$ is context-free.
2. $L(A) = L((11*0+0)(0 + 1)*0*1*)$
3. For the language accepted by *A*, *A* is the minimal DFA.
4. *A* accepts all strings over {0, 1} of length at least 2.

(A) 1 and 3 only (B) 2 and 4 only (C) 2 and 3 only (D) 3 and 4 only

Q.45 A shared variable *x*, initialized to zero, is operated on by four concurrent processes *W*, *X*, *Y*, *Z* as follows. Each of the processes *W* and *X* reads *x* from memory, increments by one, stores it to memory, and then terminates. Each of the processes *Y* and *Z* reads *x* from memory, decrements by two, stores it to memory, and then terminates. Each process before reading *x* invokes the *P* operation (i.e., *wait*) on a counting semaphore *S* and invokes the *V* operation (i.e., *signal*) on the semaphore *S* after storing *x* to memory. Semaphore *S* is initialized to two. What is the maximum possible value of *x* after all processes complete execution?

(A) –2 (B) –1 (C) 1 (D) 2

Q.46 Consider the following relational schema.

Students(rollno: integer, sname: string)
Courses(courseno: integer, cname: string)
Registration(rollno: integer, courseno: integer, percent: real)

Which of the following queries are equivalent to this query in English?
"Find the distinct names of all students who score more than 90% in the course numbered 107"

(I) SELECT DISTINCT S.sname
FROM Students as S, Registration as R
WHERE R.rollno=S.rollno AND R.courseno=107 AND R.percent >90

(II) $\prod_{sname}(\sigma_{courseno=107 \wedge percent>90}$ (Registration $\bowtie$ Students))

(III) $\{T \mid \exists S \in$ Students, $\exists R \in$ Registration (S.rollno=R.rollno $\wedge$

R.courseno=107 $\wedge$ R.percent>90 $\wedge$T.sname=S.sname)}

(IV) $\{<S_N> \mid \exists S_R \exists R_P$ ($<S_R, S_N> \in$ Students $\wedge <S_R, 107, R_P> \in$ Registration $\wedge R_P>90)\}$

(A) I, II, III and IV (B) I, II and III only
(C) I, II and IV only (D) II, III and IV only

Q.47 Determine the maximum length of the cable (in km) for transmitting data at a rate of 500 Mbps in an Ethernet LAN with frames of size 10,000 bits. Assume the signal speed in the cable to be 2,00,000 km/s.

(A) 1 (B) 2 (C) 2.5 (D) 5

Common Data Questions

Common Data for Questions 48 and 49:

The procedure given below is required to find and replace certain characters inside an input character string supplied in array `A`. The characters to be replaced are supplied in array `oldc`, while their respective replacement characters are supplied in array `newc`. Array `A` has a fixed length of five characters, while arrays `oldc` and `newc` contain three characters each. However, the procedure is flawed.

```
void find_and_replace (char *A, char *oldc, char *newc) {
        for (int i=0; i<5; i++)
            for (int j=0; j<3; j++)
                if (A[i] == oldc[j])  A[i] = newc[j];
 }
```

The procedure is tested with the following four test cases.

(1) `oldc = "abc", newc = "dab"` (2) `oldc = "cde", newc = "bcd"`
(3) `oldc = "bca", newc = "cda"` (4) `oldc = "abc", newc = "bac"`

Q.48 If array `A` is made to hold the string `"abcde"`, which of the above four test cases will be successful in exposing the flaw in this procedure?

(A) None (B) 2 only
(C) 3 and 4 only (D) 4 only

Q.49 The tester now tests the program on all input strings of length five consisting of characters `'a'`, `'b'`, `'c'`, `'d'` and `'e'` with duplicates allowed. If the tester carries out this testing with the four test cases given above, how many test cases will be able to capture the flaw?

(A) Only one (B) Only two (C) Only three (D) All four

Common Data for Questions 50 and 51:

The following code segment is executed on a processor which allows only register operands in its instructions. Each instruction can have atmost two source operands and one destination operand. Assume that all variables are dead after this code segment.

```
c = a + b;
d = c * a;
e = c + a;
x = c * c;
if (x > a) {
   y = a * a;
}
else {
   d = d * d;
   e = e * e;
}
```

Q.50 What is the minimum number of registers needed in the instruction set architecture of the processor to compile this code segment without any spill to memory? Do not apply any optimization other than optimizing register allocation.

(A) 3 (B) 4 (C) 5 (D) 6

Q.51 Suppose the instruction set architecture of the processor has only two registers. The only allowed compiler optimization is code motion, which moves statements from one place to another while preserving correctness. What is the minimum number of spills to memory in the compiled code?

(A) 0 (B) 1 (C) 2 (D) 3

Linked Answer Questions

Statement for Linked Answer Questions 52 and 53:

Relation R has eight attributes $ABCDEFGH$. Fields of R contain only atomic values.
$F=\{CH\rightarrow G,\ A\rightarrow BC,\ B\rightarrow CFH,\ E\rightarrow A,\ F\rightarrow EG\}$ is a set of functional dependencies (FDs) so that F^+ is exactly the set of FDs that hold for R.

Q.52 How many candidate keys does the relation R have?

(A) 3 (B) 4 (C) 5 (D) 6

Q.53 The relation R is

(A) in 1NF, but not in 2NF.
(B) in 2NF, but not in 3NF.
(C) in 3NF, but not in BCNF.
(D) in BCNF.

Statement for Linked Answer Questions 54 and 55:

A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level table (T_1), which occupies exactly one page. Each entry of T_1 stores the base address of a page of the second-level table (T_2). Each entry of T_2 stores the base address of a page of the third-level table (T_3). Each entry of T_3 stores a page table entry (PTE). The PTE is 32 bits in size. The processor used in the computer has a 1 MB 16-way set associative virtually indexed physically tagged cache. The cache block size is 64 bytes.

Q.54 What is the size of a page in KB in this computer?

(A) 2 (B) 4 (C) 8 (D) 16

Q.55 What is the minimum number of page colours needed to guarantee that no two synonyms map to different sets in the processor cache of this computer?

(A) 2 (B) 4 (C) 8 (D) 16

General Aptitude (GA) Questions
Q.56 to Q.60 carry one mark each.

Q.56 Choose the grammatically **INCORRECT** sentence:

(A) He is of Asian origin.
(B) They belonged to Africa.
(C) She is an European.
(D) They migrated from India to Australia.

Q.57 What will be the maximum sum of 44, 42, 40, ... ?

(A) 502 (B) 504 (C) 506 (D) 500

Q.58 Were you a bird, you _____ in the sky.

(A) would fly (B) shall fly (C) should fly (D) shall have flown

Q.59 Which one of the following options is the closest in meaning to the word given below?

Nadir

(A) Highest (B) Lowest (C) Medium (D) Integration

Q.60 Complete the sentence:
Universalism is to particularism as diffuseness is to _____.

(A) specificity (B) neutrality (C) generality (D) adaptation

Q.61 to Q.65 carry two marks each.

Q.61 The current erection cost of a structure is *Rs*. 13,200. If the labour wages per day increase by 1/5 of the current wages and the working hours decrease by 1/24 of the current period, then the new cost of erection in *Rs*. is

(A) 16,500 (B) 15,180 (C) 11,000 (D) 10,120

Q.62 After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again, the spider failed but that did not deter it to refrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

Which one of the following assertions is best supported by the above information?

(A) Failure is the pillar of success.
(B) Honesty is the best policy.
(C) Life begins and ends with adventures.
(D) No adversity justifies giving up hope.

Q.63 A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

(A) 36 (B) 30 (C) 24 (D) 18

Q.64 Find the sum of the expression

$$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+\ldots\ldots\ldots\ldots+\frac{1}{\sqrt{80}+\sqrt{81}}$$

(A) 7 (B) 8
(C) 9 (D) 10

Q.65 Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

(A) 13/90 (B) 12/90 (C) 78/90 (D) 77/90

END OF THE QUESTION PAPER

SESSION - 1

GATE 2014: General Instructions during Examination

1. Total duration of the GATE examination is **180** minutes.
2. The clock will be set at the server. The countdown timer at the top right corner of screen will display the remaining time available for you to complete the examination. When the timer reaches zero, the examination will end by itself. You need not terminate the examination or submit your paper.
3. Any useful data required for your paper can be viewed by clicking on the **Useful Common Data** button that appears on the screen.
4. Use the scribble pad provided to you for any rough work. Submit the scribble pad at the end of the examination.
5. You are allowed to use a non-programmable type calculator, however, sharing of calculators is not allowed.
6. The Question Palette displayed on the right side of screen will show the status of each question using one of the following symbols:

You have not visited the question yet.

You have not answered the question.

You have answered the question.

You have NOT answered the question, but have marked the question for review.

You have answered the question, but marked it for review.

The **Marked for Review** status for a question simply indicates that you would like to look at that question again. *If a question is answered, but marked for review, then the answer will be considered for evaluation unless the status is modified by the candidate.*

<u>Navigating to a Question :</u>

7. To answer a question, do the following:
    a. Click on the question number in the Question Palette to go to that question directly.
    b. Select an answer for a multiple choice type question by clicking on the bubble placed before the 4 choices, namely A, B, C and D. Use the virtual numeric keypad to enter a number as answer for a numerical type question.
    c. Click on **Save & Next** to save your answer for the current question and then go to the next question.
    d. Click on **Mark for Review & Next** to save your answer for the current question and also mark it for review, and then go to the next question.

Caution: Note that your answer for the current question will not be saved, if you navigate to another question directly by clicking on a question number without saving the answer to the previous question.

You can view all the questions by clicking on the **Question Paper** button. This feature is provided, so that if you want you can just see the entire question paper at a glance.

Answering a Question :

8. Procedure for answering a multiple choice (MCQ) type question:
    a. Choose one answer from the 4 options (A,B,C,D) given below the question, click on the bubble placed before the chosen option.
    b. To deselect your chosen answer, click on the bubble of the chosen option again or click on the **Clear Response** button.
    c. To change your chosen answer, click on the bubble of another option.
    d. To save your answer, you MUST click on the **Save & Next** button.

9. Procedure for answering a numerical answer type question:
    a. To enter a number as your answer, use the virtual numerical keypad.
    b. A fraction (e.g. -0.3 or -.3) can be entered as an answer with or without '0' before the decimal point. As many as four decimal points, e.g. 12.5435 or 0.003 or -932.6711 or 12.82 can be entered.
    c. To clear your answer, click on the **Clear Response** button.
    d. To save your answer, you MUST click on the **Save & Next** button
10. To mark a question for review, click on the **Mark for Review & Next** button. *If an answer is selected (for MCQ) or entered (for numerical answer type) for a question that is* **Marked for Review***, that answer will be considered in the evaluation unless the status is modified by the candidate.*

11. To change your answer to a question that has already been answered, first select that question for answering and then follow the procedure for answering that type of question.
12. Note that ONLY Questions for which answers are ***saved*** or ***marked for review after answering*** will be considered for evaluation.

Choosing a Section :

13. Sections in this question paper are displayed on the top bar of the screen. Questions in a Section can be viewed by clicking on the name of that Section. The Section you are currently viewing will be highlighted.
14. A checkbox is displayed for every optional Section, if any, in the Question Paper. To select the optional Section for answering, click on the checkbox for that Section.
15. If the checkbox for an optional Section is not selected, the **Save & Next** button and the **Mark for Review & Next** button will NOT be enabled for that Section. You will

only be able to see questions in this Section, but you will not be able to answer questions in the Section.

16. After clicking the **Save & Next** button for the last question in a Section, you will automatically be taken to the first question of the next Section in sequence.
17. You can move the mouse cursor over the name of a Section to view the answering status for that Section.

Changing the Optional Section :

18. After answering the chosen optional Section, partially or completely, you can change the optional Section by selecting the checkbox for a new Section that you want to attempt. A warning message will appear along with a table showing the number of questions answered in each of the previously chosen optional Sections and a checkbox against each of these Sections. Click on a checkbox against a Section that you want to reset and then click on the **RESET** button. Note that RESETTING a Section will DELETE all the answers for questions in that Section. Hence, if you think that you may want to select this Section again later, you will have to note down your answers for questions in that Section. If you do not want to reset the Section and want to continue answering the previously chosen optional Section, then click on the **BACK** button.

19. If you deselect the checkbox for an optional Section in the top bar, the following warning message will appear: "Deselecting the checkbox will DELETE all the answers for questions in this Section. Do you want to deselect this Section?" If you want to deselect, click on the **RESET** button. If you do not want to deselect, click on the **BACK** button.

20. You can shuffle between different Sections or change the optional Sections any number of times.

GATE 2014 Examination

CS: Computer Science & Information Technology

Duration: **180 minutes** *Maximum Marks*: **100**

Read the following instructions carefully.

1. To login, enter your Registration Number and password provided to you. Kindly go through the various symbols used in the test and understand their meaning before you start the examination.
2. Once you login and after the start of the examination, you can view all the questions in the question paper, by clicking on the **View All Questions** button in the screen.
3. This question paper consists of **2 sections**, General Aptitude (GA) for **15 marks** and the subject specific GATE paper for **85 marks**. Both these sections are compulsory.
   The GA section consists of **10** questions. Question numbers 1 to 5 are of 1-mark each, while question numbers 6 to 10 are of 2-mark each.
   The subject specific GATE paper section consists of **55** questions, out of which question numbers 1 to 25 are of 1-mark each, while question numbers 26 to 55 are of 2-mark each.
4. Depending upon the GATE paper, there may be useful common data that may be required for answering the questions. If the paper has such useful data, the same can be viewed by clicking on the **Useful Common Data** button that appears at the top, right hand side of the screen.
5. The computer allotted to you at the examination center runs specialized software that permits only one answer to be selected for multiple-choice questions using a mouse and to enter a suitable number for the numerical answer type questions using the virtual keyboard and mouse.
6. Your answers shall be updated and saved on a server periodically and also at the end of the examination. The examination will **stop automatically** at the end of **180 minutes**.
7. In each paper a candidate can answer a total of 65 questions carrying 100 marks.
8. The question paper may consist of questions of **multiple choice type** (MCQ) and **numerical answer type**.
9. Multiple choice type questions will have four choices against A, B, C, D, out of which only **ONE** is the correct answer. The candidate has to choose the correct answer by clicking on the bubble (○) placed before the choice.
10. For numerical answer type questions, each question will have a numerical answer and there will not be any choices**. For these questions, the answer should be entered**by using the virtual keyboard that appears on the monitor and the mouse.
11. All questions that are not attempted will result in zero marks. However, wrong answers for multiple choice type questions (MCQ) will result in **NEGATIVE** marks. For all MCQ questions a wrong answer will result in deduction of⅓ marks for a 1-mark question and ⅔ marks for a 2-mark question.
12. There is **NO NEGATIVE MARKING** for questions of **NUMERICAL ANSWER TYPE**.
13. Non-programmable type Calculator is allowed. Charts, graph sheets, and mathematical tables are **NOT** allowed in the Examination Hall. You must use the Scribble pad provided to you at the examination centre for all your rough work. The Scribble Pad has to be returned at the end of the examination.

Declaration by the candidate:

"I have read and understood all the above instructions. I have also read and understood clearly the instructions given on the admit card and shall follow the same. I also understand that in case I am found to violate any of these instructions, my candidature is liable to be cancelled. I also confirm that at the start of the examination all the computer hardware allotted to me are in proper working condition".

Q. 1 – Q. 5 carry one mark each.

Q.1 Which of the following options is the closest in meaning to the phrase underlined in the sentence below?

It is fascinating to see life forms cope with varied environmental conditions.

(A) adopt to (B) adapt to (C) adept in (D) accept with

Q.2 Choose the most appropriate word from the options given below to complete the following sentence.

He could not understand the judges awarding her the first prize, because he thought that her performance was quite _____.

(A) superb (B) medium (C) mediocre (D) exhilarating

Q.3 In a press meet on the recent scam, the minister said, "The buck stops here". What did the minister convey by the statement?

(A) He wants all the money
(B) He will return the money
(C) He will assume final responsibility
(D) He will resist all enquiries

Q.4 If $(z + 1/z)^2 = 98$, compute $(z^2 + 1/z^2)$.

Q.5 The roots of $ax^2 + bx + c = 0$ are real and positive. *a*, *b* and *c* are real. Then $ax^2 + b|x| + c = 0$ has

(A) no roots
(B) 2 real roots
(C) 3 real roots
(D) 4 real roots

Q. 6 – Q. 10 carry two marks each.

Q.6 The Palghat Gap (or Palakkad Gap), a region about 30 km wide in the southern part of the Western Ghats in India, is lower than the hilly terrain to its north and south. The exact reasons for the formation of this gap are not clear. It results in the neighbouring regions of Tamil Nadu getting more rainfall from the South West monsoon and the neighbouring regions of Kerala having higher summer temperatures.

What can be inferred from this passage?

(A) The Palghat gap is caused by high rainfall and high temperatures in southern Tamil Nadu and Kerala
(B) The regions in Tamil Nadu and Kerala that are near the Palghat Gap are low-lying
(C) The low terrain of the Palghat Gap has a significant impact on weather patterns in neighbouring parts of Tamil Nadu and Kerala
(D) Higher summer temperatures result in higher rainfall near the Palghat Gap area

GA7 (GATE 2014)

Q.7 Geneticists say that they are very close to confirming the genetic roots of psychiatric illnesses such as depression and schizophrenia, and consequently, that doctors will be able to eradicate these diseases through early identification and gene therapy.

On which of the following assumptions does the statement above rely?

(A) Strategies are now available for eliminating psychiatric illnesses
(B) Certain psychiatric illnesses have a genetic basis
(C) All human diseases can be traced back to genes and how they are expressed
(D) In the future, genetics will become the only relevant field for identifying psychiatric illnesses

Q.8 Round-trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is Rs 100, a group of 5 tourists purchasing round-trip tickets will be charged Rs _____.

Q.9 In a survey, 300 respondents were asked whether they own a vehicle or not. If yes, they were further asked to mention whether they own a car or scooter or both. Their responses are tabulated below. What percent of respondents do not own a scooter?

		Men	Women
Own vehicle	Car	40	34
	Scooter	30	20
	Both	60	46
Do not own vehicle		20	50

Q.10 When a point inside of a tetrahedron (a solid with four triangular surfaces) is connected by straight lines to its corners, how many (new) internal planes are created with these lines? _____

END OF THE QUESTION PAPER

GA07 (GATE 2014)

Q. 1 – Q. 25 carry one mark each.

Q.1 Consider the statement

"Not all that glitters is gold"

Predicate $glitters(x)$ is true if x glitters and predicate $gold(x)$ is true if x is gold. Which one of the following logical formulae represents the above statement?

(A) $\forall x:\ glitters\ (x)\ \Rightarrow \neg gold(x)$

(B) $\forall x:\ gold\ (x)\ \Rightarrow glitters(x)$

(C) $\exists x:\ gold\ (x) \wedge \neg\ glitters\ (x)$

(D) $\exists x:\ glitters\ (x)\ \wedge \neg\ gold\ (x)$

Q.2 Suppose you break a stick of unit length at a point chosen uniformly at random. Then the expected length of the shorter stick is _____ .

Q.3 Let *G=(V,E)* be a directed graph where *V* is the set of vertices and *E* the set of edges. Then which one of the following graphs has the same strongly connected components as *G* ?

(A) $G_1 = (V, E_1)$ where $E_1 = \{(u,v)|(u,v) \notin E\}$

(B) $G_2 = (V, E_2)$ where $E_2 = \{(u,v)|(v,u) \in E\}$

(C) $G_3 = (V, E_3)$ where $E_3 = \{(u,v)|there\ is\ a\ path\ of\ length \leq 2\ from\ u\ to\ v\ in\ E\}$

(D) $G_4 = (V_4, E)$ where V_4 is the set of vertices in G which are not isolated

Q.4 Consider the following system of equations:

$$3x + 2y = 1$$
$$4x + 7z = 1$$
$$x + y + z = 3$$
$$x - 2y + 7z = 0$$

The number of solutions for this system is _____

Q.5 The value of the dot product of the eigenvectors corresponding to any pair of different eigenvalues of a 4-*by*-4 symmetric positive definite matrix is _____.

CS01 (GATE 2014)

Q.6 Let the function

$$f(\theta) = \begin{vmatrix} \sin\theta & \cos\theta & \tan\theta \\ \sin(\pi/6) & \cos(\pi/6) & \tan(\pi/6) \\ \sin(\pi/3) & \cos(\pi/3) & \tan(\pi/3) \end{vmatrix}$$

where $\theta \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$ and $f'(\theta)$ denote the derivative of f with respect to θ. Which of the following statements is/are **TRUE**?

(I) There exists $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$ such that $f'(\theta) = 0$.

(II) There exists $\theta \in (\frac{\pi}{6}, \frac{\pi}{3})$ such that $f'(\theta) \neq 0$.

(A) I only (B) II only (C) Both I and II (D) Neither I nor II

Q.7 Consider the following Boolean expression for *F*:

$$F(P, Q, R, S) = PQ + \bar{P}QR + \bar{P}Q\bar{R}S$$

The minimal sum-of-products form of *F* is

(A) $PQ + QR + QS$ (B) $P + Q + R + S$ (C) $\bar{P} + \bar{Q} + \bar{R} + \bar{S}$ (D) $\bar{P}R + \bar{P}\bar{R}S + P$

Q.8 The base (or radix) of the number system such that the following equation holds is_____.

$$\frac{312}{20} = 13.1$$

Q.9 A machine has a 32-bit architecture, with 1-word long instructions. It has 64 registers, each of which is 32 bits long. It needs to support 45 instructions, which have an immediate operand in addition to two register operands. Assuming that the immediate operand is an unsigned integer, the maximum value of the immediate operand is _____.

Q.10 Consider the following program in C language:

```
#include <stdio.h>

main()
{
 int i;
 int *pi = &i;

 scanf("%d",pi);
 printf("%d\n", i+5);
}
```

Which one of the following statements is **TRUE**?

(A) Compilation fails.

(B) Execution results in a run-time error.

(C) On execution, the value printed is `5` more than the address of variable `i`.

(D) On execution, the value printed is `5` more than the integer value entered.

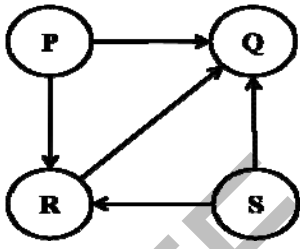
CS01 (GATE 2014)

Q.11 Let G be a graph with n vertices and m edges. What is the tightest upper bound on the running time of Depth First Search on G, when G is represented as an adjacency matrix?

(A) $\Theta(n)$ (B) $\Theta(n+m)$ (C) $\Theta(n^2)$ (D) $\Theta(m^2)$

Q.12 Consider a rooted n node binary tree represented using pointers. The best upper bound on the time required to determine the number of subtrees having exactly 4 nodes is $O(n^a \log^b n)$. Then the value of $a + 10b$ is _____.

Q.13 Consider the directed graph given below.



Which one of the following is **TRUE**?

(A) The graph does not have any topological ordering.

(B) Both PQRS and SRQP are topological orderings.

(C) Both PSRQ and SPRQ are topological orderings.

(D) PSRQ is the only topological ordering.

Q.14 Let P be a quicksort program to sort numbers in ascending order using the first element as the pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs [1 2 3 4 5] and [4 1 5 3 2] respectively. Which one of the following holds?

(A) $t_1 = 5$ (B) $t_1 < t_2$

(C) $t_1 > t_2$ (D) $t_1 = t_2$

Q.15 Which one of the following is **TRUE**?

(A) The language $L = \{a^n b^n \mid n \geq 0\}$ is regular.

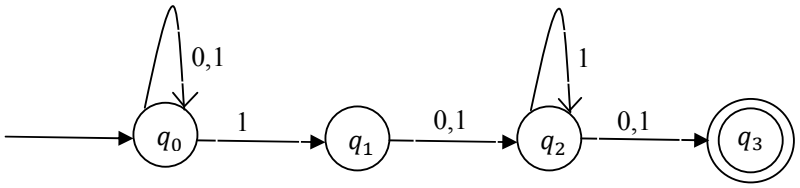
(B) The language $L = \{a^n \mid n \ is \ prime\}$ is regular.

(C) The language $L = \{w \mid w \ has \ 3k+1 \ b's \ for \ some \ k \in \mathbb{N} \ with \ \Sigma = \{a, b\}\}$ is regular.

(D) The language $L = \{ww \mid w \in \Sigma^* \ with \ \Sigma = \{0,1\}\}$ is regular.

CS01 (GATE 2014)

Q.16 Consider the finite automaton in the following figure.



What is the set of reachable states for the input string 0011?

(A) $\{q_0, q_1, q_2\}$

(B) $\{q_0, q_1\}$

(C) $\{q_0, q_1, q_2, q_3\}$

(D) $\{q_3\}$

Q.17 Which one of the following is **FALSE**?

(A) A basic block is a sequence of instructions where control enters the sequence at the beginning and exits at the end.

(B) Available expression analysis can be used for common subexpression elimination.

(C) Live variable analysis can be used for dead code elimination.

(D) $x = 4 * 5 \Rightarrow x = 20$ is an example of common subexpression elimination.

Q.18 Match the following:

1) Waterfall model	a) Specifications can be developed incrementally
2) Evolutionary model	b) Requirements compromises are inevitable
3) Component-based software engineering	c) Explicit recognition of risk
4) Spiral development	d) Inflexible partitioning of the project into stages

(A) 1-a, 2-b, 3-c, 4-d

(B) 1-d, 2-a, 3-b, 4-c

(C) 1-d, 2-b, 3-a, 4-c

(D) 1-c, 2-a, 3-b, 4-d

CS01 (GATE 2014)

Q.19 Suppose a disk has 201 cylinders, numbered from 0 to 200. At some time the disk arm is at cylinder 100, and there is a queue of disk access requests for cylinders 30, 85, 90, 100, 105, 110, 135 and 145. If Shortest-Seek Time First (SSTF) is being used for scheduling the disk access, the request for cylinder 90 is serviced after servicing _____ number of requests.

Q.20 Which one of the following is **FALSE**?

(A) User level threads are not scheduled by the kernel.

(B) When a user level thread is blocked, all other threads of its process are blocked.

(C) Context switching between user level threads is faster than context switching between kernel level threads.

(D) Kernel level threads cannot share the code segment.

Q.21 Consider the relation scheme $R = (E, F, G, H, I, J, K, L, M, N)$ and the set of functional dependencies $\{\{E, F\} \rightarrow \{G\},\ \{F\} \rightarrow \{I, J\},\ \{E, H\} \rightarrow \{K, L\},\ \{K\} \rightarrow \{M\},\ \{L\} \rightarrow \{N\}\}$ on R. What is the key for R?

(A) $\{E, F\}$ (B) $\{E, F, H\}$ (C) $\{E, F, H, K, L\}$ (D) $\{E\}$

Q.22 Given the following statements:

S1: A foreign key declaration can always be replaced by an equivalent check assertion in SQL.

S2: Given the table `R(a,b,c)` where `a` and `b` together form the primary key, the following is a valid table definition.

```
CREATE TABLE S (
      a INTEGER,
      d INTEGER,
      e INTEGER,
      PRIMARY KEY (d),
      FOREIGN KEY (a) references R)
```

Which one of the following statements is **CORRECT**?

(A) S1 is TRUE and S2 is FALSE.

(B) Both S1 and S2 are TRUE.

(C) S1 is FALSE and S2 is TRUE.

(D) Both S1 and S2 are FALSE.

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Q.23 Consider the following three statements about link state and distance vector routing protocols, for a large network with 500 network nodes and 4000 links.

[S1] The computational overhead in link state protocols is higher than in distance vector protocols.
[S2] A distance vector protocol (with split horizon) avoids persistent routing loops, but not a link state protocol.
[S3] After a topology change, a link state protocol will converge faster than a distance vector protocol.

Which one of the following is correct about S1, S2, and S3 ?

(A) S1, S2, and S3 are all true.
(B) S1, S2, and S3 are all false.
(C) S1 and S2 are true, but S3 is false.
(D) S1 and S3 are true, but S2 is false.

Q.24 Which of the following are used to generate a message digest by the network security protocols?

(P) RSA (Q) SHA-1 (R) DES (S) MD5

(A) P and R only
(B) Q and R only
(C) Q and S only
(D) R and S only

Q.25 Identify the correct order in which the following actions take place in an interaction between a web browser and a web server.

1. The web browser requests a webpage using HTTP.
2. The web browser establishes a TCP connection with the web server.
3. The web server sends the requested webpage using HTTP.
4. The web browser resolves the domain name using DNS.

(A) 4,2,1,3 (B) 1,2,3,4 (C) 4,1,2,3 (D) 2,4,1,3

Q. 26 – Q. 55 carry two marks each.

Q.26 Consider a token ring network with a length of 2 km having 10 stations including a monitoring station. The propagation speed of the signal is 2×10^8 m/s and the token transmission time is ignored. If each station is allowed to hold the token for 2 µsec, the minimum time for which the monitoring station should wait (**in µsec**)before assuming that the token is lost is _____.

Q.27 Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (**in msec**) by the TCP connection to get back to 32 KB congestion window is _____.

Q.28 Consider a selective repeat sliding window protocol that uses a frame size of 1 KB to send data on a 1.5 Mbps link with a one-way latency of 50 msec. To achieve a link utilization of 60%, the minimum number of bits required to represent the sequence number field is _____.

CS01 (GATE 2014)

Q.29 Consider the following four schedules due to three transactions (indicated by the subscript) using *read* and *write* on a data item `x`, denoted by `r(x)` and `w(x)` respectively. Which one of them is conflict serializable?

(A) $r_1(x);\ r_2(x);\ w_1(x);\ r_3(x);\ w_2(x)$

(B) $r_2(x);r_1(x);w_2(x);r_3(x);w_1(x)$

(C) $r_3(x);r_2(x);r_1(x);w_2(x);w_1(x)$

(D) $r_2(x);w_2(x);r_3(x);r_1(x);w_1(x)$

Q.30 Given the following two statements:

S1: Every table with two single-valued attributes is in 1NF, 2NF, 3NF and BCNF.
S2: AB→C, D→E, E→C is a minimal cover for the set of functional dependencies AB→C, D→E, AB→E, E→C.

Which one of the following is **CORRECT**?

(A) S1 is TRUE and S2 is FALSE.

(B) Both S1 and S2 are TRUE.

(C) S1 is FALSE and S2 is TRUE.

(D) Both S1 and S2 are FALSE.

CS01 (GATE 2014)

Q.31 An operating system uses the *Banker's algorithm* for deadlock avoidance when managing the allocation of three resource types X, Y, and Z to three processes P0, P1, and P2. The table given below presents the current system state. Here, the *Allocation* matrix shows the current number of resources of each type allocated to each process and the *Max* matrix shows the maximum number of resources of each type required by each process during its execution.

	Allocation			**Max**		
	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3
P1	3	2	0	6	2	0
P2	2	1	1	3	3	3

There are 3 units of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in a **safe** state. Consider the following independent requests for additional resources in the current state:

REQ1: P0 requests 0 units of X, 0 units of Y and 2 units of Z
REQ2: P1 requests 2 units of X, 0 units of Y and 0 units of Z

Which one of the following is **TRUE**?

(A) Only REQ1 can be permitted.

(B) Only REQ2 can be permitted.

(C) Both REQ1 and REQ2 can be permitted.

(D) Neither REQ1 nor REQ2 can be permitted.

Q.32 Consider the following set of processes that need to be scheduled on a single CPU. All the times are given in milliseconds.

Process Name	**Arrival Time**	**Execution Time**
A	0	6
B	3	2
C	5	4
D	7	6
E	10	3

Using the *shortest remaining time first* scheduling algorithm, the average process turnaround time (in msec) is _____.

Q.33 Assume that there are 3 page frames which are initially empty. If the page reference string is 1, 2, 3, 4, 2, 1, 5, 3, 2, 4, 6, the number of page faults using the *optimal replacement policy* is_____.

Q.34 A canonical set of items is given below

$$S \longrightarrow L\,.\,> R$$
$$Q \longrightarrow R\,.$$

On input symbol $<$ the set has

(A) a shift-reduce conflict and a reduce-reduce conflict.

(B) a shift-reduce conflict but not a reduce-reduce conflict.

(C) a reduce-reduce conflict but not a shift-reduce conflict.

(D) neither a shift-reduce nor a reduce-reduce conflict.

CS01 (GATE 2014)

Q.35 Let L be a language and $\bar{L}$ be its complement. Which one of the following is NOT a viable possibility?

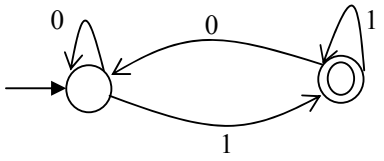
(A) Neither L nor $\bar{L}$ is recursively enumerable (r.e.).

(B) One of L and $\bar{L}$ is r.e. but not recursive; the other is not r.e.

(C) Both L and $\bar{L}$ are r.e. but not recursive.

(D) Both L and $\bar{L}$ are recursive.

Q.36 Which of the regular expressions given below represent the following DFA?



I) 0*1(1+00*1)*
II) 0*1*1+11*0*1
III) (0+1)*1

(A) I and II only (B) I and III only

(C) II and III only (D) I, II, and III

Q.37 There are 5 bags labeled 1 to 5. All the coins in a given bag have the same weight. Some bags have coins of weight 10 gm, others have coins of weight 11 gm. I pick 1, 2, 4, 8, 16 coins respectively from bags 1 to 5. Their total weight comes out to 323 gm. Then the product of the labels of the bags having 11 gm coins is ___.

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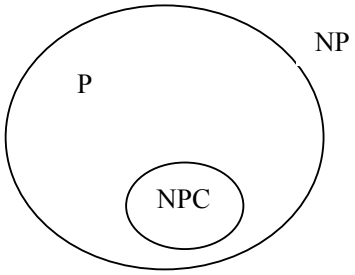
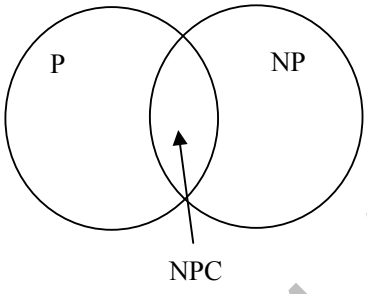
Q.38 Suppose a polynomial time algorithm is discovered that correctly computes the largest clique in a given graph. In this scenario, which one of the following represents the correct Venn diagram of the complexity classes P, NP and NP Complete (NPC)?

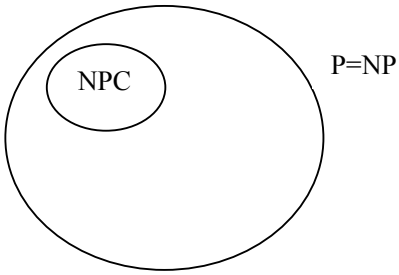
(A)



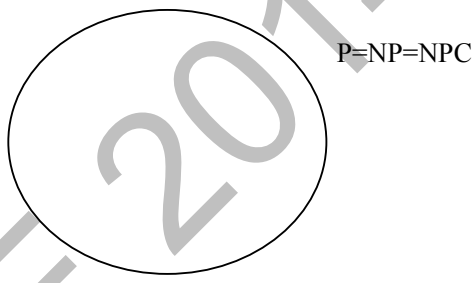
(B)



(C)



(D)



Q.39 The minimum number of comparisons required to find the minimum and the maximum of 100 numbers is _____.

Q.40 Consider a hash table with 9 slots. The hash function is $h(k) = k \bmod 9$. The collisions are resolved by chaining. The following 9 keys are inserted in the order: 5, 28, 19, 15, 20, 33, 12, 17, 10. The maximum, minimum, and average chain lengths in the hash table, respectively, are

(A) 3, 0, and 1

(B) 3, 3, and 3

(C) 4, 0, and 1

(D) 3, 0, and 2

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Q.41 Consider the following C function in which **size** is the number of elements in the array **E**:

```
int MyX(int *E, unsigned int size)
{
  int Y = 0;
  int Z;
  int i, j, k;

  for(i = 0; i < size; i++)
      Y = Y + E[i];

  for(i = 0; i < size; i++)
      for(j = i; j < size; j++)
      {
         Z = 0;
         for(k = i; k <= j; k++)
             Z = Z + E[k];
         if (Z > Y)
             Y = Z;
       }
  return Y;
}
```

The value returned by the function **MyX** is the

(A) maximum possible sum of elements in any sub-array of array **E**.

(B) maximum element in any sub-array of array **E**.

(C) sum of the maximum elements in all possible sub-arrays of array **E**.

(D) the sum of all the elements in the array **E**.

Q.42 Consider the following pseudo code. What is the total number of multiplications to be performed?

```
D = 2
for i = 1 to n do
    for j = i to n do
        for k =  j + 1 to n do
            D = D * 3
```

(A) Half of the product of the 3 consecutive integers.

(B) One-third of the product of the 3 consecutive integers.

(C) One-sixth of the product of the 3 consecutive integers.

(D) None of the above.

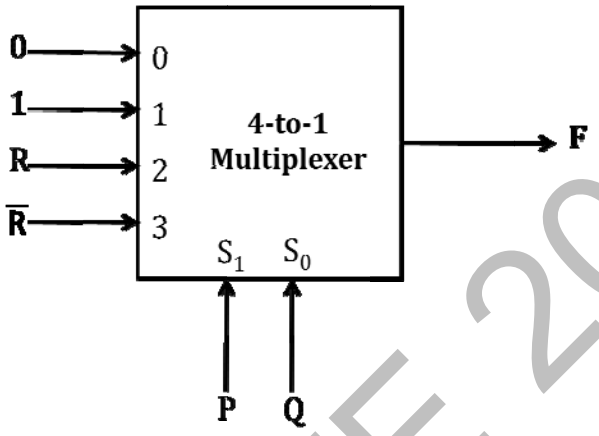
Q.43 Consider a 6-stage instruction pipeline, where all stages are perfectly balanced.Assume that there is no cycle-time overhead of pipelining. When an application is executing on this 6-stage pipeline, the speedup achieved with respect to non-pipelined execution if 25% of the instructions incur 2 pipeline stall cycles is _____.

CS01 (GATE 2014)

Q.44 An access sequence of cache block addresses is of length N and contains n unique block addresses. The number of unique block addresses between two consecutive accesses to the same block address is bounded above by k. What is the miss ratio if the access sequence is passed through a cache of associativity A ≥ k exercising least-recently-used replacement policy?

(A) *n/N* (B) 1/*N* (C) 1/*A* (D) *k/n*

Q.45 Consider the 4-to-1 multiplexer with two select lines S_1 and S_0 given below.



The minimal sum-of-products form of the Boolean expression for the output *F* of the multiplexer is

(A) $\bar{P}Q + Q\bar{R} + P\bar{Q}R$

(B) $\bar{P}Q + \bar{P}Q\bar{R} + PQ\bar{R} + P\bar{Q}R$

(C) $\bar{P}QR + \bar{P}Q\bar{R} + Q\bar{R} + P\bar{Q}R$

(D) $PQ\bar{R}$

Q.46 The function $f(x) = x \sin x$ satisfies the following equation: $f''(x) + f(x) + t\cos x = 0$. The value of t is _____.

Q.47 A function $f(x)$ is continuous in the interval [0,2]. It is known that $f(0) = f(2) = -1$ and $f(1) = 1$. Which one of the following statements must be true?

(A) There exists a y in the interval $(0,1)$ such that $f(y) = f(y+1)$

(B) For every y in the interval $(0,1)$, $f(y) = f(2-y)$

(C) The maximum value of the function in the interval $(0,2)$ is 1

(D) There exists a y in the interval $(0,1)$ such that $f(y) = -f(2-y)$

Q.48 Four fair six-sided dice are rolled. The probability that the sum of the results being 22 is $X/1296$. The value of X is _____.

CS01 (GATE 2014)

Q.49 A pennant is a sequence of numbers, each number being 1 or 2. An n-pennant is a sequence of numbers with sum equal to n. For example, (1,1,2) is a 4-pennant. The set of all possible 1-pennants is {(1)}, the set of all possible 2-pennants is {(2), (1,1)}and the set of all 3-pennants is {(2,1), (1,1,1), (1,2)}. Note that the pennant (1,2) is not the same as the pennant (2,1). The number of 10-pennants is _____.

Q.50 Let S denote the set of all functions $f: \{0,1\}^4 \rightarrow \{0,1\}$. Denote by N the number of functions from S to the set $\{0,1\}$. The value of $log_2\ log_2 N$ is _____.

Q.51 Consider an undirected graph G where self-loops are not allowed. The vertex set of G is $\{(i,j): 1 \leq i \leq 12, 1 \leq j \leq 12\}$. There is an edge between (a,b) and (c,d) if $|a-c| \leq 1$ and $|b-d| \leq 1$. The number of edges in this graph is _____.

Q.52 An ordered n-tuple $(d_1, d_2, \ldots, d_n)$ with $d_1 \geq d_2 \geq \cdots \geq d_n$ is called *graphic* if there exists a simple undirected graph with n vertices having degrees $d_1, d_2, \ldots, d_n$ respectively. Which of the following 6-tuples is NOT graphic?

(A) (1, 1, 1, 1, 1, 1)

(B) (2, 2, 2, 2, 2, 2)

(C) (3, 3, 3, 1, 0, 0)

(D) (3, 2, 1, 1, 1, 0)

Q.53 Which one of the following propositional logic formulas is TRUE when exactly two of p, q, and r are **TRUE**?

(A) $\big((p \leftrightarrow q) \wedge r\big) \vee (p \wedge q \wedge \sim r)$

(B) $(\sim (p \leftrightarrow q) \wedge r) \vee (p \wedge q \wedge \sim r)$

(C) $\big((p \rightarrow q) \wedge r\big) \vee (p \wedge q \wedge \sim r)$

(D) $(\sim (p \leftrightarrow q) \wedge r) \wedge (p \wedge q \wedge \sim r)$

CS01 (GATE 2014)

Q.54 Given the following schema:

```
employees(emp-id, first-name, last-name, hire-date,
          dept-id, salary)
departments(dept-id, dept-name, manager-id, location-id)
```

You want to display the last names and hire dates of all latest hires in their respective departments in the location ID 1700. You issue the following query:

```
SQL>SELECT last-name, hire-date
    FROM employees
    WHERE (dept-id, hire-date) IN
    (SELECT dept-id, MAX(hire-date)
    FROM employees JOIN departments USING(dept-id)
    WHERE location-id = 1700
    GROUP BY dept-id);
```

What is the outcome?

(A) It executes but does not give the correct result.

(B) It executes and gives the correct result.

(C) It generates an error because of pairwise comparison.

(D) It generates an error because the GROUP BY clause cannot be used with table joins in a sub-query.

Q.55 Consider two processors P_1 and P_2 executing the same instruction set. Assume that under identical conditions, for the same input, a program running on P_2 takes 25% less time but incurs 20% more CPI (clock cycles per instruction) as compared to the program running on P_1. If the clock frequency of P_1 is 1GHz, then the clock frequency of P_2 (in GHz) is _____.

END OF THE QUESTION PAPER

CS01 (GATE 2014)

GATE 2014

Answer Keys for CS - Computer Science and Information Technology

Section	Q. No.	SESSION - 1	
		Key / Range	Marks
GA	1	B	1
GA	2	C	1
GA	3	C	1
GA	4	96 to 96	1
GA	5	D	1
GA	6	C	2
GA	7	B	2
GA	8	850 to 850	2
GA	9	48 to 48	2
GA	10	6 to 6	2
CS	1	D	1
CS	2	0.24 to 0.27	1
CS	3	B	1
CS	4	1 to 1	1
CS	5	0 to 0	1
CS	6	C	1
CS	7	A	1
CS	8	5 to 5	1
CS	9	16383 to 16383	1
CS	10	D	1
CS	11	C	1
CS	12	1 to 1	1
CS	13	C	1
CS	14	C	1
CS	15	C	1
CS	16	A	1
CS	17	D	1
CS	18	B	1
CS	19	3 to 3	1
CS	20	D	1
CS	21	B	1
CS	22	D	1
CS	23	D	1

Section	Q. No.	SESSION - 1	
		Key / Range	Marks
CS	24	C	1
CS	25	A	1
CS	26	28 to 30	2
CS	27	1100 to 1300	2
CS	28	5 to 5	2
CS	29	D	2
CS	30	A	2
CS	31	B	2
CS	32	7.2 to 7.2	2
CS	33	7 to 7	2
CS	34	D	2
CS	35	C	2
CS	36	B	2
CS	37	12 to 12	2
CS	38	D	2
CS	39	147.1 to 148.1	2
CS	40	A	2
CS	41	A	2
CS	42	C	2
CS	43	4 to 4	2
CS	44	A	2
CS	45	A	2
CS	46	-2 to -2	2
CS	47	A	2
CS	48	10 to 10	2
CS	49	88.9 to 89.1	2
CS	50	16 to 16	2
CS	51	506 to 506	2
CS	52	C	2
CS	53	B	2
CS	54	B	2
CS	55	1.6 to 1.6	2

SESSION - 2

GATE 2014: General Instructions during Examination

1. Total duration of the GATE examination is **180** minutes.
2. The clock will be set at the server. The countdown timer at the top right corner of screen will display the remaining time available for you to complete the examination. When the timer reaches zero, the examination will end by itself. You need not terminate the examination or submit your paper.
3. Any useful data required for your paper can be viewed by clicking on the **Useful Common Data** button that appears on the screen.
4. Use the scribble pad provided to you for any rough work. Submit the scribble pad at the end of the examination.
5. You are allowed to use a non-programmable type calculator, however, sharing of calculators is not allowed.
6. The Question Palette displayed on the right side of screen will show the status of each question using one of the following symbols:

You have not visited the question yet.

You have not answered the question.

You have answered the question.

You have NOT answered the question, but have marked the question for review.

You have answered the question, but marked it for review.

The **Marked for Review** status for a question simply indicates that you would like to look at that question again. *If a question is answered, but marked for review, then the answer will be considered for evaluation unless the status is modified by the candidate.*

<u>Navigating to a Question :</u>

7. To answer a question, do the following:
    a. Click on the question number in the Question Palette to go to that question directly.
    b. Select an answer for a multiple choice type question by clicking on the bubble placed before the 4 choices, namely A, B, C and D. Use the virtual numeric keypad to enter a number as answer for a numerical type question.
    c. Click on **Save & Next** to save your answer for the current question and then go to the next question.
    d. Click on **Mark for Review & Next** to save your answer for the current question and also mark it for review, and then go to the next question.

Caution: Note that your answer for the current question will not be saved, if you navigate to another question directly by clicking on a question number without saving the answer to the previous question.

You can view all the questions by clicking on the **Question Paper** button. This feature is provided, so that if you want you can just see the entire question paper at a glance.

Answering a Question :

8. Procedure for answering a multiple choice (MCQ) type question:
    a. Choose one answer from the 4 options (A,B,C,D) given below the question, click on the bubble placed before the chosen option.
    b. To deselect your chosen answer, click on the bubble of the chosen option again or click on the **Clear Response** button.
    c. To change your chosen answer, click on the bubble of another option.
    d. To save your answer, you MUST click on the **Save & Next** button.

9. Procedure for answering a numerical answer type question:
    a. To enter a number as your answer, use the virtual numerical keypad.
    b. A fraction (e.g. -0.3 or -.3) can be entered as an answer with or without '0' before the decimal point. As many as four decimal points, e.g. 12.5435 or 0.003 or -932.6711 or 12.82 can be entered.
    c. To clear your answer, click on the **Clear Response** button.
    d. To save your answer, you MUST click on the **Save & Next** button
10. To mark a question for review, click on the **Mark for Review & Next** button. *If an answer is selected (for MCQ) or entered (for numerical answer type) for a question that is* **Marked for Review***, that answer will be considered in the evaluation unless the status is modified by the candidate.*

11. To change your answer to a question that has already been answered, first select that question for answering and then follow the procedure for answering that type of question.
12. Note that ONLY Questions for which answers are ***saved*** or ***marked for review after answering*** will be considered for evaluation.

Choosing a Section :

13. Sections in this question paper are displayed on the top bar of the screen. Questions in a Section can be viewed by clicking on the name of that Section. The Section you are currently viewing will be highlighted.
14. A checkbox is displayed for every optional Section, if any, in the Question Paper. To select the optional Section for answering, click on the checkbox for that Section.
15. If the checkbox for an optional Section is not selected, the **Save & Next** button and the **Mark for Review & Next** button will NOT be enabled for that Section. You will

only be able to see questions in this Section, but you will not be able to answer questions in the Section.

16. After clicking the **Save & Next** button for the last question in a Section, you will automatically be taken to the first question of the next Section in sequence.
17. You can move the mouse cursor over the name of a Section to view the answering status for that Section.

Changing the Optional Section :

18. After answering the chosen optional Section, partially or completely, you can change the optional Section by selecting the checkbox for a new Section that you want to attempt. A warning message will appear along with a table showing the number of questions answered in each of the previously chosen optional Sections and a checkbox against each of these Sections. Click on a checkbox against a Section that you want to reset and then click on the **RESET** button. Note that RESETTING a Section will DELETE all the answers for questions in that Section. Hence, if you think that you may want to select this Section again later, you will have to note down your answers for questions in that Section. If you do not want to reset the Section and want to continue answering the previously chosen optional Section, then click on the **BACK** button.

19. If you deselect the checkbox for an optional Section in the top bar, the following warning message will appear: "Deselecting the checkbox will DELETE all the answers for questions in this Section. Do you want to deselect this Section?" If you want to deselect, click on the **RESET** button. If you do not want to deselect, click on the **BACK** button.

20. You can shuffle between different Sections or change the optional Sections any number of times.

GATE 2014 Examination

CS: Computer Science & Information Technology

Duration: **180 minutes** *Maximum Marks*: **100**

Read the following instructions carefully.

1. To login, enter your Registration Number and password provided to you. Kindly go through the various symbols used in the test and understand their meaning before you start the examination.
2. Once you login and after the start of the examination, you can view all the questions in the question paper, by clicking on the **View All Questions** button in the screen.
3. This question paper consists of **2 sections**, General Aptitude (GA) for **15 marks** and the subject specific GATE paper for **85 marks**. Both these sections are compulsory.
   The GA section consists of **10** questions. Question numbers 1 to 5 are of 1-mark each, while question numbers 6 to 10 are of 2-mark each.
   The subject specific GATE paper section consists of **55** questions, out of which question numbers 1 to 25 are of 1-mark each, while question numbers 26 to 55 are of 2-mark each.
4. Depending upon the GATE paper, there may be useful common data that may be required for answering the questions. If the paper has such useful data, the same can be viewed by clicking on the **Useful Common Data** button that appears at the top, right hand side of the screen.
5. The computer allotted to you at the examination center runs specialized software that permits only one answer to be selected for multiple-choice questions using a mouse and to enter a suitable number for the numerical answer type questions using the virtual keyboard and mouse.
6. Your answers shall be updated and saved on a server periodically and also at the end of the examination. The examination will **stop automatically** at the end of **180 minutes**.
7. In each paper a candidate can answer a total of 65 questions carrying 100 marks.
8. The question paper may consist of questions of **multiple choice type** (MCQ) and **numerical answer type**.
9. Multiple choice type questions will have four choices against A, B, C, D, out of which only **ONE** is the correct answer. The candidate has to choose the correct answer by clicking on the bubble (○) placed before the choice.
10. For numerical answer type questions, each question will have a numerical answer and there will not be any choices**. For these questions, the answer should be entered**by using the virtual keyboard that appears on the monitor and the mouse.
11. All questions that are not attempted will result in zero marks. However, wrong answers for multiple choice type questions (MCQ) will result in **NEGATIVE** marks. For all MCQ questions a wrong answer will result in deduction of⅓ marks for a 1-mark question and ⅔ marks for a 2-mark question.
12. There is **NO NEGATIVE MARKING** for questions of **NUMERICAL ANSWER TYPE**.
13. Non-programmable type Calculator is allowed. Charts, graph sheets, and mathematical tables are **NOT** allowed in the Examination Hall. You must use the Scribble pad provided to you at the examination centre for all your rough work. The Scribble Pad has to be returned at the end of the examination.

Declaration by the candidate:

"I have read and understood all the above instructions. I have also read and understood clearly the instructions given on the admit card and shall follow the same. I also understand that in case I am found to violate any of these instructions, my candidature is liable to be cancelled. I also confirm that at the start of the examination all the computer hardware allotted to me are in proper working condition".

Q. 1 – Q. 5 carry one mark each.

Q.1 Choose the most appropriate phrase from the options given below to complete the following sentence.

India is a post-colonial country because

(A) it was a former British colony
(B) Indian Information Technology professionals have colonized the world
(C) India does not follow any colonial practices
(D) India has helped other countries gain freedom

Q.2 Who _____ was coming to see us this evening?

(A) you said (B) did you say
(C) did you say that (D) had you said

Q.3 Match the columns.

Column 1	Column 2
1) eradicate	P) misrepresent
2) distort	Q) soak completely
3) saturate	R) use
4) utilize	S) destroy utterly

(A) 1:S, 2:P, 3:Q, 4:R (B) 1:P, 2:Q, 3:R, 4:S
(C) 1:Q, 2:R, 3:S, 4:P (D) 1:S, 2:P, 3:R, 4:Q

Q.4 What is the average of all multiples of 10 from 2 to 198?

(A) 90 (B) 100 (C) 110 (D) 120

Q.5 The value of $\sqrt{12+\sqrt{12+\sqrt{12+\cdots}}}$ is

(A) 3.464 (B) 3.932 (C) 4.000 (D) 4.444

Q. 6 – Q. 10 carry two marks each.

Q.6 The old city of Koenigsberg, which had a German majority population before World War 2, is now called Kaliningrad. After the events of the war, Kaliningrad is now a Russian territory and has a predominantly Russian population. It is bordered by the Baltic Sea on the north and the countries of Poland to the south and west and Lithuania to the east respectively. Which of the statements below can be inferred from this passage?

(A) Kaliningrad was historically Russian in its ethnic make up
(B) Kaliningrad is a part of Russia despite it not being contiguous with the rest of Russia
(C) Koenigsberg was renamed Kaliningrad, as that was its original Russian name
(D) Poland and Lithuania are on the route from Kaliningrad to the rest of Russia

GA08 (GATE 2014)

Q.7 The number of people diagnosed with dengue fever (contracted from the bite of a mosquito) in north India is twice the number diagnosed last year. Municipal authorities have concluded that measures to control the mosquito population have failed in this region.

Which one of the following statements, if true, does not contradict this conclusion?

(A) A high proportion of the affected population has returned from neighbouring countries where dengue is prevalent
(B) More cases of dengue are now reported because of an increase in the Municipal Office's administrative efficiency
(C) Many more cases of dengue are being diagnosed this year since the introduction of a new and effective diagnostic test
(D) The number of people with malarial fever (also contracted from mosquito bites) has increased this year

Q.8 If x is real and $|x^2 - 2x + 3| = 11$, then possible values of $|-x^3 + x^2 - x|$ include

(A) 2, 4 (B) 2, 14 (C) 4, 52 (D) 14, 52

Q.9 The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students doubled in 2009, by what percent did the number of male students increase in 2009?

Q.10 At what time between 6 $a.m.$ and 7 $a.m.$ will the minute hand and hour hand of a clock make an angle closest to 60°?

(A) 6: 22 $a.m.$ (B) 6: 27 $a.m.$
(C) 6: 38 $a.m.$ (D) 6: 45 $a.m.$

END OF THE QUESTION PAPER

GA08 (GATE 2014)

Q. 1 – Q. 25 carry one mark each.

Q.1 The security system at an IT office is composed of 10 computers of which exactly four are working. To check whether the system is functional, the officials inspect four of the computers picked at random (without replacement). The system is deemed functional if at least three of the four computers inspected are working. Let the probability that the system is deemed functional be denoted by p. Then $100p =$ _____.

Q.2 Each of the nine words in the sentence **"The quick brown fox jumps over the lazy dog"** is written on a separate piece of paper. These nine pieces of paper are kept in a box. One of the pieces is drawn at random from the box. The *expected* length of the word drawn is _____. (The answer should be rounded to one decimal place.)

Q.3 The maximum number of edges in a bipartite graph on 12 vertices is _____.

Q.4 If the matrix A is such that

$$A = \begin{bmatrix} 2 \\ -4 \\ 7 \end{bmatrix} [1 \quad 9 \quad 5]$$

then the determinant of A is equal to _____.

Q.5 A non-zero polynomial $f(x)$ of degree 3 has roots at $x = 1, x = 2$ and $x = 3$. Which one of the following must be TRUE?

(A) $f(0)f(4) < 0$

(B) $f(0)f(4) > 0$

(C) $f(0) + f(4) > 0$

(D) $f(0) + f(4) < 0$

Q.6 The dual of a Boolean function $F(x_1, x_2, \ldots, x_n, +, \cdot, ')$, written as F^D, is the same expression as that of F with $+$ and $\cdot$ swapped. F is said to be self-dual if $F = F^D$. The number of self-dual functions with n Boolean variables is

(A) 2^n (B) 2^{n-1} (C) 2^{2^n} (D) $2^{2^{n-1}}$

CS02 (GATE 2014)

Q.7 Let $k = 2^n$. A circuit is built by giving the output of an n-bit binary counter as input to an n-to-2^n bit decoder. This circuit is equivalent to a

(A) k-bit binary up counter.

(B) k-bit binary down counter.

(C) k-bit ring counter.

(D) k-bit Johnson counter.

Q.8 Consider the equation $(123)_5 = (x8)_y$ with x and y as unknown. The number of possible solutions is _____ .

Q.9 A 4-way set-associative cache memory unit with a capacity of 16 KB is built using a block size of 8 words. The word length is 32 bits. The size of the physical address space is 4 GB. The number of bits for the TAG field is _____

Q.10 Consider the function `func` shown below:

```
int func(int num) {
int count = 0;
    while (num) {
       count++;
       num>>= 1;
    }
return (count);
}
```

The value returned by `func(435)` is _____.

Q.11 Suppose `n` and `p` are `unsigned int` variables in a C program. We wish to set `p` to ${}^{n}C_3$. If `n` is large, which one of the following statements is most likely to set `p` correctly?

```
(A)   p = n * (n-1) * (n-2) / 6;
(B)   p = n * (n-1) / 2 * (n-2) / 3;
(C)   p = n * (n-1) / 3 * (n-2) / 2;
(D)   p = n * (n-1) * (n-2) / 6.0;
```

CS02 (GATE 2014)

Q.12 A priority queue is implemented as a Max-Heap. Initially, it has 5 elements. The level-order traversal of the heap is: 10, 8, 5, 3, 2. Two new elements 1 and 7 are inserted into the heap in that order. The level-order traversal of the heap after the insertion of the elements is:

(A) 10, 8, 7, 3, 2, 1, 5
(B) 10, 8, 7, 2, 3, 1, 5
(C) 10, 8, 7, 1, 2, 3, 5
(D) 10, 8, 7, 5, 3, 2, 1

Q.13 Which one of the following correctly determines the solution of the recurrence relation with T(1) = 1?

$$T(n) = 2T\left(\frac{n}{2}\right) + \log n$$

(A) $\Theta(n)$ (B) $\Theta(n \log n)$ (C) $\Theta(n^2)$ (D) $\Theta(\log n)$

Q.14 Consider the tree arcs of a BFS traversal from a source node **W** in an unweighted, connected, undirected graph. The tree **T** formed by the tree arcs is a data structure for computing

(A) the shortest path between every pair of vertices.
(B) the shortest path from **W** to every vertex in the graph.
(C) the shortest paths from **W** to only those nodes that are leaves of **T**.
(D) the longest path in the graph.

Q.15 If $L_1 = \{a^n | n \geq 0\}$ and $L_2 = \{b^n | n \geq 0\}$, consider

(I) $L_1 \cdot L_2$ is a regular language
(II) $L_1 \cdot L_2 = \{a^n b^n | n \geq 0\}$

Which one of the following is CORRECT?

(A) Only (I)
(B) Only (II)
(C) Both (I) and (II)
(D) Neither (I) nor (II)

Q.16 Let $A \leq_m B$ denotes that language A is mapping reducible (also known as many-to-one reducible) to language B. Which one of the following is FALSE?

(A) If $A \leq_m B$ and B is recursive then A is recursive.
(B) If $A \leq_m B$ and A is undecidable then B is undecidable.
(C) If $A \leq_m B$ and B is recursively enumerable then A is recursively enumerable.
(D) If $A \leq_m B$ and B is not recursively enumerable then A is not recursively enumerable.

CS02 (GATE 2014)

Q.17 Consider the grammar defined by the following production rules, with two operators $*$ and $+$

$$S \longrightarrow T * P$$
$$T \longrightarrow U \mid T * U$$
$$P \longrightarrow Q + P \mid Q$$
$$Q \longrightarrow Id$$
$$U \longrightarrow Id$$

Which one of the following is TRUE?

(A) $+$ is left associative, while $*$ is right associative

(B) $+$ is right associative, while $*$ is left associative

(C) Both $+$ and $*$ are right associative

(D) Both $+$ and $*$ are left associative

Q.18 Which one of the following is **NOT** performed during compilation?

(A) Dynamic memory allocation (B) Type checking

(C) Symbol table management (D) Inline expansion

Q.19 Which one of the following is TRUE?

(A) The requirements document also describes how the requirements that are listed in the document are implemented efficiently.

(B) Consistency and completeness of functional requirements are always achieved in practice.

(C) Prototyping is a method of requirements validation.

(D) Requirements review is carried out to find the errors in system design.

Q.20 A FAT (file allocation table) based file system is being used and the total overhead of each entry in the FAT is 4 bytes in size. Given a 100×10^6 bytes disk on which the file system is stored and data block size is 10^3 bytes, the maximum size of a file that can be stored on this disk in units of 10^6 bytes is _____.

Q.21 The maximum number of superkeys for the relation schema `R(E,F,G,H)` with `E` as the key is _____.

CS02 (GATE 2014)

Q.22 Given an instance of the STUDENTS relation as shown below:

StudentID	***StudentName***	***StudentEmail***	***StudentAge***	***CPI***
2345	Shankar	shankar@math	X	9.4
1287	Swati	swati@ee	19	9.5
7853	Shankar	shankar@cse	19	9.4
9876	Swati	swati@mech	18	9.3
8765	Ganesh	ganesh@civil	19	8.7

For ***(StudentName, StudentAge)*** to be a key for this instance, the value X should NOT be equal to _____.

Q.23 Which one of the following is TRUE about the interior gateway routing protocols – Routing Information Protocol (RIP) and Open Shortest Path First (OSPF)?

(A) RIP uses distance vector routing and OSPF uses link state routing

(B) OSPF uses distance vector routing and RIP uses link state routing

(C) Both RIP and OSPF use link state routing

(D) Both RIP and OSPF use distance vector routing

Q.24 Which one of the following socket API functions converts an unconnected active TCP socket into a passive socket?

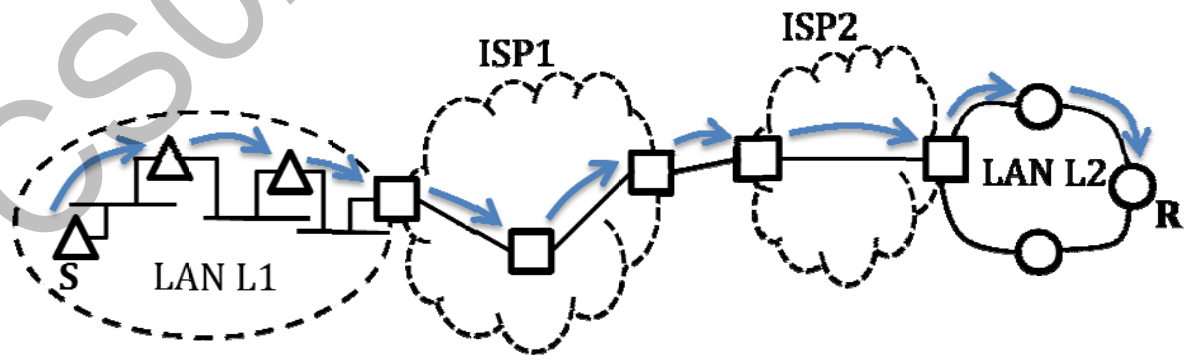
(A) connect

(B) bind

(C) listen

(D) accept

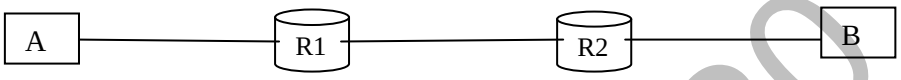
Q.25 In the diagram shown below, L1 is an Ethernet LAN and L2 is a Token-Ring LAN. An IP packet originates from sender S and traverses to R, as shown. The links within each ISP and across the two ISPs, are all point-to-point optical links. The initial value of the TTL field is 32. The maximum possible value of the TTL field when R receives the datagram is _____.



CS02 (GATE 2014)

Q. 26 – Q. 55 carry two marks each.

Q.26 Consider the store and forward packet switched network given below. Assume that the bandwidth of each link is 10^6 bytes / sec. A user on host A sends a file of size 10^3 bytes to host B through routers R1 and R2 in three different ways. In the first case a single packet containing the complete file is transmitted from A to B. In the second case, the file is split into 10 equal parts, and these packets are transmitted from A to B. In the third case, the file is split into 20 equal parts and these packets are sent from A to B. Each packet contains 100 bytes of header information along with the user data. Consider only transmission time and ignore processing, queuing and propagation delays. Also assume that there are no errors during transmission. Let T1, T2 and T3 be the times taken to transmit the file in the first, second and third case respectively. Which one of the following is CORRECT?



(A) T1 < T2 < T3

(B) T1 > T2 > T3

(C) T2 = T3, T3 < T1

(D) T1 = T3, T3 > T2

Q.27 An IP machine Q has a path to another IP machine H via three IP routers R1, R2, and R3.

Q—R1—R2—R3—H

H acts as an HTTP server, and Q connects to H via HTTP and downloads a file. Session layer encryption is used, with DES as the shared key encryption protocol. Consider the following four pieces of information:

[I1] The URL of the file downloaded by Q
[I2] The TCP port numbers at Q and H
[I3] The IP addresses of Q and H
[I4] The link layer addresses of Q and H

Which of I1, I2, I3, and I4 can an intruder learn through sniffing at R2 alone?

(A) Only I1 and I2
(B) Only I1
(C) Only I2 and I3
(D) Only I3 and I4

CS02 (GATE 2014)

Q.28 A graphical HTML browser resident at a network client machine *Q* accesses a static HTML webpage from a HTTP server *S*. The static HTML page has exactly one static embedded image which is also at *S*. Assuming no caching, which one of the following is correct about the HTML webpage loading (including the embedded image)?

(A) *Q* needs to send at least 2 HTTP requests to *S*, each necessarily in a separate TCP connection to server *S*

(B) *Q* needs to send at least 2 HTTP requests to *S*, but a single TCP connection to server *S* is sufficient

(C) A single HTTP request from *Q* to *S* is sufficient, and a single TCP connection between *Q* and *S* is necessary for this

(D) A single HTTP request from *Q* to *S* is sufficient, and this is possible without any TCP connection between *Q* and *S*

Q.29 Consider the following schedule **S** of transactions T1, T2, T3, T4:

T1	**T2**	**T3**	**T4**
	Reads(X)		
		Writes(X) Commit	
Writes(X) Commit			
	Writes(Y) Reads(Z) Commit		
			Reads(X) Reads(Y) Commit

Which one of the following statements is CORRECT?

(A) **S** is conflict-serializable but not recoverable

(B) **S** is not conflict-serializable but is recoverable

(C) **S** is both conflict-serializable and recoverable

(D) **S** is neither conflict-serializable nor is it recoverable

CS02 (GATE 2014)

Q.30 Consider a join (relation algebra) between relations `r(R)` and `s(S)` using the nested loop method. There are 3 buffers each of size equal to disk block size, out of which one buffer is reserved for intermediate results. Assuming `size(r(R))<size(s(S))`, the join will have fewer number of disk block accesses if

(A) relation `r(R)` is in the outer loop.

(B) relation `s(S)` is in the outer loop.

(C) join selection factor between `r(R)` and `s(S)` is more than 0.5.

(D) join selection factor between `r(R)` and `s(S)` is less than 0.5.

Q.31 Consider the procedure below for the *Producer-Consumer* problem which uses semaphores:

```
semaphore n = 0;
semaphore s = 1;
void producer()
{
      while(true)
      {
       produce();
       semWait(s);
       addToBuffer();
       semSignal(s);
       semSignal(n);
      }
}
```

```
void consumer()
{
      while(true)
      {
       semWait(s);
       semWait(n);
       removeFromBuffer();
       semSignal(s);
       consume();
      }
}
```

Which one of the following is **TRUE**?

(A) The producer will be able to add an item to the buffer, but the consumer can never consume it.

(B) The consumer will remove no more than one item from the buffer.

(C) Deadlock occurs if the consumer succeeds in acquiring semaphore **s** when the buffer is empty.

(D) The starting value for the semaphore **n** must be 1 and not 0 for deadlock-free operation.

CS02 (GATE 2014)

Q.32 Three processes A, B and C each execute a loop of 100 iterations. In each iteration of the loop, a process performs a single computation that requires $\mathbf{t_c}$ CPU milliseconds and then initiates a single I/O operation that lasts for $\mathbf{t_{io}}$ milliseconds. It is assumed that the computer where the processes execute has sufficient number of I/O devices and the OS of the computer assigns different I/O devices to each process. Also, the scheduling overhead of the OS is negligible. The processes have the following characteristics:

Process id	t_c	t_{io}
A	100 ms	500 ms
B	350 ms	500 ms
C	200 ms	500 ms

The processes A, B, and C are started at times 0, 5 and 10 milliseconds respectively, in a pure time sharing system (round robin scheduling) that uses a time slice of 50 milliseconds. The time in milliseconds at which process C would ***complete*** its first I/O operation is _____.

Q.33 A computer has twenty physical page frames which contain pages numbered 101 through 120. Now a program accesses the pages numbered 1, 2, …, 100 in that order, and repeats the access sequence THRICE. Which one of the following page replacement policies experiences the same number of page faults as the optimal page replacement policy for this program?

(A) Least-recently-used (B) First-in-first-out

(C) Last-in-first-out (D) Most-recently-used

Q.34 For a C program accessing **`X[i][j][k]`**, the following intermediate code is generated by a compiler. Assume that the size of an **`integer`** is 32 bits and the size of a **`character`** is 8 bits.

```
t0 = i * 1024
t1 = j * 32
t2 = k * 4
t3 = t1 + t0
t4 = t3 + t2
t5 = X[t4]
```

Which one of the following statements about the source code for the C program is **CORRECT**?

(A) **`X`** is declared as "**`int X[32][32][8]`**".

(B) **`X`** is declared as "**`int X[4][1024][32]`**".

(C) **`X`** is declared as "**`char X[4][32][8]`**".

(D) **`X`** is declared as "**`char X[32][16][2]`**".

CS02 (GATE-2014)

Q.35 Let $< M >$ be the encoding of a Turing machine as a string over $\Sigma = \{0, 1\}$. Let $L = \{ < M > \mid M \text{ is a Turing machine that accepts a string of length } 2014 \}$. Then, L is

(A) decidable and recursively enumerable

(B) undecidable but recursively enumerable

(C) undecidable and not recursively enumerable

(D) decidable but not recursively enumerable

Q.36 Let $L_1 = \{w \in \{0,1\}^* \mid w$ has at least as many occurrences of (110)'s as (011)'s$\}$. Let $L_2 = \{w \in \{0,1\}^* \mid w$ has at least as many occurrences of (000)'s as (111)'s$\}$. Which one of the following is TRUE?

(A) L_1 is regular but not L_2

(B) L_2 is regular but not L_1

(C) Both L_1 and L_2 are regular

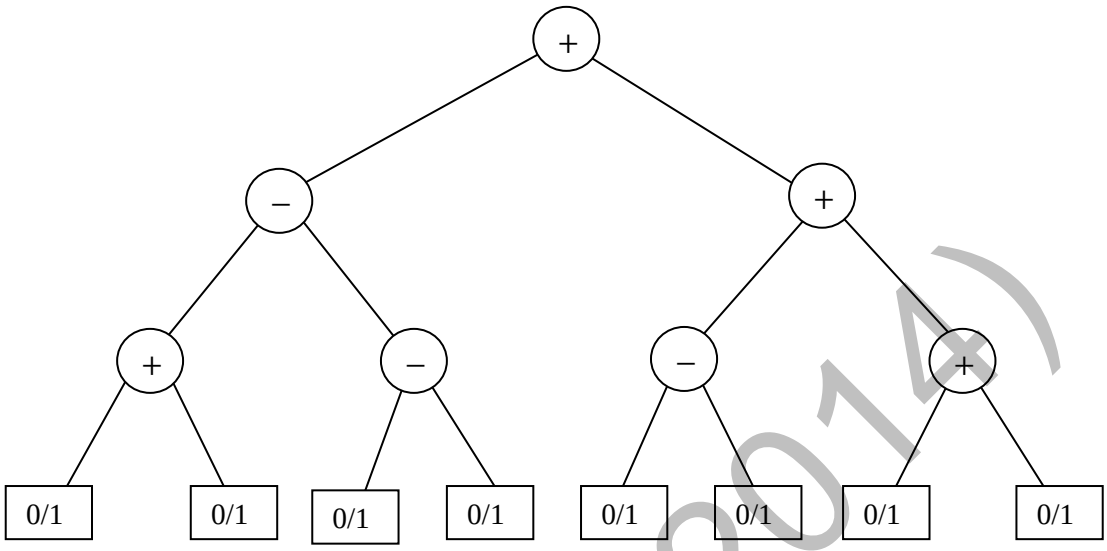
(D) Neither L_1 nor L_2 are regular

Q.37 Consider two strings $A = "qpqrr"$ and $B = "pqprqrp"$. Let x be the length of the longest common subsequence (not necessarily contiguous) between A and B and let y be the number of such longest common subsequences between A and B. Then $x + 10y$ = ___.

Q.38 Suppose P, Q, R, S, T are sorted sequences having lengths $20, 24, 30, 35, 50$ respectively. They are to be merged into a single sequence by merging together two sequences at a time. The number of comparisons that will be needed in the worst case by the optimal algorithm for doing this is ____.

CS02 (GATE 2014)

Q.39 Consider the expression tree shown. Each leaf represents a numerical value, which can either be 0 or 1. Over all possible choices of the values at the leaves, the maximum possible value of the expression represented by the tree is ___.



Q.40 Consider the following function

```
double f(double x){
   if( abs(x*x - 3) < 0.01) return x;
   else return f(x/2 + 1.5/x);
}
```

Give a value `q` (to 2 decimals) such that `f(q)` will return `q`:_____.

Q.41 Suppose a stack implementation supports an instruction REVERSE, which reverses the order of elements on the stack, in addition to the PUSH and POP instructions. Which one of the following statements is TRUE *with respect to this modified stack*?

(A) A queue cannot be implemented using this stack.

(B) A queue can be implemented where ENQUEUE takes a single instruction and DEQUEUE takes a sequence of two instructions.

(C) A queue can be implemented where ENQUEUE takes a sequence of three instructions and DEQUEUE takes a single instruction.

(D) A queue can be implemented where both ENQUEUE and DEQUEUE take a single instruction each.

CS02 (GATE 2014)

Q.42 Consider the C function given below.

```
int f(int j)
{
  static int i = 50;
  int k;
  if (i == j)
     {
       printf("something");
       k = f(i);
       return 0;
     }
  else return 0;
}
```

Which one of the following is **TRUE**?

(A) The function returns `0` for all values of `j`.

(B) The function prints the string `something` for all values of `j`.

(C) The function returns `0` when `j = 50`.

(D) The function will exhaust the runtime stack or run into an infinite loop when `j = 50`.

Q.43 In designing a computer's cache system, the cache block (or cache line) size is an important parameter. Which one of the following statements is correct in this context?

(A) A smaller block size implies better spatial locality

(B) A smaller block size implies a smaller cache tag and hence lower cache tag overhead

(C) A smaller block size implies a larger cache tag and hence lower cache hit time

(D) A smaller block size incurs a lower cache miss penalty

Q.44 If the associativity of a processor cache is doubled while keeping the capacity and block size unchanged, which one of the following is guaranteed to be NOT affected?

(A) Width of tag comparator

(B) Width of set index decoder

(C) Width of way selection multiplexor

(D) Width of processor to main memory data bus

Q.45 The value of a *float* type variable is represented using the single-precision 32-bit floating point format of IEEE-754 standard that uses 1 bit for sign, 8 bits for biased exponent and 23 bits for mantissa. A *float* type variable X is assigned the decimal value of -14.25. The representation of X in hexadecimal notation is

(A) C1640000H (B) 416C0000H (C) 41640000H (D) C16C0000H

CS02 (GATE 2014)

Q.46 In the Newton-Raphson method, an initial guess of $x_0 = 2$ is made and the sequence x_0, x_1, x_2 ... is obtained for the function

$$0.75x^3 - 2x^2 - 2x + 4 = 0$$

Consider the statements

(I) $x_3 = 0$.

(II) The method converges to a solution in a finite number of iterations.

Which of the following is TRUE?

(A) Only I (B) Only II (C) Both I and II (D) Neither I nor II

Q.47 The product of the non-zero eigenvalues of the matrix

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

is _____.

Q.48 The probability that a given positive integer lying between 1 and 100 (both inclusive) is NOT divisible by 2, 3 or 5 is _____ .

Q.49 The number of distinct positive integral factors of 2014 is _____ _____

Q.50 Consider the following relation on subsets of the set S of integers between 1 and 2014. For two distinct subsets U and V of S we say $U < V$ if the minimum element in the symmetric difference of the two sets is in U.

Consider the following two statements:

$S1$: There is a subset of S that is larger than every other subset.
$S2$: There is a subset of S that is smaller than every other subset.

Which one of the following is CORRECT?

(A) Both $S1$ and $S2$ are true

(B) $S1$ is true and $S2$ is false

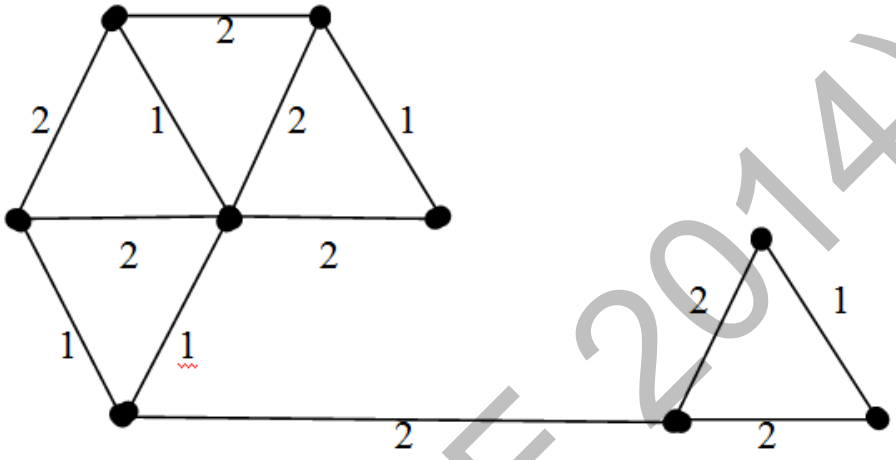
(C) $S2$ is true and $S1$ is false

(D) Neither $S1$ nor $S2$ is true

CS02 (GATE 2014)

Q.51 A cycle on n vertices is isomorphic to its complement. The value of n is _____.

Q.52 The number of distinct minimum spanning trees for the weighted graph below is _____



Q.53 Which one of the following Boolean expressions is NOT a tautology?

(A) $((a \longrightarrow b) \wedge (b \longrightarrow c)) \longrightarrow (a \longrightarrow c)$

(B) $(a \leftrightarrow c) \longrightarrow (\backsim b \longrightarrow (a \wedge c))$

(C) $(a \wedge b \wedge c) \longrightarrow (c \vee a)$

(D) $a \longrightarrow (b \longrightarrow a)$

Q.54 SQL allows duplicate tuples in relations, and correspondingly defines the multiplicity of tuples in the result of joins. Which one of the following queries always gives the same answer as the nested query shown below:

```
select * from R where a in (select S.a from S)
```

(A)
```
select R.* from R, S where R.a=S.a
```

(B)
```
select distinct R.* from R,S where R.a=S.a
```

(C)
```
select R.* from R,(select distinct a from S) as S1 where
    R.a=S1.a
```

(D)
```
select R.* from R,S where R.a=S.a and is unique R
```

CS02 (GATE 2014)

Q.55 Consider a main memory system that consists of 8 memory modules attached to the system bus, which is one word wide. When a write request is made, the bus is occupied for 100 nanoseconds (ns) by the data, address, and control signals. During the same 100 ns, and for 500 ns thereafter, the addressed memory module executes one cycle accepting and storing the data. The (internal) operation of different memory modules may overlap in time, but only one request can be on the bus at any time. The maximum number of stores (of one word each) that can be initiated in 1 millisecond is _____

END OF THE QUESTION PAPER

CS02 (GATE 2014)

GATE 2014

Answer Keys for CS - Computer Science and Information Technology

Section	Q. No.	SESSION - 2	
		Key / Range	Marks
GA	1	A	1
GA	2	B	1
GA	3	A	1
GA	4	B	1
GA	5	C	1
GA	6	B	2
GA	7	D	2
GA	8	D	2
GA	9	140 to 140	2
GA	10	A	2
CS	1	11.85 to 11.95	1
CS	2	3.8 to 3.9	1
CS	3	36 to 36	1
CS	4	0 to 0	1
CS	5	A	1
CS	6	D	1
CS	7	C	1
CS	8	3 to 3	1
CS	9	20 to 20	1
CS	10	9 to 9	1
CS	11	B	1
CS	12	A	1
CS	13	A	1
CS	14	B	1
CS	15	A	1
CS	16	D	1
CS	17	B	1
CS	18	A	1
CS	19	C	1
CS	20	99.55 to 99.65	1
CS	21	8 to 8	1
CS	22	19 to 19	1
CS	23	A	1

Section	Q. No.	SESSION - 2	
		Key / Range	Marks
CS	24	C	1
CS	25	26 to 26	1
CS	26	D	2
CS	27	C	2
CS	28	B	2
CS	29	C	2
CS	30	A	2
CS	31	C	2
CS	32	1000 to 1000	2
CS	33	D	2
CS	34	A	2
CS	35	B	2
CS	36	A	2
CS	37	34 to 34	2
CS	38	358 to 358	2
CS	39	6 to 6	2
CS	40	1.72 to 1.74	2
CS	41	C	2
CS	42	D	2
CS	43	D	2
CS	44	D	2
CS	45	A	2
CS	46	A	2
CS	47	6 to 6	2
CS	48	0.259 to 0.261	2
CS	49	8 to 8	2
CS	50	A	2
CS	51	5 to 5	2
CS	52	6 to 6	2
CS	53	B	2
CS	54	C	2
CS	55	10000 to 10000	2

SESSION - 3

GATE 2014: General Instructions during Examination

1. Total duration of the GATE examination is **180** minutes.
2. The clock will be set at the server. The countdown timer at the top right corner of screen will display the remaining time available for you to complete the examination. When the timer reaches zero, the examination will end by itself. You need not terminate the examination or submit your paper.
3. Any useful data required for your paper can be viewed by clicking on the **Useful Common Data** button that appears on the screen.
4. Use the scribble pad provided to you for any rough work. Submit the scribble pad at the end of the examination.
5. You are allowed to use a non-programmable type calculator, however, sharing of calculators is not allowed.
6. The Question Palette displayed on the right side of screen will show the status of each question using one of the following symbols:

You have not visited the question yet.

You have not answered the question.

You have answered the question.

You have NOT answered the question, but have marked the question for review.

You have answered the question, but marked it for review.

The **Marked for Review** status for a question simply indicates that you would like to look at that question again. *If a question is answered, but marked for review, then the answer will be considered for evaluation unless the status is modified by the candidate.*

<u>Navigating to a Question :</u>

7. To answer a question, do the following:
    a. Click on the question number in the Question Palette to go to that question directly.
    b. Select an answer for a multiple choice type question by clicking on the bubble placed before the 4 choices, namely A, B, C and D. Use the virtual numeric keypad to enter a number as answer for a numerical type question.
    c. Click on **Save & Next** to save your answer for the current question and then go to the next question.
    d. Click on **Mark for Review & Next** to save your answer for the current question and also mark it for review, and then go to the next question.

Caution: Note that your answer for the current question will not be saved, if you navigate to another question directly by clicking on a question number without saving the answer to the previous question.

You can view all the questions by clicking on the **Question Paper** button. This feature is provided, so that if you want you can just see the entire question paper at a glance.

Answering a Question :

8. Procedure for answering a multiple choice (MCQ) type question:
    a. Choose one answer from the 4 options (A,B,C,D) given below the question, click on the bubble placed before the chosen option.
    b. To deselect your chosen answer, click on the bubble of the chosen option again or click on the **Clear Response** button.
    c. To change your chosen answer, click on the bubble of another option.
    d. To save your answer, you MUST click on the **Save & Next** button.

9. Procedure for answering a numerical answer type question:
    a. To enter a number as your answer, use the virtual numerical keypad.
    b. A fraction (e.g. -0.3 or -.3) can be entered as an answer with or without '0' before the decimal point. As many as four decimal points, e.g. 12.5435 or 0.003 or -932.6711 or 12.82 can be entered.
    c. To clear your answer, click on the **Clear Response** button.
    d. To save your answer, you MUST click on the **Save & Next** button
10. To mark a question for review, click on the **Mark for Review & Next** button. *If an answer is selected (for MCQ) or entered (for numerical answer type) for a question that is* **Marked for Review***, that answer will be considered in the evaluation unless the status is modified by the candidate.*

11. To change your answer to a question that has already been answered, first select that question for answering and then follow the procedure for answering that type of question.
12. Note that ONLY Questions for which answers are ***saved*** or ***marked for review after answering*** will be considered for evaluation.

Choosing a Section :

13. Sections in this question paper are displayed on the top bar of the screen. Questions in a Section can be viewed by clicking on the name of that Section. The Section you are currently viewing will be highlighted.
14. A checkbox is displayed for every optional Section, if any, in the Question Paper. To select the optional Section for answering, click on the checkbox for that Section.
15. If the checkbox for an optional Section is not selected, the **Save & Next** button and the **Mark for Review & Next** button will NOT be enabled for that Section. You will

only be able to see questions in this Section, but you will not be able to answer questions in the Section.

16. After clicking the **Save & Next** button for the last question in a Section, you will automatically be taken to the first question of the next Section in sequence.
17. You can move the mouse cursor over the name of a Section to view the answering status for that Section.

Changing the Optional Section :

18. After answering the chosen optional Section, partially or completely, you can change the optional Section by selecting the checkbox for a new Section that you want to attempt. A warning message will appear along with a table showing the number of questions answered in each of the previously chosen optional Sections and a checkbox against each of these Sections. Click on a checkbox against a Section that you want to reset and then click on the **RESET** button. Note that RESETTING a Section will DELETE all the answers for questions in that Section. Hence, if you think that you may want to select this Section again later, you will have to note down your answers for questions in that Section. If you do not want to reset the Section and want to continue answering the previously chosen optional Section, then click on the **BACK** button.

19. If you deselect the checkbox for an optional Section in the top bar, the following warning message will appear: "Deselecting the checkbox will DELETE all the answers for questions in this Section. Do you want to deselect this Section?" If you want to deselect, click on the **RESET** button. If you do not want to deselect, click on the **BACK** button.

20. You can shuffle between different Sections or change the optional Sections any number of times.

GATE 2014 Examination

CS: Computer Science & Information Technology

Duration: **180 minutes** *Maximum Marks*: **100**

Read the following instructions carefully.

1. To login, enter your Registration Number and password provided to you. Kindly go through the various symbols used in the test and understand their meaning before you start the examination.
2. Once you login and after the start of the examination, you can view all the questions in the question paper, by clicking on the **View All Questions** button in the screen.
3. This question paper consists of **2 sections**, General Aptitude (GA) for **15 marks** and the subject specific GATE paper for **85 marks**. Both these sections are compulsory.
   The GA section consists of **10** questions. Question numbers 1 to 5 are of 1-mark each, while question numbers 6 to 10 are of 2-mark each.
   The subject specific GATE paper section consists of **55** questions, out of which question numbers 1 to 25 are of 1-mark each, while question numbers 26 to 55 are of 2-mark each.
4. Depending upon the GATE paper, there may be useful common data that may be required for answering the questions. If the paper has such useful data, the same can be viewed by clicking on the **Useful Common Data** button that appears at the top, right hand side of the screen.
5. The computer allotted to you at the examination center runs specialized software that permits only one answer to be selected for multiple-choice questions using a mouse and to enter a suitable number for the numerical answer type questions using the virtual keyboard and mouse.
6. Your answers shall be updated and saved on a server periodically and also at the end of the examination. The examination will **stop automatically** at the end of **180 minutes**.
7. In each paper a candidate can answer a total of 65 questions carrying 100 marks.
8. The question paper may consist of questions of **multiple choice type** (MCQ) and **numerical answer type**.
9. Multiple choice type questions will have four choices against A, B, C, D, out of which only **ONE** is the correct answer. The candidate has to choose the correct answer by clicking on the bubble (○) placed before the choice.
10. For numerical answer type questions, each question will have a numerical answer and there will not be any choices**. For these questions, the answer should be entered**by using the virtual keyboard that appears on the monitor and the mouse.
11. All questions that are not attempted will result in zero marks. However, wrong answers for multiple choice type questions (MCQ) will result in **NEGATIVE** marks. For all MCQ questions a wrong answer will result in deduction of⅓ marks for a 1-mark question and ⅔ marks for a 2-mark question.
12. There is **NO NEGATIVE MARKING** for questions of **NUMERICAL ANSWER TYPE**.
13. Non-programmable type Calculator is allowed. Charts, graph sheets, and mathematical tables are **NOT** allowed in the Examination Hall. You must use the Scribble pad provided to you at the examination centre for all your rough work. The Scribble Pad has to be returned at the end of the examination.

Declaration by the candidate:

"I have read and understood all the above instructions. I have also read and understood clearly the instructions given on the admit card and shall follow the same. I also understand that in case I am found to violate any of these instructions, my candidature is liable to be cancelled. I also confirm that at the start of the examination all the computer hardware allotted to me are in proper working condition".

Q. 1 – Q. 5 carry one mark each.

Q.1 While trying to collect (I) an envelope from under the table (II), Mr. X fell down (III) and was losing consciousness (IV).

Which one of the above underlined parts of the sentence is NOT appropriate?

(A) I (B) II (C) III (D) IV

Q.2 If she _____ how to calibrate the instrument, she _____ done the experiment.

(A) knows, will have (B) knew, had
(C) had known, could have (D) should have known, would have

Q.3 Choose the word that is opposite in meaning to the word "coherent".

(A) sticky (B) well-connected (C) rambling (D) friendly

Q.4 Which number does not belong in the series below?
2, 5, 10, 17, 26, 37, 50, 64

(A) 17 (B) 37 (C) 64 (D) 26

Q.5 The table below has question-wise data on the performance of students in an examination. The marks for each question are also listed. There is no negative or partial marking in the examination.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

What is the average of the marks obtained by the class in the examination?

(A) 1.34 (B) 1.74 (C) 3.02 (D) 3.91

Q. 6 – Q. 10 carry two marks each.

Q.6 A dance programme is scheduled for 10.00 a.m. Some students are participating in the programme and they need to come an hour earlier than the start of the event. These students should be accompanied by a parent. Other students and parents should come in time for the programme. The instruction you think that is appropriate for this is

(A) Students should come at 9.00 a.m. and parents should come at 10.00 a.m.
(B) Participating students should come at 9.00 a.m. accompanied by a parent, and other parents and students should come by 10.00 a.m.
(C) Students who are not participating should come by 10.00 a.m. and they should not bring their parents. Participating students should come at 9.00 a.m.
(D) Participating students should come before 9.00 a.m. Parents who accompany them should come at 9.00 a.m. All others should come at 10.00 a.m.

GA09 (GATE 2014)

Q.7 By the beginning of the 20th century, several hypotheses were being proposed, suggesting a paradigm shift in our understanding of the universe. However, the clinching evidence was provided by experimental measurements of the position of a star which was directly behind our sun.

Which of the following inference(s) may be drawn from the above passage?

(i) Our understanding of the universe changes based on the positions of stars
(ii) Paradigm shifts usually occur at the beginning of centuries
(iii) Stars are important objects in the universe
(iv) Experimental evidence was important in confirming this paradigm shift

(A) (i), (ii) and (iv) (B) (iii) only (C) (i) and (iv) (D) (iv) only

Q.8 The Gross Domestic Product (GDP) in Rupees grew at 7% during 2012-2013. For international comparison, the GDP is compared in US Dollars (USD) after conversion based on the market exchange rate. During the period 2012-2013 the exchange rate for the USD increased from Rs. 50/ USD to Rs. 60/ USD. India's GDP in USD during the period 2012-2013

(A) increased by 5 %
(B) decreased by 13%
(C) decreased by 20%
(D) decreased by 11%

Q.9 The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students in 2011 and 2012 is equal, what is the ratio of male students in 2012 to male students in 2011?

(A) 1:1 (B) 2:1 (C) 1.5:1 (D) 2.5:1

Q.10 Consider the equation: $(7526)_8 - (Y)_8 = (4364)_8$, where $(X)_N$ stands for X to the base N. Find Y.

(A) 1634 (B) 1737 (C) 3142 (D) 3162

END OF THE QUESTION PAPER

GA09 (GATE 2014)

Q. 1 – Q. 25 carry one mark each.

Q.1 Consider the following statements:

P: Good mobile phones are not cheap
Q: Cheap mobile phones are not good

L: P implies Q
M: Q implies P
N: P is equivalent to Q

Which one of the following about L, M, and N is **CORRECT**?

(A) Only L is TRUE.

(B) Only M is TRUE.

(C) Only N is TRUE.

(D) L, M and N are TRUE.

Q.2 Let X and Y be finite sets and $f: X \to Y$ be a function. Which one of the following statements is TRUE?

(A) For any subsets A and B of X, $|f(A \cup B)| = |f(A)| + |f(B)|$

(B) For any subsets A and B of X, $f(A \cap B) = f(A) \cap f(B)$

(C) For any subsets A and B of X, $|f(A \cap B)| = \min\{|f(A)|, |f(B)|\}$

(D) For any subsets S and T of Y, $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$

Q.3 Let G be a group with 15 elements. Let L be a subgroup of G. It is known that $L \neq G$ and that the size of L is at least 4. The size of L is _____.

Q.4 Which one of the following statements is TRUE about every $n \times n$ matrix with only real eigenvalues?

(A) If the trace of the matrix is positive and the determinant of the matrix is negative, at least one of its eigenvalues is negative.

(B) If the trace of the matrix is positive, all its eigenvalues are positive.

(C) If the determinant of the matrix is positive, all its eigenvalues are positive.

(D) If the product of the trace and determinant of the matrix is positive, all its eigenvalues are positive.

CS03 (GATE 2014)

Q.5 If V_1 and V_2 are 4-dimensional subspaces of a 6-dimensional vector space V, then the smallest possible dimension of $V_1 \cap V_2$ is _____.

Q.6 If $\int_0^{2\pi} |x \sin x| \, dx = k\pi$, then the value of k is equal to _____ .

Q.7 Consider the following minterm expression for F:

$$F(P, Q, R, S) = \sum 0, 2, 5, 7, 8, 10, 13, 15$$

The minterms 2, 7, 8 and 13 are 'do not care' terms. The minimal sum-of-products form for F is

(A) $Q\bar{S} + \bar{Q}S$

(B) $\bar{Q}\bar{S} + QS$

(C) $\bar{Q}\bar{R}\bar{S} + \bar{Q}R\bar{S} + Q\bar{R}S + QRS$

(D) $\bar{P}\bar{Q}\bar{S} + \bar{P}QS + PQS + P\bar{Q}\bar{S}$

Q.8 Consider the following combinational function block involving four Boolean variables `x, y, a, b` where `x, a, b` are inputs and `y` is the output.

```
f (x, y, a, b)
{
      if (x is 1) y = a;
      else y = b;
}
```

Which one of the following digital logic blocks is the most suitable for implementing this function?

(A) Full adder (B) Priority encoder (C) Multiplexor (D) Flip-flop

Q.9 Consider the following processors (ns stands for nanoseconds). Assume that the pipeline registers have zero latency.

P1: Four-stage pipeline with stage latencies 1 ns, 2 ns, 2 ns, 1 ns.
P2: Four-stage pipeline with stage latencies 1 ns, 1.5 ns, 1.5 ns, 1.5 ns.
P3: Five-stage pipeline with stage latencies 0.5 ns, 1 ns, 1 ns, 0.6 ns, 1 ns.
P4: Five-stage pipeline with stage latencies 0.5 ns, 0.5 ns, 1 ns, 1 ns, 1.1 ns.

Which processor has the highest peak clock frequency?

(A) P1 (B) P2 (C) P3 (D) P4

CS03 (GATE 2014)

Q.10 Let A be a square matrix of size $n \times n$. Consider the following pseudocode. What is the expected output?

```
C = 100;
for i = 1 to n do
  for j = 1 to n do
    {
     Temp = A[i][j] + C;
     A[i][j]= A[j][i];
     A[j][i] = Temp – C;
    }
for i = 1 to n do
  for j = 1 to n do
     output (A[i][j]);
```

(A) The matrix A itself

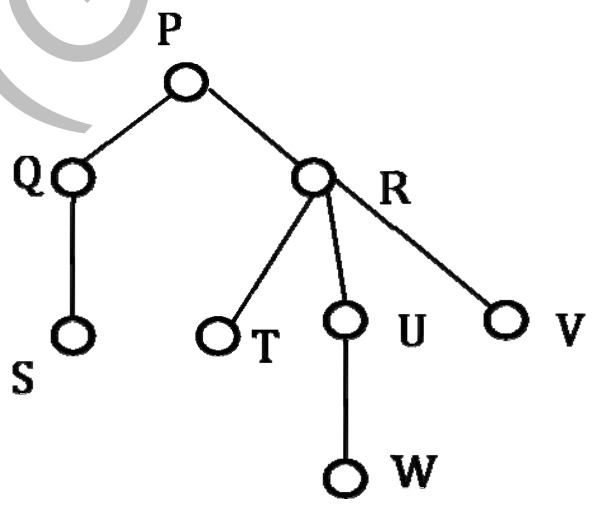
(B) Transpose of the matrix A

(C) Adding 100 to the upper diagonal elements and subtracting 100 from lower diagonal elements of A

(D) None of the above

Q.11 The minimum number of arithmetic operations required to evaluate the polynomial $P(X) = X^5 + 4X^3 + 6X + 5$ for a given value of X, using only one temporary variable is _____.

Q.12 Consider the following rooted tree with the vertex labeled P as the root:



The order in which the nodes are visited during an in-order traversal of the tree is

(A) SQPTRWUV

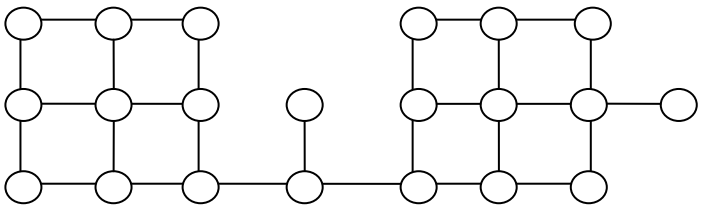
(B) SQPTUWRV

(C) SQPTWUVR

(D) SQPTRUWV

CS03 (GATE 2014)

Q.13 Suppose depth first search is executed on the graph below starting at some unknown vertex. Assume that a recursive call to visit a vertex is made only after first checking that the vertex has not been visited earlier. Then the maximum possible recursion depth (including the initial call) is _____.



Q.14 You have an array of *n* elements. Suppose you implement quicksort by always choosing the central element of the array as the pivot. Then the tightest upper bound for the worst case performance is

(A) $O(n^2)$ (B) $O(n \log n)$ (C) $\Theta(n \log n)$ (D) $O(n^3)$

Q.15 The length of the shortest string NOT in the language (over $\Sigma = \{a, b\}$) of the following regular expression is _____.

$$a^* b^* (ba)^* a^*$$

Q.16 Let Σ be a finite non-empty alphabet and let 2^{Σ^*} be the power set of Σ*. Which one of the following is **TRUE**?

(A) Both 2^{Σ^*} and Σ* are countable
(B) 2^{Σ^*} is countable and Σ* is uncountable
(C) 2^{Σ^*} is uncountable and Σ* is countable
(D) Both 2^{Σ^*} and Σ* are uncountable

Q.17 One of the purposes of using intermediate code in compilers is to

(A) make parsing and semantic analysis simpler.
(B) improve error recovery and error reporting.
(C) increase the chances of reusing the machine-independent code optimizer in other compilers.
(D) improve the register allocation.

CS03 (GATE 2014)

Q.18 Which of the following statements are CORRECT?

1) Static allocation of all data areas by a compiler makes it impossible to implement recursion.
2) Automatic garbage collection is essential to implement recursion.
3) Dynamic allocation of activation records is essential to implement recursion.
4) Both heap and stack are essential to implement recursion.

(A) 1 and 2 only (B) 2 and 3 only (C) 3 and 4 only (D) 1 and 3 only

Q.19 In the context of modular software design, which one of the following combinations is desirable?

(A) High cohesion and high coupling
(B) High cohesion and low coupling
(C) Low cohesion and high coupling
(D) Low cohesion and low coupling

Q.20 A system uses 3 page frames for storing process pages in main memory. It uses the Least Recently Used (**LRU**) page replacement policy. Assume that all the page frames are initially empty. What is the total number of page faults that will occur while processing the page reference string given below?

4, 7, 6, 1, 7, 6, 1, 2, 7, 2

Q.21 What is the optimized version of the relation algebra expression $\pi_{A1}(\pi_{A2}(\sigma_{F1}(\sigma_{F2}(r))))$, where $A1, A2$ are sets of attributes in r with $A1 \subset A2$ and $F1, F2$ are Boolean expressions based on the attributes in r?

(A) $\pi_{A1}(\sigma_{(F1 \wedge F2)}(r))$
(B) $\pi_{A1}(\sigma_{(F1 \vee F2)}(r))$
(C) $\pi_{A2}(\sigma_{(F1 \wedge F2)}(r))$
(D) $\pi_{A2}(\sigma_{(F1 \vee F2)}(r))$

Q.22 A *prime attribute* of a relation scheme R is an attribute that appears

(A) in all candidate keys of R.
(B) in some candidate key of R.
(C) in a foreign key of R.
(D) only in the primary key of R.

CS03 (GATE 2014)

Q.23 In the following pairs of OSI protocol layer/sub-layer and its functionality, the **INCORRECT** pair is

(A) Network layer and Routing

(B) Data Link Layer and Bit synchronization

(C) Transport layer and End-to-end process communication

(D) Medium Access Control sub-layer and Channel sharing

Q.24 A bit-stuffing based framing protocol uses an 8-bit delimiter pattern of 01111110. If the output bit-string after stuffing is 01111100101, then the input bit-string is

(A) 0111110100

(B) 0111110101

(C) 0111111101

(D) 0111111111

Q.25 Host A (on TCP/IP v4 network A) sends an IP datagram D to host B (also on TCP/IP v4 network B). Assume that no error occurred during the transmission of D. When D reaches B, which of the following IP header field(s) may be different from that of the original datagram D?

(i) TTL (ii) Checksum (iii) Fragment Offset

(A) (i) only

(B) (i) and (ii) only

(C) (ii) and (iii) only

(D) (i), (ii) and (iii)

Q. 26 – Q. 55 carry two marks each.

Q.26 An IP router implementing Classless Inter-domain Routing (CIDR) receives a packet with address 131.23.151.76. The router's routing table has the following entries:

Prefix	Output Interface Identifier
131.16.0.0/ 12	3
131.28.0.0/ 14	5
131.19.0.0/ 16	2
131.22.0.0/ 15	1

The identifier of the output interface on which this packet will be forwarded is _____.

CS03 (GATE 2014)

Q.27 Every host in an IPv4 network has a 1-second resolution real-time clock with battery backup. Each host needs to generate up to 1000 unique identifiers per second. Assume that each host has a globally unique IPv4 address. Design a 50-bit globally unique ID for this purpose. After what period (in seconds) will the identifiers generated by a host wrap around?

Q.28 An IP router with a Maximum Transmission Unit (MTU) of 1500 bytes has received an IP packet of size 4404 bytes with an IP header of length 20 bytes. The values of the relevant fields in the header of the third IP fragment generated by the router for this packet are

(A) MF bit: 0, Datagram Length: 1444; Offset: 370

(B) MF bit: 1, Datagram Length: 1424; Offset: 185

(C) MF bit: 1, Datagram Length: 1500; Offset: 370

(D) MF bit: 0, Datagram Length: 1424; Offset: 2960

Q.29 Consider the transactions **T1**, **T2**, and **T3** and the schedules **S1** and **S2** given below.

```
T1: r1(X); r1(Z); w1(X); w1(Z)
T2: r2(Y); r2(Z); w2(Z)
T3: r3(Y); r3(X); w3(Y)

S1: r1(X); r3(Y); r3(X); r2(Y); r2(Z); w3(Y); w2(Z); r1(Z); w1(X); w1(Z)
S2: r1(X); r3(Y); r2(Y); r3(X); r1(Z); r2(Z); w3(Y); w1(X); w2(Z); w1(Z)
```

Which one of the following statements about the schedules is **TRUE**?

(A) Only **S1** is conflict-serializable.

(B) Only **S2** is conflict-serializable.

(C) Both **S1** and **S2** are conflict-serializable.

(D) Neither **S1** nor **S2** is conflict-serializable.

Q.30 Consider the relational schema given below, where **eId** of the relation **dependent** is a foreign key referring to **empId** of the relation **employee**. Assume that every employee has at least one associated dependent in the **dependent** relation.

```
employee (empId, empName, empAge)
dependent(depId, eId, depName, depAge)
```

Consider the following relational algebra query:

$$\prod_{empId}(employee) - \prod_{empId}(employee \bowtie_{(empId = eID) \wedge (empAge \leq depAge)} dependent)$$

The above query evaluates to the set of **empId**s of employees whose age is greater than that of

(A) some dependent.

(B) all dependents.

(C) some of his/her dependents.

(D) all of his/her dependents.

CS03 (GATE 2014)

Q.31 A system contains three programs and each requires three tape units for its operation. The minimum number of tape units which the system must have such that deadlocks never arise is _____.

Q.32 An operating system uses *shortest remaining time first* scheduling algorithm for pre-emptive scheduling of processes. Consider the following set of processes with their arrival times and CPU burst times (in milliseconds):

Process	Arrival Time	Burst Time
P1	0	12
P2	2	4
P3	3	6
P4	8	5

The average waiting time (in milliseconds) of the processes is _____.

Q.33 Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is _____.

Q.34 Consider the basic block given below.

```
a = b + c
c = a + d
d = b + c
e = d - b
a = e + b
```

The minimum number of nodes and edges present in the DAG representation of the above basic block respectively are

(A) 6 and 6 (B) 8 and 10 (C) 9 and 12 (D) 4 and 4

Q.35 Which one of the following problems is undecidable?

(A) Deciding if a given context-free grammar is ambiguous.

(B) Deciding if a given string is generated by a given context-free grammar.

(C) Deciding if the language generated by a given context-free grammar is empty.

(D) Deciding if the language generated by a given context-free grammar is finite.

CS03 (GATE 2014)

Q.36 Consider the following languages over the alphabet $\Sigma = \{0,1,c\}$:

$L_1 = \{0^n 1^n \mid n \geq 0\}$
$L_2 = \{wcw^r \mid w \in \{0,1\}^*\}$
$L_3 = \{ww^r \mid w \in \{0,1\}^*\}$

Here, w^r is the reverse of the string w. Which of these languages are deterministic Context-free languages?

(A) None of the languages

(B) Only L_1

(C) Only L_1 and L_2

(D) All the three languages

Q.37 Suppose you want to move from 0 to 100 on the number line. In each step, you either move right by a unit distance or you take a *shortcut.* A shortcut is simply a pre-specified pair of integers i, j with $i < j$. Given a shortcut i, j if you are at position i on the number line, you may directly move to j. Suppose $T(k)$ denotes the smallest number of steps needed to move from k to 100. Suppose further that there is at most 1 shortcut involving any number, and in particular from 9 there is a shortcut to 15. Let y and z be such that $T(9) = 1 + \min(T(y), T(z))$. Then the value of the product yz is _____.

Q.38 Consider the decision problem *2CNFSAT* defined as follows:

{ Φ | Φ is a satisfiable propositional formula in CNF with at most two literals per clause }

For example, $\Phi = (x_1 \vee x_2) \wedge (x_1 \vee \overline{x_3}) \wedge (x_2 \vee x_4)$ is a Boolean formula and it is in *2CNFSAT*.

The decision problem *2CNFSAT* is

(A) NP-Complete.

(B) solvable in polynomial time by reduction to directed graph reachability.

(C) solvable in constant time since any input instance is satisfiable.

(D) NP-hard, but not NP-complete.

Q.39 Suppose we have a balanced binary search tree T holding n numbers. We are given two numbers L and H and wish to sum up all the numbers in T that lie between L and H. Suppose there are m such numbers in T. If the tightest upper bound on the time to compute the sum is $O(n^a \log^b n + m^c \log^d n)$, the value of $a + 10b + 100c + 1000d$ is _____.

CS03 (GATE 2014)

Q.40 Consider a hash table with 100 slots. Collisions are resolved using chaining. Assuming simple uniform hashing, what is the probability that the first 3 slots are unfilled after the first 3 insertions?

(A) $(97 \times 97 \times 97)/100^3$ (B) $(99 \times 98 \times 97)/100^3$

(C) $(97 \times 96 \times 95)/100^3$ (D) $(97 \times 96 \times 95)/(3! \times 100^3)$

Q.41 Consider the pseudocode given below. The function **DoSomething()** takes as argument a pointer to the root of an arbitrary tree represented by the *leftMostChild-rightSibling* representation. Each node of the tree is of type **treeNode**.

```
typedef struct treeNode* treeptr;

struct treeNode
{
       treeptr leftMostChild, rightSibling;
};

int DoSomething (treeptr tree)
{
   int value=0;
   if (tree != NULL) {
     if (tree->leftMostChild == NULL)
         value = 1;
     else
         value =  DoSomething(tree->leftMostChild);
     value = value + DoSomething(tree->rightSibling);
   }
   return(value);
}
```

When the pointer to the root of a tree is passed as the argument to **DoSomething**, the value returned by the function corresponds to the

(A) number of internal nodes in the tree.

(B) height of the tree.

(C) number of nodes without a right sibling in the tree.

(D) number of leaf nodes in the tree.

CS03 (GATE 2014)

Q.42 Consider the C function given below. Assume that the array **listA** contains **n** (> 0) elements, sorted in ascending order.

```
int ProcessArray(int *listA, int x, int n)
{
  int i, j, k;
  i = 0;
  j = n-1;
  do {
       k = (i+j)/2;
       if (x <= listA[k])
           j = k-1;
       if (listA[k] <= x)
           i = k+1;
      }while (i <= j);

  if (listA[k] == x)
      return(k);
  else
      return -1;
}
```

Which one of the following statements about the function **ProcessArray** is **CORRECT**?

(A) It will run into an infinite loop when **x** is not in **listA**.

(B) It is an implementation of binary search.

(C) It will always find the maximum element in **listA**.

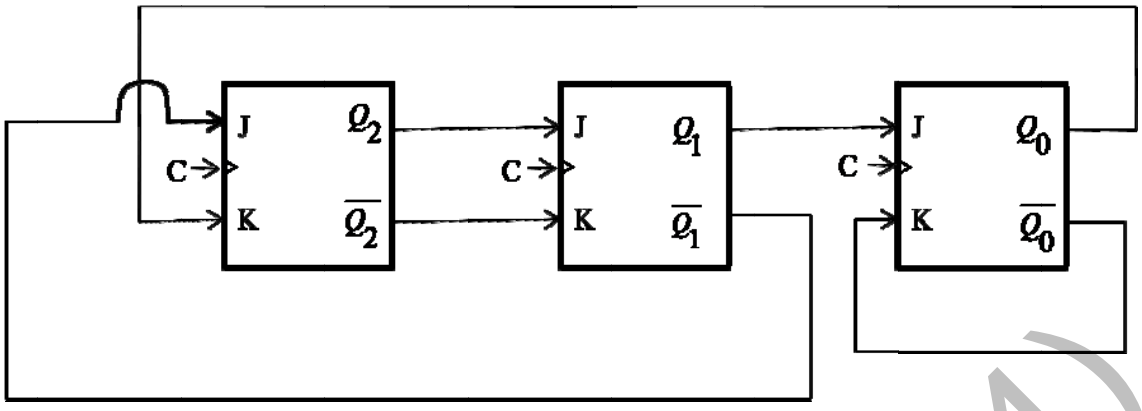
(D) It will return −1 even when **x** is present in **listA**.

Q.43 An instruction pipeline has five stages, namely, instruction fetch (IF), instruction decode and register fetch (ID/RF), instruction execution (EX), memory access (MEM), and register writeback (WB) with stage latencies 1 ns, 2.2 ns, 2 ns, 1 ns, and 0.75 ns, respectively (ns stands for nanoseconds). To gain in terms of frequency, the designers have decided to split the ID/RF stage into three stages (ID, RF1, RF2) each of latency 2.2/3 ns. Also, the EX stage is split into two stages (EX1, EX2) each of latency 1 ns. The new design has a total of eight pipeline stages. A program has 20% branch instructions which execute in the EX stage and produce the next instruction pointer at the end of the EX stage in the old design and at the end of the EX2 stage in the new design. The IF stage stalls after fetching a branch instruction until the next instruction pointer is computed. All instructions other than the branch instruction have an average CPI of one in both the designs. The execution times of this program on the old and the new design are P and Q nanoseconds, respectively. The value of P/Q is _____.

Q.44 The memory access time is 1 nanosecond for a read operation with a hit in cache, 5 nanoseconds for a read operation with a miss in cache, 2 nanoseconds for a write operation with a hit in cache and 10 nanoseconds for a write operation with a miss in cache. Execution of a sequence of instructions involves 100 instruction fetch operations, 60 memory operand read operations and 40 memory operand write operations. The cache hit-ratio is 0.9. The average memory access time (in nanoseconds) in executing the sequence of instructions is _____.

CS08 (GATE 2014)

Q.45



The above synchronous sequential circuit built using JK flip-flops is initialized with $Q_2Q_1Q_0 = 000$. The state sequence for this circuit for the next 3 clock cycles is

(A) 001, 010, 011

(B) 111, 110, 101

(C) 100, 110, 111

(D) 100, 011, 001

Q.46 With respect to the numerical evaluation of the definite integral, $K = \int_a^b x^2\,dx$, where a and b are given, which of the following statements is/are TRUE?

I) The value of K obtained using the trapezoidal rule is always greater than or equal to the exact value of the definite integral.

II) The value of K obtained using the Simpson's rule is always equal to the exact value of the definite integral.

(A) I only (B) II only (C) Both I and II (D) Neither I nor II

Q.47 The value of the integral given below is

$$\int_0^{\pi} x^2 \cos x\,dx$$

(A) -2π (B) π (C) $-\pi$ (D) 2π

Q.48 Let S be a sample space and two mutually exclusive events A and B be such that $A \cup B = S$. If $P(\cdot)$ denotes the probability of the event, the maximum value of $P(A)P(B)$ is _____.

CS03 (GATE 2014)

Q.49 Consider the set of all functions $f: \{0,1,\ldots,2014\} \rightarrow \{0,1,\ldots,2014\}$ such that $f(f(i)) = i$, for all $0 \leq i \leq 2014$. Consider the following statements:

P. For each such function it must be the case that for every $i, f(i) = i$.
Q. For each such function it must be the case that for some $i, f(i) = i$.
R. Each such function must be onto.

Which one of the following is CORRECT?

(A) P, Q and R are true

(B) Only Q and R are true

(C) Only P and Q are true

(D) Only R is true

Q.50 There are two elements x, y in a group $(G,*)$ such that every element in the group can be written as a product of some number of x's and y's in some order. It is known that

$$x * x = y * y = x * y * x * y = y * x * y * x = e$$

where e is the identity element. The maximum number of elements in such a group is _____.

Q.51 If G is a forest with n vertices and k connected components, how many edges does G have?

(A) $\lfloor n/k \rfloor$ (B) $\lceil n/k \rceil$ (C) $n-k$ (D) $n-k+1$

Q.52 Let δ denote the minimum degree of a vertex in a graph. For all planar graphs on n vertices with $\delta \geq 3$, which one of the following is **TRUE**?

(A) In any planar embedding, the number of faces is at least $\frac{n}{2}+2$

(B) In any planar embedding, the number of faces is less than $\frac{n}{2}+2$

(C) There is a planar embedding in which the number of faces is less than $\frac{n}{2}+2$

(D) There is a planar embedding in which the number of faces is at most $\frac{n}{\delta+1}$

CS03 (GATE 2014)

Q.53 The CORRECT formula for the sentence, "not all rainy days are cold" is

(A) ∀d (Rainy(d) ∧~Cold(d))

(B) ∀d (~Rainy(d) → Cold(d))

(C) ∃d (~Rainy(d) → Cold(d))

(D) ∃d (Rainy(d) ∧~Cold(d))

Q.54 Consider the following relational schema:

employee(empId,empName,empDept)
customer(custId,custName,salesRepId,rating)

salesRepId is a foreign key referring to **empId** of the `employee` relation. Assume that each employee makes a sale to at least one customer. What does the following query return?

```
SELECT empName
FROM employee E
WHERE NOT EXISTS (SELECT custId
                  FROM customer C
                  WHERE C.salesRepId = E.empId
                  AND C.rating <> 'GOOD');
```

(A) Names of all the employees with at least one of their customers having a 'GOOD' rating.

(B) Names of all the employees with at most one of their customers having a 'GOOD' rating.

(C) Names of all the employees with none of their customers having a 'GOOD' rating.

(D) Names of all the employees with all their customers having a 'GOOD' rating.

Q.55 Let $\oplus$ denote the Exclusive OR (XOR) operation. Let '1' and '0' denote the binary constants. Consider the following Boolean expression for F over two variables P and Q:

$$F(P,Q) = \big((1 \oplus P) \oplus (P \oplus Q)\big) \oplus ((P \oplus Q) \oplus (Q \oplus 0))$$

The equivalent expression for F is

(A) $P + Q$ (B) $\overline{P + Q}$ (C) $P \oplus Q$ (D) $\overline{P \oplus Q}$

END OF THE QUESTION PAPER

CS03 (GATE 2014)

GATE 2014

Answer Keys for CS - Computer Science and Information Technology

Section	Q. No.	SESSION - 3	
		Key / Range	Marks
GA	1	D	1
GA	2	C	1
GA	3	C	1
GA	4	C	1
GA	5	C	1
GA	6	B	2
GA	7	D	2
GA	8	D	2
GA	9	C	2
GA	10	C	2
CS	1	D	1
CS	2	D	1
CS	3	5 to 5	1
CS	4	A	1
CS	5	2 to 2	1
CS	6	4 to 4	1
CS	7	B	1
CS	8	C	1
CS	9	C	1
CS	10	A	1
CS	11	7 to 7	1
CS	12	A	1
CS	13	19 to 19	1
CS	14	A	1
CS	15	3 to 3	1
CS	16	C	1
CS	17	C	1
CS	18	D	1
CS	19	B	1
CS	20	6 to 6	1
CS	21	A	1
CS	22	B	1
CS	23	B	1

Section	Q. No.	SESSION - 3	
		Key / Range	Marks
CS	24	B	1
CS	25	D	1
CS	26	1 to 1	2
CS	27	256 to 256	2
CS	28	A	2
CS	29	A	2
CS	30	D	2
CS	31	7 to 7	2
CS	32	5.5 to 5.5	2
CS	33	122 to 122	2
CS	34	A	2
CS	35	A	2
CS	36	C	2
CS	37	150 to 150	2
CS	38	B	2
CS	39	110 to 110	2
CS	40	A	2
CS	41	D	2
CS	42	B	2
CS	43	1.50 to 1.60	2
CS	44	1.68 to 1.68	2
CS	45	C	2
CS	46	C	2
CS	47	A	2
CS	48	0.25 to 0.25	2
CS	49	B	2
CS	50	4 to 4	2
CS	51	C	2
CS	52	A	2
CS	53	D	2
CS	54	D	2
CS	55	D	2

SESSION - 1

Graduate Aptitude Test in Engineering

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Question Paper Name: CS: COMPUTER SCIENCE AND INFORMATION TECHNOLOGY 7th Feb Shift1
Number of Questions: 65
Total Marks: 100.0

Wrong answer for MCQ will result in negative marks, (-1/3) for 1 mark Questions and (-2/3) for 2 marks Questions.

General Aptitude

Number of Questions: 10
Section Marks: 15.0

Q.1 to Q.5 carry 1 mark each & Q.6 to Q.10 carry 2 marks each.

Question Number : 1 Question Type : MCQ

Didn't you buy _____ when you went shopping?

(A) any paper (B) much paper (C) no paper (D) a few paper

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 2 Question Type : MCQ

Which of the following options is the closest in meaning to the sentence below?

She enjoyed herself immensely at the party.

(A) She had a terrible time at the party

(B) She had a horrible time at the party

(C) She had a terrific time at the party

(D) She had a terrifying time at the party

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 3 Question Type : MCQ

Which one of the following combinations is incorrect?

(A) Acquiescence - Submission
(B) Wheedle - Roundabout
(C) Flippancy - Lightness
(D) Profligate - Extravagant

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 4 Question Type : MCQ

Based on the given statements, select the most appropriate option to solve the given question.

If two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from the first floor to the second floor of the building?

Statements:

(I) Each step is 3/4 foot high.
(II) Each step is 1 foot wide.

(A) Statement I alone is sufficient, but statement II alone is not sufficient.
(B) Statement II alone is sufficient, but statement I alone is not sufficient.
(C) Both statements together are sufficient, but neither statement alone is sufficient.
(D) Statement I and II together are not sufficient.

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 5 Question Type : MCQ

Given Set A = {2, 3, 4, 5} and Set B = {11, 12, 13, 14, 15}, two numbers are randomly selected, one from each set. What is the probability that the sum of the two numbers equals 16?

(A) 0.20 (B) 0.25 (C) 0.30 (D) 0.33

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 6 Question Type : MCQ

Select the alternative meaning of the underlined part of the sentence.

The chain snatchers took to their heels when the police party arrived.

(A) took shelter in a thick jungle
(B) open indiscriminate fire
(C) took to flight
(D) unconditionally surrendered

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 7 Question Type : MCQ

The given statement is followed by some courses of action. Assuming the statement to be true, decide the correct option.

Statement:

There has been a significant drop in the water level in the lakes supplying water to the city.

Course of action:

(I) The water supply authority should impose a partial cut in supply to tackle the situation.
(II) The government should appeal to all the residents through mass media for minimal use of water.
(III) The government should ban the water supply in lower areas.

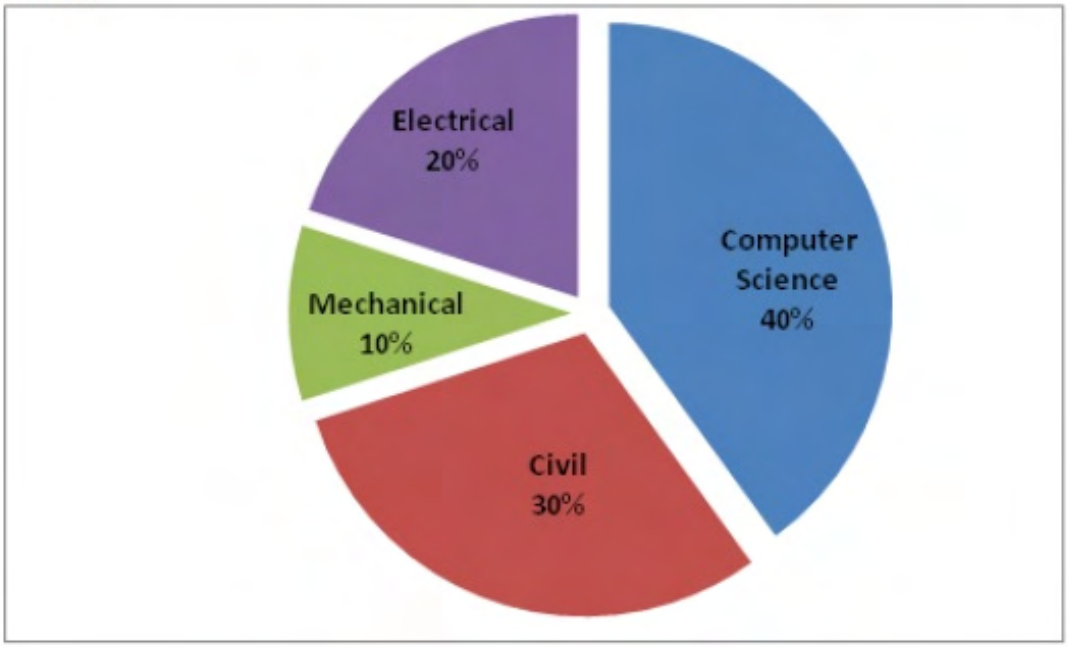
(A) Statements I and II follow.
(B) Statements I and III follow.
(C) Statements II and III follow.
(D) All statements follow.

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 8 Question Type : NAT

The pie chart below has the breakup of the number of students from different departments in an engineering college for the year 2012. The proportion of male to female students in each department is 5:4. There are 40 males in Electrical Engineering. What is the difference between the numbers of female students in the Civil department and the female students in the Mechanical department?



Correct Answer:
32

Question Number : 9 Question Type : MCQ

The probabilities that a student passes in Mathematics, Physics and Chemistry are m, p, and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m, p, c:

(I) $p + m + c = 27/20$
(II) $p + m + c = 13/20$
(III) $(p) \times (m) \times (c) = 1/10$

(A) Only relation I is true.
(B) Only relation II is true.
(C) Relations II and III are true.
(D) Relations I and III are true.

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 10 Question Type : MCQ

The number of students in a class who have answered correctly, wrongly, or not attempted each question in an exam, are listed in the table below. The marks for each question are also listed. There is no negative or partial marking.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

What is the average of the marks obtained by the class in the examination?

(A) 2.290 (B) 2.970 (C) 6.795 (D) 8.795

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Computer Science and Information Technology

Number of Questions:	55
Section Marks:	85.0

Q.11 to Q.35 carry 1 mark each & Q.36 to Q.65 carry 2 marks each.

Question Number : 11 Question Type : MCQ

If $g(x)=1-x$ and $h(x)=\frac{x}{x-1}$, then $\frac{g(h(x))}{h(g(x))}$ is:

(A) $\frac{h(x)}{g(x)}$ (B) $\frac{-1}{x}$ (C) $\frac{g(x)}{h(x)}$ (D) $\frac{x}{(1-x)^2}$

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 12 Question Type : MCQ

$\lim_{x\to\infty} x^{1/x}$ is

(A) ∞ (B) 0 (C) 1 (D) Not defined

Options :

1. ✖ A

2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 13 Question Type : MCQ

Match the following:

(P) Prim's algorithm for minimum spanning tree	(i) Backtracking
(Q) Floyd-Warshall algorithm for all pairs shortest paths	(ii) Greedy method
(R) Mergesort	(iii) Dynamic programming
(S) Hamiltonian circuit	(iv) Divide and conquer

(A) P-iii, Q-ii, R-iv, S-i
(B) P-i, Q-ii, R-iv, S-iii
(C) P-ii, Q-iii, R-iv, S-i
(D) P-ii, Q-i, R-iii, S-iv

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 14 Question Type : MCQ

Which one of the following is the recurrence equation for the worst case time complexity of the Quicksort algorithm for sorting n (≥ 2) numbers? In the recurrence equations given in the options below, c is a constant.

(A) $T(n) = 2\,T(n/2) + cn$
(B) $T(n) = T(n-1) + T(1) + cn$
(C) $T(n) = 2T(n-1) + cn$
(D) $T(n) = T(n/2) + cn$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 15 Question Type : MCQ

The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 5 are

(A) 63 and 6, respectively
(B) 64 and 5, respectively
(C) 32 and 6, respectively
(D) 31 and 5, respectively

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 16 Question Type : MCQ

Match the following:

(P) Condition coverage	(i) Black-box testing
(Q) Equivalence class partitioning	(ii) System testing
(R) Volume testing	(iii) White-box testing
(S) Alpha testing	(iv) Performance testing

(A) P-ii, Q-iii, R-i, S-iv
(B) P-iii, Q-iv, R-ii, S- i
(C) P-iii, Q-i, R-iv, S-ii
(D) P-iii, Q-i, R-ii, S-iv

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 17 Question Type : MCQ

Which of the following is/are correct inorder traversal sequence(s) of binary search tree(s)?

I. 3, 5, 7, 8, 15, 19, 25
II. 5, 8, 9, 12, 10, 15, 25
III. 2, 7, 10, 8, 14, 16, 20
IV. 4, 6, 7, 9 18, 20, 25

(A) I and IV only (B) II and III only (C) II and IV only (D) II only

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 18 Question Type : MCQ

Which one of the following is TRUE at any valid state in shift-reduce parsing?

(A) Viable prefixes appear only at the bottom of the stack and not inside
(B) Viable prefixes appear only at the top of the stack and not inside
(C) The stack contains only a set of viable prefixes
(D) The stack never contains viable prefixes

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 19 Question Type : MCQ

Which one of the following is NOT equivalent to $p \leftrightarrow q$?

(A) $(\neg p \vee q) \wedge (p \vee \neg q)$
(B) $(\neg p \vee q) \wedge (q \rightarrow p)$
(C) $(\neg p \wedge q) \vee (p \wedge \neg q)$
(D) $(\neg p \wedge \neg q) \vee (p \wedge q)$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 20 Question Type : MCQ

For a set A, the power set of A is denoted by 2^A. If $A = \{5, \{6\}, \{7\}\}$, which of the following options are TRUE?

I. $\emptyset \in 2^A$ II. $\emptyset \subseteq 2^A$ III. $\{5, \{6\}\} \in 2^A$ IV. $\{5, \{6\}\} \subseteq 2^A$

(A) I and III only
(B) II and III only
(C) I, II and III only
(D) I, II and IV only

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 21 Question Type : MCQ

Consider a 4-bit Johnson counter with an initial value of 0000. The counting sequence of this counter is

(A) 0, 1, 3, 7, 15, 14, 12, 8, 0
(B) 0, 1, 3, 5, 7, 9, 11, 13, 15, 0
(C) 0, 2, 4, 6, 8, 10, 12, 14, 0
(D) 0, 8, 12, 14, 15, 7, 3, 1, 0

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 22 Question Type : MCQ

For computers based on three-address instruction formats, each address field can be used to specify which of the following:

(S1) A memory operand
(S2) A processor register
(S3) An implied accumulator register

(A) Either S1 or S2
(B) Either S2 or S3
(C) Only S2 and S3
(D) All of S1, S2 and S3

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 23 Question Type : MCQ

Suppose two hosts use a TCP connection to transfer a large file. Which of the following statements is/are FALSE with respect to the TCP connection?

I. If the sequence number of a segment is m, then the sequence number of the subsequent segment is always $m+1$.
II. If the estimated round trip time at any given point of time is t sec, the value of the retransmission timeout is always set to greater than or equal to t sec.
III. The size of the advertised window never changes during the course of the TCP connection.
IV. The number of unacknowledged bytes at the sender is always less than or equal to the advertised window.

(A) III only (B) I and III only (C) I and IV only (D) II and IV only

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 24 Question Type : MCQ

Suppose that everyone in a group of N people wants to communicate secretly with the N-1 others using symmetric key cryptographic system. The communication between any two persons should not be decodable by the others in the group. The number of keys required in the system as a whole to satisfy the confidentiality requirement is

(A) $2N$ (B) $N(N-1)$ (C) $N(N-1)/2$ (D) $(N-1)^2$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 25 Question Type : MCQ

Which of the following statements is/are FALSE?

I. XML overcomes the limitations in HTML to support a structured way of organizing content.
II. XML specification is not case sensitive while HTML specification is case sensitive.
III. XML supports user defined tags while HTML uses pre-defined tags.
IV. XML tags need not be closed while HTML tags must be closed.

(A) II only (B) I only (C) II and IV only (D) III and IV only

Options :

1. ✖ A

2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 26 Question Type : MCQ

Which one of the following fields of an IP header is NOT modified by a typical IP router?

(A) Checksum (B) Source address
(C) Time to Live (TTL) (D) Length

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 27 Question Type : MCQ

In one of the pairs of protocols given below, both the protocols can use multiple TCP connections between the same client and the server. Which one is that?

(A) HTTP, FTP (B) HTTP, TELNET (C) FTP, SMTP (D) HTTP, SMTP

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 28 Question Type : MCQ

For any two languages L_1 and L_2 such that L_1 is context-free and L_2 is recursively enumerable but not recursive, which of the following is/are necessarily true?

I. $\bar{L}_1$ (complement of L_1) is recursive
II. $\bar{L}_2$ (complement of L_2) is recursive
III. $\bar{L}_1$ is context-free
IV. $\bar{L}_1 \cup L_2$ is recursively enumerable

(A) I only (B) III only (C) III and IV only (D) I and IV only

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 29 Question Type : NAT

Consider a system with byte-addressable memory, 32-bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. The size of the page table in the system in megabytes is _____ .

Correct Answer :
4

Question Number : 30 Question Type : NAT

The following two functions `P1` and `P2` that share a variable `B` with an initial value of 2 execute concurrently.

```
P1( ) {                          P2( ) {
        C = B - 1;                       D = 2 * B;
        B = 2 * C;                       B = D - 1;
}                                }
```

The number of distinct values that `B` can possibly take after the execution is_____.

Correct Answer :
3

Question Number : 31 Question Type : MCQ

SELECT operation in SQL is equivalent to

(A) the selection operation in relational algebra
(B) the selection operation in relational algebra, except that SELECT in SQL retains duplicates
(C) the projection operation in relational algebra
(D) the projection operation in relational algebra, except that SELECT in SQL retains duplicates

Options :
1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 32 Question Type : MCQ

A file is organized so that the ordering of data records is the same as or close to the ordering of data entries in some index. Then that index is called

(A) Dense (B) Sparse (C) Clustered (D) Unclustered

Options :
1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 33 Question Type : NAT

In the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 \\ 4 & 9 \end{bmatrix}$, if the diagonal elements of U are both 1, then the lower diagonal entry l_{22} of L is_____.

Correct Answer :

5

Question Number : 34 Question Type : NAT

The output of the following C program is_____.

```
void f1(int a, int b) {
      int c;
      c=a; a=b; b=c;
}
void f2(int *a, int *b) {
      int c;
      c=*a; *a=*b; *b=c;
}
int main(){
      int a=4, b=5, c=6;
      f1(a,b);
      f2(&b, &c);
      printf("%d",c-a-b);
 }
```

Correct Answer :

-5

Question Number : 35 Question Type : MCQ

What are the worst-case complexities of insertion and deletion of a key in a binary search tree?

(A) $\theta(\log n)$ for both insertion and deletion

(B) $\theta(n)$ for both insertion and deletion

(C) $\theta(n)$ for insertion and $\theta(\log n)$ for deletion

(D) $\theta(\log n)$ for insertion and $\theta(n)$ for deletion

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 36 Question Type : NAT

Suppose that the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Assume that the transmission time for the acknowledgement and the processing time at nodes are negligible. Then the minimum frame size in bytes to achieve a link utilization of at least 50% is_____.

Correct Answer :

160

Question Number : 37 Question Type : MCQ

Consider a max heap, represented by the array: 40, 30, 20, 10, 15, 16, 17, 8, 4.

Array Index	1	2	3	4	5	6	7	8	9
Value	40	30	20	10	15	16	17	8	4

Now consider that a value 35 is inserted into this heap. After insertion, the new heap is

(A) 40, 30, 20, 10, 15, 16, 17, 8, 4, 35
(B) 40, 35, 20, 10, 30, 16, 17, 8, 4, 15
(C) 40, 30, 20, 10, 35, 16, 17, 8, 4, 15
(D) 40, 35, 20, 10, 15, 16, 17, 8, 4, 30

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 38 Question Type : NAT

Consider the following C program segment.

```
while(first <= last)
{
      if (array[middle] < search)
         first = middle + 1;
      else if (array[middle] == search)
                found = TRUE;
           else last = middle - 1;
      middle = (first + last)/2;
}
if (first > last)  notPresent = TRUE;
```

The cyclomatic complexity of the program segment is _____.

Correct Answer :

5

Question Number : 39 Question Type : NAT

Consider a LAN with four nodes S_1, S_2, S_3 and S_4. Time is divided into fixed-size slots, and a node can begin its transmission only at the beginning of a slot. A collision is said to have occurred if more than one node transmit in the same slot. The probabilities of generation of a frame in a time slot by S_1, S_2, S_3 and S_4 are 0.1, 0.2, 0.3 and 0.4, respectively. The probability of sending a frame in the first slot without any collision by any of these four stations is _____.

Correct Answer :

0.40 to 0.46

Question Number : 40 Question Type : MCQ

The binary operator $\neq$ is defined by the following truth table.

p	q	$p \neq q$
0	0	0
0	1	1
1	0	1
1	1	0

Which one of the following is true about the binary operator $\neq$?

(A) Both commutative and associative
(B) Commutative but not associative
(C) Not commutative but associative
(D) Neither commutative nor associative

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 41 Question Type : NAT

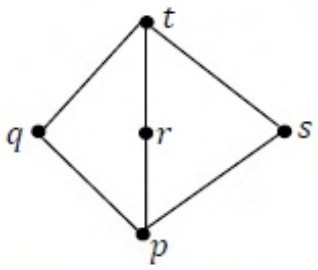
$$\sum_{x=1}^{99} \frac{1}{x(x+1)} = _____.$$

Correct Answer :

0.99

Question Number : 42 Question Type : MCQ

Suppose $\mathcal{L} = \{p, q, r, s, t\}$ is a lattice represented by the following Hasse diagram:



For any $x, y \in \mathcal{L}$, not necessarily distinct, $x \vee y$ and $x \wedge y$ are join and meet of x, y, respectively. Let $\mathcal{L}^3 = \{(x, y, z): x, y, z \in \mathcal{L}\}$ be the set of all ordered triplets of the elements of $\mathcal{L}$. Let p_r be the probability that an element $(x, y, z) \in \mathcal{L}^3$ chosen equiprobably satisfies $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$. Then

(A) $p_r = 0$
(B) $p_r = 1$
(C) $0 < p_r \leq \frac{1}{5}$
(D) $\frac{1}{5} < p_r < 1$

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 43 Question Type : MCQ

Consider the operations

$f(X, Y, Z) = X'YZ + XY' + Y'Z'$ and $g(X, Y, Z) = X'YZ + X'YZ' + XY$.

Which one of the following is correct?

(A) Both $\{f\}$ and $\{g\}$ are functionally complete
(B) Only $\{f\}$ is functionally complete
(C) Only $\{g\}$ is functionally complete
(D) Neither $\{f\}$ nor $\{g\}$ is functionally complete

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 44 Question Type : NAT

Let G be a connected planar graph with 10 vertices. If the number of edges on each face is three, then the number of edges in G is _____.

Correct Answer :
24

Question Number : 45 Question Type : MCQ

Let a_n represent the number of bit strings of length n containing two consecutive 1s. What is the recurrence relation for a_n ?

(A) $a_{n-2} + a_{n-1} + 2^{n-2}$ (B) $a_{n-2} + 2a_{n-1} + 2^{n-2}$

(C) $2a_{n-2} + a_{n-1} + 2^{n-2}$ (D) $2a_{n-2} + 2a_{n-1} + 2^{n-2}$

Options :
1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

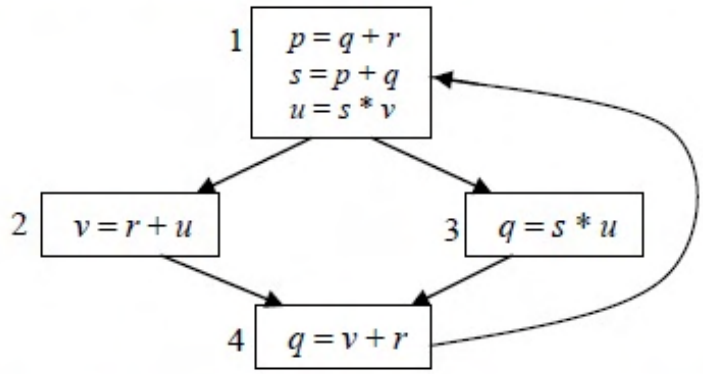
Question Number : 46 Question Type : MCQ

A variable x is said to be live at a statement S_i in a program if the following three conditions hold simultaneously:

i. There exists a statement S_j that uses x

ii. There is a path from S_i to S_j in the flow graph corresponding to the program

iii. The path has no intervening assignment to x including at S_i and S_j



The variables which are live both at the statement in basic block 2 and at the statement in basic block 3 of the above control flow graph are

(A) p, s, u (B) r, s, u (C) r, u (D) q, v

Options :
1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 47 Question Type : NAT

The least number of temporary variables required to create a three-address code in static single assignment form for the expression $q + r / 3 + s - t * 5 + u * v / w$ is _____.

Correct Answer :
8

Question Number : 48 Question Type : NAT

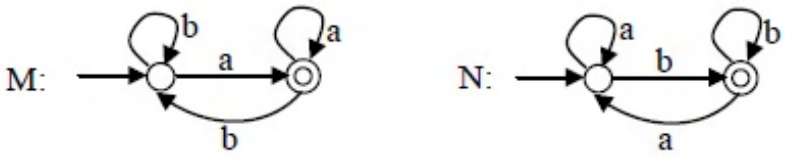
Consider an Entity-Relationship (ER) model in which entity sets E_1 and E_2 are connected by an $m : n$ relationship R_{12}. E_1 and E_3 are connected by a $1 : n$ (1 on the side of E_1 and n on the side of E_3) relationship R_{13}.

E_1 has two single-valued attributes a_{11} and a_{12} of which a_{11} is the key attribute. E_2 has two single-valued attributes a_{21} and a_{22} of which a_{21} is the key attribute. E_3 has two single-valued attributes a_{31} and a_{32} of which a_{31} is the key attribute. The relationships do not have any attributes.

If a relational model is derived from the above ER model, then the minimum number of relations that would be generated if all the relations are in 3NF is _____.

Correct Answer :
4

Question Number : 49 Question Type : NAT

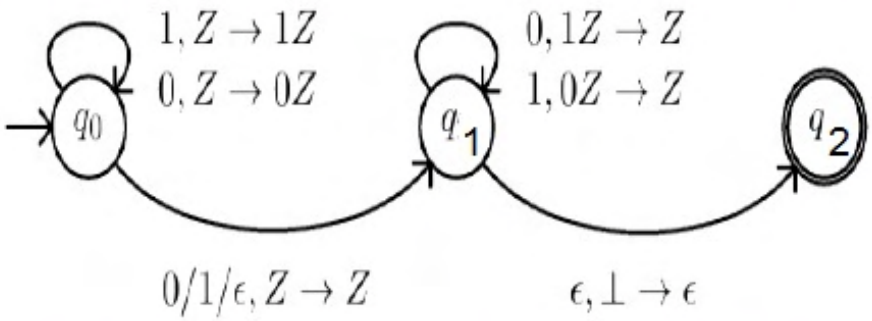


Consider the DFAs M and N given above. The number of states in a minimal DFA that accepts the language L(M) ∩ L(N) is_____

Correct Answer :
1

Question Number : 50 Question Type : MCQ

Consider the NPDA $\langle Q = \{q_o, q_1, q_2\}, \sum = \{0,1\}, \Gamma = \{0,1,\perp\}, \delta, q_0, \perp, F = \{q_2\}\rangle$, where (as per usual convention) Q is the set of states, $\sum$ is the input alphabet, Γ is the stack alphabet, δ is the state transition function, q_0 is the initial state, $\perp$ is the initial stack symbol, and F is the set of accepting states. The state transition is as follows:



Which one of the following sequences must follow the string 101100 so that the overall string is accepted by the automaton?

(A) 10110 (B) 10010 (C) 01010 (D) 01001

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 51 Question Type : MCQ

Let $G = (V, E)$ be a simple undirected graph, and s be a particular vertex in it called the source. For $x \in V$, let $d(x)$ denote the shortest distance in G from s to x. A breadth first search (BFS) is performed starting at s. Let T be the resultant BFS tree. If (u,v) is an edge of G that is not in T, then which one of the following CANNOT be the value of $d(u) - d(v)$?

(A) -1 (B) 0 (C) 1 (D) 2

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 52 Question Type : NAT

Consider a uniprocessor system executing three tasks T_1, T_2 and T_3, each of which is composed of an infinite sequence of jobs (or instances) which arrive periodically at intervals of 3, 7 and 20 milliseconds, respectively. The priority of each task is the inverse of its period, and the available tasks are scheduled in order of priority, with the highest priority task scheduled first. Each instance of T_1, T_2 and T_3 requires an execution time of 1, 2 and 4 milliseconds, respectively. Given that all tasks initially arrive at the beginning of the 1^{st} millisecond and task preemptions are allowed, the first instance of T_3 completes its execution at the end of _____ milliseconds.

Correct Answer :

12

Question Number : 53 Question Type : MCQ

A positive edge-triggered D flip-flop is connected to a positive edge-triggered JK flip-flop as follows. The Q output of the D flip-flop is connected to both the J and K inputs of the JK flip-flop, while the Q output of the JK flip-flop is connected to the input of the D flip-flop. Initially, the output of the D flip-flop is set to logic one and the output of the JK flip-flop is cleared. Which one of the following is the bit sequence (including the initial state) generated at the Q output of the JK flip-flop when the flip-flops are connected to a free-running common clock? Assume that J = K = 1 is the toggle mode and J = K = 0 is the state-holding mode of the JK flip-flop. Both the flip-flops have non-zero propagation delays.

(A) 0110110...
(B) 0100100...
(C) 011101110...
(D) 011001100...

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 54 Question Type : NAT

Consider a disk pack with a seek time of 4 milliseconds and rotational speed of 10000 rotations per minute (RPM). It has 600 sectors per track and each sector can store 512 bytes of data. Consider a file stored in the disk. The file contains 2000 sectors. Assume that every sector access necessitates a seek, and the average rotational latency for accessing each sector is half of the time for one complete rotation. The total time (in milliseconds) needed to read the entire file is _____.

Correct Answer:

14020

Question Number : 55 Question Type : NAT

Consider a non-pipelined processor with a clock rate of 2.5 gigahertz and average cycles per instruction of four. The same processor is upgraded to a pipelined processor with five stages; but due to the internal pipeline delay, the clock speed is reduced to 2 gigahertz. Assume that there are no stalls in the pipeline. The speed up achieved in this pipelined processor is_____.

Correct Answer :

3.2

Question Number : 56 Question Type : NAT

Suppose the following disk request sequence (track numbers) for a disk with 100 tracks is given: 45, 20, 90, 10, 50, 60, 80, 25, 70. Assume that the initial position of the R/W head is on track 50. The additional distance that will be traversed by the R/W head when the Shortest Seek Time First (SSTF) algorithm is used compared to the SCAN (Elevator) algorithm (assuming that SCAN algorithm moves towards 100 when it starts execution) is_____ tracks.

Correct Answer :

10

Question Number : 57 Question Type : MCQ

Consider a main memory with five page frames and the following sequence of page references: 3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3. Which one of the following is true with respect to page replacement policies First In First Out (FIFO) and Least Recently Used (LRU)?

(A) Both incur the same number of page faults
(B) FIFO incurs 2 more page faults than LRU
(C) LRU incurs 2 more page faults than FIFO
(D) FIFO incurs 1 more page faults than LRU

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 58 Question Type : NAT

$$\int_{1/\pi}^{2/\pi} \frac{\cos(1/x)}{x^2}\,dx = _____.$$

Correct Answer:

-1

Question Number : 59 Question Type : MCQ

Consider the following 2×2 matrix A where two elements are unknown and are marked by a and b. The eigenvalues of this matrix are -1 and 7. What are the values of a and b?

$$A = \begin{pmatrix} 1 & 4 \\ b & a \end{pmatrix}.$$

(A) $a = 6, b = 4$

(B) $a = 4, b = 6$

(C) $a = 3, b = 5$

(D) $a = 5, b = 3$

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 60 Question Type : MCQ

An algorithm performs $(\log N)^{1/2}$ find operations, N insert operations, $(\log N)^{1/2}$ delete operations, and $(\log N)^{1/2}$ decrease-key operations on a set of data items with keys drawn from a linearly ordered set. For a delete operation, a pointer is provided to the record that must be deleted. For the decrease-key operation, a pointer is provided to the record that has its key decreased. Which one of the following data structures is the most suited for the algorithm to use, if the goal is to achieve the best total asymptotic complexity considering all the operations?

(A) Unsorted array
(B) Min-heap
(C) Sorted array
(D) Sorted doubly linked list

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 61 Question Type : NAT

Consider the following relations:

Student

Roll_No	Student_Name
1	Raj
2	Rohit
3	Raj

Performance

Roll_No	Course	Marks
1	Math	80
1	English	70
2	Math	75
3	English	80
2	Physics	65
3	Math	80

Consider the following SQL query.

```
SELECT S.Student_Name, sum(P.Marks)
FROM Student S, Performance P
WHERE S.Roll_No = P.Roll_No
GROUP BY S.Student_Name
```

The number of rows that will be returned by the SQL query is _____.

Correct Answer :

2

Question Number : 62 Question Type : MCQ

What is the output of the following C code? Assume that the address of x is 2000 (in decimal) and an integer requires four bytes of memory.

```
int main () {
        unsigned int x[4][3] =
         {{1,2,3},{4,5,6},{7,8,9},{10,11,12}};
        printf("%u, %u, %u", x+3, *(x+3), *(x+2)+3);
}
```

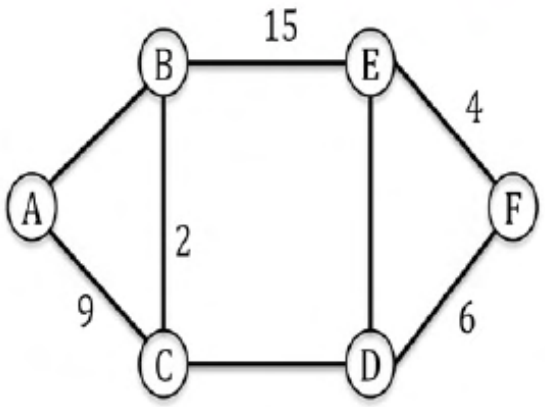
(A) 2036, 2036, 2036 (B) 2012, 4, 2204 (C) 2036, 10, 10 (D) 2012, 4, 6

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 63 Question Type : NAT

The graph shown below has 8 edges with distinct integer edge weights. The minimum spanning tree (MST) is of weight 36 and contains the edges: {(A, C), (B, C), (B, E), (E, F), (D, F)}. The edge weights of only those edges which are in the MST are given in the figure shown below. The minimum possible sum of weights of all 8 edges of this graph is _____.



Correct Answer :
69

Question Number : 64 Question Type : MCQ

Consider the following C function.

```
int fun1(int n){
      int i,j,k,p,q=0;
      for (i=1; i<n; ++i) {
            p=0;
            for (j=n; j>1; j=j/2)
                  ++p;
            for (k=1; k<p; k=k*2)
                  ++q;
      }
      return q;
}
```

Which one of the following most closely approximates the return value of the function `fun1`?

(A) n^3 (B) $n(\log n)^2$ (C) $n \log n$ (D) $n \log(\log n)$

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 65 Question Type : MCQ

Consider the following pseudo code, where x and y are positive integers.

```
begin
  q := 0
  r := x
  while r ≥ y do
    begin
       r := r - y
       q := q + 1
    end
end
```

The post condition that needs to be satisfied after the program terminates is

(A) $\{r = qx + y \wedge r < y\}$
(B) $\{x = qy + r \wedge r < y\}$
(C) $\{y = qx + r \wedge 0 < r < y\}$
(D) $\{q + 1 < r - y \wedge y > 0\}$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

SESSION - 2

Graduate Aptitude Test in Engineering

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Question Paper Name: CS: COMPUTER SCIENCE AND INFORMATION TECHNOLOGY 7th Feb Shift2
Number of Questions: 65
Total Marks: 100.0

Wrong answer for MCQ will result in negative marks, (-1/3) for 1 mark Questions and (-2/3) for 2 marks Questions.

General Aptitude

Number of Questions: 10
Section Marks: 15.0

Q.1 to Q.5 carry 1 mark each & Q.6 to Q.10 carry 2 marks each.

Question Number : 1 Question Type : MCQ

We _____ our friend's birthday and we _____ how to make it up to him.

(A) completely forgot --- don't just know
(B) forgot completely --- don't just know
(C) completely forgot --- just don't know
(D) forgot completely --- just don't know

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 2 Question Type : MCQ

Choose the statement where underlined word is used correctly.

(A) The industrialist had a personnel jet.
(B) I write my experience in my personnel diary.
(C) All personnel are being given the day off.
(D) Being religious is a personnel aspect.

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 3 Question Type : MCQ

A generic term that includes various items of clothing such as a skirt, a pair of trousers and a shirt is

(A) fabric (B) textile (C) fibre (D) apparel

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 4 Question Type : MCQ

Based on the given statements, select the most appropriate option to solve the given question.

What will be the total weight of 10 poles each of same weight?

Statements:
(I) One fourth of the weight of a pole is 5 Kg.
(II) The total weight of these poles is 160 kg more than the total weight of two poles.

(A) Statement I alone is not sufficient.
(B) Statement II alone is not sufficient.
(C) Either I or II alone is sufficient.
(D) Both statements I and II together are not sufficient.

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 5 Question Type : MCQ

Consider a function $f(x) = 1 - |x|$ on $-1 \leq x \leq 1$. The value of x at which the function attains a maximum, and the maximum value of the function are:

(A) 0, −1 (B) −1, 0 (C) 0, 1 (D) −1, 2

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 6 Question Type : MCQ

Out of the following four sentences, select the most suitable sentence with respect to grammar and usage:

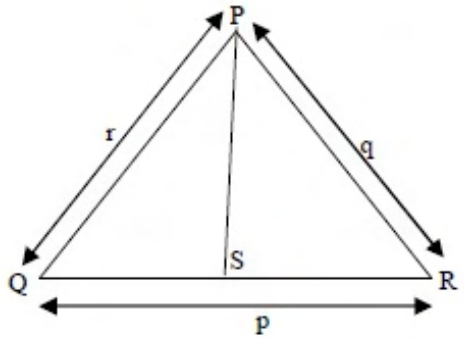
(A) Since the report lacked needed information, it was of no use to them.
(B) The report was useless to them because there were no needed information in it.
(C) Since the report did not contain the needed information, it was not real useful to them.
(D) Since the report lacked needed information, it would not had been useful to them.

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 7 Question Type : MCQ

In a triangle PQR, PS is the angle bisector of ∠QPR and ∠QPS = 60°. What is the length of PS?



(A) $\frac{(q+r)}{qr}$

(B) $\frac{qr}{(q+r)}$

(C) $\sqrt{(q^2+r^2)}$

(D) $\frac{(q+r)^2}{qr}$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 8 Question Type : NAT

If p, q, r, s are distinct integers such that:
f (p, q, r, s) = max (p, q, r, s)
g (p, q, r, s) = min (p, q, r, s)
h (p, q, r, s) = remainder of (p × q) / (r × s) if (p × q) > (r × s) or remainder of (r × s) / (p × q) if (r × s) > (p × q)

Also a function fgh (p, q, r, s) = f (p, q, r, s) × g (p, q, r, s) × h (p, q, r, s)
Also the same operations are valid with two variable functions of the form f (p, q).
What is the value of fg (h (2,5,7,3), 4,6,8)?

Correct Answer :

8

Question Number : 9 Question Type : MCQ

If the list of letters, P, R, S, T, U is an arithmetic sequence, which of the following are also in arithmetic sequence?

I. 2P, 2R, 2S, 2T, 2U
II. P−3, R−3, S−3, T−3, U−3
III. P^2, R^2, S^2, T^2, U^2

(A) I only
(B) I and II
(C) II and III
(D) I and III

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 10 Question Type : MCQ

Four branches of a company are located at M, N, O, and P. M is north of N at a distance of 4 km; P is south of O at a distance of 2 km; N is southeast of O by 1 km. What is the distance between M and P in km?

(A) 5.34 (B) 6.74 (C) 28.5 (D) 45.49

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Computer Science and Information Technology

Number of Questions:	55
Section Marks:	85.0

Q.11 to Q.35 carry 1 mark each & Q.36 to Q.65 carry 2 marks each.

Question Number : 11 Question Type : MCQ

Consider the following two statements.

*S*1: If a candidate is known to be corrupt, then he will not be elected
*S*2: If a candidate is kind, he will be elected

Which one of the following statements follows from *S*1 and *S*2 as per sound inference rules of logic?

(A) If a person is known to be corrupt, he is kind
(B) If a person is not known to be corrupt, he is not kind
(C) If a person is kind, he is not known to be corrupt
(D) If a person is not kind, he is not known to be corrupt

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 12 Question Type : NAT

The cardinality of the power set of { 0, 1, 2, …, 10 } is _____.

Correct Answer:

2048

Question Number : 13 Question Type : MCQ

Let R be the relation on the set of positive integers such that aRb if and only if a and b are distinct and have a common divisor other than 1. Which one of the following statements about R is true?

(A) R is symmetric and reflexive but not transitive
(B) R is reflexive but not symmetric and not transitive
(C) R is transitive but not reflexive and not symmetric
(D) R is symmetric but not reflexive and not transitive

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 14 Question Type : NAT

The number of divisors of 2100 is _____ .

Correct Answer :

36

Question Number : 15 Question Type : NAT

The larger of the two eigenvalues of the matrix $\begin{bmatrix} 4 & 5 \\ 2 & 1 \end{bmatrix}$ is _____.

Correct Answer :
6

Question Number : 16 Question Type : MCQ

An unordered list contains n distinct elements. The number of comparisons to find an element in this list that is neither maximum nor minimum is

(A) $\Theta(n \log n)$ (B) $\Theta(n)$ (C) $\Theta(\log n)$ (D) $\Theta(1)$

Options :
1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 17 Question Type : NAT

The minimum number of JK flip-flops required to construct a synchronous counter with the count sequence (0,0,1,1,2,2,3,3,0,0,...) is _____ .

Correct Answer:
3

Question Number : 18 Question Type : NAT

Assume that for a certain processor, a read request takes 50 nanoseconds on a cache miss and 5 nanoseconds on a cache hit. Suppose while running a program, it was observed that 80% of the processor's read requests result in a cache hit. The average read access time in nanoseconds is _____.

Correct Answer :
14

Question Number : 19 Question Type : NAT

A computer system implements a 40-bit virtual address, page size of 8 kilobytes, and a 128-entry translation look-aside buffer (TLB) organized into 32 sets each having four ways. Assume that the TLB tag does not store any process id. The minimum length of the TLB tag in bits is _____.

Correct Answer:
22

Question Number : 20 Question Type : MCQ

Consider the following statements.

I. The complement of every Turing decidable language is Turing decidable
II. There exists some language which is in NP but is not Turing decidable
III. If L is a language in NP, L is Turing decidable

Which of the above statements is/are true?

(A) Only II (B) Only III (C) Only I and II (D) Only I and III

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 21 Question Type : MCQ

Consider the following function written in the C programming language.

```
void foo(char *a){
    if ( *a && *a != ' '){
        foo(a+1);
        putchar(*a);
    }
}
```

The output of the above function on input "ABCD EFGH" is

(A) ABCD EFGH (B) ABCD (C) HGFE DCBA (D) DCBA

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 22 Question Type : MCQ

Consider a complete binary tree where the left and the right subtrees of the root are max-heaps. The lower bound for the number of operations to convert the tree to a heap is

(A) $\Omega(\log n)$ (B) $\Omega(n)$ (C) $\Omega(n \log n)$ (D) $\Omega(n^2)$

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 23 Question Type : NAT

A binary tree T has 20 leaves. The number of nodes in T having two children is _____.

Correct Answer :

19

Question Number : 24 Question Type : NAT

Consider the following C function.

```
int fun(int n){
      int x=1,k;
      if (n==1) return x;
      for (k=1; k<n; ++k)
            x = x + fun(k) * fun(n-k);
      return x;
}
```

The return value of `fun(5)` is _____.

Correct Answer :

51

Question Number : 25 Question Type : MCQ

A software requirements specification (SRS) document should avoid discussing which one of the following?

(A) User interface issues
(B) Non-functional requirements
(C) Design specification
(D) Interfaces with third party software

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 26 Question Type : MCQ

Consider two decision problems Q_1, Q_2 such that Q_1 reduces in polynomial time to 3-SAT and 3-SAT reduces in polynomial time to Q_2. Then which one of the following is consistent with the above statement?

(A) Q_1 is in NP, Q_2 is NP hard.
(B) Q_2 is in NP, Q_1 is NP hard.
(C) Both Q_1 and Q_2 are in NP .
(D) Both Q_1 and Q_2 are NP hard.

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 27 Question Type : MCQ

Match the following:

P. Lexical analysis	1. Graph coloring
Q. Parsing	2. DFA minimization
R. Register allocation	3. Post-order traversal
S. Expression evaluation	4. Production tree

(A) P-2, Q-3, R-1, S-4
(B) P-2, Q-1, R-4, S-3
(C) P-2, Q-4, R-1, S-3
(D) P-2, Q-3, R-4, S-1

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 28 Question Type : MCQ

In the context of abstract-syntax-tree (AST) and control-flow-graph (CFG), which one of the following is TRUE?

(A) In both AST and CFG, let node N_2 be the successor of node N_1. In the input program, the code corresponding to N_2 is present after the code corresponding to N_1
(B) For any input program, neither AST nor CFG will contain a cycle
(C) The maximum number of successors of a node in an AST and a CFG depends on the input program
(D) Each node in AST and CFG corresponds to at most one statement in the input program

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 29 Question Type : MCQ

Consider the basic COCOMO model where E is the effort applied in person-months, D is the development time in chronological months, $KLOC$ is the estimated number of delivered lines of code (in thousands) and a_b, b_b, c_b, d_b have their usual meanings. The basic COCOMO equations are of the form

(A) $E = a_b(KLOC)\exp(b_b), D = c_b(E)\exp(d_b)$
(B) $D = a_b(KLOC)\exp(b_b), E = c_b(D)\exp(d_b)$
(C) $E = a_b\exp(b_b), D = c_b(KLOC)\exp(d_b)$
(D) $E = a_b\exp(d_b), D = c_b(KLOC)\exp(b_b)$

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 30 Question Type : MCQ

A system has 6 identical resources and N processes competing for them. Each process can request atmost 2 resources. Which one of the following values of N could lead to a deadlock?

(A) 1 (B) 2 (C) 3 (D) 4

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 31 Question Type : MCQ

Consider the following transaction involving two bank accounts x and y.

```
read(x); x := x - 50; write(x); read(y); y:= y + 50; write(y)
```

The constraint that the sum of the accounts x and y should remain constant is that of

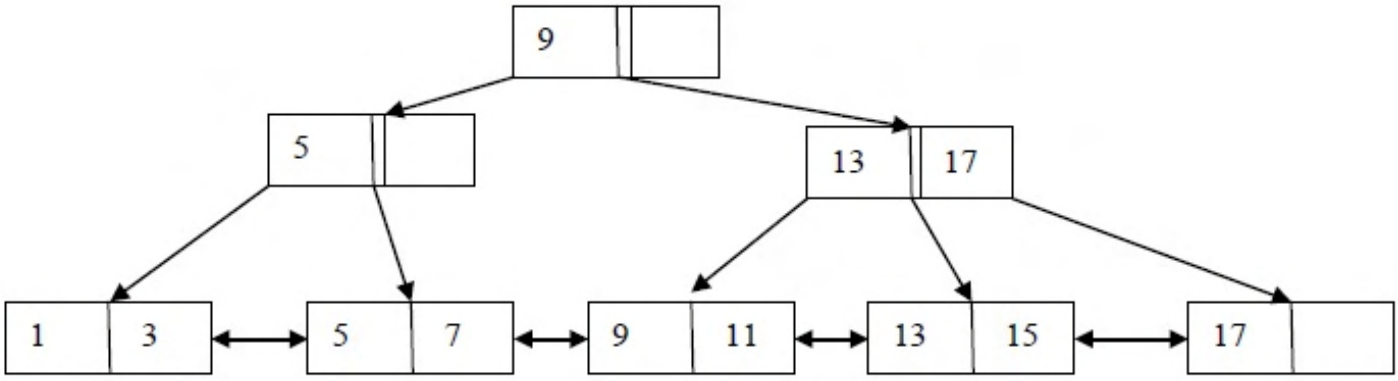
(A) Atomicity
(B) Consistency
(C) Isolation
(D) Durability

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 32 Question Type : NAT

With reference to the B+ tree index of order 1 shown below, the minimum number of nodes (including the Root node) that must be fetched in order to satisfy the following query: "Get all records with a search key greater than or equal to 7 and less than 15" is _____.



Correct Answer :
5

Question Number : 33 Question Type : MCQ

Identify the correct order in which a server process must invoke the function calls `accept`, `bind`, `listen`, and `recv` according to UNIX socket API.

(A) listen, accept, bind, recv
(B) bind, listen, accept, recv
(C) bind, accept, listen, recv
(D) accept, listen, bind, recv

Options :
1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 34 Question Type : NAT

A link has a transmission speed of 10^6 bits/sec. It uses data packets of size 1000 bytes each. Assume that the acknowledgment has negligible transmission delay, and that its propagation delay is the same as the data propagation delay. Also assume that the processing delays at nodes are negligible. The efficiency of the stop-and-wait protocol in this setup is exactly 25%. The value of the one-way propagation delay (in milliseconds) is _____.

Correct Answer :
12

Question Number : 35 Question Type : MCQ

Which one of the following statements is NOT correct about HTTP cookies?

(A) A cookie is a piece of code that has the potential to compromise the security of an Internet user
(B) A cookie gains entry to the user's work area through an HTTP header
(C) A cookie has an expiry date and time
(D) Cookies can be used to track the browsing pattern of a user at a particular site

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 36 Question Type : MCQ

Consider the following routing table at an IP router:

Network No.	Net Mask	Next Hop
128.96.170.0	255.255.254.0	Interface 0
128.96.168.0	255.255.254.0	Interface 1
128.96.166.0	255.255.254.0	R2
128.96.164.0	255.255.252.0	R3
0.0.0.0	Default	R4

For each IP address in Group I identify the correct choice of the next hop from Group II using the entries from the routing table above.

Group I
i) 128.96.171.92
ii) 128.96.167.151
iii) 128.96.163.151
iv) 128.96.165.121

Group II
a) Interface 0
b) Interface 1
c) R2
d) R3
e) R4

(A) i-a, ii-c, iii-e, iv-d
(B) i-a, ii-d, iii-b, iv-e
(C) i-b, ii-c, iii-d, iv-e
(D) i-b, ii-c, iii-e, iv-d

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 37 Question Type : MCQ

Host A sends a UDP datagram containing 8880 bytes of user data to host B over an Ethernet LAN. Ethernet frames may carry data up to 1500 bytes (i.e. MTU=1500 bytes). Size of UDP header is 8 bytes and size of IP header is 20 bytes. There is no option field in IP header. How many total number of IP fragments will be transmitted and what will be the contents of offset field in the last fragment?

(A) 6 and 925
(B) 6 and 7400
(C) 7 and 1110
(D) 7 and 8880

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 38 Question Type : MCQ

Assume that the bandwidth for a TCP connection is 1048560 bits /sec. Let α be the value of RTT in milliseconds (rounded off to the nearest integer) after which the TCP window scale option is needed. Let β be the maximum possible window size with window scale option. Then the values of α and β are

(A) 63 milliseconds, 65535×2^{14}
(B) 63 milliseconds, 65535×2^{16}
(C) 500 milliseconds, 65535×2^{14}
(D) 500 milliseconds, 65535×2^{16}

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 39 Question Type : MCQ

Consider a simple checkpointing protocol and the following set of operations in the log.

(start, T4); (write, T4, *y*, 2, 3); (start, T1); (commit, T4); (write, T1, *z*, 5, 7);
(checkpoint);
(start, T2); (write, T2, *x*, 1, 9); (commit, T2); (start, T3), (write, T3, *z*, 7, 2);

If a crash happens now and the system tries to recover using both undo and redo operations, what are the contents of the undo list and the redo list?

(A) Undo: T3, T1; Redo: T2
(B) Undo: T3, T1; Redo: T2, T4
(C) Undo: none; Redo: T2, T4, T3, T1
(D) Undo: T3, T1, T4; Redo: T2

Options :

1. ✔ A
2. ✖ B
3. ✖ C

4. ✖ D

Question Number : 40 Question Type : MCQ

Consider two relations $R_1(A,B)$ with the tuples (1,5), (3,7) and $R_2(A,C) = (1,7), (4,9)$. Assume that R(A,B,C) is the full natural outer join of R_1 and R_2. Consider the following tuples of the form (A,B,C): $a = (1,5,null)$, $b = (1,null,7)$, $c = (3, null, 9)$, $d = (4,7,null)$, $e = (1,5,7)$, $f = (3,7,null)$, $g = (4,null,9)$. Which one of the following statements is correct?

(A) R contains *a, b, e, f, g* but not *c, d.*
(B) R contains all of *a, b, c, d, e, f, g.*
(C) R contains *e, f, g* but not *a, b.*
(D) R contains *e* but not *f, g.*

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 41 Question Type : MCQ

Consider six memory partitions of sizes 200 KB, 400 KB, 600 KB, 500 KB, 300 KB and 250 KB, where KB refers to kilobyte. These partitions need to be allotted to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. If the best fit algorithm is used, which partitions are NOT allotted to any process?

(A) 200 KB and 300 KB
(B) 200 KB and 250 KB
(C) 250 KB and 300 KB
(D) 300 KB and 400 KB

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 42 Question Type : NAT

Consider a typical disk that rotates at 15000 rotations per minute (RPM) and has a transfer rate of 50×10^6 bytes/sec. If the average seek time of the disk is twice the average rotational delay and the controller's transfer time is 10 times the disk transfer time, the average time (in milliseconds) to read or write a 512-byte sector of the disk is _____.

Correct Answer :
6.1 to 6.2

Question Number : 43 Question Type : NAT

A computer system implements 8 kilobyte pages and a 32-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the virtual address supported by the system is _____ bits.

Correct Answer :
36

Question Number : 44 Question Type : MCQ

Consider the intermediate code given below.

```
(1)   i = 1
(2)   j = 1
(3)   t1 = 5 * i
(4)   t2 = t1 + j
(5)   t3 = 4 * t2
(6)   t4 = t3
(7)   a[t4] = -1
(8)   j = j + 1
(9)   if j<=5 goto (3)
(10)  i=i+1
(11)  if i<5 goto (2)
```

The number of nodes and edges in the control-flow-graph constructed for the above code, respectively, are

(A) 5 and 7 (B) 6 and 7 (C) 5 and 5 (D) 7 and 8

Options :
1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 45 Question Type : NAT

The number of states in the minimal deterministic finite automaton corresponding to the regular expression $(0+1)^*(10)$ is _____.

Correct Answer :
3

Question Number : 46 Question Type : MCQ

Which of the following languages is/are regular?

L_1: $\{wxw^R \mid w, x \in \{a, b\}^*$ and $|w|, |x| > 0\}$, w^R is the reverse of string w
L_2: $\{a^n b^m \mid m \neq n$ and $m, n \geq 0\}$
L_3: $\{a^p b^q c^r \mid p, q, r \geq 0\}$

(A) L_1 and L_3 only (B) L_2 only (C) L_2 and L_3 only (D) L_3 only

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 47 Question Type : MCQ

Given below are some algorithms, and some algorithm design paradigms.

1. Dijkstra's Shortest Path	i. Divide and Conquer
2. Floyd-Warshall algorithm to compute all pairs shortest path	ii. Dynamic Programming
3. Binary search on a sorted array	iii. Greedy design
4. Backtracking search on a graph	iv. Depth-first search
	v. Breadth-first search

Match the above algorithms on the left to the corresponding design paradigm they follow.

(A) 1-i, 2-iii, 3-i, 4-v.
(B) 1-iii, 2-iii, 3-i, 4-v.
(C) 1-iii, 2-ii, 3-i, 4-iv.
(D) 1-iii, 2-ii, 3-i, 4-v.

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 48 Question Type : NAT

A Young tableau is a 2D array of integers increasing from left to right and from top to bottom. Any unfilled entries are marked with ∞, and hence there cannot be any entry to the right of, or below a ∞. The following Young tableau consists of unique entries.

1	2	5	14
3	4	6	23
10	12	18	25
31	∞	∞	∞

When an element is removed from a Young tableau, other elements should be moved into its place so that the resulting table is still a Young tableau (unfilled entries may be filled in with a ∞). The minimum number of entries (other than 1) to be shifted, to remove 1 from the given Young tableau is _____.

Correct Answer :

5

Question Number : 49 Question Type : MCQ

Suppose you are provided with the following function declaration in the C programming language.

```
int partition(int a[], int n);
```

The function treats the first element of `a[]` as a pivot, and rearranges the array so that all elements less than or equal to the pivot is in the left part of the array, and all elements greater than the pivot is in the right part. In addition, it moves the pivot so that the pivot is the last element of the left part. The return value is the number of elements in the left part.

The following partially given function in the C programming language is used to find the k^{th} smallest element in an array `a[]` of size `n` using the partition function. We assume $k \leq n$.

```
int kth_smallest(int a[], int n, int k)
{
   int left_end = partition(a,n);

   if ( left_end+1 == k ){
       return a[left_end];
   }

   if ( left_end+1 > k ){
       return kth_smallest( _____________________ );
   } else {
       return kth_smallest( _____________________ );
   }
}
```

The missing argument lists are respectively

(A) `(a, left_end, k)` and `(a+left_end+1, n-left_end-1, k-left_end-1)`
(B) `(a, left_end, k)` and `(a, n-left_end-1, k-left_end-1)`
(C) `(a+left_end+1, n-left_end-1, k-left_end-1)` and `(a, left_end, k)`
(D) `(a, n-left_end-1, k-left_end-1)` and `(a, left_end, k)`

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 50 Question Type : MCQ

Which one of the following hash functions on integers will distribute keys most uniformly over 10 buckets numbered 0 to 9 for i ranging from 0 to 2020?

(A) $h(i) = i^2 \bmod 10$
(B) $h(i) = i^3 \bmod 10$
(C) $h(i) = (11 * i^2) \bmod 10$
(D) $h(i) = (12 * i) \bmod 10$

Options :

1. ✖ A
2. ✔ B

3. ✖ C
4. ✖ D

Question Number : 51 Question Type : MCQ

The secant method is used to find the root of an equation f(x) = 0. It is started from two distinct estimates x_a and x_b for the root. It is an iterative procedure involving linear interpolation to a root. The iteration stops if $f(x_b)$ is very small and then x_b is the solution. The procedure is given below. Observe that there is an expression which is missing and is marked by ?. Which is the suitable expression that is to be put in place of ? so that it follows all steps of the secant method?

Secant

```
Initialize: xa, xb, ε, N        // ε = convergence indicator
                                 // N = maximum no. of iterations
     fb = f(xb)
     i = 0
     while (i < N and  |fb| >  ε) do
          i = i + 1               // update counter
          xt = ?                  // missing expression for
                                  // intermediate value
          xa = xb                 // reset xa
          xb = xt                 //  reset xb
          fb = f(xb)              // function value at new xb
     end while
     if  |fb| >  ε then           // loop is terminated with i=N
          write "Non-convergence"
     else
           write "return xb"
     end if
```

(A) $x_b - (f_b - f(x_a))\, f_b / (x_b - x_a)$
(B) $x_a - (f_a - f(x_a))\, f_a / (x_b - x_a)$
(C) $x_b - (x_b - x_a)\, f_b / (f_b - f(x_a))$
(D) $x_a - (x_b - x_a)\, f_a / (f_b - f(x_a))$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 52 Question Type : NAT

Consider the C program below.

```
#include <stdio.h>
int *A, stkTop;

int stkFunc(int opcode, int val)
{
      static int size=0, stkTop=0;

      switch (opcode) {
            case -1: size = val; break;
            case  0: if (stkTop < size) A[stkTop++] = val; break;
            default: if (stkTop) return A[--stkTop];
      }
      return -1;
}

int main()
{
      int B[20]; A = B; stkTop = -1;

      stkFunc (-1, 10);
      stkFunc ( 0, 5);
      stkFunc ( 0, 10);
      printf ("%d\n", stkFunc(1, 0) + stkFunc(1, 0));
}
```

The value printed by the above program is _____.

Correct Answer :

15

Question Number : 53 Question Type : NAT

Consider the sequence of machine instructions given below:

```
MUL    R5, R0, R1
DIV    R6, R2, R3
ADD    R7, R5, R6
SUB    R8, R7, R4
```

In the above sequence, R0 to R8 are general purpose registers. In the instructions shown, the first register stores the result of the operation performed on the second and the third registers. This sequence of instructions is to be executed in a pipelined instruction processor with the following 4 stages: (1) Instruction Fetch and Decode (IF), (2) Operand Fetch (OF), (3) Perform Operation (PO) and (4) Write back the result (WB). The IF, OF and WB stages take 1 clock cycle each for any instruction. The PO stage takes 1 clock cycle for ADD or SUB instruction, 3 clock cycles for MUL instruction and 5 clock cycles for DIV instruction. The pipelined processor uses operand forwarding from the PO stage to the OF stage. The number of clock cycles taken for the execution of the above sequence of instructions is _____.

Correct Answer :

13

Question Number : 54 Question Type : MCQ

Consider a processor with byte-addressable memory. Assume that all registers, including Program Counter (PC) and Program Status Word (PSW), are of size 2 bytes. A stack in the main memory is implemented from memory location $(0100)_{16}$ and it grows upward. The stack pointer (SP) points to the top element of the stack. The current value of SP is $(016E)_{16}$. The CALL instruction is of two words, the first word is the op-code and the second word is the starting address of the subroutine (one word = 2 bytes). The CALL instruction is implemented as follows:

- Store the current value of PC in the stack
- Store the value of PSW register in the stack
- Load the starting address of the subroutine in PC

The content of PC just before the fetch of a CALL instruction is $(5FA0)_{16}$. After execution of the CALL instruction, the value of the stack pointer is

(A) $(016A)_{16}$ (B) $(016C)_{16}$ (C) $(0170)_{16}$ (D) $(0172)_{16}$

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 55 Question Type : NAT

The number of min-terms after minimizing the following Boolean expression is _____.

$$[D' + AB' + A'C + AC'D + A'C'D]'$$

Correct Answer :
1

Question Number : 56 Question Type : MCQ

Let $f(x) = x^{-(1/3)}$ and A denote the area of the region bounded by $f(x)$ and the X-axis, when x varies from -1 to 1. Which of the following statements is/are TRUE?

I) f is continuous in $[-1, 1]$
II) f is not bounded in $[-1, 1]$
III) A is nonzero and finite

(A) II only
(B) III only
(C) II and III only
(D) I, II and III

Options :
1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 57 Question Type : NAT

Perform the following operations on the matrix $\begin{bmatrix} 3 & 4 & 45 \\ 7 & 9 & 105 \\ 13 & 2 & 195 \end{bmatrix}$.

(i) Add the third row to the second row
(ii) Subtract the third column from the first column.

The determinant of the resultant matrix is_____.

Correct Answer :
0

Question Number : 58 Question Type : NAT

The number of onto functions (surjective functions) from set $X = \{1, 2, 3, 4\}$ to set $Y = \{a, b, c\}$ is _____.

Correct Answer :
36

Question Number : 59 Question Type : NAT

Let X and Y denote the sets containing 2 and 20 distinct objects respectively and F denote the set of all possible functions defined from X to Y. Let f be randomly chosen from F. The probability of f being one-to-one is _____.

Correct Answer :
0.95

Question Number : 60 Question Type : MCQ

Consider the alphabet $\sum = \{0, 1\}$, the null/empty string λ and the sets of strings X_0, X_1, and X_2 generated by the corresponding non-terminals of a regular grammar. X_0, X_1, and X_2 are related as follows.

$$X_0 = 1\,X_1$$
$$X_1 = 0\,X_1 + 1\,X_2$$
$$X_2 = 0\,X_1 + \{\lambda\}$$

Which one of the following choices precisely represents the strings in X_0?

(A) 10(0* + (10)*)1
(B) 10(0* + (10)*)*1
(C) 1(0 + 10)*1
(D) 10(0 + 10)*1 + 110(0 + 10)*1

Options :
1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 61 Question Type : MCQ

A graph is self-complementary if it is isomorphic to its complement. For all self-complementary graphs on n vertices, n is

(A) A multiple of 4
(B) Even
(C) Odd
(D) Congruent to 0 *mod* 4, or, 1 *mod* 4.

Options :
1. ✖ A

2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 62 Question Type : MCQ

In a connected graph, a bridge is an edge whose removal disconnects a graph. Which one of the following statements is true?

(A) A tree has no bridges
(B) A bridge cannot be part of a simple cycle
(C) Every edge of a clique with size ≥ 3 is a bridge (A clique is any complete subgraph of a graph)
(D) A graph with bridges cannot have a cycle

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 63 Question Type : MCQ

Which one of the following well formed formulae is a tautology?

(A) $\forall x\, \exists y\, R(x,y) \leftrightarrow \exists y\, \forall x\, R(x,y)$
(B) $(\forall x\, [\exists y\, R(x,y) \rightarrow S(x,y)]) \rightarrow \forall x \exists y\, S(x,y)$
(C) $[\forall x\, \exists y\, (P(x,y) \rightarrow R(x,y)] \leftrightarrow [\forall x\, \exists y\, (\neg P(x,y) \vee R(x,y)]$
(D) $\forall x\, \forall y\, P(x,y) \rightarrow \forall x\, \forall y\, P(y,x)$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 64 Question Type : MCQ

Which one of the following assertions concerning code inspection and code walkthrough is true?

(A) Code inspection is carried out once the code has been unit tested
(B) Code inspection and code walkthrough are synonyms
(C) Adherence to coding standards is checked during code inspection
(D) Code walkthrough is usually carried out by an independent test team

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 65 Question Type : NAT

A half adder is implemented with XOR and AND gates. A full adder is implemented with two half adders and one OR gate. The propagation delay of an XOR gate is twice that of an AND/OR gate. The propagation delay of an AND/OR gate is 1.2 microseconds. A 4-bit ripple-carry binary adder is implemented by using four full adders. The total propagation time of this 4-bit binary adder in microseconds is_____.

Correct Answer:

19.2

SESSION - 3

Graduate Aptitude Test in Engineering

Notations :

1.Options shown in green color and with ✔ icon are correct.

2.Options shown in red color and with ✖ icon are incorrect.

Question Paper Name: CS: COMPUTER SCIENCE AND INFORMATION TECHNOLOGY 8th Feb Shift1
Number of Questions: 65
Total Marks: 100.0

Wrong answer for MCQ will result in negative marks, (-1/3) for 1 mark Questions and (-2/3) for 2 marks Questions.

General Aptitude

Number of Questions: 10
Section Marks: 15.0

Q.1 to Q.5 carry 1 mark each & Q.6 to Q.10 carry 2 marks each.

Question Number : 1 Question Type : MCQ

Extreme focus on syllabus and studying for tests has become such a dominant concern of Indian students that they close their minds to anything _____ to the requirements of the exam.

(A) related (B) extraneous (C) outside (D) useful

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 2 Question Type : MCQ

Select the pair that best expresses a relationship similar to that expressed in the pair:

Children : Pediatrician

(A) Adult : Orthopaedist
(B) Females : Gynaecologist
(C) Kidney : Nephrologist
(D) Skin : Dermatologist

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 3 Question Type : MCQ

The Tamil version of _____ John Abraham-starrer *Madras Cafe* _____cleared by the Censor Board with no cuts last week, but the film's distributors _____ no takers among the exhibitors for a release in Tamil Nadu _____ this Friday.

(A) Mr., was, found, on
(B) a, was, found, at
(C) the, was, found, on
(D) a, being, find at

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 4 Question Type : MCQ

If ROAD is written as URDG, then SWAN should be written as:

(A) VXDQ
(B) VZDQ
(C) VZDP
(D) UXDQ

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 5 Question Type : MCQ

A function f(x) is linear and has a value of 29 at $x = -2$ and 39 at $x = 3$. Find its value at $x = 5$.

(A) 59 (B) 45 (C) 43 (D) 35

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 6 Question Type : MCQ

Alexander turned his attention towards India, since he had conquered Persia.

Which one of the statements below is logically valid and can be inferred from the above sentence?

(A) Alexander would not have turned his attention towards India had he not conquered Persia.

(B) Alexander was not ready to rest on his laurels, and wanted to march to India.

(C) Alexander was completely in control of his army and could command it to move towards India.

(D) Since Alexander's kingdom extended to Indian borders after the conquest of Persia, he was keen to move further.

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 7 Question Type : MCQ

Most experts feel that in spite of possessing all the technical skills required to be a batsman of the highest order, he is unlikely to be so due to lack of requisite temperament. He was guilty of throwing away his wicket several times after working hard to lay a strong foundation. His critics pointed out that until he addressed this problem, success at the highest level will continue to elude him.

Which of the statement(s) below is/are logically valid and can be inferred from the above passage?

(i) He was already a successful batsman at the highest level.
(ii) He has to improve his temperament in order to become a great batsman.
(iii) He failed to make many of his good starts count.
(iv) Improving his technical skills will guarantee success.

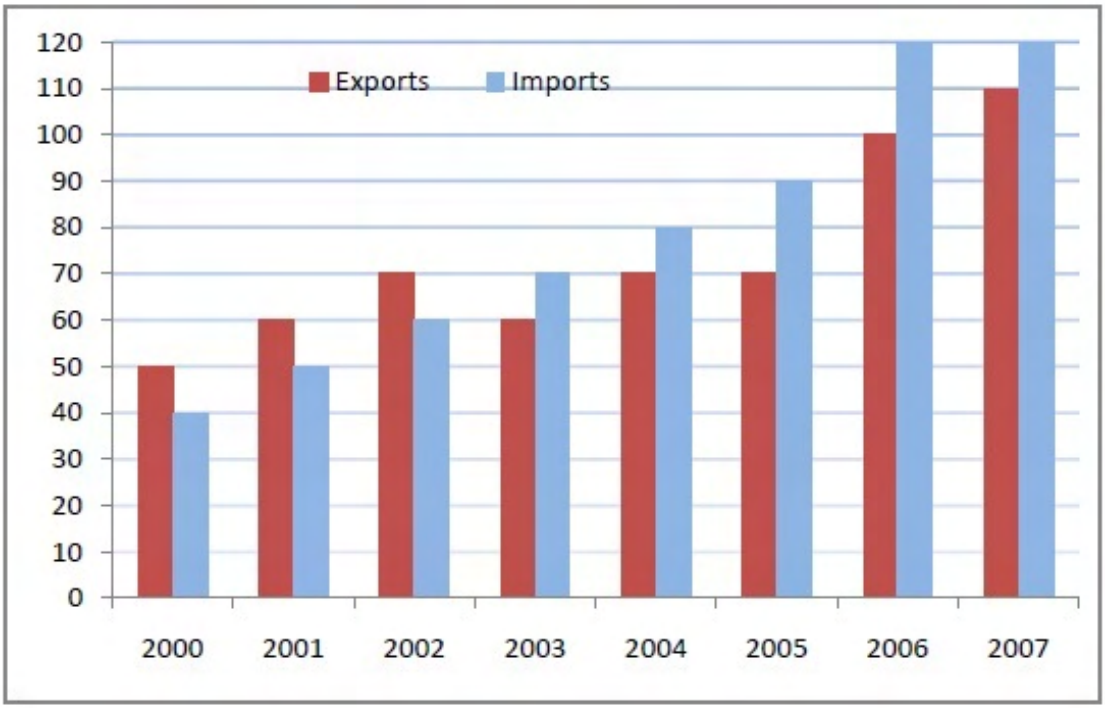
(A) (iii) and (iv)
(B) (ii) and (iii)
(C) (i), (ii) and (iii)
(D) (ii) only

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 8 Question Type : NAT

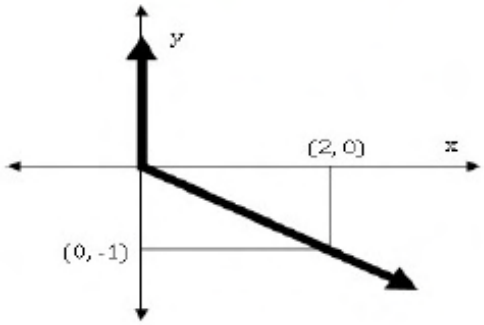
The exports and imports (in crores of Rs.) of a country from the year 2000 to 2007 are given in the following bar chart. In which year is the combined percentage increase in imports and exports the highest?



Correct Answer:
2006

Question Number : 9 Question Type : MCQ

Choose the most appropriate equation for the function drawn as a thick line, in the plot below.



(A) $x = y - |y|$ (B) $x = -(y - |y|)$ (C) $x = y + |y|$ (D) $x = -(y + |y|)$

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 10 Question Type : MCQ

The head of a newly formed government desires to appoint five of the six selected members P, Q, R, S, T, and U to portfolios of Home, Power, Defense, Telecom, and Finance. U does not want any portfolio if S gets one of the five. R wants either Home or Finance or no portfolio. Q says that if S gets either Power or Telecom, then she must get the other one. T insists on a portfolio if P gets one.

Which is the valid distribution of portfolios?

(A) P-Home, Q-Power, R-Defense, S-Telecom, T-Finance
(B) R-Home, S-Power, P-Defense, Q-Telecom, T-Finance
(C) P-Home, Q-Power, T-Defense, S-Telecom, U-Finance
(D) Q-Home, U-Power, T-Defense, R-Telecom, P-Finance

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Computer Science and Information Technology

Number of Questions:	55
Section Marks:	85.0

Q.11 to Q.35 carry 1 mark each & Q.36 to Q.65 carry 2 marks each.

Question Number : 11 Question Type : MCQ

Consider the following C program segment.

```
#include <stdio.h>

int main()
{
      char s1[7] = "1234", *p;
      p = s1 + 2;
      *p = '0';
      printf("%s", s1);
}
```

What will be printed by the program?

(A) 12 (B) 120400 (C) 1204 (D) 1034

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 12 Question Type : MCQ

Suppose U is the power set of the set $S = \{1,2,3,4,5,6\}$. For any $T \in U$, let $|T|$ denote the number of elements in T and T' denote the complement of T. For any $T, R \in U$, let $T \setminus R$ be the set of all elements in T which are not in R. Which one of the following is true?

(A) $\forall X \in U\ (|X| = |X'|)$
(B) $\exists X \in U\ \exists Y \in U\ (|X| = 5, |Y| = 5 \text{ and } X \cap Y = \emptyset)$
(C) $\forall X \in U\ \forall Y \in U\ (|X| = 2, |Y| = 3 \text{ and } X \setminus Y = \emptyset)$
(D) $\forall X \in U\ \forall Y \in U\ (X \setminus Y = Y' \setminus X')$

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 13 Question Type : MCQ

Consider the relation $X(P, Q, R, S, T, U)$ with the following set of functional dependencies

$$F = \{ \{P,R\} \rightarrow \{S,T\}, \{P,S,U\} \rightarrow \{Q,R\} \}$$

Which of the following is the trivial functional dependency in F^+, where F^+ is closure of F ?

(A) $\{P,R\} \rightarrow \{S,T\}$ (B) $\{P,R\} \rightarrow \{R,T\}$ (C) $\{P,S\} \rightarrow \{S\}$ (D) $\{P,S,U\} \rightarrow \{Q\}$

Options :

1. ✖ A

2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 14 Question Type : MCQ

The maximum number of processes that can be in *Ready* state for a computer system with n CPUs is

(A) n (B) n^2 (C) 2^n (D) Independent of n

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 15 Question Type : MCQ

Among simple LR (SLR) , canonical LR, and look-ahead LR (LALR), which of the following pairs identify the method that is very easy to implement and the method that is the most powerful , in that order?

(A) SLR, LALR
(B) Canonical LR, LALR
(C) SLR, canonical LR
(D) LALR, canonical LR

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 16 Question Type : MCQ

Let # be a binary operator defined as

$X \# Y = X' + Y'$ where X and Y are Boolean variables.

Consider the following two statements.

(S1) $(P \# Q)\#R = P\#(Q \# R)$
(S2) $Q\#R = R\#Q$

Which of the following is/are true for the Boolean variables P, Q and R?

(A) Only S1 is true
(B) Only S2 is true
(C) Both S1 and S2 are true
(D) Neither S1 nor S2 are true

Options :

1. ✖ A

2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 17 Question Type : NAT

Consider a software project with the following information domain characteristics for calculation of function point metric.

Number of external inputs (I) = 30
Number of external outputs (O) = 60
Number of external inquiries (E) = 23
Number of files (F) = 08
Number of external interfaces (N) = 02

It is given that the complexity weighting factors for I, O, E, F and N are 4, 5, 4, 10 and 7, respectively. It is also given that, out of fourteen value adjustment factors that influence the development effort, four factors are not applicable, each of the other four factors have value 3, and each of the remaining factors have value 4. The computed value of function point metric is _____.

Correct Answer:
612 to 613

Question Number : 18 Question Type : MCQ

In a web server, ten WebPages are stored with the URLs of the form http://www.yourname.com/*var*.html; where, *var* is a different number from 1 to 10 for each Webpage. Suppose, the client stores the Webpage with *var* = 1 (say W1) in local machine, edits and then tests. Rest of the WebPages remains on the web server. W1 contains several relative URLs of the form "*var*.html" referring to the other WebPages. Which one of the following statements needs to be added in W1, so that all the relative URLs in W1 refer to the appropriate WebPages on the web server?

(A) <a href: "http://www.yourname.com/", href: "...*var*.html">
(B) <base href: "http://www.yourname.com/">
(C) <a href: "http://www.yourname.com/">
(D) <base href: "http://www.yourname.com/", range:"…*var*.html">

Options :
1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 19 Question Type : MCQ

Consider the following statements.

I. TCP connections are full duplex
II. TCP has no option for selective acknowledgement
III. TCP connections are message streams

(A) Only I is correct
(B) Only I and III are correct
(C) Only II and III are correct
(D) All of I, II and III are correct

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 20 Question Type : MCQ

Consider the equality $\sum_{i=0}^{n} i^3 = X$ and the following choices for X

I. $\Theta(n^4)$
II. $\Theta(n^5)$
III. $O(n^5)$
IV. $\Omega(n^3)$

The equality above remains correct if X is replaced by

(A) Only I
(B) Only II
(C) I or III or IV but not II
(D) II or III or IV but not I

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 21 Question Type : NAT

Consider a binary tree T that has 200 leaf nodes. Then, the number of nodes in T that have exactly two children are _____.

Correct Answer :

199

Question Number : 22 Question Type : NAT

Given a hash table T with 25 slots that stores 2000 elements, the load factor α for T is _____.

Correct Answer :
80

Question Number : 23 Question Type : MCQ

In the given matrix $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}$, one of the eigenvalues is 1. The eigenvectors corresponding to the eigenvalue 1 are

(A) $\{\alpha(4,2,1) | \alpha \neq 0, \alpha \in \mathbb{R}\}$
(B) $\{\alpha(-4,2,1) | \alpha \neq 0, \alpha \in \mathbb{R}\}$
(C) $\{\alpha(\sqrt{2},0,1) | \alpha \neq 0, \alpha \in \mathbb{R}\}$
(D) $\{\alpha(-\sqrt{2},0,1) | \alpha \neq 0, \alpha \in \mathbb{R}\}$

Options :
1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 24 Question Type : MCQ

The value of $\lim_{x \to \infty} (1 + x^2)^{e^{-x}}$ is

(A) 0 (B) $\frac{1}{2}$ (C) 1 (D) ∞

Options :
1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 25 Question Type : NAT

The number of 4 digit numbers having their digits in non-decreasing order (from left to right) constructed by using the digits belonging to the set {1, 2, 3} is_____.

Correct Answer :
15

Question Number : 26 Question Type : MCQ

In a room there are only two types of people, namely Type 1 and Type 2. Type 1 people always tell the truth and Type 2 people always lie. You give a fair coin to a person in that room, without knowing which type he is from and tell him to toss it and hide the result from you till you ask for it. Upon asking, the person replies the following

"The result of the toss is head if and only if I am telling the truth."

Which of the following options is correct?

(A) The result is head
(B) The result is tail
(C) If the person is of Type 2, then the result is tail
(D) If the person is of Type 1, then the result is tail

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 27 Question Type : MCQ

While inserting the elements 71, 65, 84, 69, 67, 83 in an empty binary search tree (BST) in the sequence shown, the element in the lowest level is

(A) 65 (B) 67 (C) 69 (D) 83

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 28 Question Type : MCQ

The result evaluating the postfix expression 10 5 + 60 6 / * 8 − is

(A) 284 (B) 213 (C) 142 (D) 71

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 29 Question Type : MCQ

Consider the following relation

Cinema(*theater*, *address*, *capacity*)

Which of the following options will be needed at the end of the SQL query

```
SELECT P1.address
FROM Cinema P1
```

such that it always finds the addresses of theaters with maximum capacity?

(A) WHERE P1.*capacity* >= All (select P2.*capacity* from Cinema P2)
(B) WHERE P1.*capacity* >= Any (select P2.*capacity* from Cinema P2)
(C) WHERE P1.*capacity* > All (select max(P2.*capacity*) from Cinema P2)
(D) WHERE P1.*capacity* > Any (select max(P2.*capacity*) from Cinema P2)

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 30 Question Type : MCQ

Consider the following array of elements.

$\langle 89, 19, 50, 17, 12, 15, 2, 5, 7, 11, 6, 9, 100 \rangle$

The minimum number of interchanges needed to convert it into a max-heap is

(A) 4 (B) 5 (C) 2 (D) 3

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 31 Question Type : MCQ

Two processes X and Y need to access a critical section. Consider the following synchronization construct used by both the processes

Process X	Process Y

`/* other code for process X */`
`while(true)`
`{`
`  varP = true;`
`  while(varQ == true)`
`  {`
`  /* Critical Section */`
`  varP = false;`
`  }`
`}`
`/* other code for process X */`
	`/* other code for process Y */`
	`while(true)`
	`{`
	`  varQ = true;`
	`  while(varP == true)`
	`  {`
	`  /* Critical Section */`
	`  varQ = false;`
	`  }`
	`}`
	`/* other code for process Y */`

Here, `varP` and `varQ` are shared variables and both are initialized to false. Which one of the following statements is true?

(A) The proposed solution prevents deadlock but fails to guarantee mutual exclusion
(B) The proposed solution guarantees mutual exclusion but fails to prevent deadlock
(C) The proposed solution guarantees mutual exclusion and prevents deadlock
(D) The proposed solution fails to prevent deadlock and fails to guarantee mutual exclusion

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 32 Question Type : MCQ

Let L be the language represented by the regular expression $\Sigma^*0011\Sigma^*$ where $\Sigma = \{0, 1\}$. What is the minimum number of states in a DFA that recognizes $\bar{L}$ (complement of L)?

(A) 4 (B) 5 (C) 6 (D) 8

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 33 Question Type : NAT

Consider a software program that is artificially seeded with 100 faults. While testing this program, 159 faults are detected, out of which 75 faults are from those artificially seeded faults. Assuming that both real and seeded faults are of same nature and have same distribution, the estimated number of undetected real faults is _____.

Correct Answer :
28

Question Number : 34 Question Type : MCQ

Consider a machine with a byte addressable main memory of 2^{20} bytes, block size of 16 bytes and a direct mapped cache having 2^{12} cache lines. Let the addresses of two consecutive bytes in main memory be $(E201F)_{16}$ and $(E2020)_{16}$. What are the tag and cache line address (in hex) for main memory address $(E201F)_{16}$?

(A) E, 201 (B) F, 201 (C) E, E20 (D) 2, 01F

Options :
1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 35 Question Type : MCQ

Consider a CSMA/CD network that transmits data at a rate of 100 Mbps (10^8 bits per second) over a 1 km (kilometer) cable with no repeaters. If the minimum frame size required for this network is 1250 bytes, what is the signal speed (km/sec) in the cable?

(A) 8000 (B) 10000 (C) 16000 (D) 20000

Options :
1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 36 Question Type : NAT

The velocity v (in kilometer/minute) of a motorbike which starts from rest, is given at fixed intervals of time t (in minutes) as follows:

t	2	4	6	8	10	12	14	16	18	20
v	10	18	25	29	32	20	11	5	2	0

The approximate distance (in kilometers) rounded to two places of decimals covered in 20 minutes using Simpson's $1/3^{rd}$ rule is _____.

Correct Answer :
308 to 310

Question Number : 37 Question Type : MCQ

Assume that a mergesort algorithm in the worst case takes 30 seconds for an input of size 64. Which of the following most closely approximates the maximum input size of a problem that can be solved in 6 minutes?

(A) 256 (B) 512 (C) 1024 (D) 2048

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 38 Question Type : MCQ

Consider the following recursive C function.

```
void get(int n)
 {
      if (n<1) return;
      get(n-1);
      get(n-3);
      printf("%d", n);
 }
```

If `get(6)` function is being called in `main()` then how many times will the `get()` function be invoked before returning to the `main()`?

(A) 15 (B) 25 (C) 35 (D) 45

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 39 Question Type : NAT

Consider a B+ tree in which the search key is 12 bytes long, block size is 1024 bytes, record pointer is 10 bytes long and block pointer is 8 bytes long. The maximum number of keys that can be accommodated in each non-leaf node of the tree is _____.

Correct Answer :

50

Question Number : 40 Question Type : MCQ

Given the function $F = P' + QR$, where F is a function in three Boolean variables P, Q and R and $P' = !P$, consider the following statements.

(S1) $F = \sum(4,5,6)$
(S2) $F = \sum(0,1,2,3,7)$
(S3) $F = \prod(4,5,6)$
(S4) $F = \prod(0,1,2,3,7)$

Which of the following is true?

(A) (S1)- False, (S2)- True, (S3)- True, (S4)- False
(B) (S1)- True, (S2)- False, (S3)- False, (S4)- True
(C) (S1)- False, (S2)- False, (S3)- True, (S4)- True
(D) (S1)- True, (S2)- True, (S3)- False, (S4)- False

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 41 Question Type : MCQ

Language L_1 is polynomial time reducible to language L_2. Language L_3 is polynomial time reducible to L_2, which in turn is polynomial time reducible to language L_4. Which of the following is/are true?

I. if $L_4 \in P$, then $L_2 \in P$
II. if $L_1 \in P$ or $L_3 \in P$, then $L_2 \in P$
III. $L_1 \in P$, if and only if $L_3 \in P$
IV. if $L_4 \in P$, then $L_1 \in P$ and $L_3 \in P$

(A) II only
(B) III only
(C) I and IV only
(D) I only

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 42 Question Type : NAT

Consider the following C program.

```
#include<stdio.h>
int f1(void);
int f2(void);
int f3(void);
int x = 10;

int main( )
 {
   int x = 1;
   x += f1( ) + f2( ) + f3( ) + f2( );
   printf("%d", x);
   return 0;
 }

int f1() { int x = 25; x++; return x;}
int f2() { static int x = 50; x++; return x;}
int f3() { x *= 10; return x};
```

The output of the program is _____.

Correct Answer :
230

Question Number : 43 Question Type : NAT

Consider the following C program.

```
#include<stdio.h>
int main( )
{
   static int a[ ] = {10, 20, 30, 40, 50};
   static int *p[ ] = {a, a+3, a+4, a+1, a+2};
   int **ptr = p;
   ptr++;
   printf("%d%d", ptr-p,**ptr);
}
```

The output of the program is _____.

Correct Answer :
140

Question Number : 44 Question Type : MCQ

Which of the following languages are context-free?

$L_1 = \{a^m b^n a^n b^m \mid m, n \geq 1\}$
$L_2 = \{a^m b^n a^m b^n \mid m, n \geq 1\}$
$L_3 = \{a^m b^n \mid m = 2n + 1\}$

(A) L_1 and L_2 only (B) L_1 and L_3 only (C) L_2 and L_3 only (D) L_3 only

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 45 Question Type : MCQ

Consider the following policies for preventing deadlock in a system with mutually exclusive resources.

I. Processes should acquire all their resources at the beginning of execution. If any resource is not available, all resources acquired so far are released
II. The resources are numbered uniquely, and processes are allowed to request for resources only in increasing resource numbers
III. The resources are numbered uniquely, and processes are allowed to request for resources only in decreasing resource numbers
IV. The resources are numbered uniquely. A process is allowed to request only for a resource with resource number larger than its currently held resources

Which of the above policies can be used for preventing deadlock?

(A) Any one of I and III but not II or IV
(B) Any one of I, III, and IV but not II
(C) Any one of II and III but not I or IV
(D) Any one of I, II, III, and IV

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 46 Question Type : NAT

In the network 200.10.11.144/27, the fourth octet (in decimal) of the last IP address of the network which can be assigned to a host is _____.

Correct Answer:

158

Question Number : 47 Question Type : NAT

Consider a network connecting two systems located 8000 kilometers apart. The bandwidth of the network is 500×10^6 bits per second. The propagation speed of the media is 4×10^6 meters per second. It is needed to design a Go-Back-N sliding window protocol for this network. The average packet size is 10^7 bits. The network is to be used to its full capacity. Assume that processing delays at nodes are negligible. Then, the minimum size in bits of the sequence number field has to be _____.

Correct Answer:

8

Question Number : 48 Question Type :NAT

Consider the following reservation table for a pipeline having three stages S_1, S_2 and S_3.

	Time → 1	2	3	4	5
S_1	X				X
S_2		X		X	
S_3			X		

The minimum average latency (MAL) is _____.

Correct Answer :

3

Question Number : 49 Question Type : MCQ

Consider the following code sequence having five instructions I_1 to I_5. Each of these instructions has the following format.

OP Ri, Rj, Rk

where operation OP is performed on contents of registers Rj and Rk and the result is stored in register Ri.

I_1: ADD R1, R2, R3
I_2: MUL R7, R1, R3
I_3: SUB R4, R1, R5
I_4: ADD R3, R2, R4
I_5: MUL R7, R8, R9

Consider the following three statements.

S1: There is an anti-dependence between instructions I_2 and I_5
S2: There is an anti-dependence between instructions I_2 and I_4
S3: Within an instruction pipeline an anti-dependence always creates one or more stalls

Which one of above statements is/are correct?

(A) Only S1 is true
(B) Only S2 is true
(C) Only S1 and S3 are true
(D) Only S2 and S3 are true

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 50 Question Type : MCQ

Consider the following two C code segments. Y and X are one and two dimensional arrays of size n and $n \times n$ respectively, where $2 \leq n \leq 10$. Assume that in both code segments, elements of Y are initialized to 0 and each element X[i][j] of array X is initialized to i+j. Further assume that when stored in main memory all elements of X are in same main memory page frame.

Code segment 1:

```
//initialize elements of Y to 0
//initialize elements X[i][j] of X to i+j

   for(i = 0; i < n; i++)
       Y[i] += X[0][i];
```

Code Segment 2:

```
//initialize elements of Y to 0
//initialize elements X[i][j] of X to i+j

     for(i = 0; i < n; i++)
       Y[i] += X[i][0];
```

Which of the following statements is/are correct?

S1: Final contents of array Y will be same in both code segments
S2: Elements of array X accessed inside the for loop shown in code segment 1 are contiguous in main memory
S3: Elements of array X accessed inside the for loop shown in code segment 2 are contiguous in main memory

(A) Only S2 is correct
(B) Only S3 is correct
(C) Only S1 and S2 are correct
(D) Only S1 and S3 are correct

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 51 Question Type : MCQ

Consider the following partial Schedule *S* involving two transactions *T1* and *T2*. Only the *read* and the *write* operations have been shown. The *read* operation on data item *P* is denoted by *read(P)* and the *write* operation on data item *P* is denoted by *write(P)*.

Time instance	Transaction-id	
	T1	*T2*
1	*read(A)*	
2	*write(A)*	
3		*read(C)*
4		*write(C)*
5		*read(B)*
6		*write(B)*
7		*read(A)*
8		*commit*
9	*read(B)*	

Schedule *S*

Suppose that the transaction *T1* fails immediately after time instance 9. Which one of the following statements is correct?

(A) *T2* must be aborted and then both *T1* and *T2* must be re-started to ensure transaction atomicity
(B) Schedule *S* is non-recoverable and cannot ensure transaction atomicity
(C) Only *T2* must be aborted and then re-started to ensure transaction atomicity
(D) Schedule *S* is recoverable and can ensure atomicity and nothing else needs to be done

Options :

1. ✖ A
2. ✔ B
3. ✖ C
4. ✖ D

Question Number : 52 Question Type : MCQ

If the following system has non-trivial solution,

$$px + qy + rz = 0$$
$$qx + ry + pz = 0$$
$$rx + py + qz = 0,$$

then which one of the following options is TRUE?

(A) $p - q + r = 0$ or $p = q = -r$
(B) $p + q - r = 0$ or $p = -q = r$
(C) $p + q + r = 0$ or $p = q = r$
(D) $p - q + r = 0$ or $p = -q = -r$

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 53 Question Type : NAT

Consider the following C program:

```
#include<stdio.h>
int main( )
{
    int i, j, k = 0;
    j = 2 * 3 / 4 + 2.0 / 5 + 8 / 5;
    k -= --j;
    for(i = 0; i < 5; i++)
      {
         switch(i + k)
         {
            case 1:
            case 2: printf("\n%d", i+k);
            case 3: printf("\n%d", i+k);
            default: printf("\n%d", i+k);
         }
      }
    return 0;
}
```

The number of times printf statement is executed is _____.

Correct Answer:

10

Question Number : 54 Question Type : MCQ

If for non-zero x, $af(x)+bf\left(\frac{1}{x}\right)=\frac{1}{x}-25$ where $a \neq b$ then $\int_1^2 f(x)dx$ is

(A) $\frac{1}{a^2-b^2}\left[a(\ln 2 - 25)+\frac{47b}{2}\right]$

(B) $\frac{1}{a^2-b^2}\left[a(2\ln 2 - 25)-\frac{47b}{2}\right]$

(C) $\frac{1}{a^2-b^2}\left[a(2\ln 2 - 25)+\frac{47b}{2}\right]$

(D) $\frac{1}{a^2-b^2}\left[a(\ln 2 - 25)-\frac{47b}{2}\right]$

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 55 Question Type : NAT

Let G be a connected undirected graph of 100 vertices and 300 edges. The weight of a minimum spanning tree of G is 500. When the weight of each edge of G is increased by five, the weight of a minimum spanning tree becomes _____.

Correct Answer :

995

Question Number : 56 Question Type : NAT

Two hosts are connected via a packet switch with 10^7 bits per second links. Each link has a propagation delay of 20 microseconds. The switch begins forwarding a packet 35 microseconds after it receives the same. If 10000 bits of data are to be transmitted between the two hosts using a packet size of 5000 bits, the time elapsed between the transmission of the first bit of data and the reception of the last bit of the data in microseconds is_____.

Correct Answer :

1575

Question Number : 57 Question Type : MCQ

For the processes listed in the following table, which of the following scheduling schemes will give the lowest average turnaround time?

Process	Arrival Time	Processing Time
A	0	3
B	1	6
C	4	4
D	6	2

(A) First Come First Serve
(B) Non-preemptive Shortest Job First
(C) Shortest Remaining Time
(D) Round Robin with Quantum value two

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

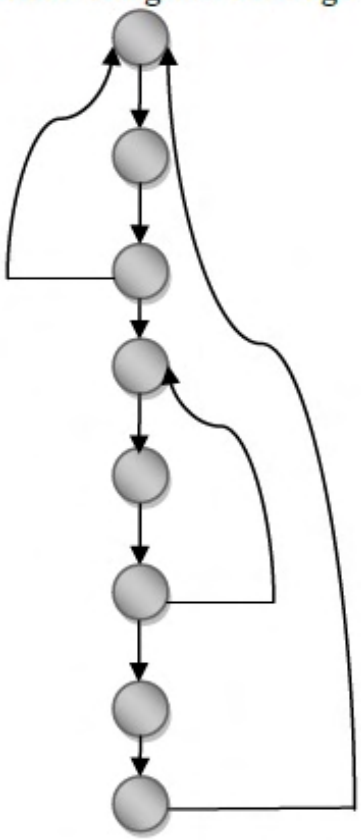
Question Number : 58 Question Type : MCQ

Consider three software items: Program-*X*, Control Flow Diagram of Program-*Y* and Control Flow Diagram of Program-*Z* as shown below

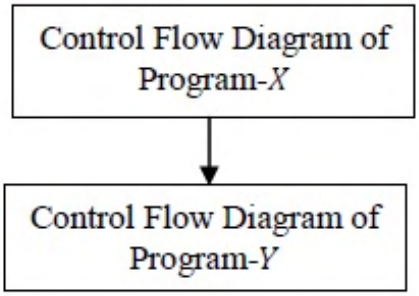
Program-*X*:

```
sumcal(int maxint, int value)
{
     int result=0, i=0;
     if (value <0)
     {
       value = -value;
     }
     while((i<value) AND (result
<= maxint))
     {
        i=i+1;
       result = result + 1;
     }
     if(result <= maxint)
     {
        printf(result);
     }
     else
     {
       printf("large");
     }
     printf("end of program");
}
```

Control Flow Diagram of Program-*Y*:



Control Flow Diagram of Program-*Z*:



The values of McCabe's Cyclomatic complexity of Program-*X*, Program-*Y*, and Program-*Z* respectively are

(A) 4, 4, 7 (B) 3, 4, 7 (C) 4, 4, 8 (D) 4, 3, 8

Options :

1. ✔ A
2. ✖ B
3. ✖ C
4. ✖ D

Question Number : 59 Question Type : NAT

Consider the equation $(43)_x = (y3)_8$ where x and y are unknown. The number of possible solutions is _____

Correct Answer :

5

Question Number : 60 Question Type : MCQ

Let R be a relation on the set of ordered pairs of positive integers such that $((p,q),(r,s)) \in R$ if and only if $p-s=q-r$. Which one of the following is true about R?

(A) Both reflexive and symmetric
(B) Reflexive but not symmetric
(C) Not reflexive but symmetric
(D) Neither reflexive nor symmetric

Options :

1. ✖ A
2. ✖ B
3. ✔ C
4. ✖ D

Question Number : 61 Question Type : NAT

Suppose X_i for $i = 1,2,3$ are independent and identically distributed random variables whose probability mass functions are $\Pr[X_i = 0] = \Pr[X_i = 1] = 1/2$ for $i = 1, 2, 3$. Define another random variable $Y = X_1 X_2 \oplus X_3$, where $\oplus$ denotes XOR. Then

$\Pr[Y = 0 | X_3 = 0] =$ _____.

Correct Answer :

0.75

Question Number : 62 Question Type : NAT

The total number of prime implicants of the function $f(w, x, y, z) = \sum(0, 2, 4, 5, 6, 10)$ is _____.

Correct Answer:

3

Question Number : 63 Question Type : NAT

Suppose $c = \langle c[0], \ldots, c[k-1] \rangle$ is an array of length k, where all the entries are from the set $\{0, 1\}$. For any positive integers a and n, consider the following pseudocode.

```
DOSOMETHING(c, a, n)
z ← 1
for i ← 0 to k − 1
    do z ← z^2 mod n
     if c[i] = 1
         then z ← (z × a) mod n
return z
```

If $k = 4$, $c = \langle 1, 0, 1, 1 \rangle$, $a = 2$ and $n = 8$, then the output of DOSOMETHING(c, a, n) is _____.

Correct Answer :
0

Question Number : 64 Question Type : MCQ

Let $f(n) = n$ and $g(n) = n^{(1+\sin n)}$, where n is a positive integer. Which of the following statements is/are correct?

I. $f(n) = O(g(n))$
II. $f(n) = \Omega(g(n))$

(A) Only I
(B) Only II
(C) Both I and II
(D) Neither I nor II

Options :
1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Question Number : 65 Question Type : MCQ

Consider the following grammar G

$$
\begin{aligned}
S &\rightarrow F \mid H \\
F &\rightarrow p \mid c \\
H &\rightarrow d \mid c
\end{aligned}
$$

where S, F, and H are non-terminal symbols, p, d, and c are terminal symbols. Which of the following statement(s) is/are correct?

S1. LL(1) can parse all strings that are generated using grammar G
S2. LR(1) can parse all strings that are generated using grammar G

(A) Only S1 (B) Only S2 (C) Both S1 and S2 (D) Neither S1 nor S2

Options :

1. ✖ A
2. ✖ B
3. ✖ C
4. ✔ D

Q. 1 – Q. 5 carry one mark each.

Q.1 Out of the following four sentences, select the most suitable sentence with respect to grammar and usage.

(A) I will not leave the place until the minister does not meet me.

(B) I will not leave the place until the minister doesn't meet me.

(C) I will not leave the place until the minister meet me.

(D) I will not leave the place until the minister meets me.

Q.2 A rewording of something written or spoken is a _____.

(A) paraphrase (B) paradox (C) paradigm (D) paraffin

Q.3 Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world."

The sentence above is an example of a _____ statement.

(A) figurative (B) collateral

(C) literal (D) figurine

Q.4 If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?

(A) zentaga (B) tagafer (C) tagazen (D) relffer

Q.5 A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.

(A) 56 (B) 64 (C) 72 (D) 96

Q. 6 – Q. 10 carry two marks each.

Q.6 A shaving set company sells 4 different types of razors, Elegance, Smooth, Soft and Executive. Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78 and Executive at Rs. 173 per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter \ Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

(A) Elegance (B) Executive (C) Smooth (D) Soft

Q.7 Indian currency notes show the denomination indicated in at least seventeen languages. If this is not an indication of the nation's diversity, nothing else is.

Which of the following can be logically inferred from the above sentences?

(A) India is a country of exactly seventeen languages.

(B) Linguistic pluralism is the only indicator of a nation's diversity.

(C) Indian currency notes have sufficient space for all the Indian languages.

(D) Linguistic pluralism is strong evidence of India's diversity.

Q.8 Consider the following statements relating to the level of poker play of four players **P**, **Q**, **R** and **S**.

I. **P** always beats **Q**
II. **R** always beats **S**
III. **S** loses to **P** only sometimes
IV. **R** always loses to **Q**

Which of the following can be logically inferred from the above statements?

(i) **P** is likely to beat all the three other players
(ii) **S** is the absolute worst player in the set

(A) (i) only (B) (ii) only (C) (i) and (ii) (D) neither (i) nor (ii)

Q.9 If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of f(x)?

(A) (x^3+8) (B) (x-1) (C) (2x-5) (D) (x+1)

Q.10 In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is _____.

(A) 40.00 (B) 46.02 (C) 60.01 (D) 92.02

END OF THE QUESTION PAPER

Q. 1 - Q. 25 carry one mark each.

Q.1 Let p, q, r, s represent the following propositions.

p: $x \in \{8, 9, 10, 11, 12\}$

q: x is a composite number

r: x is a perfect square

s: x is a prime number

The integer $x \geq 2$ which satisfies $\neg((p \Rightarrow q) \wedge (\neg r \vee \neg s))$ is _____ .

Q.2 Let a_n be the number of n-bit strings that do NOT contain two consecutive 1s. Which one of the following is the recurrence relation for a_n?

(A) $a_n = a_{n-1} + 2a_{n-2}$

(B) $a_n = a_{n-1} + a_{n-2}$

(C) $a_n = 2a_{n-1} + a_{n-2}$

(D) $a_n = 2a_{n-1} + 2a_{n-2}$

Q.3

$$\lim_{x \to 4} \frac{\sin(x-4)}{x-4} = _____.$$

Q.4 A probability density function on the interval $[a, 1]$ is given by $1/x^2$ and outside this interval the value of the function is zero. The value of a is _____ .

Q.5 Two eigenvalues of a 3×3 real matrix P are $(2 + \sqrt{-1})$ and 3. The determinant of P is _____ .

Q.6 Consider the Boolean operator # with the following properties:
$x\#0 = x$, $x\#1 = \bar{x}$, $x\#x = 0$ and $x\#\bar{x} = 1$. Then $x\#y$ is equivalent to

(A) $x\bar{y} + \bar{x}y$

(B) $x\bar{y} + \bar{x}\bar{y}$

(C) $\bar{x}y + xy$

(D) $xy + \bar{x}\bar{y}$

Q.7 The 16-bit 2's complement representation of an integer is 1111 1111 1111 0101; its decimal representation is _____ .

Q.8 We want to design a synchronous counter that counts the sequence 0-1-0-2-0-3 and then repeats. The minimum number of J-K flip-flops required to implement this counter is _____ .

Q.9 A processor can support a maximum memory of 4 GB, where the memory is word-addressable (a word consists of two bytes). The size of the address bus of the processor is at least _____ bits.

Q.10 A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is **CORRECT** (n refers to the number of items in the queue)?

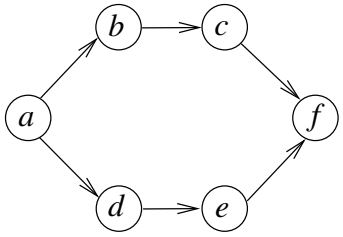
(A) Both operations can be performed in $O(1)$ time

(B) At most one operation can be performed in $O(1)$ time but the worst case time for the other operation will be $\Omega(n)$

(C) The worst case time complexity for both operations will be $\Omega(n)$

(D) Worst case time complexity for both operations will be $\Omega(\log n)$

Q.11 Consider the following directed graph:



The number of different topological orderings of the vertices of the graph is _____ .

Q.12 Consider the following C program.

```
void f(int, short);
void main()
{
     int i = 100;
     short s = 12;
     short *p = &s;
     __________ ;    // call to f()
}
```

Which one of the following expressions, when placed in the blank above, will **NOT** result in a type checking error?

(A) `f(s,*s)`

(B) `i = f(i,s)`

(C) `f(i,*s)`

(D) `f(i,*p)`

Q.13 The worst case running times of *Insertion sort*, *Merge sort* and *Quick sort*, respectively, are:

(A) $\Theta(n \log n)$, $\Theta(n \log n)$, and $\Theta(n^2)$

(B) $\Theta(n^2)$, $\Theta(n^2)$, and $\Theta(n \log n)$

(C) $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n \log n)$

(D) $\Theta(n^2)$, $\Theta(n \log n)$, and $\Theta(n^2)$

Q.14 Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are **TRUE**?

P: Minimum spanning tree of G does not change

Q: Shortest path between any pair of vertices does not change

(A) P only

(B) Q only

(C) Neither P nor Q

(D) Both P and Q

Q.15 Consider the following C program.

```
#include<stdio.h>
void mystery(int *ptra, int *ptrb) {
      int *temp;
      temp = ptrb;
      ptrb = ptra;
      ptra = temp;
}
int main() {
      int a=2016, b=0, c=4, d=42;
      mystery(&a, &b);
      if (a < c)
            mystery(&c, &a);
      mystery(&a, &d);
      printf("%d\n", a);
}
```

The output of the program is _____ .

Q.16 Which of the following languages is generated by the given grammar?

$$S \longrightarrow aS \mid bS \mid \varepsilon$$

(A) $\{a^n b^m \mid n, m \geq 0\}$

(B) $\{w \in \{a,b\}^* \mid w$ has equal number of a's and b's $\}$

(C) $\{a^n \mid n \geq 0\} \cup \{b^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$

(D) $\{a,b\}^*$

Q.17 Which of the following decision problems are undecidable?

I. Given NFAs N_1 and N_2, is $L(N_1) \cap L(N_2) = \Phi$?

II. Given a CFG $G = (N, \Sigma, P, S)$ and a string $x \in \Sigma^*$, does $x \in L(G)$?

III. Given CFGs G_1 and G_2, is $L(G_1) = L(G_2)$?

IV. Given a TM M, is $L(M) = \Phi$?

(A) I and IV only

(B) II and III only

(C) III and IV only

(D) II and IV only

Q.18 Which one of the following regular expressions represents the language: *the set of all binary strings having two consecutive* 0*s and two consecutive* 1*s*?

(A) $(0+1)^*0011(0+1)^* + (0+1)^*1100(0+1)^*$

(B) $(0+1)^*(00(0+1)^*11 + 11(0+1)^*00)(0+1)^*$

(C) $(0+1)^*00(0+1)^* + (0+1)^*11(0+1)^*$

(D) $00(0+1)^*11 + 11(0+1)^*00$

Q.19 Consider the following code segment.

```
x = u - t;
y = x * v;
x = y + w;
y = t - z;
y = x * y;
```

The minimum number of *total* variables required to convert the above code segment to *static single assignment* form is _____ .

Q.20 Consider an arbitrary set of CPU-bound processes with unequal CPU burst lengths submitted at the same time to a computer system. Which one of the following process scheduling algorithms would minimize the average waiting time in the ready queue?

(A) Shortest remaining time first

(B) Round-robin with time quantum less than the shortest CPU burst

(C) Uniform random

(D) Highest priority first with priority proportional to CPU burst length

Q.21 Which of the following is **NOT** a superkey in a relational schema with attributes V, W, X, Y, Z and primary key $V\,Y$?

(A) $VXYZ$

(B) $VWXZ$

(C) $VWXY$

(D) $VWXYZ$

Q.22 Which one of the following is **NOT** a part of the ACID properties of database transactions?

(A) Atomicity

(B) Consistency

(C) Isolation

(D) Deadlock-freedom

Q.23 A database of research articles in a journal uses the following schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, YEAR, PRICE)

The primary key is (VOLUME, NUMBER, STARTPAGE, ENDPAGE) and the following functional dependencies exist in the schema.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE) $\rightarrow$ TITLE
(VOLUME, NUMBER) $\rightarrow$ YEAR
(VOLUME, NUMBER, STARTPAGE, ENDPAGE) $\rightarrow$ PRICE

The database is redesigned to use the following schemas.

(VOLUME, NUMBER, STARTPAGE, ENDPAGE, TITLE, PRICE)
(VOLUME, NUMBER, YEAR)

Which is the weakest normal form that the new database satisfies, but the old one does not?

(A) 1NF
(B) 2NF
(C) 3NF
(D) BCNF

Q.24 Which one of the following protocols is **NOT** used to resolve one form of address to another one?

(A) DNS
(B) ARP
(C) DHCP
(D) RARP

Q.25 Which of the following is/are example(s) of stateful application layer protocols?

(i) HTTP
(ii) FTP
(iii) TCP
(iv) POP3

(A) (i) and (ii) only
(B) (ii) and (iii) only
(C) (ii) and (iv) only
(D) (iv) only

Q. 26 - Q. 55 carry two marks each.

Q.26 The coefficient of x^{12} in $(x^3+x^4+x^5+x^6+\cdots)^3$ is _____ .

Q.27 Consider the recurrence relation $a_1 = 8$, $a_n = 6n^2 + 2n + a_{n-1}$. Let $a_{99} = K \times 10^4$. The value of K is _____ .

Q.28 A function $f : \mathbb{N}^+ \to \mathbb{N}^+$, defined on the set of positive integers $\mathbb{N}^+$, satisfies the following properties:

$$f(n) = f(n/2) \quad \textbf{if } n \text{ is even}$$
$$f(n) = f(n+5) \quad \textbf{if } n \text{ is odd}$$

Let $R = \{i \mid \exists j : f(j) = i\}$ be the set of distinct values that f takes. The maximum possible size of R is _____ .

Q.29 Consider the following experiment.

Step 1. Flip a fair coin twice.

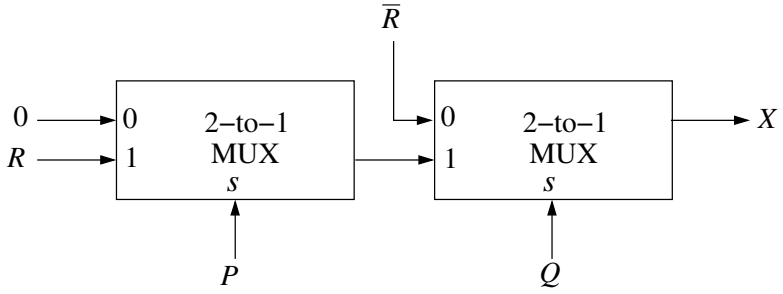
Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop.

Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop.

Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1.

The probability that the output of the experiment is Y is (up to two decimal places) _____ .

Q.30 Consider the two cascaded 2-to-1 multiplexers as shown in the figure.



The minimal sum of products form of the output X is

(A) $\bar{P}\bar{Q}+PQR$

(B) $\bar{P}Q+QR$

(C) $PQ+\bar{P}\bar{Q}R$

(D) $\bar{Q}\bar{R}+PQR$

Q.31 The size of the data count register of a DMA controller is 16 bits. The processor needs to transfer a file of 29,154 kilobytes from disk to main memory. The memory is byte addressable. The minimum number of times the DMA controller needs to get the control of the system bus from the processor to transfer the file from the disk to main memory is _____ .

Q.32 The stage delays in a 4-stage pipeline are 800, 500, 400 and 300 picoseconds. The first stage (with delay 800 picoseconds) is replaced with a functionally equivalent design involving two stages with respective delays 600 and 350 picoseconds. The throughput increase of the pipeline is _____ percent.

Q.33 Consider a carry lookahead adder for adding two n-bit integers, built using gates of fan-in at most two. The time to perform addition using this adder is

(A) $\Theta(1)$

(B) $\Theta(\log(n))$

(C) $\Theta(\sqrt{n})$

(D) $\Theta(n)$

Q.34 The following function computes the maximum value contained in an integer array `p[]` of size `n (n >= 1)`.

```
int max(int *p, int n) {
    int a=0, b=n-1;

    while (__________) {
        if (p[a] <= p[b]) { a = a+1; }
        else              { b = b-1; }
    }

    return p[a];
}
```

The missing loop condition is

(A) `a != n`

(B) `b != 0`

(C) `b > (a + 1)`

(D) `b != a`

Q.35 What will be the output of the following C program?

```
void count(int n){
     static int d=1;

     printf("%d ", n);
     printf("%d ", d);
     d++;
     if(n>1) count(n-1);
     printf("%d ", d);
}

void main(){
     count(3);
}
```

(A) 3 1 2 2 1 3 4 4 4

(B) 3 1 2 1 1 1 2 2 2

(C) 3 1 2 2 1 3 4

(D) 3 1 2 1 1 1 2

Q.36 What will be the output of the following pseudo-code when parameters are passed by reference and dynamic scoping is assumed?

```
a=3;
void n(x) {x = x * a; print(x);}
void m(y) {a = 1; a = y - a; n(a); print(a);}
void main() {m(a);}
```

(A) 6, 2

(B) 6, 6

(C) 4, 2

(D) 4, 4

Q.37 An operator `delete`(i) for a binary heap data structure is to be designed to delete the item in the i-th node. Assume that the heap is implemented in an array and i refers to the i-th index of the array. If the heap tree has depth d (number of edges on the path from the root to the farthest leaf), then what is the time complexity to re-fix the heap efficiently after the removal of the element?

(A) $O(1)$

(B) $O(d)$ but not $O(1)$

(C) $O(2^d)$ but not $O(d)$

(D) $O(d\,2^d)$ but not $O(2^d)$

Q.38 Consider the weighted undirected graph with 4 vertices, where the weight of edge $\{i, j\}$ is given by the entry W_{ij} in the matrix W.

$$W = \begin{bmatrix} 0 & 2 & 8 & 5 \\ 2 & 0 & 5 & 8 \\ 8 & 5 & 0 & \text{x} \\ 5 & 8 & \text{x} & 0 \end{bmatrix}$$

The largest possible integer value of x, for which at least one shortest path between some pair of vertices will contain the edge with weight x is _____ .

Q.39 Let G be a complete undirected graph on 4 vertices, having 6 edges with weights being 1, 2, 3, 4, 5, and 6. The maximum possible weight that a minimum weight spanning tree of G can have is _____ .

Q.40 $G = (V, E)$ is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G. Which of the following statements about the minimum spanning trees (MSTs) of G is/are **TRUE**?

I. If e is the lightest edge of some cycle in G, then every MST of G includes e

II. If e is the heaviest edge of some cycle in G, then every MST of G excludes e

(A) I only (B) II only (C) both I and II (D) neither I nor II

Q.41 Let Q denote a queue containing sixteen numbers and S be an empty stack. `Head`(Q) returns the element at the head of the queue Q **without** removing it from Q. Similarly `Top`(S) returns the element at the top of S **without** removing it from S. Consider the algorithm given below.

```
while Q is not Empty do
    if S is Empty OR Top(S) ≤ Head(Q) then
        x := Dequeue(Q);
        Push(S,x);
    else
        x := Pop(S);
        Enqueue(Q,x);
    end
end
```

The maximum possible number of iterations of the **while** loop in the algorithm is _____ .

Q.42 Consider the following context-free grammars:

G_1: $S \to aS|B$, $B \to b|bB$

G_2: $S \to aA|bB$, $A \to aA|B|\varepsilon$, $B \to bB|\varepsilon$

Which one of the following pairs of languages is generated by G_1 and G_2, respectively?

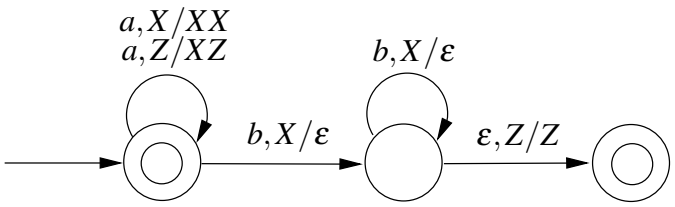
(A) $\{a^m b^n | m > 0 \text{ or } n > 0\}$ and $\{a^m b^n | m > 0 \text{ and } n > 0\}$

(B) $\{a^m b^n | m > 0 \text{ and } n > 0\}$ and $\{a^m b^n | m > 0 \text{ or } n \geq 0\}$

(C) $\{a^m b^n | m \geq 0 \text{ or } n > 0\}$ and $\{a^m b^n | m > 0 \text{ and } n > 0\}$

(D) $\{a^m b^n | m \geq 0 \text{ and } n > 0\}$ and $\{a^m b^n | m > 0 \text{ or } n > 0\}$

Q.43 Consider the transition diagram of a PDA given below with input alphabet $\Sigma = \{a, b\}$ and stack alphabet $\Gamma = \{X, Z\}$. Z is the initial stack symbol. Let L denote the language accepted by the PDA.



Which one of the following is **TRUE**?

(A) $L = \{a^n b^n | n \geq 0\}$ and is not accepted by any finite automata

(B) $L = \{a^n | n \geq 0\} \cup \{a^n b^n | n \geq 0\}$ and is not accepted by any deterministic PDA

(C) L is not accepted by any Turing machine that halts on every input

(D) $L = \{a^n | n \geq 0\} \cup \{a^n b^n | n \geq 0\}$ and is deterministic context-free

Q.44 Let X be a recursive language and Y be a recursively enumerable but not recursive language. Let W and Z be two languages such that $\overline{Y}$ reduces to W, and Z reduces to $\overline{X}$ (reduction means the standard many-one reduction). Which one of the following statements is **TRUE**?

(A) W can be recursively enumerable and Z is recursive.

(B) W can be recursive and Z is recursively enumerable.

(C) W is not recursively enumerable and Z is recursive.

(D) W is not recursively enumerable and Z is not recursive.

Q.45 The attributes of three arithmetic operators in some programming language are given below.

Operator	Precedence	Associativity	Arity
$+$	High	Left	Binary
$-$	Medium	Right	Binary
$*$	Low	Left	Binary

The value of the expression $\mathbf{2-5+1-7*3}$ in this language is _____ .

Q.46 Consider the following Syntax Directed Translation Scheme (SDTS), with non-terminals {S, A} and terminals {**a**, **b**}.

S $\longrightarrow$ **a**A { print 1 }
S $\longrightarrow$ **a** { print 2 }
A $\longrightarrow$ S**b** { print 3 }

Using the above SDTS, the output printed by a bottom-up parser, for the input **aab** is:

(A) 1 3 2

(B) 2 2 3

(C) 2 3 1

(D) syntax error

Q.47 Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then the size of the per-process page table is _____ megabytes.

Q.48 Consider a disk queue with requests for I/O to blocks on cylinders 47, 38, 121, 191, 87, 11, 92, 10. The C-LOOK scheduling algorithm is used. The head is initially at cylinder number 63, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. The total head movement (in number of cylinders) incurred while servicing these requests is _____ .

Q.49 Consider a computer system with ten physical page frames. The system is provided with an access sequence $(a_1, a_2, \ldots, a_{20}, a_1, a_2, \ldots, a_{20})$, where each a_i is a distinct virtual page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is _____ .

Q.50 Consider the following proposed solution for the critical section problem. There are n processes: $P_0 \ldots P_{n-1}$. In the code, function `pmax` returns an integer not smaller than any of its arguments. For all i, `t[i]` is initialized to zero.

Code for P_i:

```
do {
     c[i]=1; t[i] = pmax(t[0],...,t[n-1])+1; c[i]=0;
     for every j ≠ i in {0,...,n-1} {
          while (c[j]);
          while (t[j] != 0 && t[j]<=t[i]);
     }
     Critical Section;
     t[i]=0;
     Remainder Section;
} while (true);
```

Which one of the following is **TRUE** about the above solution?

(A) At most one process can be in the critical section at any time

(B) The bounded wait condition is satisfied

(C) The progress condition is satisfied

(D) It cannot cause a deadlock

Q.51 Consider the following two phase locking protocol. Suppose a transaction T accesses (for read or write operations), a certain set of objects $\{O_1, \ldots, O_k\}$. This is done in the following manner:

Step 1. T acquires exclusive locks to $O_1, \ldots, O_k$ in increasing order of their addresses.

Step 2. The required operations are performed.

Step 3. All locks are released.

This protocol will

(A) guarantee serializability and deadlock-freedom

(B) guarantee neither serializability nor deadlock-freedom

(C) guarantee serializability but not deadlock-freedom

(D) guarantee deadlock-freedom but not serializability

Q.52 Consider that B wants to send a message m that is digitally signed to A. Let the pair of private and public keys for A and B be denoted by K_x^- and K_x^+ for $x = A, B$, respectively. Let $K_x(m)$ represent the operation of encrypting m with a key K_x and $H(m)$ represent the message digest. Which one of the following indicates the **CORRECT** way of sending the message m along with the digital signature to A?

(A) $\{m, K_B^+(H(m))\}$ (B) $\{m, K_B^-(H(m))\}$ (C) $\{m, K_A^-(H(m))\}$ (D) $\{m, K_A^+(m)\}$

Q.53 An IP datagram of size 1000 bytes arrives at a router. The router has to forward this packet on a link whose MTU (maximum transmission unit) is 100 bytes. Assume that the size of the IP header is 20 bytes.

The number of fragments that the IP datagram will be divided into for transmission is _____ .

Q.54 For a host machine that uses the token bucket algorithm for congestion control, the token bucket has a capacity of 1 megabyte and the maximum output rate is 20 megabytes per second. Tokens arrive at a rate to sustain output at a rate of 10 megabytes per second. The token bucket is currently full and the machine needs to send 12 megabytes of data. The minimum time required to transmit the data is _____ seconds.

Q.55 A sender uses the Stop-and-Wait ARQ protocol for reliable transmission of frames. Frames are of size 1000 bytes and the transmission rate at the sender is 80 Kbps (1Kbps = 1000 bits/second). Size of an acknowledgement is 100 bytes and the transmission rate at the receiver is 8 Kbps. The one-way propagation delay is 100 milliseconds.

Assuming no frame is lost, the sender throughput is _____ bytes/second.

Q. No	Type	Section	Key	Marks
1	MCQ	GA	D	1
2	MCQ	GA	A	1
3	MCQ	GA	A	1
4	MCQ	GA	C	1
5	MCQ	GA	D	1
6	MCQ	GA	B	2
7	MCQ	GA	D	2
8	MCQ	GA	D	2
9	MCQ	GA	B	2
10	MCQ	GA	B	2
1	NAT	CS-1	11.0 : 11.0	1
2	MCQ	CS-1	B	1
3	NAT	CS-1	1.0 : 1.0	1
4	NAT	CS-1	0.5 : 0.5	1
5	NAT	CS-1	15.0 : 15.0	1
6	MCQ	CS-1	A	1
7	NAT	CS-1	-11.0 : -11.0	1
8	NAT	CS-1	3.0 : 4.0	1
9	NAT	CS-1	31.0 : 31.0	1
10	MCQ	CS-1	A	1
11	NAT	CS-1	6.0 : 6.0	1
12	MCQ	CS-1	D	1
13	MCQ	CS-1	D	1
14	MCQ	CS-1	A	1
15	NAT	CS-1	2016.0 : 2016.0	1
16	MCQ	CS-1	D	1
17	MCQ	CS-1	C	1
18	MCQ	CS-1	B	1
19	NAT	CS-1	10.0 : 10.0	1
20	MCQ	CS-1	A	1
21	MCQ	CS-1	B	1
22	MCQ	CS-1	D	1
23	MCQ	CS-1	B	1
24	MCQ	CS-1	C	1
25	MCQ	CS-1	C	1
26	NAT	CS-1	10.0 : 10.0	2
27	NAT	CS-1	197.9 : 198.1	2
28	NAT	CS-1	2.0 : 2.0	2
29	NAT	CS-1	0.33 : 0.34	2
30	MCQ	CS-1	D	2
31	NAT	CS-1	456.0 : 456.0	2
32	NAT	CS-1	33.0 : 34.0	2
33	MCQ	CS-1	B	2
34	MCQ	CS-1	D	2
35	MCQ	CS-1	A	2
36	MCQ	CS-1	D	2
37	MCQ	CS-1	B	2
38	NAT	CS-1	12.0 : 12.0	2
39	NAT	CS-1	7.0 : 7.0	2

40	MCQ	CS-1	B	2
41	NAT	CS-1	256.0 : 256.0	2
42	MCQ	CS-1	D	2
43	MCQ	CS-1	D	2
44	MCQ	CS-1	C	2
45	NAT	CS-1	9.0 : 9.0	2
46	MCQ	CS-1	C	2
47	NAT	CS-1	384.0 : 384.0	2
48	NAT	CS-1	346.0 : 346.0	2
49	NAT	CS-1	1.0 : 1.0	2
50	MCQ	CS-1	A	2
51	MCQ	CS-1	A	2
52	MCQ	CS-1	B	2
53	NAT	CS-1	13.0 : 13.0	2
54	NAT	CS-1	1.1 : 1.1	2
55	NAT	CS-1	2500.0 : 2500.0	2

Q. 1 – Q. 5 carry one mark each.

Q.1 The man who is now Municipal Commissioner worked as _____.

(A) the security guard at a university

(B) a security guard at the university

(C) a security guard at university

(D) the security guard at the university

Q.2 Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia.

Choose the option which is closest in meaning to the underlined phrase in the above sentence.

(A) put up with (B) put in with (C) put down to (D) put up against

Q.3 Find the odd one in the following group of words.

mock, deride, praise, jeer

(A) mock (B) deride (C) praise (D) jeer

Q.4 Pick the odd one from the following options.

(A) CADBE (B) JHKIL (C) XVYWZ (D) ONPMQ

Q.5 In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$

(A) n^4 (B) 4^n (C) 2^{2n-1} (D) 4^{n-1}

Q. 6 – Q. 10 carry two marks each.

Q.6 Among 150 faculty members in an institute, 55 are connected with each other through Facebook® and 85 are connected through WhatsApp®. 30 faculty members do not have Facebook® or WhatsApp® accounts. The number of faculty members connected only through Facebook® accounts is _____.

(A) 35 (B) 45 (C) 65 (D) 90

Q.7 Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required.

Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?

(i) The author believes that computers are not good for us.
(ii) Mobile computers and the internet are both intended inventions

(A) (i) only (B) (ii) only (C) both (i) and (ii) (D) neither (i) nor (ii)

Q.8 All hill-stations have a lake. Ooty has two lakes.

Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

(i) Ooty is not a hill-station.
(ii) No hill-station can have more than one lake.

(A) (i) only

(B) (ii) only

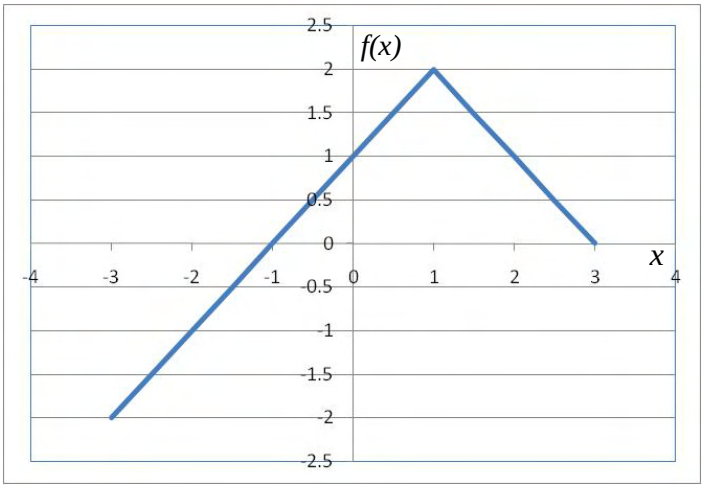
(C) both (i) and (ii)

(D) neither (i) nor (ii)

Q.9 In a 2 × 4 rectangle grid shown below, each cell is a rectangle. How many rectangles can be observed in the grid?

(A) 21 (B) 27 (C) 30 (D) 36

Q.10



Choose the correct expression for *f(x)* given in the graph.

(A) $f(x) = 1 - |x - 1|$

(B) $f(x) = 1 + |x - 1|$

(C) $f(x) = 2 - |x - 1|$

(D) $f(x) = 2 + |x - 1|$

END OF THE QUESTION PAPER

Q. 1 - Q. 25 carry one mark each.

Q.1 Consider the following expressions:

(i) *false*

(ii) Q

(iii) *true*

(iv) $P \vee Q$

(v) $\neg Q \vee P$

The number of expressions given above that are logically implied by $P \wedge (P \Rightarrow Q)$ is _____ .

Q.2 Let $f(x)$ be a polynomial and $g(x) = f'(x)$ be its derivative. If the degree of $(f(x) + f(-x))$ is 10, then the degree of $(g(x) - g(-x))$ is _____ .

Q.3 The minimum number of colours that is sufficient to vertex-colour any planar graph is _____ .

Q.4 Consider the systems, each consisting of m linear equations in n variables.

I. If $m < n$, then all such systems have a solution

II. If $m > n$, then none of these systems has a solution

III. If $m = n$, then there exists a system which has a solution

Which one of the following is **CORRECT**?

(A) I, II and III are true

(B) Only II and III are true

(C) Only III is true

(D) None of them is true

Q.5 Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is _____ .

Q.6 Suppose that the eigenvalues of matrix A are $1, 2, 4$. The determinant of $(A^{-1})^T$ is _____ .

Q.7 Consider an eight-bit ripple-carry adder for computing the sum of A and B, where A and B are integers represented in 2's complement form. If the decimal value of A is one, the decimal value of B that leads to the longest latency for the sum to stabilize is _____ .

Q.8 Let, $x_1 \oplus x_2 \oplus x_3 \oplus x_4 = 0$ where x_1, x_2, x_3, x_4 are Boolean variables, and $\oplus$ is the XOR operator. Which one of the following must always be **TRUE**?

(A) $x_1 x_2 x_3 x_4 = 0$

(B) $x_1 x_3 + x_2 = 0$

(C) $\bar{x}_1 \oplus \bar{x}_3 = \bar{x}_2 \oplus \bar{x}_4$

(D) $x_1 + x_2 + x_3 + x_4 = 0$

Q.9 Let X be the number of distinct 16-bit integers in 2's complement representation. Let Y be the number of distinct 16-bit integers in sign magnitude representation.
Then $X - Y$ is _____ .

Q.10 A processor has 40 distinct instructions and 24 general purpose registers. A 32-bit instruction word has an opcode, two register operands and an immediate operand. The number of bits available for the immediate operand field is _____ .

Q.11 Breadth First Search (BFS) is started on a binary tree beginning from the root vertex. There is a vertex t at a distance four from the root. If t is the n-th vertex in this BFS traversal, then the maximum possible value of n is _____ .

Q.12 The value printed by the following program is _____ .

```
void f(int* p, int m){
   m = m + 5;
   *p = *p + m;
   return;
}

void main(){
   int i=5, j=10;

   f(&i, j);
   printf("%d", i+j);
}
```

Q.13 Assume that the algorithms considered here sort the input sequences in ascending order. If the input is already in ascending order, which of the following are **TRUE**?

I. Quicksort runs in $\Theta(n^2)$ time

II. Bubblesort runs in $\Theta(n^2)$ time

III. Mergesort runs in $\Theta(n)$ time

IV. Insertion sort runs in $\Theta(n)$ time

(A) I and II only

(B) I and III only

(C) II and IV only

(D) I and IV only

Q.14 The Floyd-Warshall algorithm for all-pair shortest paths computation is based on

(A) Greedy paradigm.

(B) Divide-and-Conquer paradigm.

(C) Dynamic Programming paradigm.

(D) neither Greedy nor Divide-and-Conquer nor Dynamic Programming paradigm.

Q.15 N items are stored in a sorted doubly linked list. For a *delete* operation, a pointer is provided to the record to be deleted. For a *decrease-key* operation, a pointer is provided to the record on which the operation is to be performed.

An algorithm performs the following operations on the list in this order: $\Theta(N)$ *delete*, $O(\log N)$ *insert*, $O(\log N)$ *find*, and $\Theta(N)$ *decrease-key*. What is the time complexity of all these operations put together?

(A) $O(\log^2 N)$ (B) $O(N)$ (C) $O(N^2)$ (D) $\Theta(N^2 \log N)$

Q.16 The number of states in the minimum sized DFA that accepts the language defined by the regular expression

$$(0+1)^*(0+1)(0+1)^*$$

is _____ .

Q.17 Language L_1 is defined by the grammar: $S_1 \rightarrow aS_1b|\varepsilon$
Language L_2 is defined by the grammar: $S_2 \rightarrow abS_2|\varepsilon$

Consider the following statements:

P: L_1 is regular
Q: L_2 is regular

Which one of the following is **TRUE**?

(A) Both P and Q are true

(B) P is true and Q is false

(C) P is false and Q is true

(D) Both P and Q are false

Q.18 Consider the following types of languages: L_1 : Regular, L_2 : Context-free, L_3 : Recursive, L_4 : Recursively enumerable. Which of the following is/are **TRUE**?

I. $\overline{L_3} \cup L_4$ is recursively enumerable

II. $\overline{L_2} \cup L_3$ is recursive

III. $L_1^* \cap L_2$ is context-free

IV. $L_1 \cup \overline{L_2}$ is context-free

(A) I only

(B) I and III only

(C) I and IV only

(D) I, II and III only

Q.19 Match the following:

(P) Lexical analysis	(i) Leftmost derivation
(Q) Top down parsing	(ii) Type checking
(R) Semantic analysis	(iii) Regular expressions
(S) Runtime environments	(iv) Activation records

(A) P $\leftrightarrow$ i, Q $\leftrightarrow$ ii, R $\leftrightarrow$ iv, S $\leftrightarrow$ iii

(B) P $\leftrightarrow$ iii, Q $\leftrightarrow$ i, R $\leftrightarrow$ ii, S $\leftrightarrow$ iv

(C) P $\leftrightarrow$ ii, Q $\leftrightarrow$ iii, R $\leftrightarrow$ i, S $\leftrightarrow$ iv

(D) P $\leftrightarrow$ iv, Q $\leftrightarrow$ i, R $\leftrightarrow$ ii, S $\leftrightarrow$ iii

Q.20 In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frames increases?

(A) LRU (Least Recently Used)

(B) OPT (Optimal Page Replacement)

(C) MRU (Most Recently Used)

(D) FIFO (First In First Out)

Q.21 B+ Trees are considered **BALANCED** because

(A) the lengths of the paths from the root to all leaf nodes are all equal.

(B) the lengths of the paths from the root to all leaf nodes differ from each other by at most 1.

(C) the number of children of any two non-leaf sibling nodes differ by at most 1.

(D) the number of records in any two leaf nodes differ by at most 1.

Q.22 Suppose a database schedule S involves transactions $T_1, \ldots, T_n$. Construct the precedence graph of S with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?

(A) Topological order

(B) Depth-first order

(C) Breadth-first order

(D) Ascending order of transaction indices

Q.23 Anarkali digitally signs a message and sends it to Salim. Verification of the signature by Salim requires

(A) Anarkali's public key.

(B) Salim's public key.

(C) Salim's private key.

(D) Anarkali's private key.

Q.24 In an Ethernet local area network, which one of the following statements is **TRUE**?

(A) A station stops to sense the channel once it starts transmitting a frame.

(B) The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size.

(C) A station continues to transmit the packet even after the collision is detected.

(D) The exponential backoff mechanism reduces the probability of collision on retransmissions.

Q.25 Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

(A) HTTP GET request, DNS query, TCP SYN

(B) DNS query, HTTP GET request, TCP SYN

(C) DNS query, TCP SYN, HTTP GET request

(D) TCP SYN, DNS query, HTTP GET request

Q. 26 - Q. 55 carry two marks each.

Q.26 A binary relation R on $\mathbb{N} \times \mathbb{N}$ is defined as follows: $(a,b)R(c,d)$ if $a \leq c$ or $b \leq d$. Consider the following propositions:

P: R is reflexive
Q: R is transitive

Which one of the following statements is **TRUE**?

(A) Both P and Q are true.

(B) P is true and Q is false.

(C) P is false and Q is true.

(D) Both P and Q are false.

Q.27 Which one of the following well-formed formulae in predicate calculus is **NOT** valid?

(A) $(\forall x\, p(x) \Rightarrow \forall x\, q(x)) \Rightarrow (\exists x \neg p(x) \vee \forall x\, q(x))$

(B) $(\exists x\, p(x) \vee \exists x\, q(x)) \Rightarrow \exists x\, (p(x) \vee q(x))$

(C) $\exists x\, (p(x) \wedge q(x)) \Rightarrow (\exists x\, p(x) \wedge \exists x\, q(x))$

(D) $\forall x\, (p(x) \vee q(x)) \Rightarrow (\forall x\, p(x) \vee \forall x\, q(x))$

Q.28 Consider a set U of 23 different compounds in a Chemistry lab. There is a subset S of U of 9 compounds, each of which reacts with exactly 3 compounds of U. Consider the following statements:

I. Each compound in $U \setminus S$ reacts with an odd number of compounds.

II. At least one compound in $U \setminus S$ reacts with an odd number of compounds.

III. Each compound in $U \setminus S$ reacts with an even number of compounds.

Which one of the above statements is **ALWAYS TRUE**?

(A) Only I

(B) Only II

(C) Only III

(D) None

Q.29 The value of the expression $13^{99} \pmod{17}$, in the range 0 to 16, is _____ .

Q.30 Suppose the functions F and G can be computed in 5 and 3 nanoseconds by functional units U_F and U_G, respectively. Given two instances of U_F and two instances of U_G, it is required to implement the computation $F(G(X_i))$ for $1 \leq i \leq 10$. Ignoring all other delays, the minimum time required to complete this computation is _____ nanoseconds.

Q.31 Consider a processor with 64 registers and an instruction set of size twelve. Each instruction has five distinct fields, namely, opcode, two source register identifiers, one destination register identifier, and a twelve-bit immediate value. Each instruction must be stored in memory in a byte-aligned fashion. If a program has 100 instructions, the amount of memory (in bytes) consumed by the program text is _____ .

Q.32 The width of the physical address on a machine is 40 bits. The width of the tag field in a 512 KB 8-way set associative cache is _____ bits.

Q.33 Consider a 3 GHz (gigahertz) processor with a three-stage pipeline and stage latencies τ_1, τ_2, and τ_3 such that $\tau_1 = 3\tau_2/4 = 2\tau_3$. If the longest pipeline stage is split into two pipeline stages of equal latency, the new frequency is _____ GHz, ignoring delays in the pipeline registers.

Q.34 A complete binary min-heap is made by including each integer in $[1, 1023]$ exactly once. The depth of a node in the heap is the length of the path from the root of the heap to that node. Thus, the root is at depth 0. The maximum depth at which integer 9 can appear is _____ .

Q.35 The following function computes $\texttt{X}^{\texttt{Y}}$ for positive integers `X` and `Y`.

```
int exp(int X, int Y) {
    int res = 1, a = X, b = Y;

    while ( b != 0 ){
        if ( b%2 == 0) { a = a*a; b = b/2; }
        else           { res = res*a; b = b-1; }
    }
    return res;
}
```

Which one of the following conditions is **TRUE** before every iteration of the loop?

(A) $\texttt{X}^{\texttt{Y}} = \texttt{a}^{\texttt{b}}$

(B) $(\texttt{res} * \texttt{a})^{\texttt{Y}} = (\texttt{res} * \texttt{X})^{\texttt{b}}$

(C) $\texttt{X}^{\texttt{Y}} = \texttt{res} * \texttt{a}^{\texttt{b}}$

(D) $\texttt{X}^{\texttt{Y}} = (\texttt{res} * \texttt{a})^{\texttt{b}}$

Q.36 Consider the following `New-order` strategy for traversing a binary tree:

- Visit the root;
- Visit the right subtree using `New-order`;
- Visit the left subtree using `New-order`;

The `New-order` traversal of the expression tree corresponding to the reverse polish expression `3 4 * 5 - 2 ^ 6 7 * 1 + -` is given by:

(A) `+ - 1 6 7 * 2 ^ 5 - 3 4 *`

(B) `- + 1 * 6 7 ^ 2 - 5 * 3 4`

(C) `- + 1 * 7 6 ^ 2 - 5 * 4 3`

(D) `1 7 6 * + 2 5 4 3 * - ^ -`

Q.37 Consider the following program:

```
int f(int *p, int n)
{
   if (n <= 1) return 0;
   else return max(f(p+1,n-1),p[0]-p[1]);
}

int main()
{
   int a[] = {3,5,2,6,4};
   printf("%d", f(a,5));
}
```

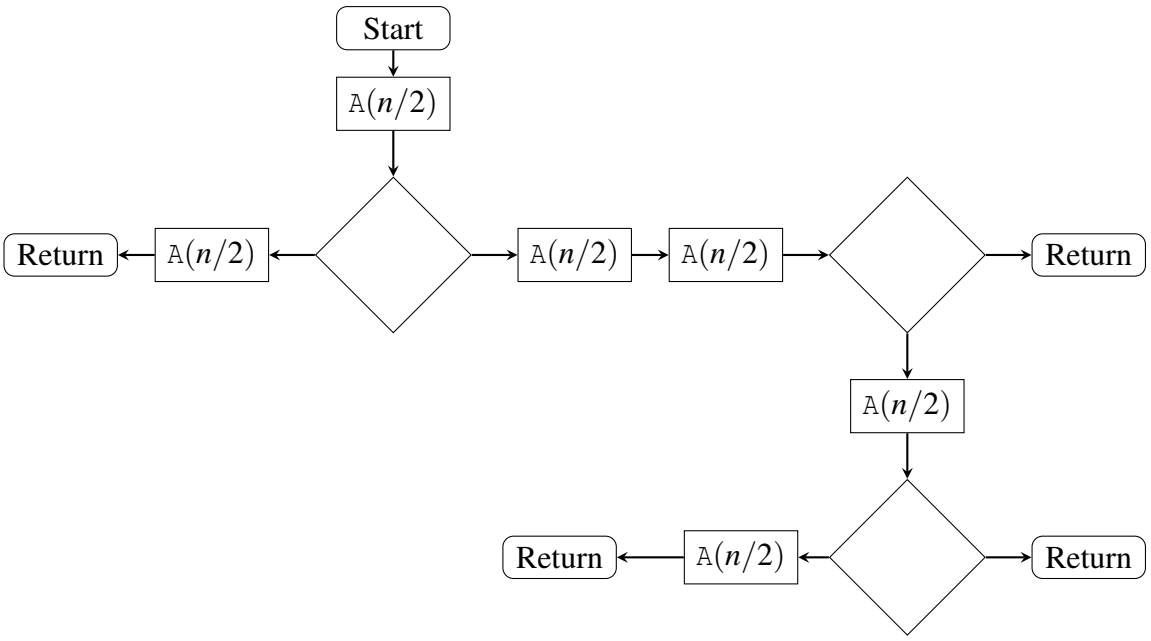
Note: `max(x,y)` *returns the maximum of* `x` *and* `y`.

The value printed by this program is _____ .

Q.38 Let A_1, A_2, A_3, and A_4 be four matrices of dimensions $10 \times 5, 5 \times 20, 20 \times 10$, and 10×5, respectively. The minimum number of scalar multiplications required to find the product $A_1A_2A_3A_4$ using the basic matrix multiplication method is _____ .

Q.39 The given diagram shows the flowchart for a recursive function $\mathtt{A}(n)$. Assume that all statements, except for the recursive calls, have $O(1)$ time complexity. If the worst case time complexity of this function is $O(n^{\alpha})$, then the least possible value (accurate up to two decimal positions) of α is _____ .

Flowchart for Recursive Function $\mathtt{A}(n)$



Q.40 The number of ways in which the numbers 1, 2, 3, 4, 5, 6, 7 can be inserted in an empty binary search tree, such that the resulting tree has height 6, is _____ .

Note: The height of a tree with a single node is 0.

Q.41 In an adjacency list representation of an undirected simple graph $G = (V, E)$, each edge (u, v) has two adjacency list entries: $[v]$ in the adjacency list of u, and $[u]$ in the adjacency list of v. These are called twins of each other. A twin pointer is a pointer from an adjacency list entry to its twin. If $|E| = m$ and $|V| = n$, and the memory size is not a constraint, what is the time complexity of the most efficient algorithm to set the twin pointer in each entry in each adjacency list?

(A) $\Theta(n^2)$

(B) $\Theta(n+m)$

(C) $\Theta(m^2)$

(D) $\Theta(n^4)$

Q.42 Consider the following two statements:

I. If all states of an NFA are accepting states then the language accepted by the NFA is Σ^*.

II. There exists a regular language A such that for all languages B, $A \cap B$ is regular.

Which one of the following is **CORRECT**?

(A) Only **I** is true

(B) Only **II** is true

(C) Both **I** and **II** are true

(D) Both **I** and **II** are false

Q.43 Consider the following languages:

$$L_1 = \{a^n b^m c^{n+m} : m, n \geq 1\}$$
$$L_2 = \{a^n b^n c^{2n} : n \geq 1\}$$

Which one of the following is **TRUE**?

(A) Both L_1 and L_2 are context-free.

(B) L_1 is context-free while L_2 is not context-free.

(C) L_2 is context-free while L_1 is not context-free.

(D) Neither L_1 nor L_2 is context-free.

Q.44 Consider the following languages.

$$\begin{aligned} L_1 &= \{\langle M\rangle \mid M \text{ takes at least 2016 steps on some input}\}, \\ L_2 &= \{\langle M\rangle \mid M \text{ takes at least 2016 steps on all inputs}\} \text{ and} \\ L_3 &= \{\langle M\rangle \mid M \text{ accepts } \varepsilon\}, \end{aligned}$$

where for each Turing machine M, $\langle M\rangle$ denotes a specific encoding of M. Which one of the following is **TRUE**?

(A) L_1 is recursive and L_2, L_3 are not recursive

(B) L_2 is recursive and L_1, L_3 are not recursive

(C) L_1, L_2 are recursive and L_3 is not recursive

(D) L_1, L_2, L_3 are recursive

Q.45 Which one of the following grammars is free from *left recursion*?

(A)
$$\begin{aligned} S &\to AB \\ A &\to Aa \mid b \\ B &\to c \end{aligned}$$

(B)
$$\begin{aligned} S &\to Ab \mid Bb \mid c \\ A &\to Bd \mid \varepsilon \\ B &\to e \end{aligned}$$

(C)
$$\begin{aligned} S &\to Aa \mid B \\ A &\to Bb \mid Sc \mid \varepsilon \\ B &\to d \end{aligned}$$

(D)
$$\begin{aligned} S &\to Aa \mid Bb \mid c \\ A &\to Bd \mid \varepsilon \\ B &\to Ae \mid \varepsilon \end{aligned}$$

Q.46 A student wrote two context-free grammars **G1** and **G2** for generating a single C-like array declaration. The dimension of the array is at least one. For example,

```
int a[10][3];
```

The grammars use D as the start symbol, and use six terminal symbols **int ; id [ ] num**.

Grammar **G1**	Grammar **G2**
$D \rightarrow \textbf{int}\, L;$	$D \rightarrow \textbf{int}\, L;$
$L \rightarrow \textbf{id}\, [E$	$L \rightarrow \textbf{id}\, E$
$E \rightarrow \textbf{num}]$	$E \rightarrow E[\textbf{num}]$
$E \rightarrow \textbf{num}]\, [E$	$E \rightarrow [\textbf{num}]$

Which of the grammars correctly generate the declaration mentioned above?

(A) Both **G1** and **G2**

(B) Only **G1**

(C) Only **G2**

(D) Neither **G1** nor **G2**

Q.47 Consider the following processes, with the arrival time and the length of the CPU burst given in milliseconds. The scheduling algorithm used is preemptive shortest remaining-time first.

Process	Arrival Time	Burst Time
P_1	0	10
P_2	3	6
P_3	7	1
P_4	8	3

The average turn around time of these processes is _____ milliseconds.

Q.48 Consider the following two-process synchronization solution.

```
Process 0                              Process 1
---------                              ----------
Entry: loop while (turn == 1);         Entry: loop while (turn == 0);
  (critical section)                     (critical section)
Exit:  turn = 1;                       Exit:  turn = 0;
```

The shared variable `turn` is initialized to zero. Which one of the following is **TRUE**?

(A) This is a correct two-process synchronization solution.

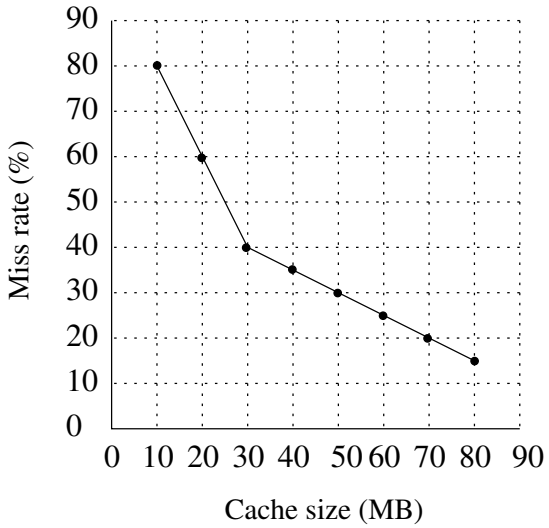
(B) This solution violates mutual exclusion requirement.

(C) This solution violates progress requirement.

(D) This solution violates bounded wait requirement.

Q.49 Consider a non-negative counting semaphore S. The operation $P(S)$ decrements S, and $V(S)$ increments S. During an execution, 20 $P(S)$ operations and 12 $V(S)$ operations are issued in some order. The largest initial value of S for which at least one $P(S)$ operation will remain blocked is _____ .

Q.50 A file system uses an in-memory cache to cache disk blocks. The miss rate of the cache is shown in the figure. The latency to read a block from the cache is 1 ms and to read a block from the disk is 10 ms. Assume that the cost of checking whether a block exists in the cache is negligible. Available cache sizes are in multiples of 10 MB.



The smallest cache size required to ensure an average read latency of less than 6 ms is _____ MB.

Q.51 Consider the following database schedule with two transactions, T_1 and T_2.

$$S = r_2(X);\ r_1(X);\ r_2(Y);\ w_1(X);\ r_1(Y);\ w_2(X);\ a_1;\ a_2$$

where $r_i(Z)$ denotes a *read* operation by transaction T_i on a variable Z, $w_i(Z)$ denotes a *write* operation by T_i on a variable Z and a_i denotes an *abort* by transaction T_i.

Which one of the following statements about the above schedule is **TRUE**?

(A) S is non-recoverable

(B) S is recoverable, but has a cascading abort

(C) S does not have a cascading abort

(D) S is strict

Q.52 Consider the following database table named *water_schemes* :

water_schemes		
scheme_no	**district_name**	**capacity**
1	Ajmer	20
1	Bikaner	10
2	Bikaner	10
3	Bikaner	20
1	Churu	10
2	Churu	20
1	Dungargarh	10

The number of tuples returned by the following SQL query is _____ .

```
with total(name, capacity) as
    select district_name, sum(capacity)
    from water_schemes
    group by district_name
with total_avg(capacity) as
    select avg(capacity)
    from total
select name
    from total, total_avg
    where total.capacity ≥ total_avg.capacity
```

Q.53 A network has a data transmission bandwidth of 20×10^6 bits per second. It uses CSMA/CD in the MAC layer. The maximum signal propagation time from one node to another node is 40 microseconds. The minimum size of a frame in the network is _____ bytes.

Q.54 For the IEEE 802.11 MAC protocol for wireless communication, which of the following statements is/are **TRUE**?

I. At least three non-overlapping channels are available for transmissions.

II. The RTS-CTS mechanism is used for collision detection.

III. Unicast frames are ACKed.

(A) All I, II, and III

(B) I and III only

(C) II and III only

(D) II only

Q.55 Consider a 128×10^3 bits/second satellite communication link with one way propagation delay of 150 milliseconds. Selective retransmission (repeat) protocol is used on this link to send data with a frame size of 1 kilobyte. Neglect the transmission time of acknowledgement. The minimum number of bits required for the sequence number field to achieve 100% utilization is _____ .

Q. No	Type	Section	Key	Marks
1	MCQ	GA	B	1
2	MCQ	GA	A	1
3	MCQ	GA	C	1
4	MCQ	GA	D	1
5	MCQ	GA	B	1
6	MCQ	GA	A	2
7	MCQ	GA	D	2
8	MCQ	GA	D	2
9	MCQ	GA	C	2
10	MCQ	GA	C	2
1	NAT	CS-2	4.0 : 4.0	1
2	NAT	CS-2	9.0 : 9.0	1
3	NAT	CS-2	4.0 : 4.0	1
4	MCQ	CS-2	C	1
5	NAT	CS-2	0.55 : 0.55	1
6	NAT	CS-2	0.124 : 0.126	1
7	NAT	CS-2	-1.0 : -1.0	1
8	MCQ	CS-2	C	1
9	NAT	CS-2	1.0 : 1.0	1
10	NAT	CS-2	16.0 : 16.0	1
11	NAT	CS-2	31.0 : 31.0	1
12	NAT	CS-2	30.0 : 30.0	1
13	MCQ	CS-2	D	1
14	MCQ	CS-2	C	1
15	MCQ	CS-2	C	1
16	NAT	CS-2	2.0 : 2.0	1
17	MCQ	CS-2	C	1
18	MCQ	CS-2	D	1
19	MCQ	CS-2	B	1
20	MCQ	CS-2	D	1
21	MCQ	CS-2	A	1
22	MCQ	CS-2	A	1
23	MCQ	CS-2	A	1
24	MCQ	CS-2	D	1
25	MCQ	CS-2	C	1
26	MCQ	CS-2	B	2
27	MCQ	CS-2	D	2
28	MCQ	CS-2	B	2
29	NAT	CS-2	4.0 : 4.0	2
30	NAT	CS-2	28.0 : 28.0	2
31	NAT	CS-2	500.0 : 500.0	2
32	NAT	CS-2	24.0 : 24.0	2
33	NAT	CS-2	3.9 : 4.1	2
34	NAT	CS-2	8.0 : 8.0	2
35	MCQ	CS-2	C	2
36	MCQ	CS-2	C	2
37	NAT	CS-2	3.0 : 3.0	2
38	NAT	CS-2	1500.0 : 1500.0	2
39	NAT	CS-2	2.2 : 2.4	2

40	NAT	CS-2	64.0 : 64.0	2
41	MCQ	CS-2	B	2
42	MCQ	CS-2	B	2
43	MCQ	CS-2	B	2
44	MCQ	CS-2	C	2
45	MCQ	CS-2	B	2
46	MCQ	CS-2	A	2
47	NAT	CS-2	8.2 : 8.3	2
48	MCQ	CS-2	C	2
49	NAT	CS-2	7.0 : 7.0	2
50	NAT	CS-2	30.0 : 30.0	2
51	MCQ	CS-2	C	2
52	NAT	CS-2	2.0 : 2.0	2
53	NAT	CS-2	200.0 : 200.0	2
54	MCQ	CS-2	B	2
55	NAT	CS-2	4.0 : 4.0	2

Q. 1 – Q. 5 carry one mark each.

Q.1 "From where are they bringing their books? _____ bringing _____ books from _____."

The words that best fill the blanks in the above sentence are

(A) Their, they're, there
(B) They're, their, there
(C) There, their, they're
(D) They're, there, there

Q.2 "A _____ investigation can sometimes yield new facts, but typically organized ones are more successful."

The word that best fills the blank in the above sentence is

(A) meandering (B) timely (C) consistent (D) systematic

Q.3 The area of a square is d. What is the area of the circle which has the diagonal of the square as its diameter?

(A) πd (B) πd^2 (C) $\frac{1}{4}\pi d^2$ (D) $\frac{1}{2}\pi d$

Q.4 What would be the smallest natural number which when divided either by 20 or by 42 or by 76 leaves a remainder of 7 in each case?

(A) 3047 (B) 6047 (C) 7987 (D) 63847

Q.5 What is the missing number in the following sequence?

2, 12, 60, 240, 720, 1440, _____, 0

(A) 2880 (B) 1440 (C) 720 (D) 0

Q. 6 – Q. 10 carry two marks each.

Q.6 In appreciation of the social improvements completed in a town, a wealthy philanthropist decided to gift Rs 750 to each male senior citizen in the town and Rs 1000 to each female senior citizen. Altogether, there were 300 senior citizens eligible for this gift. However, only 8/9th of the eligible men and 2/3rd of the eligible women claimed the gift. How much money (in Rupees) did the philanthropist give away in total?

(A) 1,50,000
(B) 2,00,000
(C) 1,75,000
(D) 1,51,000

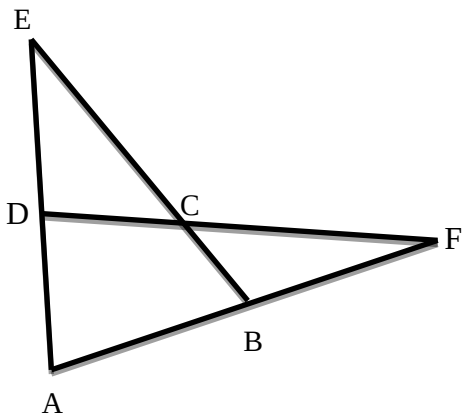
Q.7 If $pqr \neq 0$ and $p^{-x} = \frac{1}{q}, q^{-y} = \frac{1}{r}, r^{-z} = \frac{1}{p}$, what is the value of the product xyz?

(A) -1
(B) $\frac{1}{pqr}$
(C) 1
(D) pqr

Q.8 In a party, 60% of the invited guests are male and 40% are female. If 80% of the invited guests attended the party and if all the invited female guests attended, what would be the ratio of males to females among the attendees in the party?

(A) 2:3
(B) 1:1
(C) 3:2
(D) 2:1

Q.9 In the figure below, $\angle DEC + \angle BFC$ is equal to _____ .



(A) $\angle BCD - \angle BAD$
(B) $\angle BAD + \angle BCF$
(C) $\angle BAD + \angle BCD$
(D) $\angle CBA + \angle ADC$

Q.10 A six sided unbiased die with four green faces and two red faces is rolled seven times. Which of the following combinations is the most likely outcome of the experiment?

(A) Three green faces and four red faces.
(B) Four green faces and three red faces.
(C) Five green faces and two red faces.
(D) Six green faces and one red face.

END OF THE QUESTION PAPER

Q. 1 – Q. 25 carry one mark each.

Q.1 Which one of the following is a closed form expression for the generating function of the sequence $\{a_n\}$, where $a_n = 2n + 3$ for all $n = 0, 1, 2, \ldots$?

(A) $\frac{3}{(1-x)^2}$ (B) $\frac{3x}{(1-x)^2}$ (C) $\frac{2-x}{(1-x)^2}$ (D) $\frac{3-x}{(1-x)^2}$

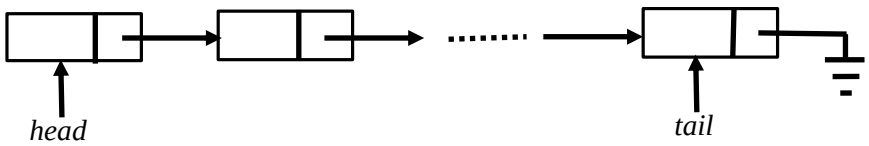
Q.2 Consider the following C program.

```
#include<stdio.h>
struct Ournode{
    char x,y,z;
};

int main(){
    struct Ournode p = {'1', '0', 'a'+2};
    struct Ournode *q = &p;
    printf ("%c, %c", *((char*)q+1), *((char*)q+2));
    return 0;
}
```

The output of this program is:

(A) 0, c (B) 0, a+2 (C) '0', 'a+2' (D) '0', 'c'

Q.3 A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let *n* denote the number of nodes in the queue. Let `enqueue` be implemented by inserting a new node at the head, and `dequeue` be implemented by deletion of a node from the tail.



Which one of the following is the time complexity of the most time-efficient implementation of `enqueue` and `dequeue`, respectively, for this data structure?

(A) $\theta(1), \theta(1)$ (B) $\theta(1), \theta(n)$ (C) $\theta(n), \theta(1)$ (D) $\theta(n), \theta(n)$

Q.4 Let $\oplus$ and $\odot$ denote the Exclusive OR and Exclusive NOR operations, respectively. Which one of the following is NOT CORRECT?

(A) $\overline{P \oplus Q} = P \odot Q$
(B) $\bar{P} \oplus Q = P \odot Q$
(C) $\bar{P} \oplus \bar{Q} = P \oplus Q$
(D) $(P \oplus \bar{P}) \oplus Q = (P \odot \bar{P}) \odot \bar{Q}$

Q.5 Consider the following processor design characteristics.

I. Register-to-register arithmetic operations only
II. Fixed-length instruction format
III. Hardwired control unit

Which of the characteristics above are used in the design of a RISC processor?

(A) I and II only (B) II and III only (C) I and III only (D) I, II and III

Q.6 Let *N* be an NFA with *n* states. Let *k* be the number of states of a minimal DFA which is equivalent to *N*. Which one of the following is necessarily true?

(A) $k \geq 2^n$ (B) $k \geq n$ (C) $k \leq n^2$ (D) $k \leq 2^n$

Q.7 The set of all recursively enumerable languages is

(A) closed under complementation.
(B) closed under intersection.
(C) a subset of the set of all recursive languages.
(D) an uncountable set.

Q.8 Which one of the following statements is FALSE?

(A) Context-free grammar can be used to specify both lexical and syntax rules.
(B) Type checking is done before parsing.
(C) High-level language programs can be translated to different Intermediate Representations.
(D) Arguments to a function can be passed using the program stack.

Q.9 The following are some events that occur after a device controller issues an interrupt while process *L* is under execution.

(P) The processor pushes the process status of *L* onto the control stack.
(Q) The processor finishes the execution of the current instruction.
(R) The processor executes the interrupt service routine.
(S) The processor pops the process status of *L* from the control stack.
(T) The processor loads the new PC value based on the interrupt.

Which one of the following is the correct order in which the events above occur?

(A) QPTRS (B) PTRSQ (C) TRPQS (D) QTPRS

Q.10 Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is *M* units if the corresponding memory page is available in memory, and *D* units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is *X* units.

Which one of the following is the correct expression for the page fault rate experienced by the process?

(A) $(D - M) / (X - M)$ (B) $(X - M) / (D - M)$
(C) $(D - X) / (D - M)$ (D) $(X - M) / (D - X)$

Q.11 In an Entity-Relationship (ER) model, suppose *R* is a many-to-one relationship from entity set E1 to entity set E2. Assume that E1 and E2 participate totally in *R* and that the cardinality of E1 is greater than the cardinality of E2.

Which one of the following is true about *R*?

(A) Every entity in E1 is associated with exactly one entity in E2.
(B) Some entity in E1 is associated with more than one entity in E2.
(C) Every entity in E2 is associated with exactly one entity in E1.
(D) Every entity in E2 is associated with at most one entity in E1.

Q.12 Consider the following two tables and four queries in SQL.

Book (isbn, bname), Stock (isbn, copies)

Query 1:
```
SELECT B.isbn, S.copies
FROM Book B INNER JOIN Stock S
ON B.isbn = S.isbn;
```

Query 2:
```
SELECT B.isbn, S.copies
FROM Book B LEFT OUTER JOIN Stock S
ON B.isbn = S.isbn;
```

Query 3:
```
SELECT B.isbn, S.copies
 FROM Book B RIGHT OUTER JOIN Stock S
 ON B.isbn = S.isbn;
```

Query 4:
```
SELECT B.isbn, S.copies
FROM Book B FULL OUTER JOIN Stock S
ON B.isbn = S.isbn;
```

Which one of the queries above is certain to have an output that is a superset of the outputs of the other three queries?

(A) Query 1 (B) Query 2 (C) Query 3 (D) Query 4

Q.13 Match the following:

Field	Length in bits
P. UDP Header's Port Number	I. 48
Q. Ethernet MAC Address	II. 8
R. IPv6 Next Header	III. 32
S. TCP Header's Sequence Number	IV. 16

(A) P-III, Q-IV, R-II, S-I
(B) P-II, Q-I, R-IV, S-III
(C) P-IV, Q-I, R-II, S-III
(D) P-IV, Q-I, R-III, S-II

Q.14 Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that *cwnd* stands for the TCP congestion window and MSS denotes the Maximum Segment Size.

(i) The *cwnd* increases by 2 MSS on every successful acknowledgment.
(ii) The *cwnd* approximately doubles on every successful acknowledgement.
(iii) The *cwnd* increases by 1 MSS every round trip time.
(iv) The *cwnd* approximately doubles every round trip time.

Which one of the following is correct?

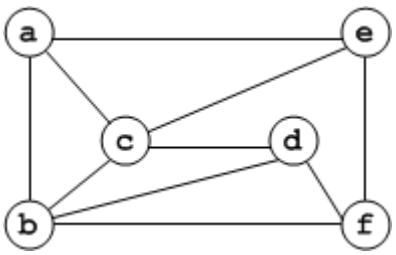
(A) Only (ii) and (iii) are true
(B) Only (i) and (iii) are true
(C) Only (iv) is true
(D) Only (i) and (iv) are true

Q.15 Two people, P and Q, decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q. Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is _____.

Q.16 The value of $\int_0^{\pi/4} x\cos(x^2)dx$ correct to three decimal places (assuming that $\pi = 3.14$) is _____.

Q.17 Consider a matrix $A = uv^T$ where $u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Note that v^T denotes the transpose of v. The largest eigenvalue of A is _____.

Q.18 The chromatic number of the following graph is _____.



Q.19 Let G be a finite group on 84 elements. The size of a largest possible proper subgroup of G is _____.

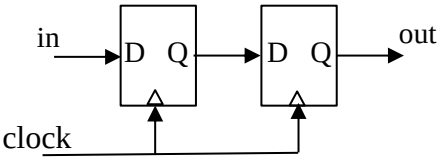
Q.20 The postorder traversal of a binary tree is 8,9,6,7,4,5,2,3,1. The inorder traversal of the same tree is 8,6,9,4,7,2,5,1,3. The height of a tree is the length of the longest path from the root to any leaf. The height of the binary tree above is _____.

Q.21 Consider the following C program:

```
#include <stdio.h>

int counter = 0;

int calc (int a, int b) {
    int c;

    counter++;
    if (b==3) return (a*a*a);
    else {
        c = calc(a, b/3);
        return (c*c*c);
    }
}

int main (){
    calc(4, 81);
    printf ("%d", counter);
}
```

The output of this program is _____.

Q.22 Consider the sequential circuit shown in the figure, where both flip-flops used are positive edge-triggered D flip-flops.



The number of states in the state transition diagram of this circuit that have a transition back to the same state on some value of "in" is _____.

Q.23 A 32-bit wide main memory unit with a capacity of 1 GB is built using 256M × 4-bit DRAM chips. The number of rows of memory cells in the DRAM chip is 2^{14}. The time taken to perform one refresh operation is 50 nanoseconds. The refresh period is 2 milliseconds. The percentage (rounded to the closest integer) of the time available for performing the memory read/write operations in the main memory unit is _____.

Q.24 Consider a system with 3 processes that share 4 instances of the same resource type. Each process can request a maximum of K instances. Resource instances can be requested and released only one at a time. The largest value of K that will always avoid deadlock is ____.

Q.25 Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps (= 10^9 bits-per-second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is _____.

Q. 26 – Q. 55 carry two marks each.

Q.26 Consider a matrix $\boldsymbol{P}$ whose only eigenvectors are the multiples of $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

Consider the following statements.

(I) $\boldsymbol{P}$ does not have an inverse
(II) $\boldsymbol{P}$ has a repeated eigenvalue
(III) $\boldsymbol{P}$ cannot be diagonalized

Which one of the following options is correct?

(A) Only I and III are necessarily true
(B) Only II is necessarily true
(C) Only I and II are necessarily true
(D) Only II and III are necessarily true

Q.27 Let N be the set of natural numbers. Consider the following sets.

P: Set of Rational numbers (positive and negative)
Q: Set of functions from $\{0, 1\}$ to N
R: Set of functions from N to $\{0, 1\}$
S: Set of finite subsets of N.

Which of the sets above are countable?

(A) Q and S only (B) P and S only (C) P and R only (D) P, Q and S only

Q.28 Consider the first-order logic sentence

$$\varphi \equiv \exists s \exists t \exists u \forall v \forall w \forall x \forall y \, \psi(s, t, u, v, w, x, y)$$

where $\psi(s, t, u, v, w, x, y)$ is a quantifier-free first-order logic formula using only predicate symbols, and possibly equality, but no function symbols. Suppose φ has a model with a universe containing 7 elements.

Which one of the following statements is necessarily true?

(A) There exists at least one model of φ with universe of size less than or equal to 3.
(B) There exists no model of φ with universe of size less than or equal to 3.
(C) There exists no model of φ with universe of size greater than 7.
(D) Every model of φ has a universe of size equal to 7.

Q.29 Consider the following C program:

```
#include<stdio.h>

void fun1(char *s1, char *s2){
     char *tmp;
     tmp = s1;
     s1 = s2;
     s2 = tmp;
}
void fun2(char **s1, char **s2){
     char *tmp;
     tmp = *s1;
     *s1 = *s2;
     *s2 = tmp;
}
int main(){
     char *str1 = "Hi", *str2 = "Bye";
     fun1(str1, str2);   printf("%s %s ", str1, str2);
     fun2(&str1, &str2); printf("%s %s", str1, str2);
     return 0;
}
```

The output of the program above is

(A) `Hi Bye Bye Hi` (B) `Hi Bye Hi Bye`
(C) `Bye Hi Hi Bye` (D) `Bye Hi Bye Hi`

Q.30 Let G be a simple undirected graph. Let T_D be a depth first search tree of G. Let T_B be a breadth first search tree of G. Consider the following statements.

(I) No edge of G is a cross edge with respect to T_D. (A cross edge in G is between two nodes neither of which is an ancestor of the other in T_D.)
(II) For every edge (u,v) of G, if u is at depth i and v is at depth j in T_B, then $|i - j| = 1$.

Which of the statements above must necessarily be true?

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

Q.31 Assume that multiplying a matrix G_1 of dimension $p \times q$ with another matrix G_2 of dimension $q \times r$ requires pqr scalar multiplications. Computing the product of n matrices $G_1G_2G_3...G_n$ can be done by parenthesizing in different ways. Define $G_i\ G_{i+1}$ as an **explicitly computed pair** for a given paranthesization if they are directly multiplied. For example, in the matrix multiplication chain $G_1G_2G_3G_4G_5G_6$ using parenthesization $(G_1(G_2G_3))(G_4(G_5G_6))$, G_2G_3 and G_5G_6 are the only explicitly computed pairs.

Consider a matrix multiplication chain $F_1F_2F_3F_4F_5$, where matrices F_1, F_2, F_3, F_4 and F_5 are of dimensions 2×25, 25×3, 3×16, 16×1 and 1×1000, respectively. In the parenthesization of $F_1F_2F_3F_4F_5$ that minimizes the total number of scalar multiplications, the explicitly computed pairs is/are

(A) F_1F_2 and F_3F_4 only
(B) F_2F_3 only
(C) F_3F_4 only
(D) F_1F_2 and F_4F_5 only

Q.32 Consider the following C code. Assume that `unsigned long int` type length is 64 bits.

```
unsigned long int fun(unsigned long int n){
    unsigned long int i, j = 0, sum = 0;
    for (i = n; i > 1; i = i/2) j++;
    for ( ; j > 1; j = j/2) sum++;
    return(sum);
}
```

The value returned when we call `fun` with the input 2^{40} is

(A) 4 (B) 5 (C) 6 (D) 40

Q.33 Consider the unsigned 8-bit fixed point binary number representation below,

$b_7\ b_6\ b_5\ b_4\ b_3\ .\ b_2\ b_1\ b_0$

where the position of the binary point is between b_3 and b_2. Assume b_7 is the most significant bit. Some of the decimal numbers listed below **cannot** be represented **exactly** in the above representation:

(i) 31.500 (ii) 0.875 (iii) 12.100 (iv) 3.001

Which one of the following statements is true?

(A) None of (i), (ii), (iii), (iv) can be exactly represented
(B) Only (ii) cannot be exactly represented
(C) Only (iii) and (iv) cannot be exactly represented
(D) Only (i) and (ii) cannot be exactly represented

Q.34 The size of the physical address space of a processor is 2^P bytes. The word length is 2^W bytes. The capacity of cache memory is 2^N bytes. The size of each cache block is 2^M words. For a K-way set-associative cache memory, the length (in number of bits) of the tag field is

(A) $P - N - \log_2 K$
(B) $P - N + \log_2 K$
(C) $P - N - M - W - \log_2 K$
(D) $P - N - M - W + \log_2 K$

Q.35 Consider the following languages:

I. $\{a^m b^n c^p d^q \mid m + p = n + q, \text{ where } m, n, p, q \geq 0\}$
II. $\{a^m b^n c^p d^q \mid m = n \text{ and } p = q, \text{ where } m, n, p, q \geq 0\}$
III. $\{a^m b^n c^p d^q \mid m = n = p \text{ and } p \neq q, \text{ where } m, n, p, q \geq 0\}$
IV. $\{a^m b^n c^p d^q \mid mn = p + q, \text{ where } m, n, p, q \geq 0\}$

Which of the languages above are context-free?

(A) I and IV only
(B) I and II only
(C) II and III only
(D) II and IV only

Q.36 Consider the following problems. $L(G)$ denotes the language generated by a grammar G. $L(M)$ denotes the language accepted by a machine M.

(I) For an unrestricted grammar G and a string w, whether $w \in L(G)$
(II) Given a Turing machine M, whether $L(M)$ is regular
(III) Given two grammars G_1 and G_2, whether $L(G_1) = L(G_2)$
(IV) Given an NFA N, whether there is a deterministic PDA P such that N and P accept the same language.

Which one of the following statements is correct?

(A) Only I and II are undecidable
(B) Only III is undecidable
(C) Only II and IV are undecidable
(D) Only I, II and III are undecidable

Q.37 A lexical analyzer uses the following patterns to recognize three tokens T_1, T_2, and T_3 over the alphabet $\{a,b,c\}$.

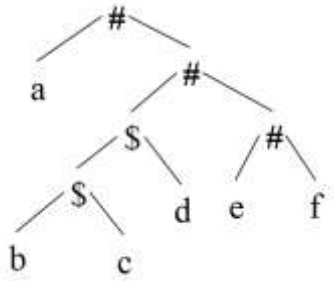
T_1: $a?\,(b|c)^*a$
T_2: $b?\,(a|c)^*b$
T_3: $c?\,(b|a)^*c$

Note that 'x?' means 0 or 1 occurrence of the symbol x. Note also that the analyzer outputs the token that matches the longest possible prefix.

If the string $bbaacabc$ is processed by the analyzer, which one of the following is the sequence of tokens it outputs?

(A) $T_1T_2T_3$ (B) $T_1T_1T_3$ (C) $T_2T_1T_3$ (D) T_3T_3

Q.38 Consider the following parse tree for the expression `a#b$c$d#e#f`, involving two binary operators $ and #.



Which one of the following is correct for the given parse tree?

(A) $ has higher precedence and is left associative; # is right associative
(B) # has higher precedence and is left associative; $ is right associative
(C) $ has higher precedence and is left associative; # is left associative
(D) # has higher precedence and is right associative; $ is left associative

Q.39 In a system, there are three types of resources: *E*, *F* and *G*. Four processes P_0, P_1, P_2 and P_3 execute concurrently. At the outset, the processes have declared their maximum resource requirements using a matrix named Max as given below. For example, Max[P_2,*F*] is the maximum number of instances of *F* that P_2 would require. The number of instances of the resources allocated to the various processes at any given state is given by a matrix named Allocation.

Consider a state of the system with the Allocation matrix as shown below, and in which 3 instances of *E* and 3 instances of *F* are the only resources available.

Allocation			
	E	*F*	*G*
P_0	1	0	1
P_1	1	1	2
P_2	1	0	3
P_3	2	0	0

Max			
	E	*F*	*G*
P_0	4	3	1
P_1	2	1	4
P_2	1	3	3
P_3	5	4	1

From the perspective of deadlock avoidance, which one of the following is true?

(A) The system is in *safe* state.
(B) The system is not in *safe* state, but would be *safe* if one more instance of *E* were available
(C) The system is not in *safe* state, but would be *safe* if one more instance of *F* were available
(D) The system is not in *safe* state, but would be *safe* if one more instance of *G* were available

Q.40 Consider the following solution to the producer-consumer synchronization problem. The shared buffer size is N. Three semaphores *empty*, *full* and `mutex` are defined with respective initial values of 0, N and 1. Semaphore *empty* denotes the number of available slots in the buffer, for the consumer to read from. Semaphore *full* denotes the number of available slots in the buffer, for the producer to write to. The placeholder variables, denoted by `P, Q, R,` and `S`, in the code below can be assigned either *empty* or *full*. The valid semaphore operations are: `wait()` and `signal()`.

Producer:	Consumer:

do{
wait(P);
wait(mutex);
//Add item to buffer
signal(mutex);
signal(Q);
}while(1);
	do{
	wait(R);
	wait(mutex);
	//Consume item from buffer
	signal(mutex);
	signal(S);
	}while(1);

Which one of the following assignments to `P, Q, R` and `S` will yield the correct solution?

(A) `P`: *full*, `Q`: *full*, `R`: *empty*, `S`: *empty*
(B) `P`: *empty*, `Q`: *empty*, `R`: *full*, `S`: *full*
(C) `P`: *full*, `Q`: *empty*, `R`: *empty*, `S`: *full*
(D) `P`: *empty*, `Q`: *full*, `R`: *full*, `S`: *empty*

Q.41 Consider the relations $r(A, B)$ and $s(B, C)$, where $s.B$ is a primary key and $r.B$ is a foreign key referencing $s.B$. Consider the query

Q: $r \bowtie (\sigma_{B<5}(s))$

Let *LOJ* denote the natural left outer-join operation. Assume that r and s contain no null values.

Which one of the following queries is NOT equivalent to Q?

(A) $\sigma_{B<5}(r \bowtie s)$
(B) $\sigma_{B<5}(r\ LOJ\ s)$
(C) $r\ LOJ\ (\sigma_{B<5}(s))$
(D) $\sigma_{B<5}(r)\ LOJ\ s$

Q.42 Consider the following four relational schemas. For each schema, all non-trivial functional dependencies are listed. The underlined attributes are the respective primary keys.

Schema I*: Registration (rollno, courses)*

Field '*courses*' is a set-valued attribute containing the set of courses a student has registered for.

Non-trivial functional dependency:

rollno $\rightarrow$ *courses*

Schema II*: Registration (rollno, courseid, email)*

Non-trivial functional dependencies:

rollno, courseid $\rightarrow$ *email*

email $\rightarrow$ *rollno*

Schema III: *Registration (rollno, courseid, marks, grade)*

Non-trivial functional dependencies:

rollno, courseid $\rightarrow$ *marks, grade*

marks $\rightarrow$ *grade*

Schema IV: *Registration (rollno, courseid, credit)*

Non-trivial functional dependencies:

rollno, courseid $\rightarrow$ *credit*

courseid $\rightarrow$ *credit*

Which one of the relational schemas above is in 3NF but not in BCNF?

(A) Schema I (B) Schema II (C) Schema III (D) Schema IV

Q.43 Let G be a graph with $100!$ vertices, with each vertex labelled by a distinct permutation of the numbers $1,2, \ldots, 100$. There is an edge between vertices u and v if and only if the label of u can be obtained by swapping two adjacent numbers in the label of v. Let y denote the degree of a vertex in G, and z denote the number of connected components in G.
Then, $y + 10z =$ _____.

Q.44 Consider Guwahati (G) and Delhi (D) whose temperatures can be classified as high (H), medium (M) and low (L). Let $P(H_G)$ denote the probability that Guwahati has high temperature. Similarly, $P(M_G)$ and $P(L_G)$ denotes the probability of Guwahati having medium and low temperatures respectively. Similarly, we use $P(H_D)$, $P(M_D)$ and $P(L_D)$ for Delhi.

The following table gives the conditional probabilities for Delhi's temperature given Guwahati's temperature.

	H_D	M_D	L_D
H_G	0.40	0.48	0.12
M_G	0.10	0.65	0.25
L_G	0.01	0.50	0.49

Consider the first row in the table above. The first entry denotes that if Guwahati has high temperature (H_G) then the probability of Delhi also having a high temperature (H_D) is 0.40; i.e., $P(H_D|H_G) = 0.40$. Similarly, the next two entries are $P(M_D|H_G) = 0.48$ and $P(L_D|H_G) = 0.12$. Similarly for the other rows.

If it is known that $P(H_G) = 0.2$, $P(M_G) = 0.5$, and $P(L_G) = 0.3$, then the probability (correct to two decimal places) that Guwahati has high temperature given that Delhi has high temperature is _____.

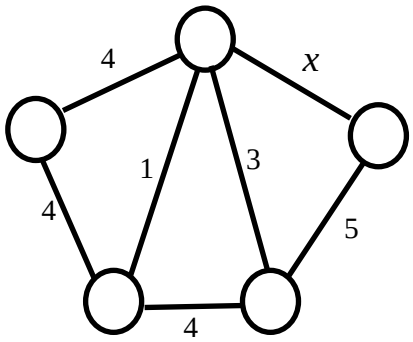
Q.45 Consider the following program written in pseudo-code. Assume that `x` and `y` are integers.

```
Count(x,y) {
    if (y != 1){
        if (x != 1) {
            print("*");
            Count(x/2, y);
        }
        else {
            y = y-1;
            Count(1024, y);
        }
    }
}
```

The number of times that the `print` statement is executed by the call `Count(1024,1024)` is _____.

Q.46 The number of possible min-heaps containing each value from {1, 2, 3, 4, 5, 6, 7} exactly once is _____.

Q.47 Consider the following undirected graph G:



Choose a value for *x* that will maximize the number of minimum weight spanning trees (MWSTs) of G. The number of MWSTs of G for this value of *x* is _____.

Q.48 Consider the weights and values of items listed below. Note that there is only one unit of each item.

Item number	Weight (in Kgs)	Value (in Rupees)
1	10	60
2	7	28
3	4	20
4	2	24

The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt}. A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy}.

The value of $V_{\text{opt}} - V_{\text{greedy}}$ is _____.

Q.49 Consider the minterm list form of a Boolean function F given below.

$$F(P, Q, R, S) = \sum m(0, 2, 5, 7, 9, 11) + d(3, 8, 10, 12, 14)$$

Here, m denotes a minterm and d denotes a don't care term. The number of essential prime implicants of the function F is _____.

Q.50 The instruction pipeline of a RISC processor has the following stages: Instruction Fetch (IF), Instruction Decode (ID), Operand Fetch (OF), Perform Operation (PO) and Writeback (WB). The IF, ID, OF and WB stages take 1 clock cycle each for every instruction. Consider a sequence of 100 instructions. In the PO stage, 40 instructions take 3 clock cycles each, 35 instructions take 2 clock cycles each, and the remaining 25 instructions take 1 clock cycle each. Assume that there are no data hazards and no control hazards.

The number of clock cycles required for completion of execution of the sequence of instructions is _____.

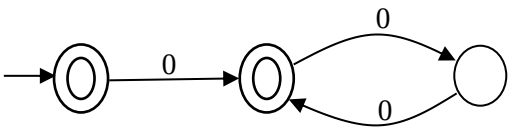
Q.51 A processor has 16 integer registers (R0, R1, .. , R15) and 64 floating point registers (F0, F1,… , F63). It uses a 2-byte instruction format. There are four categories of instructions: Type-1, Type-2, Type-3, and Type-4. Type-1 category consists of four instructions, each with 3 integer register operands (3Rs). Type-2 category consists of eight instructions, each with 2 floating point register operands (2Fs). Type-3 category consists of fourteen instructions, each with one integer register operand and one floating point register operand (1R+1F). Type-4 category consists of N instructions, each with a floating point register operand (1F).

The maximum value of N is _____.

Q.52 Given a language L, define L^i as follows:

$$L^0 = \{\varepsilon\}$$
$$L^i = L^{i-1} \cdot L \quad for\ all\ i > 0$$

The order of a language L is defined as the smallest *k* such that $L^k = L^{k+1}$.
Consider the language L_1 (over alphabet 0) accepted by the following automaton.



The order of L_1 is _____.

Q.53 Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, … , 199), and 256 sectors per track (numbered as 0, 1, … , 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2] , [180, 134, 1] , [60, 20, 0] , [212, 86, 3] , [56, 116, 2] , [118, 16, 1]

Currently the head is positioned at sector number 100 of cylinder 80, and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible.

The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is _____.

Q.54 Consider an IP packet with a length of 4,500 bytes that includes a 20-byte IPv4 header and a 40-byte TCP header. The packet is forwarded to an IPv4 router that supports a Maximum Transmission Unit (MTU) of 600 bytes. Assume that the length of the IP header in all the outgoing fragments of this packet is 20 bytes. Assume that the fragmentation offset value stored in the first fragment is 0.

The fragmentation offset value stored in the third fragment is _____.

Q.55 Consider a simple communication system where multiple nodes are connected by a shared broadcast medium (like Ethernet or wireless). The nodes in the system use the following carrier-sense based medium access protocol. A node that receives a packet to transmit will carrier-sense the medium for 5 units of time. If the node does not detect any other transmission in this duration, it starts transmitting its packet in the next time unit. If the node detects another transmission, it waits until this other transmission finishes, and then begins to carrier-sense for 5 time units again. Once they start to transmit, nodes do not perform any collision detection and continue transmission even if a collision occurs. All transmissions last for 20 units of time. Assume that the transmission signal travels at the speed of 10 meters per unit time in the medium.

Assume that the system has two nodes P and Q, located at a distance *d* meters from each other. P starts transmitting a packet at time *t=0* after successfully completing its carrier-sense phase. Node Q has a packet to transmit at time *t=0* and begins to carrier-sense the medium.

The maximum distance *d* (in meters, rounded to the closest integer) that allows Q to successfully avoid a collision between its proposed transmission and P's ongoing transmission is _____.

END OF THE QUESTION PAPER

Q.No.	Type	Section	Key/Range	Marks
1	MCQ	GA	B	1
2	MCQ	GA	A	1
3	MCQ	GA	D	1
4	MCQ	GA	C	1
5	MCQ	GA	B	1
6	MCQ	GA	B	2
7	MCQ	GA	C	2
8	MCQ	GA	B	2
9	MCQ	GA	A	2
10	MCQ	GA	C	2
1	MCQ	CS	D	1
2	MCQ	CS	A	1
3	MCQ	CS	B	1
4	MCQ	CS	D	1
5	MCQ	CS	D	1
6	MCQ	CS	D	1
7	MCQ	CS	B	1
8	MCQ	CS	B	1
9	MCQ	CS	A	1
10	MCQ	CS	B	1
11	MCQ	CS	A	1
12	MCQ	CS	D	1
13	MCQ	CS	C	1
14	MCQ	CS	C	1

Q.No.	Type	Section	Key/Range	Marks
15	NAT	CS	0.021 to 0.024	1
16	NAT	CS	0.27 to 0.30	1
17	NAT	CS	3 to 3	1
18	NAT	CS	3 to 3	1
19	NAT	CS	42 to 42	1
20	NAT	CS	4 to 4	1
21	NAT	CS	4 to 4	1
22	NAT	CS	2 to 2	1
23	NAT	CS	59.0 to 60.0	1
24	NAT	CS	2 to 2	1
25	NAT	CS	34 to 35	1
26	MCQ	CS	D	2
27	MCQ	CS	D	2
28	MCQ	CS	A	2
29	MCQ	CS	A	2
30	MCQ	CS	A	2
31	MCQ	CS	C	2
32	MCQ	CS	B	2
33	MCQ	CS	C	2
34	MCQ	CS	B	2
35	MCQ	CS	B	2
36	MCQ	CS	D	2
37	MCQ	CS	D	2
38	MCQ	CS	A	2

Q.No.	Type	Section	Key/Range	Marks
39	MCQ	CS	A	2
40	MCQ	CS	C	2
41	MCQ	CS	C	2
42	MCQ	CS	B	2
43	NAT	CS	109 to 109	2
44	NAT	CS	0.60 to 0.62	2
45	NAT	CS	10230 to 10230	2
46	NAT	CS	80 to 80	2
47	NAT	CS	4 to 4	2
48	NAT	CS	16 to 16	2
49	NAT	CS	3 to 3	2
50	NAT	CS	219 to 219	2
51	NAT	CS	32 to 32	2
52	NAT	CS	2 to 2	2
53	NAT	CS	85 to 85	2
54	NAT	CS	144 to 144	2
55	NAT	CS	50 to 50	2

Q. 1 – Q. 5 carry one mark each.

Q.1 The expenditure on the project _____ as follows: equipment Rs.20 lakhs, salaries Rs.12 lakhs, and contingency Rs.3 lakhs.

(A) break down (B) break (C) breaks down (D) breaks

Q.2 The search engine's business model _____ around the fulcrum of trust.

(A) revolves (B) plays (C) sinks (D) bursts

Q.3 Two cars start at the same time from the same location and go in the same direction. The speed of the first car is 50 km/h and the speed of the second car is 60 km/h. The number of hours it takes for the distance between the two cars to be 20 km is ___.

(A) 1 (B) 2 (C) 3 (D) 6

Q.4 Ten friends planned to share equally the cost of buying a gift for their teacher. When two of them decided not to contribute, each of the other friends had to pay Rs 150 more. The cost of the gift was Rs. ____.

(A) 666 (B) 3000 (C) 6000 (D) 12000

Q.5 A court is to a judge as _____ is to a teacher.

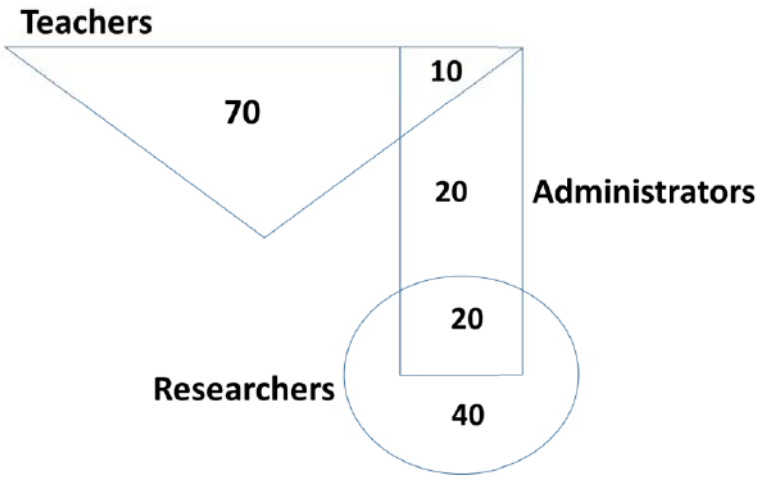
(A) a student (B) a punishment (C) a syllabus (D) a school

Q. 6 – Q. 10 carry two marks each.

Q.6 The police arrested four criminals – P, Q, R and S. The criminals knew each other. They made the following statements:
P says "Q committed the crime."
Q says "S committed the crime."
R says "I did not do it."
S says "What Q said about me is false."
Assume only one of the arrested four committed the crime and only one of the statements made above is true. Who committed the crime?

(A) P (B) R (C) S (D) Q

Q.7 In the given diagram, teachers are represented in the triangle, researchers in the circle and administrators in the rectangle. Out of the total number of the people, the percentage of administrators shall be in the range of _____.



(A) 0 to 15 (B) 16 to 30 (C) 31 to 45 (D) 46 to 60

Q.8 "A recent High Court judgement has sought to dispel the idea of begging as a disease — which leads to its stigmatization and criminalization — and to regard it as a symptom. The underlying disease is the failure of the state to protect citizens who fall through the social security net."

Which one of the following statements can be inferred from the given passage?

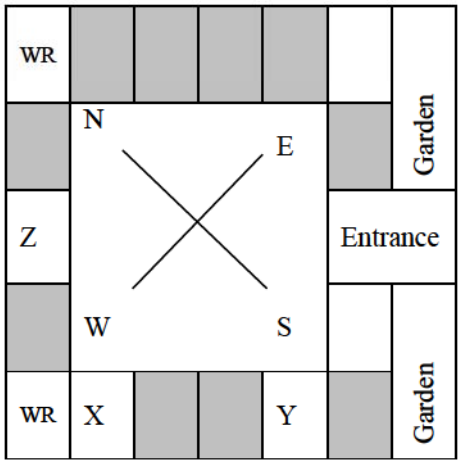
(A) Beggars are lazy people who beg because they are unwilling to work
(B) Beggars are created because of the lack of social welfare schemes
(C) Begging is an offence that has to be dealt with firmly
(D) Begging has to be banned because it adversely affects the welfare of the state

Q.9 In a college, there are three student clubs. Sixty students are only in the Drama club, 80 students are only in the Dance club, 30 students are only in the Maths club, 40 students are in both Drama and Dance clubs, 12 students are in both Dance and Maths clubs, 7 students are in both Drama and Maths clubs, and 2 students are in all the clubs. If 75% of the students in the college are not in any of these clubs, then the total number of students in the college is _____.

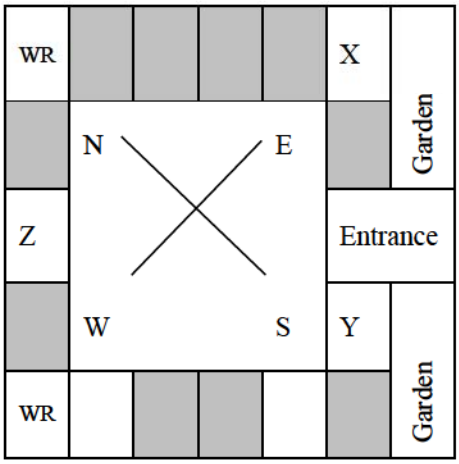
(A) 1000 (B) 975 (C) 900 (D) 225

Q.10 Three of the five students allocated to a hostel put in special requests to the warden. Given the floor plan of the vacant rooms, select the allocation plan that will accommodate all their requests.
Request by X: Due to pollen allergy, I want to avoid a wing next to the garden.
Request by Y: I want to live as far from the washrooms as possible, since I am very sensitive to smell.
Request by Z: I believe in Vaastu and so want to stay in the South-west wing.
The shaded rooms are already occupied. WR is washroom.

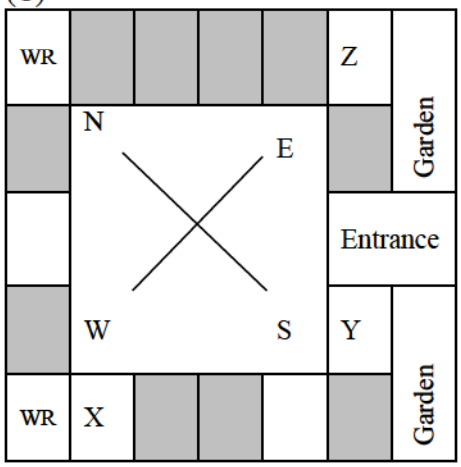
(A)



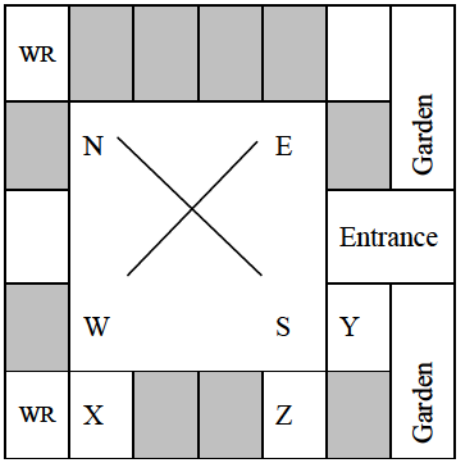
(B)



(C)



(D)

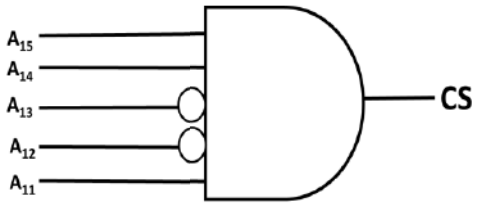


END OF THE QUESTION PAPER

Q.1 A certain processor uses a fully associative cache of size 16 kB. The cache block size is 16 bytes. Assume that the main memory is byte addressable and uses a 32-bit address. How many bits are required for the *Tag* and the *Index* fields respectively in the addresses generated by the processor?

(A) 24 bits and 0 bits
(B) 28 bits and 4 bits
(C) 24 bits and 4 bits
(D) 28 bits and 0 bits

Q.2 The chip select logic for a certain DRAM chip in a memory system design is shown below. Assume that the memory system has 16 address lines denoted by A_{15} to A_0. What is the range of addresses (in hexadecimal) of the memory system that can get enabled by the chip select (CS) signal?



(A) C800 to CFFF
(B) CA00 to CAFF
(C) C800 to C8FF
(D) DA00 to DFFF

Q.3 Which one of the following kinds of derivation is used by LR parsers?

(A) Leftmost
(B) Leftmost in reverse
(C) Rightmost
(D) Rightmost in reverse

Q.4 In 16-bit 2's complement representation, the decimal number −28 is:

(A) 1111 1111 0001 1100
(B) 0000 0000 1110 0100
(C) 1111 1111 1110 0100
(D) 1000 0000 1110 0100

Q.5 Let $U = \{1,2, \ldots, n\}$. Let $A = \{(x, X) | x \in X, X \subseteq U\}$. Consider the following two statements on $|A|$.

I. $|A| = n2^{n-1}$
II. $|A| = \sum_{k=1}^{n} k\binom{n}{k}$

Which of the above statements is/are TRUE?

(A) Only I
(B) Only II
(C) Both I and II
(D) Neither I nor II

Q.6 Which one of the following is NOT a valid identity?

(A) $(x \oplus y) \oplus z = x \oplus (y \oplus z)$
(B) $(x + y) \oplus z = x \oplus (y + z)$
(C) $x \oplus y = x + y$, if $xy = 0$
(D) $x \oplus y = (xy + x'y')'$

Q.7 If L is a regular language over $\Sigma = \{a, b\}$, which one of the following languages is NOT regular ?

(A) $L \cdot L^R = \{xy \mid x \in L, y^R \in L\}$
(B) $\{ww^R \mid w \in L\}$
(C) Prefix $(L) = \{x \in \Sigma^* | \exists y \in \Sigma^*$ such that $xy \subset L\}$
(D) Suffix $(L) = \{y \in \Sigma^* | \exists x \in \Sigma^*$ such that $xy \in L\}$

Q.8 Consider Z = X – Y, where X, Y and Z are all in sign-magnitude form. X and Y are each represented in n bits. To avoid overflow, the representation of Z would require a minimum of:

(A) n bits
(B) $n - 1$ bits
(C) $n + 1$ bits
(D) $n + 2$ bits

Q.9 Let X be a square matrix. Consider the following two statements on X.

I. X is invertible.
II. Determinant of X is non-zero.

Which one of the following is TRUE?

(A) I implies II; II does not imply I.
(B) II implies I; I does not imply II.
(C) I does not imply II; II does not imply I.
(D) I and II are equivalent statements.

Q.10 Let G be an arbitrary group. Consider the following relations on G:

R_1: $\forall a, b \in G$, $a\ R_1 b$ if and only if $\exists g \in G$ such that $a = g^{-1}bg$
R_2: $\forall a, b \in G$, $a\ R_2 b$ if and only if $a = b^{-1}$

Which of the above is/are equivalence relation/relations?

(A) R_1 and R_2 (B) R_1 only (C) R_2 only (D) Neither R_1 nor R_2

Q.11 Consider the following two statements about database transaction schedules:

I. Strict two-phase locking protocol generates conflict serializable schedules that are also recoverable.

II. Timestamp-ordering concurrency control protocol with Thomas' Write Rule can generate view serializable schedules that are not conflict serializable.

Which of the above statements is/are TRUE?

(A) I only (B) II only (C) Both I and II (D) Neither I nor II

Q.12 Let G be an undirected complete graph on n vertices, where $n > 2$. Then, the number of different Hamiltonian cycles in G is equal to

(A) $n!$ (B) $(n-1)!$ (C) 1 (D) $\frac{(n-1)!}{2}$

Q.13 Compute $\lim_{x \to 3} \frac{x^4-81}{2x^2-5x-3}$

(A) 1 (B) 53/12

(C) 108/7 (D) Limit does not exist

Q.14 Which one of the following statements is NOT correct about the B+ tree data structure used for creating an index of a relational database table?

(A) B+ Tree is a height-balanced tree
(B) Non-leaf nodes have pointers to data records
(C) Key values in each node are kept in sorted order
(D) Each leaf node has a pointer to the next leaf node

Q.15 For $\Sigma = \{a, b\}$, let us consider the regular language $L = \{ x \mid x = a^{2+3k}$ or $x = b^{10+12k}$, $k \geq 0\}$. Which one of the following can be a pumping length (the constant guaranteed by the pumping lemma) for L ?

(A) 3 (B) 5 (C) 9 (D) 24

Q.16 Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

(A) IMAP, POP3 (B) SMTP, POP3
(C) SMTP, MIME (D) IMAP, SMTP

Q.17 The following C program is executed on a Unix/Linux system:

```
#include <unistd.h>
int main()
{
        int i;
        for (i=0; i<10; i++)
                if (i%2 == 0) fork();
        return 0;
}
```

The total number of child processes created is _____.

Q.18 Consider the following C program:

```
#include <stdio.h>
int jumble(int x, int y){
        x=2*x+y;
        return x;
}
int main(){
        int x=2, y=5;
        y=jumble(y,x);
        x=jumble(y,x);
        printf("%d \n", x);
        return 0;
}
```

The value printed by the program is _____.

Q.19 Consider the grammar given below:

$S \rightarrow Aa$
$A \rightarrow BD$
$B \rightarrow b \mid \epsilon$
$D \rightarrow d \mid \epsilon$

Let a, b, d, and $ be indexed as follows:

a	b	d	$
3	2	1	0

Compute the FOLLOW set of the non-terminal B and write the index values for the symbols in the FOLLOW set in the descending order. (For example, if the FOLLOW set is {a, b, d, $}, then the answer should be 3210)

Answer: _____.

Q.20 An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is _____.

Q.21 The value of $3^{51} \bmod 5$ is _____.

Q.22 Two numbers are chosen independently and uniformly at random from the set $\{1, 2, \ldots, 13\}$. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is _____.

Q.23 Consider three concurrent processes P1, P2 and P3 as shown below, which access a shared variable D that has been initialized to 100.

P1	P2	P3
:	:	:
:	:	:
D = D + 20	D = D – 50	D = D + 10
:	:	:
:	:	:

The processes are executed on a uniprocessor system running a time-shared operating system. If the minimum and maximum possible values of D after the three processes have completed execution are X and Y respectively, then the value of Y – X is _____.

Q.24 Consider the following C program:

```
#include <stdio.h>
int main(){
        int arr[]={1,2,3,4,5,6,7,8,9,0,1,2,5}, *ip=arr+4;
        printf("%d\n", ip[1]);
        return 0;
}
```

The number that will be displayed on execution of the program is_____ .

Q.25 Consider a sequence of 14 elements: A = [−5, −10, 6, 3, −1, −2, 13, 4, −9, −1, 4, 12, −3, 0]. The subsequence sum $S(i,j) = \sum_{k=i}^{j} A[k]$. Determine the maximum of $S(i,j)$, where $0 \leq i \leq j < 14$. (Divide and conquer approach may be used.)

Answer: _____.

Q.26 Consider the following C function.

```
void convert(int n){
        if(n<0)
          printf("%d",n);
        else {
            convert(n/2);
            printf("%d",n%2);
        }
}
```

Which one of the following will happen when the function `convert` is called with any positive integer `n` as argument?

(A) It will print the binary representation of `n` and terminate
(B) It will print the binary representation of `n` in the reverse order and terminate
(C) It will print the binary representation of `n` but will not terminate
(D) It will not print anything and will not terminate

Q.27 Consider the following C program:

```
#include <stdio.h>
int r(){
      static int num=7;
      return num--;
}

int main(){
      for (r();r();r())
            printf("%d",r());
      return 0;
}
```

Which one of the following values will be displayed on execution of the programs?

(A) 41 (B) 52 (C) 63 (D) 630

Q.28 Consider three machines M, N, and P with IP addresses 100.10.5.2, 100.10.5.5, and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all the three machines. Which one of the following is true?

(A) M, N, and P all belong to the same subnet
(B) Only M and N belong to the same subnet
(C) Only N and P belong to the same subnet
(D) M, N, and P belong to three different subnets

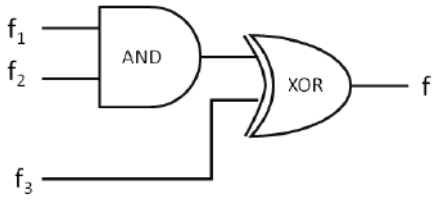
Q.29 Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

(A) X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X
(B) X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X
(C) X sends an ARP request packet with broadcast MAC address in its local subnet
(D) X sends an ARP request packet with broadcast IP address in its local subnet

Q.30 Consider three 4-variable functions f_1, f_2, and f_3, which are expressed in sum-of-minterms as

$f_1 = \Sigma(0, 2, 5, 8, 14)$, $f_2 = \Sigma(2, 3, 6, 8, 14, 15)$, $f_3 = \Sigma(2, 7, 11, 14)$

For the following circuit with one AND gate and one XOR gate, the output function f can be expressed as:



(A) $\Sigma(7, 8, 11)$
(B) $\Sigma(2, 7, 8, 11, 14)$
(C) $\Sigma(2, 14)$
(D) $\Sigma(0, 2, 3, 5, 6, 7, 8, 11, 14, 15)$

Q.31 Which one of the following languages over $\Sigma = \{a, b\}$ is NOT context-free?

(A) $\{ww^R \mid w \in \{a, b\}^*\}$
(B) $\{wa^n b^n w^R \mid w \in \{a, b\}^*, n \geq 0\}$
(C) $\{wa^n w^R b^n \mid w \in \{a, b\}^*, n \geq 0\}$
(D) $\{a^n b^i \mid i \in \{n, 3n, 5n\}, n \geq 0\}$

Q.32 Let the set of functional dependencies F = {QR → S, R → P, S → Q} hold on a relation schema X = (PQRS). X is not in BCNF. Suppose X is decomposed into two schemas Y and Z, where Y = (PR) and Z = (QRS).

Consider the two statements given below.

I. Both Y and Z are in BCNF
II. Decomposition of X into Y and Z is dependency preserving and lossless

Which of the above statements is/are correct?

(A) Both I and II (B) I only (C) II only (D) Neither I nor II

Q.33 Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressible. The page size is 8 kB and the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?

(A) 16×2^{10} (B) 256×2^{10} (C) 4×2^{20} (D) 8×2^{20}

Q.34 Consider the following sets:

S1. Set of all recursively enumerable languages over the alphabet {0,1}
S2. Set of all syntactically valid C programs
S3. Set of all languages over the alphabet {0,1}
S4. Set of all non-regular languages over the alphabet {0,1}

Which of the above sets are uncountable?

(A) S1 and S2 (B) S3 and S4 (C) S2 and S3 (D) S1 and S4

Q.35 Consider the first order predicate formula φ:
$\forall x\,[(\forall z\ z|x \Rightarrow ((z = x) \vee (z = 1))) \Rightarrow \exists w\ (w > x) \wedge (\forall z\ z|w \Rightarrow ((w = z) \vee (z = 1)))]$
Here '$a|b$' denotes that 'a divides b', where a and b are integers. Consider the following sets:

S1. {1,2,3, ..., 100}
S2. Set of all positive integers
S3. Set of all integers

Which of the above sets satisfy φ?

(A) S1 and S2 (B) S1 and S3 (C) S2 and S3 (D) S1, S2 and S3

Q.36 Consider the following grammar and the semantic actions to support the inherited type declaration attributes. Let X_1, X_2, X_3, X_4, X_5, and X_6 be the placeholders for the non-terminals D, T, L or L_1 in the following table:

Production rule	Semantic action
D → T L	X_1.type = X_2.type
T → int	T.type = int
T → float	T.type = float
L → L_1 , id	X_3.type = X_4.type addType(id.entry, X_5.type)
L → id	addType(id.entry, X_6.type)

Which one of the following are the appropriate choices for X_1, X_2, X_3 and X_4?

(A) $X_1 = L , X_2 = T, X_3 = L_1, X_4 = L$
(B) $X_1 = T , X_2 = L, X_3 = L_1, X_4 = T$
(C) $X_1 = L , X_2 = L, X_3 = L_1, X_4 = T$
(D) $X_1 = T , X_2 = L, X_3 = T, X_4 = L_1$

Q.37 There are n unsorted arrays: $A_1, A_2, \ldots, A_n$. Assume that n is odd. Each of $A_1, A_2, \ldots, A_n$ contains n distinct elements. There are no common elements between any two arrays. The worst-case time complexity of computing the median of the medians of $A_1, A_2, \ldots, A_n$ is

(A) $O(n)$ (B) $O(n \log n)$ (C) $O(n^2)$ (D) $\Omega(n^2 \log n)$

Q.38 Let G be any connected, weighted, undirected graph.

I. G has a unique minimum spanning tree, if no two edges of G have the same weight.
II. G has a unique minimum spanning tree, if, for every cut of G, there is a unique minimum-weight edge crossing the cut.

Which of the above two statements is/are TRUE?

(A) I only (B) II only (C) Both I and II (D) Neither I nor II

Q.39 Consider the following snapshot of a system running n concurrent processes. Process i is holding X_i instances of a resource R, $1 \leq i \leq n$. Assume that all instances of R are currently in use. Further, for all i, process i can place a request for at most Y_i additional instances of R while holding the X_i instances it already has. Of the n processes, there are exactly two processes p and q such that $Y_p = Y_q = 0$. Which one of the following conditions guarantees that no other process apart from p and q can complete execution?

(A) $X_p + X_q <$ Min $\{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
(B) $X_p + X_q <$ Max $\{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
(C) Min $(X_p, X_q) \geq$ Min $\{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$
(D) Min $(X_p, X_q) \leq$ Max $\{Y_k \mid 1 \leq k \leq n, k \neq p, k \neq q\}$

Q.40 Consider the following statements:

I. The smallest element in a max-heap is always at a leaf node
II. The second largest element in a max-heap is always a child of the root node
III. A max-heap can be constructed from a binary search tree in $\Theta(n)$ time
IV. A binary search tree can be constructed from a max-heap in $\Theta(n)$ time

Which of the above statements are TRUE?

(A) I, II and III (B) I, II and IV (C) I, III and IV (D) II, III and IV

Q.41 Consider the following four processes with arrival times (in milliseconds) and their length of CPU bursts (in milliseconds) as shown below:

Process	P1	P2	P3	P4
Arrival time	0	1	3	4
CPU burst time	3	1	3	Z

These processes are run on a single processor using preemptive Shortest Remaining Time First scheduling algorithm. If the average waiting time of the processes is 1 millisecond, then the value of Z is_____.

Q.42 The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointers. The disk block size is 4 kB, and the disk block address is 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) _____ GB.

Q.43 Consider the augmented grammar given below:

$S' \rightarrow S$
$S \rightarrow \langle L \rangle \mid id$
$L \rightarrow L,S \mid S$

Let I_0 = CLOSURE ({[S' → •S]}). The number of items in the set GOTO (I_0 , ⟨) is: _____.

Q.44 Consider the following matrix:

$$R = \begin{bmatrix} 1 & 2 & 4 & 8 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \\ 1 & 5 & 25 & 125 \end{bmatrix}$$

The absolute value of the product of Eigen values of R is _____.

Q.45 A certain processor deploys a single-level cache. The cache block size is 8 words and the word size is 4 bytes. The memory system uses a 60-MHz clock. To service a cache miss, the memory controller first takes 1 cycle to accept the starting address of the block, it then takes 3 cycles to fetch all the eight words of the block, and finally transmits the words of the requested block at the rate of 1 word per cycle. The maximum bandwidth for the memory system when the program running on the processor issues a series of read operations is _____ × 10^6 bytes/sec.

Q.46 Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.) Suppose two leaves a and b of T are chosen uniformly and independently at random. The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places) _____.

Q.47 Suppose Y is distributed uniformly in the open interval (1,6). The probability that the polynomial $3x^2 + 6xY + 3Y + 6$ has only real roots is (rounded off to 1 decimal place) _____.

Q.48 Let Σ be the set of all bijections from $\{1, \ldots, 5\}$ to $\{1, \ldots, 5\}$, where id denotes the identity function, i.e. $id(j) = j, \forall j$. Let $\circ$ denote composition on functions. For a string $x = x_1\, x_2 \cdots x_n \in \Sigma^n, n \geq 0$, let $\pi(x) = x_1 \circ x_2 \circ \cdots \circ x_n$.
Consider the language $L = \{x \in \Sigma^* \mid \pi(x) = id\}$. The minimum number of states in any DFA accepting L is _____.

Q.49 Consider that 15 machines need to be connected in a LAN using 8-port Ethernet switches. Assume that these switches do not have any separate uplink ports. The minimum number of switches needed is_____.

Q.50 What is the minimum number of 2-input NOR gates required to implement a 4-variable function expressed in sum-of-minterms form as $f = \Sigma (0, 2, 5, 7, 8, 10, 13, 15)$? Assume that all the inputs and their complements are available. Answer: _____.

Q.51 A relational database contains two tables Student and Performance as shown below:

Student

Roll_no.	Student_name
1	Amit
2	Priya
3	Vinit
4	Rohan
5	Smita

Performance

Roll_no.	Subject_code	Marks
1	A	86
1	B	95
1	C	90
2	A	89
2	C	92
3	C	80

The primary key of the Student table is Roll_no. For the Performance table, the columns Roll_no. and Subject_code together form the primary key. Consider the SQL query given below:

```
SELECT S.Student_name, sum(P.Marks)
FROM Student S, Performance P
WHERE P.Marks > 84
GROUP BY S.Student_name;
```

The number of rows returned by the above SQL query is _____.

Q.52 Consider the following C program:

```
#include <stdio.h>
int main(){
   float sum = 0.0, j = 1.0, i = 2.0;
   while (i/j > 0.0625){
         j = j + j;
         sum = sum + i/j;
         printf("%f\n", sum);
   }
   return 0;
}
```

The number of times the variable `sum` will be printed, when the above program is executed, is _____.

Q.53 Consider the following C program:

```
#include <stdio.h>
int main()
{
   int a[] = {2, 4, 6, 8, 10};
   int i, sum = 0, *b = a + 4;
   for (i = 0; i < 5; i++)
         sum = sum + (*b – i) – *(b – i);
   printf ("%d\n", sum);
   return 0;
}
```

The output of the above C program is _____.

Q.54 In an RSA cryptosystem, the value of the public modulus parameter n is 3007. If it is also known that $\varphi(n) = 2880$, where $\varphi()$ denotes Euler's Totient Function, then the prime factor of n which is greater than 50 is _____.

Q.55 Consider the following relations P(X,Y,Z), Q(X,Y,T) and R(Y,V).

P		
X	**Y**	**Z**
X1	Y1	Z1
X1	Y1	Z2
X2	Y2	Z2
X2	Y4	Z4

Q		
X	**Y**	**T**
X2	Y1	2
X1	Y2	5
X1	Y1	6
X3	Y3	1

R	
Y	**V**
Y1	V1
Y3	V2
Y2	V3
Y2	V2

How many tuples will be returned by the following relational algebra query?

$$\prod_X \left(\sigma_{(P.Y=R.Y \wedge R.V=\text{V2})}(P \times R)\right) - \prod_X \left(\sigma_{(Q.Y=R.Y \wedge Q.T>2)}(Q \times R)\right)$$

Answer: _____.

Q.No.	Type	Section	Key	Marks
1	MCQ	GA	C	1
2	MCQ	GA	A	1
3	MCQ	GA	B	1
4	MCQ	GA	C	1
5	MCQ	GA	D	1
6	MCQ	GA	B	2
7	MCQ	GA	C	2
8	MCQ	GA	B	2
9	MCQ	GA	C	2
10	MCQ	GA	D	2
1	MCQ	CS	D	1
2	MCQ	CS	A	1
3	MCQ	CS	D	1
4	MCQ	CS	C	1
5	MCQ	CS	C	1
6	MCQ	CS	B	1
7	MCQ	CS	B	1
8	MCQ	CS	C	1
9	MCQ	CS	D	1
10	MCQ	CS	B	1
11	MCQ	CS	C	1
12	MCQ	CS	C OR D	1
13	MCQ	CS	C	1

Q.No.	Type	Section	Key	Marks
14	MCQ	CS	B	1
15	MCQ	CS	D	1
16	MCQ	CS	B	1
17	NAT	CS	31 to 31	1
18	NAT	CS	26 to 26	1
19	NAT	CS	31 to 31	1
20	NAT	CS	0.08 to 0.08	1
21	NAT	CS	2 to 2	1
22	NAT	CS	0.502 to 0.504	1
23	NAT	CS	80 to 80	1
24	NAT	CS	6 to 6	1
25	NAT	CS	29 to 29	1
26	MCQ	CS	D	2
27	MCQ	CS	B	2
28	MCQ	CS	C	2
29	MCQ	CS	C	2
30	MCQ	CS	A	2
31	MCQ	CS	C	2
32	MCQ	CS	C	2
33	MCQ	CS	B	2
34	MCQ	CS	B	2
35	MCQ	CS	C	2
36	MCQ	CS	A	2

Q.No.	Type	Section	Key	Marks
37	MCQ	CS	C	2
38	MCQ	CS	C	2
39	MCQ	CS	A	2
40	MCQ	CS	A	2
41	NAT	CS	2 to 2	2
42	NAT	CS	3.7 to 3.8 OR 4.0 to 4.1	2
43	NAT	CS	5 to 5	2
44	NAT	CS	12 to 12	2
45	NAT	CS	160 to 160	2
46	NAT	CS	4.25 to 4.25	2
47	NAT	CS	0.8 to 0.8	2
48	NAT	CS	120 to 120	2
49	NAT	CS	3 to 3	2
50	NAT	CS	3 to 3	2
51	NAT	CS	5 to 5	2
52	NAT	CS	5 to 5	2
53	NAT	CS	10 to 10	2
54	NAT	CS	97 to 97	2
55	NAT	CS	1 to 1	2

CS: Computer Sc. and Information Technology

GA - General Aptitude

Q1 - Q5 carry one mark each.

Q.No. 1 Raman is confident of speaking English_____six months as he has been practising regularly_____the last three weeks.

(A) during, for

(B) for, since

(C) for, in

(D) within, for

Q.No. 2 His knowledge of the subject was excellent but his classroom performance was_____.

(A) extremely poor

(B) good

(C) desirable

(D) praiseworthy

Q.No. 3 Select the word that fits the analogy:

Cook : Cook :: Fly : _____

(A) Flyer

(B) Flying

(C) Flew

(D) Flighter

Q.No. 4 The dawn of the 21st century witnessed the melting glaciers oscillating between giving too much and too little to billions of people who depend on them for fresh water. The UN climate report estimates that without deep cuts to man-made emissions, at least 30% of the northern hemisphere's surface permafrost could melt by the end of the century. Given this situation of imminent global exodus of billions of people displaced by rising seas, nation-states need to rethink their carbon footprint for political concerns, if not for environmental ones.

Which one of the following statements can be inferred from the given passage?

(A) Nation-states do not have environmental concerns.

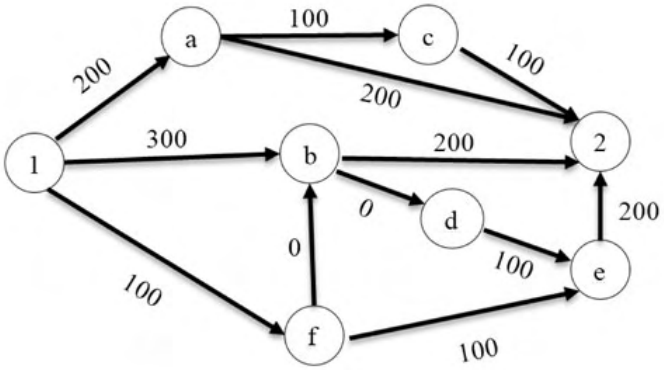
(B) Nation-states are responsible for providing fresh water to billions of people.

(C) Billions of people are responsible for man-made emissions.

(D) Billions of people are affected by melting glaciers.

Q.No. 5

There are multiple routes to reach from node 1 to node 2, as shown in the network.



The cost of travel on an edge between two nodes is given in rupees. Nodes 'a', 'b', 'c', 'd', 'e', and 'f' are toll booths. The toll price at toll booths marked 'a' and 'e' is Rs. 200, and is Rs. 100 for the other toll booths. Which is the cheapest route from node 1 to node 2?

(A) 1-a-c-2

(B) 1-f-b-2

(C) 1-b-2

(D) 1-f-e-2

Q6 - Q10 carry two marks each.

Q.No. 6 Goods and Services Tax (GST) is an indirect tax introduced in India in 2017 that is imposed on the supply of goods and services, and it subsumes all indirect taxes except few. It is a destination-based tax imposed on goods and services used, and it is not imposed at the point of origin from where goods come. GST also has a few components specific to state governments, central government and Union Territories (UTs).

Which one of the following statements can be inferred from the given passage?

(A) GST is imposed on the production of goods and services.

(B) GST includes all indirect taxes.

(C) GST does not have a component specific to UT.

(D) GST is imposed at the point of usage of goods and services.

Q.No. 7 If P = 3, R = 27, T = 243, then Q + S = _____.

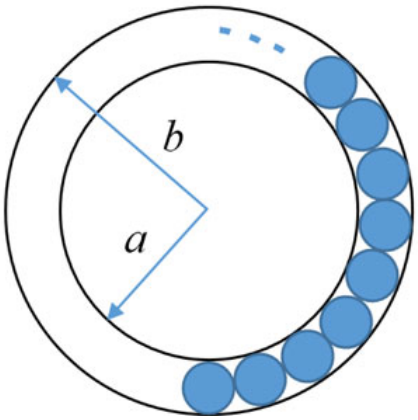
(A) 40

(B) 80

(C) 90

(D) 110

Q.No. 8 The figure below shows an annular ring with outer and inner radii as b and a, respectively. The annular space has been painted in the form of blue colour circles touching the outer and inner periphery of annular space. If maximum n number of circles can be painted, then the unpainted area available in annular space is_____.



(A) $\pi\left[(b^2-a^2)-\frac{n}{4}(b-a)^2\right]$

(B) $\pi[(b^2-a^2)-n(b-a)^2]$

(C) $\pi\left[(b^2-a^2)+\frac{n}{4}(b-a)^2\right]$

(D) $\pi[(b^2-a^2)+n(b-a)^2]$

Q.No. 9 Two straight lines are drawn perpendicular to each other in X-Y plane. If α and β are the acute angles the straight lines make with the X-axis, then $\alpha+\beta$ is_____.

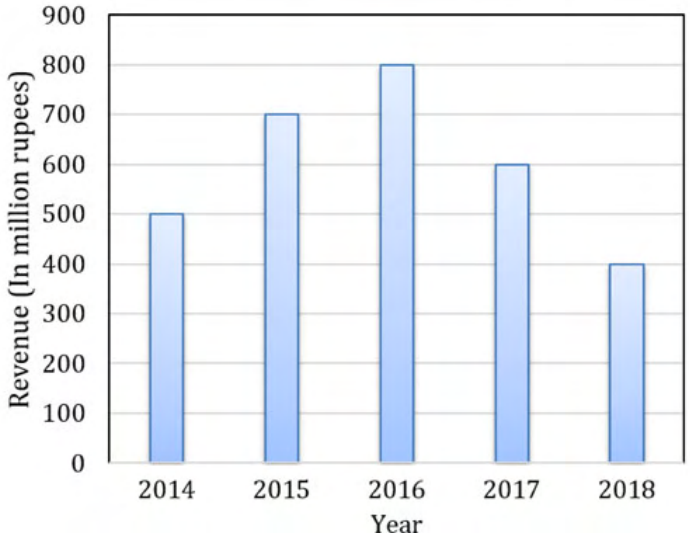
(A) $60°$

(B) $90°$

(C) $120°$

(D) $180°$

Q.No. 10 The total revenue of a company during 2014-2018 is shown in the bar graph. If the total expenditure of the company in each year is 500 million rupees, then the aggregate profit or loss (in percentage) on the total expenditure of the company during 2014-2018 is ____.



(A) 16.67 % profit

(B) 16.67 % loss

(C) 20 % profit

(D) 20 % loss

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Q1 - Q25 carry one mark each.

Q.No. 1 Consider the functions

I. e^{-x}

II. $x^2-\sin x$

III. $\sqrt{x^3+1}$

Which of the above functions is/are increasing everywhere in [0,1]?

(A) III only

(B) II only

(C) II and III only

(D) I and III only

Q.No. 2

For parameters a and b, both of which are $\omega(1)$, $T(n) = T(n^{1/a}) + 1$, and $T(b) = 1$.

Then $T(n)$ is

(A) $\Theta(\log_a \log_b n)$

(B) $\Theta(\log_{ab} n)$

(C) $\Theta(\log_b \log_a n)$

(D) $\Theta(\log_2 \log_2 n)$

Q.No. 3 Consider the following statements.

I. Daisy chaining is used to assign priorities in attending interrupts.

II. When a device raises a vectored interrupt, the CPU does polling to identify the source of interrupt.

III. In polling, the CPU periodically checks the status bits to know if any device needs its attention.

IV. During DMA, both the CPU and DMA controller can be bus masters at the same time.

Which of the above statements is/are TRUE?

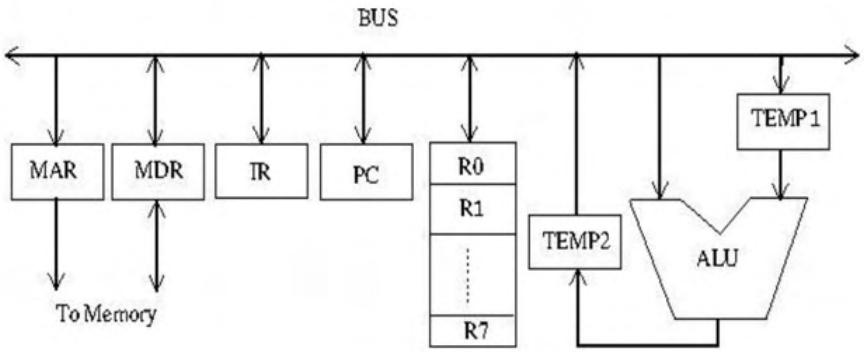
(A) I and II only

(B) I and IV only

(C) I and III only

(D) III only

Q.No. 4 Consider the following data path diagram.



Consider an instruction: R0 ← R1 + R2. The following steps are used to execute it over the given data path. Assume that PC is incremented appropriately. The subscripts r and w indicate read and write operations, respectively.

1. $R2_r$, $TEMP1_r$, ALU_{add}, $TEMP2_w$
2. $R1_r$, $TEMP1_w$
3. PC_r, MAR_w, MEM_r
4. $TEMP2_r$, $R0_w$
5. MDR_r, IR_w

Which one of the following is the correct order of execution of the above steps?

(A) 2, 1, 4, 5, 3

(B) 1, 2, 4, 3, 5

(C) 3, 5, 2, 1, 4

(D) 3, 5, 1, 2, 4

Q.No. 5 The preorder traversal of a binary search tree is 15, 10, 12, 11, 20, 18, 16, 19. Which one of the following is the postorder traversal of the tree?

(A) 10, 11, 12, 15, 16, 18, 19, 20

(B) 11, 12, 10, 16, 19, 18, 20, 15

(C) 20, 19, 18, 16, 15, 12, 11, 10

(D) 19, 16, 18, 20, 11, 12, 10, 15

Q.No. 6 What is the worst case time complexity of inserting n^2 elements into an AVL-tree with n elements initially?

(A) $\Theta(n^4)$

(B) $\Theta(n^2)$

(C) $\Theta(n^2 \log n)$

(D) $\Theta(n^3)$

Q.No. 7 Which one of the following regular expressions represents the set of all binary strings with an odd number of 1's?

(A) $((0+1)^*1(0+1)^*1)^*10^*$

(B) $(0^*10^*10^*)^*0^*1$

(C) $10^*(0^*10^*10^*)^*$

(D) $(0^*10^*10^*)^*10^*$

Q.No. 8 Consider the following statements.

I. If $L_1 \cup L_2$ is regular, then both L_1 and L_2 must be regular.

II. The class of regular languages is closed under infinite union.

Which of the above statements is/are TRUE?

(A) I only

(B) II only

(C) Both I and II

(D) Neither I nor II

Q.No. 9 Consider the following statements.

I. Symbol table is accessed only during lexical analysis and syntax analysis.

II. Compilers for programming languages that support recursion necessarily need heap storage for memory allocation in the run-time environment.

III. Errors violating the condition '*any variable must be declared before its use*' are detected during syntax analysis.

Which of the above statements is/are TRUE?

(A) I only

(B) I and III only

(C) II only

(D) None of I, II, and III

Q.No. 10

Consider the language $L = \{a^n \mid n \geq 0\} \cup \{a^n b^n \mid n \geq 0\}$ and the following statements.

I. L is deterministic context-free.

II. L is context-free but not deterministic context-free.

III. L is not LL(k) for any k.

Which of the above statements is/are TRUE?

(A) I only

(B) II only

(C) I and III only

(D) III only

Q.No. 11 Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statements is TRUE?

(A) The hole created by first fit is always larger than the hole created by next fit.

(B) The hole created by worst fit is always larger than the hole created by first fit.

(C) The hole created by best fit is never larger than the hole created by first fit.

(D) The hole created by next fit is never larger than the hole created by best fit.

Q.No. 12 Consider the following statements about process state transitions for a system using preemptive scheduling.

I. A running process can move to ready state.

II. A ready process can move to running state.

III. A blocked process can move to running state.

IV. A blocked process can move to ready state.

Which of the above statements are TRUE?

(A) I, II, and III only

(B) II and III only

(C) I, II, and IV only

(D) I, II, III, and IV

Q.No. 13

Consider a relational database containing the following schemas.

Catalogue

sno	pno	cost
S1	P1	150
S1	P2	50
S1	P3	100
S2	P4	200
S2	P5	250
S3	P1	250
S3	P2	150
S3	P5	300
S3	P4	250

Suppliers

sno	sname	location
S1	M/s Royal furniture	Delhi
S2	M/s Balaji furniture	Bangalore
S3	M/s Premium furniture	Chennai

Parts

pno	pname	part_spec
P1	Table	Wood
P2	Chair	Wood
P3	Table	Steel
P4	Almirah	Steel
P5	Almirah	Wood

The primary key of each table is indicated by underlining the constituent fields.

```
SELECT s.sno, s.sname
FROM   Suppliers s, Catalogue c
WHERE  s.sno = c.sno AND
       cost > (SELECT AVG (cost)
               FROM Catalogue
               WHERE pno = 'P4'
               GROUP BY pno);
```

The number of rows returned by the above SQL query is

(A) 4
(B) 5
(C) 0
(D) 2

Q.No. 14 Which one of the following is used to represent the supporting many-one relationships of a weak entity set in an entity-relationship diagram?

(A) Diamonds with double/bold border
(B) Rectangles with double/bold border
(C) Ovals with double/bold border
(D) Ovals that contain underlined identifiers

Q.No. 15 Consider the following statements about the functionality of an IP based router.

I. A router does not modify the IP packets during forwarding.
II. It is not necessary for a router to implement any routing protocol.
III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet.

Which of the above statements is/are TRUE?

(A) I and II only
(B) I only
(C) II and III only
(D) II only

Q.No. 16 What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be maintained in sorted order?

(A) $\Theta(n)$

(B) $\Theta(n \log n)$

(C) $\Theta(n^2)$

(D) $\Theta(1)$

Q.No. 17 Let $\mathcal{R}$ be the set of all binary relations on the set $\{1,2,3\}$. Suppose a relation is chosen from $\mathcal{R}$ at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is _____.

Q.No. 18 Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is _____.

Q.No. 19 A multiplexer is placed between a group of 32 registers and an accumulator to regulate data movement such that at any given point in time the content of only one register will move to the accumulator. The minimum number of select lines needed for the multiplexer is _____.

Q.No. 20 If there are m input lines and n output lines for a decoder that is used to uniquely address a byte addressable 1 KB RAM, then the minimum value of $m + n$ is ____ .

Q.No. 21 A direct mapped cache memory of 1 MB has a block size of 256 bytes. The cache has an access time of 3 ns and a hit rate of 94%. During a cache miss, it takes 20 ns to bring the first word of a block from the main memory, while each subsequent word takes 5 ns. The word size is 64 bits. The average memory access time in ns (round off to 1 decimal place) is _____.

Q.No. 22 Consider the following C program.

```
#include <stdio.h>
int main() {
    int a[4][5]={{1, 2, 3, 4, 5},
                 {6, 7, 8, 9, 10},
                 {11, 12, 13, 14, 15},
                 {16, 17, 18 , 19, 20}};
    printf("%d\n", *(*(a+**a+2)+3));
    return(0);
}
```

The output of the program is _____.

Q.No. 23 Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 23$, and the secondary hash function is $h_2(k) = 1 + (k \bmod 19)$. Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value $k = 90$ is _____.

Q.No. 24

Consider the following grammar.

$S \rightarrow aSB \mid d$

$B \rightarrow b$

The number of reduction steps taken by a bottom-up parser while accepting the string *aaadbbb* is _____.

Q.No. 25 Assume that you have made a request for a web page through your web browser to a web server. Initially the browser cache is empty. Further, the browser is configured to send HTTP requests in non-persistent mode. The web page contains text and five very small images. The minimum number of TCP connections required to display the web page completely in your browser is _____.

Q26 - Q55 carry two marks each.

Q.No. 26 Which of the following languages are undecidable? Note that $\langle M \rangle$ indicates encoding of the Turing machine M.

$L_1 = \{\langle M \rangle | \ L(M) = \emptyset\}$

$L_2 = \{\langle M, w, q \rangle | \ M \text{ on input } w \text{ reaches state } q \text{ in exactly 100 steps}\}$

$L_3 = \{\langle M \rangle | \ L(M) \text{ is not recursive}\}$

$L_4 = \{\langle M \rangle | \ L(M) \text{ contains at least 21 members}\}$

(A) L_1, L_3, and L_4 only
(B) L_1 and L_3 only
(C) L_2 and L_3 only
(D) L_2, L_3, and L_4 only

Q.No. 27 Let A and B be two $n \times n$ matrices over real numbers. Let $\text{rank}(M)$ and $\det(M)$ denote the rank and determinant of a matrix M, respectively. Consider the following statements.

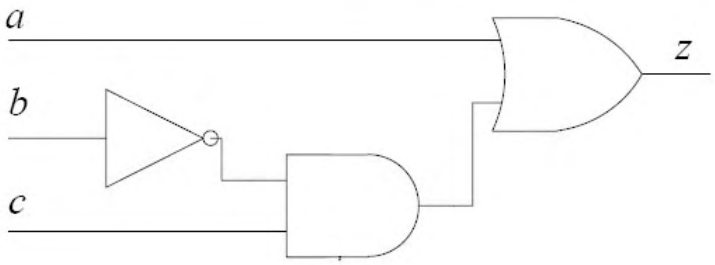
I. $\text{rank}(AB) = \text{rank}(A)\ \text{rank}(B)$
II. $\det(AB) = \det(A)\det(B)$
III. $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$
IV. $\det(A + B) \leq \det(A) + \det(B)$

Which of the above statements are TRUE?

(A) I and II only
(B) I and IV only
(C) II and III only
(D) III and IV only

Q.No. 28

Consider the Boolean function $z(a, b, c)$.



Which one of the following minterm lists represents the circuit given above?

(A) $z = \sum(0, 1, 3, 7)$

(B) $z = \sum(1, 4, 5, 6, 7)$

(C) $z = \sum(2, 4, 5, 6, 7)$

(D) $z = \sum(2, 3, 5)$

Q.No. 29 Consider three registers R1, R2, and R3 that store numbers in IEEE-754 single precision floating point format. Assume that R1 and R2 contain the values (in hexadecimal notation) 0x42200000 and 0xC1200000, respectively.

If $R3 = \frac{R1}{R2}$, what is the value stored in R3?

(A) 0x40800000

(B) 0xC0800000

(C) 0x83400000

(D) 0xC8500000

Q.No. 30 A computer system with a word length of 32 bits has a 16 MB byte-addressable main memory and a 64 KB, 4-way set associative cache memory with a block size of 256 bytes. Consider the following four physical addresses represented in hexadecimal notation.

A1 = 0x42C8A4, A2 = 0x546888, A3 = 0x6A289C, A4 = 0x5E4880

Which one of the following is TRUE?

(A) A1 and A4 are mapped to different cache sets.

(B) A2 and A3 are mapped to the same cache set.

(C) A3 and A4 are mapped to the same cache set.

(D) A1 and A3 are mapped to the same cache set.

Q.No. 31 Let $G = (V, E)$ be a weighted undirected graph and let T be a Minimum Spanning Tree (MST) of G maintained using adjacency lists. Suppose a new weighted edge $(u, v) \in V \times V$ is added to G. The worst case time complexity of determining if T is still an MST of the resultant graph is

(A) $\Theta(|E| + |V|)$

(B) $\Theta(|E||V|)$

(C) $\Theta(|E| \log |V|)$

(D) $\Theta(|V|)$

Q.No. 32 Consider the following languages.

$$L_1 = \{wxyx \mid w, x, y \in (0+1)^+\}$$

$$L_2 = \{xy \mid x, y \in (a+b)^*, |x| = |y|, x \neq y\}$$

Which one of the following is TRUE?

(A) L_1 is regular and L_2 is context-free.

(B) L_1 is context-free but not regular and L_2 is context-free.

(C) Neither L_1 nor L_2 is context-free.

(D) L_1 is context-free but L_2 is not context-free.

Q.No. 33 Consider the productions A → PQ and A→ XY. Each of the five non-terminals A, P, Q, X, and Y has two attributes: *s* is a synthesized attribute, and *i* is an inherited attribute. Consider the following rules.

Rule 1: P.*i* = A.*i* + 2, Q.*i* = P.*i* + A.*i*, and A.*s* = P.*s* + Q.*s*

Rule 2: X.*i* = A.*i* + Y.*s* and Y.*i* = X.*s* + A.*i*

Which one of the following is TRUE?

(A) Both Rule 1 and Rule 2 are L-attributed.

(B) Only Rule 1 is L-attributed.

(C) Only Rule 2 is L-attributed.

(D) Neither Rule 1 nor Rule 2 is L-attributed.

Q.No. 34 Each of a set of n processes executes the following code using two semaphores `a` and `b` initialized to 1 and 0, respectively. Assume that `count` is a shared variable initialized to 0 and not used in CODE SECTION P.

CODE SECTION P

```
wait(a); count=count+1;
if (count==n) signal(b);
signal(a); wait(b); signal(b);
```

CODE SECTION Q

What does the code achieve?

(A) It ensures that no process executes CODE SECTION Q before every process has finished CODE SECTION P.

(B) It ensures that at most two processes are in CODE SECTION Q at any time.

(C) It ensures that all processes execute CODE SECTION P mutually exclusively.

(D) It ensures that at most $n-1$ processes are in CODE SECTION P at any time.

Q.No. 35

Consider the following five disk access requests of the form (request id, cylinder number) that are present in the disk scheduler queue at a given time.

(P, 155), (Q, 85), (R, 110), (S, 30), (T, 115)

Assume the head is positioned at cylinder 100. The scheduler follows Shortest Seek Time First scheduling to service the requests.

Which one of the following statements is FALSE?

(A) T is serviced before P.

(B) Q is serviced after S, but before T.

(C) The head reverses its direction of movement between servicing of Q and P.

(D) R is serviced before P.

Q.No. 36 Consider a relational table R that is in 3NF, but not in BCNF. Which one of the following statements is TRUE?

(A) R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a prime attribute.

(B) R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a non-prime attribute and X is not a proper subset of any key.

(C) R has a nontrivial functional dependency $X \rightarrow A$, where X is not a superkey and A is a non-prime attribute and X is a proper subset of some key.

(D) A cell in R holds a set instead of an atomic value.

Q.No. 37 Consider a schedule of transactions T_1 and T_2:

T_1	RA			RC		WD		WB	Commit	
T_2		RB	WB		RD		WC			Commit

Here, RX stands for "Read(X)" and WX stands for "Write(X)". Which one of the following schedules is conflict equivalent to the above schedule?

(A)

T_1				RA	RC	WD	WB		Commit	
T_2	RB	WB	RD					WC		Commit

(B)

T_1	RA	RC	WD	WB					Commit	
T_2					RB	WB	RD	WC		Commit

(C)

T_1	RA	RC	WD				WB		Commit	
T_2				RB	WB	RD		WC		Commit

(D)

T_1					RA	RC	WD	WB	Commit	
T_2	RB	WB	RD	WC						Commit

Q.No. 38

An organization requires a range of IP addresses to assign one to each of its 1500 computers. The organization has approached an Internet Service Provider (ISP) for this task. The ISP uses CIDR and serves the requests from the available IP address space 202.61.0.0/17. The ISP wants to assign an address space to the organization which will minimize the number of routing entries in the ISP's router using route aggregation. Which of the following address spaces are potential candidates from which the ISP can allot any one to the organization?

I. 202.61.84.0/21
II. 202.61.104.0/21
III. 202.61.64.0/21
IV. 202.61.144.0/21

(A) I and II only
(B) II and III only
(C) III and IV only
(D) I and IV only

Q.No. 39 Which one of the following predicate formulae is NOT logically valid?

Note that W is a predicate formula without any free occurrence of x.

(A) $\forall x(p(x) \vee W) \equiv \forall x\, p(x) \vee W$
(B) $\exists x(p(x) \wedge W) \equiv \exists x\, p(x) \wedge W$
(C) $\forall x(p(x) \rightarrow W) \equiv \forall x\, p(x) \rightarrow W$
(D) $\exists x(p(x) \rightarrow W) \equiv \forall x\, p(x) \rightarrow W$

Q.No. 40 Let $G = (V, E)$ be a directed, weighted graph with weight function $w: E \rightarrow \mathbb{R}$. For some function $f: V \rightarrow \mathbb{R}$, for each edge $(u, v) \in E$, define $w'(u, v)$ as $w(u, v) + f(u) - f(v)$.

Which one of the options completes the following sentence so that it is TRUE?
"The shortest paths in G under w are shortest paths under w' too, _____".

(A) for every $f: V \rightarrow \mathbb{R}$
(B) if and only if $\forall u \in V$, $f(u)$ is positive
(C) if and only if $\forall u \in V$, $f(u)$ is negative
(D) if and only if $f(u)$ is the distance from s to u in the graph obtained by adding a new vertex s to G and edges of zero weight from s to every vertex of G

Q.No. 41 In a balanced binary search tree with n elements, what is the worst case time complexity of reporting all elements in range $[a, b]$? Assume that the number of reported elements is k.

(A) $\Theta(\log n)$
(B) $\Theta(\log n + k)$
(C) $\Theta(k \log n)$
(D) $\Theta(n \log k)$

Q.No. 42 The number of permutations of the characters in LILAC so that no character appears in its original position, if the two L's are indistinguishable, is _____.

Q.No. 43 Consider a non-pipelined processor operating at 2.5 GHz. It takes 5 clock cycles to complete an instruction. You are going to make a 5-stage pipeline out of this processor. Overheads associated with pipelining force you to operate the pipelined processor at 2 GHz. In a given program, assume that 30% are memory instructions, 60% are ALU instructions and the rest are branch instructions. 5% of the memory instructions cause stalls of 50 clock cycles each due to cache misses and 50% of the branch instructions cause stalls of 2 cycles each. Assume that there are no stalls associated with the execution of ALU instructions. For this program, the speedup achieved by the pipelined processor over the non-pipelined processor (round off to 2 decimal places) is _____.

Q.No. 44 A processor has 64 registers and uses 16-bit instruction format. It has two types of instructions: I-type and R-type. Each I-type instruction contains an opcode, a register name, and a 4-bit immediate value. Each R-type instruction contains an opcode and two register names. If there are 8 distinct I-type opcodes, then the maximum number of distinct R-type opcodes is _____.

Q.No. 45 For $n > 2$, let $a \in \{0,1\}^n$ be a non-zero vector. Suppose that x is chosen uniformly at random from $\{0,1\}^n$. Then, the probability that $\sum_{i=1}^{n} a_i x_i$ is an odd number is _____.

Q.No. 46 Consider the following C functions.

```
int fun1(int n) {
  static int i = 0;
  if (n > 0) {
     ++i;
     fun1(n-1);
  }
  return(i);
}
```

```
int fun2(int n) {
  static int i = 0;
  if (n > 0) {
     i = i + fun1(n);
     fun2(n-1);
  }
  return(i);
}
```

The return value of `fun2(5)` is _____.

Q.No. 47 Consider the array representation of a binary min-heap containing 1023 elements. The minimum number of comparisons required to find the maximum in the heap is _____.

Q.No. 48

Consider the following C functions.

```
int tob(int b, int* arr){
   int i;
   for(i=0; b>0; i++){
      if(b%2) arr[i]=1;
      else    arr[i]=0;
      b = b/2;
   }
   return(i);

}
```

```
int pp(int a,int b) {
    int arr[20];
    int i, tot = 1, ex, len;
    ex = a;
    len = tob(b,arr);
    for(i=0; i<len; i++){
       if(arr[i]==1)
          tot = tot * ex;
       ex = ex * ex;
    }
    return(tot);
}
```

The value returned by `pp(3,4)` is _____.

Q.No. 49 Consider a graph $G = (V, E)$, where $V = \{v_1, v_2, \ldots, v_{100}\}$, $E = \{(v_i, v_j) | 1 \leq i < j \leq 100\}$, and weight of the edge (v_i, v_j) is $|i - j|$. The weight of minimum spanning tree of G is _____.

Q.No. 50 Consider the following set of processes, assumed to have arrived at time 0. Consider the CPU scheduling algorithms Shortest Job First (SJF) and Round Robin (RR). For RR, assume that the processes are scheduled in the order P_1, P_2, P_3, P_4.

Processes	P_1	P_2	P_3	P_4
Burst time (in ms)	8	7	2	4

If the time quantum for RR is 4 ms, then the absolute value of the difference between the average turnaround times (in ms) of SJF and RR (round off to 2 decimal places) is _____.

Q.No. 51 Consider the following language.

$L = \{x \in \{a, b\}^* |$ number of $a's$ in x is divisible by 2 but not divisible by 3$\}$

The minimum number of states in a DFA that accepts L is _____.

Q.No. 52 Graph G is obtained by adding vertex s to $K_{3,4}$ and making s adjacent to every vertex of $K_{3,4}$. The minimum number of colours required to edge-colour G is _____.

Q.No. 53

Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read in from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is _____.

Q.No. 54 Consider a database implemented using B+ tree for file indexing and installed on a disk drive with block size of 4 KB. The size of search key is 12 bytes and the size of tree/disk pointer is 8 bytes. Assume that the database has one million records. Also assume that no node of the B+ tree and no records are present initially in main memory. Consider that each record fits into one disk block. The minimum number of disk accesses required to retrieve any record in the database is _____.

Q.No. 55 Consider a TCP connection between a client and a server with the following specifications: the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time $t = 0$. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time $t + 60$ ms after all acknowledgements are processed is _____.

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Answer Key - CS: Computer Sc. and Information Technology

Q.No.	Session	Que.Type	Sec. Name	Key	Marks
1	6	MCQ	GA	D	1
2	6	MCQ	GA	A	1
3	6	MCQ	GA	A	1
4	6	MCQ	GA	D	1
5	6	MCQ	GA	B	1
6	6	MCQ	GA	D	2
7	6	MCQ	GA	C	2
8	6	MCQ	GA	A	2
9	6	MCQ	GA	B	2
10	6	MCQ	GA	C	2
1	6	MCQ	CS	A	1
2	6	MCQ	CS	A	1
3	6	MCQ	CS	C	1
4	6	MCQ	CS	C	1
5	6	MCQ	CS	B	1
6	6	MCQ	CS	C	1
7	6	MCQ	CS	MTA	1
8	6	MCQ	CS	D	1
9	6	MCQ	CS	D	1
10	6	MCQ	CS	C	1
11	6	MCQ	CS	C	1
12	6	MCQ	CS	C	1
13	6	MCQ	CS	A	1
14	6	MCQ	CS	A	1
15	6	MCQ	CS	D	1
16	6	MCQ	CS	C	1
17	6	NAT	CS	0.125 to 0.125	1
18	6	NAT	CS	7 to 7	1
19	6	NAT	CS	5 to 5	1
20	6	NAT	CS	1034 to 1034	1
21	6	NAT	CS	13.3 to 13.3 OR 13.5 to 13.5	1
22	6	NAT	CS	19 to 19	1
23	6	NAT	CS	13 to 13	1
24	6	NAT	CS	7 to 7	1
25	6	NAT	CS	6 to 6	1
26	6	MCQ	CS	A	2
27	6	MCQ	CS	C	2
28	6	MCQ	CS	B	2
29	6	MCQ	CS	B	2
30	6	MCQ	CS	B	2
31	6	MCQ	CS	D	2
32	6	MCQ	CS	A	2

33	6	MCQ	CS	B	2
34	6	MCQ	CS	A	2
35	6	MCQ	CS	B	2
36	6	MCQ	CS	A	2
37	6	MCQ	CS	A	2
38	6	MCQ	CS	B	2
39	6	MCQ	CS	C	2
40	6	MCQ	CS	A	2
41	6	MCQ	CS	B	2
42	6	NAT	CS	12 to 12	2
43	6	NAT	CS	2.15 to 2.18	2
44	6	NAT	CS	14 to 14	2
45	6	NAT	CS	0.5 to 0.5	2
46	6	NAT	CS	55 to 55	2
47	6	NAT	CS	511 to 511	2
48	6	NAT	CS	81 to 81	2
49	6	NAT	CS	99 to 99	2
50	6	NAT	CS	5.25 to 5.25	2
51	6	NAT	CS	6 to 6	2
52	6	NAT	CS	7 to 7	2
53	6	NAT	CS	154.5 to 155.5	2
54	6	NAT	CS	4 to 4	2
55	6	NAT	CS	44 to 44	2

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GATE 2022 General Aptitude (GA)

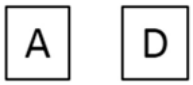
Q.1 – Q.5 Carry ONE mark each.

Q.1	The _____ is too high for it to be considered _____.
(A)	fair / fare
(B)	faer / fair
(C)	fare / fare
(D)	fare / fair

Q.2	A function $y(x)$ is defined in the interval $[0, 1]$ on the x-axis as $$y(x) = \begin{cases} 2 & \text{if } 0 \le x < \frac{1}{3} \\ 3 & \text{if } \frac{1}{3} \le x < \frac{3}{4} \\ 1 & \text{if } \frac{3}{4} \le x \le 1 \end{cases}$$ Which one of the following is the area under the curve for the interval $[0, 1]$ on the x-axis?
(A)	$\frac{5}{6}$
(B)	$\frac{6}{5}$
(C)	$\frac{13}{6}$
(D)	$\frac{6}{13}$

Q.3	Let r be a root of the equation $x^2 + 2x + 6 = 0$. Then the value of the expression $(r + 2)(r + 3)(r + 4)(r + 5)$ is
(A)	51
(B)	-51
(C)	126
(D)	-126

Q.4	Given below are four statements. Statement 1: All students are inquisitive. Statement 2: Some students are inquisitive. Statement 3: No student is inquisitive. Statement 4: Some students are not inquisitive. From the given four statements, find the two statements that **CANNOT BE TRUE** simultaneously, assuming that there is at least one student in the class.
(A)	Statement 1 and Statement 3
(B)	Statement 1 and Statement 2
(C)	Statement 2 and Statement 4
(D)	Statement 3 and Statement 4

Q.5	A palindrome is a word that reads the same forwards and backwards. In a game of words, a player has the following two plates painted with letters.  From the additional plates given in the options, which one of the combinations of additional plates would allow the player to construct a five-letter palindrome. The player should use all the five plates exactly once. The plates can be rotated in their plane.
(A)	
(B)	
(C)	
(D)	

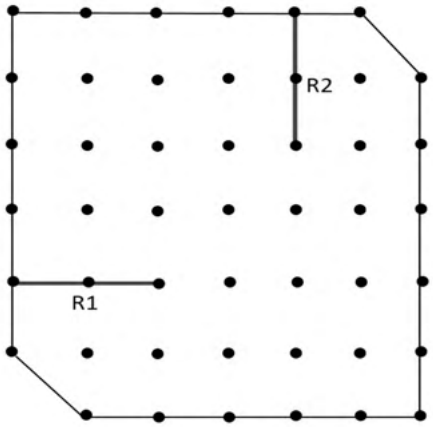
Q. 6 – Q. 10 Carry TWO marks each.

Q.6	Some people believe that "what gets measured, improves". Some others believe that "what gets measured, gets gamed". One possible reason for the difference in the beliefs is the work culture in organizations. In organizations with good work culture, metrics help improve outcomes. However, the same metrics are counterproductive in organizations with poor work culture. Which one of the following is the CORRECT logical inference based on the information in the above passage?
(A)	Metrics are useful in organizations with poor work culture
(B)	Metrics are useful in organizations with good work culture
(C)	Metrics are always counterproductive in organizations with good work culture
(D)	Metrics are never useful in organizations with good work culture

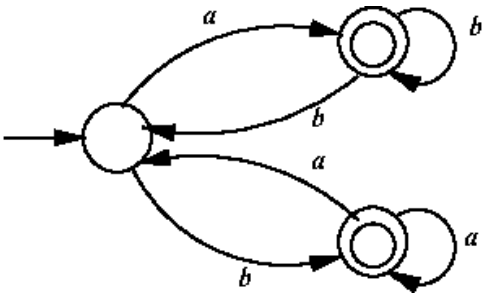
Q.7	In a recently conducted national entrance test, boys constituted 65% of those who appeared for the test. Girls constituted the remaining candidates and they accounted for 60% of the qualified candidates. Which one of the following is the correct logical inference based on the information provided in the above passage?
(A)	Equal number of boys and girls qualified
(B)	Equal number of boys and girls appeared for the test
(C)	The number of boys who appeared for the test is less than the number of girls who appeared
(D)	The number of boys who qualified the test is less than the number of girls who qualified

Q.8	A box contains five balls of same size and shape. Three of them are green coloured balls and two of them are orange coloured balls. Balls are drawn from the box one at a time. If a green ball is drawn, it is not replaced. If an orange ball is drawn, it is replaced with another orange ball. First ball is drawn. What is the probability of getting an orange ball in the next draw?
(A)	$\frac{1}{2}$
(B)	$\frac{8}{25}$
(C)	$\frac{19}{50}$
(D)	$\frac{23}{50}$

Q.9	The corners and mid-points of the sides of a triangle are named using the distinct letters P, Q, R, S, T and U, but not necessarily in the same order. Consider the following statements: • The line joining P and R is parallel to the line joining Q and S. • P is placed on the side opposite to the corner T. • S and U cannot be placed on the same side. Which one of the following statements is correct based on the above information?
(A)	P cannot be placed at a corner
(B)	S cannot be placed at a corner
(C)	U cannot be placed at a mid-point
(D)	R cannot be placed at a corner

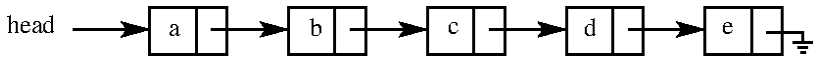
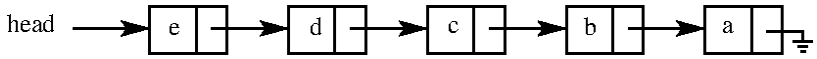
Q.10	A plot of land must be divided between four families. They want their individual plots to be similar in shape, not necessarily equal in area. The land has equally spaced poles, marked as dots in the below figure. Two ropes, R1 and R2, are already present and cannot be moved. What is the least number of **additional** straight ropes needed to create the desired plots? A single rope can pass through three poles that are aligned in a straight line. 
(A)	2
(B)	4
(C)	5
(D)	3

Q.11 – Q.22 Multiple Choice Questions (MCQ), carry ONE mark each.

Q.11	Which one of the following statements is TRUE for all positive functions $f(n)$?
(A)	$f(n^2) = \theta(f(n)^2)$, when $f(n)$ is a polynomial
(B)	$f(n^2) = o(f(n)^2)$
(C)	$f(n^2) = O(f(n)^2)$, when $f(n)$ is an exponential function
(D)	$f(n^2) = \Omega(f(n)^2)$
Q.12	Which one of the following regular expressions correctly represents the language of the finite automaton given below?
	
(A)	$ab^*bab^* + ba^*aba^*$
(B)	$(ab^*b)^*ab^* + (ba^*a)^*ba^*$
(C)	$(ab^*b + ba^*a)^*(a^* + b^*)$
(D)	$(ba^*a + ab^*b)^*(ab^* + ba^*)$

Q.13	Which one of the following statements is TRUE?
(A)	The $LALR(1)$ parser for a grammar G cannot have reduce-reduce conflict if the $LR(1)$ parser for G does not have reduce-reduce conflict.
(B)	Symbol table is accessed only during the lexical analysis phase.
(C)	Data flow analysis is necessary for run-time memory management.
(D)	$LR(1)$ parsing is sufficient for deterministic context-free languages.
Q.14	In a relational data model, which one of the following statements is TRUE?
(A)	A relation with only two attributes is always in BCNF.
(B)	If all attributes of a relation are prime attributes, then the relation is in BCNF.
(C)	Every relation has at least one non-prime attribute.
(D)	BCNF decompositions preserve functional dependencies.

GATE 2022 Computer Science and Information Technology (CS)

Q.15	Consider the problem of reversing a singly linked list. To take an example, given the linked list below,  the reversed linked list should look like  Which one of the following statements is TRUE about the time complexity of algorithms that solve the above problem in $O(1)$ space?
(A)	The best algorithm for the problem takes $\theta(n)$ time in the worst case.
(B)	The best algorithm for the problem takes $\theta(n \log n)$ time in the worst case.
(C)	The best algorithm for the problem takes $\theta(n^2)$ time in the worst case.
(D)	It is not possible to reverse a singly linked list in $O(1)$ space.
Q.16	Suppose we are given n keys, m hash table slots, and two simple uniform hash functions h_1 and h_2. Further suppose our hashing scheme uses h_1 for the odd keys and h_2 for the even keys. What is the expected number of keys in a slot?
(A)	$\frac{m}{n}$
(B)	$\frac{n}{m}$
(C)	$\frac{2n}{m}$
(D)	$\frac{n}{2m}$

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Q.17	Which one of the following facilitates transfer of bulk data from hard disk to main memory with the highest throughput?
(A)	DMA based I/O transfer
(B)	Interrupt driven I/O transfer
(C)	Polling based I/O transfer
(D)	Programmed I/O transfer
Q.18	Let R1 and R2 be two 4-bit registers that store numbers in 2's complement form. For the operation R1+R2, which one of the following values of R1 and R2 gives an arithmetic overflow?
(A)	R1 = 1011 and R2 = 1110
(B)	R1 = 1100 and R2 = 1010
(C)	R1 = 0011 and R2 = 0100
(D)	R1 = 1001 and R2 = 1111

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Q.19 Consider the following threads, T_1, T_2, and T_3 executing on a single processor, synchronized using three binary semaphore variables, S_1, S_2, and S_3, operated upon using standard `wait()` and `signal()`. The threads can be context switched in any order and at any time.

T_1	T_2	T_3

`while(true){`
`  wait(S3);`
`  print("C");`
`  signal(S2); }`
	`while(true){`
	`  wait(S1);`
	`  print("B");`
	`  signal(S3); }`
		`while(true){`
		`  wait(S2);`
		`  print("A");`
		`  signal(S1); }`

Which initialization of the semaphores would print the sequence `BCABCABCA`….?

(A) $S_1 = 1; S_2 = 1; S_3 = 1$

(B) $S_1 = 1; S_2 = 1; S_3 = 0$

(C) $S_1 = 1; S_2 = 0; S_3 = 0$

(D) $S_1 = 0; S_2 = 1; S_3 = 1$

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Q.20	Consider the following two statements with respect to the matrices $A_{m\times n}$, $B_{n\times m}$, $C_{n\times n}$ and $D_{n\times n}$. Statement 1: *tr*(AB) = *tr*(BA) Statement 2: *tr*(CD) = *tr*(DC) where *tr*() represents the trace of a matrix. Which one of the following holds?
(A)	Statement 1 is correct and Statement 2 is wrong.
(B)	Statement 1 is wrong and Statement 2 is correct.
(C)	Both Statement 1 and Statement 2 are correct.
(D)	Both Statement 1 and Statement 2 are wrong.

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Q.21	What is printed by the following ANSI C program?
	`#include<stdio.h>`
	`int main(int argc, char *argv[])`
	`{`
	`int x = 1, z[2] = {10, 11};`
	`int *p = NULL;`
	`p = &x;`
	`*p = 10;`
	`p = &z[1];`
	`*(&z[0] + 1) += 3;`
	`printf("%d, %d, %d\n", x, z[0], z[1]);`
	`return 0;`
	`}`

(A)	`1, 10, 11`
(B)	`1, 10, 14`
(C)	`10, 14, 11`
(D)	`10, 10, 14`

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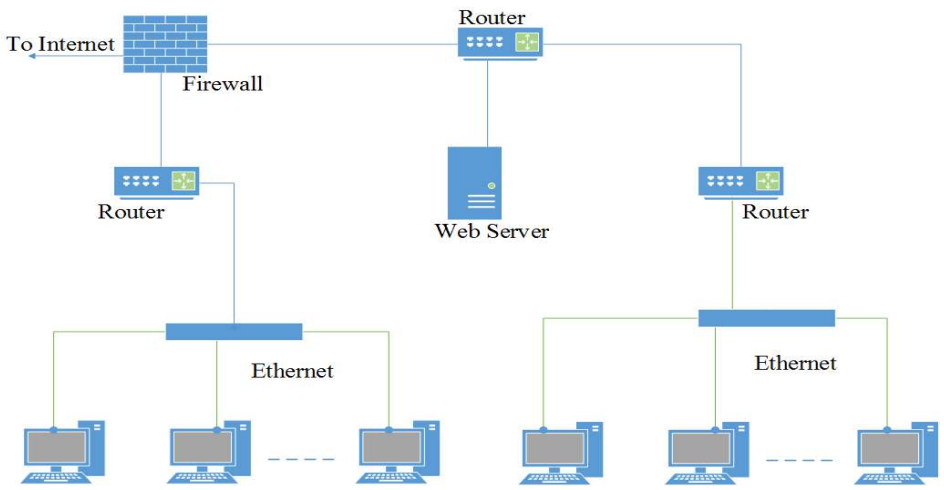
Q.22	Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via three routers as shown below.
	 What is the number of subnets inside the enterprise network?
(A)	3
(B)	12
(C)	6
(D)	8

Q.23 – Q.27 Multiple Select Questions (MSQ), carry ONE mark each.

Q.23	Which of the following statements is/are TRUE?
(A)	Every subset of a recursively enumerable language is recursive.
(B)	If a language L and its complement $\overline{L}$ are both recursively enumerable, then L must be recursive.
(C)	Complement of a context-free language must be recursive.
(D)	If L_1 and L_2 are regular, then $L_1 \cap L_2$ must be deterministic context-free.
Q.24	Let WB and WT be two set associative cache organizations that use LRU algorithm for cache block replacement. WB is a write back cache and WT is a write through cache. Which of the following statements is/are FALSE?
(A)	Each cache block in WB and WT has a dirty bit.
(B)	Every write hit in WB leads to a data transfer from cache to main memory.
(C)	Eviction of a block from WT will not lead to data transfer from cache to main memory.
(D)	A read miss in WB will never lead to eviction of a dirty block from WB.

Q.25	Consider the following three relations in a relational database. $Employee(\underline{eId}, Name)$, $Brand(\underline{bId}, bName)$, $Own(\underline{eId}, \underline{bId})$ Which of the following relational algebra expressions return the set of *eId*s who own all the brands?
(A)	$\Pi_{eId}(\Pi_{eId,bId}(Own) / \Pi_{bId}(Brand))$
(B)	$\Pi_{eId}(Own) - \Pi_{eId}\left((\Pi_{eId}(Own) \times \Pi_{bId}(Brand)) - \Pi_{eId,bId}(Own)\right)$
(C)	$\Pi_{eId}(\Pi_{eId,bId}(Own) / \Pi_{bId}(Own))$
(D)	$\Pi_{eId}\left((\Pi_{eId}(Own) \times \Pi_{bId}(Own)) / \Pi_{bId}(Brand)\right)$
Q.26	Which of the following statements is/are TRUE with respect to deadlocks?
(A)	Circular wait is a necessary condition for the formation of deadlock.
(B)	In a system where each resource has more than one instance, a cycle in its wait-for graph indicates the presence of a deadlock.
(C)	If the current allocation of resources to processes leads the system to unsafe state, then deadlock will necessarily occur.
(D)	In the resource-allocation graph of a system, if every edge is an assignment edge, then the system is not in deadlock state.

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Q.27	Which of the following statements is/are TRUE for a group G ?
(A)	If for all $x, y \in G$, $(xy)^2 = x^2y^2$, then G is commutative.
(B)	If for all $x \in G$, $x^2 = 1$, then G is commutative. Here, 1 is the identity element of G.
(C)	If the order of G is 2, then G is commutative.
(D)	If G is commutative, then a subgroup of G need not be commutative.

Q.28 – Q.35 Numerical Answer Type (NAT), carry ONE mark each.

Q.28	Suppose a binary search tree with 1000 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with 0, the 3rd largest element of the tree is stored at index_____.
Q.29	Consider the augmented grammar with $\{+, *, (,), id\}$ as the set of terminals. $S' \rightarrow S$ $S \rightarrow S + R \mid R$ $R \rightarrow R * P \mid P$ $P \rightarrow (S) \mid id$ If I_0 is the set of two $LR(0)$ items $\{[S' \rightarrow S_{\bullet}], [S \rightarrow S_{\bullet} + R]\}$, then $goto(closure(I_0), +)$ contains exactly _____ items.
Q.30	Consider a simple undirected graph of 10 vertices. If the graph is disconnected, then the maximum number of edges it can have is _____.
Q.31	Consider a relation $R(A, B, C, D, E)$ with the following three functional dependencies. $AB \rightarrow C;\ \ BC \rightarrow D;\ \ C \rightarrow E;$ The number of superkeys in the relation R is _____.
Q.32	The number of arrangements of six identical balls in three identical bins is_____.

Q.33	A cache memory that has a hit rate of 0.8 has an access latency 10 ns and miss penalty 100 ns. An optimization is done on the cache to reduce the miss rate. However, the optimization results in an increase of cache access latency to 15 ns, whereas the miss penalty is not affected. The minimum hit rate (*rounded off to two decimal places*) needed after the optimization such that it should not increase the average memory access time is _____.
Q.34	The value of the following limit is _____. $$\lim_{x \to 0+} \frac{\sqrt{x}}{1-e^{2\sqrt{x}}}$$
Q.35	Consider the resolution of the domain name `www.gate.org.in` by a DNS resolver. Assume that no resource records are cached anywhere across the DNS servers and that iterative query mechanism is used in the resolution. The number of DNS query-response pairs involved in completely resolving the domain name is_____.

Q.36 – Q.45 Multiple Choice Questions (MCQ), carry TWO marks each.

Q.36	Which one of the following is the closed form for the generating function of the sequence $\{a_n\}_{n\geq 0}$ defined below? $a_n = \begin{cases} n+1, & n \text{ is odd} \\ 1, & \text{otherwise} \end{cases}$
(A)	$\frac{x\left(1+x^2\right)}{\left(1-x^2\right)^2}+\frac{1}{1-x}$
(B)	$\frac{x(3-x^2)}{\left(1-x^2\right)^2}+\frac{1}{1-x}$
(C)	$\frac{2x}{\left(1-x^2\right)^2}+\frac{1}{1-x}$
(D)	$\frac{x}{(1-x^2)^2}+\frac{1}{1-x}$
Q.37	Consider a simple undirected unweighted graph with at least three vertices. If A is the adjacency matrix of the graph, then the number of 3-cycles in the graph is given by the trace of
(A)	A^3
(B)	A^3 divided by 2
(C)	A^3 divided by 3
(D)	A^3 divided by 6

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Q.38	Which one of the following statements is FALSE?
(A)	The TLB performs an associative search in parallel on all its valid entries using page number of incoming virtual address.
(B)	If the virtual address of a word given by CPU has a TLB hit, but the subsequent search for the word results in a cache miss, then the word will always be present in the main memory.
(C)	The memory access time using a given inverted page table is always same for all incoming virtual addresses.
(D)	In a system that uses hashed page tables, if two distinct virtual addresses V1 and V2 map to the same value while hashing, then the memory access time of these addresses will not be the same.
Q.39	Let $R_i(z)$ and $W_i(z)$ denote read and write operations on a data element z by a transaction T_i, respectively. Consider the schedule $\boldsymbol{S}$ with four transactions. $\boldsymbol{S}$: $R_4(x)R_2(x)R_3(x)R_1(y)W_1(y)W_2(x)W_3(y)R_4(y)$ Which one of the following serial schedules is conflict equivalent to $\boldsymbol{S}$?
(A)	$T_1 \rightarrow T_3 \rightarrow T_4 \rightarrow T_2$
(B)	$T_1 \rightarrow T_4 \rightarrow T_3 \rightarrow T_2$
(C)	$T_4 \rightarrow T_1 \rightarrow T_3 \rightarrow T_2$
(D)	$T_3 \rightarrow T_1 \rightarrow T_4 \rightarrow T_2$

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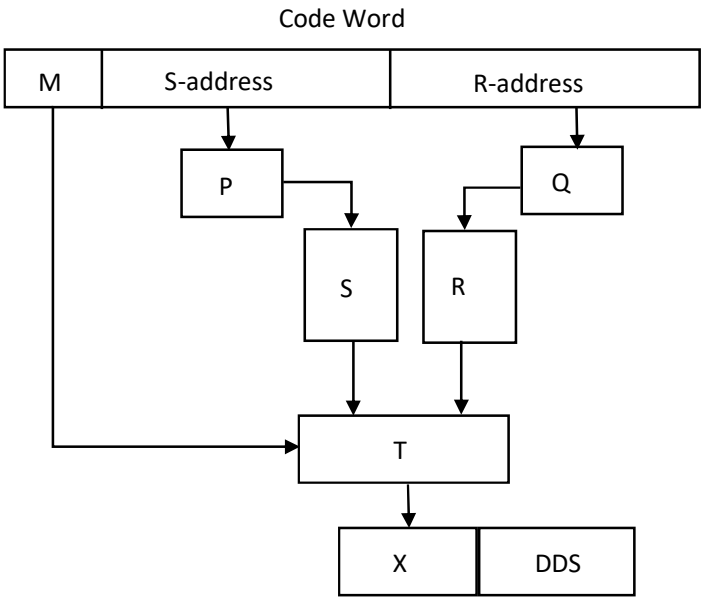
Q.40	Consider a digital display system (DDS) shown in the figure that displays the contents of register X. A 16-bit code word is used to load a word in X, either from S or from R. S is a 1024-word memory segment and R is a 32-word register file. Based on the value of mode bit M, T selects an input word to load in X. P and Q interface with the corresponding bits in the code word to choose the addressed word. Which one of the following represents the functionality of P, Q, and T?
	
(A)	P is 10:1 multiplexer; Q is 5:1 multiplexer; T is 2:1 multiplexer
(B)	P is 10:2^{10} decoder; Q is 5:2^5 decoder; T is 2:1 encoder
(C)	P is 10:2^{10} decoder; Q is 5:2^5 decoder; T is 2:1 multiplexer
(D)	P is 1:10 de-multiplexer; Q is 1:5 de-multiplexer; T is 2:1 multiplexer

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Q.41 Consider three floating point numbers A, B and C stored in registers R_A, R_B and R_C, respectively as per IEEE-754 single precision floating point format. The 32-bit content stored in these registers (in hexadecimal form) are as follows.

R_A= 0xC1400000	R_B = 0x42100000	R_C = 0x41400000

Which one of the following is FALSE?

(A) $A + C = 0$

(B) $C = A + B$

(C) $B = 3C$

(D) $(B - C) > 0$

Q.42 Consider four processes P, Q, R, and S scheduled on a CPU as per round robin algorithm with a time quantum of 4 units. The processes arrive in the order P, Q, R, S, all at time t = 0. There is exactly one context switch from S to Q, exactly one context switch from R to Q, and exactly two context switches from Q to R. There is no context switch from S to P. Switching to a ready process after the termination of another process is also considered a context switch. Which one of the following is NOT possible as CPU burst time (in time units) of these processes?

(A) P = 4, Q = 10, R = 6, S = 2

(B) P = 2, Q = 9, R = 5, S = 1

(C) P = 4, Q = 12, R = 5, S = 4

(D) P = 3, Q = 7, R = 7, S = 3

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Q.43 What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[])
{
          int a[3][3][3] =
          {{1, 2, 3, 4, 5, 6, 7, 8, 9},
           {10, 11, 12, 13, 14, 15, 16, 17, 18},
           {19, 20, 21, 22, 23, 24, 25, 26, 27}};
     int i = 0, j = 0, k = 0;
     for( i = 0; i < 3; i++ ){
          for(k = 0; k < 3; k++ )
               printf("%d ", a[i][j][k]);
          printf("\n");
     }
     return 0;
}
```

(A)

```
1 2 3
10 11 12
19 20 21
```

(B)

```
1 4 7
10 13 16
19 22 25
```

(C)

```
1 2 3
4 5 6
7 8 9
```

(D)

```
1 2 3
13 14 15
25 26 27
```

Q.44 What is printed by the following ANSI C program?

```
#include<stdio.h>
int main(int argc, char *argv[]){
     char a = 'P';
     char b = 'x';
     char c = (a & b) + '*';
     char d = (a | b) - '-';
     char e = (a ^ b) + '+';
     printf("%c %c %c\n", c, d, e);
     return 0;
}
```

ASCII encoding for relevant characters is given below

A	B	C	...	Z
65	66	67	...	90

a	b	c	...	z
97	98	99	...	122

*	+	–
42	43	45

(A) `z K S`

(B) `122 75 83`

(C) `* - +`

(D) `P x +`

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Q.45 Consider solving the following system of simultaneous equations using LU decomposition.

$$\begin{aligned} x_1 + x_2 - 2x_3 &= 4 \\ x_1 + 3x_2 - x_3 &= 7 \\ 2x_1 + x_2 - 5x_3 &= 7 \end{aligned}$$

where L and U are denoted as

$$L = \begin{pmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{pmatrix}, \quad U = \begin{pmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{pmatrix}$$

Which one of the following is the correct combination of values for L_{32}, U_{33}, and x_1?

(A) $L_{32} = 2,\ U_{33} = -\frac{1}{2},\ x_1 = -1$

(B) $L_{32} = 2,\ U_{33} = 2,\ x_1 = -1$

(C) $L_{32} = -\frac{1}{2},\ U_{33} = 2,\ x_1 = 0$

(D) $L_{32} = -\frac{1}{2},\ U_{33} = -\frac{1}{2},\ x_1 = 0$

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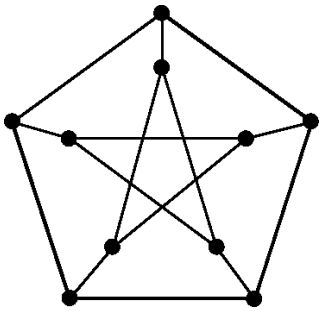
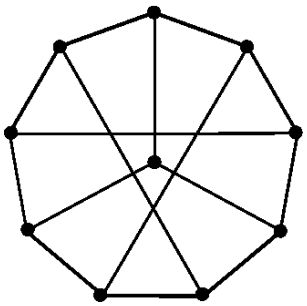
Q.46 – Q.555 Multiple Select Questions (MSQ), carry TWO marks each.

Q.46	Which of the following is/are undecidable?
(A)	Given two Turing machines M_1 and M_2, decide if $L(M_1) = L(M_2)$.
(B)	Given a Turing machine M, decide if $L(M)$ is regular.
(C)	Given a Turing machine M, decide if M accepts all strings.
(D)	Given a Turing machine M, decide if M takes more than 1073 steps on every string.
Q.47	Consider the following languages: $L_1 = \{a^n w a^n \mid w \in \{a,b\}^*\}$ $L_2 = \{wxw^R \mid w,x \in \{a,b\}^*, \mid w \mid, \mid x \mid > 0\}$ Note that w^R is the reversal of the string w. Which of the following is/are TRUE?
(A)	L_1 and L_2 are regular.
(B)	L_1 and L_2 are context-free.
(C)	L_1 is regular and L_2 is context-free.
(D)	L_1 and L_2 are context-free but not regular.

Q.48	Consider the following languages: $L_1 = \{ww \mid w \in \{a,b\}^*\}$ $L_2 = \{a^n b^n c^m \mid m,n \geq 0\}$ $L_3 = \{a^m b^n c^n \mid m,n \geq 0\}$ Which of the following statements is/are FALSE?
(A)	L_1 is not context-free but L_2 and L_3 are deterministic context-free.
(B)	Neither L_1 nor L_2 is context-free.
(C)	L_2, L_3 and $L_2 \cap L_3$ all are context-free.
(D)	Neither L_1 nor its complement is context-free.
Q.49	Consider a simple undirected weighted graph G, all of whose edge weights are distinct. Which of the following statements about the minimum spanning trees of G is/are TRUE?
(A)	The edge with the second smallest weight is always part of any minimum spanning tree of G.
(B)	One or both of the edges with the third smallest and the fourth smallest weights are part of any minimum spanning tree of G.
(C)	Suppose $S \subseteq V$ be such that $S \neq \phi$ and $S \neq V$. Consider the edge with the minimum weight such that one of its vertices is in S and the other in $V \setminus S$. Such an edge will always be part of any minimum spanning tree of G.
(D)	G can have multiple minimum spanning trees.

Q.50	The following simple undirected graph is referred to as the Peterson graph.  Which of the following statements is/are TRUE?
(A)	The chromatic number of the graph is 3.
(B)	The graph has a Hamiltonian path.
(C)	The following graph is isomorphic to the Peterson graph. 
(D)	The size of the largest independent set of the given graph is 3. (A subset of vertices of a graph form an independent set if no two vertices of the subset are adjacent.)

Q.51	Consider the following recurrence: $f(1) = 1;$ $f(2n) = 2f(n)-1, \quad \text{for } n \geq 1;$ $f(2n+1) = 2f(n)+1, \quad \text{for } n \geq 1.$ Then, which of the following statements is/are TRUE?
(A)	$f(2^n-1) = 2^n-1$
(B)	$f(2^n) = 1$
(C)	$f(5 \cdot 2^n) = 2^{n+1}+1$
(D)	$f(2^n+1) = 2^n+1$
Q.52	Which of the properties hold for the adjacency matrix A of a simple undirected unweighted graph having n vertices?
(A)	The diagonal entries of A^2 are the degrees of the vertices of the graph.
(B)	If the graph is connected, then none of the entries of $A^{n-1}+I_n$ can be zero.
(C)	If the sum of all the elements of A is at most $2(n-1)$, then the graph must be acyclic.
(D)	If there is at least a 1 in each of A's rows and columns, then the graph must be connected.

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Q.53	Which of the following is/are the eigenvector(s) for the matrix given below? $\begin{pmatrix} -9 & -6 & -2 & -4 \\ -8 & -6 & -3 & -1 \\ 20 & 15 & 8 & 5 \\ 32 & 21 & 7 & 12 \end{pmatrix}$
(A)	$\begin{pmatrix} -1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$
(B)	$\begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}$
(C)	$\begin{pmatrix} -1 \\ 0 \\ 2 \\ 2 \end{pmatrix}$
(D)	$\begin{pmatrix} 0 \\ 1 \\ -3 \\ 0 \end{pmatrix}$

Q.54	Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (*i.e.,* PQRSPQRS…). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?
(A)	Every access to S is a hit.
(B)	Once P is brought to the cache it is never evicted.
(C)	At the end of the execution only R and S reside in the cache.
(D)	Every access to R evicts Q from the cache.

Q.55 Consider routing table of an organization's router shown below:

Subnet Number	Subnet Mask	Next Hop
12.20.164.0	255.255.252.0	R1
12.20.170.0	255.255.254.0	R2
12.20.168.0	255.255.254.0	Interface 0
12.20.166.0	255.255.254.0	Interface 1
default		R3

Which of the following prefixes in CIDR notation can be collectively used to correctly aggregate all of the subnets in the routing table?

(A) 12.20.164.0/20

(B) 12.20.164.0/22

(C) 12.20.164.0/21

(D) 12.20.168.0/22

Q.56 – Q.65 Numerical Answer Type (NAT), carry TWO marks each.

Q.56

Consider the relational database with the following four schemas and their respective instances.

`Student(sNo, sName, dNo)` `Dept(dNo, dName)`

`Course(cNo, cName, dNo)` `Register(sNo, cNo)`

Student		
sNo	sName	dNo
S01	James	D01
S02	Rocky	D01
S03	Jackson	D02
S04	Jane	D01
S05	Milli	D02

Dept	
dNo	dName
D01	CSE
D02	EEE

Course		
cNo	cName	dNo
C11	DS	D01
C12	OS	D01
C21	DE	D02
C22	PT	D02
C23	CV	D03

Register	
sNo	cNo
S01	C11
S01	C12
S02	C11
S03	C21
S03	C22
S03	C23
S04	C11
S04	C12
S05	C11
S05	C21

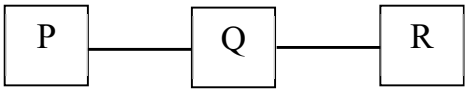
SQL Query:

```
SELECT * FROM Student AS S WHERE NOT EXIST
 (SELECT cNo FROM Course WHERE dNo = "D01"
     EXCEPT
  SELECT cNo FROM Register WHERE sNo = S.sNo)
```

The number of rows returned by the above SQL query is_____.

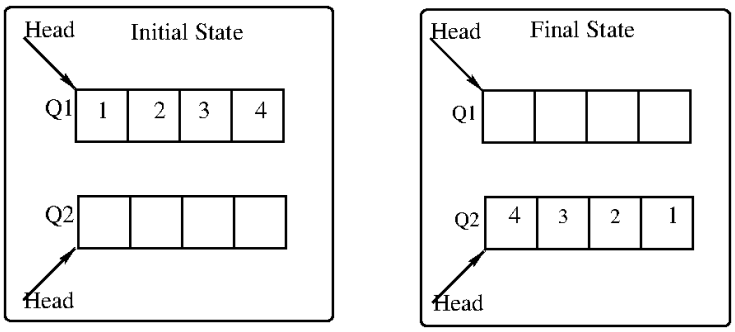
Q.57

Consider a network with three routers P, Q, R shown in the figure below. All the links have cost of unity.



The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q−R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation (*rounded off to one decimal place*) between P and Q, leading to count-to-infinity problem, is_____.

Q.58	Let $G(V, E)$ be a directed graph, where $V = \{1,2,3,4,5\}$ is the set of vertices and E is the set of directed edges, as defined by the following adjacency matrix A. $$A[i][j] = \begin{cases} 1, & 1 \le j \le i \le 5 \\ 0, & otherwise \end{cases}$$ $A[i][j] = 1$ indicates a directed edge from node i to node j. A *directed spanning tree* of G, rooted at $r \in V$, is defined as a subgraph T of G such that the undirected version of T is a tree, and T contains a directed path from r to every other vertex in V. The number of such directed spanning trees rooted at vertex 5 is_____.
Q.59	Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at a speed of 3×10^8 m/s. The time taken (in milliseconds, *rounded off to two decimal places*) for the receiver to completely receive a packet of 1000 bytes transmitted by the sender is_____.
Q.60	Consider the data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL is_____.
Q.61	A processor X_1 operating at 2 GHz has a standard 5-stage RISC instruction pipeline having a base CPI (cycles per instruction) of one without any pipeline hazards. For a given program P that has 30% branch instructions, control hazards incur 2 cycles stall for every branch. A new version of the processor X_2 operating at same clock frequency has an additional branch predictor unit (BPU) that completely eliminates stalls for correctly predicted branches. There is neither any savings nor any additional stalls for wrong predictions. There are no structural hazards and data hazards for X_1 and X_2. If the BPU has a prediction accuracy of 80%, the speed up (*rounded off to two decimal places*) obtained by X_2 over X_1 in executing P is_____.

Q.62	Consider the queues Q_1 containing four elements and Q_2 containing none (shown as the Initial State in the figure). The only operations allowed on these two queues are `Enqueue(Q,element)` and `Dequeue(Q)`. The minimum number of `Enqueue` operations on Q_1 required to place the elements of Q_1 in Q_2 in reverse order (shown as the Final State in the figure) without using any additional storage is_____.
	
Q.63	Consider two files systems A and B, that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50^{th} and 51^{st} block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B, respectively, then the value of $n_A + n_B$ is_____.
Q.64	Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string $7,2,7,3,2,5,3,4,6,7,7,1,5,6,1$ the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (*rounded off to one decimal place*) is_____.

Q.65	Consider the following grammar along with translation rules. $S \rightarrow S_1 \# T \quad \{S_{\bullet val} = S_{1\bullet val} * T_{\bullet val}\}$ $S \rightarrow T \quad \{S_{\bullet val} = T_{\bullet val}\}$ $T \rightarrow T_1 \% R \quad \{T_{\bullet val} = T_{1\bullet val} \div R_{\bullet val}\}$ $T \rightarrow R \quad \{T_{\bullet val} = R_{\bullet val}\}$ $R \rightarrow id \quad \{R_{\bullet val} = id_{\bullet val}\}$ Here # and % are operators and *id* is a token that represents an integer and $id_{\bullet val}$ represents the corresponding integer value. The set of non-terminals is $\{S, T, R, P\}$ and a subscripted non-terminal indicates an instance of the non-terminal. Using this translation scheme, the computed value of $S_{\bullet val}$ for root of the parse tree for the expression $20\#10\%5\#8\%2\%2$ is _____.

General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

Q.1	We reached the station late, and _____ missed the train.
(A)	near
(B)	nearly
(C)	utterly
(D)	mostly
Q.2	Kind : _____ : : Often : Frequently (By word meaning)
(A)	Mean
(B)	Type
(C)	Cruel
(D)	Kindly

Q.3	A series of natural numbers $F_1,\ F_2,\ F_3,\ F_4,\ F_5,\ F_6,\ F_7,$ ... obeys $F_{n+1} = F_n + F_{n-1}$ for all integers $n \geq 2$. If $F_6 = 37$, and $F_7 = 60$, then what is F_1 ?
(A)	4
(B)	5
(C)	8
(D)	9

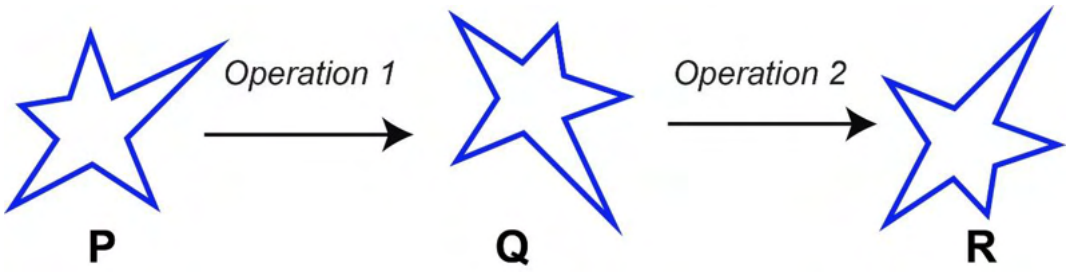
Q.4	A survey for a certain year found that 90% of pregnant women received medical care at least once before giving birth. Of these women, 60% received medical care from doctors, while 40% received medical care from other healthcare providers. Given this information, which one of the following statements can be inferred with *certainty*?
(A)	More than half of the pregnant women received medical care at least once from a doctor.
(B)	Less than half of the pregnant women received medical care at least once from a doctor.
(C)	More than half of the pregnant women received medical care at most once from a doctor.
(D)	Less than half of the pregnant women received medical care at most once from a doctor.
Q.5	Looking at the surface of a smooth 3-dimensional object from the outside, which one of the following options is TRUE?
(A)	The surface of the object must be concave everywhere.
(B)	The surface of the object must be convex everywhere.
(C)	The surface of the object may be concave in some places and convex in other places.
(D)	The object can have edges, but no corners.

Q.6 – Q.10 Carry TWO marks Each

Q.6	The country of Zombieland is in distress since more than 75% of its working population is suffering from serious health issues. Studies conducted by competent health experts concluded that a complete lack of physical exercise among its working population was one of the leading causes of their health issues. As one of the measures to address the problem, the Government of Zombieland has decided to provide monetary incentives to those who ride bicycles to work. Based only on the information provided above, which one of the following statements can be logically inferred with *certainty*?
(A)	All the working population of Zombieland will henceforth ride bicycles to work.
(B)	Riding bicycles will ensure that all of the working population of Zombieland is free of health issues.
(C)	The health experts suggested to the Government of Zombieland to declare riding bicycles as mandatory.
(D)	The Government of Zombieland believes that riding bicycles is a form of physical exercise.

Q.7	Consider two functions of time (t), $$f(t) = 0.01\, t^2$$ $$g(t) = 4\, t$$ where $0 < t < \infty$. Now consider the following two statements: (i) For some $t > 0$, $g(t) > f(t)$. (ii) There exists a T, such that $f(t) > g(t)$ for all $t > T$. Which one of the following options is TRUE?
(A)	only (i) is correct
(B)	only (ii) is correct
(C)	both (i) and (ii) are correct
(D)	neither (i) nor (ii) is correct

Q.8	Which one of the following sentence sequences creates a coherent narrative? (i) Once on the terrace, on her way to her small room in the corner, she notices the man right away. (ii) She begins to pant by the time she has climbed all the stairs. (iii) Mina has bought vegetables and rice at the market, so her bags are heavy. (iv) He was leaning against the parapet, watching the traffic below.
(A)	(i), (ii), (iv), (iii)
(B)	(ii), (iii), (i), (iv)
(C)	(iv), (ii), (i), (iii)
(D)	(iii), (ii), (i), (iv)
Q.9	$f(x)$ and $g(y)$ are functions of x and y, respectively, and $f(x)=g(y)$ for all real values of x and y. Which one of the following options is *necessarily* TRUE for all x and y?
(A)	$f(x)=0$ and $g(y)=0$
(B)	$f(x)=g(y)=$ constant
(C)	$f(x)\neq$ constant and $g(y)\neq$ constant
(D)	$f(x)+g(y)=f(x)-g(y)$

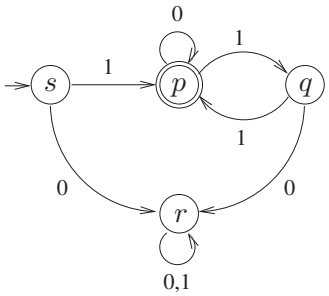
Q.10	Which one of the options best describes the transformation of the 2-dimensional figure **P** to **Q**, and then to **R**, as shown?
	
(A)	*Operation 1*: A clockwise rotation by 90º about an axis perpendicular to the plane of the figure *Operation 2*: A reflection along a horizontal line
(B)	*Operation 1*: A counter clockwise rotation by 90º about an axis perpendicular to the plane of the figure *Operation 2*: A reflection along a horizontal line
(C)	*Operation 1*: A clockwise rotation by 90º about an axis perpendicular to the plane of the figure *Operation 2*: A reflection along a vertical line
(D)	*Operation 1*: A counter clockwise rotation by 180º about an axis perpendicular to the plane of the figure *Operation 2*: A reflection along a vertical line

Q.11 – Q.35 Carry ONE mark each.

Q.11	Consider the following statements regarding the front-end and back-end of a compiler. **S1**: The front-end includes phases that are independent of the target hardware. **S2**: The back-end includes phases that are specific to the target hardware. **S3**: The back-end includes phases that are specific to the programming language used in the source code. Identify the CORRECT option.
(A)	Only **S1** is TRUE.
(B)	Only **S1** and **S2** are TRUE.
(C)	**S1**, **S2**, and **S3** are all TRUE.
(D)	Only **S1** and **S3** are TRUE.

Q.12	Which one of the following sequences when stored in an array at locations $A[1], \ldots, A[10]$ forms a max-heap?
(A)	23, 17, 10, 6, 13, 14, 1, 5, 7, 12
(B)	23, 17, 14, 7, 13, 10, 1, 5, 6, 12
(C)	23, 17, 14, 6, 13, 10, 1, 5, 7, 15
(D)	23, 14, 17, 1, 10, 13, 16, 12, 7, 5

Q.13	Let `SLLdel` be a function that deletes a node in a singly-linked list given a pointer to the node and a pointer to the head of the list. Similarly, let `DLLdel` be another function that deletes a node in a doubly-linked list given a pointer to the node and a pointer to the head of the list. Let n denote the number of nodes in each of the linked lists. Which one of the following choices is TRUE about the worst-case time complexity of `SLLdel` and `DLLdel`?
(A)	`SLLdel` is $O(1)$ and `DLLdel` is $O(n)$
(B)	Both `SLLdel` and `DLLdel` are $O(\log(n))$
(C)	Both `SLLdel` and `DLLdel` are $O(1)$
(D)	`SLLdel` is $O(n)$ and `DLLdel` is $O(1)$

Q.14	Consider the Deterministic Finite-state Automaton (DFA) $\mathcal{A}$ shown below. The DFA runs on the alphabet $\{0, 1\}$, and has the set of states $\{s, p, q, r\}$, with s being the start state and p being the only final state.  Which one of the following regular expressions correctly describes the language accepted by $\mathcal{A}$?
(A)	$1(0^*11)^*$
(B)	$0(0+1)^*$
(C)	$1(0+11)^*$
(D)	$1(110^*)^*$

Q.15	The Lucas sequence L_n is defined by the recurrence relation: $$L_n = L_{n-1} + L_{n-2}, \quad \text{for} \quad n \geq 3,$$ with $L_1 = 1$ and $L_2 = 3$. Which one of the options given is TRUE?
(A)	$L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{2}\right)^n$
(B)	$L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{3}\right)^n$
(C)	$L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{1-\sqrt{5}}{3}\right)^n$
(D)	$L_n = \left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n$

Q.16	Which one of the options given below refers to the degree (or arity) of a relation in relational database systems?
(A)	Number of attributes of its relation schema.
(B)	Number of tuples stored in the relation.
(C)	Number of entries in the relation.
(D)	Number of distinct domains of its relation schema.

Q.17	Suppose two hosts are connected by a point-to-point link and they are configured to use `Stop-and-Wait` protocol for reliable data transfer. Identify in which one of the following scenarios, the utilization of the link is the lowest.
(A)	Longer link length and lower transmission rate
(B)	Longer link length and higher transmission rate
(C)	Shorter link length and lower transmission rate
(D)	Shorter link length and higher transmission rate

Q.18	Let $$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \end{bmatrix}$$ and $$B = \begin{bmatrix} 3 & 4 & 1 & 2 \\ 4 & 1 & 2 & 3 \\ 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{bmatrix}.$$ Let $\texttt{det}(A)$ and $\texttt{det}(B)$ denote the determinants of the matrices A and B, respectively. Which one of the options given below is TRUE?
(A)	$\texttt{det}(A) = \texttt{det}(B)$
(B)	$\texttt{det}(B) = -\texttt{det}(A)$
(C)	$\texttt{det}(A) = 0$
(D)	$\texttt{det}(AB) = \texttt{det}(A) + \texttt{det}(B)$

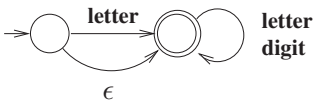
Q.19

Consider the following definition of a lexical token **id** for an identifier in a programming language, using extended regular expressions:

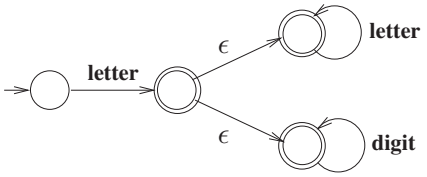
$$\begin{aligned} \textbf{letter} &\rightarrow \texttt{[A-Za-z]} \\ \textbf{digit} &\rightarrow \texttt{[0-9]} \\ \textbf{id} &\rightarrow \textbf{letter}\ (\textbf{letter} \mid \textbf{digit})^* \end{aligned}$$

Which one of the following Non-deterministic Finite-state Automata with ϵ-transitions accepts the set of valid identifiers? (A double-circle denotes a final state)

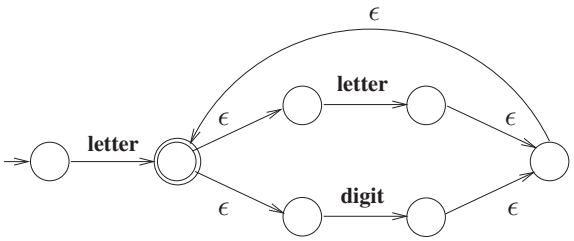
(A)



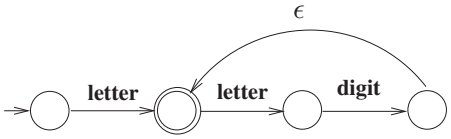
(B)



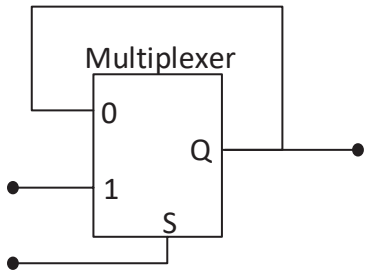
(C)



(D)



Q.20	An algorithm has to store several keys generated by an adversary in a hash table. The adversary is malicious who tries to maximize the number of collisions. Let k be the number of keys, m be the number of slots in the hash table, and $k > m$. Which one of the following is the best hashing strategy to counteract the adversary?
(A)	Division method, i.e., use the hash function $h(k) = k \mod m$.
(B)	Multiplication method, i.e., use the hash function $h(k) = \lfloor m(kA - \lfloor kA \rfloor) \rfloor$, where A is a carefully chosen constant.
(C)	Universal hashing method.
(D)	If k is a prime number, use Division method. Otherwise, use Multiplication method.

Q.21	The output of a 2-input multiplexer is connected back to one of its inputs as shown in the figure.  Match the functional equivalence of this circuit to one of the following options.
(A)	D Flip-flop
(B)	D Latch
(C)	Half-adder
(D)	Demultiplexer

Q.22	Which one or more of the following need to be saved on a context switch from one thread (T1) of a process to another thread (T2) of the same process?
(A)	Page table base register
(B)	Stack pointer
(C)	Program counter
(D)	General purpose registers

Q.23	Which one or more of the following options guarantee that a computer system will transition from user mode to kernel mode?
(A)	Function Call
(B)	`malloc` Call
(C)	Page Fault
(D)	System Call

Q.24	Which of the following statements is/are CORRECT?
(A)	The intersection of two regular languages is regular.
(B)	The intersection of two context-free languages is context-free.
(C)	The intersection of two recursive languages is recursive.
(D)	The intersection of two recursively enumerable languages is recursively enumerable.

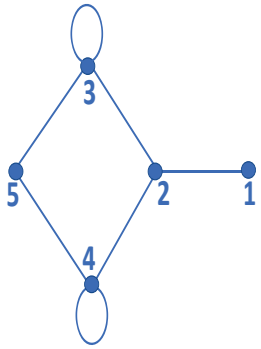
Q.25	Which of the following statements is/are INCORRECT about the `OSPF (Open Shortest Path First)` routing protocol used in the Internet?
(A)	`OSPF` implements Bellman-Ford algorithm to find shortest paths.
(B)	`OSPF` uses Dijkstra's shortest path algorithm to implement least-cost path routing.
(C)	`OSPF` is used as an inter-domain routing protocol.
(D)	`OSPF` implements hierarchical routing.

Q.26	Geetha has a conjecture about integers, which is of the form $$\forall x\Big(P(x) \implies \exists y Q(x,y)\Big),$$ where P is a statement about integers, and Q is a statement about pairs of integers. Which of the following (one or more) option(s) would *imply* Geetha's conjecture?
(A)	$\exists x\Big(P(x) \wedge \forall y Q(x,y)\Big)$
(B)	$\forall x \forall y Q(x,y)$
(C)	$\exists y \forall x\Big(P(x) \implies Q(x,y)\Big)$
(D)	$\exists x\Big(P(x) \wedge \exists y Q(x,y)\Big)$

Q.27	Which one or more of the following CPU scheduling algorithms can potentially cause starvation?
(A)	First-in First-Out
(B)	Round Robin
(C)	Priority Scheduling
(D)	Shortest Job First

Q.28	Let $$f(x) = x^3 + 15x^2 - 33x - 36$$ be a real-valued function. Which of the following statements is/are TRUE?
(A)	$f(x)$ does not have a local maximum.
(B)	$f(x)$ has a local maximum.
(C)	$f(x)$ does not have a local minimum.
(D)	$f(x)$ has a local minimum.

Q.29	Let f and g be functions of natural numbers given by $f(n) = n$ and $g(n) = n^2$. Which of the following statements is/are TRUE?
(A)	$f \in O(g)$
(B)	$f \in \Omega(g)$
(C)	$f \in o(g)$
(D)	$f \in \Theta(g)$

Q.30	Let A be the adjacency matrix of the graph with vertices $\{1, 2, 3, 4, 5\}$.  Let λ_1, λ_2, λ_3, λ_4, and λ_5 be the five eigenvalues of A. Note that these eigenvalues need not be distinct. The value of $\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 + \lambda_5 =$ _____.

Q.31	The value of the definite integral $$\int_{-3}^{3}\int_{-2}^{2}\int_{-1}^{1}\left(4x^2y - z^3\right)\,\mathrm{d}z\,\mathrm{d}y\,\mathrm{d}x$$ is _____ . (Rounded off to the nearest integer)

Q.32	A particular number is written as 132 in radix-4 representation. The same number in radix-5 representation is _____.

Q.33	Consider a 3-stage pipelined processor having a delay of 10 ns (nanoseconds), 20 ns, and 14 ns, for the first, second, and the third stages, respectively. Assume that there is no other delay and the processor does not suffer from any pipeline hazards. Also assume that one instruction is fetched every cycle. The total execution time for executing 100 instructions on this processor is _____ ns.

Q.34	A keyboard connected to a computer is used at a rate of 1 keystroke per second. The computer system polls the keyboard every 10 ms (milli seconds) to check for a keystroke and consumes 100 μs (micro seconds) for each poll. If it is determined after polling that a key has been pressed, the system consumes an additional 200 μs to process the keystroke. Let T_1 denote the fraction of a second spent in polling and processing a keystroke. In an alternative implementation, the system uses interrupts instead of polling. An interrupt is raised for every keystroke. It takes a total of 1 ms for servicing an interrupt and processing a keystroke. Let T_2 denote the fraction of a second spent in servicing the interrupt and processing a keystroke. The ratio $\frac{T_1}{T_2}$ is _____. (Rounded off to one decimal place)

Q.35 The integer value printed by the ANSI-C program given below is _____.

```
#include<stdio.h>

int funcp(){
        static int x = 1;
        x++;
        return x;
}

int main(){
        int x,y;
        x = funcp();
        y = funcp()+x;
        printf("%d\n", (x+y));
        return 0;
}
```

Q.36 – Q.65 Carry TWO mark each.

Q.36

Consider the following program:

```
int main()
{
   f1();
   f2(2);
   f3();
   return(0);
}
```

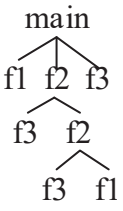
```
int f1()
{
  return(1);
}
```

```
int f2(int X)
{
   f3();
  if (X==1)
    return f1();
  else
    return (X*f2(X-1));
}
```

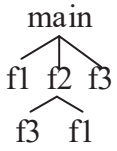
```
int f3()
{
  return(5);
}
```

Which one of the following options represents the activation tree corresponding to the main function?

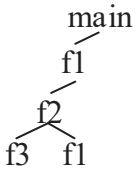
(A)



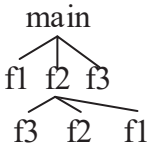
(B)



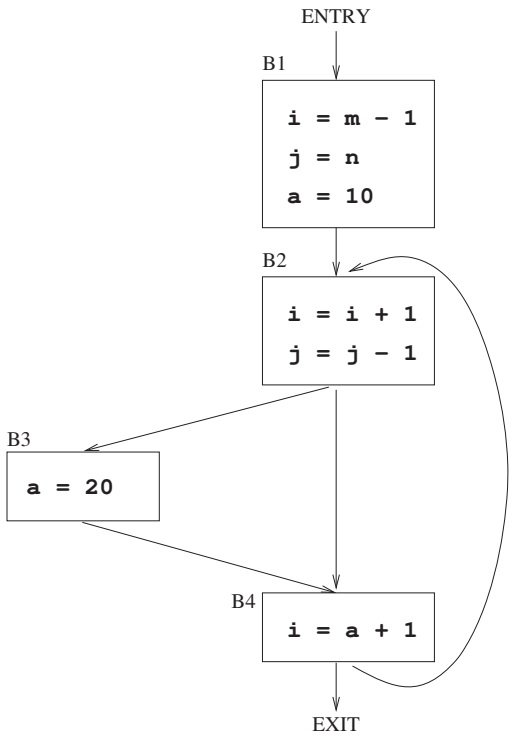
(C)



(D)



Q.37 Consider the control flow graph shown.



Which one of the following choices correctly lists the set of *live* variables at the exit point of each basic block?

(A) B1: {}, B2: {a}, B3: {a}, B4: {a}

(B) B1: {i, j}, B2: {a}, B3: {a}, B4: {i}

(C) B1: {a, i, j}, B2: {a, i, j}, B3: {a, i}, B4: {a}

(D) B1: {a, i, j}, B2: {a, j}, B3: {a, j}, B4: {a, i, j}

Q.38

Consider the two functions `incr` and `decr` shown below.

```
incr(){
 wait(s);
 X = X+1;
 signal(s);
}
```

```
decr(){
 wait(s);
 X = X-1;
 signal(s);
}
```

There are 5 threads each invoking `incr` once, and 3 threads each invoking `decr` once, on the same shared variable `X`. The initial value of `X` is 10.

Suppose there are two implementations of the semaphore `s`, as follows:

I-1: `s` is a binary semaphore initialized to 1.

I-2: `s` is a counting semaphore initialized to 2.

Let V1, V2 be the values of `X` at the end of execution of all the threads with implementations **I-1**, **I-2**, respectively.

Which one of the following choices corresponds to the minimum possible values of V1, V2, respectively?

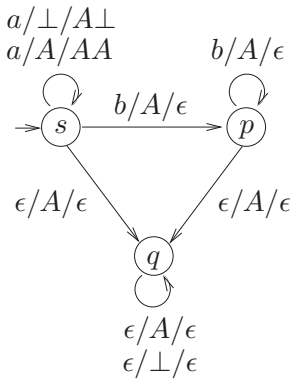
(A) 15, 7

(B) 7, 7

(C) 12, 7

(D) 12, 8

Q.39	Consider the context-free grammar G below $$S \rightarrow aSb \mid X$$ $$X \rightarrow aX \mid Xb \mid a \mid b \ ,$$ where S and X are non-terminals, and a and b are terminal symbols. The starting non-terminal is S. Which one of the following statements is CORRECT?
(A)	The language generated by G is $(a+b)^*$
(B)	The language generated by G is $a^*(a+b)b^*$
(C)	The language generated by G is $a^*b^*(a+b)$
(D)	The language generated by G is not a regular language

Q.40	Consider the pushdown automaton (PDA) P below, which runs on the input alphabet $\{a, b\}$, has stack alphabet $\{\perp, A\}$, and has three states $\{s, p, q\}$, with s being the start state. A transition from state u to state v, labelled $c/X/\gamma$, where c is an input symbol or ϵ, X is a stack symbol, and γ is a string of stack symbols, represents the fact that in state u, the PDA can read c from the input, with X on the top of its stack, pop X from the stack, push in the string γ on the stack, and go to state v. In the initial configuration, the stack has only the symbol $\perp$ in it. The PDA accepts by empty stack.  Which one of the following options correctly describes the language accepted by P?
(A)	$\{a^m b^n \mid 1 \leq m \text{ and } n < m\}$
(B)	$\{a^m b^n \mid 0 \leq n \leq m\}$
(C)	$\{a^m b^n \mid 0 \leq m \text{ and } 0 \leq n\}$
(D)	$\{a^m \mid 0 \leq m\} \cup \{b^n \mid 0 \leq n\}$

Q.41

Consider the given C-code and its corresponding assembly code, with a few operands U1–U4 being unknown. Some useful information as well as the semantics of each unique assembly instruction is annotated as inline comments in the code. The memory is byte-addressable.

```
//C-code                    |    ;assembly-code (; indicates comments)
                            |    ;r1-r5 are 32-bit integer registers
                            |    ;initialize r1=0, r2=10
                            |    ;initialize r3, r4 with base address of a, b
                            |
int a[10], b[10], i;        |     L01: jeq r1, r2, end  ;if(r1==r2) goto end
// int is 32-bit            |     L02: lw  r5, 0(r4)    ;r5 <- Memory[r4+0]
for (i=0; i<10;i++)         |     L03: shl r5, r5, U1   ;r5 <- r5 << U1
    a[i] = b[i] * 8;        |     L04: sw  r5, 0(r3)    ;Memory[r3+0] <- r5
                            |     L05: add r3, r3, U2   ;r3 <- r3+U2
                            |     L06: add r4, r4, U3
                            |     L07: add r1, r1, 1
                            |     L08: jmp U4           ;goto U4
                            |     L09: end
```

Which one of the following options is a CORRECT replacement for operands in the position (U1, U2, U3, U4) in the above assembly code?

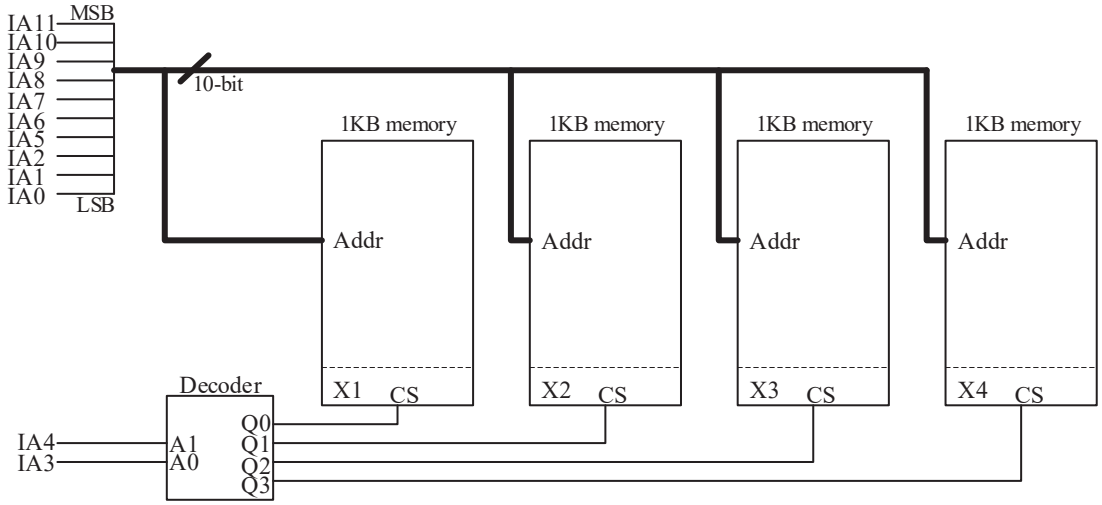
(A) (8, 4, 1, L02)

(B) (3, 4, 4, L01)

(C) (8, 1, 1, L02)

(D) (3, 1, 1, L01)

Q.42 A 4 kilobyte (KB) byte-addressable memory is realized using four 1 KB memory blocks. Two input address lines (IA4 and IA3) are connected to the chip select (CS) port of these memory blocks through a decoder as shown in the figure. The remaining ten input address lines from IA11–IA0 are connected to the address port of these blocks. The chip select (CS) is active high.



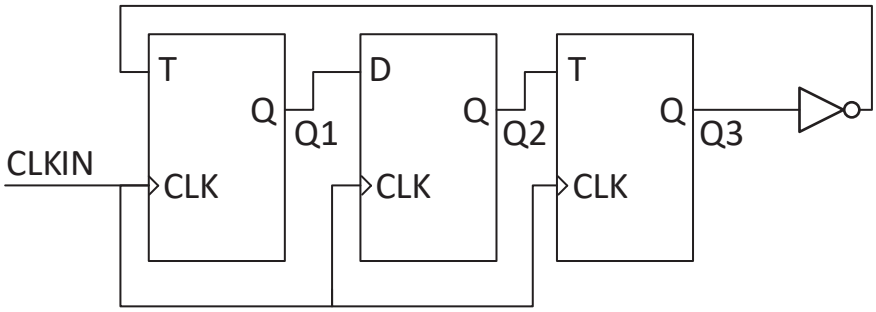
The input memory addresses (IA11–IA0), in decimal, for the starting locations (Addr=0) of each block (indicated as X1, X2, X3, X4 in the figure) are among the options given below. Which one of the following options is CORRECT?

(A) (0, 1, 2, 3)

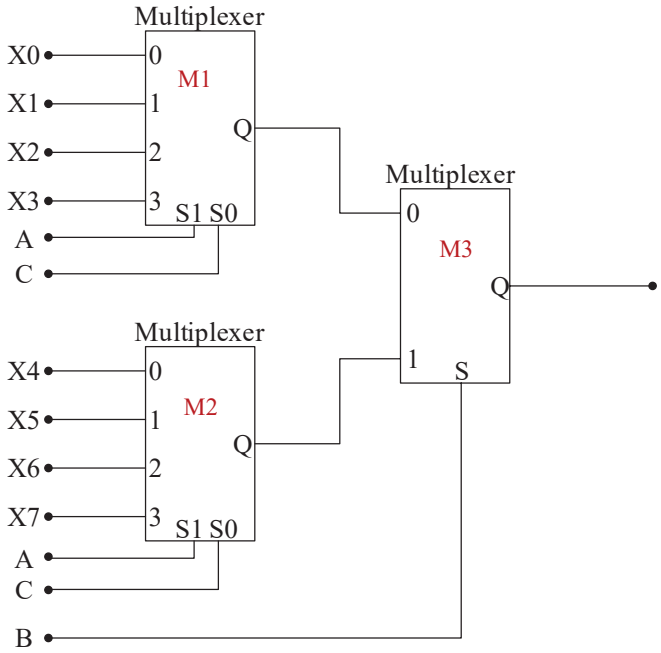
(B) (0, 1024, 2048, 3072)

(C) (0, 8, 16, 24)

(D) (0, 0, 0, 0)

Q.43	Consider a sequential digital circuit consisting of T flip-flops and D flip-flops as shown in the figure. CLKIN is the clock input to the circuit. At the beginning, Q1, Q2 and Q3 have values 0, 1 and 1, respectively.  Which one of the given values of (Q1, Q2, Q3) can NEVER be obtained with this digital circuit?
(A)	(0, 0, 1)
(B)	(1, 0, 0)
(C)	(1, 0, 1)
(D)	(1, 1, 1)

Q.44 A Boolean digital circuit is composed using two 4-input multiplexers (M1 and M2) and one 2-input multiplexer (M3) as shown in the figure. X0–X7 are the inputs of the multiplexers M1 and M2 and could be connected to either 0 or 1. The select lines of the multiplexers are connected to Boolean variables A, B and C as shown.



Which one of the following set of values of (X0, X1, X2, X3, X4, X5, X6, X7) will realise the Boolean function $\overline{A} + \overline{A}.\overline{C} + A.\overline{B}.C$?

(A) (1, 1, 0, 0, 1, 1, 1, 0)

(B) (1, 1, 0, 0, 1, 1, 0, 1)

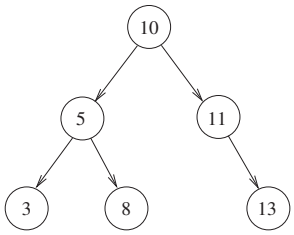
(C) (1, 1, 0, 1, 1, 1, 0, 0)

(D) (0, 0, 1, 1, 0, 1, 1, 1)

Q.45	Consider the IEEE-754 single precision floating point numbers P=`0xC1800000` and Q=`0x3F5C2EF4`. Which one of the following corresponds to the product of these numbers (i.e., P $\times$ Q), represented in the IEEE-754 single precision format?
(A)	`0x404C2EF4`
(B)	`0x405C2EF4`
(C)	`0xC15C2EF4`
(D)	`0xC14C2EF4`

Q.46	Let A be a priority queue for maintaining a set of elements. Suppose A is implemented using a max-heap data structure. The operation EXTRACT-MAX(A) extracts and deletes the maximum element from A. The operation INSERT(A,key) inserts a new element key in A. The properties of a max-heap are preserved at the end of each of these operations. When A contains n elements, which one of the following statements about the worst case running time of these two operations is TRUE?
(A)	Both EXTRACT-MAX(A) and INSERT(A,key) run in $O(1)$.
(B)	Both EXTRACT-MAX(A) and INSERT(A,key) run in $O(\log(n))$.
(C)	EXTRACT-MAX(A) runs in $O(1)$ whereas INSERT(A,key) runs in $O(n)$.
(D)	EXTRACT-MAX(A) runs in $O(1)$ whereas INSERT(A,key) runs in $O(\log(n))$.

Q.47 Consider the C function `foo` and the binary tree shown.




```
typedef struct node {
  int val;
  struct node *left, *right;
} node;

int foo(node *p) {
  int retval;
  if (p == NULL)
    return 0;
  else {
    retval = p->val + foo(p->left) + foo(p->right);
    printf("%d ", retval);
    return retval;
  }
}
```

When `foo` is called with a pointer to the root node of the given binary tree, what will it print?

(A) 3 8 5 13 11 10

(B) 3 5 8 10 11 13

(C) 3 8 16 13 24 50

(D) 3 16 8 50 24 13

Q.48	Let $U = \{1, 2, \ldots, n\}$, where n is a large positive integer greater than 1000. Let k be a positive integer less than n. Let A, B be subsets of U with $\|A\| = \|B\| = k$ and $A \cap B = \emptyset$. We say that a permutation of U separates A from B if one of the following is true. - All members of A appear in the permutation before any of the members of B. - All members of B appear in the permutation before any of the members of A. How many permutations of U separate A from B?
(A)	$n!$
(B)	$\binom{n}{2k}(n-2k)!$
(C)	$\binom{n}{2k}(n-2k)!\,(k!)^2$
(D)	$2\binom{n}{2k}(n-2k)!\,(k!)^2$

Q.49	Let $f : A \to B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation $\sim$ on the set A as $$a_1 \sim a_2 \text{ if } f(a_1) = f(a_2) ,$$ where $a_1, a_2 \in A$. Let $\mathcal{E} = \{[x] : x \in A\}$ be the set of all the equivalence classes under $\sim$. Define a new mapping $F : \mathcal{E} \to B$ as $$F([x]) = f(x), \quad \text{for all the equivalence classes } [x] \text{ in } \mathcal{E}.$$ Which of the following statements is/are TRUE?
(A)	F is NOT well-defined.
(B)	F is an onto (or surjective) function.
(C)	F is a one-to-one (or injective) function.
(D)	F is a bijective function.

Q.50	Suppose you are asked to design a new reliable byte-stream transport protocol like TCP. This protocol, named `myTCP`, runs over a 100 Mbps network with Round Trip Time of 150 milliseconds and the maximum segment lifetime of 2 minutes. Which of the following is/are valid lengths of the **Sequence Number** field in the `myTCP` header?
(A)	30 bits
(B)	32 bits
(C)	34 bits
(D)	36 bits

Q.51	Let X be a set and 2^X denote the powerset of X. Define a binary operation Δ on 2^X as follows: $$A\Delta B = (A - B) \cup (B - A) \ .$$ Let $H = (2^X, \Delta)$. Which of the following statements about H is/are correct?
(A)	H is a group.
(B)	Every element in H has an inverse, but H is NOT a group.
(C)	For every $A \in 2^X$, the inverse of A is the complement of A.
(D)	For every $A \in 2^X$, the inverse of A is A.

Q.52	Suppose in a web browser, you click on the `www.gate-2023.in` URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers. Let RTT denote the round trip time between your local host and DNS servers in the DNS hierarchy. The round trip time between the local host and the web server hosting `www.gate-2023.in` is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server. Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?
(A)	7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
(B)	5 RTTs, in case of persistent HTTP with pipelining.
(C)	9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.
(D)	6 RTTs, in case of persistent HTTP with pipelining.

Q.53	Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on the first throw, and C be the event that denotes HEAD on the second throw. Which of the following statements is/are TRUE?
(A)	A and B are independent.
(B)	A and C are independent.
(C)	B and C are independent.
(D)	$\text{Prob}(B\|C) = \text{Prob}(B)$

Q.54

Consider functions **Function_1** and **Function_2** expressed in pseudocode as follows:

Function_1

```
while n > 1 do
    for i = 1 to n do
        x = x + 1;
    end for
    n = ⌊n/2⌋;
end while
```

Function_2

```
for i = 1 to 100 * n do
    x = x + 1;
end for
```

Let $f_1(n)$ and $f_2(n)$ denote the number of times the statement "$x = x + 1$" is executed in **Function_1** and **Function_2**, respectively.

Which of the following statements is/are TRUE?

(A) $f_1(n) \in \Theta(f_2(n))$

(B) $f_1(n) \in o(f_2(n))$

(C) $f_1(n) \in \omega(f_2(n))$

(D) $f_1(n) \in O(n)$

Q.55	Let G be a simple, finite, undirected graph with vertex set $\{v_1, \ldots, v_n\}$. Let $\Delta(G)$ denote the maximum degree of G and let $\mathbb{N} = \{1, 2, \ldots\}$ denote the set of all possible colors. Color the vertices of G using the following greedy strategy: for $i = 1, \ldots, n$ $$\text{color}(v_i) \leftarrow \min\{j \in \mathbb{N} : \text{ no neighbour of } v_i \text{ is colored } j\}$$ Which of the following statements is/are TRUE?
(A)	This procedure results in a proper vertex coloring of G.
(B)	The number of colors used is at most $\Delta(G) + 1$.
(C)	The number of colors used is at most $\Delta(G)$.
(D)	The number of colors used is equal to the chromatic number of G.

Q.56	Let $U = \{1, 2, 3\}$. Let 2^U denote the powerset of U. Consider an undirected graph G whose vertex set is 2^U. For any $A, B \in 2^U$, (A, B) is an edge in G if and only if (i) $A \neq B$, and (ii) either $A \subsetneq B$ or $B \subsetneq A$. For any vertex A in G, the set of all possible orderings in which the vertices of G can be visited in a Breadth First Search (BFS) starting from A is denoted by $\mathcal{B}(A)$. If $\emptyset$ denotes the empty set, then the cardinality of $\mathcal{B}(\emptyset)$ is _____.

Q.57

Consider the following two-dimensional array `D` in the C programming language, which is stored in row-major order:

```
int D[128][128];
```

Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array `D`. The Least Recently Used (LRU) page-replacement policy is used by the operating system. A total of 30 physical page frames are allocated to a process which executes the following code snippet:

```
for (int i = 0; i < 128; i++)
   for (int j = 0; j < 128; j++)
         D[j][i] *= 10;
```

The number of page faults generated during the execution of this code snippet is _____.

Q.58	Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with `L` levels for virtual to physical address translation. The page size is 4 KB (1 KB = 1024 B) and a page table entry at any of the levels occupies 8 bytes. The value of `L` is _____.

Q.59	Consider a sequence a of elements $a_0 = 1$, $a_1 = 5$, $a_2 = 7$, $a_3 = 8$, $a_4 = 9$, and $a_5 = 2$. The following operations are performed on a stack S and a queue Q, both of which are initially empty. I: `push` the elements of a from a_0 to a_5 in that order into S. II: `enqueue` the elements of a from a_0 to a_5 in that order into Q. III: `pop` an element from S. IV: `dequeue` an element from Q. V: `pop` an element from S. VI: `dequeue` an element from Q. VII: `dequeue` an element from Q and `push` the same element into S. VIII: Repeat operation VII three times. IX: `pop` an element from S. X: `pop` an element from S. The `top` element of S after executing the above operations is _____.

Q.60

Consider the syntax directed translation given by the following grammar and semantic rules. Here N, I, F and B are non-terminals. N is the starting non-terminal, and #, **0** and **1** are lexical tokens corresponding to input letters "#", "0" and "1", respectively. $X.val$ denotes the synthesized attribute (a numeric value) associated with a non-terminal X. I_1 and F_1 denote occurrences of I and F on the right hand side of a production, respectively. For the tokens **0** and **1**, $\mathbf{0}.val = 0$ and $\mathbf{1}.val = 1$.

$$
\begin{array}{lcll}
N & \rightarrow & I \,\#\, F & N.val = I.val + F.val \\
I & \rightarrow & I_1 \, B & I.val = (2 \; I_1.val) + B.val \\
I & \rightarrow & B & I.val = B.val \\
F & \rightarrow & B \, F_1 & F.val = \frac{1}{2}(B.val + F_1.val) \\
F & \rightarrow & B & F.val = \frac{1}{2}B.val \\
B & \rightarrow & \mathbf{0} & B.val = \mathbf{0}.val \\
B & \rightarrow & \mathbf{1} & B.val = \mathbf{1}.val
\end{array}
$$

The value computed by the translation scheme for the input string

10#011

is _____. (Rounded off to three decimal places)

Q.61

Consider the following table named `Student` in a relational database. The primary key of this table is rollNum.

`Student`

rollNum	name	gender	marks
1	Naman	M	62
2	Aliya	F	70
3	Aliya	F	80
4	James	M	82
5	Swati	F	65

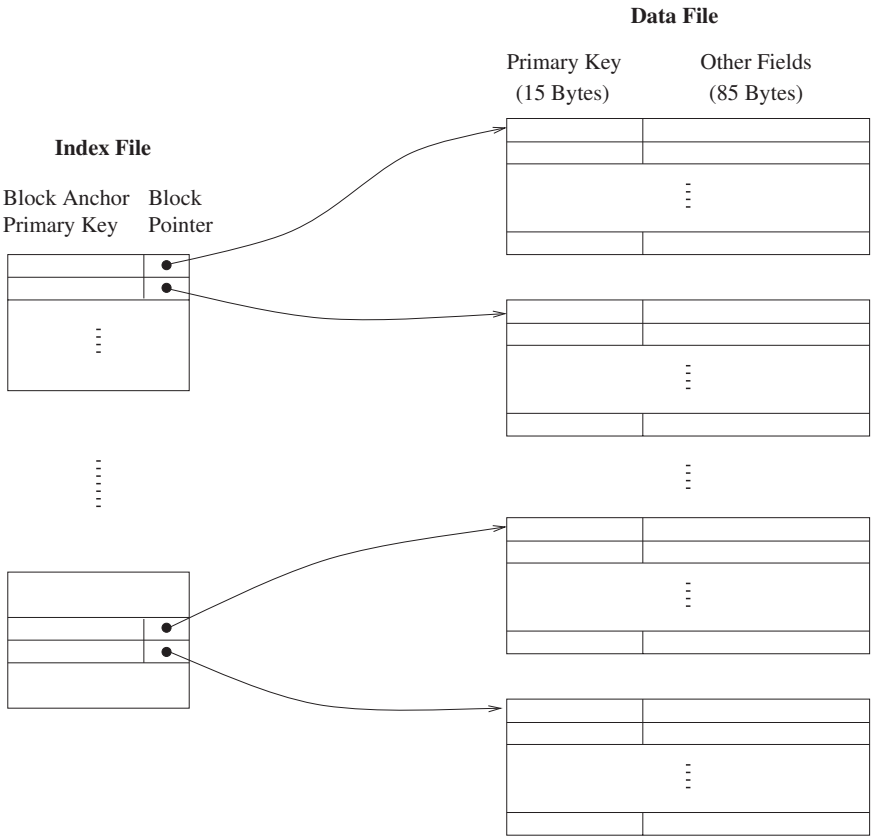
The SQL query below is executed on this database.

```
SELECT *
FROM Student
WHERE gender = 'F' AND
   marks > 65;
```

The number of rows returned by the query is _____.

Q.62

Consider a database of fixed-length records, stored as an ordered file. The database has 25,000 records, with each record being 100 bytes, of which the primary key occupies 15 bytes. The data file is block-aligned in that each data record is fully contained within a block. The database is indexed by a primary index file, which is also stored as a block-aligned ordered file. The figure below depicts this indexing scheme.



Suppose the block size of the file system is 1024 bytes, and a pointer to a block occupies 5 bytes. The system uses binary search on the index file to search for a record with a given key. You may assume that a binary search on an index file of b blocks takes $\lceil \log_2 b \rceil$ block accesses in the worst case.

Given a key, the number of block accesses required to identify the block in the data file that may contain a record with the key, in the worst case, is _____.

Q.63	Consider the language L over the alphabet $\{0, 1\}$, given below: $L = \{w \in \{0,1\}^* \mid w$ does not contain three or more consecutive 1's$\}$. The minimum number of states in a Deterministic Finite-State Automaton (DFA) for L is _____.

Q.64	An 8-way set associative cache of size 64 KB (1 KB = 1024 bytes) is used in a system with 32-bit address. The address is sub-divided into TAG, INDEX, and BLOCK OFFSET. The number of bits in the TAG is _____.

Q.65

The forwarding table of a router is shown below.

Subnet Number	Subnet Mask	Interface ID
200.150.0.0	255.255.0.0	1
200.150.64.0	255.255.224.0	2
200.150.68.0	255.255.255.0	3
200.150.68.64	255.255.255.224	4
Default		0

A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID _____.

General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

Q.1 If '→' denotes increasing order of intensity, then the meaning of the words [dry → arid → parched] is analogous to [diet → fast → _____].

Which one of the given options is appropriate to fill the blank?

(A) starve

(B) reject

(C) feast

(D) deny

Q.2 If two distinct non-zero real variables x and y are such that $(x + y)$ is proportional to $(x - y)$ then the value of $\frac{x}{y}$

(A) depends on xy

(B) depends only on x and not on y

(C) depends only on y and not on x

(D) is a constant

GATE 2024

Q.3 Consider the following sample of numbers:

9, 18, 11, 14, 15, 17, 10, 69, 11, 13

The median of the sample is

(A) 13.5

(B) 14

(C) 11

(D) 18.7

Q.4 The number of coins of ₹1, ₹5, and ₹10 denominations that a person has are in the ratio 5:3:13. Of the total amount, the percentage of money in ₹5 coins is

(A) 21%

(B) $14\frac{2}{7}\%$

(C) 10%

(D) 30%

GATE 2024

Q.5 For positive non-zero real variables p and q, if

$$\log\,(p^2 + q^2) = \log p + \log q + 2\log 3\,,$$

then, the value of $\frac{p^4+q^4}{p^2q^2}$ is

(A) 79

(B) 81

(C) 9

(D) 83

GATE 2024

Q.6 – Q.10 Carry TWO marks Each

Q.6 In the given text, the blanks are numbered (i)–(iv). Select the best match for all the blanks.

Steve was advised to keep his head _____(i)_____ before heading _____(ii)_____ to bat; for, while he had a head _____(iii)_____ batting, he could only do so with a cool head _____(iv)_____ his shoulders.

(A)	(i) down	(ii) down	(iii) on	(iv) for
(B)	(i) on	(ii) down	(iii) for	(iv) on
(C)	(i) down	(ii) out	(iii) for	(iv) on
(D)	(i) on	(ii) out	(iii) on	(iv) for

GATE 2024

Q.7 A rectangular paper sheet of dimensions 54 cm × 4 cm is taken. The two longer edges of the sheet are joined together to create a cylindrical tube. A cube whose surface area is equal to the area of the sheet is also taken.

Then, the ratio of the volume of the cylindrical tube to the volume of the cube is

(A) $1/\pi$

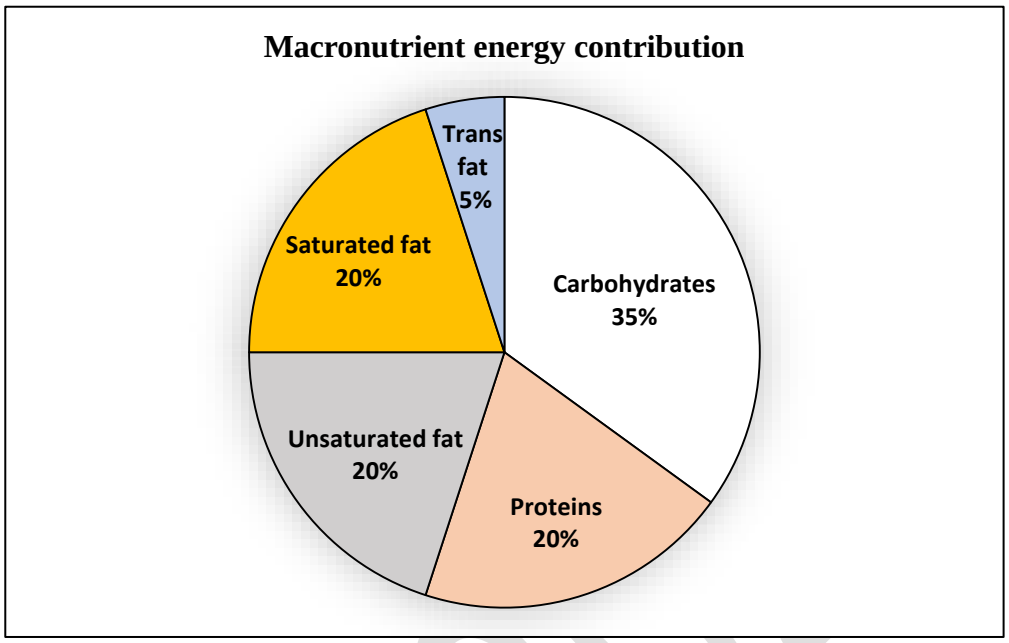
(B) $2/\pi$

(C) $3/\pi$

(D) $4/\pi$

GATE 2024

Q.8 The pie chart presents the percentage contribution of different macronutrients to a typical 2,000 kcal diet of a person.



The typical energy density (kcal/g) of these macronutrients is given in the table.

Macronutrient	Energy density (kcal/g)
Carbohydrates	4
Proteins	4
Unsaturated fat	9
Saturated fat	9
Trans fat	9

The total fat (all three types), in grams, this person consumes is

(A) 44.4

(B) 77.8

(C) 100

(D) 3,600

Q.9 A rectangular paper of 20 cm × 8 cm is folded 3 times. Each fold is made along the line of symmetry, which is perpendicular to its long edge. The perimeter of the final folded sheet (in cm) is

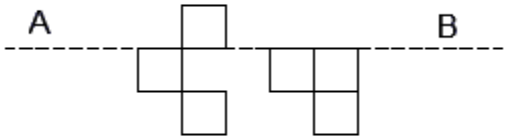
(A) 18

(B) 24

(C) 20

(D) 21

Q.10 The least number of squares to be added in the figure to make AB a line of symmetry is



(A) 6

(B) 4

(C) 5

(D) 7

Q.11 – Q.35 Carry ONE mark Each

Q.11 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) = \max\{x, x^3\}$, $x \in \mathbb{R}$, where $\mathbb{R}$ is the set of all real numbers. The set of all points where $f(x)$ is NOT differentiable is

(A) $\{-1, 1, 2\}$

(B) $\{-2, -1, 1\}$

(C) $\{0, 1\}$

(D) $\{-1, 0, 1\}$

Q.12 The product of all eigenvalues of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is

(A) -1

(B) 0

(C) 1

(D) 2

GATE 2024

Q.13 Consider a system that uses 5 bits for representing signed integers in 2's complement format. In this system, two integers A and B are represented as A=01010 and B=11010. Which one of the following operations will result in either an arithmetic overflow or an arithmetic underflow?

(A) $A + B$

(B) $A - B$

(C) $B - A$

(D) $2 * B$

Q.14 Consider a permutation sampled uniformly at random from the set of all permutations of $\{1, 2, 3, \cdots, n\}$ for some $n \geq 4$. Let X be the event that 1 occurs before 2 in the permutation, and Y the event that 3 occurs before 4. Which one of the following statements is TRUE?

(A) The events X and Y are mutually exclusive

(B) The events X and Y are independent

(C) Either event X or Y must occur

(D) Event X is more likely than event Y

GATE 2024

Q.15 Which one of the following statements is FALSE?

(A) In the cycle stealing mode of DMA, one word of data is transferred between an I/O device and main memory in a stolen cycle

(B) For bulk data transfer, the burst mode of DMA has a higher throughput than the cycle stealing mode

(C) Programmed I/O mechanism has a better CPU utilization than the interrupt driven I/O mechanism

(D) The CPU can start executing an interrupt service routine faster with vectored interrupts than with non-vectored interrupts

GATE 2024

Q.16 A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.

(i) HTTP GET request for the index page
(ii) DNS request to resolve the web server's name to its IP address
(iii) HTTP GET request for an image object
(iv) TCP SYN to open a connection to the web server

Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer ?

(A) (iv), (ii), (iii), (i)

(B) (ii), (iv), (iii), (i)

(C) (ii), (iv), (i), (iii)

(D) (iv), (ii), (i), (iii)

Q.17 Given an integer array of size N, we want to check if the array is sorted (in either ascending or descending order). An algorithm solves this problem by making a single pass through the array and comparing each element of the array only with its adjacent elements. The worst-case time complexity of this algorithm is

(A) both $O(N)$ and $\Omega(N)$

(B) $O(N)$ but not $\Omega(N)$

(C) $\Omega(N)$ but not $O(N)$

(D) neither $O(N)$ nor $\Omega(N)$

Q.18 Consider the following C program:

```
#include <stdio.h>
int main(){
   int a = 6;
   int b = 0;
   while(a < 10) {
      a = a / 12 + 1;
      a += b;}
   printf("%d", a);
   return 0;}
```

Which one of the following statements is CORRECT?

(A) The program prints `9` as output

(B) The program prints `10` as output

(C) The program gets stuck in an infinite loop

(D) The program prints `6` as output

GATE 2024

Q.19 Consider the following C program:

```
#include <stdio.h>
void fX();
int main(){
   fX();
   return 0;}
```

```
void fX(){
 char a;
 if((a=getchar()) != '\n')
     fX();
 if(a != '\n')
     putchar(a);}
```

Assume that the input to the program from the command line is `1234` followed by a newline character. Which one of the following statements is CORRECT?

(A) The program will not terminate

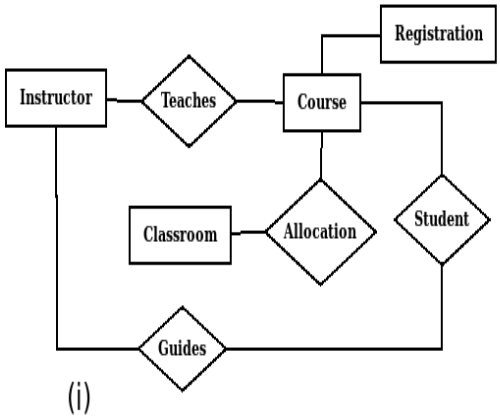
(B) The program will terminate with no output

(C) The program will terminate with `4321` as output

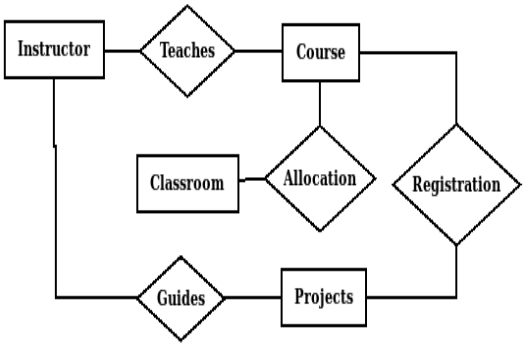
(D) The program will terminate with `1234` as output

GATE 2024

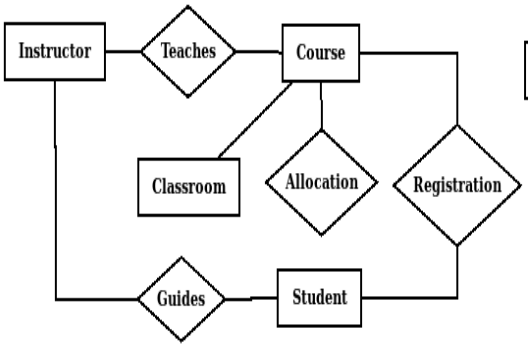
Q.20 Let S be the specification: "Instructors teach courses. Students register for courses. Courses are allocated classrooms. Instructors guide students." Which one of the following ER diagrams CORRECTLY represents S?



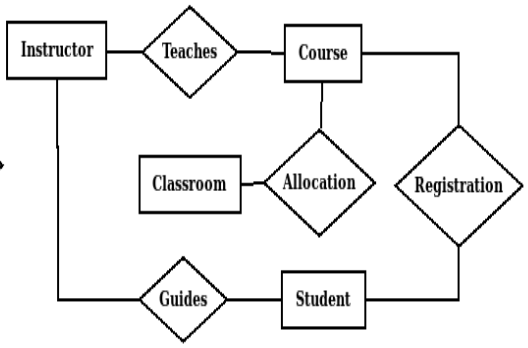
(i)



(ii)



(iii)



(iv)

(A) (i)

(B) (ii)

(C) (iii)

(D) (iv)

Q.21 In a B+ tree, the requirement of at least half-full (50%) node occupancy is relaxed for which one of the following cases?

(A) Only the root node

(B) All leaf nodes

(C) All internal nodes

(D) Only the leftmost leaf node

Q.22 Which of the following statements about a relation **R** in first normal form (1NF) is/are TRUE ?

(A) **R** can have a multi-attribute key

(B) **R** cannot have a foreign key

(C) **R** cannot have a composite attribute

(D) **R** cannot have more than one candidate key

GATE 2024

Q.23 Let L_1, L_2 be two regular languages and L_3 a language which is not regular. Which of the following statements is/are *always* TRUE?

(A) $L_1 = L_2$ if and only if $L_1 \cap \overline{L_2} = \phi$

(B) $L_1 \cup L_3$ is not regular

(C) $\overline{L_3}$ is not regular

(D) $\overline{L_1} \cup \overline{L_2}$ is regular

Q.24 Which of the following statements about threads is/are TRUE?

(A) Threads can only be implemented in kernel space

(B) Each thread has its own file descriptor table for open files

(C) All the threads belonging to a process share a common stack

(D) Threads belonging to a process are by default not protected from each other

GATE 2024

Q.25 Which of the following process state transitions is/are NOT possible?

(A) Running to Ready

(B) Waiting to Running

(C) Ready to Waiting

(D) Running to Terminated

Q.26 Which of the following is/are Bottom-Up Parser(s)?

(A) Shift-reduce Parser

(B) Predictive Parser

(C) LL(1) Parser

(D) LR Parser

GATE 2024

Q.27 Let A and B be two events in a probability space with $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cap B) = 0.1$. Which of the following statements is/are TRUE?

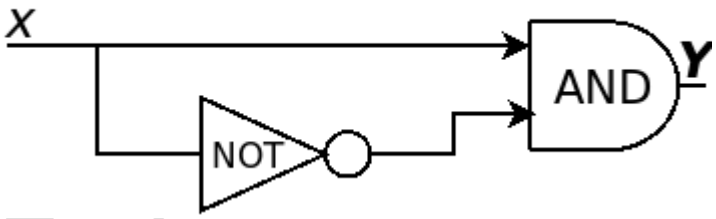
(A) The two events A and B are independent

(B) $P(A \cup B) = 0.7$

(C) $P(A \cap B^c) = 0.2$, where B^c is the complement of the event B

(D) $P(A^c \cap B^c) = 0.4$, where A^c and B^c are the complements of the events A and B, respectively

Q.28 Consider the circuit shown below where the gates may have propagation delays. Assume that all signal transitions occur instantaneously and that wires have no delays. Which of the following statements about the circuit is/are CORRECT?



(A) With no propagation delays, the output ***Y*** is always logic Zero

(B) With no propagation delays, the output ***Y*** is always logic One

(C) With propagation delays, the output ***Y*** can have a transient logic One after ***X*** transitions from logic Zero to logic One

(D) With propagation delays, the output ***Y*** can have a transient logic Zero after ***X*** transitions from logic One to logic Zero

Q.29 TCP client P successfully establishes a connection to TCP server Q. Let N_P denote the sequence number in the SYN sent from P to Q. Let N_Q denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

(A) The sequence number N_P is chosen randomly by P

(B) The sequence number N_P is always 0 for a new connection

(C) The acknowledgement number N_Q is equal to N_P

(D) The acknowledgement number N_Q is equal to $N_P + 1$

Q.30 Consider a 5-stage pipelined processor with Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Register Writeback (WB) stages. Which of the following statements about *forwarding* is/are CORRECT?

(A) In a pipelined execution, forwarding means the result from a source stage of an earlier instruction is passed on to the destination stage of a later instruction

(B) In forwarding, data from the output of the MEM stage can be passed on to the input of the EX stage of the next instruction

(C) Forwarding cannot prevent all pipeline stalls

(D) Forwarding does not require any extra hardware to retrieve the data from the pipeline stages

Q.31 Which of the following fields is/are modified in the IP header of a packet going out of a network address translation (NAT) device from an internal network to an external network?

(A) Source IP

(B) Destination IP

(C) Header Checksum

(D) Total Length

Q.32 Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from $A \times A$ to $A \cup B$. The number of possible values of $|A|$ is _____

Q.33 Consider the operator precedence and associativity rules for the *integer* arithmetic operators given in the table below.

Operator	Precedence	Associativity
$+$	Highest	Left
$-$	High	Right
$*$	Medium	Right
$/$	Low	Right

The value of the expression $3 + 1 + 5 * 2 / 7 + 2 - 4 - 7 - 6 / 2$ as per the above rules is _____

Q.34 The number of spanning trees in a *complete* graph of 4 vertices labelled A, B, C, and D is _____

Q.35 Consider the following two relations, ***R(A,B)*** and ***S(A,C)***:

R	
A	***B***
10	20
20	30
30	40
30	50
50	95

S	
A	***C***
10	90
30	45
40	80

The total number of tuples obtained by evaluating the following expression

$$\sigma_{B<C}(R \bowtie_{R.A=S.A} S)$$

is _____

GATE 2024

Q.36 – Q.65 Carry TWO marks Each

Q.36 Consider a network path P—Q—R between nodes P and R via router Q. Node P sends a file of size 10^6 bytes to R via this path by splitting the file into chunks of 10^3 bytes each. Node P sends these chunks one after the other without any wait time between the successive chunk transmissions. Assume that the size of extra headers added to these chunks is negligible, and that the chunk size is less than the MTU.

Each of the links P—Q and Q—R has a bandwidth of 10^6 bits/sec, and negligible propagation latency. Router Q immediately transmits every packet it receives from P to R, with negligible processing and queueing delays. Router Q can simultaneously receive on link P—Q and transmit on link Q—R.

Assume P starts transmitting the chunks at time $t = 0$.
Which one of the following options gives the time (*in seconds, rounded off to 3 decimal places*) at which R receives all the chunks of the file?

(A) 8.000

(B) 8.008

(C) 15.992

(D) 16.000

GATE 2024

Q.37 Consider the following syntax-directed definition (SDD).

$S \rightarrow DHTU$	$\{ S.val = D.val + H.val + T.val + U.val; \}$
$D \rightarrow$ "M"D_1	$\{ D.val = 5 + D_1.val; \}$
$D \rightarrow \epsilon$	$\{ D.val = -5; \}$
$H \rightarrow$ "L"H_1	$\{ H.val = 5 * 10 + H_1.val; \}$
$H \rightarrow \epsilon$	$\{ H.val = -10; \}$
$T \rightarrow$ "C"T_1	$\{ T.val = 5 * 100 + T_1.val; \}$
$T \rightarrow \epsilon$	$\{ T.val = -5; \}$
$U \rightarrow$ "K"	$\{ U.val = 5; \}$

Given "MMLK" as the input, which one of the following options is the CORRECT value computed by the SDD (in the attribute $S.val$)?

(A) 45

(B) 50

(C) 55

(D) 65

GATE 2024

Q.38 Consider the following grammar G, with S as the start symbol. The grammar G has three incomplete productions denoted by (1), (2), and (3).

$$S \rightarrow daT \mid \underline{\quad (1) \quad}$$
$$T \rightarrow aS \mid bT \mid \underline{\quad (2) \quad}$$
$$R \rightarrow \underline{\quad (3) \quad} \mid \epsilon$$

The set of terminals is $\{a, b, c, d, f\}$. The FIRST and FOLLOW sets of the different non-terminals are as follows.

$$\text{FIRST}(S) = \{c, d, f\},\ \text{FIRST}(T) = \{a, b, \epsilon\},\ \text{FIRST}(R) = \{c, \epsilon\}$$

$$\text{FOLLOW}(S) = \text{FOLLOW}(T) = \{c, f, \$\},\ \text{FOLLOW}(R) = \{f\}$$

Which one of the following options CORRECTLY fills in the incomplete productions?

(A) (1) $S \rightarrow Rf$ (2) $T \rightarrow \epsilon$ (3) $R \rightarrow cTR$

(B) (1) $S \rightarrow fR$ (2) $T \rightarrow \epsilon$ (3) $R \rightarrow cTR$

(C) (1) $S \rightarrow fR$ (2) $T \rightarrow cT$ (3) $R \rightarrow cR$

(D) (1) $S \rightarrow Rf$ (2) $T \rightarrow cT$ (3) $R \rightarrow cR$

GATE 2024

Q.39 Consider the following pseudo-code.

$L1$: $t1 = -1$
$L2$: $t2 = 0$
$L3$: $t3 = 0$
$L4$: $t4 = 4 * t3$
$L5$: $t5 = 4 * t2$
$L6$: $t6 = t5 * M$
$L7$: $t7 = t4 + t6$
$L8$: $t8 = a[t7]$
$L9$: **if** $t8 <= max$ **goto** $L11$
$L10$: $t1 = t8$
$L11$: $t3 = t3 + 1$
$L12$: **if** $t3 < M$ **goto** $L4$
$L13$: $t2 = t2 + 1$
$L14$: **if** $t2 < N$ **goto** $L3$
$L15$: $max = t1$

Which one of the following options CORRECTLY specifies the number of basic blocks and the number of instructions in the largest basic block, respectively ?

(A) 6 and 6

(B) 6 and 7

(C) 7 and 7

(D) 7 and 6

GATE 2024

Q.40 Consider the following two threads T1 and T2 that update two shared variables `a` and `b`. Assume that initially `a = b = 1`. Though context switching between threads can happen at any time, each statement of T1 or T2 is executed atomically without interruption.

T1	T2
`a = a + 1;`	`b = 2 * b;`
`b = b + 1;`	`a = 2 * a;`

Which one of the following options lists all the possible combinations of values of `a` and `b` after both T1 and T2 finish execution?

(A) `(a = 4, b = 4); (a = 3, b = 3); (a = 4, b = 3)`

(B) `(a = 3, b = 4); (a = 4, b = 3); (a = 3, b = 3)`

(C) `(a = 4, b = 4); (a = 4, b = 3); (a = 3, b = 4)`

(D) `(a = 2, b = 2); (a = 2, b = 3); (a = 3, b = 4)`

Q.41 An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?

(A) 82, 90, 101

(B) 82, 11, 93

(C) 131, 11, 93

(D) 131, 111, 90

Q.42 Consider the following recurrence relation:

$$T(n) = \begin{cases} \sqrt{n}T\left(\sqrt{n}\right) + n & \text{for } n \geq 1, \\ 1 & \text{for } n = 1. \end{cases}$$

Which one of the following options is CORRECT?

(A) $T(n) = \Theta(n \log\log n)$

(B) $T(n) = \Theta(n \log n)$

(C) $T(n) = \Theta(n^2 \log n)$

(D) $T(n) = \Theta(n^2 \log\log n)$

Q.43 Consider a binary min-heap containing 105 distinct elements. Let k be the index (in the underlying array) of the maximum element stored in the heap. The number of possible values of k is

(A) 53

(B) 52

(C) 27

(D) 1

GATE 2024

Q.44 The symbol $\rightarrow$ indicates functional dependency in the context of a relational database. Which of the following options is/are TRUE?

(A) $(X, Y) \rightarrow (Z, W)$ implies $X \rightarrow (Z, W)$

(B) $(X, Y) \rightarrow (Z, W)$ implies $(X, Y) \rightarrow Z$

(C) $\big((X, Y) \rightarrow Z$ and $W \rightarrow Y\big)$ implies $(X, W) \rightarrow Z$

(D) $(X \rightarrow Y$ and $Y \rightarrow Z)$ implies $X \rightarrow Z$

Q.45 Let G be a directed graph and T a depth first search (DFS) spanning tree in G that is rooted at a vertex v. Suppose T is also a breadth first search (BFS) tree in G, rooted at v. Which of the following statements is/are TRUE for *every* such graph G and tree T ?

(A) There are no back-edges in G with respect to the tree T

(B) There are no cross-edges in G with respect to the tree T

(C) There are no forward-edges in G with respect to the tree T

(D) The only edges in G are the edges in T

Q.46 Consider the following read-write schedule S over three transactions T_1, T_2, and T_3, where the subscripts in the schedule indicate transaction IDs:

$$S: r_1(z);\ w_1(z);\ r_2(x);\ r_3(y);\ w_3(y);\ r_2(y);\ w_2(x);\ w_2(y);$$

Which of the following transaction schedules is/are conflict equivalent to S ?

(A) $T_1T_2T_3$

(B) $T_1T_3T_2$

(C) $T_3T_2T_1$

(D) $T_3T_1T_2$

Q.47 Consider a Boolean expression given by $F(X,Y,Z) = \sum(3,5,6,7)$.

Which of the following statements is/are CORRECT?

(A) $F(X,Y,Z) = \Pi(0,1,2,4)$

(B) $F(X,Y,Z) = XY + YZ + XZ$

(C) $F(X,Y,Z)$ is independent of input Y

(D) $F(X,Y,Z)$ is independent of input X

GATE 2024

Q.48 Consider the following C function definition.

```
int f(int x, int y) {
  for (int i=0; i<y; i++) {
    x=x+x+y;
  }
  return x;
}
```

Which of the following statements is/are TRUE about the above function?

(A) If the inputs are `x`=20, `y`=10, then the return value is greater than 2^{20}

(B) If the inputs are `x`=20, `y`=20, then the return value is greater than 2^{20}

(C) If the inputs are `x`=20, `y`=10, then the return value is less than 2^{10}

(D) If the inputs are `x`=10, `y`=20, then the return value is greater than 2^{20}

Q.49 Let A be any $n \times m$ matrix, where $m > n$. Which of the following statements is/are TRUE about the system of linear equations $Ax = \mathbf{0}$?

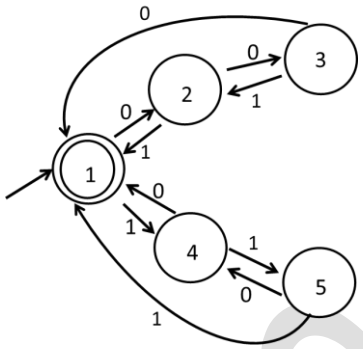
(A) There exist at least $m - n$ linearly independent solutions to this system

(B) There exist $m - n$ linearly independent vectors such that every solution is a linear combination of these vectors

(C) There exists a non-zero solution in which at least $m - n$ variables are 0

(D) There exists a solution in which at least n variables are non-zero

Q.50 Consider the 5-state DFA M accepting the language $L(M) \subset (0+1)^*$ shown below. For any string $w \in (0+1)^*$ let $n_0(w)$ be the number of $0's$ in w and $n_1(w)$ be the number of $1's$ in w.



Which of the following statements is/are FALSE?

(A) States 2 and 4 are distinguishable in M

(B) States 3 and 4 are distinguishable in M

(C) States 2 and 5 are distinguishable in M

(D) Any string w with $n_0(w) = n_1(w)$ is in $L(M)$

GATE 2024

Q.51 The chromatic number of a graph is the minimum number of colours used in a *proper* colouring of the graph. Let G be any graph with n vertices and chromatic number k. Which of the following statements is/are always TRUE?

(A) G contains a complete subgraph with k vertices

(B) G contains an independent set of size at least n/k

(C) G contains at least $k(k-1)/2$ edges

(D) G contains a vertex of degree at least k

Q.52 Consider the operators $\Diamond$ and $\square$ defined by $a \Diamond b = a + 2b$, $a \square b = ab$, for positive integers. Which of the following statements is/are TRUE?

(A) Operator $\Diamond$ obeys the associative law

(B) Operator $\square$ obeys the associative law

(C) Operator $\Diamond$ over the operator $\square$ obeys the distributive law

(D) Operator $\square$ over the operator $\Diamond$ obeys the distributive law

GATE 2024

Q.53 Consider two set-associative cache memory architectures: **WBC**, which uses the write back policy, and **WTC**, which uses the write through policy. Both of them use the LRU (*Least Recently Used*) block replacement policy. The cache memory is connected to the main memory. Which of the following statements is/are TRUE?

(A) A read miss in **WBC** never evicts a dirty block

(B) A read miss in **WTC** never triggers a write back operation of a cache block to main memory

(C) A write hit in **WBC** can modify the value of the dirty bit of a cache block

(D) A write miss in **WTC** always writes the victim cache block to main memory before loading the missed block to the cache

Q.54 Consider a 512 GB hard disk with 32 storage surfaces. There are 4096 sectors per track and each sector holds 1024 bytes of data. The number of cylinders in the hard disk is _____

Q.55 The baseline execution time of a program on a 2 GHz single core machine is 100 nanoseconds (*ns*). The code corresponding to 90% of the execution time can be fully parallelized. The overhead for using an additional core is 10 *ns* when running on a multicore system. Assume that all cores in the multicore system run their share of the parallelized code for an equal amount of time.

The number of cores that minimize the execution time of the program is _____

Q.56 A given program has 25% load/store instructions. Suppose the ideal CPI (cycles per instruction) without any memory stalls is 2. The program exhibits 2% miss rate on instruction cache and 8% miss rate on data cache. The miss penalty is 100 cycles. The speedup (*rounded off to two decimal places*) achieved with a perfect cache (i.e., with NO data or instruction cache misses) is _____

Q.57 Consider the following code snippet using the `fork()` and `wait()` system calls. Assume that the code compiles and runs correctly, and that the system calls run successfully without any errors.

```
int x = 3;
while(x > 0) {
  fork();
  printf("hello");
  wait(NULL);
  x--;
}
```

The total number of times the `printf` statement is executed is _____

Q.58 Consider the entries shown below in the forwarding table of an IP router. Each entry consists of an IP prefix and the corresponding next hop router for packets whose destination IP address matches the prefix. The notation "/N" in a prefix indicates a subnet mask with the most significant N bits set to 1.

Prefix	**Next hop router**
10.1.1.0/24	R1
10.1.1.128/25	R2
10.1.1.64/26	R3
10.1.1.192/26	R4

This router forwards 20 packets each to 5 hosts. The IP addresses of the hosts are 10.1.1.16, 10.1.1.72, 10.1.1.132, 10.1.1.191, and 10.1.1.205 . The number of packets forwarded via the next hop router R2 is _____

Q.59 Let $G = (V, \Sigma, S, P)$ be a context-free grammar in Chomsky Normal Form with $\Sigma = \{a, b, c\}$ and V containing 10 variable symbols including the start symbol S. The string $w = a^{30}b^{30}c^{30}$ is derivable from S. The number of steps (application of rules) in the derivation $S \rightarrow^* w$ is _____

Q.60 The number of edges present in the forest generated by the DFS traversal of an undirected graph G with 100 vertices is 40. The number of connected components in G is _____

Q.61 Consider the following two regular expressions over the alphabet {0,1}:

$$r = 0^* + 1^*$$

$$s = 01^* + 10^*$$

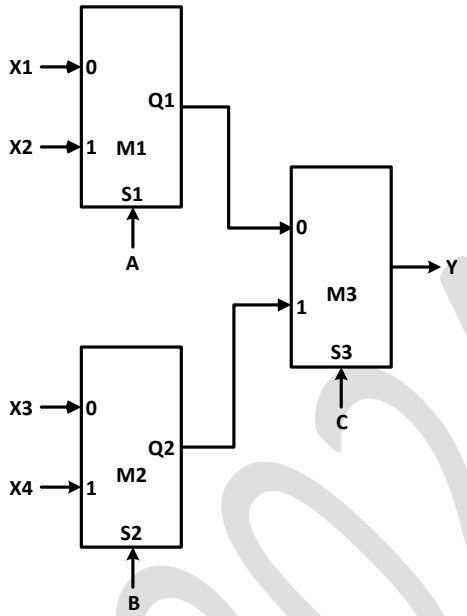
The total number of strings of length less than or equal to 5, which are neither in r nor in s, is _____

Q.62 Consider a memory management system that uses a page size of 2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0, 1, 2, and 3 are stored in the page frames 1, 3, 2, and 0, respectively. The physical address (*in decimal format*) corresponding to the virtual address 2500 (*in decimal format*) is _____

Q.63 A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (*rounded off to 3 decimal places*) that both balls drawn are red is _____

GATE 2024

Q.64 Consider a digital logic circuit consisting of three 2-to-1 multiplexers M1, M2, and M3 as shown below. X1 and X2 are inputs of M1. X3 and X4 are inputs of M2. A, B, and C are select lines of M1, M2, and M3, respectively.



For an instance of inputs **X1=1, X2=1, X3=0**, and **X4=0**, the number of combinations of A, B, C that give the output **Y=1** is _____

Q.65 Consider sending an IP datagram of size 1420 bytes (including 20 bytes of IP header) from a sender to a receiver over a path of two links with a router between them. The first link (sender to router) has an MTU (Maximum Transmission Unit) size of 542 bytes, while the second link (router to receiver) has an MTU size of 360 bytes. The number of fragments that would be delivered at the receiver is _____

General Aptitude

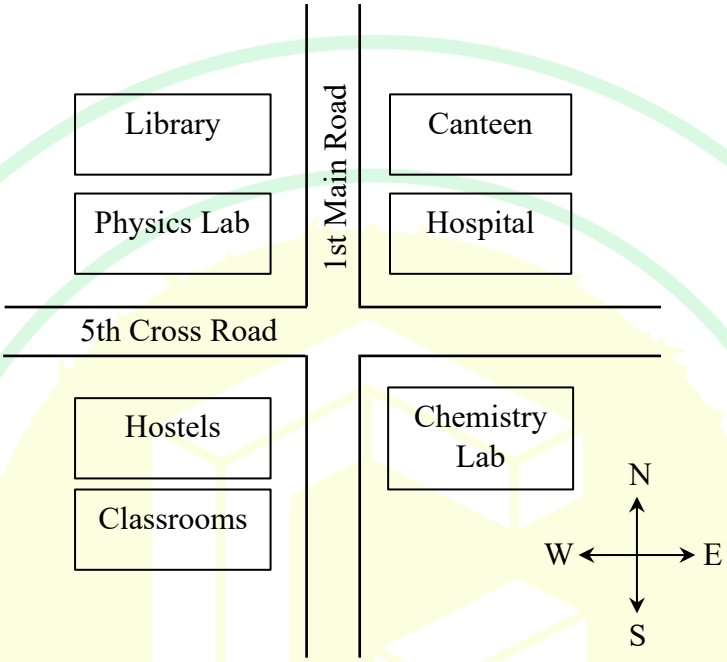
Q.1 – Q.5 Carry ONE mark Each

Q.1	Ravi had _____ younger brother who taught at _____ university. He was widely regarded as _____ honorable man. Select the option with the correct sequence of articles to fill in the blanks.
(A)	a; a; an
(B)	the; an; a
(C)	a; an; a
(D)	an; an; a

Q.2	The CEO's decision to downsize the workforce was considered myopic because it sacrificed long-term stability to accommodate short-term gains. Select the most appropriate option that can replace the word "myopic" without changing the meaning of the sentence.
(A)	visionary
(B)	shortsighted
(C)	progressive
(D)	innovative

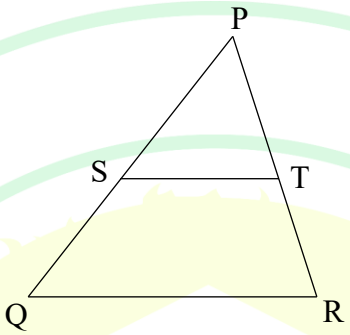
Q.3	The average marks obtained by a class in an examination were calculated as 30.8. However, while checking the marks entered, the teacher found that the marks of one student were entered incorrectly as 24 instead of 42. After correcting the marks, the average becomes 31.4. How many students does the class have?
(A)	25
(B)	28
(C)	30
(D)	32

Q.4	Consider the relationships among P, Q, R, S, and T: • P is the brother of Q. • S is the daughter of Q. • T is the sister of S. • R is the mother of Q. The following statements are made based on the relationships given above. (1) R is the grandmother of S. (2) P is the uncle of S and T. (3) R has only one son. (4) Q has only one daughter. Which one of the following options is correct?
(A)	Both (1) and (2) are true.
(B)	Both (1) and (3) are true.
(C)	Only (3) is true.
(D)	Only (4) is true.

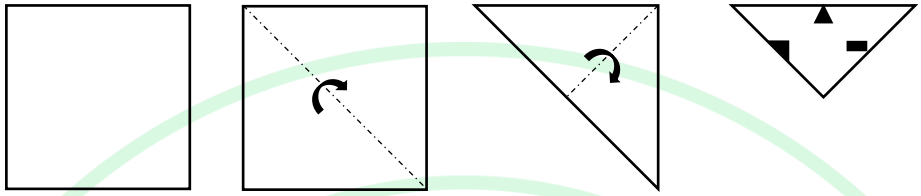
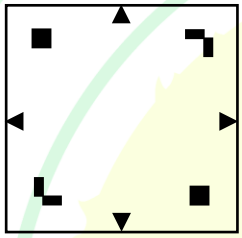
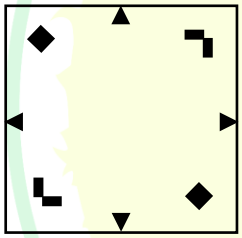
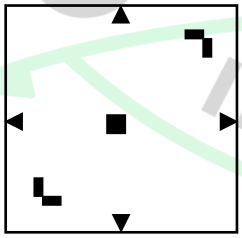
Q.5	According to the map shown in the figure, which one of the following statements is correct? Note: The figure shown is representative.
	
(A)	The library is located to the northwest of the canteen.
(B)	The hospital is located to the east of the chemistry lab.
(C)	The chemistry lab is to the southeast of physics lab.
(D)	The classrooms and canteen are next to each other.

Q.6 – Q.10 Carry TWO marks Each

Q.6	"I put the brown paper in my pocket along with the chalks, and possibly other things. I suppose every one must have reflected how primeval and how poetical are the things that one carries in one's pocket: the pocket-knife, for instance the type of all human tools, the infant of the sword. Once I planned to write a book of poems entirely about the things in my pocket. But I found it would be too long: and the age of the great epics is past." (From G.K. Chesterton's "A Piece of Chalk") Based only on the information provided in the above passage, which one of the following statements is true?
(A)	The author of the passage carries a mirror in his pocket to reflect upon things.
(B)	The author of the passage had decided to write a poem on epics.
(C)	The pocket-knife is described as the infant of the sword.
(D)	Epics are described as too inconvenient to write.

Q.7	In the diagram, the lines QR and ST are parallel to each other. The shortest distance between these two lines is half the shortest distance between the point P and line QR. What is the ratio of the area of the triangle PST to the area of the trapezium SQRT? Note: The figure shown is representative.
	
(A)	$\frac{1}{3}$
(B)	$\frac{1}{4}$
(C)	$\frac{2}{5}$
(D)	$\frac{1}{2}$

Q.8	A fair six-faced dice, with the faces labelled ‘1’, ‘2’, ‘3’, ‘4’, ‘5’, and ‘6’, is rolled thrice. What is the probability of rolling ‘6’ exactly once?
(A)	$\frac{75}{216}$
(B)	$\frac{1}{6}$
(C)	$\frac{1}{18}$
(D)	$\frac{25}{216}$

Q.9	A square paper, shown in figure (I), is folded along the dotted lines as shown in the figures (II) and (III). Then a few cuts are made as shown in figure (IV). Which one of the following patterns will be obtained when the paper is unfolded? Note: The figures shown are representative.
	 (I) (II) (III) (IV)
(A)	
(B)	
(C)	
(D)	

Q.10	A shop has 4 distinct flavors of ice-cream. One can purchase any number of scoops of any flavor. **The order in which the scoops are purchased is inconsequential.** If one wants to purchase 3 scoops of ice-cream, in how many ways can one make that purchase?
(A)	4
(B)	20
(C)	24
(D)	48

Q.11 – Q.35 Carry ONE mark Each

Q.11	Suppose a program is running on a non-pipelined single processor computer system. The computer is connected to an external device that can interrupt the processor asynchronously. The processor needs to execute the interrupt service routine (ISR) to serve this interrupt. The following steps (not necessarily in order) are taken by the processor when the interrupt arrives: (i) The processor saves the content of the program counter. (ii) The program counter is loaded with the start address of the ISR. (iii) The processor finishes the present instruction. Which ONE of the following is the CORRECT sequence of steps?
(A)	(iii), (i), (ii)
(B)	(i), (iii), (ii)

(C)	(i), (ii), (iii)
(D)	(iii), (ii), (i)

Q.12	Which ONE of the following statements is **FALSE** regarding the symbol table?
(A)	Symbol table is responsible for keeping track of the scope of variables.
(B)	Symbol table can be implemented using a binary search tree.
(C)	Symbol table is not required after the parsing phase.
(D)	Symbol table is created during the lexical analysis phase.

Q.13	Which ONE of the following techniques used in compiler code optimization uses live variable analysis?
(A)	Run-time function call management
(B)	Register assignment to variables
(C)	Strength reduction
(D)	Constant folding

Q.14	Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?
(A)	2^{21}
(B)	2^{20}
(C)	2^{22}
(D)	2^{24}

Q.15	A schedule of three database transactions T_1, T_2, and T_3 is shown. $R_i(A)$ and $W_i(A)$ denote read and write of data item A by transaction $T_i, i = 1,2,3$. The transaction T_1 aborts at the end. Which other transaction(s) will be required to be rolled back?
	$R_1(X)\ W_1(Y)\ R_2(X)\ R_2(Y)\ R_3(Y)\ ABORT(T_1)$
(A)	Only T_2
(B)	Only T_3
(C)	Both T_2 and T_3
(D)	Neither T_2 nor T_3

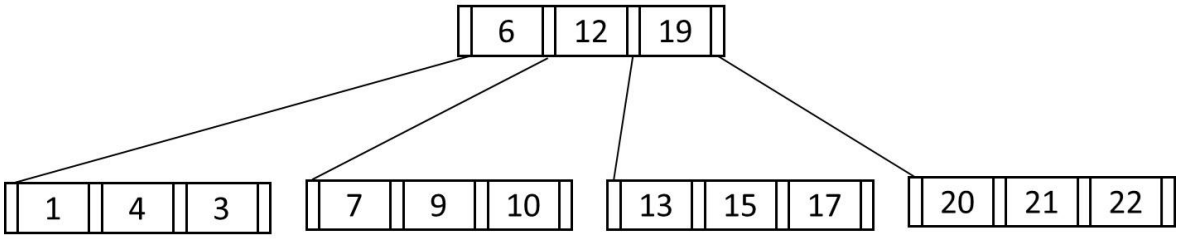
<table>
<tr><td>Q.16</td><td>Identify the ONE CORRECT matching between the OSI layers and their corresponding functionalities as shown.</td></tr>
<tr><td></td><td>
<table>
<tr><th><u>OSI Layers</u></th><th><u>Functionalities</u></th></tr>
<tr><td>(a) Network layer</td><td>(I) Packet routing</td></tr>
<tr><td>(b) Transport layer</td><td>(II) Framing and error handling</td></tr>
<tr><td>(c) Datalink layer</td><td>(III) Host to host communication</td></tr>
</table>
</td></tr>
<tr><td>(A)</td><td>(a)-(I), (b)-(II), (c)-(III)</td></tr>
<tr><td>(B)</td><td>(a)-(I), (b)-(III), (c)-(II)</td></tr>
<tr><td>(C)</td><td>(a)-(II), (b)-(I), (c)-(III)</td></tr>
<tr><td>(D)</td><td>(a)-(III), (b)-(II), (c)-(I)</td></tr>
</table>

Q.17	$g(.)$ is a function from A to B, $f(.)$ is a function from B to C, and their composition defined as $f(g(.))$ is a mapping from A to C. If $f(.)$ and $f(g(.))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(.)$?
(A)	$g(.)$ must be an onto (surjective) function.
(B)	$g(.)$ must be a one-to-one (injective) function.
(C)	$g(.)$ must be a bijective function, that is, both one-to-one and onto.
(D)	$g(.)$ is not required to be a one-to-one or onto function.

Q.18	Let G be any undirected graph with positive edge weights, and T be a minimum spanning tree of G. For any two vertices, u and v, let $d_1(u,v)$ and $d_2(u,v)$ be the shortest distances between u and v in G and T, respectively. Which ONE of the options is CORRECT for all possible G, T, u and v?
(A)	$d_1(u,v) = d_2(u,v)$
(B)	$d_1(u,v) \leq d_2(u,v)$
(C)	$d_1(u,v) \geq d_2(u,v)$
(D)	$d_1(u,v) \neq d_2(u,v)$

Q.19	Consider the following context-free grammar G, where S, A, and B are the variables (non-terminals), a and b are the terminal symbols, S is the start variable, and the rules of G are described as: $S \rightarrow aaB \mid Abb$ $A \rightarrow a \mid aA$ $B \rightarrow b \mid bB$ Which ONE of the languages $L(G)$ is accepted by G?
(A)	$L(G) = \{a^2 b^n \mid n \geq 1\} \cup \{a^n b^2 \mid n \geq 1\}$
(B)	$L(G) = \{a^n b^{2n} \mid n \geq 1\} \cup \{a^{2n} b^n \mid n \geq 1\}$
(C)	$L(G) = \{a^n b^n \mid n \geq 1\}$
(D)	$L(G) = \{a^{2n} b^{2n} \mid n \geq 1\}$

Q.20	Consider the following recurrence relation: $T(n) = 2T(n-1) + n2^n$ for $n > 0, \quad T(0) = 1.$ Which ONE of the following options is CORRECT?
(A)	$T(n) = \Theta(n^2 2^n)$
(B)	$T(n) = \Theta(n2^n)$
(C)	$T(n) = \Theta((\log n)^2\, 2^n)$
(D)	$T(n) = \Theta(4^n)$

Q.21	Consider the following B^+ tree with 5 nodes, in which a node can store at most 3 key values. The value 23 is now inserted in the B^+ tree. Which of the following options(s) is/are CORRECT?
	
(A)	None of the nodes will split.
(B)	At least one node will split and redistribute.
(C)	The total number of nodes will remain same.
(D)	The height of the tree will increase.

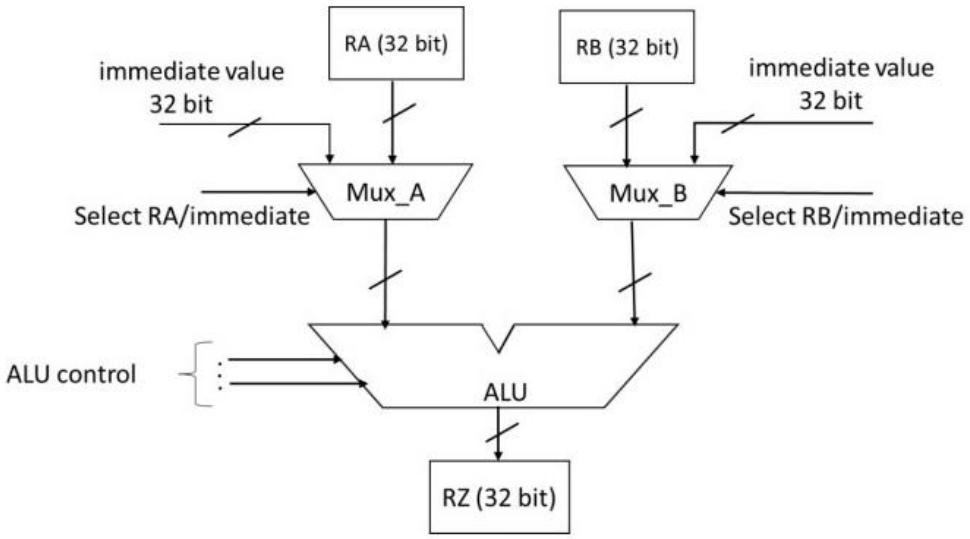
Q.22	Consider the 3-way handshaking protocol for TCP connection establishment. Let the three packets exchanged during the connection establishment be denoted as P1, P2, and P3, in order. Which of the following option(s) is/are TRUE with respect to TCP header flags that are set in the packets?
(A)	P3: SYN = 1, ACK = 1
(B)	P2: SYN = 1, ACK = 1
(C)	P2: SYN = 0, ACK = 1
(D)	P1: SYN = 1

Q.23	Consider the given system of linear equations for variables x and y, where k is a real-valued constant. Which of the following option(s) is/are CORRECT?
	$$x + ky = 1$$ $$kx + y = -1$$
(A)	There is exactly one value of k for which the above system of equations has no solution.
(B)	There exist an infinite number of values of k for which the system of equations has no solution.
(C)	There exists exactly one value of k for which the system of equations has exactly one solution.
(D)	There exists exactly one value of k for which the system of equations has an infinite number of solutions.

Q.24	Let X be a 3-variable Boolean function that produces output as '1' when at least two of the input variables are '1'. Which of the following statement(s) is/are CORRECT, where a, b, c, d, e are Boolean variables?
(A)	$X(a, b, X(c, d, e)) = X(X(a, b, c), d, e)$
(B)	$X(a, b, X(a, b, c)) = X(a, b, c)$
(C)	$X(a, b, X(a, c, d)) = (X(a, b, a)$ AND $X(c, d, c))$
(D)	$X(a, b, c) = X(a, X(a, b, c), X(a, c, c))$

Q.25	The number -6 can be represented as 1010 in 4-bit 2's complement representation. Which of the following is/are CORRECT 2's complement representation(s) of -6?
(A)	1000 1010 in 8-bits
(B)	1111 1010 in 8-bits
(C)	1000 0000 0000 1010 in 16-bits
(D)	1111 1111 1111 1010 in 16-bits

Q.26	Which of the following statement(s) is/are TRUE for any binary search tree (BST) having n distinct integers?
(A)	The maximum length of a path from the root node to any other node is $(n-1)$.
(B)	An inorder traversal will always produce a sorted sequence of elements.
(C)	Finding an element takes $O(\log_2 n)$ time in the worst case.
(D)	Every BST is also a Min-Heap.

Q.27	A partial data path of a processor is given in the figure, where RA, RB, and RZ are 32-bit registers. Which option(s) is/are CORRECT related to arithmetic operations using the data path as shown?
	
(A)	The data path can implement arithmetic operations involving two registers.
(B)	The data path can implement arithmetic operations involving one register and one immediate value.
(C)	The data path can implement arithmetic operations involving two immediate values.
(D)	The data path can only implement arithmetic operations involving one register and one immediate value.

Q.28	A regular language L is accepted by a non-deterministic finite automaton (NFA) with n states. Which of the following statement(s) is/are FALSE?
(A)	L may have an accepting NFA with $< n$ states.
(B)	L may have an accepting DFA with $< n$ states.
(C)	There exists a DFA with $\leq 2^n$ states that accepts L.
(D)	Every DFA that accepts L has $> 2^n$ states.

Q.29	Suppose in a multiprogramming environment, the following C program segment is executed. A process goes into I/O queue whenever an I/O related operation is performed. Assume that there will always be a context switch whenever a process requests for an I/O, and also whenever the process returns from an I/O. The number of times the process will enter the ready queue during its lifetime (not counting the time the process enters the ready queue when it is run initially) is _____. (Answer in integer)

```
int main()
{
    int x=0,i=0;
    scanf("%d",&x);
    for(i=0; i<20; i++)
    {
        x = x+20;
        printf("%d\n",x);
    }
    return 0;
}
```

Q.30	Let S be the set of all ternary strings defined over the alphabet $\{a, b, c\}$. Consider all strings in S that contain at least one occurrence of two consecutive symbols, that is, “aa”, “bb” or “cc”. The number of such strings of length 5 that are possible is _____. (Answer in integer)

Q.31	Consider the given function $f(x)$. $$f(x) = \begin{cases} ax + b & \text{for } x < 1 \\ x^3 + x^2 + 1 & \text{for } x \geq 1 \end{cases}$$ If the function is differentiable everywhere, the value of b must be _____. (rounded off to one decimal place)

Q.32	A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, the probability $P(head) = 0.5$ and for a fake coin, $P(head) = 1$. You pick a coin at random and toss it twice, and get two heads. The probability that the coin you have chosen is the fake coin is _____. (rounded off to two decimal places)

Q.33

The pseudocode of a function **fun()** is given below:

```
fun(int A[0,…,n-1]){
    for i=0 to n-2
        for j=0 to n-i-2
            if (A[j]>A[j+1])
                then swap A[j] and A[j+1]

}
```

Let $A[0, \ldots, 29]$ be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function **fun()** is called with $A[0, \ldots, 29]$ as argument, is _____. (Answer in integer)

Q.34

```
#include <stdio.h>
void foo(int *p, int x){
    *p=x;
}
int main(){
    int *z;
    int a = 20, b = 25;
    z = &a;
    foo(z,b);
    printf("%d",a);
    return 0;
}
```

The output of the given C program is _____. (Answer in integer)

Q.35	The height of any rooted tree is defined as the maximum number of edges in the path from the root node to any leaf node. Suppose a Min-Heap T stores 32 keys. The height of T is _____. (Answer in integer)

Q.36 – Q.65 Carry TWO marks Each

Q.36	Consider a memory system with 1M bytes of main memory and 16K bytes of cache memory. Assume that the processor generates 20-bit memory address, and the cache block size is 16 bytes. If the cache uses direct mapping, how many bits will be required to store all the *tag* values? [Assume memory is byte addressable, $1K=2^{10}$, $1M=2^{20}$.]
(A)	6×2^{10}
(B)	8×2^{10}
(C)	2^{12}
(D)	2^{14}

Q.37	A processor has 64 general-purpose registers and 50 distinct instruction types. An instruction is encoded in 32-bits. What is the maximum number of bits that can be used to store the immediate operand for the given instruction? `ADD R1, #25 // R1 = R1 + 25`
(A)	16
(B)	20
(C)	22
(D)	24

Q.38	A computer has two processors, M_1 and M_2. Four processes P_1, P_2, P_3, P_4 with CPU bursts of 20, 16, 25, and 10 milliseconds, respectively, arrive at the same time and these are the only processes in the system. The scheduler uses non-preemptive priority scheduling, with priorities decided as follows: • M_1 uses priority of execution for the processes as, $P_1 > P_3 > P_2 > P_4$, i.e., P_1 and P_4 have highest and lowest priorities, respectively. • M_2 uses priority of execution for the processes as, $P_2 > P_3 > P_4 > P_1$, i.e., P_2 and P_1 have highest and lowest priorities, respectively. A process P_i is scheduled to a processor M_k, if the processor is free and no other process P_j is waiting with higher priority. At any given point of time, a process can be allocated to any one of the free processors without violating the execution priority rules. Ignore the context switch time. What will be the average waiting time of the processes in milliseconds?
(A)	9.00
(B)	8.75
(C)	6.50
(D)	7.50

Q.39	Consider two relations describing $teams$ and $players$ in a sports league: • $teams(tid, tname)$: tid, $tname$ are team-id and team-name, respectively • $players(pid, pname, tid)$: pid, $pname$, and tid denote player-id, player-name and the team-id of the player, respectively Which ONE of the following tuple relational calculus queries returns the name of the players who play for the team having $tname$ as $'MI'$?
(A)	$\{p.pname \mid p \in players \wedge \exists t\ (t \in teams \wedge p.tid = t.tid \wedge t.tname = 'MI')\}$
(B)	$\{p.pname \mid p \in teams \wedge \exists t\ (t \in players \wedge p.tid = t.tid \wedge t.tname = 'MI')\}$
(C)	$\{p.pname \mid p \in players \wedge \exists t\ (t \in teams \wedge t.tname = 'MI')\}$
(D)	$\{p.pname \mid p \in teams \wedge \exists t\ (t \in players \wedge t.tname = 'MI')\}$

Q.40 A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?

Subnet Address	Subnet Mask (in CIDR notation)	Interface
145.36.0.0	/16	E1
145.36.128.0	/17	E2
145.36.64.0	/18	E3
145.36.255.0	/24	E4
Default	--	E5

(A) E3

(B) E1

(C) E2

(D) E5

Q.41	Let A be a 2×2 matrix as given. $$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$ What are the eigenvalues of the matrix A^{13} ?
(A)	$1,\ -1$
(B)	$2\sqrt{2},\ -2\sqrt{2}$
(C)	$4\sqrt{2},\ -4\sqrt{2}$
(D)	$64\sqrt{2},\ -64\sqrt{2}$

Q.42	Consider the following four variable Boolean function in sum-of-product form $$F(b_3, b_2, b_1, b_0) = \sum(0, 2, 4, 8, 10, 11, 12).$$ where the value of the function is computed by considering $b_3b_2b_1b_0$ as a 4-bit binary number, where b_3 denotes the most significant bit and b_0 denotes the least significant bit. Note that there are no don't care terms. Which ONE of the following options is the CORRECT minimized Boolean expression for F?
(A)	$\bar{b}_1\bar{b}_0 + \bar{b}_2\bar{b}_0 + b_1\,\bar{b}_2b_3$
(B)	$\bar{b}_1\,\bar{b}_0 + \bar{b}_2\,\bar{b}_0$
(C)	$\bar{b}_2\bar{b}_0 + b_1b_2b_3$
(D)	$\bar{b}_0\bar{b}_2 + \bar{b}_3$

Q.43	Let $G(V, E)$ be an undirected and unweighted graph with 100 vertices. Let $d(u, v)$ denote the number of edges in a shortest path between vertices u and v in V. Let the maximum value of $d(u, v)$, $u, v \in V$ such that $u \neq v$, be 30. Let T be any breadth-first-search tree of G. Which ONE of the given options is CORRECT for every such graph G?
(A)	The height of T is exactly 15.
(B)	The height of T is exactly 30.
(C)	The height of T is at least 15.
(D)	The height of T is at least 30.

Q.44	Consider the following two languages over the alphabet $\{a, b\}$: $L_1 = \{ \alpha\beta\alpha \mid \alpha \in \{a,b\}^+ \text{ AND } \beta \in \{a,b\}^+ \}$ $L_2 = \{ \alpha\beta\alpha \mid \alpha \in \{a\}^+ \text{ AND } \beta \in \{a,b\}^+ \}$ Which ONE of the following statements is CORRECT?
(A)	Both L_1 and L_2 are regular languages.
(B)	L_1 is a regular language but L_2 is not a regular language.
(C)	L_1 is not a regular language but L_2 is a regular language.
(D)	Neither L_1 nor L_2 is a regular language.

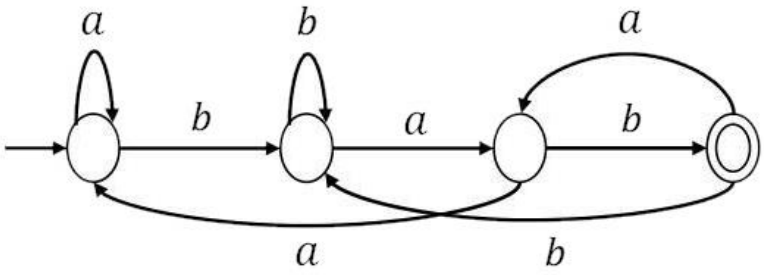
Q.45	Consider the following two languages over the alphabet $\{a, b, c\}$, where m and n are natural numbers. $$L_1 = \{a^m b^m c^{m+n} \mid m, n \geq 1\}$$ $$L_2 = \{a^m b^n c^{m+n} \mid m, n \geq 1\}$$ Which ONE of the following statements is CORRECT?
(A)	Both L_1 and L_2 are context-free languages.
(B)	L_1 is a context-free language but L_2 is not a context-free language.
(C)	L_1 is not a context-free language but L_2 is a context-free language.
(D)	Neither L_1 nor L_2 are context-free languages.

Q.46	Which of the following statement(s) is/are TRUE while computing *First* and *Follow* during top down parsing by a compiler?
(A)	For a production $A \rightarrow \epsilon$, ϵ will be added to $First(A)$.
(B)	If there is any input right end marker, it will be added to $First(S)$, where S is the start symbol.
(C)	For a production $A \rightarrow \epsilon$, ϵ will be added to $Follow(A)$.
(D)	If there is any input right end marker, it will be added to $Follow(S)$, where S is the start symbol.

Q.47	Consider a relational schema $team(name, city, owner)$, with functional dependencies $\{name \rightarrow city,\ name \rightarrow owner\}$. The relation $team$ is decomposed into two relations, $t1(name, city)$ and $t2(name, owner)$. Which of the following statement(s) is/are TRUE?
(A)	The relation $team$ is NOT in BCNF.
(B)	The relations $t1$ and $t2$ are in BCNF.
(C)	The decomposition constitutes a lossless join.
(D)	The relation $team$ is NOT in 3NF.

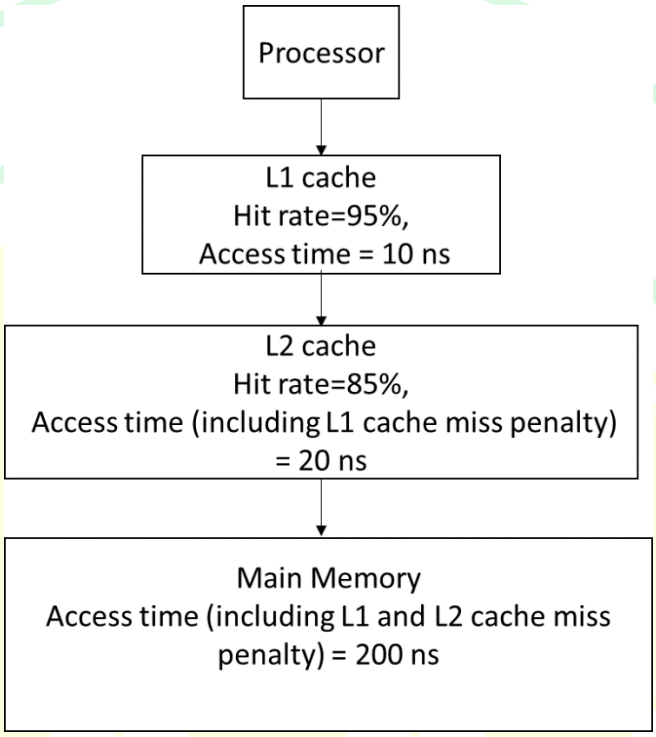
Q.48	Which of the following predicate logic formulae/formula is/are CORRECT representation(s) of the statement: "*Everyone has exactly one mother*"? The meanings of the predicates used are: • $mother(y, x)$: y is the mother of x • $noteq(x, y)$: x and y are not equal
(A)	$\forall x \exists y \exists z \big(mother(y, x) \wedge \neg mother(z, x)\big)$
(B)	$\forall x \exists y \big[mother(y, x) \wedge \forall z \big(noteq(z, y) \rightarrow \neg mother(z, x)\big)\big]$
(C)	$\forall x \forall y [mother(y, x) \rightarrow \exists z (mother(z, x) \wedge \neg noteq(z, y))]$
(D)	$\forall x \exists y \big[mother(y, x) \wedge \neg \exists z \big(noteq(z, y) \wedge mother(z, x)\big)\big]$

Q.49	$A = \{0, 1, 2, 3, \ldots\}$ is the set of non-negative integers. Let F be the set of functions from A to itself. For any two functions, $f_1, f_2 \in$ F, we define $$(f_1 \odot f_2)(n) = f_1(n) + f_2(n)$$ for every number n in A. Which of the following is/are CORRECT about the mathematical structure $(\text{F}, \odot)$?
(A)	$(\text{F}, \odot)$ is an Abelian group.
(B)	$(\text{F}, \odot)$ is an Abelian monoid.
(C)	$(\text{F}, \odot)$ is a non-Abelian group.
(D)	$(\text{F}, \odot)$ is a non-Abelian monoid.

Q.50	Consider the following deterministic finite automaton (DFA) defined over the alphabet, $\Sigma = \{a, b\}$. Identify which of the following language(s) is/are accepted by the given DFA.
	
(A)	The set of all strings containing an even number of b's.
(B)	The set of all strings containing the pattern bab.
(C)	The set of all strings ending with the pattern bab.
(D)	The set of all strings not containing the pattern aba.

Q.51	A disk of size 512M bytes is divided into blocks of 64K bytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a *NULL* pointer (also of 4 bytes). Suppose a file of 1M bytes needs to be stored in the disk. Assume, 1K = 2^{10} and 1M = 2^{20}. The amount of space in bytes that will be wasted due to internal fragmentation is _____. (Answer in integer)

Q.52	Refer to the given 3-address code sequence. This code sequence is split into basic blocks. The number of basic blocks is _____. (Answer in integer)
	1001: i = 1 1002: j = 1 1003: t1 = 10*i 1004: t2 = t1+j 1005: t3 = 8*t2 1006: t4 = t3-88 1007: a[t4] = 0.0 1008: j = j+1 1009: if j <= 10 goto 1003 1010: i = i+1 1011: if i <= 10 goto 1002 1012: i = 1 1013: t5 = i-1 1014: t6 = 88*t5 1015: a[t6] = 1.0 1016: i = i+1 1017: if i <= 10 goto 1013

Q.53	A computer has a memory hierarchy consisting of two-level cache (L1 and L2) and a main memory. If the processor needs to access data from memory, it first looks into L1 cache. If the data is not found in L1 cache, it goes to L2 cache. If it fails to get the data from L2 cache, it goes to main memory, where the data is definitely available. *Hit rates* and *access times* of various memory units are shown in the figure. The *average memory access time* in nanoseconds (*ns*) is _____. (rounded off to two decimal places)
	

Q.54	In optimal page replacement algorithm, information about all future page references is available to the operating system (OS). A modification of the optimal page replacement algorithm is as follows: *The OS correctly predicts only up to next 4 page references (including the current page) at the time of allocating a frame to a page.* A process accesses the pages in the following order of page numbers: $1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4.$ If the system has three memory frames that are initially empty, the number of page faults that will occur during execution of the process is _____ . (Answer in integer)

Q.55 Consider the following database tables of a sports league.

```
player(pid,pname,age)          team(tid,tname,city,cid)
coach(cid,cname)               members(pid,tid)
```

An instance of the table and an SQL query are given.

player

pid	pname	age
1	Jasprit	31
2	Atharva	24
3	Ishan	26
4	Axar	30

coach

cid	cname
101	Ricky
102	Mark
103	Trevor

team

tid	tname	city	cid
10	MI	Mumbai	102
20	DC	Delhi	101
30	PK	Mohali	103

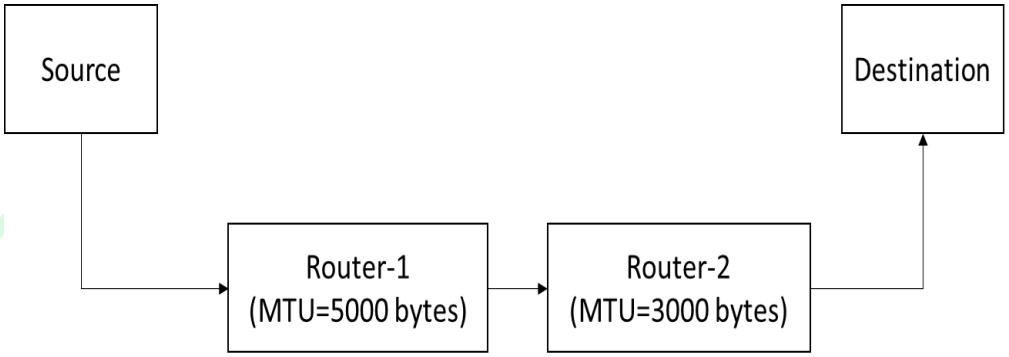
members

pid	tid
1	10
2	30
3	10
4	20

```
SELECT MIN(P.age)
FROM player P
WHERE P.pid IN (
    SELECT M.pid
    FROM team T, coach C, members M
    WHERE C.cname = 'Mark'
          AND T.cid = C.cid
          AND M.tid = T.tid
    )
```

The value returned by the given SQL query is _____ . (Answer in integer)

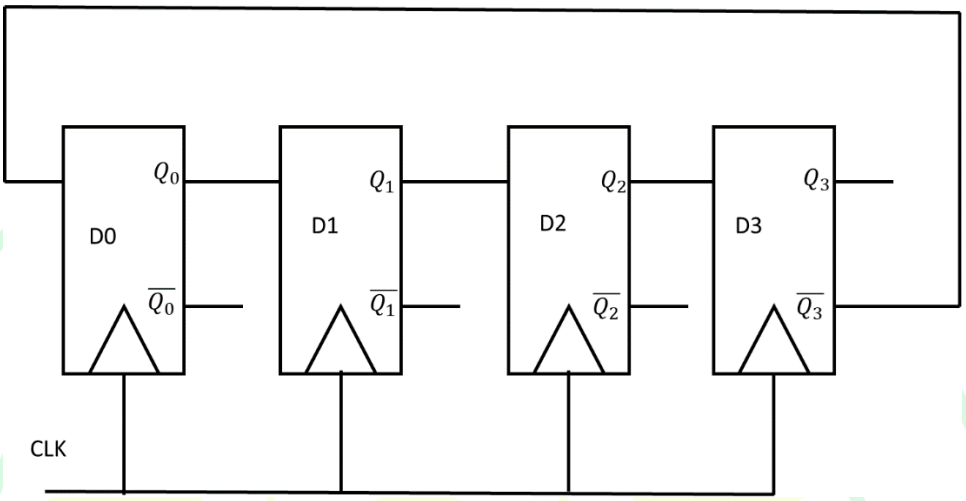
Q.56	Suppose a 5-bit message is transmitted from a source to a destination through a noisy channel. The probability that a bit of the message gets flipped during transmission is 0.01. Flipping of each bit is independent of one another. The probability that the message is delivered error-free to the destination is _____ . (rounded off to three decimal places)

Q.57	Suppose a message of size 15000 bytes is transmitted from a source to a destination using IPv4 protocol via two routers as shown in the figure. Each router has a defined maximum transmission unit (MTU) as shown in the figure, including IP header. The number of fragments that will be delivered to the destination is _____ . (Answer in integer)
	

Q.58	Consider a probability distribution given by the density function $P(x)$. $$P(x) = \begin{cases} Cx^2, & \text{for } 1 \leq x \leq 4 \\ 0, & \text{for } x < 1 \text{ or } x > 4 \end{cases}$$ The probability that x lies between 2 and 3, i.e., $P(2 \leq x \leq 3)$ is _____. (rounded off to three decimal places)

Q.59 Consider a finite state machine (FSM) with one input X and one output f, represented by the given state transition table. The minimum number of states required to realize this FSM is _____. (Answer in integer)

Present state	Next state		Output f	
	$X = 0$	$X = 1$	$X = 0$	$X = 1$
A	F	B	0	0
B	D	C	0	0
C	F	E	0	0
D	G	A	1	0
E	D	C	0	0
F	F	B	1	1
G	G	H	0	1
H	G	A	1	0

Q.60	Consider the given sequential circuit designed using D-Flip-flops. The circuit is initialized with some value (initial state). The number of distinct states the circuit will go through before returning back to the initial state is _____ . (Answer in integer)
	

Q.61

```
#include <stdio.h>
int foo(int S[],int size){
    if(size == 0) return 0;
    if(size == 1) return 1;
    if(S[0] != S[1]) return 1+foo(S+1,size-1);
    return foo(S+1,size-1);
}
int main(){
    int A[]={0,1,2,2,2,0,0,1,1};
    printf("%d",foo(A,9));
    return 0;
}
```

The value printed by the given C program is _____ . (Answer in integer)

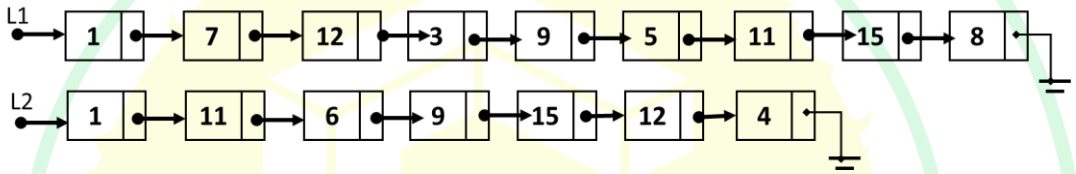
Q.62

Let **LIST** be a datatype for an implementation of linked list defined as follows:

```
typedef struct list {
    int data;
    struct list *next;
} LIST;
```

Suppose a program has created two linked lists, L1 and L2, whose contents are given in the figure below (code for creating L1 and L2 is not provided here). L1 contains 9 nodes, and L2 contains 7 nodes.

Consider the following C program segment that modifies the list L1. The number of nodes that will be there in L1 after the execution of the code segment is _____ . (Answer in integer)




```
int find (int query, LIST *list) {
    while (list != NULL){
        if(list->data == query) return 1;
        list = list->next;
    }
    return 0;
}
int main () {
     … … …
    ptr1=L1; ptr2=L2;
    while (ptr1->next != NULL){
        query = ptr1->next->data;
        if (find (query, L2))
            ptr1->next = ptr1->next->next;
        else ptr1 = ptr1->next;
    }
     … … …
    return 0;
}
```

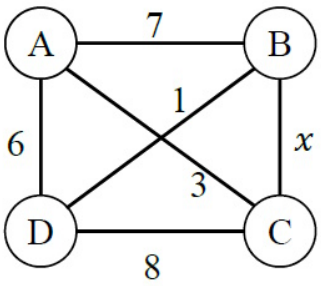
Q.63

Consider the following C program:

```
#include <stdio.h>
int gate (int n) {
    int d, t, newnum, turn;
    newnum = turn = 0; t=1;
    while (n>=t) t *= 10;
    t /=10;
    while (t>0) {
        d = n/t;
        n = n%t;
        t /= 10;
        if (turn) newnum = 10*newnum + d;
        turn = (turn + 1) %2;
    }
    return newnum;
}
int main () {
    printf ("%d", gate(14362));
    return 0;

}
```

The value printed by the given C program is _____ . (Answer in integer)

Q.64	The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is _____ . (answer in integer)
	

Q.65	In a double hashing scheme, $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 7)$ are the auxiliary hash functions. The size m of the hash table is 11. The hash function for the i-th probe in the open address table is $[h_1(k) + i\, h_2(k)] \bmod m$. The following keys are inserted in the given order: 63, 50, 25, 79, 67, 24. The slot at which key 24 gets stored is _____. (Answer in integer)

General Aptitude (GA)

Q.1 – Q.5 Carry ONE mark Each

Q.1 If '→' denotes increasing order of intensity, then the meaning of the words

[walk → jog → sprint] is analogous to [bothered → _____ → daunted].

Which one of the given options is appropriate to fill the blank?

(A) phased

(B) phrased

(C) fazed

(D) fused

Q.2 Two wizards try to create a spell using all the four elements, *water*, *air*, *fire*, and *earth*. For this, they decide to mix all these elements in all possible orders. They also decide to work independently. After trying all possible combination of elements, they conclude that the spell does not work.

How many attempts does each wizard make before coming to this conclusion, independently?

(A) 24

(B) 48

(C) 16

(D) 12

GATE 2024

Q.3 In an engineering college of 10,000 students, 1,500 like neither their core branches nor other branches. The number of students who like their core branches is $1/4^{th}$ of the number of students who like other branches. The number of students who like both their core and other branches is 500.

The number of students who like their core branches is

(A) 1,800

(B) 3,500

(C) 1,600

(D) 1,500

Q.4 For positive non-zero real variables x and y, if

$$\ln\left(\frac{x+y}{2}\right) = \frac{1}{2}[\ln(x) + \ln(y)]$$

then, the value of $\frac{x}{y} + \frac{y}{x}$ is

(A) 1

(B) 1/2

(C) 2

(D) 4

GATE 2024

Q.5 In the sequence 6, 9, 14, x, 30, 41, a possible value of x is

(A) 25

(B) 21

(C) 18

(D) 20

GATE 2024

Q.6 – Q.10 Carry TWO marks Each

Q.6 Sequence the following sentences in a coherent passage.

P: This fortuitous geological event generated a colossal amount of energy and heat that resulted in the rocks rising to an average height of 4 km across the contact zone.

Q: Thus, the geophysicists tend to think of the Himalayas as an active geological event rather than as a static geological feature.

R: The natural process of the cooling of this massive edifice absorbed large quantities of atmospheric carbon dioxide, altering the earth's atmosphere and making it better suited for life.

S: Many millennia ago, a breakaway chunk of bedrock from the Antarctic Plate collided with the massive Eurasian Plate.

(A) QPSR

(B) QSPR

(C) SPRQ

(D) SRPQ

GATE 2024

Q.7 A person sold two different items at the same price. He made 10% profit in one item, and 10% loss in the other item. In selling these two items, the person made a total of

(A) 1% profit

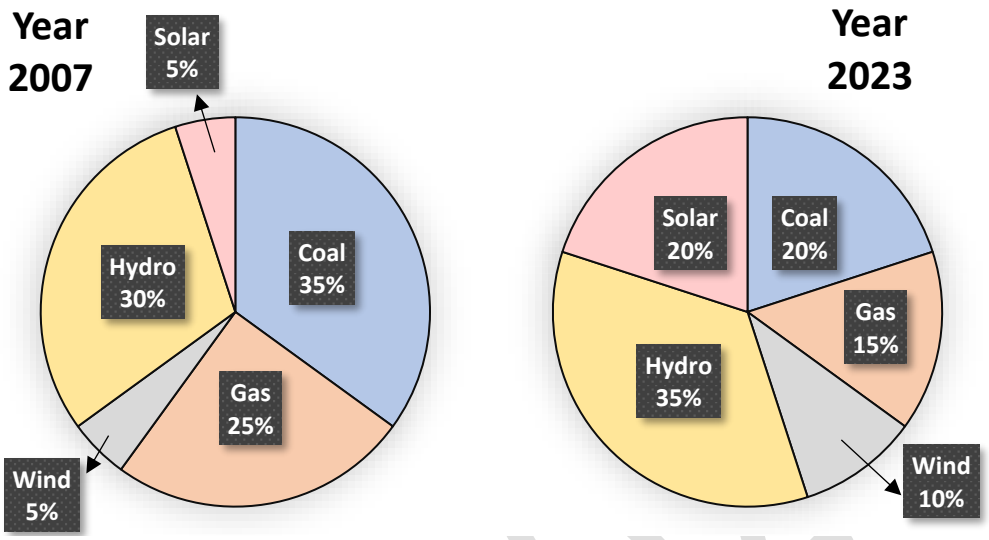
(B) 2% profit

(C) 1% loss

(D) 2% loss

GATE 2024

Q.8 The pie charts depict the shares of various power generation technologies in the total electricity generation of a country for the years 2007 and 2023.



The renewable sources of electricity generation consist of Hydro, Solar and Wind. Assuming that the total electricity generated remains the same from 2007 to 2023, what is the percentage increase in the share of the renewable sources of electricity generation over this period?

(A) 25%

(B) 50%

(C) 77.5%

(D) 62.5%

Q.9 A cube is to be cut into 8 pieces of equal size and shape. Here, each cut should be straight and it should not stop till it reaches the other end of the cube.

The minimum number of such cuts required is

(A) 3

(B) 4

(C) 7

(D) 8

GATE 2024

Q.10 In the 4 × 4 array shown below, each cell of the first three rows has either a cross (X) or a number.

1	X	4	3
X	5	5	4
3	X	6	X

The number in a cell represents the count of the immediate neighboring cells (left, right, top, bottom, diagonals) NOT having a cross (X). Given that the last row has no crosses (X), the sum of the four numbers to be filled in the last row is

(A) 11

(B) 10

(C) 12

(D) 9

GATE 2024

Q.11 – Q.35 Carry ONE mark Each

Q.11 Consider a computer with a 4 MHz processor. Its DMA controller can transfer 8 bytes in 1 cycle from a device to main memory through cycle stealing at regular intervals. Which one of the following is the data transfer rate (*in bits per second*) of the DMA controller if 1% of the processor cycles are used for DMA?

(A) 2,56,000

(B) 3,200

(C) 25,60,000

(D) 32,000

Q.12 Let p and q be the following propositions:

p: Fail grade can be given.

q: Student scores more than 50% marks.

Consider the statement: "*Fail grade cannot be given when student scores more than 50% marks.*"

Which one of the following is the CORRECT representation of the above statement in propositional logic?

(A) $q \rightarrow \neg\, p$

(B) $q \rightarrow p$

(C) $p \rightarrow q$

(D) $\neg\, p \rightarrow q$

Q.13 Consider the following C program. Assume parameters to a function are evaluated from *right to left*.

```
#include <stdio.h>

int g(int p) { printf("%d", p); return p; }
int h(int q) { printf("%d", q); return q; }

void f(int x, int y) {
  g(x);
  h(y);
}

int main() {
  f(g(10),h(20));
}
```

Which one of the following options is the CORRECT output of the above C program?

(A) 20101020

(B) 10202010

(C) 20102010

(D) 10201020

GATE 2024

Q.14 The format of a single-precision floating-point number as per the **IEEE 754** standard is:

Sign (1bit)	Exponent (8 bits)	Mantissa (23 bits)

Choose the largest floating-point number among the following options.

(A)

Sign	Exponent	Mantissa
0	0111 1111	1111 1111 1111 1111 1111 111

(B)

Sign	Exponent	Mantissa
0	1111 1110	1111 1111 1111 1111 1111 111

(C)

Sign	Exponent	Mantissa
0	1111 1111	1111 1111 1111 1111 1111 111

(D)

Sign	Exponent	Mantissa
0	0111 1111	0000 0000 0000 0000 0000 000

GATE 2024

Q.15 Let $T(n)$ be the recurrence relation defined as follows:

$$T(0) = 1,$$
$$T(1) = 2, \text{ and}$$
$$T(n) = 5T(n-1) - 6T(n-2) \quad \text{for } n \geq 2$$

Which one of the following statements is TRUE?

(A) $T(n) = \Theta(2^n)$

(B) $T(n) = \Theta(n2^n)$

(C) $T(n) = \Theta(3^n)$

(D) $T(n) = \Theta(n3^n)$

Q.16 Let $f(x)$ be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) = 1 - f(2 - x)$$

Which one of the following options is the CORRECT value of $\int_0^2 f(x)dx$?

(A) 0

(B) 1

(C) 2

(D) -1

GATE 2024

Q.17 Let A be the adjacency matrix of a simple undirected graph G. Suppose A is its own inverse. Which one of the following statements is *always* TRUE?

(A) G is a cycle

(B) G is a perfect matching

(C) G is a complete graph

(D) There is no such graph G

Q.18 When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (*i.e.*, 1, 2, 3, 4, 5, and 6) is

(A) $\frac{1}{324}$

(B) $\frac{5}{324}$

(C) $\frac{7}{324}$

(D) $\frac{11}{324}$

GATE 2024

Q.19 Once the DBMS informs the user that a transaction has been successfully completed, its effect should persist even if the system crashes before all its changes are reflected on disk. This property is called

(A) durability

(B) atomicity

(C) consistency

(D) isolation

Q.20 In the context of owner and weak entity sets in the ER (Entity-Relationship) data model, which one of the following statements is TRUE?

(A) The weak entity set MUST have total participation in the identifying relationship

(B) The owner entity set MUST have total participation in the identifying relationship

(C) Both weak and owner entity sets MUST have total participation in the identifying relationship

(D) Neither weak entity set nor owner entity set MUST have total participation in the identifying relationship

GATE 2024

Q.21 Consider the following two sets:

Set *X*	**Set *Y***
P. Lexical Analyzer	1. Abstract Syntax Tree
Q. Syntax Analyzer	2. Token
R. Intermediate Code Generator	3. Parse Tree
S. Code Optimizer	4. Constant Folding

Which one of the following options is the CORRECT match from **Set *X*** to **Set *Y*** ?

(A) P – 4; Q – 1; R – 3; S – 2

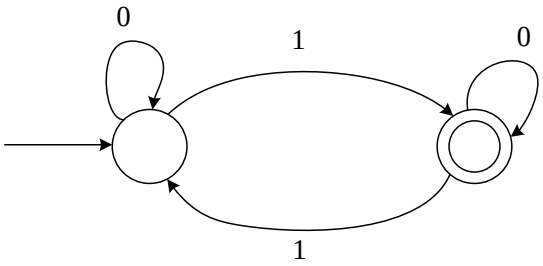
(B) P – 2; Q – 3; R – 1; S – 4

(C) P – 2; Q – 1; R – 3; S – 4

(D) P – 4; Q – 3; R – 2; S – 1

GATE 2024

Q.22 Which one of the following regular expressions is equivalent to the language accepted by the DFA given below?



(A) $0^*1(0 + 10^*1)^*$

(B) $0^*(10^*11)^*0^*$

(C) $0^*1(010^*1)^*0^*$

(D) $0(1 + 0^*10^*1)^*0^*$

GATE 2024

Q.23 Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection.

Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

(A) The destination IP address is the IP address of R

(B) The destination IP address is the IP address of Y

(C) The destination MAC address is the MAC address of S

(D) The destination MAC address is the MAC address of Y

Q.24 Which of the following tasks is/are the responsibility/responsibilities of the memory management unit (MMU) in a system with paging-based memory management?

(A) Allocate a new page table for a newly created process

(B) Translate a virtual address to a physical address using the page table

(C) Raise a trap when a virtual address is not found in the page table

(D) Raise a trap when a process tries to write to a page marked with read-only permission in the page table

Q.25 Consider a process P running on a CPU. Which one or more of the following events will *always* trigger a context switch by the OS that results in process P moving to a non-running state (e.g., ready, blocked)?

(A) P makes a blocking system call to read a block of data from the disk

(B) P tries to access a page that is in the swap space, triggering a page fault

(C) An interrupt is raised by the disk to deliver data requested by some other process

(D) A timer interrupt is raised by the hardware

Q.26 Which of the following file organizations is/are I/O efficient for the scan operation in DBMS?

(A) Sorted

(B) Heap

(C) Unclustered tree index

(D) Unclustered hash index

GATE 2024

Q.27 Which of the following statements about the Two Phase Locking (2PL) protocol is/are TRUE?

(A) 2PL permits only serializable schedules

(B) With 2PL, a transaction always locks the data item being read or written just before every operation and always releases the lock just after the operation

(C) With 2PL, once a lock is released on any data item inside a transaction, no more locks on any data item can be obtained inside that transaction

(D) A deadlock is possible with 2PL

Q.28 Which of the following statements about IPv4 fragmentation is/are TRUE?

(A) The fragmentation of an IP datagram is performed *only* at the source of the datagram

(B) The fragmentation of an IP datagram is performed at any IP router which finds that the size of the datagram to be transmitted exceeds the MTU

(C) The reassembly of fragments is performed *only* at the destination of the datagram

(D) The reassembly of fragments is performed at all intermediate routers along the path from the source to the destination

GATE 2024

Q.29 Which of the following statements is/are FALSE?

(A) An attribute grammar is a syntax-directed definition (SDD) in which the functions in the semantic rules have no side effects

(B) The attributes in a L-attributed definition cannot always be evaluated in a depth-first order

(C) Synthesized attributes can be evaluated by a bottom-up parser as the input is parsed

(D) All L-attributed definitions based on LR(1) grammar can be evaluated using a bottom-up parsing strategy

Q.30 For a Boolean variable x, which of the following statements is/are FALSE?

(A) $x.1 = x$

(B) $x + 1 = x$

(C) $x.x = 0$

(D) $x + \bar{x} = 1$

GATE 2024

Q.31 An instruction format has the following structure:

Instruction Number: *Opcode destination reg, source reg-1, source reg-2*

Consider the following sequence of instructions to be executed in a pipelined processor:

`I1`: DIV R3, R1, R2

`I2`: SUB R5, R3, R4

`I3`: ADD R3, R5, R6

`I4`: MUL R7, R3, R8

Which of the following statements is/are TRUE?

(A) There is a RAW dependency on R3 between `I1` and `I2`

(B) There is a WAR dependency on R3 between `I1` and `I3`

(C) There is a RAW dependency on R3 between `I2` and `I3`

(D) There is a WAW dependency on R3 between `I3` and `I4`

GATE 2024

Q.32 Which of the following fields of an IP header is/are *always* modified by any router before it forwards the IP packet?

(A) Source IP Address

(B) Protocol

(C) Time to Live (TTL)

(D) Header Checksum

Q.33 Consider the following C function definition.

```
int fX(char *a){
    char *b = a;
    while(*b)
        b++;
    return b - a;}
```

Which of the following statements is/are TRUE?

(A) The function call `fX("abcd")` will always return a value

(B) Assuming a character array `c` is declared as `char c[] = "abcd"` in `main()`, the function call `fX(c)` will always return a value

(C) The code of the function will not compile

(D) Assuming a character pointer `c` is declared as `char *c = "abcd"` in `main()`, the function call `fX(c)` will always return a value

Q.34 Let P be the partial order defined on the set $\{1,2,3,4\}$ as follows

$$P = \{(x, x) \mid x \in \{1,2,3,4\}\} \cup \{(1,2), (3,2), (3,4)\}$$

The number of total orders on $\{1,2,3,4\}$ that contain P is _____

Q.35 Let A be an array containing integer values. The distance of A is defined as the minimum number of elements in A that must be replaced with another integer so that the resulting array is sorted in non-decreasing order. The distance of the array $[2, 5, 3, 1, 4, 2, 6]$ is _____

GATE 2024

Q.36 – Q.65 Carry TWO marks Each

Q.36 What is the output of the following C program?

```
#include <stdio.h>
int main() {
    double a[2]={20.0, 25.0}, *p, *q;
    p = a;
    q = p + 1;
    printf("%d,%d", (int)(q - p), (int)(*q - *p));
    return 0;}
```

(A) 4,8

(B) 1,5

(C) 8,5

(D) 1,8

GATE 2024

Q.37 Consider a single processor system with four processes A, B, C, and D, represented as given below, where for each process the first value is its arrival time, and the second value is its CPU burst time.

A (0, 10), B (2, 6), C (4, 3), and D (6, 7).

Which one of the following options gives the average waiting times when preemptive Shortest Remaining Time First (SRTF) and Non-Preemptive Shortest Job First (NP-SJF) CPU scheduling algorithms are applied to the processes?

(A) SRTF = 6, NP-SJF = 7

(B) SRTF = 6, NP-SJF = 7.5

(C) SRTF = 7, NP-SJF = 7.5

(D) SRTF = 7, NP-SJF = 8.5

Q.38 Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

(A) 10.12.2.0/23

(B) 10.12.2.0/24

(C) 10.12.0.0/22

(D) 10.12.2.0/22

GATE 2024

Q.39 You are given a set V of distinct integers. A binary search tree T is created by inserting all elements of V one by one, starting with an empty tree. The tree T follows the convention that, at each node, all values stored in the left subtree of the node are smaller than the value stored at the node. You are not aware of the sequence in which these values were inserted into T, and you do not have access to T.

Which one of the following statements is TRUE?

(A) Inorder traversal of T can be determined from V

(B) Root node of T can be determined from V

(C) Preorder traversal of T can be determined from V

(D) Postorder traversal of T can be determined from V

GATE 2024

Q.40 Consider the following context-free grammar where the start symbol is S and the set of terminals is $\{a,b,c,d\}$.

$$S \rightarrow AaAb \mid BbBa$$
$$A \rightarrow cS \mid \epsilon$$
$$B \rightarrow dS \mid \epsilon$$

The following is a partially-filled LL(1) parsing table.

	a	b	c	d	\$
S	$S \rightarrow AaAb$	$S \rightarrow BbBa$	(1)	(2)	
A	$A \rightarrow \epsilon$	(3)	$A \rightarrow cS$		
B	(4)	$B \rightarrow \epsilon$		$B \rightarrow dS$	

Which one of the following options represents the CORRECT combination for the numbered cells in the parsing table?

Note: In the options, "blank" denotes that the corresponding cell is empty.

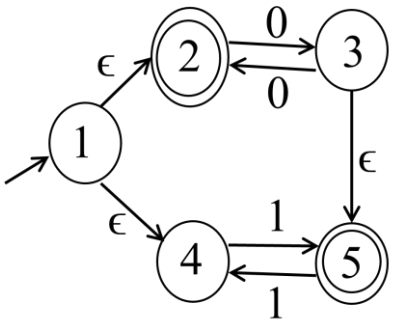
(A) (1) $S \rightarrow AaAb$ (2) $S \rightarrow BbBa$ (3) $A \rightarrow \epsilon$ (4) $B \rightarrow \epsilon$

(B) (1) $S \rightarrow BbBa$ (2) $S \rightarrow AaAb$ (3) $A \rightarrow \epsilon$ (4) $B \rightarrow \epsilon$

(C) (1) $S \rightarrow AaAb$ (2) $S \rightarrow BbBa$ (3) blank (4) blank

(D) (1) $S \rightarrow BbBa$ (2) $S \rightarrow AaAb$ (3) blank (4) blank

Q.41 Let M be the 5-state NFA with ϵ-transitions shown in the diagram below.



Which one of the following regular expressions represents the language accepted by M ?

(A) $(00)^* + 1(11)^*$

(B) $0^* + (1 + 0(00)^*)(11)^*$

(C) $(00)^* + (1 + (00)^*)(11)^*$

(D) $0^+ + 1(11)^* + 0(11)^*$

GATE 2024

Q.42 Consider an array `X` that contains `n` positive integers. A subarray of `X` is defined to be a sequence of array locations with consecutive indices.

The C code snippet given below has been written to compute the length of the longest subarray of `X` that contains at most two distinct integers. The code has two missing expressions labelled (P) and (Q).

```
int first=0, second=0, len1=0, len2=0, maxlen=0;
for (int i=0; i < n; i++) {
  if (X[i] == first) {
    len2++; len1++;
  } else if (X[i] == second) {
    len2++;
    len1 = (P) ;
    second = first;
  } else {
    len2 = (Q) ;
    len1 = 1; second = first;
  }
  if (len2 > maxlen) {
    maxlen = len2;
  }
  first = X[i];
}
```

Which one of the following options gives the CORRECT missing expressions?

(**Hint:** At the end of the `i`-th iteration, the value of `len1` is the length of the longest subarray ending with `X[i]` that contains all equal values, and `len2` is the length of the longest subarray ending with `X[i]` that contains at most two distinct values.)

(A) (P) `len1+1` (Q) `len2+1`

(B) (P) `1` (Q) `len1+1`

(C) (P) `1` (Q) `len2+1`

(D) (P) `len2+1` (Q) `len1+1`

Q.43 Consider the following expression: $x[i] = (p + r) * -s[i] + u/w$. The following sequence shows the list of triples representing the given expression, with entries missing for triples (1), (3), and (6).

(0)	+	p	r
(1)			
(2)	uminus	(1)	
(3)			
(4)	/	u	w
(5)	+	(3)	(4)
(6)			
(7)	=	(6)	(5)

Which one of the following options fills in the missing entries CORRECTLY?

(A) (1) =[] s i (3) * (0) (2) (6) []= x i

(B) (1) []= s i (3) – (0) (2) (6) =[] x (5)

(C) (1) =[] s i (3) * (0) (2) (6) []= x (5)

(D) (1) []= s i (3) – (0) (2) (6) =[] x i

Q.44 Let x and y be random variables, not necessarily independent, that take real values in the interval [0,1]. Let $z = xy$ and let the mean values of x, y, z be $\bar{x}, \bar{y}, \bar{z}$, respectively. Which one of the following statements is TRUE?

(A) $\bar{z} = \bar{x}\bar{y}$

(B) $\bar{z} \leq \bar{x}\bar{y}$

(C) $\bar{z} \geq \bar{x}\,\bar{y}$

(D) $\bar{z} \leq \bar{x}$

Q.45 The relation schema, `Person(pid,city)`, describes the city of residence for every person uniquely identified by `pid`. The following relational algebra operators are available: `selection, projection, cross product`, and `rename`.

To find the list of cities where at least 3 persons reside, using the above operators, the minimum number of `cross product` operations that must be used is

(A) 1

(B) 2

(C) 3

(D) 4

GATE 2024

Q.46 Consider a multi-threaded program with two threads `T1` and `T2`. The threads share two semaphores: `s1` (initialized to 1) and `s2` (initialized to 0). The threads also share a global variable `x` (initialized to 0). The threads execute the code shown below.

```
// code of T1          // code of T2
wait(s1);              wait(s1);
x = x+1;               x = x+1;
print(x);              print(x);
wait(s2);              signal(s2);
signal(s1);            signal(s1);
```

Which of the following outcomes is/are possible when threads `T1` and `T2` execute concurrently?

(A) `T1` runs first and prints 1, `T2` runs next and prints 2

(B) `T2` runs first and prints 1, `T1` runs next and prints 2

(C) `T1` runs first and prints 1, `T2` does not print anything (deadlock)

(D) `T2` runs first and prints 1, `T1` does not print anything (deadlock)

Q.47 Let A be an $n \times n$ matrix over the set of all real numbers $\mathbb{R}$. Let B be a matrix obtained from A by swapping two rows. Which of the following statements is/are TRUE?

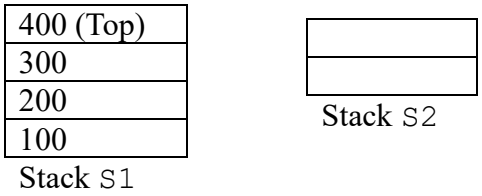
(A) The determinant of B is the negative of the determinant of A

(B) If A is invertible, then B is also invertible

(C) If A is symmetric, then B is also symmetric

(D) If the trace of A is zero, then the trace of B is also zero

Q.48 Let `S1` and `S2` be two stacks. `S1` has capacity of 4 elements. `S2` has capacity of 2 elements. `S1` already has 4 elements: 100, 200, 300, and 400, whereas `S2` is empty, as shown below.



Only the following three operations are available:

`PushToS2`: Pop the top element from `S1` and push it on `S2`.
`PushToS1`: Pop the top element from `S2` and push it on `S1`.
`GenerateOutput`: Pop the top element from `S1` and output it to the user.

Note that the pop operation is not allowed on an empty stack and the push operation is not allowed on a full stack.

Which of the following output sequences can be generated by using the above operations?

(A) 100, 200, 400, 300

(B) 200, 300, 400, 100

(C) 400, 200, 100, 300

(D) 300, 200, 400, 100

GATE 2024

Q.49 Which of the following is/are EQUAL to 224 in radix-5 (*i.e.*, base-5) notation?

(A) 64 in radix-10

(B) 100 in radix-8

(C) 50 in radix-16

(D) 121 in radix-7

GATE 2024

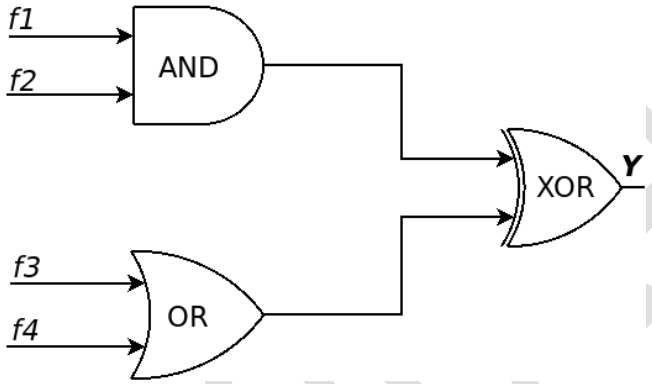
Q.50 Consider 4-variable functions *f1*, *f2*, *f3*, *f4* expressed in sum-of-minterms form as given below.

$f1 = \sum(0,2,3,5,7,8,11,13)$

$f2 = \sum(1,3,5,7,11,13,15)$

$f3 = \sum(0,1,4,11)$

$f4 = \sum(0,2,6,13)$



With respect to the circuit given above, which of the following options is/are CORRECT?

(A) $\boldsymbol{Y} = \sum(0,1,2,11,13)$

(B) $\boldsymbol{Y} = \Pi(3,4,5,6,7,8,9,10,12,14,15)$

(C) $\boldsymbol{Y} = \sum(0,1,2,3,4,5,6,7)$

(D) $\boldsymbol{Y} = \Pi(8,9,10,11,12,13,14,15)$

Q.51 Let G be an undirected connected graph in which every edge has a positive integer weight. Suppose that every spanning tree in G has *even* weight. Which of the following statements is/are TRUE for *every* such graph G ?

(A) All edges in G have even weight

(B) All edges in G have even weight **OR** all edges in G have odd weight

(C) In each cycle C in G, all edges in C have even weight

(D) In each cycle C in G, either all edges in C have even weight **OR** all edges in C have odd weight

Q.52 Consider a context-free grammar G with the following 3 rules.

$$S \rightarrow aS,\ S \rightarrow aSbS,\ S \rightarrow c$$

Let $w \in L(G)$. Let $n_a(w),\ n_b(w),\ n_c(w)$ denote the number of times a, b, c occur in w, respectively. Which of the following statements is/are TRUE?

(A) $n_a(w) > n_b(w)$

(B) $n_a(w) > n_c(w) - 2$

(C) $n_c(w) = n_b(w) + 1$

(D) $n_c(w) = n_b(w) * 2$

Q.53 Consider a disk with the following specifications: rotation speed of 6000 RPM, average seek time of 5 milliseconds, 500 sectors/track, 512-byte sectors. A file has content stored in 3000 sectors located *randomly* on the disk. Assuming average rotational latency, the total time (*in seconds, rounded off to 2 decimal places*) to read the entire file from the disk is _____

Q.54 Consider a TCP connection operating at a point of time with the congestion window of size 12 MSS (Maximum Segment Size), when a timeout occurs due to packet loss. Assuming that all the segments transmitted in the next two RTTs (Round Trip Time) are acknowledged correctly, the congestion window size (*in MSS*) during the third RTT will be _____

Q.55 Consider an Ethernet segment with a transmission speed of 10^8 bits/sec and a maximum segment length of 500 meters. If the speed of propagation of the signal in the medium is 2×10^8 meters/sec, then the minimum frame size (*in bits*) required for collision detection is _____

Q.56 A functional dependency $F: X \rightarrow Y$ is termed as a *useful functional dependency* if and only if it satisfies all the following three conditions:

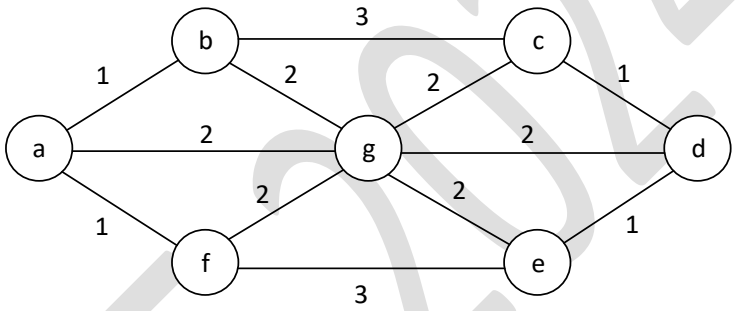
- X is not the empty set.
- Y is not the empty set.
- Intersection of X and Y is the empty set.

For a relation R with 4 attributes, the total number of possible useful functional dependencies is _____

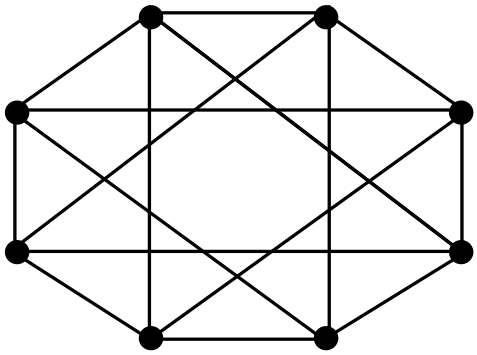
Q.57 A processor with 16 general purpose registers uses a 32-bit instruction format. The instruction format consists of an opcode field, an addressing mode field, two register operand fields, and a 16-bit scalar field. If 8 addressing modes are to be supported, the maximum number of unique opcodes possible for every addressing mode is _____

Q.58 A non-pipelined instruction execution unit operating at 2 GHz takes an average of 6 cycles to execute an instruction of a program P. The unit is then redesigned to operate on a 5-stage pipeline at 2 GHz. Assume that the ideal throughput of the pipelined unit is 1 instruction per cycle. In the execution of program P, 20% instructions incur an average of 2 cycles stall due to data hazards and 20% instructions incur an average of 3 cycles stall due to control hazards. The speedup (*rounded off to one decimal place*) obtained by the pipelined design over the non-pipelined design is _____

Q.59 The number of distinct minimum-weight spanning trees of the following graph is _____



Q.60 The chromatic number of a graph is the minimum number of colours used in a *proper* colouring of the graph. The chromatic number of the following graph is _____



Q.61 A processor uses a 32-bit instruction format and supports byte-addressable memory access. The ISA of the processor has 150 distinct instructions. The instructions are equally divided into two types, namely R-type and I-type, whose formats are shown below.

R-type Instruction Format:

OPCODE	UNUSED	DST Register	SRC Register1	SRC Register 2

I-type Instruction Format:

OPCODE	DST Register	SRC Register	# Immediate value/address

In the OPCODE, 1 bit is used to distinguish between I-type and R-type instructions and the remaining bits indicate the operation. The processor has 50 architectural registers, and all register fields in the instructions are of equal size.

Let X be the number of bits used to encode the UNUSED field, Y be the number of bits used to encode the OPCODE field, and Z be the number of bits used to encode the immediate value/address field. The value of $X + 2Y + Z$ is _____

Q.62 Let L_1 be the language represented by the regular expression $b^*ab^*(ab^*ab^*)^*$ and $L_2 = \{ w \in (a+b)^* \mid |w| \leq 4\}$, where $|w|$ denotes the length of string w. The number of strings in L_2 which are also in L_1 is _____

Q.63 Let Z_n be the group of integers $\{0, 1, 2, \ldots, n-1\}$ with addition modulo n as the group operation. The number of elements in the group $Z_2 \times Z_3 \times Z_4$ that are their own inverses is _____

Q.64 Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume 1 KB = 2^{10} bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X+Y$ is _____

Q.65 Consider the following augmented grammar, which is to be parsed with a SLR parser. The set of terminals is $\{a,b,c,d,\#,@\}$

$$S' \rightarrow S$$
$$S \rightarrow SS \mid Aa \mid bAc \mid Bc \mid bBa$$
$$A \rightarrow d\#$$
$$B \rightarrow @$$

Let $I_0 = CLOSURE(\{S' \rightarrow \bullet S\})$. The number of items in the set $GOTO(I_0,\ S)$ is _____

GATE 2024

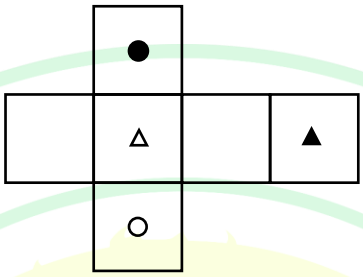
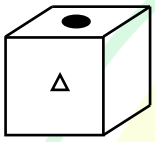
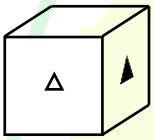
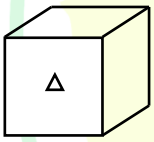
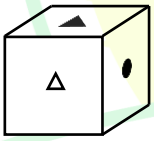
General Aptitude

Q.1 – Q.5 Carry ONE mark Each

Q.1	Despite his initial hesitation, Rehman's _____ to contribute to the success of the project never wavered. Select the most appropriate option to complete the above sentence.
(A)	ambivalence
(B)	satisfaction
(C)	resolve
(D)	revolve

Q.2	Bird : Nest :: Bee : _____ Select the correct option to complete the analogy.
(A)	Kennel
(B)	Hammock
(C)	Hive
(D)	Lair

Q.3	If $Pe^{x} = Qe^{-x}$ for all real values of x, which one of the following statements is true?
(A)	$P = Q = 0$
(B)	$P = Q = 1$
(C)	$P = 1;\ Q = -1$
(D)	$\frac{P}{Q} = 0$

Q.4	The paper as shown in the figure is folded to make a cube where each square corresponds to a particular face of the cube. Which one of the following options correctly represents the cube? Note: The figures shown are representative.
	
(A)	
(B)	
(C)	
(D)	

Q.5	Let p_1 and p_2 denote two arbitrary prime numbers. Which one of the following statements is correct for all values of p_1 and p_2?
(A)	$p_1 + p_2$ is not a prime number.
(B)	$p_1 p_2$ is not a prime number.
(C)	$p_1 + p_2 + 1$ is a prime number.
(D)	$p_1 p_2 + 1$ is a prime number.

Q.6 – Q.10 Carry TWO marks Each

Q.6	Based only on the conversation below, identify the logically correct inference: “Even if I had known that you were in the hospital, I would not have gone there to see you”, Ramya told Josephine.
(A)	Ramya knew that Josephine was in the hospital.
(B)	Ramya did not know that Josephine was in the hospital.
(C)	Ramya and Josephine were once close friends; but now, they are not.
(D)	Josephine was in the hospital due to an injury to her leg.

Q.7	If IMAGE and FIELD are coded as FHBNJ and EMFJG respectively then, which one among the given options is the most appropriate code for BEACH ?
(A)	CEADP
(B)	IDBFC
(C)	JGIBC
(D)	IBCEC

Q.8 Which one of the following options is correct for the given data in the table?

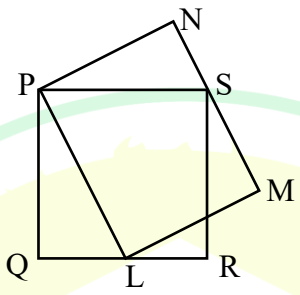
Iteration (i)	0	1	2	3
Input (I)	20	-4	10	15
Output (X)	20	16	26	41
Output (Y)	20	-80	-800	-12000

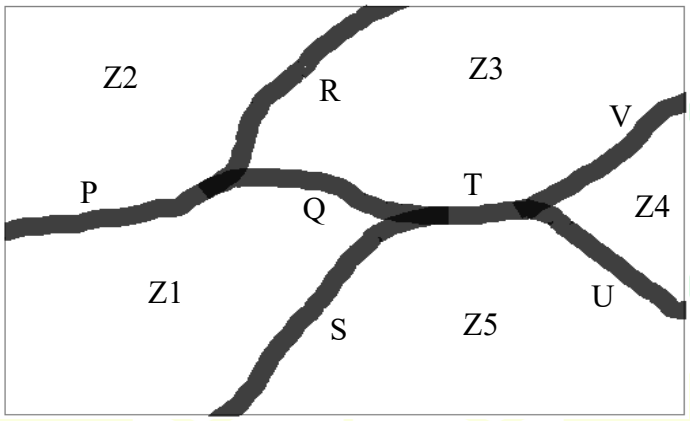
(A) $X(i) = X(i-1) + I(i); \quad Y(i) = Y(i-1)I(i); \quad i > 0$

(B) $X(i) = X(i-1)I(i); \quad Y(i) = Y(i-1) + I(i); \quad i > 0$

(C) $X(i) = X(i-1)I(i); \quad Y(i) = Y(i-1)I(i); \quad i > 0$

(D) $X(i) = X(i-1) + I(i); \quad Y(i) = Y(i-1)I(i-1); \quad i > 0$

Q.9	In the given figure, PQRS is a square of side 2 cm and PLMN is a rectangle. The corner L of the rectangle is on the side QR. Side MN of the rectangle passes through the corner S of the square. What is the area (in cm^2) of the rectangle PLMN? Note: The figure shown is representative.
	
(A)	$2\sqrt{2}$
(B)	2
(C)	8
(D)	4

Q.10	The diagram below shows a river system consisting of 7 segments, marked P, Q, R, S, T, U, and V. It splits the land into 5 zones, marked Z1, Z2, Z3, Z4, and Z5. We need to connect these zones using the least number of bridges. Out of the following options, which one is correct? Note: The figure shown is representative.
	
(A)	Bridges on P, Q, and T
(B)	Bridges on P, Q, S, and T
(C)	Bridges on Q, R, T, and V
(D)	Bridges on P, Q, S, U, and V

Q.11 – Q.35 Carry ONE mark Each

Q.11	If $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$, then which ONE of the following is A^8 ?
(A)	$\begin{pmatrix} 25 & 0 \\ 0 & 25 \end{pmatrix}$
(B)	$\begin{pmatrix} 125 & 0 \\ 0 & 125 \end{pmatrix}$
(C)	$\begin{pmatrix} 625 & 0 \\ 0 & 625 \end{pmatrix}$
(D)	$\begin{pmatrix} 3125 & 0 \\ 0 & 3125 \end{pmatrix}$
Q.12	The value of x such that $x > 1$, satisfying the equation $\int_1^x t \ln t \, dt = \frac{1}{4}$ is
(A)	$\sqrt{e}$
(B)	e
(C)	e^2
(D)	$e - 1$

Q.13	Consider a binary tree T in which every node has either zero or two children. Let $n > 0$ be the number of nodes in T. Which ONE of the following is the number of nodes in T that have exactly two children?
(A)	$\frac{n-2}{2}$
(B)	$\frac{n-1}{2}$
(C)	$\frac{n}{2}$
(D)	$\frac{n+1}{2}$

Q.14	Let $L, M,$ and N be non-singular matrices of order 3 satisfying the equations $L^2 = L^{-1}$, $M = L^8$ and $N = L^2$. Which ONE of the following is the value of the determinant of $(M - N)$?
(A)	0
(B)	1
(C)	2
(D)	3
Q.15	Let $P(x)$ be an arbitrary predicate over the domain of natural numbers. Which ONE of the following statements is TRUE?
(A)	$\big(P(0) \wedge (\forall x\,[P(x) \Rightarrow P(x+1)])\big) \Rightarrow \big(\forall x\,P(x)\big)$
(B)	$\big(P(0) \wedge (\forall x\,[P(x) \Rightarrow P(x-1)])\big) \Rightarrow \big(\forall x\,P(x)\big)$
(C)	$\big(P(1000) \wedge (\forall x\,[P(x) \Rightarrow P(x-1)])\big) \Rightarrow \big(\forall x\,P(x)\big)$
(D)	$\big(P(1000) \wedge (\forall x\,[P(x) \Rightarrow P(x+1)])\big) \Rightarrow \big(\forall x\,P(x)\big)$

Q.16	Consider the following statements: (i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address. (ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S. Which ONE of the following is CORRECT?
(A)	Both (i) and (ii) are TRUE
(B)	(i) is TRUE and (ii) is FALSE
(C)	(i) is FALSE and (ii) is TRUE
(D)	Both (i) and (ii) are FALSE

<table>
<tr><td>Q.17</td><td>Consider the routing protocols given in List I and the names given in List II:
<table><tr><th colspan="2">List I</th><th colspan="2">List II</th></tr><tr><td>(i)</td><td>Distance vector routing</td><td>(a)</td><td>Bellman-Ford</td></tr><tr><td>(ii)</td><td>Link state routing</td><td>(b)</td><td>Dijkstra</td></tr></table>For matching of items in List I with those in List II, which ONE of the following options is CORRECT?</td></tr>
<tr><td></td><td></td></tr>
<tr><td>(A)</td><td>(i) – (a) and (ii) – (b)</td></tr>
<tr><td>(B)</td><td>(i) – (a) and (ii) – (a)</td></tr>
<tr><td>(C)</td><td>(i) – (b) and (ii) – (a)</td></tr>
<tr><td>(D)</td><td>(i) – (b) and (ii) – (b)</td></tr>
<tr><td></td><td></td></tr>
</table>

Q.18	A machine receives an IPv4 datagram. The protocol field of the IPv4 header has the protocol number of a protocol X. Which ONE of the following is NOT a possible candidate for X?
(A)	Internet Control Message Protocol (ICMP)
(B)	Internet Group Management Protocol (IGMP)
(C)	Open Shortest Path First (OSPF)
(D)	Routing Information Protocol (RIP)

Q.19 Consider the following C program:

```
#include <stdio.h>
void stringcopy(char *, char *);
int main(){
      char a[30] = "@#Hello World!";
      stringcopy(a, a + 2);
      printf("%s\n", a);
      return 0;
}
void stringcopy(char *s, char *t) {
      while(*t)
            *s++ = *t++;
}
```

Which ONE of the following will be the output of the program?

(A) `@#Hello World!`

(B) `Hello World!`

(C) `ello World!`

(D) `Hello World!d!`

Q.20	Consider an unordered list of N distinct integers. What is the minimum number of element comparisons required to find an integer in the list that is NOT the largest in the list?
(A)	1
(B)	$N-1$
(C)	N
(D)	$2N-1$
Q.21	Consider the following statements about the use of backpatching in a compiler for intermediate code generation: (I) Backpatching can be used to generate code for Boolean expression in one pass. (II) Backpatching can be used to generate code for flow-of-control statements in one pass. Which ONE of the following options is CORRECT?
(A)	Only (I) is correct.
(B)	Only (II) is correct.
(C)	Both (I) and (II) are correct.
(D)	Neither (I) nor (II) is correct.

Q.22	Given the following syntax directed translation rules: Rule 1: $R \rightarrow AB \{B.i = R.i - 1; A.i = B.i; R.i = A.i + 1;\}$ Rule 2: $P \rightarrow CD \{P.i = C.i + D.i; D.i = C.i + 2;\}$ Rule 3: $Q \rightarrow EF \{Q.i = E.i + F.i;\}$ Which ONE is the CORRECT option among the following?
(A)	Rule 1 is S-attributed and L-attributed; Rule 2 is S-attributed and not L-attributed; Rule 3 is neither S-attributed nor L-attributed
(B)	Rule 1 is neither S-attributed nor L-attributed; Rule 2 is S-attributed and L-attributed; Rule 3 is S-attributed and L-attributed
(C)	Rule 1 is neither S-attributed nor L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed
(D)	Rule 1 is S-attributed and not L-attributed; Rule 2 is not S-attributed and is L-attributed; Rule 3 is S-attributed and L-attributed

Q.23	Consider a network that uses Ethernet and IPv4. Assume that IPv4 headers do not use any options field. Each Ethernet frame can carry a maximum of 1500 bytes in its data field. A UDP segment is transmitted. The payload (data) in the UDP segment is 7488 bytes. Which ONE of the following choices has the CORRECT total number of fragments transmitted and the size of the last fragment including IPv4 header?
(A)	5 fragments, 1488 bytes
(B)	6 fragments, 88 bytes
(C)	6 fragments, 108 bytes
(D)	6 fragments, 116 bytes
Q.24	Which ONE of the following languages is accepted by a deterministic pushdown automaton?
(A)	Any regular language.
(B)	Any context-free language.
(C)	Any language accepted by a non-deterministic pushdown automaton.
(D)	Any decidable language.

Q.25	Let G_1, G_2 be Context Free Grammars (CFGs) and R be a regular expression. For a grammar G, let $L(G)$ denote the language generated by G. Which ONE among the following questions is decidable?
(A)	Is $L(G_1) = L(G_2)$?
(B)	Is $L(G_1) \cap L(G_2) = \emptyset$?
(C)	Is $L(G_1) = L(R)$?
(D)	Is $L(G_1) = \emptyset$?
Q.26	Processes $P1$, $P2$, $P3$, $P4$ arrive in that order at times 0, 1, 2, and 8 milliseconds respectively, and have execution times of 10, 13, 6, and 9 milliseconds respectively. Shortest Remaining Time First (SRTF) algorithm is used as the CPU scheduling policy. Ignore context switching times. Which ONE of the following correctly gives the average turnaround time of the four processes in milliseconds?
(A)	22
(B)	15
(C)	37
(D)	19

Q.27	An audit of a banking transactions system has found that on an earlier occasion, two joint holders of account A attempted simultaneous transfers of Rs. 10000 each from account A to account B. Both transactions read the same value, Rs. 11000, as the initial balance in A and were allowed to go through. B was credited Rs. 10000 twice. A was debited only once and ended up with a balance of Rs. 1000. Which of the following properties is/are certain to have been violated by the system?
(A)	Atomicity
(B)	Consistency
(C)	Isolation
(D)	Durability
Q.28	Which of the following is/are part of an Instruction Set Architecture of a processor?
(A)	The size of the cache memory
(B)	The clock frequency of the processor
(C)	The number of cache memory levels
(D)	The total number of registers

Q.29	Which of the following statements regarding Breadth First Search (BFS) and Depth First Search (DFS) on an undirected simple graph G is/are TRUE?
(A)	A DFS tree of G is a Shortest Path tree of G.
(B)	Every non-tree edge of G with respect to a DFS tree is a forward/back edge.
(C)	If (u, v) is a non-tree edge of G with respect to a BFS tree, then the distances from the source vertex s to u and v in the BFS tree are within ± 1 of each other.
(D)	Both BFS and DFS can be used to find the connected components of G.

Q.30 Consider the two lists **List I** and **List II** given below:

List I	List II
(i) Context free languages	(a) Closed under union
(ii) Recursive languages	(b) Not closed under complementation
(iii) Regular languages	(c) Closed under intersection

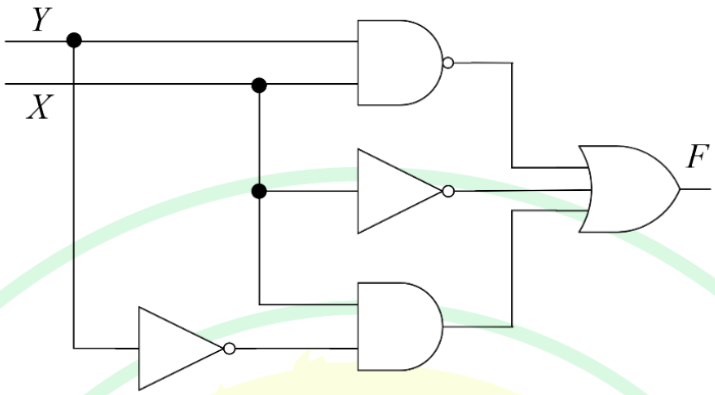
For matching of items in **List I** with those in **List II,** which of the following option(s) is/are CORRECT?

(A) (i) – (a), (ii) – (b), and (iii) – (c)

(B) (i) – (b), (ii) – (a), and (iii) – (c)

(C) (i) – (b), (ii) – (c), and (iii) – (a)

(D) (i) – (a), (ii) – (c), and (iii) – (b)

Q.31	Consider the following logic circuit diagram.  Which is/are the CORRECT option(s) for the output function F?
(A)	$\overline{X\,Y}$
(B)	$\bar{X} + \bar{Y} + X\bar{Y}$
(C)	$\overline{X\,Y} + \bar{X} + X\bar{Y}$
(D)	$X + \bar{Y}$
Q.32	The following two signed 2's complement numbers (multiplicand M and multiplier Q) are being multiplied using Booth's algorithm: M: 1100 1101 1110 1101 and Q: 1010 0100 1010 1010 The total number of addition and subtraction operations to be performed is _____. (Answer in integer)

Q.33	```
	int x=126,y=105;
	do {
	if(x>y) x=x-y;
	else y=y-x;
	} while(x!=y);
	
	printf("%d",x);
	``
	The output of the given C code segment is _____. (Answer in integer)

Q.34	In a 4-bit ripple counter, if the period of the waveform at the last flip-flop is 64 microseconds, then the frequency of the ripple counter in kHz is _____. (Answer in integer)
Q.35	Suppose the values $10, -4, 15, 30, 20, 5, 60, 19$ are inserted in that order into an initially empty binary search tree. Let T be the resulting binary search tree. The number of edges in the path from the node containing 19 to the root node of T is _____. (Answer in integer)

Q.36 – Q.65 Carry TWO marks Each

Q.36	Suppose we are transmitting frames between two nodes using Stop-and-Wait protocol. The frame size is 3000 bits. The transmission rate of the channel is 2000 bps (bits/second) and the propagation delay between the two nodes is 100 milliseconds. Assume that the processing times at the source and destination are negligible. Also, assume that the size of the acknowledgement packet is negligible. Which ONE of the following most accurately gives the channel utilization for the above scenario in percentage?
(A)	88.23
(B)	93.75
(C)	85.44
(D)	66.67

Q.37	Let G be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant α is added to the weight of every edge. Which ONE of the following statements is TRUE about the minimum spanning trees (MSTs) and shortest paths (SPs) in G before and after the edge weight update?
(A)	Every MST remains an MST, and every SP remains an SP.
(B)	MSTs need not remain MSTs, and every SP remains an SP.
(C)	Every MST remains an MST, and SPs need not remain SPs.
(D)	MSTs need not remain MSTs, and SPs need not remain SPs.

Q.38	A meld operation on two instances of a data structure combines them into one single instance of the same data structure. Consider the following data structures: P: Unsorted doubly linked list with pointers to the head node and tail node of the list. Q: Min-heap implemented using an array. R: Binary Search Tree. Which ONE of the following options gives the worst-case time complexities for meld operation on instances of size n of these data structures?
(A)	P: $\Theta(1)$, Q: $\Theta(n)$, R: $\Theta(n)$
(B)	P: $\Theta(1)$, Q: $\Theta(n \log n)$, R: $\Theta(n)$
(C)	P: $\Theta(n)$, Q: $\Theta(n \log n)$, R: $\Theta(n^2)$
(D)	P: $\Theta(1)$, Q: $\Theta(n)$, R: $\Theta(n \log n)$

Q.39	For a direct-mapped cache, 4 bits are used for the tag field and 12 bits are used to index into a cache block. The size of each cache block is one byte. Assume that there is no other information stored for each cache block. Which ONE of the following is the CORRECT option for the sizes of the main memory and the cache memory in this system (byte addressable), respectively?
(A)	64 KB and 4 KB
(B)	128 KB and 16 KB
(C)	64 KB and 8 KB
(D)	128 KB and 6 KB
Q.40	Given a Context-Free Grammar G as follows: $S \rightarrow Aa \mid bAc \mid dc \mid bda$ $A \rightarrow d$ Which ONE of the following statements is TRUE?
(A)	G is neither LALR(1) nor SLR(1)
(B)	G is CLR(1), not LALR(1)
(C)	G is LALR(1), not SLR(1)
(D)	G is LALR(1), also SLR(1)

Q.41	An array A of length n with distinct elements is said to be bitonic if there is an index $1 \leq i \leq n$ such that $A[1..i]$ is sorted in the non-decreasing order and $A[i+1..n]$ is sorted in the non-increasing order. Which ONE of the following represents the best possible asymptotic bound for the worst-case number of comparisons by an algorithm that searches for an element in a bitonic array A?
(A)	$\Theta(n)$
(B)	$\Theta(1)$
(C)	$\Theta(\log^2 n)$
(D)	$\Theta(\log n)$
Q.42	Let $\mathcal{F}$ be the set of all functions from $\{1, \ldots, n\}$ to $\{0,1\}$. Define the binary relation $\preccurlyeq$ on $\mathcal{F}$ as follows: $\forall f, g \in \mathcal{F}$, $f \preccurlyeq g$ if and only if $\forall x \in \{1, \ldots, n\}$, $f(x) \leq g(x)$, where $0 \leq 1$. Which of the following statement(s) is/are TRUE?
(A)	$\preccurlyeq$ is a symmetric relation
(B)	$(\mathcal{F}, \preccurlyeq)$ is a partial order
(C)	$(\mathcal{F}, \preccurlyeq)$ is a lattice
(D)	$\preccurlyeq$ is an equivalence relation

Q.43 Given the following Karnaugh Map for a Boolean function $F(w, x, y, z)$:

wx \ yz				
	1	0	0	1
	0	1	1	0
	0	1	1	0
	1	0	0	1

Which one or more of the following Boolean expression(s) represent(s) F?

(A) $\bar{w}\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}y\bar{z} + xz$

(B) $\bar{w}\bar{x}\bar{y}\bar{z} + \bar{w}\bar{x}y\bar{z} + w\bar{x}yz + xz$

(C) $\bar{w}\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}\bar{z} + w\bar{x}\bar{y}z + xz$

(D) $\bar{x}\bar{z} + xz$

Q.44	Consider a system of linear equations $PX = Q$ where $P \in \mathbb{R}^{3\times3}$ and Q $\in \mathbb{R}^{3\times1}$. Suppose P has an LU decomposition, $P = LU$, where $L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix}$ and $U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$. Which of the following statement(s) is/are TRUE?
(A)	The system $PX = Q$ can be solved by first solving $LY = Q$ and then $UX = Y$.
(B)	If P is invertible, then both L and U are invertible.
(C)	If P is singular, then at least one of the diagonal elements of U is zero.
(D)	If P is symmetric, then both L and U are symmetric.

Q.45	Consider a stack data structure into which we can PUSH and POP records. Assume that each record pushed in the stack has a positive integer key and that all keys are distinct. We wish to augment the stack data structure with an $O(1)$ time MIN operation that returns a pointer to the record with smallest key present in the stack 1) without deleting the corresponding record, and 2) without increasing the complexities of the standard stack operations. Which one or more of the following approach(es) can achieve it?
(A)	Keep with every record in the stack, a pointer to the record with the smallest key below it.
(B)	Keep a pointer to the record with the smallest key in the stack.
(C)	Keep an auxiliary array in which the key values of the records in the stack are maintained in sorted order.
(D)	Keep a Min-Heap in which the key values of the records in the stack are maintained.

Q.46	Consider the following relational schema along with all the functional dependencies that hold on them. *R1(A, B, C, D, E):* $\{D \rightarrow E,\ EA \rightarrow B,\ EB \rightarrow C\}$ *R2(A, B, C, D):* $\{A \rightarrow D,\ A \rightarrow B,\ C \rightarrow A\}$ Which of the following statement(s) is/are TRUE?
(A)	*R1* is in 3NF
(B)	*R2* is in 3NF
(C)	*R1* is NOT in 3NF
(D)	*R2* is NOT in 3NF

Q.47

Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. The contents of the frames are as follows initially and after each reference (from left to right):

initially	*after*												
-	1*	2*	3*	4*	5*	4	1	6*	4	5	1*	3*	2*
-	1	1	1	1	1	1	1	6	6	6	6	6	2
-	-	2	2	4	4	4	4	4	4	4	1	1	1
-	-	-	3	3	5	5	5	5	5	5	5	3	3

The *-marked references cause page replacements.

Which one or more of the following could be the page replacement policy/policies in use?

(A) Least Recently Used page replacement policy

(B) Least Frequently Used page replacement policy

(C) Most Frequently Used page replacement policy

(D) Optimal page replacement policy

Q.48	$P = \{P_1, P_2, P_3, P_4\}$ consists of all active processes in an operating system. $R = \{R_1, R_2, R_3, R_4\}$ consists of single instances of distinct types of resources in the system. The resource allocation graph has the following assignment and claim edges. Assignment edges: $R_1 \rightarrow P_1$, $R_2 \rightarrow P_2$, $R_3 \rightarrow P_3$, $R_4 \rightarrow P_4$ (the assignment edge $R_1 \rightarrow P_1$ means resource R_1 is assigned to process P_1 , and so on for others) Claim edges: $P_1 \rightarrow R_2$, $P_2 \rightarrow R_3$, $P_3 \rightarrow R_1$, $P_2 \rightarrow R_4$, $P_4 \rightarrow R_2$ (the claim edge $P_1 \rightarrow R_2$ means process P_1 is waiting for resource R_2 , and so on for others) Which of the following statement(s) is/are CORRECT?
(A)	Aborting P_1 makes the system deadlock free.
(B)	Aborting P_3 makes the system deadlock free.
(C)	Aborting P_2 makes the system deadlock free.
(D)	Aborting P_1 and P_4 makes the system deadlock free.

Q.49	Three floating point numbers X, Y, and Z are stored in three registers R_X, R_Y, and R_Z, respectively in IEEE 754 single precision format as given below in hexadecimal: R_X = 0xC1100000, R_Y = 0x40C00000, and R_Z = 0x41400000 Which of the following option(s) is/are CORRECT?
(A)	$4(X + Y) + Z = 0$
(B)	$2Y - Z = 0$
(C)	$4X + 3Z = 0$
(D)	$X + Y + Z = 0$
Q.50	Which of the following Boolean algebraic equation(s) is/are CORRECT?
(A)	$\bar{A}\,B\,C + A\,\bar{B}\,\bar{C} + \bar{A}\,\bar{B}\,\bar{C} + A\,\bar{B}\,C + A\,B\,C = B\,C + \bar{B}\,\bar{C} + \bar{A}\,\bar{B}$
(B)	$A\,B + \bar{A}\,C + B\,C = A\,B + \bar{A}\,C$
(C)	$(A + C)\,(\bar{A} + B) = A\,B + \bar{A}\,C$
(D)	$\overline{(A + \bar{B} + \bar{D})\,(C + D)\,(\bar{A} + C + D)\,(A + B + \bar{D})} = \bar{A}\,D + \bar{C}\,\bar{D}$

Q.51	Consider two grammars G_1 and G_2 with the production rules given below: $G_1: S \rightarrow if\ \ E\ then\ \ S \mid if\ E\ then\ S\ else\ S \mid a$ $E \rightarrow b$ $G_2: S \rightarrow if\ E\ then\ S \mid M$ $M \rightarrow if\ E\ then\ M\ else\ S \mid c$ $E \rightarrow b$ where $if, then, else, a, b, c$ are the terminals. Which of the following option(s) is/are CORRECT?
(A)	G_1 is not $LL(1)$ and G_2 is $LL(1)$.
(B)	G_1 is $LL(1)$ and G_2 is not $LL(1)$.
(C)	G_1 and G_2 are not $LL(1)$.
(D)	G_1 and G_2 are ambiguous.

Q.52	Let $\Sigma = \{a, b, c\}$. For $x \in \Sigma^*$, and $\alpha \in \Sigma$, let $\#_\alpha(x)$ denote the number of occurrences of α in x. Which one or more of the following option(s) define(s) regular language(s)?
(A)	$\{a^m b^n \mid m, n \geq 0\}$
(B)	$\{a, b\}^* \cap \{a^m b^n c^{m-n} \mid m \geq n \geq 0\}$
(C)	$\{w \mid w \in \{a, b\}^*, \#_a(w) \equiv 2 \pmod 7, \text{and } \#_b(w) \equiv 3 \pmod 9\}$
(D)	$\{w \mid w \in \{a, b\}^*, \#_a(w) \equiv 2 \pmod 7, \text{and } \#_a(w) = \#_b(w)\}$

Q.53	Consider the database transactions T1 and T2, and data items X and Y. Which of the schedule(s) is/are conflict serializable?
	Transaction T1 R1(X) W1(Y) R1(X) W1(X) COMMIT(T1) **Transaction T2** W2(X) W2(Y) COMMIT(T2)
(A)	R1(X), W2(X), W1(Y), W2(Y), R1(X), W1(X), COMMIT(T2), COMMIT(T1)
(B)	W2(X), R1(X), W2(Y), W1(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)
(C)	R1(X), W1(Y), W2(X), W2(Y), R1(X), W1(X), COMMIT(T1), COMMIT(T2)
(D)	W2(X), R1(X), W1(Y), W2(Y), R1(X), COMMIT(T2), W1(X), COMMIT(T1)

Q.54 Consider the following relational schema:

Students (rollno: integer, name: string, age: integer, cgpa: real)

Courses (courseno: integer, cname: string, credits: integer)

Enrolled (rollno: integer, courseno: integer, grade: string)

Which of the following options is/are correct SQL query/queries to retrieve the names of the students enrolled in course number (i.e., courseno) 1470?

(A)

```
SELECT S.name
FROM Students S
WHERE EXISTS (SELECT * FROM Enrolled E
              WHERE E.courseno = 1470
                    AND E.rollno = S.rollno);
```

(B)

```
SELECT S.name
FROM Students S
WHERE SIZEOF (SELECT * FROM Enrolled E
              WHERE E.courseno = 1470
                    AND E.rollno = S.rollno) > 0;
```

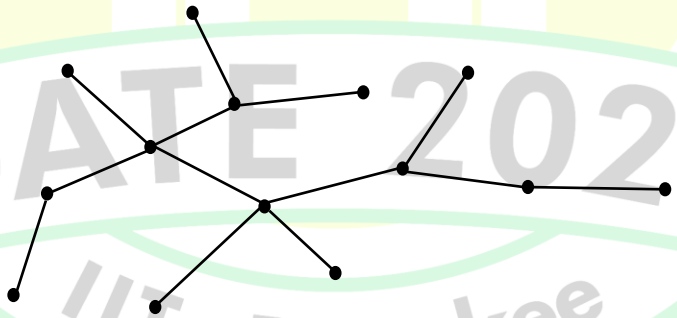
(C)

```
SELECT S.name
FROM Students S
WHERE  0 < (SELECT COUNT(*)
            FROM Enrolled E
            WHERE E.courseno = 1470
                  AND E.rollno = S.rollno);
```

(D)

```
SELECT S.name
FROM Students S NATURAL JOIN Enrolled E
WHERE E.courseno = 1470;
```

Q.55	Given a computing system with two levels of cache (L1 and L2) and a main memory. The first level (L1) cache access time is 1 nanosecond (ns) and the "hit rate" for L1 cache is 90% while the processor is accessing the data from L1 cache. Whereas, for the second level (L2) cache, the "hit rate" is 80% and the "miss penalty" for transferring data from L2 cache to L1 cache is 10 ns. The "miss penalty" for the data to be transferred from main memory to L2 cache is 100 ns. Then the average memory access time in this system in nanoseconds is _____ . (rounded off to one decimal place)
Q.56	A 5-stage instruction pipeline has stage delays of 180, 250, 150, 170, and 250, respectively, in nanoseconds. The delay of an inter-stage latch is 10 nanoseconds. Assume that there are no pipeline stalls due to branches and other hazards. The time taken to process 1000 instructions in microseconds is _____ . (rounded off to two decimal places)
Q.57	In a B^+- tree where each node can hold at most four key values, a root to leaf path consists of the following nodes: A = (49, 77, 83, -), B = (7, 19, 33, 44), C = (20*, 22*, 25*, 26*) The *-marked keys signify that these are data entries in a leaf. Assume that a pointer between keys k_1 and k_2 points to a subtree containing keys in $[k_1, k_2)$, and that when a leaf is created, the smallest key in it is copied up into its parent. A record with key value 23 is inserted into the B^+- tree. The smallest key value in the parent of the leaf that contains 25* is _____. (Answer in integer)

Q.58	A computer system supports a logical address space of 2^{32} bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b-bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size. The value of b is _____ . (Answer in integer)
Q.59	Consider the following algorithm `someAlgo` that takes an undirected graph G as input. `someAlgo(`G`)` 1. Let v be any vertex in G. Run BFS on G starting at v. Let u be a vertex in G at maximum distance from v as given by the BFS. 2. Run BFS on G again with u as the starting vertex. Let z be the vertex at maximum distance from u as given by the BFS. 3. Output the distance between u and z in G. The output of `someAlgo(`T`)` for the tree shown in the given figure is _____. (Answer in integer)
	

Q.60

Let $\Sigma = \{1,2,3,4\}$. For $x \in \Sigma^*$, let $prod(x)$ be the product of symbols in x modulo 7. We take $prod(\epsilon) = 1$, where ϵ is the null string.

For example, $prod(124) = (1 \times 2 \times 4) \bmod 7 = 1$.

Define $L = \{x \in \Sigma^* \mid prod(x) = 2\}$.

The number of states in a minimum state DFA for L is _____. (Answer in integer)

Q.61

An application executes 6.4×10^8 number of instructions in 6.3 seconds. There are four types of instructions, the details of which are given in the table. The duration of a clock cycle in nanoseconds is _____. (rounded off to one decimal place)

Instruction type	Clock cycles required per instruction (CPI)	Number of instructions executed
Branch	2	2.25×10^8
Load	5	1.20×10^8
Store	4	1.65×10^8
Arithmetic	3	1.30×10^8

Q.62

Consider the following C program:

```
#include <stdio.h>
int main(){
      int a;
      int arr[5] = {30,50,10};
      int *ptr;
      ptr = &arr[0] + 1;
      a = *ptr;
      (*ptr)++;
      ptr++;
      printf("%d", a + (*ptr) + arr[1]);
      return 0;
}
```

The output of the above program is _____. (Answer in integer)

Q.63

Consider the following C program:

```
#include <stdio.h>

int g(int n) {
    return (n+10);
}

int f(int n) {
    return g(n*2);
}

int main() {
    int sum, n;
    sum=0;
    for (n=1; n<3; n++)
        sum += g(f(n));
    printf ("%d", sum);
    return 0;

}
```

The output of the given C program is _____. (Answer in integer)

Q.64

A quadratic polynomial $(x-\alpha)(x-\beta)$ over complex numbers is said to be square invariant if $(x-\alpha)(x-\beta) = (x-\alpha^2)(x-\beta^2)$. Suppose from the set of all square invariant quadratic polynomials we choose one at random.

The probability that the roots of the chosen polynomial are equal is _____. (rounded off to one decimal place)

Q.65	The unit interval $(0,1)$ is divided at a point chosen uniformly distributed over $(0,1)$ in $\mathbb{R}$ into two disjoint subintervals. The expected length of the subinterval that contains 0.4 is _____. (rounded off to two decimal places)

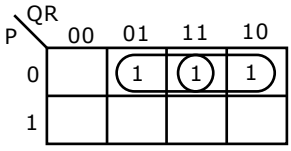
Q. No. 1 – 25 Carry One Mark Each

1. The simplified SOP (Sum of Product) form of the Boolean expression $(P+\overline{Q}+\overline{R}).(P+\overline{Q}+R).(P+Q+\overline{R})$ is

(A) $(\overline{P}Q+\overline{R})$ (B) $(P+\overline{Q}\overline{R})$ (C) $(\overline{P}Q+R)$ (D) $(PQ+R)$

Answer: - (B)

Exp: -



$f=(P+\overline{R})(P+\overline{Q})$

$=P+\overline{Q}\overline{R}$

Alternate method

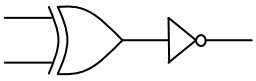
$(P+\overline{Q}+\overline{R}).(P+\overline{Q}+R).(P+Q+\overline{R})=\overline{\overline{(P+\overline{Q}+\overline{R}).(P+\overline{Q}+R).(P+Q+\overline{R})}}$

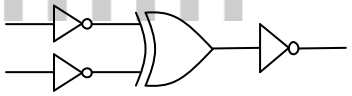
$=\overline{\overline{P}QR+\overline{P}Q\overline{R}+\overline{P}\overline{Q}R}=\overline{\overline{P}Q(R+\overline{R})+\overline{P}\overline{Q}R}=\overline{\overline{P}Q+\overline{P}\overline{Q}R}=\overline{\overline{P}(Q+\overline{Q}R)}$

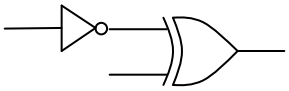
$=\overline{\overline{P}(Q+R)}=P+\overline{Q}\,\overline{R}$

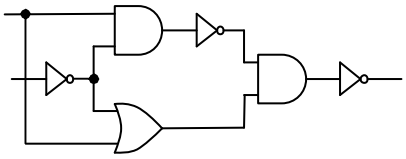
GATEFORUM Engineering Success

2. Which one of the following circuits is NOT equivalent to a 2-input XNOR (exclusive NOR) gate?

(A) 

(B) 

(C) 

(D) 

Answer: - (D)

Exp: - All options except option 'D' gives EX-NOR gates

3. The minimum number of D flip-flops needed to design a mod-258 counter is

(A) 9 (B) 8 (C) 512 (D) 258

Answer: - (A)

Exp: - $2^n \geq 258 \Rightarrow n=9$

4. A thread is usually defined as a 'light weight process' because an operating system (OS) maintains smaller data structures for a thread than for a process. In relation to this, which of the followings is TRUE?

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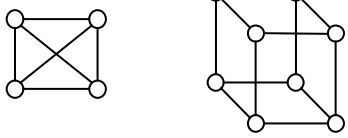
(A) On per-thread basis, the OS maintains only CPU register state
(B) The OS does not maintain a separate stack for each thread
(C) On per-thread basis, the OS does not maintain virtual memory state
(D) On per thread basis, the OS maintains only scheduling and accounting information

Answer: - (A)

5. K4 and Q3 are graphs with the following structures

K4 Q3

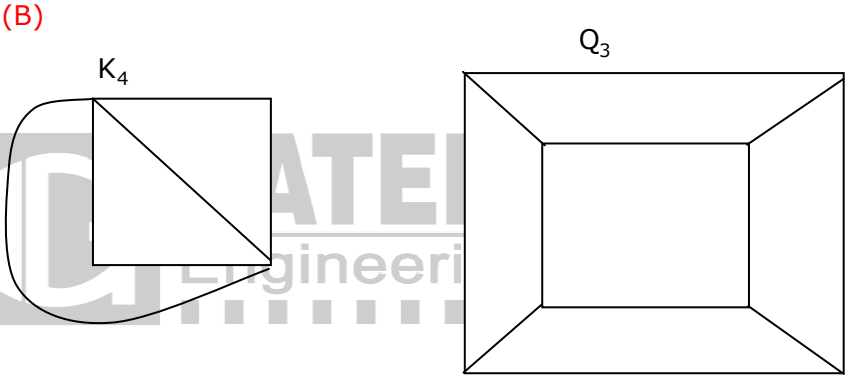


Which one of the following statements is TRUE in relation to these graphs?

(A) K4 is planar while Q3 is not
(B) Both K4 and Q3 are planar
(C) Q3 is planar while K4 is not
(D) Neither K4 not Q3 is planar

Answer: - (B)

Exp: -



$\therefore$ Both K_4 and Q_3 are planar

6. If the difference between the expectation of the square of random variable $\left(E\left[X^2\right]\right)$ and the square of the expectation of the random variable $\left(E\left[X^2\right]\right)$ is denoted by R then

(A) $R = 0$ (B) $R<0$ (C) $R \geq 0$ (D) $R>0$

Answer: - (C)

7. The lexical analysis for a modern computer language such as Java needs the power of which one of the following machine models in a necessary and sufficient sense?

(A) Finite state automata

(B) Deterministic pushdown automata

(C) Non-Deterministic pushdown automata

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(D) Turing machine

Answer: - (A)

Exp: - Lexical Analysis is implemented by finite automata

8. Let the page fault service time be 10ms in a computer with average memory access time being 20ns. If one page fault is generated for every 10^6 memory accesses, what is the effective access time for the memory?

(A) 21ns (B) 30ns (C) 23ns (D) 35ns

Answer: - (B)

Exp: - P = page fault rate

EA = p × page fault service time

$+(1-p)\times$ Memory access time

$$=\frac{1}{10^6}\times 10\times 10^6+\left(1-\frac{1}{10^6}\right)\times 20\cong 29.9\,\text{ns}$$

9. Consider a hypothetical processor with an instruction of type LW R1, 20(R2), which during execution reads a 32-bit word from memory and stores it in a 32-bit register R1. The effective address of the memory location is obtained by the addition of constant 20 and the contents of register R2. Which of the following best reflects the addressing mode implemented by this instruction for the operand in memory?

(A) Immediate Addressing (B) Register Addressing

(C) Register Indirect Scaled Addressing (D) Base Indexed Addressing

Answer: - (D)

Exp: - Here 20 will act as base and content of R_2 will be index

10. What does the following fragment of C-program print?

```
char c[ ]= "GATE2011";
char *p =c;
printf("%s", p+p[3]-p[1]);
```

(A) GATE2011 (B) E2011 (C) 2011 (D) 011

Answer: - (C)

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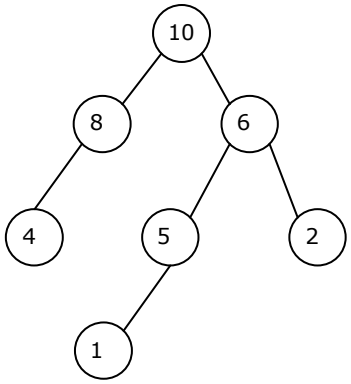
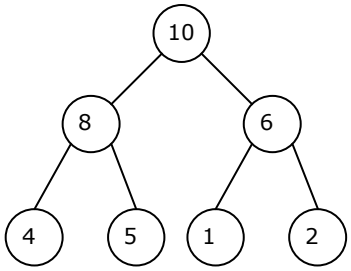
11. A max-heap is a heap where the value of each parent is greater than or equal to the value of its children. Which of the following is a max-heap?

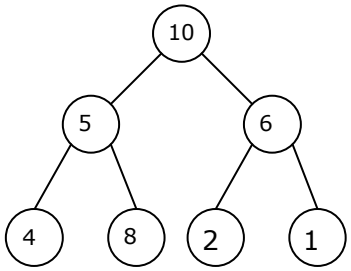
(A)



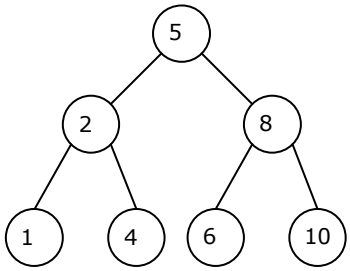
(B)



(C)



(D)



Answer: - (B)

Exp: - Heap is a complete binary tree

12. An algorithm to find the length of the longest monotonically increasing sequence of numbers in an array $A[0:n-1]$ is given below.

Let L_i denote the length of the longest monotonically increasing sequence starting at index i in the array

Initialize $L_{n-1} = 1$

For all i such that $0 \le i \le n-2$

$$L_i = \begin{cases} 1 + L_{i+1} & \text{if } A[i] < A[i+1] \\ 1 & \text{Otherwise} \end{cases}$$

Finally the length of the longest monotonically increasing sequence is $\text{Max}(L_0, L_1, \ldots, L_{n-1})$. Which of the following statements is TRUE?

(A) The algorithm uses dynamic programming paradigm

(B) The algorithm has a linear complexity and uses branch and bound paradigm

(C) The algorithm has a non-linear polynomial complexity and uses branch and bound paradigm

(D) The algorithm uses divide and conquer paradigm.

Answer: - (A)

13. Let P be a regular language and Q be a context free language such that $Q \subseteq P$. (For example, let P be the language represented by the regular expression p*q* and Q be $\{p^n q^n \mid n \in N\}$). Then which of the following is ALWAYS regular?

(A) $P \cap Q$ (B) $P - Q$ (C) $\Sigma^* - P$ (D) $\Sigma^* - Q$

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Answer: - (C)

Exp: - $\Sigma^* - P$ is the complement of P so it is always regular,

since regular languages are closed under complementation

14. In a compiler, keywords of a language are recognized during
(A) parsing of the program (B) the code generation
(C) the lexical analysis of the program (D) dataflow analysis

Answer: - (C)

Exp: - Any identifier is also a token so it is recognized in lexical Analysis

15. A layer-4 firewall (a device that can look at all protocol headers up to the transport layer) CANNOT
(A) block entire HTTP traffic during 9:00PM and 5:00AM
(B) block all ICMP traffic
(C) stop incoming traffic from a specific IP address but allow outgoing traffic to the same IP address
(D) block TCP traffic from a specific user on a multi-user system during 9:00PM and 5:00AM

Answer: - (A)

Exp: - Since it is a layer 4 firewall it cannot block application layer protocol like HTTP.

16. If two fair coins are flipped and at least one of the outcomes is known to be a head, what is the probability that both outcomes are heads?
(A) 1/3 (B) 1/4 (C) 1/2 (D) 2/3

Answer: - (A)

Exp: - Sample space = $\{HH, HT, TH\}$

Required probability $= \frac{1}{3}$

17. Consider different activities related to email.
m1: Send an email from a mail client to a mail server
m2: Download an email from mailbox server to a mail client
m3: Checking email in a web browser
Which is the application level protocol used in each activity?
(A) m1:HTTP m2:SMTP m3:POP (B) m1:SMTP m2:FTP m3:HTTP
(C) m1: SMTP m2: POP m3: HTTP (D) m1: POP m2: SMTP m3:IMAP

Answer: - (C)

Exp: - Sending an email will be done through user agent and message transfer agent by SMTP, downloading an email from mail box is done through POP, checking email in a web browser is done through HTTP

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18. A company needs to develop a strategy for software product development for which it has a choice of two programming languages L1 and L2. The number of lines of code (LOC) developed using L2 is estimated to be twice the LOC developed with L1. the product will have to be maintained for five years. Various parameters for the company are given in the table below.

Parameter	Language L1	Language L2
Man years needed for development	LOC / 10000	LOC / 10000
Development Cost per year	Rs. 10,00,000	Rs. 7,50,000
Maintenance time	5 years	5 years
Cost of maintenance per year	Rs. 1,00,000	Rs. 50,000

Total cost of the project includes cost of development and maintenance. What is the LOC for L1 for which the cost of the project using L1 is equal to the cost of the project using L2?

(A) 4000 (B) 5000 (C) 4333 (D) 4667

Answer: - (B)

Exp: - LOC $L_1 = x$

$L_2 = 2x$

Total cost of project

$$\frac{x}{10000}\times 1000000 + 5\times 100000 = \frac{2x}{10000}\times 750000 + 50000\times 5$$

$$100x + 500000 = 150x + 250000$$

$$\Rightarrow 50x = 500000 - 250000$$

$$\therefore \quad x = \frac{250000}{50} \Rightarrow x = 5000$$

19. Let the time taken to switch between user and kernel modes of execution be t_1 while the time taken to switch between two processes be t_2. Which of the following is TRUE?

(A) $t_1 > t_2$

(B) $t_1 = t_2$

(C) $t_1 < t_2$

(D) Nothing can be said about the relation between t_1 and t_2

Answer: - (C)

Exp: - Process switching also involves mode changing.

20. A company needs to develop digital signal processing software for one of its newest inventions. The software is expected to have 40000 lines of code. The company needs to determine the effort in person-months needed to develop this software using the basic COCOMO model. The multiplicative factor for this model is given as 2.8 for the software development on embedded systems, while the exponentiation factor is given as 1.20. What is the estimated effort in person-months?

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(A) 234.25 (B) 932.50 (C) 287.80 (D) 122.40

Answer: - (A)

Exp: - Effort person per month

$= \alpha.(\text{kDSI})^{B}$

KDSI = Kilo LOC

$= 2.8 \times (40)^{1.20}$

$= 2.8 \times 83.6511$

$= 234.22$ person per month

21. Which of the following pairs have DIFFERENT expressive power?

(A) Deterministic finite automata (DFA) and Non-deterministic finite automata (NFA)

(B) Deterministic push down automata (DPDA) and Non-deterministic push down automata (NPDA)

(C) Deterministic single-tape Turing machine and Non-deterministic single tape Turing machine

(D) Single-tape Turing machine and multi-tape Turing machine

Answer: - (B)

Exp: - NPDA is more powerful than DPDA.

Hence answer is (B)

22. HTML (Hyper Text Markup Language) has language elements which permit certain actions other than describing the structure of the web document. Which one of the following actions is NOT supported by pure HTML (without any server or client side scripting) pages?

(A) Embed web objects from different sites into the same page

(B) Refresh the page automatically after a specified interval

(C) Automatically redirect to another page upon download

(D) Display the client time as part of the page

Answer: - (D)

23. Which of the following is NOT desired in a good Software Requirement Specifications (SRS) document?

(A) Functional Requirements

(B) Non Functional Requirements

(C) Goals of Implementation

(D) Algorithms for Software Implementation

Answer: - (D)

24. A computer handles several interrupt sources of which the following are relevant for this question.

Interrupt from CPU temperature sensor

Interrupt from Mouse

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Interrupt from Keyboard
Interrupt from Hard Disk

(A) Interrupt from Hard Disk
(B) Interrupt from Mouse
(C) Interrupt from Keyboard
(D) Interrupt from CPU temp sensor

Answer: - (D)

25. Consider a relational table with a single record for each registered student with the following attributes.

1. Registration_Number: Unique registration number for each registered student
2. UID: Unique Identity number, unique at the national level for each citizen
3. BankAccount_Number: Unique account number at the bank. A student can have multiple accounts or joint accounts. This attributes stores the primary account number
4. Name: Name of the Student
5. Hostel_Room: Room number of the hostel

Which of the following options is INCORRECT?

(A) BankAccount_Number is a candidate key
(B) Registration_Number can be a primary key
(C) UID is a candidate key if all students are from the same country
(D) If S is a superkey such that $S \cap UID$ is NULL then $S \cup UID$ is also a superkey

Answer: - (A)

Exp: - In case two students hold joint account then BankAccount_Num will not uniquely determine other attributes.

Q. No. 26 – 51 Carry Two Marks Each

26. Which of the given options provides the increasing order of asymptotic complexityoffunctionsf_1,f_2,$f_3$andf_4?

$f_1(n) = 2^n$; $f_2(n) = n^{3/2}$; $f_3(n) = n\log_2 n$; $f_4(n) = n^{\log_2 n}$

(A) f_3, f_2, f_4,f_1 (B) f_3, f_2, f_1,f_4 (C) f_2, f_3, f_1,f_4 (D) f_2, f_3, f_4,f_1

Answer: - (A)

Let $n = 1024$

$f_1(n) = 2^{1024}$

$f_2(n) = 2^{15}$

$f_3(n) = 10 \times 2^{10}$

$f_4(n) = 1024^{10} = 2^{100}$

$\therefore f_3, f_2, f_4, f_1$ is the rquired increasing order

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27. Four matrices M_1, M_2, M_3 and M_4 are dimensions $p \times q$, $q \times r$, $r \times s$ and $s \times t$ respectively can be multiplied in several ways with different number of total scalar multiplications. For example When multiplied as $((M_1 \times M_2) \times (M_3 \times M_4))$ the total number of scalar multiplications is pqr+rst+prt. When multiplied as $(((M_1 \times M_2) \times M_3) \times M_4)$, the total number of scalar multiplications is pqr+prs+pst.

If p=10, q=100, r=20, s=5 and t=80, then the minimum number of scalar multiplications needed is

(A) 248000 (B) 44000 (C) 19000 (D) 25000

Answer: - (C)

Exp: - Multiply as $(M_1 \times (M_2 \times M_3)) \times M_4$

The total number of scalar multiplication is

$= qrs + pqs + pst$

$= 10000 + 5000 + 4000 = 19000$

28. Consider a relational table r with sufficient number of records, having attributes A_1, A_2,..., A_n and let $1 \le p \le n$. Two queries Q1 and Q2 are given below.

Q1 : $\pi_{A1...A_n}(\sigma_{A_p=c}(r))$ where c is a const

Q2: $\pi_{A1...A_n}(\sigma_{c_1 \le A_p \le c_2}(r))$ where c_1 and c_2 are constants

The database can be configured to do ordered indexing on A_p or hashing on A_p. Which of the following statements is TRUE?

(A) Ordered indexing will always outperform hashing for both queries

(B) Hashing will always outperform ordered indexing for both queries

(C) Hashing will outperform ordered indexing on Q1, but not on Q2

(D) Hashing will outperform ordered indexing on Q2, but not on Q1.

Answer: - (C)

29. Consider the matrix as given below.

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 7 \\ 0 & 0 & 3 \end{bmatrix}$$

Which one of the following provides the CORRECT values of eigenvalues of the matrix?

(A) 1,4,3 (B) 3,7,3 (C) 7,3,2 (D) 1,2,3

Answer: - (A)

Exp: - Given matrix is upper triangular matrix and its diagonal elements are its eigen values = 1, 4, 3

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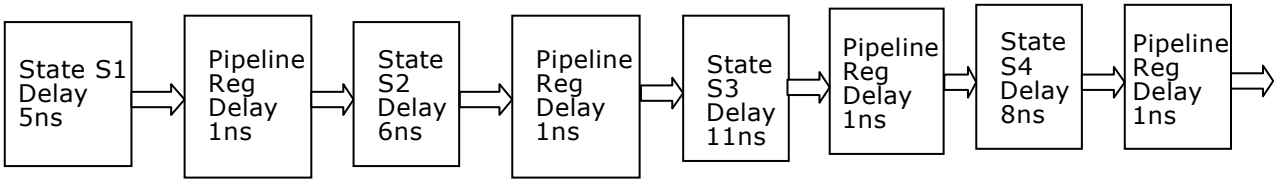
30. Consider an instruction pipeline with four stages (S1, S2, S3 and S4) each with combinational circuit only. The pipeline registers are required between each stage and at the end of the last stage. Delays for the stages and for the pipeline registers are as given in the figure.



What is the approximate speed up of the pipeline in steady state under ideal conditions when compared to the corresponding non-pipeline implementation?

(A) 4.0 (B) 2.5 (C) 1.1 (D) 3.0

Answer: - (B)

Exp: - $\frac{(5+6+11+8)}{(11+1)} = \frac{30}{12} = 2.5$

31. Definition of a language L with alphabet {a} is given as following

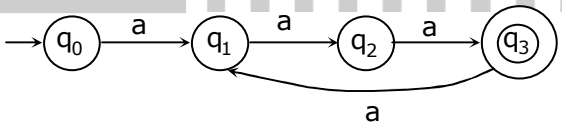
$L = \{a^{nk} \mid k > 0,$ and n is a positive integer constant$\}$

What is the minimum number of states needed in a DFA to recognize L?

(A) k+1 (B) n+1 (C) 2^{n+1} (D) 2^{k+1}

Answer: - (B)

Exp: - Let n = 3 and k=1



$(n+1)$ states

32. An 8KB direct mapped write-back cache is organized as multiple blocks, each of size 32-bytes. The processor generates 32-bit addresses. The cache controller maintains the tag information for each cache block comprising of the following.

1 Valid bit

1 Modified bit

As many bits as the minimum needed to identify the memory block mapped in the cache.

What is the total size of memory needed at the cache controller to store meta-data (tags) for the cache?

(A) 4864 bits (B) 6144bits (C) 6656bits (D) 5376bits

Answer: - (D)

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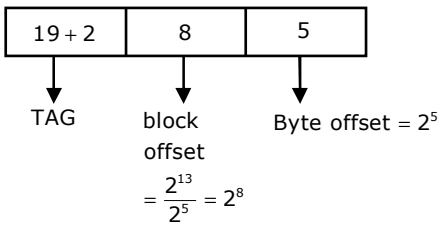
Exp: -



Required answer $= 256 \times (19 + 2) = 5376$ bits

33. An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10ms. Rotational speed of disk is 6000rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected)

(A) 0.50s (B) 1.50s (C) 1.25s (D) 1.00s

Answer: - (B)

Exp: - 6000 rotations _____ 60 sec

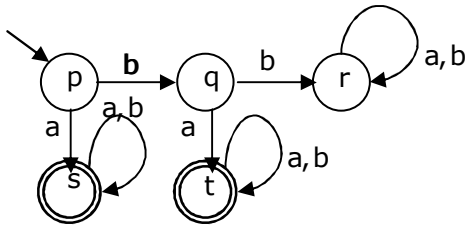
1 rotation _____ 10 ms

$\therefore$ Rotational latency = 5ms

Time for one disk access = 15 ms

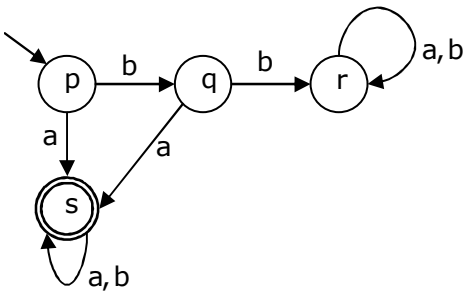
Time to load all libraries $= 15 \times 100 = 1500\,\text{ms} = 1.5\,\text{sec}$

34. A deterministic finite automation (DFA) D with alphabet $\Sigma = \{a, b\}$ is given below

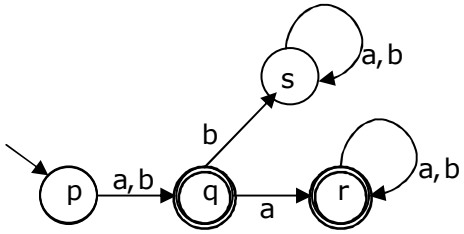


Which of the following finite state machines is a valid minimal DFA which accepts the same language as D?

(A)



(B)



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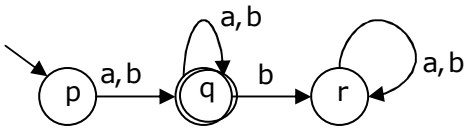
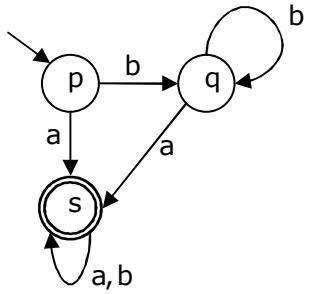
(C)



(D)



Answer: - (A)

Exp: - Options B and C will accept the string b

Option – D will accept the string "bba"

Both are invalid strings.

So the minimized DFA is option A

35. The following is comment written for a C function

```
/* This function computes the roots of a quadratic equation
a.x^2+b.x+c=0. The function stores two real roots
in *root1 and *root2 and returns the status of validity of
roots. It handles four different kinds of cases.
(i) When coefficient a is zero irrespective of discriminant
(ii) When discriminant is positive
(iii) When discrimanant is zero
(iv) When discrimanant is negative
Only in cases (ii) and (iii), the stored roots are valid.
Otherwise 0 is stored in the roots. the function returns 0 when
the roots are valid and -1 otherwise.
The functin also ensures root1>=root2.
int get_QuadRoots(float a, float b, float c, float *root1, float *root2);
*/
```

A software test engineer is assigned the job of doing black box testing. He comes up with the following test cases, many of which are redundant.

Test Case	Input set			Expected Output set		
	a	b	C	Root1	Root2	Return Value
T1	0.0	0.0	7.0	0.0	0.0	-1
T2	0.0	1.0	3.0	0.0	0.0	-1
T3	1.0	2.0	1.0	-1.0	-1.0	0
T4	4.0	-12.0	9.0	1.5	1.5	0
T5	1.0	-2.0	-3.0	3.0	-1.0	0
T6	1.0	1.0	4.0	0.0	0.0	-1

Which one of the following options provide the set of non-redundant tests using equivalence class partitioning approach from input perspective for black box testing?

(A) T1,T2,T3,T6 (B) T1,T3,T4,T5 (C) T2,T4,T5,T6 (D) T2,T3,T4,T5

Answer: - (C)

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Exp: - T_1 and T_2 checking same condition a = 0 hence, any one of T_1 and T_2 is redundant.

T_3, T_4: in both case discriminant $(D)=b^2-4ac=0$. Hence any one of it is redundant.

T_5 : D>0

T_6 : D<0

36. Database table by name Loan_Records is given below.

Borrower	Bank_Manager	Loan_ Amount
Ramesh	Sunderajan	10000.00
Suresh	Ramgopal	5000.00
Mahesh	Sunderajan	7000.00

What is the output of the following SQL query?

```
SELECT count(*)
FROM(
(SELECT Borrower. Bank_Manager FROM Loan_Records) AS S
NATURAL JOIN
(SELECT Bank_Manager, Loan_Amount FROM Loan_Records) AS T
);
```

(A) 3 (B) 9 (C) 5 (D) 6

Answer: - (C)

Exp: -

S

Borrower	Bank _ Manager
Ramesh	Sunderajan
Suresh	Ramgqpal
Mahesh	Sunderjan

T

Bank _ Manager	Loan _ Amount
Sunderajan	10000.00
Ramgopal	5000.00
Sunderjan	7000.00

After executing the given query, the output would be

Borrower	Bank_Manager	Load_Amount
Ramesh	Sunderajan	10000.00
Ramesh	Sunderajan	7000.00
Suresh	Ramgopal	5000.00
Mahesh	Sunderajan	10000.00
Mahesh	Sunderajan	7000.00

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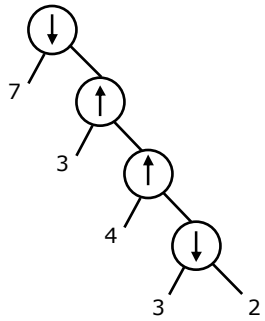
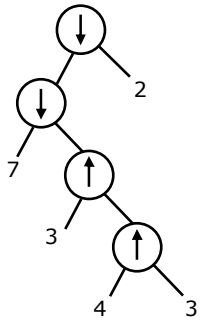
37. Consider two binary operators '$\uparrow$' and '$\downarrow$' with the precedence of operator $\downarrow$ being lower than that of the operator $\uparrow$. Operator $\uparrow$ is right associative while operator $\downarrow$, is left associative. Which one of the following represents the parse tree for expression $(7 \downarrow 3 \uparrow 4 \uparrow 3 \downarrow 2)$?

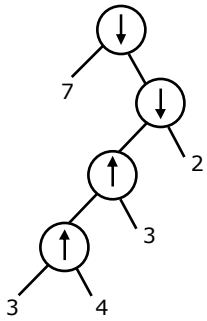
(A)



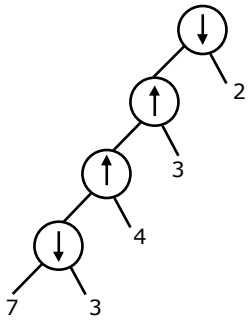
(B)



(C)



(D)



Answer: - (B)

Exp: - $7 \downarrow 3 \uparrow 4 \uparrow 3 \downarrow 2$

$\Rightarrow 7 \downarrow 3 \uparrow (4 \uparrow 3) \downarrow 2$ as $\uparrow$ is right associative

$\Rightarrow 7 \downarrow (3 \uparrow (4 \uparrow 3)) \downarrow 2$

$\Rightarrow (7 \downarrow (3 \uparrow (4 \uparrow 3))) \downarrow 2$ as $\downarrow$ is left associative

38. Consider the languages L1, L2 and L3 as given below

$L1 = \{0^p 1^q \mid p, q \in N\}$

$L2 = \{0^p 1^q \mid p, q \in N \text{ and } p = q\}$ and

$L3 = \{0^p 1^q 0^r \mid p, q, r \in N \text{ and } p = q = r\}$

Which of the following statements is NOT TRUE?

(A) Push Down Automata (PDA) can be used to recognize L1 and L2

(B) L1 is a regular language

(C) All the three languages are context free

(D) Turing machines can be used to recognize all the languages

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Answer: - (C)

Exp: - L1: regular language

L2: context free language

L3: context sensitive language

39. On a non-pipelined sequential processor, a program segment, which is a part of the interrupt service routine, is given to transfer 500 bytes from an I/O device to memory.

Initialize the address register

Initialize the count to 500

LOOP: Load a byte from device

Store in memory at address given by address register

Increment the address register

Decrement the count

If count != 0 go to LOOP

Assume that each statement in this program is equivalent to a machine instruction which takes one clock cycle to execute if it is a non-load/store instruction. The load-store instructions take two clock cycles to execute.

The designer of the system also has an alternate approach of using the DMA controller to implement the same transfer. The DMA controller requires 20 clock cycles for initialization and other overheads. Each DMA transfer cycle takes two clock cycles to transfer one byte of data from the device to the memory.

What is the approximate speedup when the DMA controller based design is used in place of the interrupt driven program based input-output?

(A) 3.4 (B) 4.4 (C) 5.1 (D) 6.7

Answer: - (A)

Exp: - No. of clock cycles required by using load-store approach = 2 + 500 × 7 = 3502 and that of by using DMA = 20 + 500 × 2 = 1020

Required speed up $= \frac{3502}{1020} = 3.4$

40. We are given a set of n distinct elements and an unlabeled binary tree with n nodes. In how many ways can we populate the tree with the given set so that it becomes a binary search tree?

(A) 0 (B) 1 (C) n! (D) $\frac{1}{n+1}.{}^{2n}C_n$

Answer: - (D)

41. Which one of the following options is CORRECT given three positive integers x, y and z, and a predicate

$$P(x) = \neg(x = 1) \wedge \forall y\left(\exists z(x = y * z) \Rightarrow (y = x) \vee (y = 1)\right)$$

(A) P(x) being true means that x is a prime number

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(B) P(x) being true means that x is a number other than 1

(C) P(x) is always true irrespective of the value of x

(D) P(x) being true means that x has exactly two factors other than 1 and x

Answer: - (A)

42. Given $i = \sqrt{-1}$, what will be the evaluation of the definite integral $\int_0^{\pi/2} \frac{\cos x + i \sin x}{\cos x - i \sin x} dx$?

(A) 0 (B) 2 (C) –i (D) i

Answer: - (D)

Exp: - $\int_0^{\pi/2} \frac{e^{ix}}{e^{-ix}} dx = \int_0^{\pi/2} e^{2ix} dx$

$$= \left(\frac{e^{2ix}}{2i} \right)_0^{\pi/2} = \frac{1}{2i} \left[e^{i\pi} - 1 \right] = \frac{1}{2i} \left[\cos \pi + i \sin \pi - 1 \right] = \frac{1}{2i} \left[-1 + 0 - 1 \right] = \frac{-2}{2i} = \frac{-1}{i} \times \frac{i}{i} = \frac{-i}{-1} = i$$

43. Consider a database table T containing two columns X and Y each of type integer. After the creation of the table, one record (X= 1, Y=l) is inserted in the table.

Let MX and MY denote the respective maximum values of X and Y among all records in the table at any point in time. Using MX and MY, new records are inserted in the table 128 times with X and Y values being MX+1, 2*MY+1 respectively. It may be noted that each time after the insertion, values of MX and MY change.

What will be the output of the following SQL query after the steps mentioned above are carried out?

SELECT Y FROM T WHERE X=7;

(A) 127 (B) 255 (C) 129 (D) 257

Answer: - (A)

Exp: -

X	Y
1	1
2	3
3	7
4	15
5	31
6	63
7	127

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44. Consider a finite sequence of random values $X = [x_1, x_2, \ldots x_n]$. Let μ_x be the mean and σ_x be the standard deviation of X .Let another finite sequence Y of equal length be derived from this as $y_i = a * x_i + b$, where a and b are positive constants. Let μ_y be the mean and σ_y be the standard deviation of this sequence. Which one of the following statements is INCORRECT?

(A) Index position of mode of X in X is the same as the index position of mode of Y in Y.

(B) Index position of median of X in X is the same as the index position of median of Y in Y.

(C) $\mu_y = a\mu_x + b$

(D) $\sigma_y = a\sigma_x + b$

Answer: - (D)

45. A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at a time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?

(A) 1/5 (B) 4/25 (C) 1/4 (D) 2/5

Answer: - (A)

Exp: - (2,1), (3,2), (4,3), (5,4)

Required probability = $\frac{4}{5 \times 4} = \frac{4}{20} = \frac{1}{5}$

GATEFORUM Engineering Success

46. Consider the following table of arrival time and burst time for three processes P0, P1 and P2.

Process	Arrival time	Burst Time
P0	0 ms	9 ms
P1	1 ms	4ms
P2	2 ms	9ms

The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of processes. What is the average waiting time for the three processes?

(A) 5.0 ms (B) 4.33 ms (C) 6.33 ms (D) 7.33 ms

Answer: - (A)

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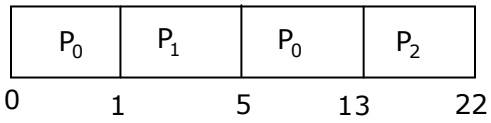
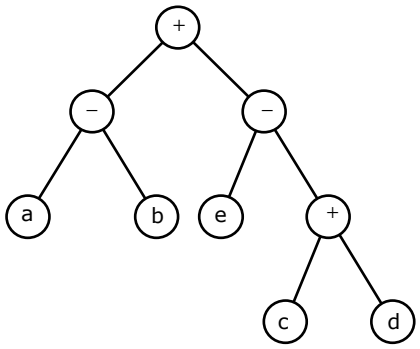
Exp: -



Average waiting time $= \frac{4+11}{3} = 5$ ms

47. Consider evaluating the following expression tree on a machine with load-store architecture in which memory can be accessed only through load and store instructions. The variables a, b, c, d and e are initially stored in memory. The binary operators used in this expression tree can be evaluated by the machine only when the operands are in registers. The instructions produce result only in a register. If no intermediate results can be stored in memory, what is the minimum number of registers needed to evaluate this expression?



(A) 2 (B) 9 (C) 5 (D) 3

Answer: - (D)

Exp: - Load R_1, a ; $R_1 \leftarrow M[a]$

Load R_2, b ; $R_2 \leftarrow M[b]$

Sub R_1, R_2 ; $R_1 \leftarrow R_1 - R_2$

Load R_2, c ; $R_2 \leftarrow M[c]$

Load R_3, d ; $R_3 \leftarrow M[d]$

Add R_2, R_3 ; $R_2 \leftarrow R_2 + R_3$

Load R_3, e ; $R_3 \leftarrow M[e]$

Sub R_3, R_2 : $R_3 \leftarrow R_3 - R_2$

Add R_1, R_3 ; $R_1 \leftarrow R_1 + R_3$

Total 3 Registers are required minimum

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Common Data Questions: 48 & 49

Consider the following recursive C function that takes two arguments

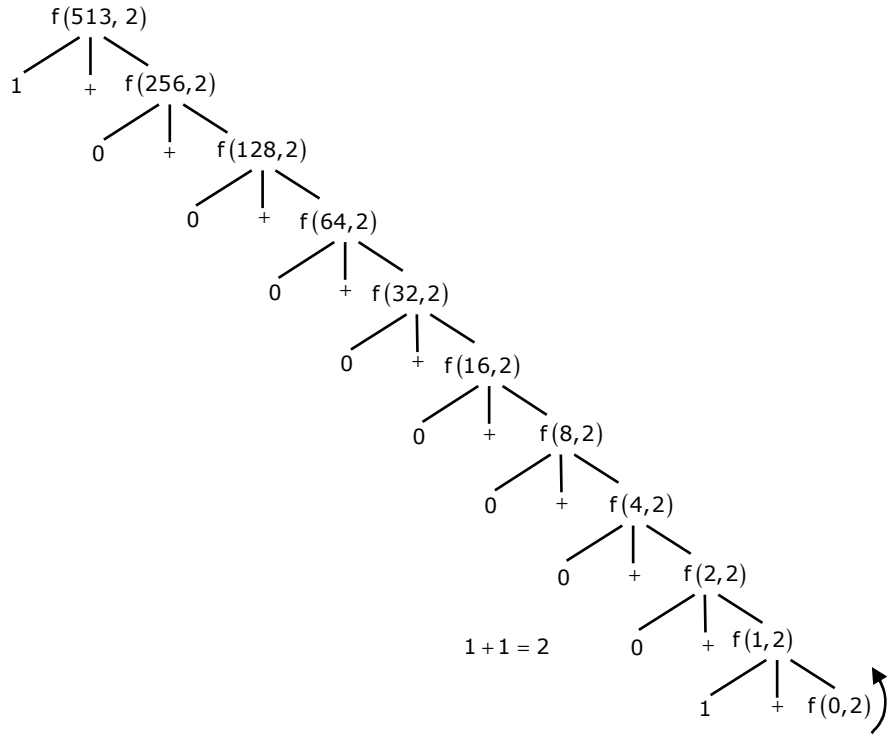
```
unsigned int foo(unsigned int n, unsigned int r) {
    if (n > 0) return (n%r) + foo (n/r, r));
    else return 0;
}
```

48. What is the return value of the function foo when it is called as foo (513, 2)?

(A) 9 (B) 8 (C) 5 (D) 2

Answer: - (D)

Exp: -

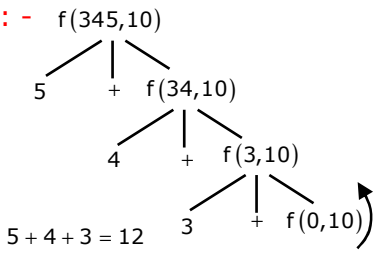


49 What is the return value of the function foo when it is called as foo (345, 10) ?

(A) 345 (B) 12 (C) 5 (D) 3

Answer: - (B)

Exp: -



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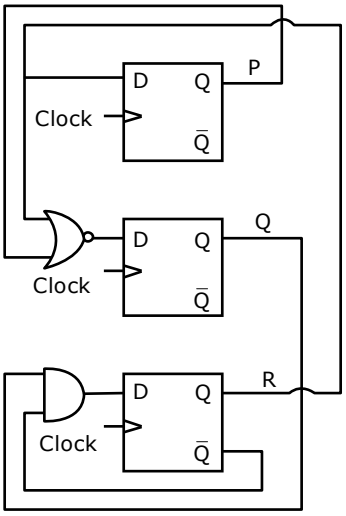
Common Data Questions: 50 & 51

Consider the following circuit involving three D-type flip-flops used in a certain type of counter configuration.



50. If all the flip-flops were reset to 0 at power on, what is the total number of distinct outputs (states) represented by PQR generated by the counter?

(A) 3 (B) 4 (C) 5 (D) 6

Answer: - (B)

Exp: -

CLOCK	Inputs			Outputs		
	$D_1 = R$	$D_2 = \overline{(P + R)}$	$D_3 = Q\,\overline{R}$	P	Q	R
1	0	1	0	0	1	0
2	0	1	1	0	1	1
3	1	0	0	1	0	0
4	0	0	0	0	0	0

So Total number of distinct outputs is 4

51. If at some instance prior to the occurrence of the clock edge, P. Q and R have a value 0, 1 and 0 respectively, what shall be the value of PQR after the clock edge?

(A) 000 (B) 001 (C) 010 (D) 011

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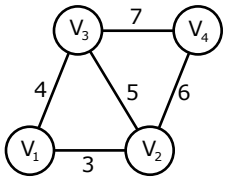
Answer: - (D)

Exp: -From the Table Shown in the explanation of question 50, if first state is 010 next State is 011

Linked Answer Questions: Q.52 to Q.55 Carry Two Marks Each

Statement for Linked Answer Questions: 52 & 53

An undirected graph G(V,E) contains $n\,(n>2)$ nodes named $v_1, v_2, \ldots v_n$. Two nodes v_i, v_j are connected if and only if $0 < |i-j| \le 2$. Each edge (v_i, v_j) is assigned a weight i + j. A sample graph with n = 4 is shown below



52. What will be the cost of the minimum spanning tree (MST) of such a graph with n nodes?

(A) $\frac{1}{12}(11n^2 - 5n)$ (B) $n^2 - n + 1$

(C) $6n - 11$ (D) $2n + 1$

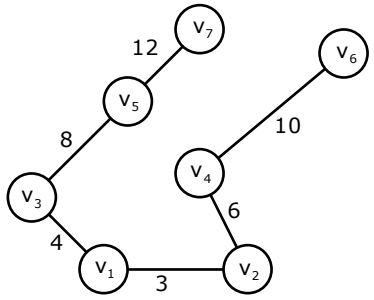
Answer: - (B)

53. The length of the path from v_5 to v_6 in the MST of previous question with n = 10 is

(A) 11 (B) 25 (C) 31 (D) 41

Answer: - (C)

Exp: -



$12 + 8 + 4 + 3 + 6 + 10 = 31$

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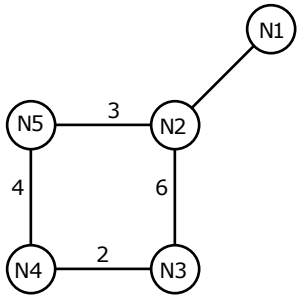
Statement for Linked Answer Questions: 54 & 55

Consider a network with five nodes, N1 to N5, as shown below



The net work uses a Distance Vector Routing protocol. Once the routes have stabilized, the distance vectors at different nodes are as following

N1 : (0, 1, 7, 8, 4)

N2 : (1, 0, 6, 7, 3)

N3 : (7, 6, 0, 2, 6)

N4 : (8, 7, 2, 0, 4)

N5 : (4, 3, 6, 4, 0)

Each distance vector is the distance of the best known path at that instance to nodes, N1 to N5, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors

54. The cost of link N2-N3 reduces to 2 in (both directions). After the next round of updates, what will be the new distance vector at node, N3?

(A) (3. 2, 0, 2, 5) (B) (3, 2, 0, 2, 6)

(C) (7, 2, 0, 2, 5) (D) (7, 2, 0, 2, 6)

Answer: - (A)

Exp: -

	N_3	
N_1	3	$\rightarrow 1+2$
N_2	2	
N_3	0	
N_4	2	
N_5	5	$\rightarrow 2+3$

55. After the update in the previous question, the link N1-N2 goes down. N2 will reflect this change immediately in its distance vector as cost, ∞. After the NEXT ROUND of update, what will be the cost to N1 in the distance vector of N3?

(A) 3 (B) 9 (C) 10 (D) ∞

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Answer: - (C)

Exp: - N_3 has neighbors N_2 and N_4

N_2 has made entry ∞

N_4 has the distance of 8 to N_1

N_3 has the distance of 2 to N_4

So 2 + 8 = 10

Q. No. 56 – 60 Carry One Mark Each

56. If Log (P) = (1/2)Log (Q) = (1/3) Log (R), then which of the following options is TRUE?

(A) $P^2 = Q^3R^2$ (B) $Q^2 = PR$ (C) $Q^2 = R^3P$ (D) $R = P^2Q^2$

Answer: - (B)

Exp:- $\log P = \frac{1}{2}\log Q = \frac{1}{3}\log(R) = k$

$\therefore P = b^k, Q = b^{2k}, R = b^{3k}$

Now, $Q^2 = b^{4k} = b^{3k}\, b^k = PR$

57. Choose the most appropriate word(s) from the options given below to complete the following sentence.

I contemplated_____Singapore for my vacation but decided against it.

(A) To visit (B) having to visit (C) visiting (D)for a visit

Answer: - (C)

Exp: - Contemplate is a transitive verb and hence is followed by a gerund Hence the correct usage of contemplate is verb+ ing form.

58. Choose the most appropriate word from the options given below to complete the following sentence.

If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or_____.

(A) Hyperbolic (B) Restrained

(C) Argumentative (D) Indifferent

Answer: - (B)

Exp: - The tone of the sentence clearly indicates a word that is similar to understated is needed for the blank. Alternatively, the word should be antonym of strong (fail to make strong impression). Therefore, the best choice is restrained which means controlled/reserved/timid.

59. Choose the word from the options given below that is most nearly opposite in meaning to the given word: **Amalgamate**

(A) Merge (B) Split (C) Collect (D) Separate

Answer: - (B)

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Exp: - Amalgamate means combine or unite to form one organization or structure. So the best option here is split. Separate on the other hand, although a close synonym, it is too general to be the best antonym in the given question while Merge is the synonym; Collect is not related.

60. Which of the following options is the closest in the meaning to the word below:

Inexplicable

(A) Incomprehensible (B) Indelible

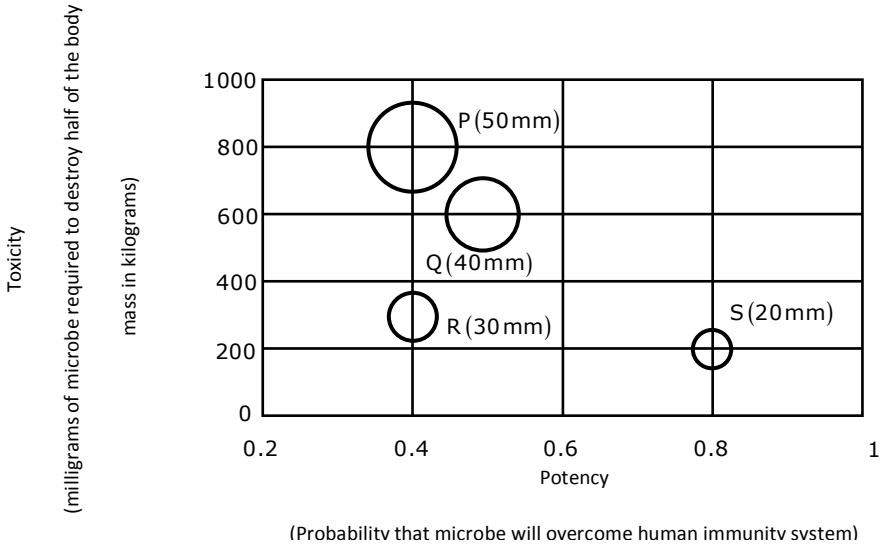
(C) Inextricable (D) Infallible

Answer: - (A)

Exp: - Inexplicable means not explicable; that cannot be explained, understood, or accounted for. So the best synonym here is incomprehensible.

Q. No. 61 – 65 Carry Two Marks Each

61. P, Q, R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below



A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

(A) P (B) Q (C) R (D) S

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Answer: - (D)

Exp: - By observation of the table, we can say S

	P	Q	R	S
Requirement	800	600	300	200
Potency	0.4	0.5	0.4	0.8

62. The variable cost (V) of manufacturing a product varies according to the equation V= 4q, where q is the quantity produced. The fixed cost (F) of production of same product reduces with q according to the equation F = 100/q. How many units should be produced to minimize the total cost (V+F)?

(A) 5 (B) 4 (C) 7 (D) 6

Answer: (A)

Exp: - Checking with all options in formula: (4q+100/q) i.e. (V+F). Option A gives the minimum cost.

63. A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders. Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10th day. What is the minimum number of trucks required so that there will be no pending order at the end of the 5th day?

(A) 4 (B) 5 (C) 6 (D) 7

Answer: - (C)

Exp: - Let each truck carry 100 units.

$2800 = 4n + e$ n = normal

$3000 = 10n + e$ e = excess/pending

$\therefore n = \frac{100}{3}, e = \frac{8000}{3}$

$5\text{days} \Rightarrow 500x = \frac{5.100}{3} + \frac{8000}{3}$

$\Rightarrow 500x = \frac{8500}{3} 17 \Rightarrow x > 5$

Minimum possible = 6

64. A container originally contains 10 litres of pure spirit. From this container 1 litre of spirit is replaced with 1 litre of water. Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?

(A) 7.58 litres (B) 7.84 litres (C) 7 litres (D) 7.29 litres

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Answer: - (D)

Exp:- $10\left(\frac{10-1}{10}\right)^3 = 10\left(\frac{9}{10}\right)^3 = \frac{729}{1000}$

$\therefore \frac{729}{1000} \times 1 = 7.29$ litres

65. **Few school curricula include a unit on how to deal with bereavement and grief, and yet all students at some point in their lives suffer from losses through death and parting.**

Based on the above passage which topic would not be included in a unit on bereavement?

(A) how to write a letter of condolence

(B) what emotional stages are passed through in the healing process

(C) what the leading causes of death are

(D) how to give support to a grieving friend

Answer: - (C)

Exp: - The given passage clearly deals with how to deal with bereavement and grief and so after the tragedy occurs and not about precautions. Therefore, irrespective of the causes of death, a school student rarely gets into details of causes—which is beyond the scope of the context. Rest all are important in dealing with grief.

GATEFORUM Engineering Success

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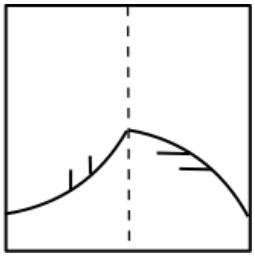
General Aptitude (GA)

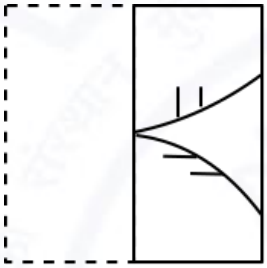
Q.1 – Q.5 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

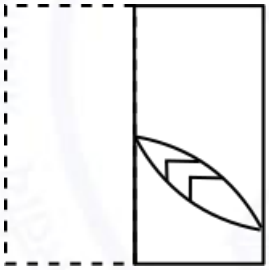
Q.1	Gauri said that she can play the keyboard _____ her sister.
(A)	as well as
(B)	as better as
(C)	as nicest as
(D)	as worse as

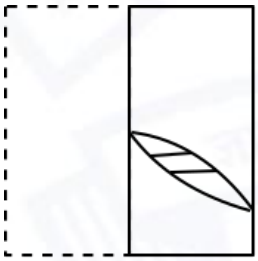
Q.2

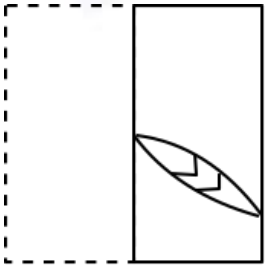


A transparent square sheet shown above is folded along the dotted line. The folded sheet will look like _____.

(A) 

(B) 

(C) 

(D) 

Q.3	If θ is the angle, in degrees, between the longest diagonal of the cube and any one of the edges of the cube, then, cos θ =
(A)	$\frac{1}{2}$
(B)	$\frac{1}{\sqrt{3}}$
(C)	$\frac{1}{\sqrt{2}}$
(D)	$\frac{\sqrt{3}}{2}$

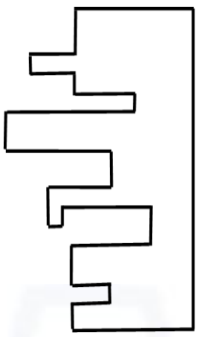
Q.4	If $\left(x-\frac{1}{2}\right)^2-\left(x-\frac{3}{2}\right)^2=x+2$, then the value of x is:
(A)	2
(B)	4
(C)	6
(D)	8

Q.5	Pen : Write :: Knife : _____ Which one of the following options maintains a similar logical relation in the above?
(A)	Vegetables
(B)	Sharp
(C)	Cut
(D)	Blunt

Q. 6 – Q. 10 Multiple Choice Question (MCQ), carry TWO marks each (for each wrong answer: – 2/3).

Q.6	Listening to music during exercise improves exercise performance and reduces discomfort. Scientists researched whether listening to music while studying can help students learn better and the results were inconclusive. Students who needed external stimulation for studying fared worse while students who did not need any external stimulation benefited from music. Which one of the following statements is the CORRECT inference of the above passage?
(A)	Listening to music has no effect on learning and a positive effect on physical exercise.
(B)	Listening to music has a clear positive effect both on physical exercise and on learning.
(C)	Listening to music has a clear positive effect on physical exercise. Music has a positive effect on learning only in some students.
(D)	Listening to music has a clear positive effect on learning in all students. Music has a positive effect only in some students who exercise.

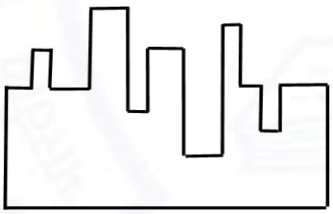
Q.7



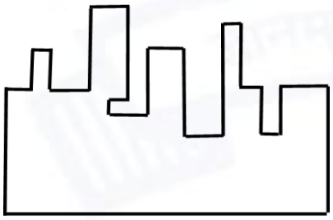
A jigsaw puzzle has 2 pieces. One of the pieces is shown above. Which one of the given options for the missing piece when assembled will form a rectangle? The piece can be moved, rotated or flipped to assemble with the above piece.

(A)

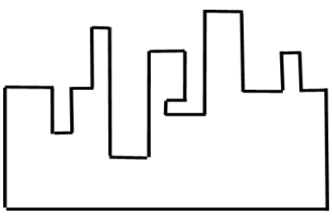
(B)



(C)

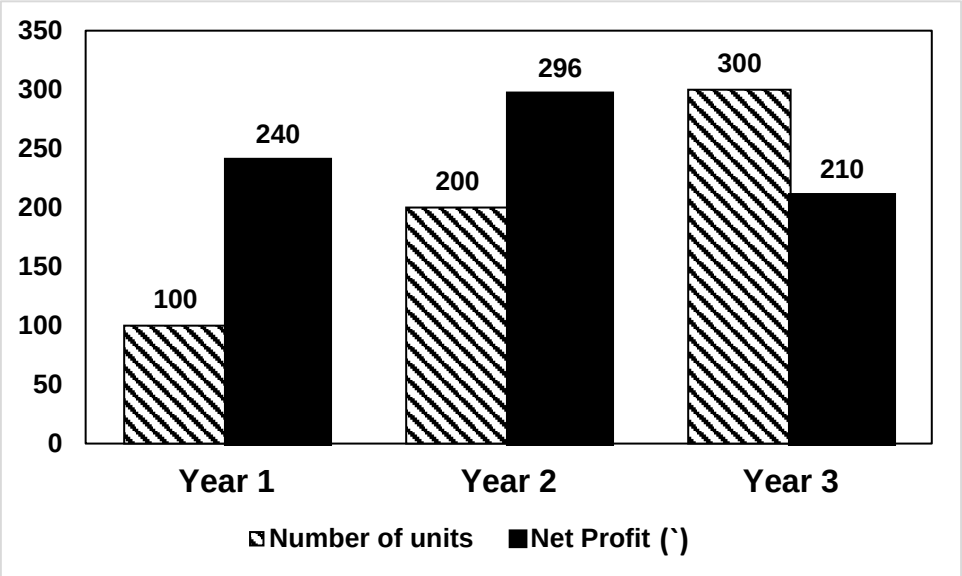


(D)



Q.8	The number of students in three classes is in the ratio 3:13:6. If 18 students are added to each class, the ratio changes to 15:35:21. The total number of students in all the three classes in the beginning was:
(A)	22
(B)	66
(C)	88
(D)	110

Q.9



The number of units of a product sold in three different years and the respective net profits are presented in the figure above. The cost/unit in Year 3 was `1, which was half the cost/unit in Year 2. The cost/unit in Year 3 was one-third of the cost/unit in Year 1. Taxes were paid on the selling price at 10%, 13% and 15% respectively for the three years. Net profit is calculated as the difference between the selling price and the sum of cost and taxes paid in that year.

The ratio of the selling price in Year 2 to the selling price in Year 3 is _____.

A)	4:3
(B)	1:1
(C)	3:4
(D)	1:2

Q.10	Six students P, Q, R, S, T and U, with distinct heights, compare their heights and make the following observations. Observation I: S is taller than R. Observation II: Q is the shortest of all. Observation III: U is taller than only one student. Observation IV: T is taller than S but is not the tallest. The number of students that are taller than R is the same as the number of students shorter than _____.
(A)	T
(B)	R
(C)	S
(D)	P

Computer Science and Information Technology (CS, Set-2)

Q.1 – Q.10 Multiple Choice Question (MCQ), carry ONE mark each (for each wrong answer: – 1/3).

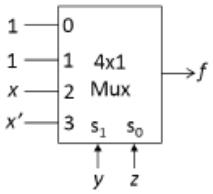
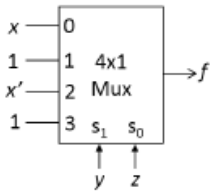
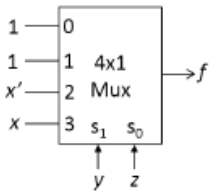
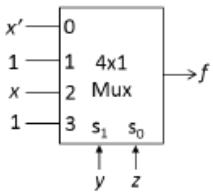
Q.1	Let G be a connected undirected weighted graph. Consider the following two statements. S_1 : There exists a minimum weight edge in G which is present in every minimum spanning tree of G. S_2 : If every edge in G has distinct weight, then G has a unique minimum spanning tree. Which one of the following options is correct?
(A)	Both S_1 and S_2 are true.
(B)	S_1 is true and S_2 is false.
(C)	S_1 is false and S_2 is true.
(D)	Both S_1 and S_2 are false.

Q.2	Let H be a binary min-heap consisting of n elements implemented as an array. What is the worst case time complexity of an optimal algorithm to find the maximum element in H?
(A)	$\Theta(1)$
(B)	$\Theta(\log n)$
(C)	$\Theta(n)$
(D)	$\Theta(n \log n)$

Q.3	Consider the following ANSI C program:
	`int main() {`
	`  Integer x;`
	`  return 0;`
	`}`
	Which one of the following phases in a seven-phase C compiler will throw an error?

(A)	Lexical analyzer
(B)	Syntax analyzer
(C)	Semantic analyzer
(D)	Machine dependent optimizer

Q.4	The format of the single-precision floating-point representation of a real number as per the IEEE 754 standard is as follows: sign | exponent | mantissa Which one of the following choices is correct with respect to the *smallest* normalized positive number represented using the standard?
(A)	exponent = 00000000 and mantissa = 00000000000000000000000
(B)	exponent = 00000000 and mantissa = 00000000000000000000001
(C)	exponent = 00000001 and mantissa = 00000000000000000000000
(D)	exponent = 00000001 and mantissa = 00000000000000000000001

Q.5	Which one of the following circuits implements the Boolean function given below? $f(x,y,z) = m_0 + m_1 + m_3 + m_4 + m_5 + m_6$, where m_i is the i^{th} minterm.
(A)	
(B)	
(C)	
(D)	

Q.6	Consider the following statements S1 and S2 about the relational data model: S1: A relation scheme can have at most one foreign key. S2: A foreign key in a relation scheme R cannot be used to refer to tuples of R. Which one of the following choices is correct?
(A)	Both S1 and S2 are true.
(B)	S1 is true and S2 is false.
(C)	S1 is false and S2 is true.
(D)	Both S1 and S2 are false.

Q.7	Consider the three-way handshake mechanism followed during TCP connection establishment between hosts P and Q. Let X and Y be two random 32-bit starting sequence numbers chosen by P and Q respectively. Suppose P sends a TCP connection request message to Q with a TCP segment having SYN bit = 1, SEQ number = X, and ACK bit = 0. Suppose Q accepts the connection request. Which one of the following choices represents the information present in the TCP segment header that is sent by Q to P?
(A)	SYN bit = 1, SEQ number = X+1, ACK bit = 0, ACK number = Y, FIN bit = 0
(B)	SYN bit = 0, SEQ number = X+1, ACK bit = 0, ACK number = Y, FIN bit = 1
(C)	SYN bit = 1, SEQ number = Y, ACK bit = 1, ACK number = X+1, FIN bit = 0
(D)	SYN bit = 1, SEQ number = Y, ACK bit = 1, ACK number = X, FIN bit = 0

Q.8	What is the worst-case number of arithmetic operations performed by recursive binary search on a sorted array of size n?
(A)	$\Theta(\sqrt{n})$
(B)	$\Theta(\log_2(n))$
(C)	$\Theta(n^2)$
(D)	$\Theta(n)$

Q.9	Let $L \subseteq \{0,1\}^*$ be an arbitrary regular language accepted by a minimal DFA with k states. Which one of the following languages must necessarily be accepted by a minimal DFA with k states?
(A)	$L - \{01\}$
(B)	$L \cup \{01\}$
(C)	$\{0,1\}^* - L$
(D)	$L \cdot L$

Q.10 Consider the following ANSI C program.

```
#include <stdio.h>
int main(){
  int arr[4][5];
  int i, j;
  for (i=0; i<4; i++){
    for (j=0; j<5; j++){
      arr[i][j] = 10*i + j;
    }
  }
  printf("%d", *(arr[1] + 9));
  return 0;
}
```

What is the output of the above program?

(A) 14

(B) 20

(C) 24

(D) 30

Q.11 – Q.15 Multiple Select Question (MSQ), carry ONE mark each (no negative marks).

Q.11	Consider the following sets, where $n \geq 2$: S_1: Set of all $n \times n$ matrices with entries from the set $\{a, b, c\}$ S_2: Set of all functions from the set $\{0, 1, 2, ..., n^2 - 1\}$ to the set $\{0, 1, 2\}$ Which of the following choice(s) is/are correct?
(A)	There does not exist a bijection from S_1 to S_2.
(B)	There exists a surjection from S_1 to S_2.
(C)	There exists a bijection from S_1 to S_2.
(D)	There does not exist an injection from S_1 to S_2.

Q.12	Let L_1 be a regular language and L_2 be a context-free language. Which of the following languages is/are context-free?
(A)	$L_1 \cap \overline{L_2}$
(B)	$\overline{\overline{L_1} \cup \overline{L_2}}$
(C)	$L_1 \cup (L_2 \cup \overline{L_2})$
(D)	$(L_1 \cap L_2) \cup (\overline{L_1} \cap L_2)$

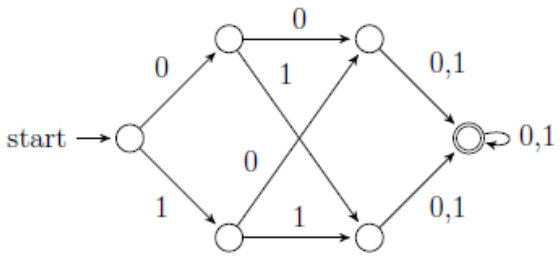
Q.13	In the context of compilers, which of the following is/are NOT an intermediate representation of the source program?
(A)	Three address code
(B)	Abstract Syntax Tree (AST)
(C)	Control Flow Graph (CFG)
(D)	Symbol table

Q.14	Which of the following statement(s) is/are correct in the context of CPU scheduling?
(A)	Turnaround time includes waiting time.
(B)	The goal is to only maximize CPU utilization and minimize throughput.
(C)	Round-robin policy can be used even when the CPU time required by each of the processes is not known apriori.
(D)	Implementing preemptive scheduling needs hardware support.

Q.15	Choose the correct choice(s) regarding the following propositional logic assertion S: $$S : ((P \wedge Q) \rightarrow R) \rightarrow ((P \wedge Q) \rightarrow (Q \rightarrow R))$$
(A)	S is neither a tautology nor a contradiction.
(B)	S is a tautology.
(C)	S is a contradiction.
(D)	The antecedent of S is logically equivalent to the consequent of S.

Q.16 – Q.25 Numerical Answer Type (NAT), carry ONE mark each (no negative marks).

Q.16	Consider a complete binary tree with 7 nodes. Let A denote the set of first 3 elements obtained by performing Breadth-First Search (BFS) starting from the root. Let B denote the set of first 3 elements obtained by performing Depth-First Search (DFS) starting from the root. The value of $\lvert A - B \rvert$ is _____.

Q.17	Consider the following deterministic finite automaton (DFA).  The number of strings of length 8 accepted by the above automaton is _____.

Q.18	If x and y are two decimal digits and $(0.1101)_2 = (0.8xy5)_{10}$, the decimal value of $x + y$ is _____.

Q.19	Consider a set-associative cache of size 2KB (1KB = 2^{10} bytes) with cache block size of 64 bytes. Assume that the cache is byte-addressable and a 32-bit address is used for accessing the cache. If the width of the tag field is 22 bits, the associativity of the cache is _____.

Q.20	Consider a computer system with DMA support. The DMA module is transferring one 8-bit character in one CPU cycle from a device to memory through *cycle stealing* at regular intervals. Consider a 2 MHz processor. If 0.5% processor cycles are used for DMA, the data transfer rate of the device is _____ bits per second.

Q.21	A data file consisting of 1,50,000 student-records is stored on a hard disk with block size of 4096 bytes. The data file is sorted on the primary key `RollNo`. The size of a record pointer for this disk is 7 bytes. Each student-record has a candidate key attribute called `ANum` of size 12 bytes. Suppose an index file with records consisting of two fields, `ANum` value and the record pointer to the corresponding student record, is built and stored on the same disk. Assume that the records of data file and index file are not split across disk blocks. The number of blocks in the index file is _____.

Q.22	For a given biased coin, the probability that the outcome of a toss is a head is 0.4. This coin is tossed 1,000 times. Let X denote the random variable whose value is the number of times that head appeared in these 1,000 tosses. The standard deviation of X (rounded to 2 decimal places) is _____.

Q.23	Consider the following ANSI C function:
	`int SomeFunction(int x, int y)`
	`{`
	`  if ((x == 1) || (y == 1)) return 1;`
	`  if (x == y) return x;`
	`  if (x > y) return SomeFunction(x - y, y);`
	`  if (y > x) return SomeFunction(x, y - x);`
	`}`
	The value returned by `SomeFunction(15, 255)` is _____.

Q.24	Suppose that P is a 4×5 matrix such that every solution of the equation $P\mathbf{x} = \mathbf{0}$ is a scalar multiple of $[2 \quad 5 \quad 4 \quad 3 \quad 1]^T$. The rank of P is _____.

Q.25	Suppose that $f: \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function on the interval [-3, 3] and a differentiable function in the interval (-3, 3) such that for every x in the interval, $f'(x) \leq 2$. If $f(-3) = 7$, then $f(3)$ is at most _____.

Q.26 – Q.39 Multiple Choice Question (MCQ), carry TWO mark each (for each wrong answer: – 2/3).

Q.26	Consider the string **abbccddeee**. Each letter in the string must be assigned a binary code satisfying the following properties: 1. For any two letters, the code assigned to one letter must not be a prefix of the code assigned to the other letter. 2. For any two letters of the same frequency, the letter which occurs earlier in the dictionary order is assigned a code whose length is at most the length of the code assigned to the other letter. Among the set of all binary code assignments which satisfy the above two properties, what is the minimum length of the encoded string?
(A)	21
(B)	23
(C)	25
(D)	30

Q.27	Assume a two-level inclusive cache hierarchy, L1 and L2, where L2 is the larger of the two. Consider the following statements. S_1 : Read misses in a write through L1 cache do not result in writebacks of dirty lines to the L2. S_2 : Write allocate policy *must* be used in conjunction with write through caches and no-write allocate policy is used with writeback caches. Which of the following statements is correct?
(A)	S_1 is true and S_2 is false
(B)	S_1 is false and S_2 is true
(C)	S_1 is true and S_2 is true
(D)	S_1 is false and S_2 is false

Q.28	Suppose we want to design a synchronous circuit that processes a string of 0's and 1's. Given a string, it produces another string by replacing the first 1 in any subsequence of consecutive 1's by a 0. Consider the following example. Input sequence: 00100011000011100 Output sequence: 00000001000001100 A *Mealy Machine* is a state machine where both the next state and the output are functions of the present state and the current input. The above mentioned circuit can be designed as a two-state Mealy machine. The states in the Mealy machine can be represented using Boolean values 0 and 1. We denote the current state, the next state, the next incoming bit, and the output bit of the Mealy machine by the variables s, t, b and y respectively. Assume the initial state of the Mealy machine is 0. What are the Boolean expressions corresponding to t and y in terms of s and b?
(A)	$t = s + b$ $y = s\,b$
(B)	$t = b$ $y = s\,b$
(C)	$t = b$ $y = s\,\overline{b}$
(D)	$t = s + b$ $y = s\,\overline{b}$

Q.29

In an examination, a student can choose the order in which two questions (`QuesA` and `QuesB`) must be attempted.

- If the first question is answered wrong, the student gets zero marks.
- If the first question is answered correctly and the second question is not answered correctly, the student gets the marks only for the first question.
- If both the questions are answered correctly, the student gets the sum of the marks of the two questions.

The following table shows the probability of correctly answering a question and the marks of the question respectively.

question	probability of answering correctly	marks
`QuesA`	0.8	10
`QuesB`	0.5	20

Assuming that the student always wants to maximize her expected marks in the examination, in which order should she attempt the questions and what is the expected marks for that order (assume that the questions are independent)?

(A) First `QuesA` and then `QuesB`. Expected marks 14.

(B) First `QuesB` and then `QuesA`. Expected marks 14.

(C) First `QuesB` and then `QuesA`. Expected marks 22.

(D) First `QuesA` and then `QuesB`. Expected marks 16.

Q.30 Consider the following ANSI C code segment:

```
z = x + 3 + y->f1 + y->f2;
for (i = 0; i < 200; i = i + 2){
 if (z > i) {
   p = p + x + 3;
   q = q + y->f1;
 } else {
   p = p + y->f2;
   q = q + x + 3;
 }
}
```

Assume that the variable `y` points to a `struct` (allocated on the heap) containing two fields `f1` and `f2`, and the local variables `x`, `y`, `z`, `p`, `q`, and `i` are allotted registers. Common sub-expression elimination (CSE) optimization is applied on the code. The number of addition and dereference operations (of the form `y->f1` or `y->f2`) in the optimized code, respectively, are:

(A) 403 and 102

(B) 203 and 2

(C) 303 and 102

(D) 303 and 2

Q.31	The relation scheme given below is used to store information about the employees of a company, where `empId` is the key and `deptId` indicates the department to which the employee is assigned. Each employee is assigned to exactly one department.
	
	`emp(empId, name, gender, salary, deptId)`
	
	Consider the following SQL query:
	
	`select deptId, count(*)`
	`from emp`
	`where gender = "female" and salary > (select avg(salary) from emp)`
	`group by deptId;`
	
	The above query gives, for each department in the company, the number of female employees whose salary is greater than the average salary of

(A)	employees in the department.
(B)	employees in the company.
(C)	female employees in the department.
(D)	female employees in the company.

Q.32	Let S be the following schedule of operations of three transactions T_1, T_2 and T_3 in a relational database system: $$R_2(Y), R_1(X), R_3(Z), R_1(Y), W_1(X), R_2(Z), W_2(Y), R_3(X), W_3(Z)$$ Consider the statements P and Q below: P: S is conflict-serializable. Q: If T_3 commits before T_1 finishes, then S is recoverable. Which one of the following choices is correct?
(A)	Both P and Q are true.
(B)	P is true and Q is false.
(C)	P is false and Q is true.
(D)	Both P and Q are false.

Q.33	A bag has r red balls and b black balls. All balls are identical except for their colours. In a trial, a ball is randomly drawn from the bag, its colour is noted and the ball is placed back into the bag along with another ball of the same colour. Note that the number of balls in the bag will increase by one, after the trial. A sequence of four such trials is conducted. Which one of the following choices gives the probability of drawing a red ball in the fourth trial?
(A)	$\frac{r}{r+b}$
(B)	$\frac{r}{r+b+3}$
(C)	$\frac{r+3}{r+b+3}$
(D)	$\left(\frac{r}{r+b}\right)\left(\frac{r+1}{r+b+1}\right)\left(\frac{r+2}{r+b+2}\right)\left(\frac{r+3}{r+b+3}\right)$

Q.34	Consider the cyclic redundancy check (CRC) based error detecting scheme having the generator polynomial X^3+X+1. Suppose the message $m_4m_3m_2m_1m_0 = 11000$ is to be transmitted. Check bits $c_2c_1c_0$ are appended at the end of the message by the transmitter using the above CRC scheme. The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$. The value of the checkbit sequence $c_2c_1c_0$ is
(A)	101
(B)	110
(C)	100
(D)	111

Q.35 Consider the following ANSI C program:

```
#include <stdio.h>
#include <stdlib.h>
struct  Node{
       int value;
       struct Node *next;};
int main(){
   struct Node  *boxE, *head, *boxN; int index = 0;
   boxE = head = (struct Node *) malloc(sizeof(struct Node));
   head->value = index;
   for (index = 1; index <= 3; index++){
       boxN = (struct Node *) malloc(sizeof(struct Node));
       boxE->next =  boxN;
       boxN->value = index;
       boxE = boxN; }
   for (index = 0; index <= 3;  index++) {
       printf("Value at index  %d is %d\n", index, head->value);
       head = head->next;
       printf("Value at index %d is %d\n", index+1, head->value); } }
```

Which one of the statements below is correct about the program?

(A) Upon execution, the program creates a linked-list of five nodes.

(B) Upon execution, the program goes into an infinite loop.

(C) It has a missing `return` which will be reported as an error by the compiler.

(D) It dereferences an uninitialized pointer that may result in a run-time error.

Q.36	Consider the following two statements about regular languages: S_1: Every infinite regular language contains an undecidable language as a subset. S_2: Every finite language is regular. Which one of the following choices is correct?
(A)	Only S_1 is true.
(B)	Only S_2 is true.
(C)	Both S_1 and S_2 are true.
(D)	Neither S_1 nor S_2 is true.

Q.37	For two n-dimensional real vectors P and Q, the operation $s(P,Q)$ is defined as follows: $$s(P,Q) = \sum_{i=1}^{n} (P[i] \cdot Q[i])$$ Let $\mathcal{L}$ be a set of 10-dimensional non-zero real vectors such that for every pair of distinct vectors $P, Q \in \mathcal{L}$, $s(P,Q) = 0$. What is the maximum cardinality possible for the set $\mathcal{L}$?
(A)	9
(B)	10
(C)	11
(D)	100

Q.38	For a statement S in a program, in the context of liveness analysis, the following sets are defined: $USE(S)$: the set of variables used in S $IN(S)$: the set of variables that are live at the entry of S $OUT(S)$: the set of variables that are live at the exit of S Consider a basic block that consists of two statements, S_1 followed by S_2. Which one of the following statements is correct?
(A)	$OUT(S_1) = IN(S_2)$
(B)	$OUT(S_1) = IN(S_1) \cup USE(S_1)$
(C)	$OUT(S_1) = IN(S_2) \cup OUT(S_2)$
(D)	$OUT(S_1) = USE(S_1) \cup IN(S_2)$

Q.39	For constants $a \geq 1$ and $b > 1$, consider the following recurrence defined on the non-negative integers: $$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$$ Which one of the following options is correct about the recurrence $T(n)$?
(A)	If $f(n)$ is $n \log_2(n)$, then $T(n)$ is $\Theta(n \log_2(n))$.
(B)	If $f(n)$ is $\frac{n}{log_2(n)}$, then $T(n)$ is $\Theta(\log_2(n))$.
(C)	If $f(n)$ is $O(n^{\log_b(a)-\epsilon})$ for some $\epsilon > 0$, then $T(n)$ is $\Theta(n^{\log_b(a)})$.
(D)	If $f(n)$ is $\Theta(n^{\log_b(a)})$, then $T(n)$ is $\Theta(n^{log_b(a)})$.

Q.40 – Q.47 Multiple Select Question (MSQ), carry TWO mark each (no negative marks).

Q.40	Suppose the following functional dependencies hold on a relation U with attributes P, Q, R, S, and T: $$P \rightarrow QR$$ $$RS \rightarrow T$$ Which of the following functional dependencies can be inferred from the above functional dependencies?
(A)	$PS \rightarrow T$
(B)	$R \rightarrow T$
(C)	$P \rightarrow R$
(D)	$PS \rightarrow Q$

Q.41	For a string w, we define w^R to be the reverse of w. For example, if $w = 01101$ then $w^R = 10110$. Which of the following languages is/are context-free?
(A)	$\{wxw^Rx^R \mid w, x \in \{0,1\}^*\}$
(B)	$\{ww^Rxx^R \mid w, x \in \{0,1\}^*\}$
(C)	$\{wxw^R \mid w, x \in \{0,1\}^*\}$
(D)	$\{wxx^Rw^R \mid w, x \in \{0,1\}^*\}$

Q.42 Consider the following multi-threaded code segment (in a mix of C and pseudo-code), invoked by two processes P1 and P2, and each of the processes spawns two threads T1 and T2:

```
int x = 0;  // global
Lock L1;    // global
main() {
 create a thread to execute foo();  // Thread T1
 create a thread to execute foo();  // Thread T2
 wait for the two threads to finish execution;
 print (x);}

foo() {
 int y = 0;
 Acquire L1;
 x = x + 1;
 y = y + 1;
 Release L1;
 print (y);}
```

Which of the following statement(s) is/are correct?

(A) Both P1 and P2 will print the value of `x` as 2.

(B) At least one of P1 and P2 will print the value of `x` as 4.

(C) At least one of the threads will print the value of `y` as 2.

(D) Both T1 and T2, in both the processes, will print the value of `y` as 1.

Q.43

Consider a computer system with multiple shared resource types, with one instance per resource type. Each instance can be owned by only one process at a time. Owning and freeing of resources are done by holding a global lock (L). The following scheme is used to own a resource instance :

```
function OwnResource(Resource R)
    Acquire lock L // a global lock
    if R is available then
        Acquire R
        Release lock L
    else
        if R is owned by another process P then
            Terminate P, after releasing all resources owned by P
            Acquire R
            Restart P
            Release lock L
        end if
    end if
end function
```

Which of the following choice(s) about the above scheme is/are correct?

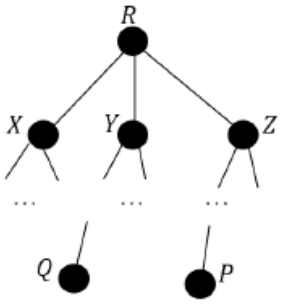
(A) The scheme ensures that deadlocks will not occur.

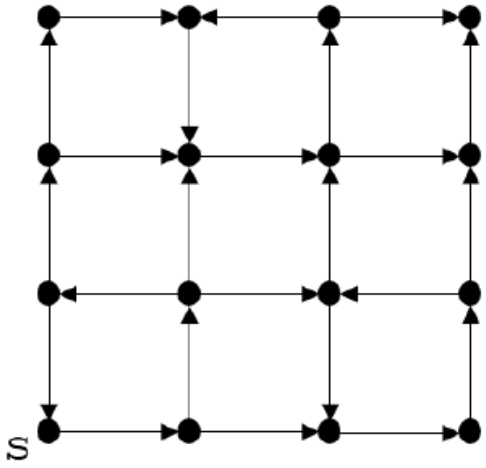
(B) The scheme may lead to live-lock.

(C) The scheme may lead to starvation.

(D) The scheme violates the mutual exclusion property.

Q.44	If the numerical value of a 2-byte unsigned integer on a little endian computer is 255 more than that on a big endian computer, which of the following choices represent(s) the unsigned integer on a little endian computer?
(A)	0x6665
(B)	0x0001
(C)	0x4243
(D)	0x0100

Q.45	Consider a computer network using the distance vector routing algorithm in its network layer. The partial topology of the network is as shown below.  The objective is to find the shortest-cost path from the router R to routers P and Q. Assume that R does not initially know the shortest routes to P and Q. Assume that R has three neighbouring routers denoted as X, Y, and Z. During one iteration, R measures its distance to its neighbours X, Y, and Z as 3, 2, and 5, respectively. Router R gets routing vectors from its neighbours that indicate that the distance to router P from routers X, Y, and Z are 7, 6, and 5, respectively. The routing vector also indicates that the distance to router Q from routers X, Y, and Z are 4, 6, and 8, respectively. Which of the following statement(s) is/are correct with respect to the new routing table of R, after updation during this iteration?
(A)	The distance from R to P will be stored as 10.
(B)	The distance from R to Q will be stored as 7.
(C)	The next hop router for a packet from R to P is Y.
(D)	The next hop router for a packet from R to Q is Z.

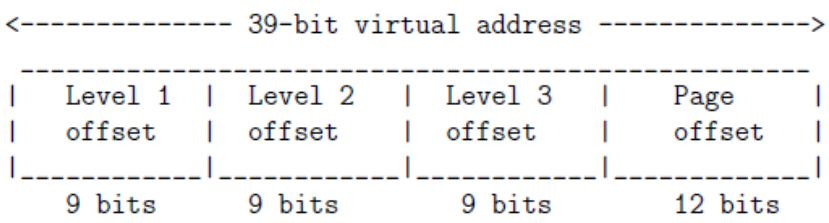
Q.46	Consider the following directed graph:  Which of the following is/are correct about the graph?
(A)	The graph does not have a topological order.
(B)	A depth-first traversal starting at vertex S classifies three directed edges as back edges.
(C)	The graph does not have a strongly connected component.
(D)	For each pair of vertices u and v, there is a directed path from u to v.

Q.47	Which of the following regular expressions represent(s) the set of all binary numbers that are divisible by three? Assume that the string ϵ is divisible by three.
(A)	$(0 + 1(01^*0)^*1)^*$
(B)	$(0 + 11 + 10(1 + 00)^*01)^*$
(C)	$(0^*(1(01^*0)^*1)^*)^*$
(D)	$(0 + 11 + 11(1 + 00)^*00)^*$

Q.48 – Q.55 Numerical Answer Type (NAT), carry TWO mark each (no negative marks).

Q.48

Consider a three-level page table to translate a 39-bit virtual address to a physical address as shown below.



The page size is 4KB (1KB = 2^{10} bytes) and page table entry size at every level is 8 bytes. A process P is currently using 2GB (1GB = 2^{30} bytes) virtual memory which is mapped to 2GB of physical memory. The minimum amount of memory required for the page table of P across all levels is _____ KB.

Q.49

Consider the following ANSI C program.

```
#include <stdio.h>

int foo(int x, int y, int q)
{
    if ((x <= 0) && (y <= 0))
        return q;
    if (x <= 0)
        return foo(x, y-q, q);
    if (y <= 0)
        return foo(x-q, y, q);
    return foo(x, y-q, q) + foo(x-q, y, q);
}

int main()
{
    int r = foo(15,15,10);
    printf("%d", r);
    return 0;
}
```

The output of the program upon execution is _____.

Q.50	Let S be a set consisting of 10 elements. The number of tuples of the form (A, B) such that A and B are subsets of S, and $A \subseteq B$ is _____.

Q.51	Consider the following augmented grammar with $\{\#, @, <, >, a, b, c\}$ as the set of terminals. $S' \rightarrow S$ $S \rightarrow S \# cS$ $S \rightarrow SS$ $S \rightarrow S@$ $S \rightarrow < S >$ $S \rightarrow a$ $S \rightarrow b$ $S \rightarrow c$ Let I_0 = CLOSURE($\{S' \rightarrow \bullet S\}$). The number of items in the set GOTO(GOTO($I_0, <$), <) is _____.

Q.52	Consider a Boolean function $f(w, x, y, z)$ such that $f(w, 0, 0, z) = 1$ $f(1, x, 1, z) = x + z$ $f(w, 1, y, z) = wz + y$ The number of literals in the minimal sum-of-products expression of f is _____.

Q.53 Consider a pipelined processor with 5 stages, Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Write Back (WB). Each stage of the pipeline, except the EX stage, takes one cycle. Assume that the ID stage merely decodes the instruction and the register read is performed in the EX stage. The EX stage takes one cycle for ADD instruction and two cycles for MUL instruction. Ignore pipeline register latencies.
Consider the following sequence of 8 instructions:

```
ADD, MUL, ADD, MUL, ADD, MUL, ADD, MUL
```

Assume that every MUL instruction is data-dependent on the ADD instruction just before it and every ADD instruction (except the first ADD) is data-dependent on the MUL instruction just before it. The *Speedup* is defined as follows:

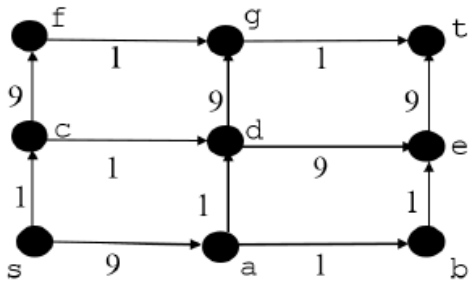
$$Speedup = \frac{\text{Execution time without operand forwarding}}{\text{Execution time with operand forwarding}}$$

The *Speedup* achieved in executing the given instruction sequence on the pipelined processor (rounded to 2 decimal places) is _____.

Q.54 Consider a network using the pure ALOHA medium access control protocol, where each frame is of length 1,000 bits. The channel transmission rate is 1 Mbps ($= 10^6$ bits per second). The aggregate number of transmissions across all the nodes (including new frame transmissions and retransmitted frames due to collisions) is modelled as a Poisson process with a rate of 1,000 frames per second. Throughput is defined as the average number of frames successfully transmitted per second. The throughput of the network (rounded to the nearest integer) is _____.

Q.55

In a directed acyclic graph with a source vertex **s**, the *quality-score* of a directed path is defined to be the product of the weights of the edges on the path. Further, for a vertex v other than **s**, the quality-score of v is defined to be the maximum among the quality-scores of all the paths from **s** to v. The quality-score of **s** is assumed to be 1.



The sum of the quality-scores of all the vertices in the graph shown above is _____.

END OF QUESTION PAPER

Graduate Aptitude Test in Engineering 2017

Question Paper Name: Computer Science and Information Technology 11th Feb 2017 Session 1
Subject Name: Computer Science and Information Technology
Duration: 180
Total Marks: 100

Organizing Institute:

Indian Institute of Technology Roorkee

Question Number : 1 **Correct : 1 Wrong : -0.33**

The statement $(\neg p) \Rightarrow (\neg q)$ is logically equivalent to which of the statements below?

I. $p \Rightarrow q$

II. $q \Rightarrow p$

III. $(\neg q) \vee p$

IV. $(\neg p) \vee q$

(A) I only (B) I and IV only (C) II only (D) II and III only

Question Number : 2 **Correct : 1 Wrong : -0.33**

Consider the first-order logic sentence $F: \forall x(\exists y R(x, y))$. Assuming non-empty logical domains, which of the sentences below are *implied* by F?

I. $\exists y(\exists x R(x, y))$

II. $\exists y(\forall x R(x, y))$

III. $\forall y(\exists x R(x, y))$

IV. $\neg\exists x(\forall y \neg R(x, y))$

(A) IV only (B) I and IV only (C) II only (D) II and III only

Question Number : 3 **Correct : 1 Wrong : -0.33**

Let $c_1,\ldots,c_n$ be scalars, not all zero, such that $\sum_{i=1}^{n} c_i a_i = 0$ where a_i are column vectors in $\mathbf{R}^n$.

Consider the set of linear equations

$$Ax = b$$

where $A=[a_1,\ldots,a_n]$ and $b=\sum_{i=1}^{n} a_i$. The set of equations has

(A) a unique solution at $x = J_n$ where J_n denotes a n-dimensional vector of all 1

(B) no solution

(C) infinitely many solutions

(D) finitely many solutions

Question Number : 4 **Correct : 1 Wrong : -0.33**

Consider the following functions from positive integers to real numbers:

$$10,\ \sqrt{n},\ n,\ \log_2 n,\ \frac{100}{n}.$$

The CORRECT arrangement of the above functions in increasing order of asymptotic complexity is:

(A) $\log_2 n,\ \frac{100}{n},\ 10,\ \sqrt{n},\ n$

(B) $\frac{100}{n},\ 10,\ \log_2 n,\ \sqrt{n},\ n$

(C) $10,\ \frac{100}{n},\ \sqrt{n},\ \log_2 n,\ n$

(D) $\frac{100}{n},\ \log_2 n,\ 10,\ \sqrt{n},\ n$

Question Number : 5 **Correct : 1 Wrong : -0.33**

Consider the following table:

Algorithms	Design Paradigms
(P) Kruskal	(i) Divide and Conquer
(Q) Quicksort	(ii) Greedy
(R) Floyd-Warshall	(iii) Dynamic Programming

Match the algorithms to the design paradigms they are based on.

(A) (P) ↔ (ii), (Q) ↔ (iii), (R) ↔ (i)

(B) (P) ↔ (iii), (Q) ↔ (i), (R) ↔ (ii)

(C) (P) ↔ (ii), (Q) ↔ (i), (R) ↔ (iii)

(D) (P) ↔ (i), (Q) ↔ (ii), (R) ↔ (iii)

Question Number : 6 **Correct : 1 Wrong : -0.33**

Let T be a binary search tree with 15 nodes. The minimum and maximum possible heights of T are:
Note: The height of a tree with a single node is 0.

(A) 4 and 15 respectively

(B) 3 and 14 respectively

(C) 4 and 14 respectively

(D) 3 and 15 respectively

Question Number : 7 **Correct : 1 Wrong : -0.33**

The n-bit fixed-point representation of an unsigned real number X uses f bits for the fraction part. Let $i = n - f$. The range of decimal values for X in this representation is

(A) 2^{-f} to 2^{i} (B) 2^{-f} to $(2^{i} - 2^{-f})$ (C) 0 to 2^{i} (D) 0 to $(2^{i} - 2^{-f})$

Question Number : 8 **Correct : 1 Wrong : -0.33**

Consider the C code fragment given below.

```
typedef struct node {
  int data;
  node* next;
} node;

void join(node* m, node* n){
  node* p = n;
  while(p->next != NULL) {
    p = p->next;
  }
  p->next = m;
}
```

Assuming that `m` and `n` point to valid `NULL`-terminated linked lists, invocation of `join` will

(A) append list `m` to the end of list `n` for all inputs.

(B) either cause a null pointer dereference or append list `m` to the end of list `n`.

(C) cause a null pointer dereference for all inputs.

(D) append list `n` to the end of list `m` for all inputs.

Question Number : 9 **Correct : 1 Wrong : -0.33**

When two 8-bit numbers $A_7 \cdots A_0$ and $B_7 \cdots B_0$ in 2's complement representation (with A_0 and B_0 as the least significant bits) are added using a **ripple-carry adder**, the sum bits obtained are $S_7 \cdots S_0$ and the carry bits are $C_7 \cdots C_0$. An overflow is said to have occurred if

(A) the carry bit C_7 is 1

(B) all the carry bits $(C_7, \cdots, C_0)$ are 1

(C) $(A_7 \cdot B_7 \cdot \overline{S_7} + \overline{A_7} \cdot \overline{B_7} \cdot S_7)$ is 1

(D) $(A_0 \cdot B_0 \cdot \overline{S_0} + \overline{A_0} \cdot \overline{B_0} \cdot S_0)$ is 1

Question Number : 10 Correct : 1 Wrong : -0.33

Consider the following context-free grammar over the alphabet $\sum = \{a, b, c\}$ with S as the start symbol:

$S \rightarrow abScT \mid abcT$
$T \rightarrow bT \mid b$

Which one of the following represents the language generated by the above grammar?

(A) $\{(ab)^n(cb)^n \mid n \geq 1\}$
(B) $\{(ab)^n cb^{m_1} cb^{m_2} \ldots cb^{m_n} \mid n, m_1, m_2, \ldots, m_n \geq 1\}$
(C) $\{(ab)^n(cb^m)^n \mid m, n \geq 1\}$
(D) $\{(ab)^n(cb^n)^m \mid m, n \geq 1\}$

Question Number : 11 Correct : 1 Wrong : -0.33

Consider the C `struct` defined below:

```
struct data {
   int marks [100];
   char grade;
   int cnumber;
};
struct data student;
```

The base address of `student` is available in register R1. The field `student.grade` can be accessed efficiently using

(A) Post-increment addressing mode, (R1)+
(B) Pre-decrement addressing mode, −(R1)
(C) Register direct addressing mode, R1
(D) Index addressing mode, X(R1), where X is an offset represented in 2's complement 16-bit representation.

Question Number : 12 **Correct : 1 Wrong : -0.33**

Consider the following intermediate program in three address code

```
p = a - b
q = p * c
p = u * v
q = p + q
```

Which one of the following corresponds to a *static single assignment* form of the above code?

(A)
```
p1 = a - b
q1 = p1 * c
p1 = u * v
q1 = p1 + q1
```

(B)
```
p3 = a - b
q4 = p3 * c
p4 = u * v
q5 = p4 + q4
```

(C)
```
p1 = a - b
q1 = p2 * c
p3 = u * v
q2 = p4 + q3
```

(D)
```
p1 = a - b
q1 = p * c
p2 = u * v
q2 = p + q
```

Question Number : 13 **Correct : 1 Wrong : -0.33**

Consider the following C code:

```
#include <stdio.h>
int *assignval(int *x, int val) {
      *x = val;
      return x;
}

void main () {
      int *x = malloc(sizeof(int));
       if(NULL == x) return;
        x = assignval(x,0);
        if(x) {
             x = (int *)malloc(sizeof(int));
             if(NULL == x) return;
              x = assignval(x,10);
        }
       printf("%d\n", *x);
       free(x);
}
```

The code suffers from which one of the following problems:

(A) compiler error as the return of `malloc` is not typecast appropriately

(B) compiler error because the comparison should be made as `x == NULL` and not as shown

(C) compiles successfully but execution may result in dangling pointer

(D) compiles successfully but execution may result in memory leak

Question Number : 14 **Correct : 1 Wrong : -0.33**

Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls `close` to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK, which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client-side TCP connection wait for the FIN from the server-side TCP?

(A) LAST-ACK
(B) TIME-WAIT
(C) FIN-WAIT-1
(D) FIN-WAIT-2

Question Number : 15 Correct : 1 Wrong : -0.33

A sender S sends a message m to receiver R, which is digitally signed by S with its private key. In this scenario, one or more of the following security violations can take place.

(I) S can launch a birthday attack to replace m with a fraudulent message.
(II) A third party attacker can launch a birthday attack to replace m with a fraudulent message.
(III) R can launch a birthday attack to replace m with a fraudulent message.

Which of the following are possible security violations?

(A) (I) and (II) only

(B) (I) only

(C) (II) only

(D) (II) and (III) only

Question Number : 16 Correct : 1 Wrong : -0.33

The following functional dependencies hold true for the relational schema $R\{V, W, X, Y, Z\}$:

$$V \rightarrow W$$
$$VW \rightarrow X$$
$$Y \rightarrow VX$$
$$Y \rightarrow Z$$

Which of the following is irreducible equivalent for this set of functional dependencies?

(A) $V \rightarrow W$, $V \rightarrow X$, $Y \rightarrow V$, $Y \rightarrow Z$

(B) $V \rightarrow W$, $W \rightarrow X$, $Y \rightarrow V$, $Y \rightarrow Z$

(C) $V \rightarrow W$, $V \rightarrow X$, $Y \rightarrow V$, $Y \rightarrow X$, $Y \rightarrow Z$

(D) $V \rightarrow W$, $W \rightarrow X$, $Y \rightarrow V$, $Y \rightarrow X$, $Y \rightarrow Z$

Question Number : 17 **Correct : 1 Wrong : -0.33**

Consider the following grammar:

$P \rightarrow xQRS$
$Q \rightarrow yz \mid z$
$R \rightarrow w \mid \varepsilon$
$S \rightarrow y$

What is FOLLOW(Q)?

(A) $\{R\}$ (B) $\{w\}$ (C) $\{w, y\}$ (D) $\{w, \$\}$

Question Number : 18 **Correct : 1 Wrong : -0.33**

Threads of a process share

(A) global variables but not heap.
(B) heap but not global variables.
(C) neither global variables nor heap.
(D) both heap and global variables.

Question Number : 19 **Correct : 1 Wrong : 0**

Let X be a Gaussian random variable with mean 0 and variance σ^2. Let $Y = max(X, 0)$ where $max(a, b)$ is the maximum of a and b. The median of Y is _____.

Question Number : 20 **Correct : 1 Wrong : 0**

Let T be a tree with 10 vertices. The sum of the degrees of all the vertices in T is _____.

Question Number : 21 Correct : 1 Wrong : 0

Consider the Karnaugh map given below, where x represents "*don't care*" and blank represents 0.

dc \ ba	00	01	11	10
00		x	x	
01	1			x
11	1			1
10		x	x	

Assume for all inputs (a, b, c, d), the respective complements ($\overline{a}, \overline{b}, \overline{c}, \overline{d}$) are also available. The above logic is implemented using 2-input NOR gates only. The minimum number of gates required is _____.

Question Number : 22 Correct : 1 Wrong : 0

Consider the language L given by the regular expression $(a+b)^* b\,(a+b)$ over the alphabet $\{a, b\}$. The smallest number of states needed in a deterministic finite-state automaton (DFA) accepting L is _____.

Question Number : 23 Correct : 1 Wrong : 0

Consider a database that has the relation schema EMP (EmpId, EmpName, and DeptName). An instance of the schema EMP and a SQL query on it are given below.

EMP		
EmpId	EmpName	DeptName
1	XYA	AA
2	XYB	AA
3	XYC	AA
4	XYD	AA
5	XYE	AB
6	XYF	AB
7	XYG	AB
8	XYH	AC
9	XYI	AC
10	XYJ	AC
11	XYK	AD
12	XYL	AD
13	XYM	AE

```
SELECT AVG(EC.Num)
FROM EC
WHERE (DeptName, Num) IN
      (SELECT DeptName, COUNT(EmpId) AS
                                  EC(DeptName, Num)
       FROM EMP
       GROUP BY DeptName)
```

The output of executing the SQL query is _____ .

Question Number : 24 Correct : 1 Wrong : 0

Consider the following CPU processes with arrival times (in milliseconds) and length of CPU bursts (in milliseconds) as given below :

Process	Arrival time	Burst time
P1	0	7
P2	3	3
P3	5	5
P4	6	2

If the pre-emptive shortest remaining time first scheduling algorithm is used to schedule the processes, then the average waiting time across all processes is _____ milliseconds.

Question Number : 25 Correct : 1 Wrong : 0

Consider a two-level cache hierarchy with L1 and L2 caches. An application incurs **1.4** memory accesses per instruction on average. For this application, the miss rate of L1 cache is **0.1**; the L2 cache experiences, on average, 7 misses per 1000 instructions. The miss rate of L2 expressed correct to two decimal places is _____.

Question Number : 26 Correct : 2 Wrong : -0.66

Let $G = (V, E)$ be *any* connected undirected edge-weighted graph. The weights of the edges in E are positive and distinct. Consider the following statements:

(I) Minimum Spanning Tree of G is always unique.
(II) Shortest path between any two vertices of G is always unique.

Which of the above statements is/are necessarily true?

(A) (I) only
(B) (II) only
(C) both (I) and (II)
(D) neither (I) nor (II)

Question Number : 27 Correct : 2 Wrong : -0.66

A multithreaded program P executes with x number of threads and uses y number of locks for ensuring mutual exclusion while operating on shared memory locations. All locks in the program are *non-reentrant*, i.e., if a thread holds a lock l, then it cannot re-acquire lock l without releasing it. If a thread is unable to acquire a lock, it blocks until the lock becomes available. The *minimum* value of x and the *minimum* value of y together for which execution of P can result in a deadlock are:

(A) $x = 1, y = 2$
(B) $x = 2, y = 1$
(C) $x = 2, y = 2$
(D) $x = 1, y = 1$

Question Number : 28 **Correct : 2 Wrong : -0.66**

The value of $\lim_{x\to 1}\frac{x^7-2x^5+1}{x^3-3x^2+2}$

(A) is 0
(B) is -1
(C) is 1
(D) does not exist

Question Number : 29 **Correct : 2 Wrong : -0.66**

Let p, q, and r be propositions and the expression $(p \to q) \to r$ be a contradiction. Then, the expression $(r \to p) \to q$ is

(A) a tautology.
(B) a contradiction.
(C) always TRUE when p is FALSE.
(D) always TRUE when q is TRUE.

Question Number : 30 **Correct : 2 Wrong : -0.66**

Let u and v be two vectors in $\mathbf{R}^2$ whose Euclidean norms satisfy $\|u\| = 2\|v\|$. What is the value of α such that $w = u + \alpha v$ bisects the angle between u and v?

(A) 2
(B) 1/2
(C) 1
(D) $-1/2$

Question Number : 31 **Correct : 2 Wrong : -0.66**

Let A be $n \times n$ real valued square symmetric matrix of rank 2 with $\sum_{i=1}^{n}\sum_{j=1}^{n} A_{ij}^2 = 50$. Consider the following statements.

(I) One eigenvalue must be in $[-5, 5]$
(II) The eigenvalue with the largest magnitude must be strictly greater than 5

Which of the above statements about eigenvalues of A is/are necessarily CORRECT?

(A) Both (I) and (II)
(B) (I) only
(C) (II) only
(D) Neither (I) nor (II)

Question Number : 32 Correct : 2 Wrong : -0.66

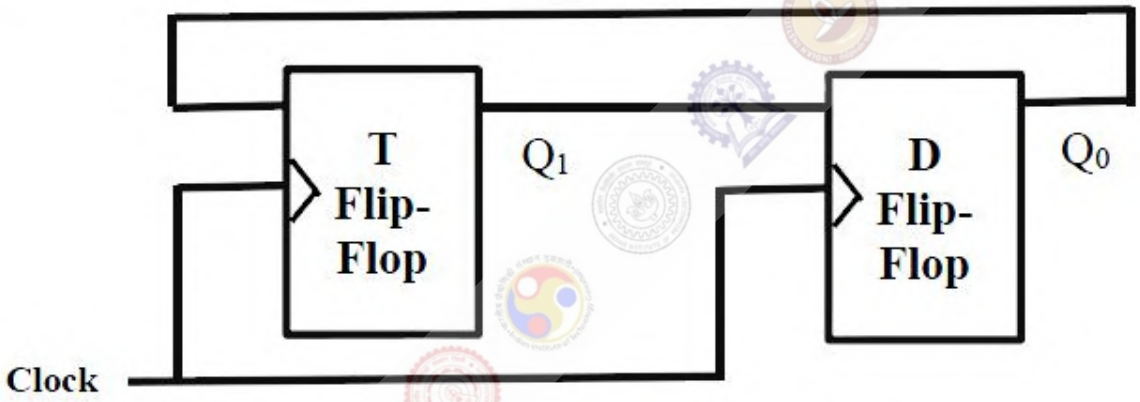
A computer network uses polynomials over *GF(2)* for error checking with 8 bits as information bits and uses $x^3 + x + 1$ as the generator polynomial to generate the check bits. In this network, the message 01011011 is transmitted as

(A) 01011011010 (B) 01011011011

(C) 01011011101 (D) 01011011100

Question Number : 33 Correct : 2 Wrong : -0.66

Consider a combination of T and D flip-flops connected as shown below. The output of the D flip-flop is connected to the input of the T flip-flop and the output of the T flip-flop is connected to the input of the D flip-flop.



Initially, both Q_0 and Q_1 are set to 1 (before the 1st clock cycle). The outputs

(A) $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 00 respectively

(B) $Q_1 Q_0$ after the 3rd cycle are 11 and after the 4th cycle are 01 respectively

(C) $Q_1 Q_0$ after the 3rd cycle are 00 and after the 4th cycle are 11 respectively

(D) $Q_1 Q_0$ after the 3rd cycle are 01 and after the 4th cycle are 01 respectively

Question Number : 34 Correct : 2 Wrong : -0.66

If G is a grammar with productions

$$S \rightarrow SaS \mid aSb \mid bSa \mid SS \mid \epsilon$$

where S is the start variable, then which one of the following strings is not generated by G?

(A) *abab* (B) *aaab* (C) *abbaa* (D) *babba*

Question Number : 35 **Correct : 2 Wrong : -0.66**

Consider the following two functions.

```
void fun1(int n) {
        if(n == 0) return;
        printf("%d", n);
        fun2(n - 2);
        printf("%d", n);
}
```

```
void fun2(int n) {
        if(n == 0) return;
        printf("%d", n);
        fun1(++n);
        printf("%d", n);
}
```

The output printed when `fun1(5)` is called is

(A) 53423122233445
(B) 53423120112233
(C) 53423122132435
(D) 53423120213243

Question Number : 36 **Correct : 2 Wrong : -0.66**

Consider the C functions `foo` and `bar` given below:

```
int foo(int val) {
   int x = 0;
   while(val > 0) {
      x = x + foo(val--);
   }
   return val;
}

int bar(int val) {
   int x = 0;
   while(val > 0) {
       x = x + bar(val-1);
   }
   return val;
}
```

Invocations of `foo(3)` and `bar(3)` will result in :

(A) Return of 6 and 6 respectively.
(B) Infinite loop and abnormal termination respectively.
(C) Abnormal termination and infinite loop respectively.
(D) Both terminating abnormally.

Question Number : 37 Correct : 2 Wrong : -0.66

Consider the context-free grammars over the alphabet $\{a,b,c\}$ given below. S and T are non-terminals.

$G_1: S \rightarrow aSb|T,\ T \rightarrow cT|\epsilon$

$G_2: S \rightarrow bSa|T,\ T \rightarrow cT|\epsilon$

The language $L(G_1) \cap L(G_2)$ is

(A) Finite.

(B) Not finite but regular.

(C) Context-Free but not regular.

(D) Recursive but not context-free.

Question Number : 38 Correct : 2 Wrong : -0.66

Consider the following languages over the alphabet $\sum = \{a,b,c\}$.
Let $L_1 = \{ a^n b^n c^m \mid m,n \geq 0\}$ and $L_2 = \{ a^m b^n c^n \mid m,n \geq 0 \}$.

Which of the following are context-free languages?

I. $L_1 \cup L_2$
II. $L_1 \cap L_2$

(A) I only

(B) II only

(C) I and II

(D) Neither I nor II

Question Number : 39 Correct : 2 Wrong : -0.66

Let A and B be finite alphabets and let $\#$ be a symbol outside both A and B. Let f be a total function from A^* to B^*. We say f is *computable* if there exists a Turing machine M which given an input x in A^*, always halts with $f(x)$ on its tape. Let L_f denote the language $\{x\#f(x) \mid x \in A^*\}$. Which of the following statements is true:

(A) f is computable if and only if L_f is recursive.

(B) f is computable if and only if L_f is recursively enumerable.

(C) If f is computable then L_f is recursive, but not conversely.

(D) If f is computable then L_f is recursively enumerable, but not conversely.

Question Number : 40 Correct : 2 Wrong : -0.66

Recall that Belady's anomaly is that the page-fault rate may *increase* as the number of allocated frames increases. Now, consider the following statements:

S1: *Random page replacement* algorithm (where a page chosen at random is replaced) suffers from Belady's anomaly
S2: *LRU page replacement* algorithm suffers from Belady's anomaly

Which of the following is CORRECT?

(A) S1 is true, S2 is true
(B) S1 is true, S2 is false
(C) S1 is false, S2 is true
(D) S1 is false, S2 is false

Question Number : 41 Correct : 2 Wrong : -0.66

Consider a database that has the relation schemas EMP(EmpId, EmpName, DeptId), and DEPT(DeptName, DeptId). Note that the DeptId can be permitted to be NULL in the relation EMP. Consider the following queries on the database expressed in tuple relational calculus.

(I) $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \forall v \in \text{DEPT}(t[\text{DeptId}] \neq v[\text{DeptId}]))\}$

(II) $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT}(t[\text{DeptId}] \neq v[\text{DeptId}]))\}$

(III) $\{t \mid \exists u \in \text{EMP}(t[\text{EmpName}] = u[\text{EmpName}] \wedge \exists v \in \text{DEPT}(t[\text{DeptId}] = v[\text{DeptId}]))\}$

Which of the above queries are safe?

(A) (I) and (II) only
(B) (I) and (III) only
(C) (II) and (III) only
(D) (I), (II) and (III)

Question Number : 42 Correct : 2 Wrong : -0.66

In a database system, unique timestamps are assigned to each transaction using Lamport's logical clock. Let $TS(T_1)$ and $TS(T_2)$ be the timestamps of transactions T_1 and T_2 respectively. Besides, T_1 holds a lock on the resource R, and T_2 has requested a conflicting lock on the same resource R. The following algorithm is used to prevent deadlocks in the database system assuming that a killed transaction is restarted with the same timestamp.

if $TS(T_2) < TS(T_1)$ then
    T_1 is killed
else T_2 waits.

Assume any transaction that is not killed terminates eventually. Which of the following is TRUE about the database system that uses the above algorithm to prevent deadlocks?

(A) The database system is both deadlock-free and starvation-free.

(B) The database system is deadlock-free, but not starvation-free.

(C) The database system is starvation-free, but not deadlock-free.

(D) The database system is neither deadlock-free nor starvation-free.

Question Number : 43 Correct : 2 Wrong : 0

Consider the following grammar:

```
stmt   -> if expr then expr else expr; stmt | ò
expr   -> term relop term | term
term   -> id | number
id     -> a | b | c
number -> [0-9]
```

where **relop** is a relational operator (e.g., <, >, ...). ò refers to the empty statement, and **if**, **then**, **else** are terminals.

Consider a program P following the above grammar containing ten **if** terminals. The number of control flow paths in P is _____. For example, the program

```
if e1 then e2 else e3
```

has 2 control flow paths, $e_1 \rightarrow e_2$ and $e_1 \rightarrow e_3$.

Question Number : 44 **Correct : 2 Wrong : 0**

In a RSA cryptosystem, a participant A uses two prime numbers $p = 13$ and $q = 17$ to generate her public and private keys. If the public key of A is 35, then the private key of A is _____.

Question Number : 45 **Correct : 2 Wrong : 0**

The values of parameters for the Stop-and-Wait ARQ protocol are as given below:

Bit rate of the transmission channel = 1 Mbps.
Propagation delay from sender to receiver = 0.75 ms.
Time to process a frame = 0.25 ms.
Number of bytes in the information frame = 1980.
Number of bytes in the acknowledge frame = 20.
Number of overhead bytes in the information frame = 20.

Assume that there are no transmission errors. Then, the transmission efficiency (expressed in percentage) of the Stop-and-Wait ARQ protocol for the above parameters is _____ (correct to 2 decimal places).

Question Number : 46 Correct : 2 Wrong : 0

Consider a database that has the relation schema CR(StudentName, CourseName). An instance of the schema CR is as given below.

CR	
StudentName	CourseName
SA	CA
SA	CB
SA	CC
SB	CB
SB	CC
SC	CA
SC	CB
SC	CC
SD	CA
SD	CB
SD	CC
SD	CD
SE	CD
SE	CA
SE	CB
SF	CA
SF	CB
SF	CC

The following query is made on the database.

$$T1 \leftarrow \pi_{CourseName}(\sigma_{StudentName='SA'}(\text{CR}))$$
$$T2 \leftarrow CR \div T1$$

The number of rows in $T2$ is _____.

Question Number : 47 Correct : 2 Wrong : 0

The number of integers between 1 and 500 (both inclusive) that are divisible by 3 or 5 or 7 is _____.

Question Number : 48 Correct : 2 Wrong : 0

Let A be an array of 31 numbers consisting of a sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i such that $A[i]$ is 1 by probing the minimum number of locations in A. The *worst case* number of probes performed by an *optimal* algorithm is _____.

Question Number : 49 Correct : 2 Wrong : 0

Consider a RISC machine where each instruction is exactly 4 bytes long. Conditional and unconditional branch instructions use PC-relative addressing mode with Offset specified in bytes to the target location of the branch instruction. Further the Offset is always with respect to the address of the next instruction in the program sequence. Consider the following instruction sequence

```
Instr. No.        Instruction
     i    :     add   R2, R3, R4
     i+1  :     sub   R5, R6, R7
     i+2  :     cmp   R1, R9, R10
     i+3  :     beq   R1,  Offset
```

If the target of the branch instruction is `i`, then the decimal value of the `Offset` is _____ .

Question Number : 50 Correct : 2 Wrong : 0

Instruction execution in a processor is divided into 5 stages, *Instruction Fetch* (IF), *Instruction Decode* (ID), *Operand Fetch* (OF), *Execute* (EX), and *Write Back* (WB). These stages take **5, 4, 20, 10,** and **3 nanoseconds (ns)** respectively. A pipelined implementation of the processor requires buffering between each pair of consecutive stages with a delay of **2 ns**. Two pipelined implementations of the processor are contemplated:

(i) a naive pipeline implementation (NP) with 5 stages and
(ii) an efficient pipeline (EP) where the OF stage is divided into stages OF1 and OF2 with execution times of **12 ns** and **8 ns** respectively.

The speedup (correct to two decimal places) achieved by EP over NP in executing 20 independent instructions with no hazards is _____ .

Question Number : 51 Correct : 2 Wrong : 0

Consider a 2-way set associative cache with 256 blocks and uses LRU replacement. Initially the cache is empty. Conflict misses are those misses which occur due to contention of multiple blocks for the same cache set. Compulsory misses occur due to first time access to the block. The following sequence of accesses to memory blocks

```
(0,128,256,128,0,128,256,128,1,129,257,129,1,129,257,129)
```

is repeated 10 times. The number of *conflict misses* experienced by the cache is _____ .

Question Number : 52 Correct : 2 Wrong : 0

Consider the expression $(a-1)*(((b+c)/3)+d\ ))$. Let X be the minimum number of registers required by an *optimal* code generation (without any register spill) algorithm for a load/store architecture, in which *(i) only load and store instructions can have memory operands* and *(ii) arithmetic instructions can have only register or immediate operands.* The value of X is _____ .

Question Number : 53 Correct : 2 Wrong : 0

Consider the following C program.

```
#include <stdio.h>
#include <string.h>

void printlength(char *s, char *t) {
   unsigned int c = 0;
   int len = ((strlen(s) - strlen(t)) > c) ? strlen(s): strlen(t);
   printf("%d\n", len);
}

void main() {
   char *x = "abc";
   char *y = "defgh";
   printlength(x,y);
}
```

Recall that `strlen` is defined in `string.h` as returning a value of type `size_t`, which is an `unsigned int`. The output of the program is _____.

Question Number : 54 Correct : 2 Wrong : 0

A cache memory unit with capacity of N words and block size of B words is to be designed. If it is designed as a direct mapped cache, the length of the TAG field is 10 bits. If the cache unit is now designed as a 16-way set-associative cache, the length of the TAG field is _____ bits.

Question Number : 55 Correct : 2 Wrong : 0

The output of executing the following C program is _____ .

```
#include <stdio.h>

int total(int v) {
   static int count = 0;
   while(v) {
      count += v&1;
      v >>= 1;
   }
   return count;
}

void main() {
   static int x = 0;
   int i = 5;
   for(; i > 0; i--) {
      x = x + total(i);
   }
   printf("%d\n", x);
}
```

Question Number : 56 **Correct : 1 Wrong : -0.33**

After Rajendra Chola returned from his voyage to Indonesia, he _____ to visit the temple in Thanjavur.

(A) was wishing (B) is wishing (C) wished (D) had wished

Question Number : 57 **Correct : 1 Wrong : -0.33**

Research in the workplace reveals that people work for many reasons _____.

(A) money beside (B) beside money (C) money besides (D) besides money

Question Number : 58 **Correct : 1 Wrong : -0.33**

Rahul, Murali, Srinivas and Arul are seated around a square table. Rahul is sitting to the left of Murali. Srinivas is sitting to the right of Arul. Which of the following pairs are seated opposite each other?

(A) Rahul and Murali
(B) Srinivas and Arul
(C) Srinivas and Murali
(D) Srinivas and Rahul

Question Number : 59 **Correct : 1 Wrong : -0.33**

Find the smallest number y such that $y \times 162$ is a perfect cube.

(A) 24 (B) 27 (C) 32 (D) 36

Question Number : 60 **Correct : 1 Wrong : -0.33**

The probability that a k-digit number does NOT contain the digits 0, 5, or 9 is

(A) 0.3^k (B) 0.6^k (C) 0.7^k (D) 0.9^k

Question Number : 61 **Correct : 2 Wrong : -0.66**

"The hold of the nationalist imagination on our colonial past is such that anything inadequately or improperly nationalist is just not history."

Which of the following statements best reflects the author's opinion?

(A) Nationalists are highly imaginative.
(B) History is viewed through the filter of nationalism.
(C) Our colonial past never happened.
(D) Nationalism has to be both adequately and properly imagined.

Question Number : 62 **Correct : 2 Wrong : -0.66**

Six people are seated around a circular table. There are at least two men and two women. There are at least three right-handed persons. Every woman has a left-handed person to her immediate right. None of the women are right-handed. The number of women at the table is

(A) 2 (B) 3 (C) 4 (D) Cannot be determined

Question Number : 63 **Correct : 2 Wrong : -0.66**

The expression $\frac{(x+y)-|x-y|}{2}$ is equal to

(A) the maximum of x and y
(B) the minimum of x and y
(C) 1
(D) none of the above

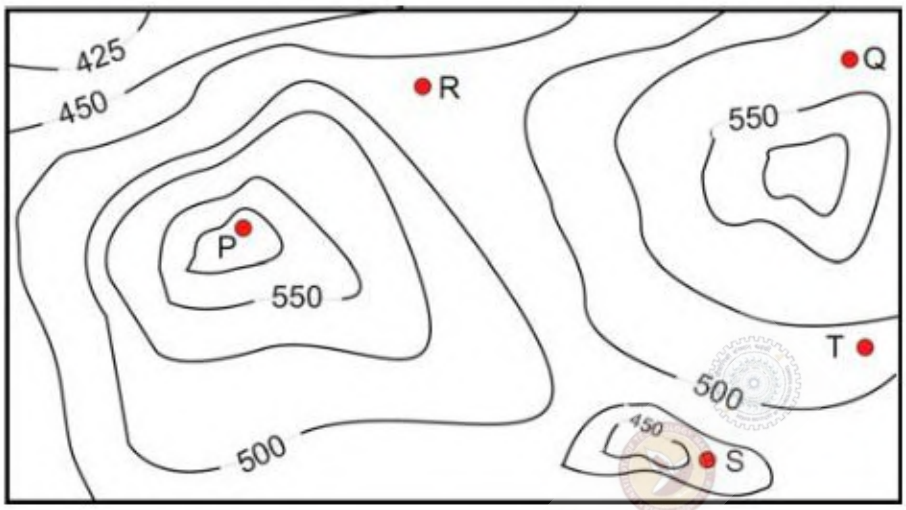
Question Number : 64 **Correct : 2 Wrong : -0.66**

Arun, Gulab, Neel and Shweta must choose one shirt each from a pile of four shirts coloured red, pink, blue and white respectively. Arun dislikes the colour red and Shweta dislikes the colour white. Gulab and Neel like all the colours. In how many different ways can they choose the shirts so that no one has a shirt with a colour he or she dislikes?

(A) 21 (B) 18 (C) 16 (D) 14

Question Number : 65 **Correct : 2 Wrong : -0.66**

A contour line joins locations having the same height above the mean sea level. The following is a contour plot of a geographical region. Contour lines are shown at 25 m intervals in this plot. If in a flood, the water level rises to 525 m, which of the villages P, Q, R, S, T get submerged?



(A) P, Q (B) P, Q, T (C) R, S, T (D) Q, R, S